

Advanced Mathematics A

Lecture Notes, Volume I

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知之者不如好之者，好之者不如乐之者 —《论语 · 雍也》

Preface

These lecture notes were prepared for the course *Advanced Mathematics A* at Peking University. They are intended primarily for students taking the course, but they may also serve as a companion for review and independent study. The aim is to present the main ideas of calculus as a coherent mathematical subject, with careful definitions, meaningful examples, and enough detail to support work beyond what can be covered in lectures.

Relationship to the Peking University Textbook

The notes stand in continuity with the third edition of Peking University's *Advanced Mathematics* (《高等数学（第三版）上册》, 李忠周建莹编著) textbook. They inherit its central subject matter, its commitment to a substantial calculus course, and many of the standard concepts and methods that generations of students have learned from it. The textbook remains an essential course resource and should be read alongside these notes.

At the same time, this text is not a line-by-line rewriting of the textbook. Some topics are reorganized to make their logical dependencies more visible; some proofs are expanded to clarify their central ideas; and additional examples, counterexamples, questions, and method summaries are included when they may improve understanding. Certain optional extensions are also added to connect elementary calculus with ideas that students will encounter later. The choice and emphasis of material therefore reflect the needs of this particular course as well as the author's own teaching experience.

A Work in Progress

Although considerable care has been taken in the preparation of these notes, their mathematical accuracy, exposition, and pedagogical design remain open to improvement. Errors, unclear arguments, infelicitous examples, and uneven choices of emphasis may still be present. Readers are warmly invited to send the author corrections, suggestions, and comments. Such feedback is valuable not only for improving later versions of the notes but also for improving the course itself.

Use and Circulation

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The author acknowledges the ElegantLaTeX template (https://github.com/ElegantLaTeX/), upon which the typesetting of these lecture notes is based.

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Introduction

Course Objectives

This course is designed to provide students with a solid foundation in the fundamental concepts, theories, and methods of calculus, while developing the ability to understand and use mathematical ideas rigorously and effectively. Students should learn not only how to carry out calculations, but also why the definitions, theorems, and methods of calculus take the forms that they do.

Throughout the course, students are expected to understand the principal ideas and concepts, master essential methods and techniques, recognize the logical structure underlying the theory, and apply mathematical knowledge to a range of problems. Computational fluency is important, but it should be supported by conceptual understanding: a correct calculation is most useful when one also knows its hypotheses, its meaning, and its limitations.

Through systematic study and practice, students should gradually develop mathematical intuition, rigorous logical reasoning, and the ability to think abstractly. They should also strengthen their capacity to formulate, analyze, and solve problems using appropriate mathematical language and tools. Ultimately, the course aims to build the mathematical foundation and analytical habits needed for further study in mathematics, science, engineering, and related disciplines.

Advice for Beginning Students

University mathematics asks for a more active kind of learning than simply following a worked calculation. The following habits will make the course more productive.

- **Be an independent learner.** Take responsibility for your own learning and do not rely solely on what is presented in class. Learn to identify what you do not yet understand, consult appropriate resources, and return to a difficult point more than once.
- **Read the textbook and the notes actively.** The textbook is an essential course resource. Assigned readings and materials remain part of the course whether or not every detail is discussed in lectures. When reading, pause to reconstruct arguments, check examples, and write down questions.
- **Do not treat lectures as an examination checklist.** Lectures and examinations are guided by the same syllabus and learning objectives, but they need not contain exactly the same material. Some ideas discussed in class may not appear on an examination, and a problem may require course material that was not worked out explicitly in class.
- **Prepare from the whole course, not from predictions.** When reviewing, pay attention to the syllabus, textbook, lecture notes, assigned work, and stated learning objectives. Trying to infer an examination solely from a list of lecture examples gives a narrow and unreliable picture of the subject.
- **Study consistently throughout the semester.** Mathematical understanding is cumulative. Regular reading, review, problem solving, and independent reflection are more effective than a short period of intensive preparation immediately before an assessment.
- **Interrogate the basic concepts.** Ask how a definition arises, what problem it resolves, why each condition

is present, and what the concept is really designed to capture. Being able to repeat a definition is not the same as understanding it.

- **Learn from examples and counterexamples.** A good example shows how a concept works; a counterexample shows where an apparently plausible statement fails. Use both positive and negative examples to test your understanding and to distinguish hypotheses from conclusions.
- **Practise with purpose.** Technique requires training, and solving exercises is indispensable. Nevertheless, indiscriminately accumulating difficult, unusual, or highly specialized problems is not the goal. After solving a problem, ask which idea made the solution possible, what alternatives were available, and how the method might apply elsewhere. Mathematical ideas and habits of thought are more fundamental than the number of problems completed.
- **Write mathematics precisely.** A solution should communicate its assumptions, reasoning, and conclusion. Practise stating definitions accurately, citing hypotheses when a theorem is used, and explaining the decisive step rather than presenting only a chain of formulas.

A Common Path to Mathematics

It is normal for serious mathematics to feel difficult. A definition may seem artificial on first reading, and a proof may remain opaque until one has examined several examples or attempted to reconstruct it independently. Temporary confusion is not evidence that one is incapable of learning the subject; it is often the stage at which deeper understanding begins.

Do not measure progress only by speed. Learn to remain with a question, to break it into smaller parts, to test simple cases, and to seek help after a genuine attempt. Over time, ideas that initially appeared isolated will begin to support one another. The goal is not to avoid ever becoming stuck, but to develop the patience, judgment, and mathematical tools needed to move forward.

Calculus rewards sustained attention. If you read thoughtfully, practise regularly, and keep asking why the theory works, you will gain more than a collection of techniques: you will acquire a language and a way of thinking that will remain useful far beyond this course.

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Chapter 1 Functions and Limits

Calculus studies change, but change can be discussed precisely only after its basic objects and limiting processes have been fixed. We therefore begin with the real number system, pass to functions as rules relating variable quantities, and then develop the limit of a sequence as our first rigorous limiting notion. Function limits lead to continuity, whose local definition yields global conclusions on closed bounded intervals.

1.1 Real Numbers and Basic Inequalities

知识点概要

- *Review rational and irrational numbers and understand why calculus needs the full real number system.*
- *Interpret completeness as the property that prevents gaps in the number line.*
- *Use intervals and neighbourhoods to describe location and distance on the real line.*
- *Use the Archimedean property and density to choose integers and rational approximations.*
- *Control errors using absolute values, AM–GM, Bernoulli's inequality, and square-root estimates.*

Calculus repeatedly asks what happens after an approximation is refined without end. For example, the decimal approximations

$$1.4, \quad 1.41, \quad 1.414, \quad 1.4142, \quad \dots$$

approach the length of the diagonal of a unit square. If our number system contained only rational numbers, that limiting length would be missing. The real numbers provide a number system in which such limiting processes can be carried out without falling into a gap.

1.1.1 From rational numbers to real numbers

Throughout these notes, the natural numbers (自然数) are

$$\mathbb{N} = \{0, 1, 2, 3, \dots\},$$

and the positive integers are denoted by $\mathbb{N}_{>0} = \{1, 2, 3, \dots\}$. We also use $\mathbb{Z}$ for the integers (整数) and $\mathbb{R}$ for the real numbers (实数).

Definition 1.1

A real number x is rational (有理数) *if*

$$x = \frac{p}{q}$$

for some $p \in \mathbb{Z}$ and $q \in \mathbb{Z} \setminus \{0\}$. The set of all rational numbers is $\mathbb{Q}$. A real number that is not rational is irrational (无理数)*; the set of irrational numbers is $\mathbb{R} \setminus \mathbb{Q}$.* ♣

A rational number has a terminating or eventually periodic decimal expansion, and conversely every terminating or eventually periodic decimal represents a rational number. Irrational numbers have nonperiodic infinite decimal expansions. This decimal description is useful for intuition, but the decisive distinction is whether the number can be represented as a ratio of integers.

Proposition 1.1

If p is a prime number, then $\sqrt{p}$ is irrational.

♠

> ***Proof Strategy.*** We take *Euclid's lemma* (欧几里得引理) as known prerequisite knowledge: if p is prime and $a, b \in \mathbb{Z}$, then
>
> $$p \mid ab \quad \Longrightarrow \quad p \mid a \text{ or } p \mid b.$$
>
> Assume that $\sqrt{p} = m/n$ is a fraction in lowest terms. After squaring, apply the lemma twice to force p to divide both m and n, contradicting the lowest-terms assumption.

Proof Suppose, to the contrary, that

$$\sqrt{p} = \frac{m}{n},$$

where $m, n \in \mathbb{N}_{>0}$ and m, n have no common factor greater than 1. Then

$$m^2 = pn^2.$$

Thus $p \mid m^2$. By Euclid's lemma applied to $m^2 = m \cdot m$, we have $p \mid m$, so $m = pk$ for some $k \in \mathbb{N}_{>0}$. Substitution gives

$$p^2k^2 = pn^2,$$

hence $n^2 = pk^2$. By Euclid's lemma again, applied to $n^2 = n \cdot n$, we obtain $p \mid n$. This contradicts the assumption that m and n have no common factor greater than 1. Consequently, $\sqrt{p}$ is irrational. □

In particular, $\sqrt{2}$ and $\sqrt{3}$ are irrational. The essential step is the contradiction with coprimality, not the particular prime chosen.

Example 1.1 A sum of two irrational numbers may require a new argument Prove that $\sqrt{2} + \sqrt{3}$ is irrational.

> ***Idea.*** The sum of two irrational numbers need not be irrational, so their irrationality alone is insufficient. Assume the sum is rational, square it, and isolate $\sqrt{6}$.

Proof Suppose that $r = \sqrt{2} + \sqrt{3}$ is rational. Then

$$r^2 = 5 + 2\sqrt{6},$$

so

$$\sqrt{6} = \frac{r^2 - 5}{2}$$

would also be rational. But $\sqrt{6}$ is irrational by the same lowest-terms argument: if $\sqrt{6} = m/n$ in lowest terms, then $m^2 = 6n^2$ forces both 2 and 3 to divide m, hence $6 \mid m$; substitution then forces $6 \mid n$, a contradiction. Therefore $\sqrt{2} + \sqrt{3}$ is irrational. □

The rational numbers are closed under addition, subtraction, multiplication, and division by a nonzero number. The real numbers have the same algebraic closure and, in addition, carry an order compatible with these operations. What distinguishes $\mathbb{R}$ from $\mathbb{Q}$ for calculus is a further property: completeness.

1.1.2 Order and completeness

The real numbers form an *ordered field* (有序域). In particular, the usual algebraic laws hold, exactly one of $a < b$, $a = b$, and $a > b$ holds for any $a, b \in \mathbb{R}$, and inequalities are compatible with addition and multiplication by positive numbers.

Completeness is best understood through bounded sets.

Definition 1.2

Let $E \subseteq \mathbb{R}$ be nonempty. A number $M \in \mathbb{R}$ is an upper bound(上界) *of E if*

$$x \le M$$

for every $x \in E$. The set E is bounded above(有上界) *if it has an upper bound. A number $s \in \mathbb{R}$ is the* supremum(上确界) *or* least upper bound *of E if*

(i) s is an upper bound of E;

(ii) every upper bound M of E satisfies $s \le M$.

It is denoted by $s = \sup E$. A lower bound(下界) *is a number m satisfying $m \le x$ for every $x \in E$. The set is* bounded below(有下界) *if such an m exists. Its* infimum(下确界)*, denoted by $\inf E$, is its greatest lower bound. A set is* bounded(有界) *if it is bounded both above and below.* ♣

Example 1.2 A supremum need not be attained The set $E = (0, 1)$ has $\inf E = 0$ and $\sup E = 1$, but neither boundary belongs to E. In contrast, $F = (0, 1]$ has the same infimum and supremum, and 1 is also its *maximum*(最大值). A maximum is an upper bound that belongs to the set; a supremum need not belong to it. The corresponding distinction holds between an infimum and a *minimum*(最小值).

Proposition 1.2

Let E be nonempty and bounded above. A number s equals $\sup E$ if and only if it is an upper bound and, for every $\varepsilon > 0$, some $x \in E$ satisfies

$$s - \varepsilon < x \le s.$$

♠

Proof If $s = \sup E$, then $s - \varepsilon$ cannot be an upper bound, so such an x exists. Conversely, suppose the stated approximation property holds. Any $M < s$ fails to be an upper bound: take $\varepsilon = s - M$ to obtain $x > M$. Thus no upper bound is smaller than s, which is the definition of $\sup E$. □

Axiom 1.1 (Least-Upper-Bound Property)

Every nonempty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$. ♡

We regard the least-upper-bound property(确界原理) as a fundamental property of $\mathbb{R}$, not as a theorem to be proved from algebra alone. It is one precise formulation of the *completeness*(完备性) of $\mathbb{R}$. It rules out gaps: whenever a nonempty set has points only up to some finite height, there is a real number marking its least possible upper boundary. Other completeness principles, such as the convergence of bounded monotone sequences and the Cauchy criterion, are equivalent to it. We shall derive the monotone sequence theorem in Section 1.3.

> ***Comment.*** The rational numbers do not satisfy this property when they are considered by themselves. For example,
>
> $$E = \{q \in \mathbb{Q} \mid q^2 < 2\}$$
>
> is nonempty and bounded above in $\mathbb{Q}$, but it has no rational least upper bound. Its missing boundary is $\sqrt{2}$.

Completeness also implies the Archimedean property(阿基米德性质): the positive integers eventually exceed every prescribed real number.

Proposition 1.3 (Archimedean Property)

For every $x \in \mathbb{R}$, there exists $n \in \mathbb{N}_{>0}$ such that $n > x$. Equivalently, for every $\varepsilon > 0$, there exists $n \in \mathbb{N}_{>0}$ such that $1/n < \varepsilon$.

> ***Idea.*** If the positive integers had a real upper bound, completeness would provide a least one. But moving one integer step forward would cross that boundary.

Proof Suppose that $\mathbb{N}_{>0}$ were bounded above, and let $s = \sup \mathbb{N}_{>0}$. The number $s - 1$ is not an upper bound, so there exists $n \in \mathbb{N}_{>0}$ with $n > s - 1$. Then $n + 1 \in \mathbb{N}_{>0}$ and $n + 1 > s$, contradicting the fact that s is an upper bound. Thus $\mathbb{N}_{>0}$ is not bounded above.

For the equivalent formulation, apply the first statement to $x = 1/\varepsilon$. Then $n > 1/\varepsilon$, and hence $1/n < \varepsilon$. □

The following density property (稠密性) says that the real line contains rational and irrational numbers at every scale.

Proposition 1.4 (Density of Rational and Irrational Numbers)

If $a, b \in \mathbb{R}$ and $a < b$, then there exist $r \in \mathbb{Q}$ and $u \in \mathbb{R} \setminus \mathbb{Q}$ such that

$$a < r < b, \qquad a < u < b.$$

Proof By Proposition 1.3, choose $n \in \mathbb{N}_{>0}$ such that

$$n(b - a) > 1.$$

Let

$$m = \lfloor na \rfloor + 1.$$

Here $\lfloor t \rfloor$ denotes the greatest integer not exceeding t. Such an integer exists by the Archimedean property and the well-ordering of the positive integers. Then $na < m \leq na + 1 < nb$, so

$$a < \frac{m}{n} < b.$$

Thus $r = m/n$ is a rational number in (a, b).

Apply the rational-density statement to a nonempty open subinterval of

$$\left(\frac{a}{\sqrt{2}}, \frac{b}{\sqrt{2}} \right)$$

that does not contain 0. It contains a nonzero rational number q. Then $u = q\sqrt{2}$ lies in (a, b) and is irrational, because $q \neq 0$ and $\sqrt{2}$ is irrational. □

> ***Comment.*** Density and completeness express different facts. The rational numbers are dense in $\mathbb{R}$, yet $\mathbb{Q}$ is not complete. Having points arbitrarily close to every location does not mean that all limiting boundary points belong to the set.

1.1.3 The real line, intervals, and neighbourhoods

After choosing an origin, a positive direction, and a unit length, every real number corresponds to one point on a line and every point corresponds to one real number. We therefore identify $\mathbb{R}$ with the *real line* (数轴). The inequality $a < b$ means that the point a lies to the left of the point b.

For $a, b \in \mathbb{R}$ with $a < b$, the open interval (开区间), closed interval (闭区间), and two half-open intervals

(半开半闭区间) are

$$(a,b) = \{x \in \mathbb{R} \mid a < x < b\},$$
$$[a,b] = \{x \in \mathbb{R} \mid a \le x \le b\},$$
$$[a,b) = \{x \in \mathbb{R} \mid a \le x < b\},$$
$$(a,b] = \{x \in \mathbb{R} \mid a < x \le b\}.$$

Infinite intervals include

$$(-\infty, a), \quad (-\infty, a], \quad (a, +\infty), \quad [a, +\infty),$$

and the whole real line is $(-\infty, +\infty)$. The symbols $+\infty$ and $-\infty$ are not real numbers and are never included as endpoints.

Definition 1.3

*Let $a \in \mathbb{R}$ and $r > 0$. The r-*neighbourhood*(邻域) of a is*

$$U_r(a) = (a - r, a + r) = \{x \in \mathbb{R} \mid |x - a| < r\}.$$

The corresponding punctured r-neighbourhood*(去心邻域) is*

$$U_r(a) \setminus \{a\} = \{x \in \mathbb{R} \mid 0 < |x - a| < r\}.$$

♣

Neighbourhood notation translates a geometric statement into an inequality. The condition $x \in U_r(a)$ says exactly that the distance from x to a is less than r. Punctured neighbourhoods will become essential for function limits, where the behaviour near a may be independent of the value at a itself.

1.1.4 Absolute value and distance estimates

Definition 1.4

For $x \in \mathbb{R}$, its absolute value*(绝对值) is*

$$|x| = \begin{cases} x, & x \ge 0, \\ -x, & x < 0. \end{cases}$$

♣

Geometrically, $|x|$ is the distance from x to 0, while $|x - y|$ is the distance between x and y. The following elementary equivalences should become automatic: for $r > 0$,

$$|x| < r \iff -r < x < r,$$
$$|x| \le r \iff -r \le x \le r,$$
$$|x - a| < r \iff a - r < x < a + r.$$

Theorem 1.1

For all $x, y \in \mathbb{R}$, the triangle inequality(三角不等式) *and reverse triangle inequality*(反三角不等式) *state that*

$$|x + y| \le |x| + |y|, \tag{1.1}$$
$$\big||x| - |y|\big| \le |x - y|. \tag{1.2}$$

More generally, for all $a, b, c \in \mathbb{R}$,

$$|a - c| \le |a - b| + |b - c|.$$

♡

> ***Idea.*** The first inequality says that travelling from 0 to $x+y$ directly cannot be longer than travelling first from 0 to x and then by the displacement y. The reverse inequality is obtained by applying the same principle in both directions.

Proof Since $-|x| \le x \le |x|$ and $-|y| \le y \le |y|$, addition gives

$$-\big(|x|+|y|\big) \le x+y \le |x|+|y|\,.$$

This is equivalent to (1.1).

Applying the triangle inequality to $x = (x-y)+y$ gives

$$|x| \le |x-y| + |y|\,,$$

so $|x|-|y| \le |x-y|$. Interchanging x and y gives $|y|-|x| \le |x-y|$. Together these inequalities yield (1.2).

Finally, set $x = a-b$ and $y = b-c$ in the triangle inequality. □

Example 1.3 Turning a distance condition into an interval Solve

$$|2x-3| < 5.$$

Solution *The inequality is equivalent to*

$$-5 < 2x-3 < 5,$$

and hence to

$$-1 < x < 4.$$

Thus the solution set is $(-1,4)$.

> ***Geometric Interpretation.*** Equivalently,
>
> $$|2x-3| = 2\left|x-\frac{3}{2}\right| < 5$$
>
> says that x lies within distance $5/2$ of $3/2$.

Example 1.4 A reusable estimate Prove that $|x| < b+|a|$ whenever $|x-a| < b$.

Proof Then

$$|x| = |x-a+a| \le |x-a| + |a| < b + |a|\,.$$

□

> ***Technique.*** This simple observation is repeatedly used in limit proofs: first restrict x or a sequence term to a convenient neighbourhood, then replace a variable factor by a constant bound.

> ***Question.*** Why is the estimate $|x-y| \ge \big||x|-|y|\big|$ called a reverse triangle inequality?

1.1.5 Selected inequalities for later use

The following estimates will reappear in sequence limits. Their value is that they replace nonlinear expressions by quantities that are easier to control.

Proposition 1.5 (Arithmetic–Geometric Mean Inequality)

For $u, v \ge 0$,

$$\sqrt{uv} \le \frac{u+v}{2},$$

with equality if and only if $u = v$. *This is the two-variable arithmetic–geometric mean inequality* (算术平均数与几何平均数不等式), *or AM–GM.* ♠

Proof The inequality $(\sqrt{u}-\sqrt{v})^2 \geq 0$ gives $u+v \geq 2\sqrt{uv}$. Equality holds exactly when $\sqrt{u}=\sqrt{v}$. □

Example 1.5 A bound for an iteration Let $a>0$. Show that the map

$$T(x)=\frac{1}{2}\left(x+\frac{a}{x}\right)$$

sends $[\sqrt{a},+\infty)$ into itself and satisfies $T(x) \leq x$ there.

Proof If $x>0$, AM–GM gives

$$\frac{1}{2}\left(x+\frac{a}{x}\right) \geq \sqrt{a}.$$

Furthermore, when $x \geq \sqrt{a}$,

$$\frac{1}{2}\left(x+\frac{a}{x}\right) \leq x.$$

□

> ***Looking Ahead.*** Thus the averaging operation sends $[\sqrt{a},+\infty)$ into itself and does not increase its input. These two observations will prove boundedness and monotonicity of a recursive sequence.

Proposition 1.6 (Bernoulli's Inequality)

For $t \geq -1$ and $n \in \mathbb{N}_{>0}$, Bernoulli's inequality(伯努利不等式) is

$$(1+t)^n \geq 1+nt.$$

♠

Proof The assertion is an equality for $n=1$. If it holds for n, multiplying by $1+t \geq 0$ gives

$$(1+t)^{n+1} \geq (1+nt)(1+t) = 1+(n+1)t+nt^2 \geq 1+(n+1)t.$$

Induction completes the proof. □

Example 1.6 Control square roots by rationalizing For $u,v \geq 0$ with $u+v>0$, prove that

$$\left|\sqrt{u}-\sqrt{v}\right| = \frac{|u-v|}{\sqrt{u}+\sqrt{v}}.$$

If $u,v \geq c>0$, then

$$\left|\sqrt{u}-\sqrt{v}\right| \leq \frac{|u-v|}{2\sqrt{c}}.$$

Even without a positive lower bound, we still have

$$\left|\sqrt{u}-\sqrt{v}\right| \leq \sqrt{|u-v|}.$$

Proof Multiplying by the conjugate gives

$$\left|\sqrt{u}-\sqrt{v}\right| = \frac{|(\sqrt{u}-\sqrt{v})(\sqrt{u}+\sqrt{v})|}{\sqrt{u}+\sqrt{v}} = \frac{|u-v|}{\sqrt{u}+\sqrt{v}}.$$

If $u,v \geq c>0$, then $\sqrt{u}+\sqrt{v} \geq 2\sqrt{c}$, which yields the first estimate.

For the last estimate, assume $u \geq v$. Then $\sqrt{u}+\sqrt{v} \geq \sqrt{u-v}$; rationalization gives the result when $u>v$, and the case $u=v$ is immediate. □

> ***Technique.*** The useful estimate depends on whether the denominator can be kept away from zero.

1.1.6 Key Concepts, Methods, and Further Remarks

This section has two complementary layers. Absolute values and inequalities provide quantitative control, while completeness and the Archimedean property provide existence. Keeping these layers distinct helps identify which part of an argument is computational and which part depends on the structure of the real numbers.

1. *Find the structural contradiction.* In an irrationality proof, put a hypothetical fraction in lowest terms before using divisibility.

2. *Translate geometry into inequalities.* Distances on the real line are absolute values, and neighbourhoods are interval conditions.
3. *Control a complicated expression by a simpler one.* Split sums by the triangle inequality, rationalize square-root differences, and use AM–GM or Bernoulli when their hypotheses fit. For a homogeneous expression with positive variables, normalization can reduce the number of variables: dividing $u + v \geq 2\sqrt{uv}$ by $v > 0$ reduces it to $t + 1 \geq 2\sqrt{t}$ with $t = u/v$.
4. *Choose an integer backward from the desired error.* To make $C/n < \varepsilon$ with $C > 0$, choose an integer $N > C/\varepsilon$. Then the estimate holds for every $n \geq N$, not just one index.
5. *Identify where completeness is used.* Algebra and order alone are shared by $\mathbb{Q}$ and $\mathbb{R}$. Existence theorems involving limits or least upper bounds require the additional completeness of $\mathbb{R}$.

> ***Caution.*** Do not treat $\pm\infty$ as real numbers. Expressions such as $a < +\infty$ are useful notation, but ordinary field operations with ∞ are not defined in $\mathbb{R}$.

> ***Looking Ahead.*** Suprema need not be attained, but they can be approached from below. This approximation property is the decisive step in proving that a bounded nondecreasing sequence converges.

Exercise 1.1

1. Prove that $\sqrt{6}$ is irrational.
2. Let $r \in \mathbb{Q}$ and $u \in \mathbb{R} \setminus \mathbb{Q}$. Prove that $r + u$ is irrational, and prove that ru is irrational when $r \neq 0$.
3. Generalize Proposition 1.1 by proving that $\sqrt{p}$ is irrational whenever p is a positive integer that is not a perfect square.

   > ***Hint.*** Use uniqueness of prime factorization: what can be said about the exponents in the prime factorization of a perfect square? If $\sqrt{p} = m/n$ with $m, n \in \mathbb{N}_{>0}$, compare prime exponents on the two sides of $m^2 = pn^2$.

4. Prove that every nonempty open interval contains infinitely many rational numbers and infinitely many irrational numbers.
5. Write each set in interval notation:
$$\{x \in \mathbb{R} \mid |x - 2| \leq 3\}, \qquad \{x \in \mathbb{R} \mid 0 < |x + 1| < 2\}.$$
6. Solve the inequalities
$$|3x + 1| \geq 2, \qquad |x - 1| < |x + 2|\,.$$
7. Prove that
$$|x + y + z| \leq |x| + |y| + |z|$$
for all $x, y, z \in \mathbb{R}$.
8. If $|x - a| < r$ and $|y - b| < s$, prove that
$$|(x + y) - (a + b)| < r + s.$$
Give a geometric interpretation.
9. Let $E \subseteq \mathbb{R}$ be nonempty and bounded above. Prove directly from the definition that $\sup E$ is unique.
10. Explain why the density of $\mathbb{Q}$ in $\mathbb{R}$ does not imply the completeness of $\mathbb{Q}$.
11. Let
$$A = \{x > 0 \mid x^2 < 5\}.$$

Determine $\sup A$, $\inf A$, and whether either extremum is attained. Justify every assertion from the definitions.

12. Let $E \subseteq \mathbb{R}$ be nonempty and bounded, let $c \in \mathbb{R}$, and let $\lambda > 0$. Prove that
$$\sup(c + E) = c + \sup E, \qquad \sup(\lambda E) = \lambda \sup E,$$
where $c + E = \{c + x : x \in E\}$ and $\lambda E = \{\lambda x : x \in E\}$. State the corresponding formulas for infima.

13. Let $a < b$. Find the minimum value of $|x - a| + |x - b|$ for $x \in \mathbb{R}$, and determine all points at which it is attained. Give both an algebraic and a geometric explanation.

14. Prove that
$$x(1 - x) \leq \frac{1}{4}$$
for every $x \in [0, 1]$, and determine the equality case.

15. Prove by rationalization that, for $t \geq 0$,
$$0 \leq \sqrt{1 + t} - 1 \leq \frac{t}{2}.$$
Explain why the constant $1/2$ cannot be replaced by a smaller positive constant valid for every sufficiently small $t > 0$.

Supplement Exercises

S1. Let $E, F \subseteq \mathbb{R}$ be nonempty and bounded above, and define $E + F = \{x + y : x \in E,\ y \in F\}$. Prove directly from the least-upper-bound property that
$$\sup(E + F) = \sup E + \sup F.$$

S2. If E and F are nonempty and bounded, prove that
$$\sup(E \cup F) = \max\{\sup E, \sup F\}, \qquad \inf(E \cup F) = \min\{\inf E, \inf F\}.$$
Which analogous formulas for $E \cap F$ fail without additional hypotheses?

S3. Prove the finite Cauchy–Schwarz inequality
$$\left(\sum_{k=1}^{n} a_k b_k\right)^2 \leq \left(\sum_{k=1}^{n} a_k^2\right)\left(\sum_{k=1}^{n} b_k^2\right)$$
by considering $\sum_{k=1}^{n} (t a_k - b_k)^2$ as a quadratic polynomial in t. Determine the equality condition.

S4. For $m \in \mathbb{N}$, prove by induction that the arithmetic–geometric mean inequality holds for 2^m positive numbers:
$$\frac{x_1 + \cdots + x_{2^m}}{2^m} \geq (x_1 \cdots x_{2^m})^{1/2^m}.$$

S5. Find the best constant C such that
$$(x + y + z)^2 \leq C(x^2 + y^2 + z^2)$$
for all $x, y, z \in \mathbb{R}$, and prove that your constant is optimal.

S6. Let
$$B = \{q \in \mathbb{Q} : q \geq 0 \text{ and } q^2 < 2\}.$$
Prove that B has a supremum as a subset of $\mathbb{R}$, but has no supremum as a subset of the ordered field $\mathbb{Q}$.

1.2 Variables and Functions

核心概念与图像方法

- ❑ *Understand a function as a mapping with a specified domain, codomain, and assignment rule.*
- ❑ *Move between formulas, tables, verbal rules, piecewise definitions, and graphs.*
- ❑ *Combine functions by arithmetic operations and composition while tracking every domain restriction.*
- ❑ *Review elementary and non-elementary functions without confusing an expression with a function.*
- ❑ *Describe monotonicity, parity, periodicity, boundedness, and graph transformations.*
- ❑ *Test injectivity before constructing an inverse and interpret inverse graphs geometrically.*

Many quantities vary, and one quantity often depends on another. The distance travelled at constant speed depends on time, the area of a circle depends on its radius, and the length of a metal rod depends on temperature. Calculus studies such dependence, so its basic object is the function.

The essential point is not that a function has a formula. The essential point is that every allowed input has one, and only one, output.

1.2.1 Functions as mappings

Definition 1.5

Let $X, Y \subseteq \mathbb{R}$. A function(函数) *$f$ from X to Y, written*

$$f\colon X \to Y,$$

is a rule that assigns to every $x \in X$ a unique number $f(x) \in Y$. The set X is the domain(定义域)*, and Y is the* codomain(陪域)*. The* range(值域)*, or* image(像)*, of f is*

$$f(X) = \{f(x) \mid x \in X\}.$$

Thus $f(X) \subseteq Y$, and equality need not hold.

♣

A function is also called a *mapping*(映射).

The input x is often called the *independent variable*(自变量).

The value $f(x)$ is the *dependent variable*(因变量). These names emphasize variation, but the definition itself is set-theoretic: a function consists of a domain together with an assignment rule.

> ***Important Distinction.*** The domain is part of the function. The formulas
>
> $$f(x) = x^2 \quad (x \in \mathbb{R}) \qquad \text{and} \qquad g(x) = x^2 \quad (x \in [0, +\infty))$$
>
> define different functions, even though the displayed expressions are identical. Their ranges are also different if the first domain is changed further.

Definition 1.6

Two functions $f\colon X \to Y$ and $g\colon X' \to Y'$ are equal if $X = X'$, $Y = Y'$, and

$$f(x) = g(x)$$

for every $x \in X$.

♣

A function may be represented by a formula, a table, a graph, an algorithm, or a verbal rule. For example, a timetable assigning a departure time to each train is a function even if no useful algebraic formula describes it.

The *graph*(图像)of $f\colon X \to \mathbb{R}$ is the set

$$\{(x, f(x)) \in \mathbb{R}^2 \mid x \in X\}.$$

A plane curve is the graph of a function of x precisely when every vertical line meets it in at most one point. This vertical-line test is simply the uniqueness condition in Definition 1.5 expressed geometrically. The horizontal projection of the graph gives the domain, and the vertical projection gives the range. Zeros are the points where the graph meets the x-axis, while the portions above and below that axis show the sign of the function.

1.2.2 Domains, ranges, and piecewise definitions

When a function is given by a formula and no domain is stated, its *natural domain*(自然定义域)is the largest subset of $\mathbb{R}$ on which the formula gives a real value. This convention must be replaced by the actual physical or mathematical domain when context imposes additional restrictions.

Example 1.7 Determining a natural domain Find the natural domain of

$$f(x) = \sqrt{\frac{1+x}{1-x}}.$$

Solution *We need* $x \neq 1$ *and*

$$\frac{1+x}{1-x} \geq 0.$$

The critical points are -1 *and* 1. *A sign analysis gives*

$$-1 \leq x < 1.$$

Thus the natural domain is $[-1, 1)$.

> ***Check.*** Notice that solving only $1 + x \geq 0$ would miss the sign of the denominator.

Definition 1.7

A piecewise-defined function(分段函数)*is a single function whose values are specified by different formulas on different parts of its domain.* ♣

For instance,

$$T(x) = \begin{cases} 0, & 0 \leq x \leq 5000, \\ 0.03(x - 5000), & 5000 < x \leq 8000, \end{cases}$$

defines one function on $[0, 8000]$. At an endpoint where two pieces might meet, the inequalities must make clear which formula applies.

Example 1.8 Range depends on the domain Let $f(x) = x^2$. Determine its range when the domain is respectively $\mathbb{R}$, $[-2, 1]$, and $(2, +\infty)$.

Solution *On* $\mathbb{R}$, *the range is* $[0, +\infty)$. *On* $[-2, 1]$, *the range is* $[0, 4]$. *On* $(2, +\infty)$, *the range is* $(4, +\infty)$.

> ***Interpretation.*** The formula alone does not determine the range; the domain matters.

1.2.3 Operations and composition

Let $f\colon X \to \mathbb{R}$ and $g\colon Z \to \mathbb{R}$. Their sum, difference, and product are defined on $X \cap Z$ by

$$(f+g)(x) = f(x) + g(x),$$
$$(f-g)(x) = f(x) - g(x),$$
$$(fg)(x) = f(x)g(x).$$

The quotient is defined by

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$$

on

$$\{x \in X \cap Z \mid g(x) \neq 0\}.$$

Definition 1.8

Let $g\colon X \to Y$ and $f\colon Y_0 \to Z$. The composition(复合函数)*of f after g is*

$$(f \circ g)(x) = f(g(x)).$$

Its domain is

$$\{x \in X \mid g(x) \in Y_0\}.$$

♣

Composition is performed from right to left: first apply g, then apply f. It is associative whenever the compositions are defined, but it is generally not commutative.

Example 1.9 Composition changes the domain Let

$$f(t) = \sqrt{t}, \qquad g(x) = x^2 - 1.$$

Determine both compositions and their domains.

Solution *Then*

$$(f \circ g)(x) = \sqrt{x^2 - 1}$$

has domain

$$(-\infty, -1] \cup [1, +\infty).$$

In the other order,

$$(g \circ f)(x) = x - 1$$

has domain $[0, +\infty)$, not all of $\mathbb{R}$.

> ***Domain Check.*** Although the final formula $x - 1$ makes sense for every real x, the composition remembers that $\sqrt{x}$ was evaluated first.

> ***Important Distinction.*** Algebraic simplification does not automatically enlarge the domain of a function. For example,
>
> $$\frac{x^2 - 1}{x - 1} = x + 1$$
>
> only for $x \neq 1$. The expression on the left and the function $x \mapsto x + 1$ on $\mathbb{R}$ are not the same function.

1.2.4 Basic elementary functions

The standard building blocks of one-variable calculus are the basic elementary functions. Their domains must always be kept in view.

Table 1.1: Basic elementary functions and their standard real domains.

Type	Typical function	Standard domain
Constant	$f(x) = c$	$\mathbb{R}$
Power	$f(x) = x^{\alpha}$	depends on α
Exponential	$f(x) = a^{x}$, $a > 0$, $a \neq 1$	$\mathbb{R}$
Logarithmic	$f(x) = \log_a x$, $a > 0$, $a \neq 1$	$(0, +\infty)$
Trigonometric	$\sin x$, $\cos x$	$\mathbb{R}$
Trigonometric	$\tan x$	$\mathbb{R} \setminus \{\frac{\pi}{2} + k\pi \mid k \in \mathbb{Z}\}$
Inverse trigonometric	$\arcsin x$, $\arccos x$	$[-1, 1]$
Inverse trigonometric	$\arctan x$	$\mathbb{R}$

For real powers, x^{α} is always defined for $x > 0$ by

$$x^{\alpha} = \exp(\alpha \ln x).$$

The maximal real domain may be larger for special exponents. For example, $x^{1/3}$ is defined on $\mathbb{R}$, $x^{1/2}$ on $[0, +\infty)$, and x^{-2} on $\mathbb{R} \setminus \{0\}$. When $\alpha = m/n$ is in lowest terms, the parity of n and the sign of m determine whether negative inputs and the input 0 are allowed.

The natural logarithm is denoted by $\ln x$. The constant e and its construction from a sequence are developed in Section 1.3. Angles in trigonometric functions are measured in radians.

Definition 1.9

An elementary function (初等函数) *is a function obtained from basic elementary functions by finitely many arithmetic operations and compositions, on the domain where the resulting expression is defined.* ♣

Two useful functions built from exponentials are the hyperbolic sine and cosine:

$$\sinh x = \frac{e^{x} - e^{-x}}{2}, \qquad \cosh x = \frac{e^{x} + e^{-x}}{2}.$$

Direct calculation gives

$$\cosh^2 x - \sinh^2 x = 1.$$

These functions will later appear naturally in differential equations and integration.

Not every function is elementary, and not every function is represented by a smooth curve. The following examples are especially useful.

Example 1.10 Floor and fractional-part functions The *floor function* (取整函数) is

$$x \longmapsto \lfloor x \rfloor,$$

where $\lfloor x \rfloor$ is the greatest integer not exceeding x. Its defining property is

$$\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1.$$

The *fractional part* (小数部分) of x is

$$\{x\} = x - \lfloor x \rfloor \in [0, 1).$$

For negative inputs, this differs from simply retaining the digits after the decimal point; for example, $\{-3.14\} = 0.86$.

Example 1.11 Sign and Dirichlet functions The *sign function*(符号函数)is

$$\operatorname{sgn} x = \begin{cases} 1, & x > 0, \\ 0, & x = 0, \\ -1, & x < 0. \end{cases}$$

The *Dirichlet function*(狄利克雷函数)is

$$D(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \in \mathbb{R} \setminus \mathbb{Q}. \end{cases}$$

By Proposition 1.4, every open interval contains both rational and irrational numbers. Thus the graph of D consists of dense points on the two horizontal lines $y = 0$ and $y = 1$; it is not a curve in the usual visual sense. This example shows why a function must be defined as a rule rather than as a formula or a drawn curve.

1.2.5 Qualitative properties and graph transformations

Formulas give exact values, but calculus also studies the overall behaviour of a function. The following properties describe that behaviour without requiring us to calculate every value.

Definition 1.10

Let $f : X \to \mathbb{R}$.

(i) On an interval $I \subseteq X$*, the function is* nondecreasing *if*

$$x_1 < x_2 \quad \Longrightarrow \quad f(x_1) \le f(x_2),$$

and nonincreasing *if the final inequality is reversed. A function of either type is a* monotone function (单调函数) *on* I*. Replacing* $\le$ *or* $\ge$ *by a strict inequality defines* strictly increasing *and* strictly decreasing *functions.*

(ii) Suppose X *is symmetric about* 0*. The function is an* even function (偶函数) *if* $f(-x) = f(x)$ *for every* $x \in X$*, and an* odd function (奇函数) *if* $f(-x) = -f(x)$ *for every* $x \in X$*.*

(iii) A number $T > 0$ *is a* period *of* f *if*

$$x \in X \iff x + T \in X, \qquad f(x + T) = f(x) \quad (x \in X).$$

A function having a period is a periodic function (周期函数).

Even functions have graphs symmetric about the y-axis, while odd functions have graphs symmetric about the origin. The function $x \mapsto |x|$ is even, nonincreasing on $(-\infty, 0]$, and nondecreasing on $[0, +\infty)$. The sine function is odd and 2π-periodic; the cosine function is even and 2π-periodic.

These properties are logically independent. For example, $\tan x$ is periodic but unbounded on its domain. The function

$$g(x) = \frac{1}{1 + x^2}$$

is bounded but not periodic: if $T > 0$, then $g(T) < g(0)$, so T cannot be a period.

Graph transformations can be read by tracking what happens to a typical point $(u, f(u))$. If the domain of f is D, then the following table records both the geometry and the new domain.

When $a < 0$, the third transformation also reflects the graph across the x-axis, and the fourth reflects it across the y-axis. The domain must be transformed together with the graph; a visually correct shift with an unchanged domain may define the wrong function.

Table 1.2: Basic transformations of the graph $y = f(x)$.

New function	Effect on the graph	Domain
$f(x) + c$	vertical shift by c	D
$f(x - c)$	horizontal shift by c	$\{u + c \mid u \in D\}$
$af(x)$	vertical scale by $\lvert a \rvert$	D
$f(ax)$, $a \neq 0$	horizontal scale by $1/\lvert a \rvert$	$\{x \mid ax \in D\}$

> ***Question.*** Does either implication hold: periodic $\Rightarrow$ bounded or bounded $\Rightarrow$ periodic? Can one implication be repaired by adding a natural hypothesis?

1.2.6 Injectivity, surjectivity, and inverse functions

Definition 1.11

Let $f : X \to Y$.

(i) The function is injective (单射) *if*

$$f(x_1) = f(x_2) \quad \Longrightarrow \quad x_1 = x_2.$$

(ii) It is surjective (满射) *if* $f(X) = Y$.

(iii) It is bijective (双射) *if it is both injective and surjective.*

♣

Injectivity says that distinct inputs never merge into the same output. Surjectivity depends on the chosen codomain: the function $x \mapsto x^2$ is surjective from $\mathbb{R}$ to $[0, +\infty)$ but not from $\mathbb{R}$ to $\mathbb{R}$.

Theorem 1.2

A function $f : X \to Y$ *has an* inverse function (反函数) $f^{-1} : Y \to X$ *if and only if* f *is bijective. In that case,*

$$f^{-1}(f(x)) = x \quad (x \in X), \qquad f(f^{-1}(y)) = y \quad (y \in Y).$$

♡

> ***Idea.*** Surjectivity guarantees that each proposed inverse input has a preimage, while injectivity guarantees that this preimage is unique. Together they make the inverse assignment a function.

Proof Suppose first that f is bijective. For every $y \in Y$, surjectivity gives at least one $x \in X$ with $f(x) = y$, and injectivity makes this x unique. Define $f^{-1}(y)$ to be that unique x. The two displayed identities follow directly.

Conversely, suppose that an inverse $f^{-1} : Y \to X$ exists. For every $y \in Y$, the element $x = f^{-1}(y)$ satisfies $f(x) = y$, so f is surjective. If $f(x_1) = f(x_2)$, applying f^{-1} gives $x_1 = x_2$, so f is injective. Thus f is bijective. □

An injective function $f : X \to Y$ is automatically a bijection from X onto its range $f(X)$, and hence has an inverse

$$f^{-1} : f(X) \to X.$$

For example, $x \mapsto x^2$ is not injective on $\mathbb{R}$, but its restriction to $[0, +\infty)$ is injective and has inverse $x \mapsto \sqrt{x}$. Likewise, $\sin x$ is not injective on $\mathbb{R}$. Its restriction to $[-\pi/2, \pi/2]$ is bijective onto $[-1, 1]$, and the inverse of this restricted function is $\arcsin x$.

Example 1.12 Constructing an inverse while tracking its domain Let

$$f : (0, +\infty) \to \mathbb{R}, \qquad f(x) = \frac{2}{x} - \frac{x}{2}.$$

Prove that f is bijective and find f^{-1}.

Proof If $0 < x_1 < x_2$, then

$$\frac{2}{x_1} > \frac{2}{x_2} \qquad \text{and} \qquad -\frac{x_1}{2} > -\frac{x_2}{2},$$

so $f(x_1) > f(x_2)$. Thus f is injective.

For any $y \in \mathbb{R}$, the equation $y = f(x)$ is equivalent to

$$x^2 + 2yx - 4 = 0.$$

Its roots are $-y \pm \sqrt{y^2+4}$. Exactly one is positive, namely

$$x = -y + \sqrt{y^2+4},$$

because $\sqrt{y^2+4} > |y|$. Hence every $y \in \mathbb{R}$ has exactly one preimage in $(0, +\infty)$, and

$$f^{-1}(y) = -y + \sqrt{y^2+4} \qquad (y \in \mathbb{R}).$$

□

The graph of an inverse is obtained by interchanging the coordinates in every ordered pair. It is therefore the reflection of the original graph across the line $y = x$.

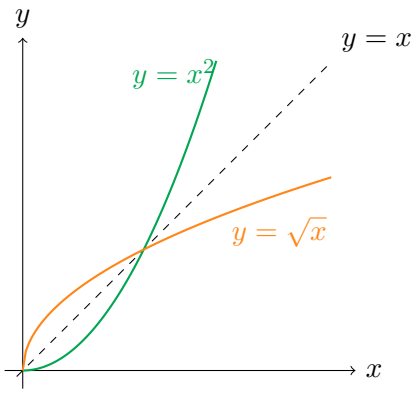


Figure 1.1: The graphs of $x \mapsto x^2$ on $[0, +\infty)$ and its inverse $x \mapsto \sqrt{x}$ are reflections across $y = x$.

> ***Comment.*** The notation f^{-1} denotes an inverse function, not a reciprocal. In general,
>
> $$f^{-1}(x) \neq \frac{1}{f(x)}.$$

1.2.7 Sequences and bounded functions

A sequence

$$a_1, a_2, a_3, \ldots$$

is a function whose domain is $\mathbb{N}_{>0}$:

$$a \colon \mathbb{N}_{>0} \to \mathbb{R}, \qquad a(n) = a_n.$$

The order of the terms matters, so a sequence is not merely the set $\{a_n \mid n \geq 1\}$. This viewpoint connects the present section directly with sequence limits in Section 1.3.

Definition 1.12

Let $f \colon X \to \mathbb{R}$. The function f is bounded above *on X if there exists $M \in \mathbb{R}$ such that*

$$f(x) \leq M$$

for every $x \in X$. It is bounded below *on X if there exists $m \in \mathbb{R}$ such that*

$$m \leq f(x)$$

for every $x \in X$. It is a bounded function(有界函数) *on X if it is bounded both above and below.*

Boundedness always refers to a specified domain. The function $f(x) = 1/x$ is bounded on $[1, +\infty)$ but unbounded on $(0, 1]$. The fact that $f(x)$ is a finite real number for every x in its domain does not supply one common bound valid for all inputs.

Proposition 1.7

A function $f\colon X \to \mathbb{R}$ is bounded if and only if there exists $C \geq 0$ such that

$$|f(x)| \leq C$$

for every $x \in X$.

Proof If f is bounded, choose $m, M \in \mathbb{R}$ such that

$$m \leq f(x) \leq M$$

for every $x \in X$. With $C = \max\{|m|, |M|\}$, we have $|f(x)| \leq C$.

Conversely, if $|f(x)| \leq C$, then

$$-C \leq f(x) \leq C$$

for every $x \in X$, so f is bounded. □

1.2.8 Section Summary and Problem-solving Strategies

The essential data of a function are its domain, codomain, and assignment rule. Formulas, graphs, and tables are representations of that data, not replacements for it. When working with a function, use the following order.

1. Identify the domain before simplifying or calculating.
2. Distinguish the codomain from the actual range.
3. For a composition, work from the inside out and impose every intermediate domain condition.
4. For an inverse, first check injectivity and determine the range that will become the inverse domain.
5. For qualitative properties, combine algebraic tests with graph symmetry, but verify conclusions on the stated domain.
6. For a graph transformation, transform the domain together with the points of the graph.
7. For boundedness, search for one constant valid for every input in the stated domain.

A compact trigonometric and hyperbolic toolkit

The formulas below are most useful when they are organized by their source rather than memorized as unrelated identities. The addition formulas generate the product-to-sum and sum-to-product formulas by addition, subtraction, and a change of variables.

For real u, v (and with nonzero denominators where tangent occurs),

Example 1.13 Derive a transformation instead of memorizing it Derive one product-to-sum formula and its corresponding sum-to-product formula.

Table 1.3: Trigonometric addition and subtraction formulas.

Function	Formula
Sine	$\sin(u \pm v) = \sin u \cos v \pm \cos u \sin v$
Cosine	$\cos(u \pm v) = \cos u \cos v \mp \sin u \sin v$
Tangent	$\tan(u \pm v) = \dfrac{\tan u \pm \tan v}{1 \mp \tan u \tan v}$

Table 1.4: Trigonometric product-to-sum and sum-to-product formulas.

Product to sum	Sum to product
$\sin u \sin v = \dfrac{\cos(u-v) - \cos(u+v)}{2}$	$\sin u + \sin v = 2 \sin \dfrac{u+v}{2} \cos \dfrac{u-v}{2}$
$\cos u \cos v = \dfrac{\cos(u-v) + \cos(u+v)}{2}$	$\sin u - \sin v = 2 \cos \dfrac{u+v}{2} \sin \dfrac{u-v}{2}$
$\sin u \cos v = \dfrac{\sin(u+v) + \sin(u-v)}{2}$	$\cos u + \cos v = 2 \cos \dfrac{u+v}{2} \cos \dfrac{u-v}{2}$
$\cos u \sin v = \dfrac{\sin(u+v) - \sin(u-v)}{2}$	$\cos u - \cos v = -2 \sin \dfrac{u+v}{2} \sin \dfrac{u-v}{2}$

Proof Adding the two sine addition formulas gives

$$\begin{aligned} \sin(u+v) + \sin(u-v) &= (\sin u \cos v + \cos u \sin v) \\ &\quad + (\sin u \cos v - \cos u \sin v) \\ &= 2 \sin u \cos v. \end{aligned}$$

Therefore

$$\sin u \cos v = \frac{\sin(u+v) + \sin(u-v)}{2}.$$

Now put $u = (x+y)/2$ and $v = (x-y)/2$. Then $u + v = x$ and $u - v = y$, so the same identity becomes

$$\sin x + \sin y = 2 \sin \frac{x+y}{2} \cos \frac{x-y}{2}.$$

□

> ***Technique.*** The other entries in Table 1.4 follow from the same two operations: add or subtract a pair of angle-sum identities, then make a half-sum and half-difference substitution.

Inverse trigonometric functions require attention to their principal ranges. For example,

$$\arcsin x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right], \quad \arccos x \in [0, \pi], \quad \arctan x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right).$$

Consequently,

$$\arcsin x + \arccos x = \frac{\pi}{2} \qquad (-1 \le x \le 1),$$

$$\arctan x + \arctan(1/x) = \begin{cases} \pi/2, & x > 0, \\ -\pi/2, & x < 0. \end{cases}$$

Although

$$\tan(\arctan x + \arctan y) = \frac{x+y}{1-xy}$$

when $xy \neq 1$, replacing the left side by a single $\arctan$ may require adding or subtracting π. This branch issue is why inverse trigonometric formulas should not be manipulated as if the inverse functions were globally periodic. There are no literal product-to-sum formulas for $\arcsin$, $\arccos$, or $\arctan$; those transformations belong to the direct trigonometric functions.

The hyperbolic functions are defined by

$$\sinh x = \frac{e^x - e^{-x}}{2}, \qquad \cosh x = \frac{e^x + e^{-x}}{2}, \qquad \tanh x = \frac{\sinh x}{\cosh x}.$$

Their addition formulas have the same algebraic pattern as the trigonometric ones, with the decisive sign change in the cosine analogue:

$$\begin{aligned}
\sinh(u \pm v) &= \sinh u \cosh v \pm \cosh u \sinh v, \\
\cosh(u \pm v) &= \cosh u \cosh v \pm \sinh u \sinh v, \\
\tanh(u \pm v) &= \frac{\tanh u \pm \tanh v}{1 \pm \tanh u \tanh v}.
\end{aligned}$$

Table 1.5: Hyperbolic product-to-sum and sum-to-product formulas.

Product to sum	Sum to product
$\sinh u \sinh v = \dfrac{\cosh(u+v) - \cosh(u-v)}{2}$	$\sinh u + \sinh v = 2 \sinh \dfrac{u+v}{2} \cosh \dfrac{u-v}{2}$
$\cosh u \cosh v = \dfrac{\cosh(u+v) + \cosh(u-v)}{2}$	$\sinh u - \sinh v = 2 \cosh \dfrac{u+v}{2} \sinh \dfrac{u-v}{2}$
$\sinh u \cosh v = \dfrac{\sinh(u+v) + \sinh(u-v)}{2}$	$\cosh u + \cosh v = 2 \cosh \dfrac{u+v}{2} \cosh \dfrac{u-v}{2}$
$\cosh u \sinh v = \dfrac{\sinh(u+v) - \sinh(u-v)}{2}$	$\cosh u - \cosh v = 2 \sinh \dfrac{u+v}{2} \sinh \dfrac{u-v}{2}$

The principal inverse hyperbolic functions have especially useful logarithmic forms:

$$\begin{aligned}
\operatorname{arsinh} x &= \ln\left(x + \sqrt{x^2+1}\right) && (x \in \mathbb{R}), \\
\operatorname{arcosh} x &= \ln\left(x + \sqrt{x^2-1}\right) && (x \geq 1), \\
\operatorname{artanh} x &= \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right) && (|x| < 1).
\end{aligned}$$

As with inverse trigonometric functions, these identities include domain and branch information; they are not obtained by formally placing "arc" in front of a direct-function identity.

Example 1.14 Why the functions are called hyperbolic Show that $(\cosh t, \sinh t)$ lies on the unit hyperbola and derive a hyperbolic addition formula directly from exponentials.

Proof From the definitions,

$$\begin{aligned}
\cosh^2 t - \sinh^2 t &= \frac{(e^t + e^{-t})^2 - (e^t - e^{-t})^2}{4} \\
&= 1.
\end{aligned}$$

Thus the point $(x, y) = (\cosh t, \sinh t)$ parametrizes the right-hand branch of

$$x^2 - y^2 = 1,$$

just as $(\cos t, \sin t)$ parametrizes the unit circle $x^2 + y^2 = 1$. Moreover,

$$\begin{aligned}
\sinh(u+v) &= \frac{e^{u+v} - e^{-(u+v)}}{2} \\
&= \frac{(e^u - e^{-u})(e^v + e^{-v}) + (e^u + e^{-u})(e^v - e^{-v})}{4} \\
&= \sinh u \cosh v + \cosh u \sinh v.
\end{aligned}$$

Adding and subtracting the formulas for $u+v$ and $u-v$ then produces Table 1.5. □

Geometric and Complex Viewpoints. The word "hyperbolic" therefore reflects geometry, not merely notation. There is also a complex-variable viewpoint: Euler's formulas give

$$\cos(ix) = \cosh x, \qquad \sin(ix) = i \sinh x.$$

Thus the circle and hyperbola identities are connected by the imaginary substitution $x \mapsto ix$; this is a useful structural explanation, although complex exponentials are not needed to use the real formulas above.

Common Pitfall. An expression and a function are not the same thing. The expression describes an assignment rule, but the function also includes its domain. Many apparent contradictions about ranges, inverses, and compositions disappear once the domain is written explicitly.

Looking Ahead. A function limit will ask what $f(x)$ does when x is near a point, possibly without using the value at that point. The distinction between formula and function remains essential because a limit can depend on which domain points are available near the point of approach.

Exercise 1.2

1. Find the natural domains of
$$\ln(x^2 - 1), \qquad \frac{\sqrt{2x+1}}{x-3}, \qquad \arccos(2\sin x).$$
2. For $f(x) = x^2 - 2x$ on each of the domains $\mathbb{R}$, $[0, 3]$, and $(2, +\infty)$, determine the range.
3. Let $f(x) = 1/(x-1)$ and $g(x) = \sqrt{x+2}$. Find formulas and domains for $f+g$, fg, f/g, $f \circ g$, and $g \circ f$.
4. Give two functions represented by the same formula but having different ranges. Explain why they are not equal as functions.
5. Prove that composition of functions is associative whenever all compositions involved are defined. Give an example showing that composition need not be commutative.
6. Determine $\lfloor x \rfloor$ and $\{x\}$ for
$$x = 3.7, \quad x = -0.2, \quad x = -4.$$
Prove that $\{x+1\} = \{x\}$ for all $x \in \mathbb{R}$.
7. Determine the parity, periods, and intervals of monotonicity that apply to
$$f(x) = x^2, \qquad g(x) = \cos(2x), \qquad h(x) = \frac{1}{1+x^2}.$$
Give a bounded nonperiodic function and a periodic unbounded function.
8. Starting from the graph of $y = f(x)$, describe the transformations and domains of
$$y = 2f(x-1) - 3, \qquad y = f(-2x).$$
9. Determine whether each map is injective, surjective, or bijective:
$$x \mapsto x^2 \colon \mathbb{R} \to [0, +\infty), \qquad x \mapsto x^3 \colon \mathbb{R} \to \mathbb{R}, \qquad x \mapsto e^x \colon \mathbb{R} \to (0, +\infty).$$
10. Find the inverse of
$$f(x) = \frac{x-1}{x+2}$$
after specifying a domain and codomain for which f is bijective.
11. Prove directly from the definitions that if f and g are bounded on the same nonempty set X, then $f+g$ and fg are bounded on X.

12. Give an example of a function that is bounded above but not bounded below, and another that is bounded on one interval but unbounded on a smaller-looking interval with a missing endpoint.
13. Explain why the Dirichlet function cannot be represented by a continuous curve, using Proposition 1.4.
14. Determine the natural domains, ranges, parity, and monotonicity of
$$\sinh x, \qquad \cosh x, \qquad \tanh x.$$
Use only their exponential definitions and elementary inequalities.
15. Starting with $y = \sinh x$ and $y = \tanh x$, solve algebraically for x and thereby verify
$$\operatorname{arsinh} y = \ln\!\left(y + \sqrt{y^2+1}\right), \qquad \operatorname{artanh} y = \frac{1}{2}\ln\!\left(\frac{1+y}{1-y}\right).$$
State the domain of each inverse function.
16. Derive the product-to-sum formulas for $\sin u \sin v$ and $\sinh u \sinh v$ from the corresponding addition formulas. Then use them to rewrite
$$\sin(3x)\sin x, \qquad \sinh(3x)\sinh x$$
as differences of two cosine-type functions.
17. Sketch the graph of
$$f(x) = |x-1| + |x+1|$$
by first writing a piecewise formula. Determine its range, parity, and intervals of monotonicity, and find all x for which $f(x) = 3$.
18. Consider $f(x) = x + 1/x$. Show that its restrictions to $(0,1]$ and $[1,+\infty)$ are one-to-one, determine their ranges, and find an inverse formula for each restriction.
19. Solve on the stated domains:
$$\arcsin(2x-1) = \frac{\pi}{6}, \qquad \arctan x + \arctan\frac{1}{x} = \frac{\pi}{2} \quad (x>0).$$
Explain why branch information is essential in the second equation.
20. Let $f(x) = \lfloor x \rfloor$ and $g(x) = \{x\}$. Determine formulas and ranges for $f \circ g$, $g \circ f$, and $g \circ g$. Which of these functions are periodic, and what are their periods?

Supplement Exercises

S1. For $x \notin 2\pi\mathbb{Z}$, prove the finite-sum identity
$$\sum_{k=1}^{n} \cos(kx) = \frac{\sin\big((2n+1)x/2\big) - \sin(x/2)}{2\sin(x/2)}.$$
Obtain it by converting each cosine into a difference of two sines so that the sum telescopes.

S2. Prove that, for $x \geq 0$,
$$2\arctan\sqrt{x} - \arcsin\frac{x-1}{x+1} = \frac{\pi}{2}.$$
Your proof must justify the ranges of all inverse trigonometric functions involved.

S3. Derive the tangent half-angle identities
$$\sin\theta = \frac{2t}{1+t^2}, \qquad \cos\theta = \frac{1-t^2}{1+t^2}, \qquad t = \tan\frac{\theta}{2},$$
and describe precisely which values of θ require separate treatment.

S4. Let D be the Dirichlet function, equal to 1 on $\mathbb{Q}$ and 0 on $\mathbb{R}\setminus\mathbb{Q}$. Determine the ranges of $xD(x)$, $D(x^2)$, and $D(D(x))$. Decide which of these functions are even, odd, periodic, or bounded.

S5. Let

$$f(x) = \frac{ax + b}{cx - a}, \qquad a^2 + bc \neq 0.$$

Choose the natural domain and codomain so that f is a bijection, and prove that $f^{-1} = f$.

> ***Optional Hint.*** Associate f with the matrix $\begin{pmatrix} a & b \\ c & -a \end{pmatrix}$ and square the matrix.

S6. Determine all real constants a, b for which

$$f(x) = \ln(a + be^x)$$

is well-defined on a nondegenerate interval I, maps I onto itself, and satisfies $f(f(x)) = x$ for every $x \in I$. State the maximal interval in each case.

1.3 Limits of Sequences

重点内容与证明方法

- ❑ *Understand a sequence as an ordered function on the positive integers.*
- ❑ *Read the quantifiers in the ε–N definition and construct direct proofs in the correct order.*
- ❑ *Establish uniqueness, boundedness, order properties, and arithmetic laws for limits.*
- ❑ *Use domination, squeeze arguments, normalization, and subsequences to prove convergence or divergence.*
- ❑ *Apply completeness through the monotone sequence theorem and define the constant e.*
- ❑ *Explore the Cauchy criterion and the Bolzano–Weierstrass theorem as completeness-based extensions.*

Limits turn approximation into an exact mathematical method. The area of a circle may be approached by areas of inscribed polygons, an irrational number by rational decimals, and a solution of an equation by repeated numerical approximations. In each case we obtain an ordered list of numbers and ask whether the terms settle near one fixed real number.

1.3.1 Sequences and the meaning of convergence

Definition 1.13

A real sequence (数列) *is a function*

$$a\colon \mathbb{N}_{>0} \to \mathbb{R}.$$

Its value at n is denoted by a_n, and the sequence is written $(a_n)_{n\geq 1}$ or simply (a_n).

♣

The index records order. For example, $(1,-1,1,-1,\ldots)$ and $(-1,1,-1,1,\ldots)$ have the same set of values but are different sequences. Some sequences are given by an explicit formula, while others are defined recursively from earlier terms.

Consider three examples:

$$a_n = \frac{1}{n}, \qquad b_n = 2 + \frac{(-1)^n}{n}, \qquad c_n = (-1)^n.$$

The first approaches 0 from one side. The second approaches 2 while alternating around it. The third continues to jump between -1 and 1 and does not approach any single number. Monotonicity is therefore not part of convergence; eventual closeness is.

Definition 1.14

Let $(a_n)_{n\geq 1}$ be a sequence and let $L \in \mathbb{R}$. We say that (a_n) converges (收敛) *to L if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}_{>0}$ such that*

$$n \geq N \quad \Longrightarrow \quad |a_n - L| < \varepsilon.$$

In this case we write

$$\lim_{n\to\infty} a_n = L \qquad \text{or} \qquad a_n \to L.$$

The number L is the limit (极限) *of the sequence. A sequence that has no real limit is called* divergent (发散).

♣

The definition has the logical form

$$\forall \varepsilon > 0 \quad \exists N \in \mathbb{N}_{>0} \quad \forall n \geq N : \quad |a_n - L| < \varepsilon.$$

The number N may depend on ε, but it must not depend on the later index n. We only need to find a suitable N; it need not be the smallest possible one.

The exact negation of the proposed limit $a_n \to L$ is

$$\exists \varepsilon_0 > 0 \quad \forall N \in \mathbb{N}_{>0} \quad \exists n \geq N : \quad |a_n - L| \geq \varepsilon_0.$$

Thus a divergence proof may fix one positive tolerance and find bad terms arbitrarily far out in the sequence. The tolerance must be independent of N; choosing a new tolerance for each tail would not negate convergence.

> ***Disproof Pattern.*** To show that a_n does not converge to a proposed value L, look for one number $\varepsilon_0 > 0$ and infinitely late indices whose terms remain at least ε_0 away from L. Two subsequences with different limits are a particularly efficient way to obtain such terms.

Geometrically, $|a_n - L| < \varepsilon$ means

$$L - \varepsilon < a_n < L + \varepsilon.$$

No matter how narrow the horizontal band around L is chosen, all sufficiently late terms must lie inside it.

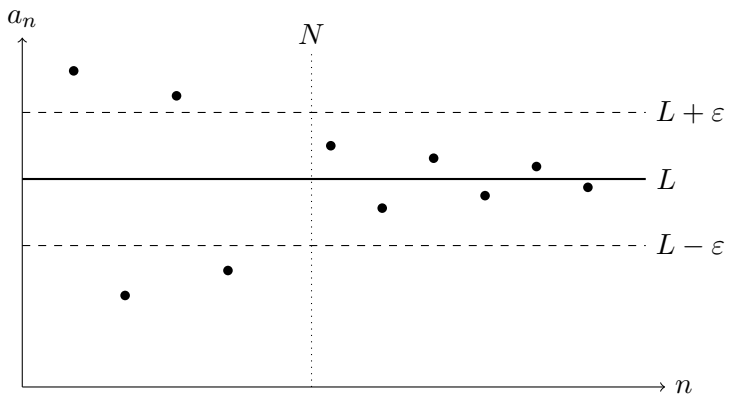


Figure 1.2: After a suitable index N, every term lies in the ε-band around L; the finitely many earlier terms are irrelevant to convergence.

> ***Question.*** Why does the definition require all $n \geq N$, rather than merely infinitely many values of n?

1.3.2 Direct proofs from the definition

An ε–N proof is usually discovered backward and written forward. We first ask what condition on n would guarantee $|a_n - L| < \varepsilon$. After determining N from this condition, the formal proof starts with an arbitrary $\varepsilon > 0$, chooses that N, and verifies the implication.

Example 1.15 A direct ε–N proof Prove that

$$a_n = 2 + \frac{(-1)^n}{n} \quad \longrightarrow \quad 2.$$

> ***Idea.*** The oscillation causes no difficulty because its magnitude is controlled by $1/n$:
>
> $$|a_n - 2| = \frac{1}{n}.$$
>
> Thus it is enough to make $n > 1/\varepsilon$.

Proof Let $\varepsilon > 0$. Choose

$$N = \left\lceil \frac{1}{\varepsilon} \right\rceil + 1.$$

If $n \geq N$, then $n > 1/\varepsilon$, and therefore

$$|a_n - 2| = \frac{|(-1)^n|}{n} = \frac{1}{n} < \varepsilon.$$

Hence $a_n \to 2$. □

Exact solution of the target inequality is often unnecessary. A simpler upper bound may reveal a workable N.

Example 1.16 Estimate first, choose N second Prove that

$$\lim_{n\to\infty} \frac{n+2}{n^2+1} = 0.$$

Proof For $n \geq 2$, we have $n + 2 \leq 2n$ and $n^2 + 1 \geq n^2$, so

$$0 \leq \frac{n+2}{n^2+1} \leq \frac{2}{n}.$$

Given $\varepsilon > 0$, choose

$$N = \max\left\{2, \left\lceil \frac{2}{\varepsilon} \right\rceil + 1\right\}.$$

Then, for $n \geq N$,

$$0 \leq \frac{n+2}{n^2+1} \leq \frac{2}{n} < \varepsilon.$$

This proves the limit directly from the definition. □

The following basic limits will be used repeatedly.

Proposition 1.8

Let $p > 0$, let $q \in \mathbb{R}$ satisfy $|q| < 1$, and let $a > 0$. Then

$$\lim_{n\to\infty} \frac{1}{n^p} = 0,$$

$$\lim_{n\to\infty} q^n = 0,$$

$$\lim_{n\to\infty} a^{1/n} = 1.$$

♠

Proof Given $\varepsilon > 0$, the Archimedean property provides N such that

$$N > \varepsilon^{-1/p}.$$

Then $n \geq N$ implies $1/n^p < \varepsilon$, proving the first limit.

The second limit is immediate when $q = 0$. If $0 < |q| < 1$, write

$$\frac{1}{|q|} = 1 + h$$

with $h > 0$. Bernoulli's inequality gives $(1 + h)^n \geq 1 + nh$, so

$$0 < |q|^n = \frac{1}{(1+h)^n} \leq \frac{1}{1+nh} < \frac{1}{nh}.$$

The first limit, with $p = 1$, now proves $q^n \to 0$.

Suppose first that $a \geq 1$, and set $h_n = a^{1/n} - 1 \geq 0$. Since

$$a = (1 + h_n)^n \geq 1 + nh_n,$$

we obtain

$$0 \leq h_n \leq \frac{a-1}{n} \to 0.$$

Thus $a^{1/n} = 1 + h_n \to 1$. If $0 < a < 1$, put $b = 1/a > 1$. The proved case gives $b^{1/n} \to 1$, and

$$0 \le 1 - a^{1/n} = \frac{b^{1/n} - 1}{b^{1/n}} \le b^{1/n} - 1 \to 0.$$

Hence $a^{1/n} \to 1$ without using the quotient law, which will be proved later. □

1.3.3 First structural consequences

Theorem 1.3

A convergent sequence has exactly one limit. ♡

> ***Idea.*** Two distinct candidate limits have disjoint sufficiently small neighbourhoods. A late term of the sequence cannot belong to both neighbourhoods at once.

Proof Suppose that $a_n \to L$ and $a_n \to M$. If $L \neq M$, set

$$\varepsilon = \frac{|L - M|}{3} > 0.$$

There exist $N_1, N_2 \in \mathbb{N}_{>0}$ such that

$$n \ge N_1 \Longrightarrow |a_n - L| < \varepsilon, \qquad n \ge N_2 \Longrightarrow |a_n - M| < \varepsilon.$$

For $n \ge \max\{N_1, N_2\}$, the triangle inequality gives

$$|L - M| \le |L - a_n| + |a_n - M| < 2\varepsilon = \frac{2}{3}|L - M|,$$

a contradiction. Therefore $L = M$. □

Proposition 1.9

Changing, adding, or deleting finitely many initial terms does not affect whether a sequence converges or what its limit is. In particular, if $a_n \to L$ and $k \in \mathbb{N}$ is fixed, then

$$a_{n+k} \to L.$$

♠

Proof Suppose two sequences agree for every $n \ge n_0$. For a given $\varepsilon > 0$, choose N for the convergent sequence and replace it by $\max\{N, n_0\}$. The same ε-estimate then holds for the other sequence. The statement about shifts follows by taking a tail and reindexing it. □

Definition 1.15

A sequence (a_n) is a bounded sequence(有界数列) *if there exists $M \ge 0$ such that*

$$|a_n| \le M$$

for every $n \ge 1$. ♣

Theorem 1.4

Every convergent sequence is bounded. ♡

Proof Suppose $a_n \to L$. Taking $\varepsilon = 1$, choose N such that

$$n \ge N \quad \Longrightarrow \quad |a_n - L| < 1.$$

For such n,

$$|a_n| \le |a_n - L| + |L| < 1 + |L|.$$

Now choose

$$M = \max\{|a_1|, \dots, |a_{N-1}|, 1 + |L|\},$$

with the finite list omitted if $N = 1$. Then $|a_n| \le M$ for every $n \ge 1$. □

The converse is false. The sequence $((-1)^n)$ is bounded but divergent. Boundedness prevents escape to infinity, but it does not prevent persistent oscillation.

1.3.4 Arithmetic laws and squeeze arguments

Theorem 1.5

Suppose that $a_n \to A$ and $b_n \to B$, and let $c \in \mathbb{R}$. Then

$$a_n + b_n \to A + B,$$
$$a_n - b_n \to A - B,$$
$$ca_n \to cA,$$
$$a_n b_n \to AB.$$

If $B \neq 0$, then $b_n \neq 0$ for all sufficiently large n and

$$\frac{a_n}{b_n} \to \frac{A}{B}.$$

♡

> ***Proof Strategy.*** For sums, split the final error into two errors. For products, add and subtract a convenient intermediate term and use boundedness of a convergent sequence. For quotients, first prove that a denominator converging to a nonzero number stays away from zero.

Proof Given $\varepsilon > 0$, choose N so that for $n \ge N$,

$$|a_n - A| < \frac{\varepsilon}{2}, \qquad |b_n - B| < \frac{\varepsilon}{2}.$$

Then

$$|(a_n + b_n) - (A + B)| \le |a_n - A| + |b_n - B| < \varepsilon.$$

The difference and constant-multiple laws follow by the same estimate.

By Theorem 1.4, choose $M \ge 0$ such that $|a_n| \le M$ for all n. Let

$$C = M + |B| + 1.$$

Choose N so that, for $n \ge N$,

$$|a_n - A| < \frac{\varepsilon}{C}, \qquad |b_n - B| < \frac{\varepsilon}{C}.$$

Then

$$\begin{aligned} |a_n b_n - AB| &= |a_n(b_n - B) + B(a_n - A)| \\ &\le M\,|b_n - B| + |B|\,|a_n - A| \\ &< \frac{M + |B|}{C}\varepsilon < \varepsilon. \end{aligned}$$

Thus $a_n b_n \to AB$.

Finally suppose $B \neq 0$. For all sufficiently large n,

$$|b_n - B| < \frac{|B|}{2},$$

so the reverse triangle inequality gives

$$|b_n| > \frac{|B|}{2} > 0.$$

For these indices,

$$\left|\frac{1}{b_n} - \frac{1}{B}\right| = \frac{|b_n - B|}{|b_n|\,|B|} \le \frac{2}{|B|^2}\,|b_n - B| \to 0.$$

Therefore $1/b_n \to 1/B$, and the quotient law follows from the product law. □

Example 1.17 An infinite repeating decimal Prove that $0.999\ldots = 1$ by interpreting the infinite decimal as the limit of its finite truncations

$$s_n = \underbrace{0.99\ldots9}_{n \text{ digits}} = \frac{9}{10} + \frac{9}{10^2} + \cdots + \frac{9}{10^n}.$$

> ***Idea.*** An infinite decimal has no last digit at which to perform a final subtraction. Instead, first ask how far the finite decimal s_n is from 1. Its digits form a finite geometric sum, so we can calculate this error exactly before taking a limit.

Proof 1: finite sums and limit laws For each positive integer n, the finite geometric-sum identity gives

$$s_n = \frac{9}{10}\,\frac{1 - 10^{-n}}{1 - 1/10} = 1 - 10^{-n}.$$

Since $(1/10)^n \to 0$ by Proposition 1.8, Theorem 1.5 yields

$$\lim_{n\to\infty} s_n = 1 - 0 = 1.$$

The notation $0.999\ldots$ denotes this limit, whose existence has now been established. Thus $0.999\ldots = 1$. □

Proof 2: direct control of the error The same finite identity gives $|s_n - 1| = 10^{-n}$. Bernoulli's inequality gives $10^n = (1+9)^n \ge 1 + 9n$. Given $\varepsilon > 0$, choose a positive integer $N > 1/(9\varepsilon)$. For every $n \ge N$,

$$|s_n - 1| = 10^{-n} \le \frac{1}{1+9n} < \frac{1}{9n} \le \frac{1}{9N} < \varepsilon.$$

This proves $s_n \to 1$ directly from the definition. □

The first proof uses the arithmetic theorem to keep the calculation short. The second gives an explicit, though not optimal, choice of N and displays the error estimate behind the limit. Both use only finite sums; neither assumes a rule for manipulating infinitely many summands. Every finite truncation is strictly less than 1, but its positive gap 10^{-n} becomes smaller than any prescribed positive number. There is no fixed positive gap left in the limit. The equality therefore identifies two decimal representations of the same real number, rather than saying that a finite truncation ever reaches 1.

> ***A third route: first identify a possible limit.*** Multiplying a finite truncation by 10 exposes its first digit and a shorter truncation. This suggests an equation for a possible limit, but taking limits requires convergence first.

Candidate proof 3: assuming convergence Assume that (s_n) converges to a finite real number L. For every $n \ge 2$, finite decimal arithmetic gives

$$10s_n = 9 + \frac{9}{10} + \cdots + \frac{9}{10^{n-1}} = 9 + s_{n-1}.$$

The shifted sequence $(s_{n-1})_{n\ge2}$ also tends to L by Proposition 1.9. Thus the arithmetic laws allow us to take limits on both sides:

$$10L = 9 + L, \qquad L = 1.$$

Under the stated convergence assumption, this proves $0.999\ldots = 1$. □

Convergence is assumed, not proved here. The later Example 1.24 supplies existence, completing a third route that separates existence from evaluation without a closed form for s_n.

Example 1.18 A rational expression Evaluate

$$\lim_{n\to\infty} \frac{n^2+3n+10}{3n^2-n+2}.$$

Proof Using the arithmetic laws,

$$\begin{aligned}\lim_{n\to\infty} \frac{n^2+3n+10}{3n^2-n+2} &= \lim_{n\to\infty} \frac{1+3/n+10/n^2}{3-1/n+2/n^2} \\ &= \frac{1}{3}.\end{aligned}$$

□

> ***Technique.*** Dividing by the highest power of n turns the expression into combinations of sequences with known limits.

Example 1.19 Rationalization reveals the dominant scale Evaluate

$$\lim_{n\to\infty} \left(\sqrt{n^2+n}-n\right).$$

> ***Idea.*** The two terms both have size n, so their leading parts cancel. Multiplying by the conjugate turns this hidden cancellation into a quotient with a nonzero limiting denominator.

Proof Rationalization gives

$$\begin{aligned}\sqrt{n^2+n}-n &= \frac{n}{\sqrt{n^2+n}+n} \\ &= \frac{1}{\sqrt{1+1/n}+1}.\end{aligned}$$

Moreover,

$$0 \le \sqrt{1+\frac{1}{n}}-1 = \frac{1/n}{\sqrt{1+1/n}+1} \le \frac{1}{2n}.$$

Hence $\sqrt{1+1/n} \to 1$ by choosing N with $\frac{1}{2N} < \epsilon$ for a fixed $\epsilon > 0$. The quotient law now yields

$$\lim_{n\to\infty} \left(\sqrt{n^2+n}-n\right) = \frac{1}{2}.$$

□

> ***Caution.*** The hypotheses in Theorem 1.5 matter. From convergence of (a_n+b_n) one cannot conclude that (a_n) and (b_n) converge. For example, $a_n = n$ and $b_n = -n$ both diverge, while $a_n + b_n = 0$. Likewise, the theorem concerns finitely many arithmetic operations; it does not justify termwise limiting of sums whose number of terms grows with n.

The following squeeze theorem(夹逼定理)turns a useful two-sided estimate into a limit.

Theorem 1.6 (Squeeze Theorem)

Suppose that for all sufficiently large n,

$$p_n \le q_n \le r_n.$$

If $p_n \to L$ and $r_n \to L$, then $q_n \to L$. ♡

Proof Let $\varepsilon > 0$. For all sufficiently large n,

$$L-\varepsilon < p_n \le q_n \le r_n < L+\varepsilon.$$

Therefore $|q_n - L| < \varepsilon$ for all sufficiently large n, which is exactly $q_n \to L$. □

Example 1.20 An oscillating sequence controlled by its size Prove that

$$\lim_{n\to\infty} \frac{\sin n}{n} = 0.$$

Proof Since

$$-\frac{1}{n} \le \frac{\sin n}{n} \le \frac{1}{n}$$

and both bounding sequences converge to 0, Theorem 1.6 gives

$$\lim_{n\to\infty} \frac{\sin n}{n} = 0.$$

□

> ***Interpretation.*** The numerator itself does not converge; the shrinking factor $1/n$ controls the oscillation.

Example 1.21 Dominate an oscillating term Prove that the sequence

$$a_n = \frac{n^{2/3} \sin n}{n+1}$$

converges to 0.

> ***Idea.*** Do not try to track the oscillation of $\sin n$; replace it by the uniform bound $|\sin n| \le 1$.

Proof

$$|a_n| \le \frac{n^{2/3}}{n+1} \le \frac{1}{n^{1/3}}.$$

The right-hand side tends to 0, so Theorem 1.6 gives $a_n \to 0$. □

> ***Technique.*** The reusable step is to replace a complicated factor by a majorant that has a known limit.

Example 1.22 Normalize a root of a sum Let $m \ge 2$ be a fixed integer and set

$$s_n = \left(2^n + 3^n + \cdots + m^n\right)^{1/n}.$$

Find $\lim_{n\to\infty} s_n$.

Proof The largest exponential supplies the natural scale:

$$m^n \le 2^n + 3^n + \cdots + m^n \le (m-1)m^n.$$

Taking positive nth roots gives

$$m \le s_n \le m(m-1)^{1/n}.$$

Since $(m-1)^{1/n} \to 1$, the squeeze theorem yields $s_n \to m$. □

> ***Technique.*** This is a typical instance of normalizing by the dominant term before estimating.

Proposition 1.10

Let $a > 1$ and let $k \in \mathbb{N}_{>0}$. Then

$$\lim_{n\to\infty} \frac{n^k}{a^n} = 0, \qquad \lim_{n\to\infty} \frac{a^n}{n!} = 0.$$

Thus exponentials eventually dominate fixed powers, while factorials eventually dominate fixed exponentials. ♠

Proof Write $a = 1 + h$ with $h > 0$. For sufficiently large n, the binomial theorem gives

$$\begin{aligned} a^n &\geq \binom{n}{k+1} h^{k+1} \\ &= \frac{n(n-1)\cdots(n-k)}{(k+1)!} h^{k+1} \\ &\geq \frac{(n/2)^{k+1}}{(k+1)!} h^{k+1}. \end{aligned}$$

Consequently,

$$0 \leq \frac{n^k}{a^n} \leq \frac{2^{k+1}(k+1)!}{h^{k+1}} \frac{1}{n} \to 0.$$

Choose $N_0 \in \mathbb{N}_{>0}$ with $N_0 > 2a$. For $n > N_0$,

$$0 \leq \frac{a^n}{n!} = \frac{a^{N_0}}{N_0!} \prod_{j=N_0+1}^{n} \frac{a}{j} \leq \frac{a^{N_0}}{N_0!} \left(\frac{1}{2}\right)^{n-N_0}.$$

The right-hand side tends to 0 by Proposition 1.8, so the squeeze theorem completes the proof. □

Corollary 1.1 (A hierarchy of dominant scales)

Let $\alpha > 0$ and $a > 1$. As $n \to \infty$,

$$\ln n = o(n^\alpha), \qquad n^\alpha = o(a^n), \qquad a^n = o(n!), \qquad n! = o(n^n),$$

where $u_n = o(v_n)$ means $u_n / v_n \to 0$. Equivalently, one may remember the scale ordering

$$\ln n \ll n^\alpha \ll a^n \ll n! \ll n^n.$$

♡

Proof For the first comparison, let $m = \lfloor \log_2 n \rfloor$. Then $2^m \leq n < 2^{m+1}$, and hence

$$0 \leq \frac{\ln n}{n^\alpha} \leq \frac{(m+1)\ln 2}{2^{\alpha m}} \longrightarrow 0$$

by Proposition 1.10, applied to the base $2^\alpha > 1$. Choose an integer $k > \alpha$. For $n \geq 1$, $n^\alpha \leq n^k$, so

$$0 \leq \frac{n^\alpha}{a^n} \leq \frac{n^k}{a^n} \longrightarrow 0.$$

The third comparison is already contained in Proposition 1.10. Finally,

$$0 \leq \frac{n!}{n^n} = \prod_{j=1}^{n} \frac{j}{n} \leq \left(\frac{1}{2}\right)^{\lfloor n/2 \rfloor} \longrightarrow 0,$$

because the first $\lfloor n/2 \rfloor$ factors are at most $1/2$ and all remaining factors are at most 1. □

> ***How to use the hierarchy.*** In a sum of positive quantities with different scales, divide by the largest plausible scale. If every other normalized term tends to 0, the dominant term determines the asymptotic behaviour. The parameter restrictions matter: the displayed hierarchy assumes $\alpha > 0$ and $a > 1$.

Example 1.23 The slowly varying sequence $n^{1/n}$ Prove that $n^{1/n} \to 1$.

Proof Put $h_n = n^{1/n} - 1 \geq 0$. For $n \geq 2$, the binomial theorem gives

$$n = (1 + h_n)^n \geq \binom{n}{2} h_n^2 = \frac{n(n-1)}{2} h_n^2.$$

Hence

$$0 \leq h_n \leq \sqrt{\frac{2}{n-1}} \to 0.$$

Therefore

$$n^{1/n} \to 1, \qquad n^{1/n} - 1 \to 0.$$

□

> ***Why a new estimate is needed.*** The exponent and the base both vary, so this conclusion does not follow from the fixed-base limit in Proposition 1.8; the binomial estimate supplies the missing control.

Proposition 1.11 (Cesàro means)

If $a_n \to L$, then the arithmetic means satisfy

$$\frac{a_1 + a_2 + \cdots + a_n}{n} \longrightarrow L.$$

♠

> ***Proof Strategy.*** The sum contains a number of terms that grows with n, so the finite-sum limit law does not apply directly. Separate a fixed initial block from a tail on which every error is uniformly small.

Proof Let $\varepsilon > 0$. Choose N such that $|a_k - L| < \varepsilon/2$ whenever $k \geq N$, and set

$$C = \sum_{k=1}^{N-1} |a_k - L|,$$

with $C = 0$ when $N = 1$. For all sufficiently large $n \geq N$, one has $C/n < \varepsilon/2$, and then

$$\begin{aligned} \left| \frac{1}{n} \sum_{k=1}^{n} a_k - L \right| &\leq \frac{1}{n} \sum_{k=1}^{n} |a_k - L| \\ &\leq \frac{C}{n} + \frac{1}{n} \sum_{k=N}^{n} \frac{\varepsilon}{2} < \varepsilon. \end{aligned}$$

Thus the means converge to L. □

1.3.5 Order and subsequences

Theorem 1.7

Suppose that $a_n \to A$ and $b_n \to B$.

(i) If $A < B$, then $a_n < b_n$ for all sufficiently large n.

(ii) If $a_n \leq b_n$ for all sufficiently large n, then $A \leq B$.

In particular, if $A > 0$, then $a_n > 0$ for all sufficiently large n.

♡

Proof For (i), let $\varepsilon = (B - A)/3$. For all sufficiently large n,

$$a_n < A + \varepsilon < B - \varepsilon < b_n.$$

For (ii), suppose instead that $A > B$. Part (i), with the two sequences interchanged, would give $a_n > b_n$ for all sufficiently large n, contradicting the hypothesis. Thus $A \leq B$. The final statement follows by comparing (a_n) with the constant zero sequence. □

Strict inequalities need not survive in the limit. Although $1/n < 2/n$ for every n, both sequences have limit 0.

Definition 1.16

Let (a_n) be a sequence, and let

$$n_1 < n_2 < n_3 < \cdots$$

be positive integers. The sequence $(a_{n_j})_{j \geq 1}$ is called a subsequence (子列) *of (a_n).*

♣

Theorem 1.8

If $a_n \to L$, then every subsequence (a_{n_j}) also converges to L.

♡

Proof Let $\varepsilon > 0$. Choose N such that $n \geq N$ implies $|a_n - L| < \varepsilon$. Since $n_j \geq j$, every $j \geq N$ satisfies $n_j \geq N$, and hence

$$\left|a_{n_j} - L\right| < \varepsilon.$$

Thus $a_{n_j} \to L$. □

The contrapositive is a powerful divergence test: if a sequence has two subsequences converging to different limits, then the original sequence diverges. For $a_n = (-1)^n$, the even subsequence is constantly 1 and the odd subsequence is constantly -1.

> ***Comment.*** Finding one convergent subsequence does not prove that the original sequence converges. The sequence $((-1)^n)$ already has two convergent subsequences. What convergence requires is that every sufficiently late term, not merely a selected infinite collection, lie close to one common limit.

1.3.6 Monotone sequences and completeness

Definition 1.17

A sequence (a_n) is nondecreasing (单调不减) *if $a_n \leq a_{n+1}$ for every n, and* nonincreasing (单调不增) *if $a_n \geq a_{n+1}$ for every n. A sequence of either type is called a* monotone sequence (单调数列).

♣

Theorem 1.9 (Monotone Sequence Theorem)

Every nondecreasing sequence that is bounded above converges. Its limit is the supremum of its set of values. Every nonincreasing sequence that is bounded below converges, with limit equal to the corresponding infimum.

♡

> ***Idea.*** Monotonicity prevents endless oscillation, while boundedness prevents escape to infinity. Completeness supplies the boundary value toward which the sequence is forced to move.

Proof Suppose that (a_n) is nondecreasing and bounded above. Let

$$S = \{a_n \mid n \geq 1\}, \qquad L = \sup S.$$

The number L exists by the least-upper-bound property.

Let $\varepsilon > 0$. Since $L - \varepsilon$ is not an upper bound of S, there exists $N \in \mathbb{N}_{>0}$ such that

$$a_N > L - \varepsilon.$$

For every $n \geq N$, monotonicity and the fact that L is an upper bound give

$$L - \varepsilon < a_N \leq a_n \leq L < L + \varepsilon.$$

Hence $|a_n - L| < \varepsilon$, so $a_n \to L$.

The nonincreasing case follows by applying the proved result to $(-a_n)$, or directly by using the infimum. □

Example 1.24 Convergence of the repeating-nine truncations For

$$s_n = \frac{9}{10} + \frac{9}{10^2} + \cdots + \frac{9}{10^n}, \qquad n \geq 1,$$

prove that (s_n) is increasing and bounded, and hence convergent. Then complete Candidate proof 3 in Example 1.17.

> ***Idea.*** Each new digit adds a positive term, suggesting monotonicity. For an upper bound, the digit-shift relation writes the next term as $(9 + s_n)/10$: if the current term lies between 0 and 1, so does the next. Induction can therefore establish a bound without assuming a limit value or using the earlier formula for s_n.

Proof For every $n \geq 1$,

$$s_{n+1} - s_n = \frac{9}{10^{n+1}} > 0,$$

so (s_n) is strictly increasing. We next prove $0 < s_n < 1$ for all $n \geq 1$ by induction. The assertion holds for $s_1 = 9/10$. If $0 < s_n < 1$, then finite decimal arithmetic gives

$$0 < s_{n+1} = \frac{9 + s_n}{10} < \frac{9 + 1}{10} = 1.$$

This completes the induction and proves boundedness.

The sequence is nondecreasing and bounded above by 1, so Theorem 1.9 now guarantees that $s_n \to L$ for some $L \in \mathbb{R}$. Only at this point do we use the conditional calculation in Example 1.17: taking limits in $10s_n = 9 + s_{n-1}$ gives $10L = 9 + L$, hence $L = 1$. This supplies the missing existence step and completes the third proof of $0.999\ldots = 1$. □

Example 1.25 A recursively defined sequence Let

$$x_1 = 1, \qquad x_{n+1} = \sqrt{2 + x_n}.$$

Prove that (x_n) converges and determine its limit.

> ***Idea.*** Establish convergence from monotonicity and boundedness before solving the fixed-point equation for a possible limit.

Proof First, induction gives $1 \leq x_n < 2$ for every n. Indeed, the assertion holds for $x_1 = 1$; if $1 \leq x_n < 2$, then

$$1 < \sqrt{3} \leq x_{n+1} = \sqrt{2 + x_n} < 2.$$

For $1 \leq x_n < 2$,

$$x_{n+1}^2 - x_n^2 = 2 + x_n - x_n^2 = (2 - x_n)(x_n + 1) > 0.$$

Since both terms are positive, $x_{n+1} > x_n$. Thus (x_n) is increasing and bounded above by 2, so Theorem 1.9 gives a limit $L \in [1, 2]$.

The shifted sequence (x_{n+1}) has the same limit by Proposition 1.9. Taking limits in

$$x_{n+1}^2 = 2 + x_n$$

and using Theorem 1.5, we obtain

$$L^2 = 2 + L.$$

Hence $L = 2$ or $L = -1$. Because $L \in [1, 2]$, the limit is $L = 2$. □

> ***Common Pitfall.*** For a recursive sequence, solving the fixed-point equation gives only possible limits. It does not prove that the sequence converges. One must first establish convergence, often by monotonicity and boundedness, and only then pass to the limit in the recurrence.

1.3.7 The natural constant e

The monotone sequence theorem produces the *natural constant* (自然常数) e, one of the central constants of mathematics.

Theorem 1.10 (The Fundamental Sequence Defining e)

The sequence

$$e_n = \left(1 + \frac{1}{n}\right)^n$$

is strictly increasing and bounded above. Consequently, it converges. Its limit is denoted by

$$e = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n. \tag{1.3}$$

Moreover, $2 < e < 3$.

♡

> ***Proof Strategy.*** The binomial expansion separates e_n into positive terms. For each fixed position in the expansion, increasing n enlarges the corresponding factor, which proves monotonicity. Replacing all those factors by 1 gives a convergent geometric upper estimate.

Proof By the binomial theorem,

$$\begin{aligned} e_n &= \sum_{k=0}^{n} \binom{n}{k} \frac{1}{n^k} \\ &= \sum_{k=0}^{n} \frac{1}{k!} \prod_{j=0}^{k-1} \left(1 - \frac{j}{n}\right), \end{aligned}$$

where the empty product for $k = 0$ equals 1. For every fixed $k \le n$, replacing n by $n+1$ increases each nonconstant factor in the product, and the expansion of e_{n+1} also has one additional positive term. Thus $e_{n+1} > e_n$.

Every factor in each product lies in $[0, 1]$, so

$$e_n \le \sum_{k=0}^{n} \frac{1}{k!}.$$

For $k \ge 3$, we have $k! \ge 6 \cdot 2^{k-3}$. Therefore

$$\begin{aligned} e_n &\le 1 + 1 + \frac{1}{2} + \sum_{k=3}^{\infty} \frac{1}{6 \cdot 2^{k-3}} \\ &= \frac{5}{2} + \frac{1}{6} \sum_{j=0}^{\infty} \frac{1}{2^j} \\ &= \frac{17}{6} < 3. \end{aligned}$$

Hence (e_n) is increasing and bounded above, so it converges by Theorem 1.9. Since $e_n > e_1 = 2$ for $n > 1$, its limit satisfies $2 < e < 3$. □

The constant e is the base of the natural logarithm. It first arose in the study of continuously compounded interest, but it now appears throughout calculus, differential equations, probability, and complex analysis.

Example 1.26 A companion limit Evaluate

$$\lim_{n\to\infty} \left(1 - \frac{1}{n}\right)^n.$$

Proof For $n \ge 2$, put $m = n - 1$. Then

$$\begin{aligned} \left(1 - \frac{1}{n}\right)^n &= \left(\frac{m}{m+1}\right)^{m+1} \\ &= \frac{1}{(1 + 1/m)^m} \cdot \frac{1}{1 + 1/m}. \end{aligned}$$

By (1.3), the shift property, and the quotient law,

$$\lim_{n\to\infty}\left(1-\frac{1}{n}\right)^n = e^{-1}.$$

□

1.3.8 Completeness extensions: monotone subsequences and Cauchy sequences

The next two results are not needed for the definition of e, but they reveal further consequences of completeness and will be useful when continuous functions on closed intervals are studied.

Lemma 1.1 (Monotone Subsequence Lemma)

Every real sequence has a monotone subsequence. ♡

Proof Call an index n a *peak* if

$$a_n \geq a_m \qquad \text{for every } m > n.$$

If there are infinitely many peaks, list them as

$$n_1 < n_2 < n_3 < \cdots.$$

Since each n_j is a peak, $a_{n_j} \geq a_{n_{j+1}}$, so (a_{n_j}) is nonincreasing.

If there are only finitely many peaks, choose n_1 larger than all of them. Then n_1 is not a peak, so some $n_2 > n_1$ satisfies $a_{n_2} > a_{n_1}$. The same argument can be repeated after n_2, producing

$$n_1 < n_2 < n_3 < \cdots, \qquad a_{n_1} < a_{n_2} < a_{n_3} < \cdots.$$

Thus the selected subsequence is strictly increasing. □

Theorem 1.11 (Bolzano–Weierstrass Theorem)

Every bounded real sequence has a convergent subsequence. ♡

> ***Idea.*** The preceding lemma supplies a monotone subsequence. Boundedness is inherited by subsequences, so completeness can be applied through the monotone sequence theorem.

Proof Let (a_n) be bounded. By Lemma 1.1, it has a monotone subsequence (a_{n_j}). This subsequence is still bounded. If it is nondecreasing, it is bounded above; if it is nonincreasing, it is bounded below. In either case, Theorem 1.9 shows that (a_{n_j}) converges. □

Definition 1.18

A sequence (a_n) is a Cauchy sequence (柯西数列) *if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}_{>0}$ such that*

$$m, n \geq N \quad \Longrightarrow \quad |a_m - a_n| < \varepsilon.$$

♣

The definition compares late terms with one another and does not require a candidate limit. The following Cauchy convergence criterion (柯西收敛准则) says that, over $\mathbb{R}$, this internal closeness is exactly convergence.

Theorem 1.12 (Cauchy Convergence Criterion)

A real sequence converges if and only if it is a Cauchy sequence. ♡

> ***Proof Strategy.*** A convergent sequence has late terms close to the same limit, so they are close to one another. Conversely, a Cauchy sequence is bounded; Bolzano–Weierstrass then gives one convergent subsequence, and the Cauchy condition pulls every late term toward that subsequential limit.

Proof Suppose first that $a_n \to L$. Given $\varepsilon > 0$, choose N such that

$$n \geq N \quad \Longrightarrow \quad |a_n - L| < \frac{\varepsilon}{2}.$$

If $m, n \geq N$, then

$$|a_m - a_n| \leq |a_m - L| + |a_n - L| < \varepsilon.$$

Thus (a_n) is Cauchy.

Conversely, suppose that (a_n) is Cauchy. Taking $\varepsilon = 1$, choose N_0 such that

$$m, n \geq N_0 \quad \Longrightarrow \quad |a_m - a_n| < 1.$$

For $n \geq N_0$, comparison with a_{N_0} gives

$$|a_n| \leq |a_{N_0}| + 1.$$

Together with the finitely many preceding terms, this proves that (a_n) is bounded.

By Theorem 1.11, some subsequence (a_{n_j}) converges to a number L. Given $\varepsilon > 0$, choose N_1 so that

$$m, n \geq N_1 \quad \Longrightarrow \quad |a_m - a_n| < \frac{\varepsilon}{2}.$$

Choose j large enough that $n_j \geq N_1$ and

$$\left|a_{n_j} - L\right| < \frac{\varepsilon}{2}.$$

Then every $n \geq N_1$ satisfies

$$|a_n - L| \leq \left|a_n - a_{n_j}\right| + \left|a_{n_j} - L\right| < \varepsilon.$$

Hence $a_n \to L$. □

> ***Question.*** Where does completeness of $\mathbb{R}$ enter the proof of the Cauchy criterion, and why can the corresponding statement fail for a sequence considered only inside $\mathbb{Q}$?

1.3.9 Key Concepts, Strategies, and Common Pitfalls

The core concepts are convergence, boundedness, subsequences, monotonicity, and completeness. The extension through Cauchy sequences shows that convergence can also be detected without knowing the limit in advance. For a proposed sequence limit, ask the following questions in order.

1. Can the target error $|a_n - L|$ be simplified or bounded by a familiar quantity?
2. Do the arithmetic laws apply, or are some component sequences divergent?
3. Can an oscillating or complicated term be squeezed by its absolute value?
4. Would two subsequences expose different limiting behaviours?
5. For a recursive sequence, can monotonicity and boundedness be proved before solving the fixed-point equation?
6. For a sum or an nth root, what is the dominant scale, and what happens after normalization by it?
7. If the number of summands grows with n, can the expression be split into a fixed head and a uniformly controlled tail?

Keep the following logical points explicit.

- In an ε–N proof, N may depend on ε but not on n.
- Finitely many exceptional terms do not affect a limit.
- Boundedness is necessary for convergence but not sufficient.
- An eventual strict inequality may become equality after taking limits.
- Limit laws for a finite number of operations do not justify an increasing number of additions or multipli-

cations.

> ***Looking Ahead.*** The definition of a function limit has the same logical architecture as sequence convergence. The discrete tail condition $n \geq N$ will be replaced by the geometric condition $0 < |x - a| < \delta$, while estimation and backward choice of parameters remain the central proof methods.

Exercise 1.3

1. Use the definition to prove
$$\lim_{n\to\infty} \frac{3n+1}{2n-3} = \frac{3}{2}.$$
Give an explicit valid choice of N in terms of ε.
2. Use the definition to prove that $n^{-p} \to 0$ for $p > 0$ without citing Proposition 1.8.
3. Prove directly from the definition that if $a_n \to L$, then $|a_n| \to |L|$. Is the converse true?
4. Prove that every convergent sequence is eventually contained in the interval $(L-1, L+1)$, but explain why this alone does not establish boundedness of the entire sequence.
5. Evaluate
$$\lim_{n\to\infty} \frac{4n^3 - n + 7}{2n^3 + 5n^2 - 1}, \qquad \lim_{n\to\infty} \left(\sqrt{n+1} - \sqrt{n}\right).$$
State which limit law or estimate is used at each step.
6. Use the squeeze theorem to find
$$\lim_{n\to\infty} \frac{\cos(n^2)}{\sqrt{n}}, \qquad \lim_{n\to\infty} \left(2^{2n} + 3^n\right)^{1/n}.$$
7. Show that $a_n = (-1)^n + 1/n$ diverges by finding two subsequences with different limits.
8. Prove that a sequence converges to L if and only if its even and odd subsequences both converge to L.
9. Let $x_1 = 2$ and
$$x_{n+1} = \frac{1}{2}\left(x_n + \frac{3}{x_n}\right).$$
Prove that (x_n) converges and determine its limit.
10. Prove that
$$\lim_{n\to\infty} \frac{n^{10}}{2^n} = 0$$
by the method of Proposition 1.10.
11. Prove that
$$\left(1 + \frac{1}{n}\right)^{n+1}$$
is decreasing and bounded below by e.
12. Explain precisely why the calculation
$$\lim_{n\to\infty} \left(\frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{2n}\right) = 0$$
cannot be justified by saying that each summand tends to 0. Find a positive lower bound for the displayed sequence.
13. Suppose that (a_n) and (b_n) are Cauchy sequences. Prove directly from the definition that $(a_n + b_n)$ is Cauchy, and explain where completeness is used when one concludes that the sum sequence converges.
14. If $a_n \to L$ and $m \in \mathbb{N}$ is fixed, prove directly from the definition that $a_{n+m} \to L$. Explain why the conclusion need not hold when the shift m is replaced by an integer sequence m_n satisfying $n + m_n \geq 1$

but for which $n + m_n$ does not tend to $+\infty$.

15. Suppose that (a_n) is bounded and $b_n \to 0$. Prove, without invoking the product law for limits, that $a_n b_n \to 0$.

16. For $a, b > 0$, prove that
$$\lim_{n\to\infty} (a^n + b^n)^{1/n} = \max\{a, b\}.$$

17. Evaluate
$$\lim_{n\to\infty} \sqrt[n]{n^2 + n}.$$
Give upper and lower bounds that make the squeeze theorem applicable.

18. Let $H_n = 1 + 1/2 + \cdots + 1/n$. Prove that
$$\lim_{n\to\infty} \sqrt[n]{H_n} = 1$$
using only the elementary bounds $1 \le H_n \le n$ and previously established basic limits.

19. If $a_n \to a$ and $b_n \to b$, prove that
$$\max\{a_n, b_n\} \to \max\{a, b\}, \qquad \min\{a_n, b_n\} \to \min\{a, b\}.$$
You may use formulas involving $|a_n - b_n|$.

20. If $a_n \to L$, prove the weighted Cesàro conclusion
$$\frac{a_1 + 2a_2 + \cdots + na_n}{1 + 2 + \cdots + n} \longrightarrow L.$$

> ***Hint.*** Apply the fixed-head/small-tail idea from Proposition 1.11 to the errors $a_k - L$. The weighted contribution of the fixed head is divided by $1 + 2 + \cdots + n$, while every tail error is bounded by the same prescribed tolerance.

Supplement Exercises

S1. For each $n \ge 2$, let $x_n \in (0, 1)$ be the unique solution of
$$x + x^2 + \cdots + x^n = 1.$$
Prove that (x_n) is monotone and bounded, and determine its limit.

S2. Define
$$r_n = \sqrt{1 + \sqrt{2 + \sqrt{3 + \cdots + \sqrt{n}}}}.$$
Prove that (r_n) is increasing and bounded above, and hence convergent. Can the simple bound used in your proof determine the limit exactly?

S3. Let $a > 0$ and choose $x_1 > \sqrt{a}$. Define
$$x_{n+1} = \frac{1}{2}\left(x_n + \frac{a}{x_n}\right).$$
Prove that $x_n \to \sqrt{a}$ and obtain an identity relating $x_{n+1} - \sqrt{a}$ to $(x_n - \sqrt{a})^2$. What does this identity say about the speed of convergence?

S4. Let $F_1 = F_2 = 1$ and $F_{n+1} = F_n + F_{n-1}$. By studying the even and odd subsequences of F_{n+1}/F_n, prove that
$$\lim_{n\to\infty} \frac{F_{n+1}}{F_n} = \frac{1 + \sqrt{5}}{2}.$$

S5. Prove the following subsequence criterion: a sequence (a_n) converges to L if and only if every subsequence of (a_n) has a further subsequence that converges to L.

> ***Hint.*** For the nontrivial direction, negate the ε–N definition and construct a subsequence that stays a fixed positive distance from L.

S6. Let

$$s_n = \sum_{k=1}^{n} \frac{1}{k^2}.$$

Prove directly that (s_n) is a Cauchy sequence by comparing $1/k^2$ with $1/(k(k-1))$ for $k \geq 2$. Explain exactly where completeness is used to conclude convergence.

S7. Let $a_1, \dots, a_m > 0$ be fixed. Prove that

$$\lim_{n\to\infty} (a_1^n + \cdots + a_m^n)^{1/n} = \max_{1\leq k\leq m} a_k.$$

1.4 Limits of Functions

重点内容与证明方法

- ❑ *Interpret the quantifiers in a function limit.*
- ❑ *Distinguish one-sided, two-sided, and infinite approaches.*
- ❑ *Prove limits using local estimates, limit laws, and sequences.*
- ❑ *Establish the fundamental trigonometric and exponential limits.*
- ❑ *Compare infinitesimals and recognize when cancellation needs finer estimates.*
- ❑ *Choose methods for direct proofs, calculations, and counterexamples.*

A sequence is already a special kind of function. Given any real sequence (a_n), define

$$f\colon \mathbb{N}_{>0} \to \mathbb{R}, \qquad f(n) = a_n.$$

The terms of the sequence are the values of this function at the positive integers. No values at noninteger inputs are needed, and the sequence alone does not specify them. In sequence convergence, the inputs follow the discrete approach $n \to \infty$. For a function on a general domain $D \subseteq \mathbb{R}$, an input may instead approach a finite point through the admissible points of D. We must control all such approaches, while respecting the domain.

Before asking what the output approaches, we must check that the proposed input approach is possible at arbitrarily small distances. For example, $f(x) = \arcsin x$ is a real-valued function on $[-1, 1]$. Every input in this domain satisfies $|x - 2| \geq 1$. In particular, there is no admissible input within distance $1/2$ of 2. Thus asking for $\lim_{x\to 2} \arcsin x$ has no meaning under the limit convention we will adopt: inputs in the domain cannot approach 2. This is a failure of the proposed approach, not an oscillation or an unbounded output near 2. By contrast, inputs in $[-1, 1]$ can approach 1 from below, so a limit question there is meaningful. We have not yet determined its value.

The limiting point need not itself be an input. For example,

$$f(x) = \frac{x^2 - 1}{x - 1} \qquad (x \neq 1)$$

has no value at 1, but its values approach 2 as x approaches 1. The equality $f(x) = x + 1$ for $x \neq 1$ explains the limiting behaviour without filling the hole in the domain.

1.4.1 Punctured neighbourhoods and the definition of a limit

Recall that $U_r(a) = \{x \in \mathbb{R} : |x - a| < r\}$. In a limit at a, the input belongs to a punctured neighbourhood: $0 < |x - a| < r$. The exclusion of a is essential. Changing the value of a function at a, or leaving that value undefined, does not affect its limit at a.

To include endpoints and restricted domains, let $f\colon D \to \mathbb{R}$, where $D \subseteq \mathbb{R}$. We call a an *accumulation point*(聚点)of D if every punctured neighbourhood of a contains a point of D. The point a need not belong to D. This condition guarantees that there are inputs that can approach a; it prevents a limit statement from becoming vacuously true. For the arcsine domain, 1 is an accumulation point but 2 is not. If we allowed $a = 2$, choosing $\delta = 1/2$ would leave no inputs to test, so the implication in the definition below would hold for every proposed L. The accumulation-point assumption rules out this empty test; it does not by itself guarantee that a limit exists. Most examples below have a domain containing a whole punctured neighbourhood of the limiting point.

Definition 1.19

Let a be an accumulation point of D and let $L \in \mathbb{R}$. We say that f has limit(函数极限) *L as $x \to a$ through D, and write*

$$\lim_{\substack{x \to a \\ x \in D}} f(x) = L$$

if for every $\varepsilon > 0$ there exists $\delta > 0$ such that, for every $x \in D$,

$$0 < |x - a| < \delta \quad \Longrightarrow \quad |f(x) - L| < \varepsilon.$$

We usually abbreviate this to $\lim_{x \to a} f(x) = L$, with the domain understood. ♣

The tolerance ε is prescribed first. We must then find one δ that works for *every* admissible input in the punctured δ-neighbourhood. The number δ may depend on ε, the function, and the fixed point a, but it may not depend on the input x subsequently chosen. The exact negation of the proposed limit is

$$\exists \varepsilon_0 > 0 \quad \forall \delta > 0 \quad \exists x \in D : \quad 0 < |x - a| < \delta, \qquad |f(x) - L| \geq \varepsilon_0.$$

Thus failure means that one fixed output error recurs at inputs arbitrarily close to a. This quantifier pattern is the source of the bad sequence constructed in Heine's criterion below. Geometrically, the graph over a sufficiently narrow punctured vertical strip must lie inside the prescribed horizontal band, as in **Figure** 1.3.

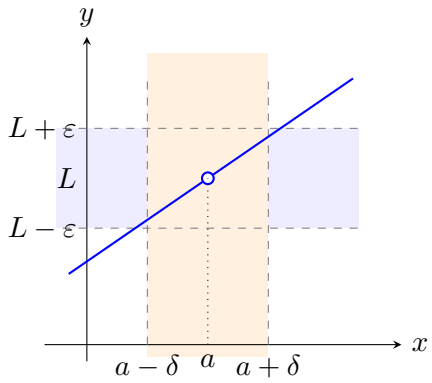


Figure 1.3: A horizontal tolerance band and a corresponding punctured input neighbourhood. The open circle does not affect the limit.

Example 1.27 Why remove the central point? Consider the function on $\mathbb{R}$ given by

$$g(x) = \begin{cases} x + 1, & x \neq 1, \\ 7, & x = 1. \end{cases}$$

Show that $\lim_{x \to 1} g(x) = 2$, and explain what would change if the definition tested all $|x - 1| < \delta$, including $x = 1$.

> ***Idea.*** There are two rules in the definition of g. The condition $0 < |x - 1|$ selects the nearby rule $g(x) = x + 1$ and excludes the single exceptional value. To test whether removing this condition matters, inspect the one input it adds: $x = 1$.

Solution *Given $\varepsilon > 0$, choose $\delta = \varepsilon$. Whenever $0 < |x - 1| < \delta$, we have $x \neq 1$, and hence*

$$|g(x) - 2| = |x + 1 - 2| = |x - 1| < \delta = \varepsilon.$$

This proves the stated limit, even though $g(1) = 7$.

If we replaced the punctured condition by $|x - 1| < \delta$, the input $x = 1$ would satisfy it for every $\delta > 0$. But for $\varepsilon = 1$ the required output estimate would fail, since $|g(1) - 2| = 5$. More generally, when $g(a)$ is defined,

including $x = a$ would require $|g(a) - L| < \varepsilon$ for every $\varepsilon > 0$, which forces $g(a) = L$. This extra equality is not part of the question about nearby values.

Changing $g(1)$ to any other real number, or deleting that value altogether, leaves the same punctured-neighbourhood proof valid. The open circle in **Figure** 1.3 illustrates the nearby rule independently of any assigned value at the centre. In the next section, continuity will require the nearby limit to agree with the assigned value. That is why a full neighbourhood can be used in a definition of continuity, whereas the limit definition keeps the two questions separate.

Example 1.28 Designing a neighbourhood Prove directly that $\lim_{x\to 1}(3x - 2) = 1$.

> ***Idea.*** Start with the desired conclusion and simplify its left side: $|(3x - 2) - 1| = 3\,|x - 1|$. The output error is exactly three times the input error.

Proof Let $\varepsilon > 0$ and choose $\delta = \varepsilon/3$. If $0 < |x - 1| < \delta$, then

$$|(3x - 2) - 1| = 3\,|x - 1| < 3\delta = \varepsilon.$$

This proves the limit. □

Example 1.29 A temporary local bound Prove directly that $\lim_{x\to 2} x^2 = 4$.

> ***Idea.*** The factorization $\left|x^2 - 4\right| = |x - 2|\,|x + 2|$ exposes the small quantity $|x - 2|$. The remaining factor still depends on x. First restrict $|x - 2| < 1$; then $|x + 2| < 5$.

Proof Given $\varepsilon > 0$, set $\delta = \min\{1, \varepsilon/5\}$. If $0 < |x - 2| < \delta$, then $|x + 2| \le |x - 2| + 4 < 5$, and therefore

$$\left|x^2 - 4\right| < 5\,|x - 2| < 5\delta \le \varepsilon.$$

□

The preliminary restriction $\delta \le 1$ is not part of the definition. It is a reusable proof device: first place all nonsmall factors under control, then choose a smaller neighbourhood to meet the requested error. The limit definition also shows that functions agreeing throughout some punctured neighbourhood have the same limit there. Unlike sequences, however, changing a function at infinitely many inputs approaching a may change its limit, even when those inputs form a very sparse set.

1.4.2 One-sided limits and different approaches

Definition 1.20

Suppose D has points arbitrarily close to a from the right. The right-hand limit (右极限) $\lim_{x\to a^+} f(x) = L$ *means that for every $\varepsilon > 0$ there is $\delta > 0$ such that*

$$x \in D, \quad 0 < x - a < \delta \quad \Longrightarrow \quad |f(x) - L| < \varepsilon.$$

The left-hand limit (左极限) $\lim_{x\to a^-} f(x) = L$ *is defined by replacing $0 < x - a < \delta$ with $0 < a - x < \delta$, assuming approach from the left is possible.*

♣

Proposition 1.12

If D has points arbitrarily close to a from both sides, then

$$\lim_{x\to a} f(x) = L \quad \Longleftrightarrow \quad \lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = L.$$

♠

Proof A two-sided δ works on each side separately. Conversely, for a given $\varepsilon > 0$, choose suitable left and right radii δ_- and δ_+. With $\delta = \min\{\delta_-, \delta_+\}$, every admissible x satisfying $0 < |x - a| < \delta$ lies on one of the two controlled sides. □

For $f(x) = |x|/x$ on $\mathbb{R} \setminus \{0\}$, the left-hand limit at 0 is -1 and the right-hand limit is 1. The two-sided limit does not exist. The value assigned at 0 cannot repair this disagreement. By contrast, for $f(x) = \sqrt{x}$ on $[0, +\infty)$, the limit at 0 relative to the domain is a right-hand limit. There is no left-hand approach in the domain, so Proposition 1.12 does not require a left-hand limit in this example.

Definition 1.21

If D is unbounded above, then $\lim_{x\to+\infty} f(x) = L$ means

$$\text{for every } \varepsilon > 0 \text{ there is } A > 0 \text{ such that } x \in D,\ x > A \implies |f(x) - L| < \varepsilon.$$

For $x \to -\infty$, assume D is unbounded below and use $x < -A$. For $|x| \to \infty$, assume D is unbounded and use $|x| > A$. When both tails of D are unbounded, the last limit exists exactly when the two tail limits exist and agree.

♣

For example, $1/x \to 0$ as $|x| \to \infty$, since $|x| > 1/\varepsilon$ implies $|1/x| < \varepsilon$. The notation $x \to \infty$ is sometimes used for $x \to +\infty$; we keep the sign explicit whenever the direction matters. In particular,

$$\lim_{x\to+\infty} \frac{x}{\sqrt{x^2+1}} = 1, \qquad \lim_{x\to-\infty} \frac{x}{\sqrt{x^2+1}} = -1.$$

The sign difference comes from $\sqrt{x^2} = |x|$, not from x. These two computations will follow from the limit laws below.

Definition 1.22

Let a be an accumulation point of D. We write $\lim_{x\to a} f(x) = +\infty$ if for every $M > 0$ there is $\delta > 0$ such that

$$x \in D, \quad 0 < |x - a| < \delta \quad \implies \quad f(x) > M.$$

The statement $\lim_{x\to a} f(x) = -\infty$ uses $f(x) < -M$ instead. One-sided infinite limits use the corresponding one-sided neighbourhoods. Infinite limits as $x \to +\infty$ or $x \to -\infty$ use a tail condition in place of $0 < |x - a| < \delta$.

♣

An *infinite limit*(无穷极限)describes eventual growth beyond every bound; it is not a finite real limit. For instance,

$$\lim_{x\to0} \frac{1}{x^2} = +\infty, \qquad \lim_{x\to0^+} \frac{1}{x} = +\infty, \qquad \lim_{x\to0^-} \frac{1}{x} = -\infty.$$

For the first statement choose $\delta = 1/\sqrt{M}$. The last two statements do not combine into a signed two-sided infinite limit for $1/x$, although $|1/x| \to +\infty$. Nor does unboundedness near a point alone imply an infinite limit: a function may repeatedly return to 0 while also taking arbitrarily large values.

Proposition 1.13 (Basic rules for infinite limits)

The following assertions hold along any one fixed approach.

(i) If $g \to +\infty$ and $f \geq g$ eventually, then $f \to +\infty$. If $g \to -\infty$ and $f \leq g$ eventually, then $f \to -\infty$.

(ii) If $f \to +\infty$ and g is bounded below eventually, then $f + g \to +\infty$. If $f \to -\infty$ and g is bounded above eventually, then $f + g \to -\infty$.

(iii) If $f \to +\infty$ and $g \to c \neq 0$, then $fg \to +\infty$ when $c > 0$ and $fg \to -\infty$ when $c < 0$. Replacing $f \to +\infty$ by $f \to -\infty$ reverses these signs.

(iv) If $|f| \to +\infty$ and g is bounded eventually, then $g/f \to 0$.

(v) If $f \to 0$ and $f > 0$ eventually, then $1/f \to +\infty$; if $f < 0$ eventually, then $1/f \to -\infty$. Conversely, $f \to +\infty$ or $f \to -\infty$ implies $1/f \to 0$, with the corresponding eventual sign.

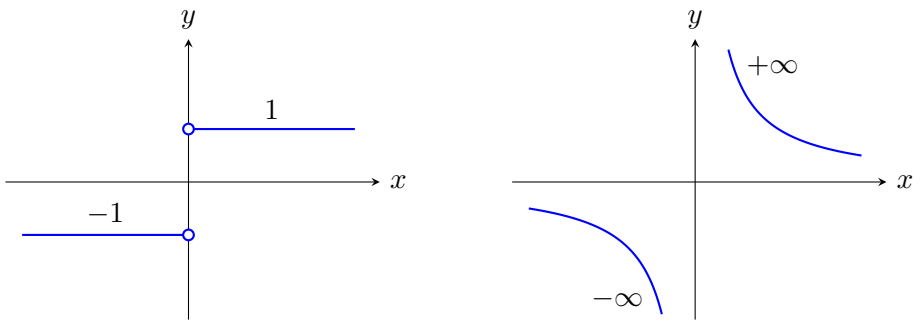
Proof The comparison statements follow immediately by applying the defining threshold M to the smaller function in the positive case and to the larger function in the negative case. For (ii), if $g \geq C$ eventually and $M > 0$, take $f > \max\{M - C, 1\}$ sufficiently far along the approach; the other sign is analogous. For (iii), when $c > 0$ one eventually has $g > c/2$, and $f > 2M/c$ then gives $fg > M$. The remaining signs follow by applying this argument to $-f$ or $-g$. For (iv), choose an eventual bound $|g| \leq C$ and then require $|f| > C/\varepsilon$; the case $C = 0$ is immediate. Finally, the inequalities

$$0 < f < \frac{1}{M} \quad \Longleftrightarrow \quad \frac{1}{f} > M, \qquad f > \frac{1}{\varepsilon} \quad \Longrightarrow \quad 0 < \frac{1}{f} < \varepsilon$$

prove (v), and the negative cases follow after changing signs. □

> ***Use sign information, not symbol pushing.*** The rules above apply only after eventual signs or one-sided bounds have been established. For example, $f \to +\infty$ and $g \to -\infty$ do not determine $f + g$; competing terms must first be compared or transformed.

In graph language, $x = a$ is a *vertical asymptote* (铅直渐近线) if at least one of the one-sided limits of f at a is $+\infty$ or $-\infty$. The line $y = L$ is a *horizontal asymptote* (水平渐近线) at a specified tail if the corresponding limit at infinity equals L. The function $1/x$ has both types of asymptote. A graph can cross a horizontal asymptote; an asymptote describes a limiting relation, not a barrier. The two graphs in **Figure** 1.4 contrast finite side limits with infinite side limits.



$f(x) = |x|/x$ $\qquad$ $g(x) = 1/x$

Figure 1.4: The two sides approach different finite levels on the left and grow with opposite signs on the right. Neither graph has a finite two-sided limit at 0.

1.4.3 Local properties and the sequential criterion

Theorem 1.13

Suppose $f(x) \to L \in \mathbb{R}$ as $x \to a$ through D. Then the following statements hold.

1. *The limit is unique.*
2. *The function is bounded on some punctured neighbourhood of a relative to D.*
3. *If $L > 0$, then $f(x) > 0$ throughout some such neighbourhood; if $L < 0$, then $f(x) < 0$ there.*
4. *If also $g(x) \to K$ on the same approach and $f(x) \leq g(x)$ for all sufficiently close admissible $x \neq a$, then $L \leq K$.*

> ***Proof Strategy.*** Choose an output tolerance adapted to the desired conclusion: a fraction of the gap between alleged limits, the fixed tolerance 1, or a fraction of the distance from L to 0. The input neighbourhood then follows from the definition.

Proof If $L \neq K$ were two limits of f, use $\varepsilon = |L-K|/3$ in both limit definitions. There is an admissible $x \neq a$ in the intersection of the resulting neighbourhoods, because a is an accumulation point. Then

$$|L-K| \leq |L-f(x)| + |f(x)-K| < \frac{2}{3}|L-K|,$$

a contradiction. For boundedness, use $\varepsilon = 1$ to obtain $|f(x)| < |L|+1$ near a. If $L > 0$, use $\varepsilon = L/2$ to obtain $f(x) > L/2 > 0$. Apply this result to $-f$ when $L < 0$. Finally, if $L > K$, use $\varepsilon = (L-K)/3$ for both f and g and restrict further to the neighbourhood where $f \leq g$. An admissible point there would satisfy

$$f(x) > L - \varepsilon > K + \varepsilon > g(x),$$

which is impossible. □

The second assertion is *local boundedness*(局部有界性): its bound need not be valid on all of D. The third is *local sign preservation*(局部保号性)when the limit is nonzero. Also, a strict inequality need not stay strict in the limit: $x^2 > 0$ for $x \neq 0$, but its limit at 0 is 0. All four arguments work on a one-sided neighbourhood or on a tail after the input condition is changed accordingly.

Theorem 1.14 (Heine's sequential criterion)

Let a be an accumulation point of D. Then $\lim_{x\to a} f(x) = L$ if and only if, for every sequence (x_n) in $D \setminus \{a\}$ satisfying $x_n \to a$, one has $f(x_n) \to L$. ♡

> ***Idea.*** A function limit controls every sufficiently close input, hence every sufficiently late term of any approaching sequence. For the converse, negate the function-limit definition and choose one bad input within distance $1/n$ for each n.

Proof Suppose first that $f(x) \to L$ and let $x_n \to a$ with $x_n \in D \setminus \{a\}$. For a prescribed $\varepsilon > 0$, choose $\delta > 0$ from the function-limit definition. For all sufficiently large n, $0 < |x_n - a| < \delta$, so $|f(x_n) - L| < \varepsilon$. Thus $f(x_n) \to L$.

Conversely, if f does not tend to L, there exists $\varepsilon_0 > 0$ such that for every $\delta > 0$ one can find $x \in D$ with

$$0 < |x-a| < \delta, \qquad |f(x) - L| \geq \varepsilon_0.$$

Choose such a point x_n with $\delta = 1/n$. Then $x_n \to a$, but $|f(x_n) - L| \geq \varepsilon_0$ for every n, contradicting the assumed sequential property. □

The criterion has one-sided and infinite-approach versions. For example, $f(x) \to L$ as $x \to +\infty$ exactly when $f(x_n) \to L$ for every sequence $x_n \in D$ tending to $+\infty$. In the converse proof, a failure supplies a bad point $x_n > n$ instead of a bad point within $1/n$ of a. For a right-hand limit use $0 < x_n - a < 1/n$, and for a negative tail use $x_n < -n$. The same selection argument gives the $|x| \to \infty$ version using $|x_n| > n$.

Example 1.30 Two ways to approach the same point Determine whether $\sin(1/x)$ and $x\sin(1/x)$ have limits as $x \to 0$.

Solution *For $n \geq 1$, set*

$$x_n = \frac{1}{2\pi n + \pi/2}, \qquad y_n = \frac{1}{2\pi n + 3\pi/2}.$$

Both sequences tend to 0, but $\sin(1/x_n) = 1$ and $\sin(1/y_n) = -1$. By Theorem 1.14, $\sin(1/x)$ has no limit at 0, even from the right. For the second function,

$$|x\sin(1/x)| \leq |x|.$$

Given $\varepsilon > 0$, the choice $\delta = \varepsilon$ proves that its limit is 0. Oscillation may persist while its amplitude tends to zero.

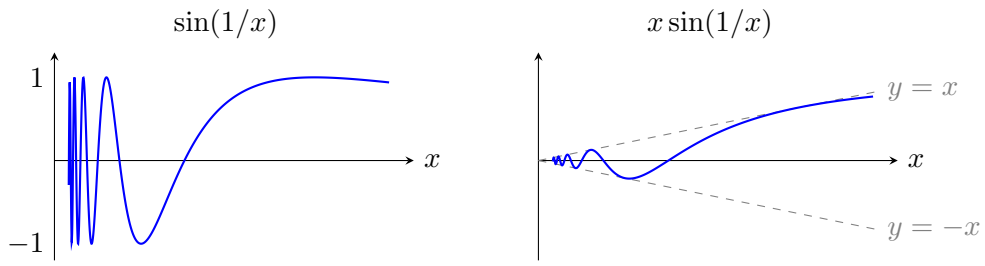


Figure 1.5: Right-hand portions of two oscillatory graphs. In the second graph, a shrinking envelope forces a limit.

The graph in **Figure** 1.5 suggests the distinction, but a finite drawing cannot display all oscillations near 0. The sequences and the estimate provide the proof. Similarly, the Dirichlet function $D(x)$ from Section 1.2 has no limit at any $a \in \mathbb{R}$. By density, choose a rational r_n and an irrational s_n in $(a, a + 1/n)$. Then both input sequences tend to a, whereas $D(r_n) = 1$ and $D(s_n) = 0$ for all n.

1.4.4 Limit laws and composition

Theorem 1.15

Suppose $f(x) \to A$ and $g(x) \to B$ along the same approach, with finite limits. For every $c \in \mathbb{R}$,

$$f(x) + g(x) \to A + B, \qquad cf(x) \to cA, \qquad f(x)g(x) \to AB.$$

If $B \neq 0$, then g is nonzero sufficiently far along the approach and

$$\frac{f(x)}{g(x)} \to \frac{A}{B}.$$

♡

Proof For every admissible approaching sequence (x_n), the sequences $f(x_n)$ and $g(x_n)$ tend to A and B by Theorem 1.14 and its corresponding approach version. The sequence arithmetic laws, Theorem 1.5, give the asserted limits along this arbitrary sequence. The converse sequential criterion gives the function limits. If $B \neq 0$, the estimate $|g(x) - B| < |B|/2$ holds eventually and implies $|g(x)| > |B|/2$. Thus the quotient is defined on the required final part of the approach. □

Theorem 1.16 (Squeeze theorem)

Suppose $u(x) \leq f(x) \leq v(x)$ eventually along an approach, and both $u(x)$ and $v(x)$ tend to the same finite limit L. Then $f(x) \to L$ along that approach.

♡

Proof Given $\varepsilon > 0$, restrict the inputs so that the two comparisons hold and

$$|u(x) - L| < \varepsilon, \qquad |v(x) - L| < \varepsilon.$$

Then $L - \varepsilon < u(x) \leq f(x) \leq v(x) < L + \varepsilon$. □

In particular, if $h(x) \to 0$ and k is bounded near the approach, then $h(x)k(x) \to 0$. Indeed, an eventual bound $|k(x)| \leq C$ yields $|h(x)k(x)| \leq C\,|h(x)|$. The bounded factor need not have a limit.

Example 1.31 Cancellation and rationalization Evaluate

$$\lim_{x \to 2} \frac{x^2 - 4}{x - 2}, \qquad \lim_{x \to 0} \frac{\sqrt{1 + x} - 1}{x}, \qquad \lim_{x \to -\infty} \left(\sqrt{x^2 + x} + x\right).$$

Solution *In the first limit, the quotient equals $x + 2$ on the punctured domain, so the answer is 4. For the second,*

rationalization gives

$$\frac{\sqrt{1+x}-1}{x}=\frac{1}{\sqrt{1+x}+1}\qquad(x\neq 0).$$

Here $\left|\sqrt{1+x}-1\right|=|x|/(\sqrt{1+x}+1)\le|x|$ *for* $x\ge -1$. *Thus* $\sqrt{1+x}\to 1$, *and the answer is* $1/2$. *For sufficiently negative* x, *the third expression is defined and*

$$\sqrt{x^2+x}+x=\frac{x}{\sqrt{x^2+x}-x}=\frac{-1}{\sqrt{1+1/x}+1}\longrightarrow -\frac{1}{2}.$$

The last equality uses $\sqrt{x^2}=|x|=-x$ *on this tail.*

The same idea applies to higher roots without searching for a special trick. If $p=\sqrt[m]{A}$ and $q=\sqrt[m]{B}$ are real, then

$$A-B=p^m-q^m=(p-q)\left(p^{m-1}+p^{m-2}q+\cdots+pq^{m-2}+q^{m-1}\right).$$

Whenever the second factor has a nonzero limit, this identity replaces the difference of two mth roots by a quotient whose small factor is $A-B$.

> ***Generalized rationalization.*** For a difference of roots, set the roots equal to new symbols and factor a difference of powers. The familiar conjugate is exactly the case $m=2$. Before dividing, check that the resulting sum in the denominator stays away from zero.

Example 1.32 Rationalizing two fifth roots Evaluate, using only algebraic identities and limit laws,

$$\lim_{x\to 0}\frac{\sqrt[5]{1+3x}-\sqrt[5]{1-2x}}{x}.$$

> ***Idea.*** The two roots both approach the same candidate value 1, so their difference hides the small factor we need to cancel. Their fifth powers have the much simpler difference $(1+3x)-(1-2x)=5x$. This suggests the factorization of p^5-q^5. First use the same identity with one root equal to 1 to justify the candidate root limits without appealing to a later continuity theorem.

Put $p=\sqrt[5]{1+3x}$ and $q=\sqrt[5]{1-2x}$. For $|x|<1/6$, both radicands and both roots are positive. The identities $p^5-1=3x$ and $q^5-1=-2x$ give

$$|p-1|=\frac{3\,|x|}{p^4+p^3+p^2+p+1}\le 3\,|x|\,,$$
$$|q-1|=\frac{2\,|x|}{q^4+q^3+q^2+q+1}\le 2\,|x|\,.$$

Each denominator is at least 1 because its terms are nonnegative. These estimates prove $p\to 1$ and $q\to 1$; for example, $\delta=\min\{1/6,\varepsilon/3\}$ controls the first error, and $\delta=\min\{1/6,\varepsilon/2\}$ controls the second.

Solution ***Method 1: factor the difference of fifth powers.*** *For* $0<|x|<1/6$, *set*

$$S=p^4+p^3q+p^2q^2+pq^3+q^4>0.$$

Since $(p-q)S=p^5-q^5=5x$, *we obtain the exact cancellation*

$$\frac{p-q}{x}=\frac{p^5-q^5}{xS}=\frac{5x}{xS}=\frac{5}{S}.$$

The already established limits $p,q\to 1$ *and the arithmetic laws give* $S\to 5$. *The limiting denominator is nonzero, so the required limit is* $5/5=1$.

> ***Another route.*** Both roots are close to 1. Inserting that shared value splits the numerator into two differences of the form "one root minus 1", each handled by the simpler identity already used in the error estimates.

Solution ***Method 2: rationalize the two changes separately.*** *For $0 < |x| < 1/6$, the same positive denominators give*

$$\begin{aligned}\frac{p-q}{x} &= \frac{p-1}{x} - \frac{q-1}{x} \\ &= \frac{3}{p^4+p^3+p^2+p+1} + \frac{2}{q^4+q^3+q^2+q+1}.\end{aligned}$$

Both denominators tend to 5, so the limit is $3/5+2/5 = 1$. The plus sign in the last term comes from subtracting $q-1$, whose rationalized numerator is $-2x$.

Both methods work on a punctured interval where the roots and the denominators are positive. Method 1 is shorter when the difference of the radicands is simple. Method 2 is useful when separate root changes can be reused in a larger expression. For $m = 5$, the rationalizing factor has five terms, not the two terms of a square-root conjugate; the difference-of-powers identity determines the factor systematically.

The quotient law cannot be applied directly when the limiting denominator is 0. The commonly occurring forms

$$\frac{0}{0}, \qquad \frac{\infty}{\infty}, \qquad \infty - \infty, \qquad 0 \cdot \infty, \qquad 1^{\infty}, \qquad 0^0, \qquad \infty^0$$

are *indeterminate forms*(未定式), not values. The notation records only the separate limiting tendencies of the visible parts; it does not determine how their rates compare. For example, x^2/x, x/x, and x/x^2 have very different limiting behaviours at 0. Likewise, as $x \to +\infty$, the three differences $x - x$, $(x+1) - x$, and $2x - x$ tend respectively to 0, 1, and $+\infty$. By contrast, a bounded numerator divided by a denominator whose absolute value tends to infinity is determined by Proposition 1.13: the quotient tends to 0. The correct response to an indeterminate form is therefore to transform or estimate the expression, not to perform arithmetic with the symbols 0 and ∞.

Theorem 1.17

Suppose $u(x) \to b$ as $x \to a$, and $v(t) \to L$ as $t \to b$ through the domain of v. Assume the composition is defined for all sufficiently close admissible $x \neq a$. Either of the following additional conditions ensures that $v(u(x)) \to L$:

1. *$u(x) \neq b$ for all sufficiently close admissible $x \neq a$;*
2. *b is in the domain of v and $v(b) = L$.*

The same statement holds for a one-sided or infinite approach of x; the values $u(x)$ must follow the approach specified for v. ♡

> ***Idea.*** The outer limit gives control when $0 < |t-b| < \eta$. The inner limit supplies $|u(x)-b| < \eta$. The only possible mismatch is $u(x) = b$, which the punctured outer limit does not control.

Proof Given $\varepsilon > 0$, choose $\eta > 0$ so that every admissible t with $0 < |t-b| < \eta$ satisfies $|v(t)-L| < \varepsilon$. Choose a sufficiently small punctured neighbourhood of a on which the composition is defined and $|u(x)-b| < \eta$. Under the first condition, restrict further so that $u(x) \neq b$; the outer estimate then applies. Under the second condition, the same estimate applies whenever $u(x) \neq b$, and when $u(x) = b$ the output equals $v(b) = L$ exactly. In either case the required error is less than ε. The proof changes only its input neighbourhood condition for other approaches. □

Neither extra condition can simply be discarded. Let $u(x) = 0$ for every x, and let $v(0) = 1$ while $v(t) = 0$

for $t \neq 0$. Then $u(x) \to 0$ and $\lim_{t\to 0} v(t) = 0$, but $v(u(x)) = 1$. In the next section, continuity of the outer function will provide precisely the second condition.

> ***Substitution Checklist.*** When setting $t = \phi(x)$ in a limit, record all four points explicitly.
>
> (1) Determine the exact approach of t: two-sided, one-sided, or infinite.
>
> (2) Rewrite the entire expression in terms of t, including restrictions inherited from the original domain.
>
> (3) Verify that $t = \phi(x)$ follows the domain and approach required by the new limit.
>
> (4) Check the composition hypothesis at the puncture; in particular, either $\phi(x) \neq b$ eventually or the outer function must be controlled at b itself.
>
> A substitution such as $t = -x$ reverses sides, while $t = 1/x$ turns a tail into a one-sided approach to 0.

1.4.5 The fundamental trigonometric limit

The basic geometric estimate is obtained from the unit circle. Angles are measured in radians throughout. For $0 < t < \pi/2$, compare an inscribed triangle, a circular sector, and a triangle formed with the tangent line in **Figure** 1.6. Their areas are respectively $\frac{1}{2}\sin t$, $\frac{1}{2}t$, and $\frac{1}{2}\tan t$. Elementary circle geometry therefore gives

$$\sin t < t < \tan t, \qquad \cos t < \frac{\sin t}{t} < 1.$$

This argument uses the geometric meaning of radian measure and the sector-area formula; it does not use differentiation or integration.

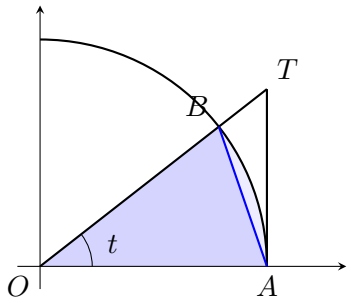


Figure 1.6: The inclusions $\triangle OAB \subset$ sector $OAB \subset \triangle OAT$ compare the sine, angle, and tangent.

For negative t, oddness of sine gives $|\sin t| \leq |t|$ as well, at least for $|t| < \pi/2$. Consequently,

$$0 \leq 1 - \cos t = 2\sin^2(t/2) \leq \frac{t^2}{2}$$

for all sufficiently small t. Hence $\cos t \to 1$ as $t \to 0$.

Theorem 1.18

With angles measured in radians,

$$\lim_{x\to 0} \frac{\sin x}{x} = 1. \tag{1.4}$$

♡

Proof For $0 < x < \pi/2$, the geometric inequalities place $\sin x/x$ between $\cos x$ and 1. Both bounds tend to 1, so Theorem 1.16 gives the right-hand limit. Since $\sin(-x)/(-x) = \sin x/x$, the left-hand limit is the same. Apply Proposition 1.12. □

To apply (1.4), match the argument of the sine with the small quantity in the denominator. The addition formulas now give, for every fixed $a \in \mathbb{R}$,

$$\lim_{x\to a} \sin x = \sin a, \qquad \lim_{x\to a} \cos x = \cos a.$$

Indeed, write $x = a + h$, use $\sin h \to 0$ and $\cos h \to 1$, and apply the arithmetic laws to the addition formulas. Where the limiting cosine is nonzero, the quotient law also gives the limit of the tangent.

Example 1.33 Matching the small argument Evaluate

$$\lim_{x\to 0} \frac{\sin(3x)}{\tan(2x)}, \qquad \lim_{x\to 0} \frac{1-\cos x}{x^2}.$$

Solution *For the first limit, separate the two standard ratios:*

$$\frac{\sin(3x)}{\tan(2x)} = \frac{3}{2}\frac{\sin(3x)}{3x}\frac{2x}{\sin(2x)}\cos(2x) \longrightarrow \frac{3}{2}.$$

All denominators are nonzero for sufficiently small nonzero x. For the second, the half-angle identity gives

$$\frac{1-\cos x}{x^2} = \frac{1}{2}\left(\frac{\sin(x/2)}{x/2}\right)^2 \longrightarrow \frac{1}{2}.$$

1.4.6 The exponential limit and related limits

In Section 1.3, the sequence $(1+1/n)^n$ defined the number e. The next step is to allow a real input. Before applying logarithms to a limit, we record the elementary limit estimates they require. We use the real exponential and logarithmic functions and their order and algebraic properties introduced in Section 1.2.

Lemma 1.2

For $b > 0$ and $c \in \mathbb{R}$,

$$\lim_{x\to c} b^x = b^c.$$

For $b > 0$, $b \neq 1$, and $t > 0$,

$$\lim_{x\to t} \log_b x = \log_b t.$$

♡

> ***Idea.*** Near exponent 0, trap a power between $b^{-1/n}$ and $b^{1/n}$ and use the root-sequence limit. For a logarithm, translate the desired output interval back to a positive input interval using monotonicity.

Proof Suppose first $b > 1$. By Proposition 1.8, $b^{1/n} \to 1$, and its reciprocal also tends to 1. For a prescribed $\varepsilon > 0$, choose a positive integer n with $b^{1/n} - 1 < \varepsilon$ and $1 - b^{-1/n} < \varepsilon$. If $|h| < 1/n$, monotonicity gives

$$b^{-1/n} < b^h < b^{1/n},$$

so $\left|b^h - 1\right| < \varepsilon$. Thus $b^h \to 1$ as $h \to 0$, and $b^{c+h} = b^c b^h \to b^c$. The case $b = 1$ is constant, and the case $0 < b < 1$ follows by writing $b^x = 1/(1/b)^x$.

For logarithms with $b > 1$, let $\varepsilon > 0$ and choose

$$\delta = \min\{t - tb^{-\varepsilon},\ tb^{\varepsilon} - t\} > 0.$$

If $|x - t| < \delta$, then $tb^{-\varepsilon} < x < tb^{\varepsilon}$, hence $\log_b t - \varepsilon < \log_b x < \log_b t + \varepsilon$. For $0 < b < 1$, use $\log_b x = -\log_{1/b} x$. □

Theorem 1.19

For the real-valued expression with $1 + x > 0$,

$$\lim_{x\to 0}(1+x)^{1/x} = e. \tag{1.5}$$

Equivalently, $\lim_{t\to+\infty}(1+1/t)^t = e$.

♡

> ***Proof Strategy.*** For a positive real t, place t between consecutive integers and trap the whole expression between two sequences already controlled by Theorem 1.10. Then transform the negative side of $x \to 0$ into a positive-side limit by an exact identity.

Proof Let $t \geq 1$ and $n = \lfloor t \rfloor$, so $n \leq t < n+1$. Monotonicity in the base and in a positive exponent gives

$$\left(1+\frac{1}{n+1}\right)^{n} \leq \left(1+\frac{1}{t}\right)^{t} \leq \left(1+\frac{1}{n}\right)^{n+1}.$$

The left bound equals $\big(1+1/(n+1)\big)^{n+1}/\big(1+1/(n+1)\big)$, and the right bound equals $\big(1+1/n\big)^{n}\big(1+1/n\big)$. Both tend to e as $n \to \infty$. Since $\lfloor t \rfloor \to \infty$ as $t \to +\infty$, their values approach e along the real input t as well: the sequence estimates for every $n \geq N$ apply whenever $t \geq N$. The squeeze theorem proves the limit at $+\infty$. The substitution $t = 1/x$ gives (1.5) from the right.

For $-1 < x < 0$, put $y = -x/(1+x) > 0$. As $x \to 0^{-}$, one has $y \to 0^{+}$, and

$$1+x = \frac{1}{1+y}, \qquad (1+x)^{1/x} = (1+y)^{(1+y)/y} = (1+y)^{1/y}(1+y) \longrightarrow e.$$

The one-sided limits agree. □

Corollary 1.2

The following limits hold:

$$\lim_{x\to 0}\frac{\ln(1+x)}{x} = 1, \qquad \lim_{x\to 0}\frac{e^{x}-1}{x} = 1. \tag{1.6}$$

More generally, for $b > 0$,

$$\lim_{x\to 0}\frac{b^{x}-1}{x} = \ln b.$$

♡

> ***Idea.*** Taking a logarithm converts the power limit into a quotient limit. Then the inverse relation between the exponential and logarithm transfers the result to $e^{x}-1$.

Proof By Lemma 1.2, the logarithm tends to its value at e. The composition theorem therefore allows us to take the logarithm in (1.5):

$$\frac{\ln(1+x)}{x} = \ln\big((1+x)^{1/x}\big) \longrightarrow \ln e = 1.$$

For the second limit, set $u = e^{x}-1$. The same lemma gives $u \to 0$, and $u \neq 0$ for $x \neq 0$ by strict monotonicity of the exponential. Since $x = \ln(1+u)$,

$$\frac{e^{x}-1}{x} = \frac{u}{\ln(1+u)} \longrightarrow 1.$$

If $b \neq 1$, substitute $t = x \ln b$ into the second limit and use $b^{x} = e^{x\ln b}$. If $b = 1$, the numerator is identically zero and the limit is $0 = \ln 1$. □

Example 1.34 Tracking direction under substitution Evaluate

$$\lim_{x\to 0^{-}}\frac{\ln(1-x)}{x}, \qquad \lim_{x\to+\infty} x\big(e^{1/x}-1\big).$$

Solution *For the first limit, set $t = -x$. Then $x \to 0^{-}$ means $t \to 0^{+}$, not merely $t \to 0$, and*

$$\frac{\ln(1-x)}{x} = -\frac{\ln(1+t)}{t} \longrightarrow -1.$$

For the second, set $t = 1/x$. The tail $x \to +\infty$ becomes $t \to 0^{+}$, and

$$x\big(e^{1/x}-1\big) = \frac{e^{t}-1}{t} \longrightarrow 1.$$

In both cases the direction and the transformed domain agree with the relevant one-sided form of (1.6).

Proposition 1.14 (Logarithmic reduction of a power)

Along a fixed approach, suppose that $F(x) > 0$ eventually and

$$G(x)\ln F(x) \longrightarrow A \in \mathbb{R}.$$

Then

$$F(x)^{G(x)} \longrightarrow e^{A}.$$

In particular, if $u(x) \to 0$, $1 + u(x) > 0$ eventually, and $u(x)v(x) \to c \in \mathbb{R}$, then

$$\big(1 + u(x)\big)^{v(x)} \longrightarrow e^{c}.$$

This conclusion remains valid when $u(x) = 0$ at some inputs.

♠

Proof The identity

$$F(x)^{G(x)} = \exp\big(G(x)\ln F(x)\big)$$

and continuity of the exponential give the first statement. For the special case, define

$$r(t) = \begin{cases} \dfrac{\ln(1+t)}{t}, & t \neq 0, \\ 1, & t = 0. \end{cases}$$

By (1.6), $r(t) \to 1$ as $t \to 0$, and the chosen value makes r continuous there. At every relevant input, including those with $u(x) = 0$,

$$v(x)\ln(1 + u(x)) = u(x)v(x)r(u(x)) \longrightarrow c.$$

Apply the first statement with $F = 1 + u$ and $G = v$. □

Example 1.35 A power with a hidden finite exponent Evaluate

$$\lim_{x\to 0}\big(1 + x\sin x\big)^{1/x^2}.$$

Solution *For sufficiently small x, one has $x\sin x \geq 0$, so the base is positive. Set $u(x) = x\sin x$ and $v(x) = 1/x^2$. Then $u(x) \to 0$ and*

$$u(x)v(x) = \frac{\sin x}{x} \longrightarrow 1.$$

Therefore Proposition 1.14 gives the limit e.

> ***Power-form decision.*** For a positive varying base, do not manipulate an apparent form such as 1^{∞}, 0^{0}, or ∞^{0} directly. Take a logarithm and study the single product $G\ln F$. The positivity of F is a domain condition, and the resulting product may still require factoring, substitution, or a rate comparison.

1.4.7 Infinitesimals and their comparison

The words "small" and "large" become meaningful only after an approach has been specified. For example, $1/x$ tends to 0 as $x \to +\infty$, but its magnitude grows without bound as $x \to 0$.

Definition 1.23

Along a fixed approach, a function α is an infinitesimal (无穷小量) *if $\alpha(x) \to 0$. It is an* infinite quantity (无穷大量) *if $|\alpha(x)| \to +\infty$.*

♣

An infinitesimal is a varying quantity with limit zero, not a particular very small positive number. A finite limit $f(x) \to L$ is equivalent to $f(x) = L + \alpha(x)$ with an infinitesimal α. Finite sums of infinitesimals are infinitesimal, as is the product of an infinitesimal and a bounded function. The quotient of two infinitesimals

requires a separate comparison. If α is eventually nonzero, then

$$\alpha(x) \to 0 \quad \Longleftrightarrow \quad \frac{1}{|\alpha(x)|} \to +\infty.$$

Indeed, the inequalities $|\alpha(x)| < 1/M$ and $1/|\alpha(x)| > M$ are equivalent. The analogous reciprocal statement holds for an infinite quantity.

Definition 1.24

Let α and β be infinitesimals along the same approach, and assume β is eventually nonzero.

1. *If $\alpha/\beta \to 0$, we write $\alpha = o(\beta)$ and say that α is of* higher order (高阶无穷小) *than β.*
2. *If $\alpha/\beta \to c \neq 0$ with $c \in \mathbb{R}$, they are of the* same order (同阶无穷小).
3. *If $\alpha/\beta \to 1$, we write $\alpha \sim \beta$ and call them* equivalent infinitesimals (等价无穷小).

♣

For example, as $x \to 0$, $x^2 = o(x)$, $3x$ and x have the same order, and $\sin x \sim x$. The comparison concerns a ratio, so signs must be retained: $-x$ and x have the same order but are not equivalent. Also, not every pair is comparable by a ratio limit; $x\sin(1/x)$ and x provide an example. The assertion $\alpha \sim \beta$ is equivalent to $\alpha = \beta(1+\eta)$ with $\eta \to 0$, or to $\alpha - \beta = o(\beta)$.

Proposition 1.15

Suppose $\alpha \sim \widetilde{\alpha}$ and $\beta \sim \widetilde{\beta}$, with all four functions nonzero sufficiently far along the approach. Then

$$\frac{\alpha\beta}{\widetilde{\alpha}\widetilde{\beta}} \to 1, \qquad \frac{\alpha/\beta}{\widetilde{\alpha}/\widetilde{\beta}} \to 1.$$

In particular, if $\widetilde{\alpha}/\widetilde{\beta} \to L \in \mathbb{R}$, then $\alpha/\beta \to L$. The same replacement preserves a signed infinite limit of the quotient.

♠

Proof The first ratio is $(\alpha/\widetilde{\alpha})(\beta/\widetilde{\beta})$, and the second is $(\alpha/\widetilde{\alpha})/(\beta/\widetilde{\beta})$. Both tend to 1 by the arithmetic laws. Multiplication by a factor tending to 1 preserves a finite limit. For an infinite limit, this factor lies between $1/2$ and $3/2$ eventually, so the definition of the corresponding infinite limit gives the conclusion. □

The trigonometric estimates and (1.6) yield the useful list, all as $x \to 0$:

$$\sin x \sim x, \qquad \tan x \sim x, \qquad 1 - \cos x \sim \frac{x^2}{2}, \qquad \ln(1+x) \sim x, \qquad e^x - 1 \sim x. \tag{1.7}$$

For a fixed $p \in \mathbb{R} \setminus \{0\}$, one also has

$$(1+x)^p - 1 \sim px.$$

To verify it, put $t = p\ln(1+x)$ and use

$$\frac{(1+x)^p - 1}{px} = \frac{e^t - 1}{t}\frac{\ln(1+x)}{x} \longrightarrow 1.$$

All expressions are taken for sufficiently small x with $1 + x > 0$. For $p = 0$, the numerator is zero; the ratio to px is undefined, so the equivalence must not be asserted.

Example 1.36 Replacement in a product Evaluate $\lim\limits_{x\to 0} \dfrac{\sin(2x)\ln(1+3x)}{1-\cos(4x)}$.

Solution *By (1.7), the three factors have equivalents $2x$, $3x$, and $8x^2$, respectively. Thus Proposition 1.15 gives*

$$\lim_{x\to 0} \frac{\sin(2x)\ln(1+3x)}{1-\cos(4x)} = \lim_{x\to 0} \frac{(2x)(3x)}{8x^2} = \frac{3}{4}.$$

Example 1.37 Cancellation changes the required precision Evaluate $\lim\limits_{x\to 0} \dfrac{\sqrt{1+x} - 1 - x/2}{x^2}$.

> ***Idea.*** The equivalence $\sqrt{1+x}-1 \sim x/2$ describes the leading term. That term is subtracted in the numerator, so the equivalence alone does not determine the remaining error. Use exact identities before passing to a limit.

Solution *For sufficiently small nonzero x, let $s=\sqrt{1+x}$. Then*

$$s-1-\frac{x}{2}=\frac{x}{s+1}-\frac{x}{2}=\frac{x(1-s)}{2(s+1)}=-\frac{x^2}{2(s+1)^2}.$$

Since $s \to 1$, the required limit is $-1/8$. Replacing $s-1$ by $x/2$ inside the difference would have erased the entire nonzero answer.

Equivalence is compatible with multiplication and division in the stated setting; it does not authorize term-by-term substitution in an arbitrary sum or difference. For the elementary counterexample $\alpha = x+x^2 \sim x$, the difference $\alpha - x = x^2$ is not equivalent to the result 0 of such a substitution. Whenever leading terms cancel, return to an identity or prove a sharper error estimate.

1.4.8 Methods, pitfalls, and further questions

A useful order of work is to determine the domain and approach, test whether direct substitution is justified, and only then choose a transformation. At a finite limiting point, first check that the domain has points arbitrarily close to it other than the point itself; the arcsine example at 2 shows why this check comes before calculation. For a direct proof, isolate a small factor and bound the other factors locally. For algebraic indeterminate forms, factor or rationalize. For higher roots, factor a difference of powers and verify the new denominator's nonzero limit, as in Example 1.32. For a bounded oscillation multiplied by a small factor, use an absolute-value estimate and squeeze. For a change of variables, track the transformed domain and the direction of approach before applying the outer limit. For a positive power with varying base and exponent, take a logarithm and study the product of the exponent with the logarithm of the base. For infinite quantities, establish eventual signs and compare rates before using the rules in Proposition 1.13. For a suspected failure of a limit, try to find two admissible approaching sequences with incompatible output limits. For products or quotients of standard small quantities, check the hypotheses of equivalent replacement.

Three distinctions deserve particular attention. The value $f(a)$ and the limit at a answer different questions. Local boundedness is weaker than existence of a limit, and unboundedness is weaker than a signed infinite limit. Finally, a finite collection of numerical samples or approaching sequences can disprove a proposed limit but cannot establish the universal requirement in Theorem 1.14.

> ***Further Questions.*** Can a function have a limit at every point while being discontinuous at some points? What additional condition makes substitution of the limiting input legitimate? The next section answers these questions by comparing the limit near a point with the value at the point itself.

Exercise 1.4

1. Write the full quantified definitions of $\lim_{x\to a^-} f(x) = L$, $\lim_{x\to -\infty} f(x) = L$, and $\lim_{x\to a^+} f(x) = -\infty$. State the corresponding assumptions on the domain.
2. Let $m, b, a \in \mathbb{R}$. Prove directly that $\lim_{x\to a}(mx + b) = ma + b$. For a fixed $\varepsilon > 0$ and $m \neq 0$, find the largest possible δ in the definition; treat $m = 0$ separately.
3. Prove directly that $\lim_{x\to 2} 1/x = 1/2$. Explain why a preliminary restriction such as $|x - 2| < 1$ is useful.
4. Determine both one-sided limits at 0 for each function, and decide whether the two-sided limit exists:
$$\frac{\sqrt{1+x}-1}{|x|}, \qquad \frac{|\sin x|}{x}, \qquad \lfloor x \rfloor, \qquad \frac{1}{x^3}.$$
Distinguish finite and infinite limits.
5. Evaluate
$$\lim_{x\to a} \frac{x^n - a^n}{x - a} \quad (n \geq 1,\ a \in \mathbb{R}), \qquad \lim_{x\to 0} \frac{\sqrt{1+2x} - \sqrt{1-x}}{x}.$$
In the first limit, handle $n = 1$ separately so that no convention for 0^0 is needed.
6. Evaluate the limits of $\sqrt{x^2 + 3x} - \sqrt{x^2 - x}$ as $x \to +\infty$ and as $x \to -\infty$. Identify where the sign of x affects the rationalized expression.
7. Find all vertical and horizontal asymptotes of $f(x) = (2x + 1)/(x - 1)$, stating the relevant limits.
8. Evaluate
$$\lim_{x\to 0} x\lfloor 1/x \rfloor, \qquad \lim_{x\to +\infty} \frac{\sin x}{x + \cos x}.$$
Use an absolute-value estimate in each case, and justify the denominator bound in the second expression.
9. Let $h(x) = \sin(\pi/x)$ for $x \neq 0$. Since $h(1/n) = 0$ for all $n \geq 1$, a proposed argument claims that $\lim_{x\to 0^+} h(x) = 0$. Explain the logical error and construct another positive input sequence that disproves the claimed limit.
10. Let $f(x) = xD(x)$, where D is the Dirichlet function. Prove that $f(x) \to 0$ as $x \to 0$, and that f has no limit at any $a \neq 0$.
11. Evaluate
$$\lim_{x\to 0} \frac{\sin(x^2)}{x \sin x}, \qquad \lim_{x\to 0} \frac{\sin(3x) - \sin x}{x}, \qquad \lim_{x\to 0} \frac{1 - \cos(3x)}{x \sin(2x)}.$$
12. Evaluate
$$\lim_{x\to 0} (1 + 2x)^{3/x}, \qquad \lim_{x\to +\infty} \left(1 - \frac{2}{x}\right)^x.$$
State where the real powers are defined before transforming them.
13. Evaluate
$$\lim_{x\to 0} \frac{\ln(1+4x)}{\sin(3x)}, \qquad \lim_{x\to 0} \frac{5^x - 1}{\tan(2x)}, \qquad \lim_{x\to 0} \frac{(1+x)^p - 1}{x} \quad (p \in \mathbb{R}).$$
14. Order x^4, $1 - \cos x$, and $\sin x$ by their relative smallness as $x \to 0$. Express the comparisons using o and give an equivalent for $1 - \cos x$.
15. Decide which statements follow from $f(x) \to L$ as $x \to a$, and give a proof or counterexample: f is bounded on its whole domain; f is bounded near a; $f(x) < L$ eventually; $|f(x)| \to |L|$. Interpret "near a" as punctured if $f(a)$ is undefined.
16. Prove the squeeze theorem using the sequential criterion and the sequence squeeze theorem. Identify the part of the argument that requires the approaching sequence to be arbitrary.
17. Let $u(x) \to 0$ as $x \to 0$, and define $v(0) = 2$ and $v(t) = 1$ for $t \neq 0$. Give one u for which $v(u(x)) \to 1$, one for which it tends to 2, and one for which it has no limit at 0.

18. Evaluate by exact rationalization
$$\lim_{x\to 0}\frac{\sqrt{1+x}+\sqrt{1-x}-2}{x^2}.$$
Explain why substituting $\sqrt{1+x}-1\sim x/2$ and $\sqrt{1-x}-1\sim -x/2$ inside the sum does not give the answer.
19. Determine all real c, d for which the function
$$f(x)=\begin{cases}(e^{cx}-1)/x, & x<0,\\ d, & x=0,\\ \ln(1+x)/\sin(2x), & x>0\end{cases}$$
has a finite limit at 0, on the domain where the displayed formulas are defined. What is that limit, and what role does d play?
20. Suppose $\alpha\sim\beta$ and $\gamma=o(\beta)$, with β eventually nonzero. Prove that $\alpha+\gamma\sim\beta$. Construct $\alpha\sim\beta$ for which $\alpha-\beta$ is not identically zero, and explain why its order cannot be recovered from the equivalence alone.

Supplement Exercises

S1. Prove the sequential criterion for $\lim_{x\to a}f(x)=+\infty$. Write out the negation of the infinite-limit definition in the converse argument.
S2. For $p>0$, study $f_p(x)=|x|^p\sin(1/x)$ as $x\to 0$. Determine its limit and find all $q>0$ for which $f_p=o(|x|^q)$.
S3. Suppose f is defined on a punctured neighbourhood of 0 and $f(x)^2\to 1$ as $x\to 0$. Show that $|f(x)|\to 1$. Must f have a limit? Give a counterexample and then a sufficient condition involving its eventual sign.
S4. Suppose $f(x)\to A$ and $f(x)g(x)\to B$ as $x\to a$, with $A\neq 0$. Prove that $g(x)\to B/A$. Show by examples that no such conclusion follows when $A=0$, even if g is bounded.
S5. Let f be increasing and bounded on (a,b), where $a<b$ are finite. Prove that its one-sided limits at the endpoints exist and equal $\inf f((a,b))$ and $\sup f((a,b))$, respectively. Here "increasing" may be read as nondecreasing.
S6. Along $x\to 0^+$, let $v(x)=1/x$. Construct choices of $u(x)\to 0$, with $1+u(x)>0$, for which $(1+u(x))^{v(x)}$ tends respectively to 1, e, and e^2. Construct one further choice for which the power has no limit.

> ***Hint.*** Use Proposition 1.14 and design the product $u(x)v(x)$ first. For the nonconvergent case, make that product oscillate between two values along suitable sequences.

S7. Construct a function on $(0,1)$ which is unbounded in every right-hand neighbourhood of 0 but has neither limit $+\infty$ nor limit $-\infty$ there. Then construct one whose absolute value tends to $+\infty$ while the function has no signed infinite limit.
S8. For integers $m, n\geq 1$, evaluate
$$\lim_{x\to 0}\frac{\sqrt{1+\sin(mx)}-\sqrt{1-\sin(mx)}}{\sin(nx)}.$$
Use only identities and limits proved in this section.
S9. Let $E=\{1/n : n\geq 1\}$ and define $f(x)=1$ on E and $f(x)=0$ on $\mathbb{R}\setminus E$. Determine every $a\in\mathbb{R}$ at which $\lim_{x\to a}f(x)$ exists. Distinguish a point of E from the accumulation point 0.
S10. Prove that $\alpha\sim\beta$ and $\beta\sim\gamma$ imply $\alpha\sim\gamma$, under the eventual nonzero conditions. Is $\alpha+\beta\sim\gamma+\eta$ implied by $\alpha\sim\gamma$ and $\beta\sim\eta$? Give a counterexample with both sums eventually nonzero, and explain how cancellation causes the failure.

1.5 Continuous Functions

重点内容与证明方法

- ❑ *Relate continuity to values, nearby limits, and domain endpoints.*
- ❑ *Apply algebra, composition, and quantitative estimates.*
- ❑ *Prove inverse continuity with the domain made explicit.*
- ❑ *Classify discontinuities and construct continuous extensions.*
- ❑ *Use sequences and dense sets to test continuity.*
- ❑ *Distinguish continuity from stronger forms of local control.*

A limit describes what the values near a point approach. Continuity asks whether that limiting behaviour agrees with the value at the point. For instance, $(x^2-1)/(x-1)$ tends to 2 at 1. Assigning the value 2 fills its hole consistently with the surrounding graph; assigning any other value creates a mismatch. This agreement, rather than any particular formula or way of drawing a graph, is the central idea.

1.5.1 Continuity at a point and on a domain

Definition 1.25

Let $f\colon D \to \mathbb{R}$ and $a \in D$. The function is continuous at a (在一点连续) *if for every $\varepsilon > 0$ there exists $\delta > 0$ such that*

$$x \in D, \quad |x-a| < \delta \quad \Longrightarrow \quad |f(x)-f(a)| < \varepsilon.$$

The function is continuous on D (在集合上连续) *if it is continuous at every point of D.* ♣

The exact negation of continuity at a is

$$\exists \varepsilon_0 > 0 \quad \forall \delta > 0 \quad \exists x \in D: \quad |x-a| < \delta, \qquad |f(x)-f(a)| \geq \varepsilon_0.$$

Unlike the negation of a punctured function limit, the selected input is allowed to equal a in principle; however, it cannot do so because the output error would then be zero. Choosing one such input for each $\delta = 1/n$ produces the bad sequence used in the sequential criterion for continuity.

This definition includes the input $x = a$. At that input the output error is zero, so no exclusion is needed. When a is an accumulation point of D, continuity is equivalent to

$$\lim_{\substack{x \to a \\ x \in D}} f(x) = f(a).$$

Thus the familiar limit test checks three things: the value $f(a)$ is defined, the finite limit exists, and that limit equals $f(a)$.

The domain is part of the statement. For example, $1/x$ is continuous at every point of $\mathbb{R} \setminus \{0\}$. It is not a function on all of $\mathbb{R}$ with the displayed formula. We may discuss its singular behaviour or possible extension at 0, but continuity at 0 requires a value there. At an isolated point of D, continuity is automatic: choose a neighbourhood containing no other point of D. The accumulation-point assumption in the function-limit definition prevents us from using a vacuous limit test at such a point.

If f is defined on $[a,b]$ with $a < b$, continuity at a means

$$\lim_{x \to a^+} f(x) = f(a),$$

and continuity at b means the corresponding equality for the left-hand limit. These are *right continuity*(右连续) and *left continuity*(左连续), respectively. Continuity on $[a, b]$ requires continuity at every interior point and these one-sided conditions at the endpoints. For half-open intervals, impose the one-sided condition only at the included endpoint.

Example 1.38 An endpoint estimate Prove that $f(x) = \sqrt{x}$ is continuous on $[0, +\infty)$.

Solution *For nonnegative x, y, assume first that $x \geq y$. Then*

$$x - y = (\sqrt{x} - \sqrt{y})(\sqrt{x} + \sqrt{y}) \geq (\sqrt{x} - \sqrt{y})^2.$$

Interchanging x, y when necessary gives

$$\left|\sqrt{x} - \sqrt{y}\right| \leq \sqrt{|x - y|}. \tag{1.8}$$

For any $a \geq 0$ and $\varepsilon > 0$, choose $\delta = \varepsilon^2$. If $x \geq 0$ and $|x - a| < \delta$, then $|\sqrt{x} - \sqrt{a}| < \varepsilon$. This argument also covers the endpoint $a = 0$.

> ***Reading a graph.*** The informal description "draw the graph without lifting the pencil" can guide intuition on an interval, but it does not specify domains, endpoints, or the required input control. For proofs, use the definition or one of the criteria established below.

1.5.2 Algebra and composition of continuous functions

Theorem 1.20

If $f, g\colon D \to \mathbb{R}$ are continuous at $a \in D$, then $f + g$, $f - g$, and fg are continuous at a. Every constant multiple of f is continuous there. If $g(a) \neq 0$, then f/g is defined on a neighbourhood of a relative to D and is continuous at a on its domain. ♡

Proof If a is an accumulation point of D, apply Theorem 1.15 to the limits $f(x) \to f(a)$ and $g(x) \to g(a)$. The resulting limits equal the values of the corresponding functions at a. When $g(a) \neq 0$, continuity gives $|g(x) - g(a)| < |g(a)|/2$ near a, so $|g(x)| > |g(a)|/2$ there. If a is isolated in D, every displayed function that is defined at a is continuous there directly from the definition. □

Restricting a continuous function to a smaller domain preserves continuity at every point retained in that domain. Indeed, the same δ controls all inputs in the smaller set. This observation lets us apply the algebraic rules on a common domain even when the original functions have different domains.

Theorem 1.21

Let $f\colon D \to E$ and $g\colon E \to \mathbb{R}$, where $D, E \subseteq \mathbb{R}$. If f is continuous at $a \in D$ and g is continuous at $f(a)$ relative to E, then $g \circ f$ is continuous at a. ♡

> ***Idea.*** Work backwards from the desired output error. Continuity of g supplies a tolerance for its input, and continuity of f turns that tolerance into a neighbourhood of a. Unlike a punctured outer limit, continuity also controls the case $f(x) = f(a)$.

Proof Let $\varepsilon > 0$. Since g is continuous at $f(a)$, there is $\eta > 0$ such that

$$y \in E, \quad |y - f(a)| < \eta \quad \Longrightarrow \quad |g(y) - g(f(a))| < \varepsilon.$$

Continuity of f at a gives $\delta > 0$ such that

$$x \in D, \quad |x - a| < \delta \quad \Longrightarrow \quad |f(x) - f(a)| < \eta.$$

Because $f(x) \in E$, substitution into the first implication proves the required estimate for $g(f(x))$. □

Example 1.39 Continuity without limits of all factors Determine all points of continuity of

$$F(x) = \begin{cases} x \sin(1/x), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Solution *At every nonzero point, continuity follows from arithmetic and composition, using the sine limits*

established in Section 1.4. At 0, the inner function $1/x$ has no finite limit, so that composition argument does not apply. Instead, $|F(x) - F(0)| \leq |x|$ proves continuity directly. Thus F is continuous on all of $\mathbb{R}$. Theorems give sufficient conditions; failure of one theorem's hypotheses does not establish discontinuity.

1.5.3 Inverse functions and elementary functions

An inverse function interchanges inputs and outputs. To establish its continuity, an input interval around a must produce a usable interval of output control around $f(a)$. Strict monotonicity provides such control without a formula for the inverse.

Theorem 1.22

Let I be an interval and let $f\colon I \to \mathbb{R}$ be strictly monotone. Then the inverse $f^{-1}\colon f(I) \to I$ is continuous relative to its domain $f(I)$ and is strictly monotone in the same direction as f. In particular, this applies when f is continuous and strictly monotone on I. ♡

> ***Proof Strategy.*** For an increasing function, points to the left and right of a have images strictly below and above $f(a)$. Keep the new output between those images; monotonicity forces its inverse image to remain near a. At an included endpoint, only the available side needs a bracket.

Proof Suppose f is strictly increasing, fix $a \in I$, and write $b = f(a)$. The inverse exists on $f(I)$ by injectivity, and order preservation of the inverse follows directly from strict increase of f. Let $\varepsilon > 0$. If I has a point less than a, the interval property allows us to choose $u \in I$ with $a - \varepsilon < u < a$. Likewise, if I has a point greater than a, choose $v \in I$ with $a < v < a + \varepsilon$. For every available bracket, the numbers $b - f(u)$ and $f(v) - b$ are strictly positive. Choose δ to be the minimum of the available positive numbers; if neither side is available, $I = \{a\}$ and the inverse is continuous automatically.

Let $y \in f(I)$ satisfy $|y - b| < \delta$, and set $x = f^{-1}(y)$. If u was chosen, then $y > f(u)$, so $x > u > a - \varepsilon$. If no left bracket was available, every point of I is at least a, which gives the same required lower bound. Similarly, either $y < f(v)$ forces $x < v < a + \varepsilon$, or every point of I is at most a. Thus $\left|f^{-1}(y) - a\right| < \varepsilon$, proving continuity at b. For strictly decreasing f, apply the increasing case to $-f$ and compose the resulting inverse with $y \mapsto -y$. □

The theorem asserts continuity on the *actual range* $f(I)$. It does not assert that this range is an interval when f is discontinuous. For example, the strictly increasing function

$$f(x) = \begin{cases} x, & x < 0, \\ x + 1, & x \geq 0 \end{cases}$$

has range $(-\infty, 0) \cup [1, +\infty)$. Its inverse is continuous on that range, including at 1 relative to the range, although f itself jumps at 0. In the next section, the intermediate value theorem will show that the range of a *continuous* function on an interval is an interval.

Corollary 1.3

Every elementary function is continuous on its natural domain. ♡

Proof Constant functions and the identity are continuous directly from the definition. The arithmetic rules give continuity of polynomials and of rational functions wherever their denominators are nonzero. The trigonometric limits in Section 1.4 give continuity of sine and cosine, and of their quotients wherever defined. The exponential and logarithmic functions are continuous by Lemma 1.2.

For $x > 0$, a real power $x^p = e^{p \ln x}$ is continuous by composition. When $p > 0$ and the value $0^p = 0$ is included, continuity at 0 follows from $0 \le x < \varepsilon^{1/p} \Longrightarrow x^p < \varepsilon$. Integer powers already follow from arithmetic. Roots on their real domains are continuous inverses of the corresponding strictly monotone power functions; odd roots include negative inputs, and even roots use the nonnegative interval. The inverse trigonometric functions are continuous by Theorem 1.22, using the standard intervals of strict monotonicity specified in Section 1.2. This also handles their included endpoints. Finally, finitely many arithmetic operations and compositions preserve continuity on the domain where the resulting expression is defined. □

Example 1.40 Determine the domain first Find the natural domain of

$$h(x) = \frac{\ln(1 - x^2)}{\sqrt{x + 2}}$$

and determine where it is continuous.

Solution *The logarithm requires* $1 - x^2 > 0$, *so* $-1 < x < 1$. *The denominator requires* $x + 2 > 0$, *which already holds on this interval. Thus the natural domain is* $(-1, 1)$, *and* h *is continuous at every point of it by Corollary 1.3. The formula has no values at* -1 *and* 1. *Endpoint extension is a separate question, to be decided by limits rather than by a direct substitution into the formula.*

1.5.4 Discontinuities and continuous extensions

When a function is defined on a neighbourhood of a but fails to be continuous there, a is a *point of discontinuity*(间断点). The same terminology is often used when the original formula is defined only on a punctured neighbourhood, in order to discuss whether it can be extended across the missing point. We will explicitly distinguish a missing value from an assigned value that fails the continuity test.

Definition 1.26

Suppose f *is defined on a punctured neighbourhood of* a.

1. *A* removable discontinuity(可去间断点) *occurs if* $\lim_{x \to a} f(x) = L \in \mathbb{R}$, *but* $f(a)$ *is missing or differs from* L.
2. *A* jump discontinuity(跳跃间断点) *occurs if both one-sided limits are finite but unequal.*
3. *An* infinite discontinuity(无穷间断点) *occurs if at least one one-sided limit is* $+\infty$ *or* $-\infty$.
4. *An* oscillatory discontinuity(振荡间断点) *describes a failure of a one-sided limit caused by persistent oscillation, as for* $\sin(1/(x - a))$.

♣

Removable and jump discontinuities are traditionally called discontinuities of the *first kind*(第一类间断点), since both one-sided finite limits exist. If at least one finite one-sided limit does not exist, the discontinuity is of the *second kind*(第二类间断点). The last two descriptions in Definition 1.26 identify common mechanisms of the second kind; a more complicated function can combine different behaviours on its two sides. Neither a jump nor persistent oscillation can be repaired by changing just $f(a)$.

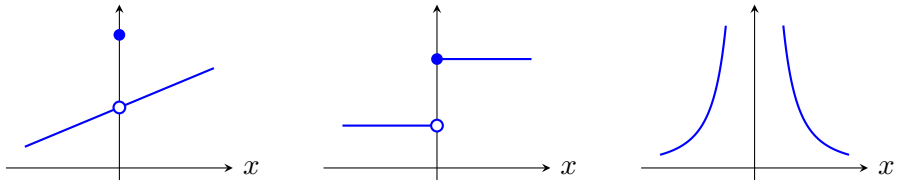
The schematics in **Figure** 1.7 separate a mismatched value, unequal side limits, and unbounded growth. The following examples make the distinctions exact.

Example 1.41 Classifying singular behaviour Classify the behaviour at 0 of the following formulas, initially defined for $x \neq 0$:

$$\frac{\sin x}{x}, \qquad \frac{|x|}{x}, \qquad \frac{1}{x^2}, \qquad \sin(1/x).$$

For each, determine whether assigning a value at 0 can make it continuous there.

Solution *The first has limit* 1, *so assigning that value gives continuity; its missing point is removable. The*



removable jump infinite

Figure 1.7: Three ways in which nearby values can fail to agree with a value at the origin. Oscillation gives another mechanism, illustrated in **Figure** 1.5.

second has side limits -1 and 1, hence a jump. The third tends to $+\infty$, so it has an infinite discontinuity under any real-valued assignment at 0. The fourth has distinct sequential output limits even from one side, hence oscillatory behaviour. Only the first admits a continuous extension.

Proposition 1.16

Let a be an accumulation point of D, with $a \notin D$, and let $f\colon D \to \mathbb{R}$. A function $F\colon D \cup \{a\} \to \mathbb{R}$ agreeing with f on D can be continuous at a if and only if $\lim_{x\to a,\, x\in D} f(x)$ exists as a finite real number. In that case the unique possible value is

$$F(a) = \lim_{\substack{x\to a \\ x\in D}} f(x).$$

If f is also continuous on D, this defines a continuous extension (连续延拓) *on $D \cup \{a\}$.*

Proof Continuity at a requires the limit to equal $F(a)$, so necessity and uniqueness follow from the definition and uniqueness of limits. Conversely, assign the finite limit to $F(a)$. The limit estimate gives continuity at a, including the trivial case $x = a$. At a point $b \in D$, restrict to a neighbourhood of radius less than $|b - a|$; within it the extension agrees with f, so any continuity at b is preserved. □

Example 1.42 Gluing three pieces Find all $A, B \in \mathbb{R}$ for which

$$f(x) = \begin{cases} \sin x/x, & x < 0, \\ A, & x = 0, \\ B + \ln(1+x)/x, & x > 0 \end{cases}$$

is continuous on $\mathbb{R}$.

Solution *Each open piece is continuous on its domain, so only 0 needs a special check. The left-hand limit is 1, and the right-hand limit is $B + 1$. Both must equal $f(0) = A$. Thus $A = 1$ and $B = 0$, and these choices do give continuity everywhere. Matching the two side limits would determine B but would leave the value A unchecked.*

Example 1.43 A floor-function composition Determine the discontinuities of $g(x) = \lfloor x^2 \rfloor$.

Solution *If x_0^2 is not an integer, then it lies strictly between two consecutive integers. Continuity of x^2 keeps nearby squared values in that same interval, so g is locally constant at x_0. At 0, one has $g(x) = 0$ for $|x| < 1$, so g is continuous there as well. For a positive integer n, at $x_0 = \sqrt{n}$ the left and right limits are $n - 1$ and n. At $x_0 = -\sqrt{n}$, their order is reversed: the limits are n and $n - 1$. Thus the discontinuities are exactly $\pm\sqrt{n}$ for $n \geq 1$, and each is a jump. An integer value of the inner function alone does not guarantee a discontinuity of the composition; the way nearby inputs reach that value matters.*

1.5.5 Sequential continuity and quantitative control

Theorem 1.23

Let $f\colon D \to \mathbb{R}$ and $a \in D$. Then f is continuous at a if and only if every sequence (x_n) in D converging to a satisfies $f(x_n) \to f(a)$. The sequence is allowed to contain the point a. ♡

> ***Idea.*** The argument resembles Heine's criterion, but the input neighbourhood is no longer punctured and the proposed output limit is fixed to be $f(a)$. Failure of continuity supplies inputs arbitrarily close to a whose outputs stay a fixed positive distance from $f(a)$.

Proof If f is continuous at a, choose δ for a given $\varepsilon > 0$. Since $x_n \to a$, eventually $|x_n - a| < \delta$, and hence $|f(x_n) - f(a)| < \varepsilon$. Conversely, if continuity fails, there is $\varepsilon_0 > 0$ such that for every $n \geq 1$ we can choose $x_n \in D$ with

$$|x_n - a| < \frac{1}{n}, \qquad |f(x_n) - f(a)| \geq \varepsilon_0.$$

This sequence tends to a but its output sequence does not tend to $f(a)$, contradicting the stated property. □

Example 1.44 A function controlled at just one point Determine all points of continuity of $f(x) = xD(x)$ on $\mathbb{R}$.

Solution *Since $|f(x)| \leq |x|$ and $f(0) = 0$, the function is continuous at 0. At $a \neq 0$, choose rational and irrational sequences both tending to a. The rational outputs tend to a, and the irrational outputs equal 0. They cannot both tend to the single value $f(a)$. Thus the only point of continuity is 0. The factor x suppresses the dense-set disagreement exactly where that factor vanishes.*

Definition 1.27

A function $f\colon D \to \mathbb{R}$ is Lipschitz continuous (Lipschitz 连续) *on D if there is a constant $K \geq 0$ such that*

$$|f(x) - f(y)| \leq K\,|x - y| \qquad (x, y \in D).$$

♣

Every Lipschitz function is continuous: if $K > 0$, choose $\delta = \varepsilon/K$ in the continuity definition at any point; if $K = 0$, the function is constant on a nonempty domain. Even a local estimate $|f(x) - f(a)| \leq C\,|x - a|$ near one point is enough for continuity at that point. Such estimates give more information than the bare existence of a suitable neighbourhood.

The reverse triangle inequality shows that $|x|$ is Lipschitz on $\mathbb{R}$ with $K = 1$. For $x, y \in [-R, R]$ with $R > 0$,

$$\left|x^2 - y^2\right| = |x - y|\,|x + y| \leq 2R\,|x - y|\,,$$

so the square function is Lipschitz on each bounded interval of this form. The square-root function is continuous at 0, but it is not Lipschitz on $[0, 1]$. Indeed, a Lipschitz estimate with $y = 0$ and $x > 0$ would require $1/\sqrt{x} \leq K$, which fails as $x \to 0^+$. The weaker estimate (1.8) still gives continuity.

1.5.6 Further reading: a dense set of discontinuities

The Dirichlet function is discontinuous everywhere because two dense sets retain a fixed output separation. Can the same rational–irrational split produce continuity at irrational inputs? The answer changes when the size of a rational value decreases with its denominator.

Example 1.45 Thomae's function On $[0,1]$, define

$$T(x) = \begin{cases} 1/q, & x = p/q \text{ in lowest terms}, \quad q \geq 1, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

Here $0 = 0/1$ and $1 = 1/1$. Prove that T is continuous at every irrational point and discontinuous at every rational point of $[0,1]$.

> ***Proof Strategy.*** For a fixed tolerance, only finitely many rationals in $[0,1]$ have denominators small enough to produce a large output. An irrational input can be separated from that finite exceptional set by a positive distance. At a rational input, an irrational sequence immediately detects the mismatch.

Proof Let $a \in [0,1]$ be irrational and let $\varepsilon > 0$. Choose a positive integer N with $1/N < \varepsilon$. There are only finitely many reduced rationals in $[0,1]$ with denominators $q \leq N$, since for each such q the numerator satisfies $0 \leq p \leq q$. Call this finite set E_N. It does not contain a, so the minimum distance from a to E_N is positive. Choose δ smaller than that distance. If $x \in [0,1]$ and $|x - a| < \delta$, then either x is irrational and $T(x) = 0$, or x has reduced denominator $q > N$ and $T(x) = 1/q < 1/N < \varepsilon$. In either case $|T(x) - T(a)| < \varepsilon$.

Now let $a = p/q$ be rational in lowest terms. By density, choose irrational points $x_n \in [0,1]$ tending to a, using the available one-sided interval at an endpoint. Then $T(x_n) = 0$ for every n, whereas $T(a) = 1/q > 0$. The sequential criterion proves discontinuity at a. □

In fact, the same finite-set argument proves that the limit of $T(x)$ at every point of $[0,1]$, through distinct domain points, is 0. At a rational a, omit a itself from E_N when choosing the distance; if no exceptional points remain, any small radius suffices. Thus each rational discontinuity is removable individually. Removing all of them produces the zero function. Continuity is local, and a complicated global pattern of exceptional points does not prevent a precise local proof.

1.5.7 Methods, pitfalls, and the local-to-global question

For a function given by elementary formulas on pieces, first identify the natural domain of each formula and the points where pieces meet. Away from those points, continuity often follows from the algebra and composition rules. At a meeting point, compute the available side limits and compare every one of them with the assigned value. At a missing point, a finite limit supplies the only possible continuous extension. When limits are difficult to compute, an error estimate or a carefully chosen sequence may settle the question directly.

Continuity of the outer function is a sufficient condition for continuity of a composition, not a necessary one. Likewise, Lipschitz control is sufficient but not necessary for continuity. Avoid turning a theorem's hypothesis into an automatic test for the opposite conclusion when that hypothesis fails. At an endpoint, all neighbourhoods are interpreted relative to the domain.

> ***Looking ahead.*** Continuity at one point controls nearby values, but the radius may depend on that point. What can be concluded when continuity holds at every point of a closed bounded interval? The next section explains why the function must attain its largest and smallest values, take every intermediate value, and admit a single input tolerance valid throughout the interval.

Exercise 1.5

1. State the definition of continuity at a for a function on $[a,b)$ with $a < b$. Explain why a left-hand limit at a is not required and why no continuity condition is imposed at the missing endpoint b.

2. Prove directly that $f(x) = x^2$ is continuous at every $a \in \mathbb{R}$. Give a possible δ in terms of a and ε.
3. Determine the natural domain of each function and justify continuity on that domain:
$$\sqrt{1 - x^2}, \qquad \ln(x^2 - 1), \qquad \frac{\arcsin x}{1 - x}, \qquad e^{1/(x-2)}.$$
4. Let $f(0) = c$ and $f(x) = \sin(3x)/x$ for $x \neq 0$. Determine the value of c giving continuity on $\mathbb{R}$. Classify the discontinuity at 0 for every other value of c.
5. Find $a, b \in \mathbb{R}$ such that the following function is continuous on $\mathbb{R}$:
$$f(x) = \begin{cases} x^2, & x \leq 1, \\ ax + b, & 1 < x < 2, \\ 3x - 1, & x \geq 2. \end{cases}$$
6. Classify every discontinuity of $\lfloor x \rfloor$, $x - \lfloor x \rfloor$, and $1/(x^2 - 1)$. For the last formula, discuss its missing points and their one-sided behaviour.
7. For the formula
$$f(x) = \frac{\sin x}{x(x - 1)} \qquad (x \in \mathbb{R} \setminus \{0, 1\}),$$
decide which missing points admit continuous extensions and find their values. Determine the largest domain in $\mathbb{R}$ to which this particular function can be continuously extended.
8. Prove that if f is continuous at a and $f(a) > 0$, then f is positive on a neighbourhood of a relative to its domain. Why is a positive value at a alone insufficient?
9. Prove that if f is continuous, then so is $|f|$. Deduce continuity of $\max\{f, g\}$ and $\min\{f, g\}$ whenever f and g are continuous on the same domain.
10. Give an example where $g \circ f$ is continuous at 0 but g is discontinuous at $f(0)$. Give another where $g \circ f$ is continuous at 0 but f is discontinuous at 0.
11. Determine all discontinuities of $f(x) = \lfloor \cos x \rfloor$ and classify each one. Pay particular attention to inputs where $\cos x$ equals 1, 0, or -1.
12. Use sequences to determine all points of continuity of
$$f(x) = \begin{cases} x^2, & x \in \mathbb{Q}, \\ -x, & x \notin \mathbb{Q}. \end{cases}$$
13. Prove that $f(x) = |2x - 1|$ is Lipschitz on $\mathbb{R}$. Prove that $g(x) = 1/x$ is Lipschitz on $[a, +\infty)$ for each fixed $a > 0$, and give a valid constant.
14. Prove that the function $x \mapsto x^{1/3}$ is continuous on $\mathbb{R}$ by applying the inverse-continuity theorem. Show that it is not Lipschitz on any interval containing 0 as an interior point.
15. Let $f\colon [-1, 1] \to \mathbb{R}$ be continuous and strictly increasing, and let $b = f(0)$. Given $0 < \varepsilon < 1$, give an explicit positive δ in terms of $f(-\varepsilon)$, $f(0)$, and $f(\varepsilon)$ which proves continuity of f^{-1} at b.
16. Let $D = \{0\} \cup [1, 2]$, define $f(0) = 100$, and set $f(x) = x$ on $[1, 2]$. Prove that f is continuous on D. Explain why continuity at the isolated point 0 should be checked using the neighbourhood definition.
17. Suppose $f\colon \mathbb{R} \to \mathbb{R}$ is continuous at a and $f(x) = 0$ for every rational x. Prove that $f(a) = 0$. Deduce that a continuous function on $\mathbb{R}$ is determined by its values on $\mathbb{Q}$.
18. Let $f\colon [-1, 1] \to \mathbb{R}$ satisfy $|f(x) - f(0)| \leq C\,|x|^p$ for some constants $C > 0$ and $p > 0$. Prove continuity at 0 and give an explicit δ for every $\varepsilon > 0$. Does this assumption imply continuity at other points? Construct a counterexample if it does not.

Supplement Exercises

S1. Let $f, g\colon \mathbb{R} \to \mathbb{R}$ be continuous, and define $h = f$ on $\mathbb{Q}$ and $h = g$ on $\mathbb{R} \setminus \mathbb{Q}$. Prove that h is continuous at a if and only if $f(a) = g(a)$.

S2. Prove that if both $f + g$ and $f - g$ are continuous at a point, then both f and g are continuous there. Is the analogous claim with $f + g$ and fg true? Give a proof or counterexample.

S3. Let $f\colon \mathbb{R} \to \mathbb{R}$ satisfy $f(x+y) = f(x)+f(y)$ for all real x, y. If f is continuous at 0, prove that $f(x) = cx$ for all x, where $c = f(1)$.

> ***Hint.*** First establish the formula on $\mathbb{Q}$ and then pass to real inputs.

S4. Suppose $f\colon (0, +\infty) \to \mathbb{R}$ is continuous at 1 and satisfies $f(x^2) = f(x)$ for every $x > 0$. Prove that f is constant, using repeated square roots and a sequence converging to 1.

S5. Prove directly that the limit of Thomae's function from Example 1.45 at each rational $a \in [0, 1]$, through distinct domain points, is 0. Explain how the finite exceptional set must be modified when it contains a.

S6. Suppose $f\colon D \to \mathbb{R}$ satisfies $|f(x) - f(y)| \le C\,|x - y|^p$ for all $x, y \in D$, where $C > 0$ and $p > 0$. Prove continuity on D. If D is an interval and $p > 1$, prove that f is constant by dividing a segment into n equal parts.

S7. Let I be an interval, and suppose $f\colon I \to \mathbb{R}$ is strictly increasing and $f(I)$ is also an interval. Prove that f is continuous. You may apply Theorem 1.22 to f^{-1}; explain why its hypotheses hold.

S8. Give a continuous injective function on a domain that is not an interval whose inverse on its range is discontinuous.

> ***Hint.*** One possible domain is $[0, 1) \cup \{2\}$; arrange for a sequence approaching the missing point 1 to have images approaching the image of 2.

S9. Let $f\colon \mathbb{R} \to \mathbb{R}$ be continuous at a and suppose $|f(a)| < 1$. Prove that some neighbourhood of a admits a constant $q < 1$ with $|f(x)| \le q$ throughout that neighbourhood. Deduce that for every sequence $x_n \to a$, one has $f(x_n)^n \to 0$.

S10. Define $F(0) = 0$ and $F(x) = x \sin(1/x^2)$ for $x \neq 0$. Prove continuity on $\mathbb{R}$ and prove that F is not Lipschitz on any neighbourhood of 0.

> ***Hint.*** For the second assertion, compare inputs $x_n = (2\pi n + \pi/2)^{-1/2}$ and $y_n = (2\pi n + 3\pi/2)^{-1/2}$.

1.6 Continuous Functions on Closed Intervals

重点内容与证明方法

- ❑ *Use intermediate values to prove existence, and test uniqueness separately.*
- ❑ *Distinguish bounds from attained extrema.*
- ❑ *Verify the hypotheses of closed-interval theorems.*
- ❑ *Study proofs using bisection and convergent subsequences.*
- ❑ *Compare continuity and uniform continuity.*
- ❑ *Prove Heine–Cantor and examine its hypotheses.*

Continuity is defined one point at a time. Nevertheless, on a closed bounded interval it forces several properties of the whole function: it takes every intermediate value, attains its extreme values, and admits a single input tolerance that controls all points at once. These conclusions connect the local language of limits with the completeness of the real line. They also make it possible to prove that equations have solutions without first finding a formula for those solutions.

Throughout this section, $a, b \in \mathbb{R}$ and $a < b$ unless stated otherwise. Continuity on $[a, b]$ includes right continuity at a and left continuity at b. We first state the main conclusions and use them in examples. Complete proofs are given in the deeper-reading subsection Section 1.6.3, using results already established in this chapter.

1.6.1 Intermediate values and zeros

Theorem 1.24 (Intermediate value theorem)

Let $f\colon [a, b] \to \mathbb{R}$ be continuous. If y lies between $f(a)$ and $f(b)$, including either endpoint value, then there exists $c \in [a, b]$ such that $f(c) = y$. If y lies strictly between $f(a)$ and $f(b)$, such a point lies in (a, b). ♡

The *intermediate value property*(介值性)expresses the absence of a gap in the values taken between two inputs. It does not assert monotonicity, and the point c need not be unique. For instance, a continuous oscillating graph may cross one horizontal level many times. The theorem applies to any continuous function on an interval by restricting it to the closed segment joining the two chosen inputs. Thus the interval need not itself be closed or bounded in order for intermediate values between two attained values to be attained.

Corollary 1.4 (Zero theorem)

If $f\colon [a, b] \to \mathbb{R}$ is continuous and $f(a)f(b) < 0$, then $f(c) = 0$ for some $c \in (a, b)$. If instead $f(a)f(b) \le 0$, then a zero exists in $[a, b]$. ♡

Proof With a strict sign change, 0 lies strictly between the endpoint values, so apply Theorem 1.24 with $y = 0$. If an endpoint value is zero, that endpoint supplies the conclusion in the nonstrict case. □

Conversely, applying the zero theorem to $g(x) = f(x) - y$ proves the intermediate value theorem for a strictly intermediate level y. Endpoint levels are already attained at the endpoints. Thus the two statements contain the same existence mechanism. The strict sign condition is convenient for locating a zero inside an interval, but it is sufficient rather than necessary: x^2 has a zero on $[-1, 1]$ although its endpoint values are both positive.

Example 1.46 Existence and uniqueness of a root Prove that $x = \cos x$ has exactly one solution in $[0, 1]$.

> ***Idea.*** To prove existence, move both sides to a continuous function with opposite endpoint signs. To prove uniqueness, use order: the left side is strictly increasing and the right side is decreasing on the interval. These are separate arguments.

Solution *Set $g(x) = x - \cos x$. It is continuous on $[0, 1]$ by the continuity rules, $g(0) = -1 < 0$, and $g(1) = 1 - \cos 1 > 0$. The last inequality follows from $1 - \cos 1 = 2\sin^2(1/2) > 0$. The zero theorem gives a solution in $(0, 1)$. If $0 \le x < y \le 1$, cosine is decreasing on $[0, 1] \subset [0, \pi]$, so $\cos x \ge \cos y$. The equalities $x = \cos x$ and $y = \cos y$ would therefore imply $x \ge y$, a contradiction. Thus the solution is unique. The elementary order property of cosine here can also be checked from $\cos x - \cos y = 2\sin((x + y)/2)\sin((y - x)/2) > 0$ for $x < y$ in this interval.*

The zero theorem supplies existence even when solving the equation exactly is impractical. It also supports a numerical procedure: once the endpoint signs are opposite, repeatedly halve the interval and keep a half whose endpoint values have opposite signs. The proof below shows why this *bisection method*(二分法)converges and gives its error bound.

1.6.2 Boundedness, extrema, and the range

Theorem 1.25

Every continuous function $f\colon [a, b] \to \mathbb{R}$ is bounded on $[a, b]$. That is, there exists $C \ge 0$ such that $|f(x)| \le C$ for every $x \in [a, b]$.

♡

Continuity at one point gives boundedness only near that point. The theorem combines the local behaviour over the entire interval. A point-dependent family of local bounds is not yet a proof of one global bound; the deeper-reading proof explains how the closed bounded domain supplies the missing step.

Definition 1.28

A function $f\colon D \to \mathbb{R}$ attains its maximum(取得最大値)*at $x_M \in D$ if $f(x) \le f(x_M)$ for every $x \in D$. It* attains its minimum(取得最小値)*at $x_m \in D$ if $f(x_m) \le f(x)$ for every $x \in D$. These are absolute extrema on the specified domain.*

♣

Theorem 1.26 (Extreme value theorem)

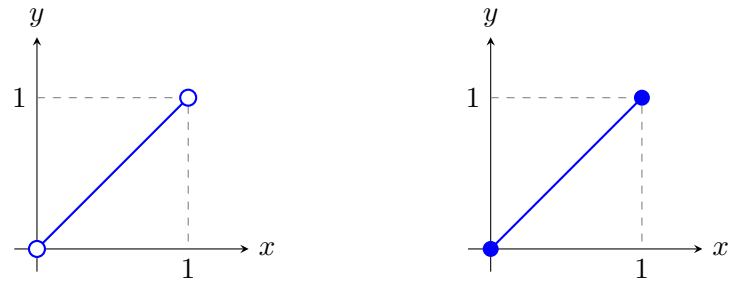
If $f\colon [a, b] \to \mathbb{R}$ is continuous, there exist $x_m, x_M \in [a, b]$ such that

$$f(x_m) \le f(x) \le f(x_M) \qquad (x \in [a, b]).$$

Thus f attains both a minimum and a maximum. The points of attainment need not be unique and may be endpoints.

♡

A supremum is a bound characterized by order; a maximum must also be a value of the function. For $f(x) = x$ on $(0, 1)$, the supremum is 1 and the infimum is 0, but neither is attained. On $[0, 1]$, both are attained. The distinction in **Figure** 1.8 is exactly what boundedness alone leaves unresolved.



$f(x) = x,\ 0 < x < 1$ $\qquad$ $f(x) = x,\ 0 \le x \le 1$

Figure 1.8: The same formula is bounded on both domains. Including the endpoints allows both extreme bounds to be attained.

Corollary 1.5

If $f\colon [a,b] \to \mathbb{R}$ is continuous and its minimum and maximum are m and M, then

$$f([a,b]) = [m, M].$$

In particular, if $f(x) > 0$ for every $x \in [a,b]$, then $f(x) \ge m > 0$ throughout the interval for some m. ♡

Proof Every value belongs to $[m, M]$ by the definitions of the extrema. If $m = M$, the function is constant and the range assertion is immediate. Otherwise, choose points x_m, x_M attaining the extrema and apply the intermediate value theorem on the segment with these two endpoints, placed in increasing order. Every number between m and M is attained there. If f is everywhere positive, its attained minimum $m = f(x_m)$ is positive. □

Pointwise positivity on an open interval gives no such uniform positive lower bound: $f(x) = x$ is positive on $(0,1)$, but its infimum is zero. On a closed bounded interval, the positive lower bound also shows that the reciprocal of an everywhere positive continuous function is bounded.

Example 1.47 Existence of a fixed point Let $f\colon [a,b] \to [a,b]$ be continuous. Prove that there is $c \in [a,b]$ with $f(c) = c$. Such a point is called a *fixed point* (不动点) of f.

> ***Idea.*** Subtract the input: the map-into-itself assumption gives the signs of $f(x) - x$ at both endpoints, even without knowing the actual endpoint values.

Solution *Let $g(x) = f(x) - x$. It is continuous on $[a,b]$, and*

$$g(a) = f(a) - a \ge 0, \qquad g(b) = f(b) - b \le 0.$$

If either endpoint value is zero, that endpoint is fixed. Otherwise the zero theorem gives a zero in (a,b). Uniqueness need not hold: the identity function fixes every point of the interval.

Proposition 1.17

If $f\colon I \to \mathbb{R}$ is continuous and injective on an interval I, then it is strictly monotone. Consequently, its inverse is continuous on the interval $f(I)$. ♠

> ***Proof Strategy.*** For three ordered inputs, the middle output must lie between the other two outputs. Otherwise an intermediate horizontal level would be attained twice, contradicting injectivity. Once this between-values property is known, the direction of increase or decrease cannot reverse.

Proof If $r < s < t$ are in I, injectivity makes their three outputs distinct. Suppose, for example, $f(s) > \max\{f(r), f(t)\}$. Choose y strictly between that maximum and $f(s)$. The intermediate value theorem on

$[r, s]$ and $[s, t]$ produces one preimage in each open segment, contradicting injectivity. The case $f(s) < \min\{f(r), f(t)\}$ is analogous. Hence $f(s)$ lies strictly between $f(r)$ and $f(t)$ for every ordered triple.

If I has at most one point, the conclusion is vacuous. Otherwise choose $u < v$ in I. Suppose $f(u) < f(v)$. For any $x < y$ in I, arrange the distinct points among u, v, x, y in increasing order. For each three consecutive points, the between-values property shows that the two consecutive output differences have the same sign. Thus all consecutive differences have one common sign. That sign is positive because $f(v) - f(u) > 0$ is a sum of some of those differences. It follows that $f(x) < f(y)$. The case $f(u) > f(v)$ similarly gives strict decrease. The intermediate value theorem shows that $f(I)$ is an interval, and Theorem 1.22 gives continuity of the inverse. □

1.6.3 Further reading: proofs from completeness

The main theorems are useful before one has mastered every detail of their proofs. The complete arguments below explain their common foundations and are intended for deeper study. They use the monotone sequence theorem, the Bolzano–Weierstrass theorem, and sequential continuity; no later calculus results are needed.

Lemma 1.3 (Nested intervals)

Let $I_n = [a_n, b_n]$, $n \geq 0$, be nonempty closed intervals with

$$I_{n+1} \subseteq I_n, \qquad b_n - a_n \to 0.$$

Then there is exactly one point c belonging to every I_n, and both endpoint sequences converge to c. ♡

> ***Idea.*** The left endpoints move right, and the right endpoints move left. They remain bounded by the first interval, so monotone convergence gives limits; shrinking lengths force those limits to agree.

Proof The sequence (a_n) is nondecreasing and bounded above by b_0, while (b_n) is nonincreasing and bounded below by a_0. By Theorem 1.9, they converge to limits α and β. Since $b_n - a_n \to 0$, the sequence limit laws give $\beta = \alpha$; write the common value as c. For each fixed n, nesting gives $a_n \leq a_k \leq b_k \leq b_n$ for all $k \geq n$. Passing to the limit yields $a_n \leq c \leq b_n$, so $c \in I_n$ for every n. If d also belongs to every I_n, then $|c - d| \leq b_n - a_n$ for every n, hence $c = d$. □

> ***Proof Strategy for the intermediate value theorem.*** It is enough to prove the zero theorem. Bisect a sign-changing interval and retain a sign-changing half. The nested intervals locate one common point, and continuity prevents that point's value from being strictly positive or strictly negative.

Proof of Theorem 1.24. First suppose $f(a) < 0 < f(b)$. Set $a_0 = a$ and $b_0 = b$. At stage n, let $m_n = (a_n + b_n)/2$. If $f(m_n) = 0$, a zero has been found. Otherwise retain $[m_n, b_n]$ when $f(m_n) < 0$, and retain $[a_n, m_n]$ when $f(m_n) > 0$. Unless the process has already found a zero, this constructs nested intervals satisfying

$$f(a_n) < 0 < f(b_n), \qquad b_n - a_n = \frac{b - a}{2^n}.$$

By Lemma 1.3, both endpoint sequences tend to a common $c \in [a, b]$. By Theorem 1.23, $f(a_n) \to f(c)$ and $f(b_n) \to f(c)$. Preservation of nonstrict order in sequence limits gives $f(c) \leq 0$ and $f(c) \geq 0$. Thus $f(c) = 0$, and the strict signs at the original endpoints imply $c \in (a, b)$. If the endpoint signs are reversed, apply the argument to $-f$.

For a level y strictly between $f(a)$ and $f(b)$, apply the established zero result to the continuous function $f - y$. If y equals an endpoint value, that endpoint supplies a preimage. This proves the full theorem. □

The proof also gives an explicit approximation guarantee. After n bisections, the retained interval has

length $(b-a)/2^n$ and contains a zero c. Its midpoint m_n therefore satisfies

$$|m_n - c| \le \frac{b-a}{2^{n+1}}.$$

This bound concerns the input error, not necessarily the output error $|f(m_n)|$. If a midpoint is already a zero, the procedure may stop early. The intervals in **Figure** 1.9 illustrate the nested structure.

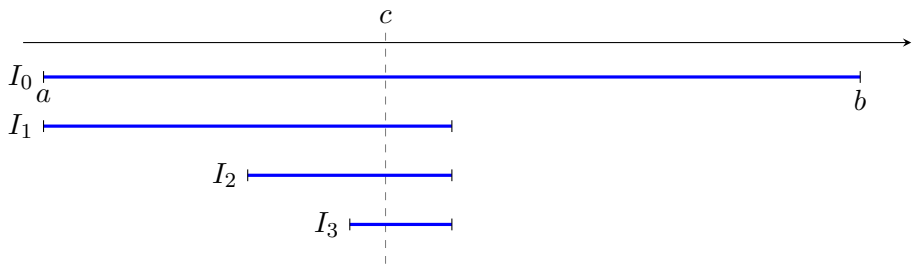


Figure 1.9: Successive sign-changing halves shrink to one point. The particular retained half depends on the sign at its midpoint.

> ***Proof Strategy for boundedness.*** Failure of a global bound would let us choose points where the outputs exceed $1, 2, 3, \ldots$ in magnitude. The inputs lie in a bounded interval, so a subsequence converges; closedness keeps its limit in the domain, where continuity contradicts the diverging outputs.

Proof of Theorem 1.25. Suppose f is not bounded on $[a,b]$. For each integer $n \ge 1$, choose $x_n \in [a,b]$ with $|f(x_n)| > n$. The sequence (x_n) is bounded, so Theorem 1.11 gives a subsequence $x_{n_k} \to c$. Because $a \le x_{n_k} \le b$, preservation of order gives $a \le c \le b$. Continuity at c implies $f(x_{n_k}) \to f(c)$, so this output subsequence is bounded by Theorem 1.4. But $|f(x_{n_k})| > n_k \ge k$, contradicting boundedness. □

Notice the three different roles in this proof. Boundedness of the interval supplies a convergent subsequence. Closedness keeps its limit in the domain. Continuity carries convergence from the inputs to the outputs. Dropping any one of these features may break the argument.

> ***Proof Strategy for attainment of extrema.*** Boundedness supplies a finite supremum M of the range. Approximate M by actual values $f(x_n)$, then extract a convergent input subsequence. Continuity turns approximation of the bound into exact attainment at the limiting input.

Proof of Theorem 1.26. The nonempty range $f([a,b])$ is bounded by Theorem 1.25. Let $M = \sup f([a,b])$. By Proposition 1.2, for every $n \ge 1$ there is $x_n \in [a,b]$ such that

$$M - \frac{1}{n} < f(x_n) \le M.$$

Thus $f(x_n) \to M$ by the squeeze theorem for sequences. By Bolzano–Weierstrass, some subsequence x_{n_k} converges to $x_M \in [a,b]$. Continuity and the subsequence limit property give

$$f(x_M) = \lim_{k\to\infty} f(x_{n_k}) = M.$$

Hence the supremum is attained. Apply the same argument to the continuous function $-f$ to obtain a point x_m at which f attains its minimum. □

The sequence (x_n) chosen in this proof is a *maximizing sequence*(极大化序列): its function values approach the supremum. The sequence itself need not converge, nor need it be monotone. Only a convergent subsequence is required. This proof pattern will recur in optimization problems beyond calculus.

1.6.4 Why the hypotheses matter

The examples below isolate different failures. On $(0,1)$, the continuous function $1/x$ is unbounded, and the continuous function x is bounded but attains neither extreme bound. On the closed unbounded interval $[0,+\infty)$, the continuous function x is unbounded, while $1/(1+x)$ is bounded but does not attain its infimum 0. Thus neither closedness alone nor boundedness of the domain alone gives the closed-interval conclusions.

Continuity is also essential. Define $f(0)=0$ and $f(x)=1/x$ for $0<x\le 1$. This real-valued function has the closed bounded domain $[0,1]$ but is unbounded because it is discontinuous at 0. For a bounded example, define

$$g(x)=\begin{cases} x, & 0\le x<1,\\ 0, & x=1. \end{cases}$$

Its supremum is 1, but no value equals 1; continuity fails at the endpoint. For failure of intermediate values, the step function equal to 0 on $[-1,0)$ and 1 on $[0,1]$ omits $1/2$ despite having endpoint values on either side of it.

Finally, the interval hypothesis is essential for the intermediate value conclusion. The function $f(x)=x$ is continuous on $[-2,-1]\cup[1,2]$, but it has no zero there. The domain has a gap through which an intermediate input would have to pass. Closed bounded domains retain subsequence properties relevant to extrema; they need not retain the interval property relevant to intermediate values.

1.6.5 Uniform continuity

For continuity at one point, the radius δ may depend on that point. The square function illustrates why this dependence can matter:

$$\left|x^2-y^2\right|=|x-y|\,|x+y|\,.$$

On a bounded interval, the second factor admits one common bound. On all of $\mathbb{R}$, inputs of very large magnitude may amplify a small input difference.

Definition 1.29

A function $f\colon D\to\mathbb{R}$ is uniformly continuous (一致连续) *on D if for every $\varepsilon>0$ there exists $\delta>0$ such that, for all $x,y\in D$,*

$$|x-y|<\delta \quad\Longrightarrow\quad |f(x)-f(y)|<\varepsilon.$$

♣

The same δ must work for every pair of points in the domain. The distinction is a change in quantifier order:

$$\begin{aligned} \text{continuity on } D:\quad & \forall a\in D\,\forall\varepsilon>0\,\exists\delta>0\,\forall x\in D,\\ & \qquad |x-a|<\delta \implies |f(x)-f(a)|<\varepsilon;\\ \text{uniform continuity on } D:\quad & \forall\varepsilon>0\,\exists\delta>0\,\forall x,y\in D,\\ & \qquad |x-y|<\delta \implies |f(x)-f(y)|<\varepsilon. \end{aligned}$$

Uniform continuity implies continuity by fixing one point of the pair. Every Lipschitz function is uniformly continuous, since $\delta=\varepsilon/K$ is independent of the base point when $K>0$. If $K=0$, any positive δ works.

Example 1.48 Different domains, different control Determine the uniform continuity of x^2 on bounded intervals and on $\mathbb{R}$, of $1/x$ on $(0,1)$, and of $\sqrt{x}$ on $[0,+\infty)$.

Solution *On $[-R,R]$ with $R>0$, the estimate $\left|x^2-y^2\right|\le 2R\,|x-y|$ gives uniform continuity of the square*

function. On $\mathbb{R}$*, take* $x_n = n$ *and* $y_n = n + 1/n$*. Then*

$$|x_n - y_n| = \frac{1}{n} \to 0, \qquad |x_n^2 - y_n^2| = 2 + \frac{1}{n^2} > 2.$$

If one radius δ *worked for* $\varepsilon = 1$*, these pairs would violate it for all sufficiently large* n*. Thus* x^2 *is not uniformly continuous on* $\mathbb{R}$*.*

For $1/x$ *on* $(0, 1)$*, use* $x_n = 1/(n + 1)$ *and* $y_n = 1/(n + 2)$ *for* $n \geq 1$*. The input differences tend to* 0*, but* $|1/x_n - 1/y_n| = 1$*. The choice* $\varepsilon = 1/2$ *rules out uniform continuity. For the square root,* (1.8) *gives the single choice* $\delta = \varepsilon^2$ *for every pair of nonnegative inputs. Hence* $\sqrt{x}$ *is uniformly continuous on the unbounded interval* $[0, +\infty)$*, even though it is unbounded and is not Lipschitz near* 0*.*

Proposition 1.18

A function $f\colon D \to \mathbb{R}$ *is uniformly continuous on* D *if and only if for every pair of sequences* $(x_n), (y_n)$ *in* D *with* $|x_n - y_n| \to 0$*, one has*

$$|f(x_n) - f(y_n)| \to 0.$$

The input sequences need not converge individually.

♠

Proof If f is uniformly continuous, choose δ for a prescribed $\varepsilon > 0$. Eventually $|x_n - y_n| < \delta$, so the output difference is less than ε. Conversely, failure of uniform continuity means that there is $\varepsilon_0 > 0$ such that for every $\delta > 0$ some pair $x, y \in D$ satisfies

$$|x - y| < \delta, \qquad |f(x) - f(y)| \geq \varepsilon_0.$$

Choose such a pair (x_n, y_n) for $\delta = 1/n$. These sequences contradict the assumed sequential property. □

Theorem 1.27 (Heine–Cantor theorem)

Every continuous function on a closed bounded interval $[a, b]$ *is uniformly continuous there.*

♡

> ***Proof Strategy.*** If no common radius worked, there would be arbitrarily close input pairs whose outputs stay a fixed positive distance apart. Extract a convergent subsequence of the first inputs; closeness makes the paired second inputs converge to the same point. Continuity at that point contradicts the output separation.

Proof Suppose the conclusion fails. By the negation in Proposition 1.18, there exist $\varepsilon_0 > 0$ and $x_n, y_n \in [a, b]$ such that

$$|x_n - y_n| < \frac{1}{n}, \qquad |f(x_n) - f(y_n)| \geq \varepsilon_0$$

for every $n \geq 1$. By Bolzano–Weierstrass, a subsequence x_{n_k} converges to some $c \in [a, b]$. The estimate

$$|y_{n_k} - c| \leq |y_{n_k} - x_{n_k}| + |x_{n_k} - c| < \frac{1}{n_k} + |x_{n_k} - c|$$

shows that $y_{n_k} \to c$ as well. Continuity gives $f(x_{n_k}) \to f(c)$ and $f(y_{n_k}) \to f(c)$, hence their difference tends to 0. This contradicts the lower bound ε_0. □

Heine–Cantor guarantees a common input radius without necessarily producing a convenient formula for it. When a direct estimate is available, it may give more quantitative information. For example, on $[a, b]$ with $0 < a < b$,

$$\left|\frac{1}{x} - \frac{1}{y}\right| = \frac{|x - y|}{xy} \leq \frac{1}{a^2}\,|x - y|\,,$$

so $\delta = a^2\varepsilon$ suffices for the reciprocal function. This choice makes the distance from 0 explicit.

> ***Connection with compactness.*** The boundedness, extreme-value, and uniform-continuity proofs use the fact that every sequence in $[a, b]$ has a subsequence converging to a point still in $[a, b]$. This is the sequential form of *compactness*(紧致性)in the real line. Later study of compact sets will organize these three arguments into one general framework. Intermediate values use a different feature: an interval contains every point between two of its points. The disconnected-domain example above explains why these two features should be kept distinct.

1.6.6 Methods, pitfalls, and the next stage of calculus

For an existence problem, first build a continuous auxiliary function whose zero expresses the desired conclusion. Choose an interval and verify the endpoint signs explicitly. For uniqueness, add an independent order argument; the intermediate value theorem alone does not provide it. For extrema, specify the full domain and check continuity at included endpoints before invoking the theorem. For uniform continuity, seek an estimate independent of the base point, or use Heine–Cantor after checking that the domain is a closed bounded interval.

The proofs introduce three reusable constructions: nested sign-changing intervals, a sequence chosen from failure of boundedness, and a maximizing sequence chosen from the supremum property. In each construction, the sequence is a tool for converting an assertion about all inputs into a limit argument at one point. The paired-sequence test serves the same purpose when disproving uniform continuity.

Chapter 1 has moved from the order and completeness of $\mathbb{R}$ to functions, sequences, limits, and continuity. The next step studies the limiting ratio of an output change to an input change. Later, uniform control on closed intervals will help compare sums formed from many small subintervals. The local estimates and global theorems developed here supply the foundations for both differentiation and integration.

Exercise 1.6

1. State the intermediate value theorem, the extreme value theorem, and the Heine–Cantor theorem with all hypotheses. For each theorem, identify whether its conclusion concerns existence of a value, attainment of a bound, or control of input errors. Outline the contradiction proof of Heine–Cantor, identifying the separate roles of boundedness, closedness, and continuity.
2. Prove that $x^3 + x - 1 = 0$ has exactly one real root and that this root lies in $(0, 1)$. Use order rather than derivatives for uniqueness.
3. Prove that $e^x = 3 - 2x$ has exactly one solution in $(0, 1)$. State separately the existence and uniqueness arguments.
4. Prove that $x^4 - 3x^2 + 1 = 0$ has at least four distinct real roots by finding four disjoint sign-changing intervals. You need not solve the equation explicitly.

   > ***Hint.*** The polynomial is even. Locate two disjoint sign changes on the positive half-line and then reflect the intervals across the origin.
5. Suppose f is continuous on $[a, b]$ and $f(a) = 2$, $f(b) = 5$. Which of the following must hold: f takes the value 3; f takes the value 0; f is strictly increasing; f has a maximum; $f([a, b]) = [2, 5]$? Prove the necessary conclusions and give counterexamples for the others.
6. Find the supremum, infimum, and any attained extrema of each function on the stated domain:
$$x \text{ on } (0, 1], \qquad x^2 \text{ on } [-1, 2], \qquad \frac{1}{1+x} \text{ on } [0, +\infty).$$
7. Let $f\colon [a, b] \to \mathbb{R}$ be continuous and satisfy $f(x) \neq 0$ everywhere. Prove that $|f(x)| \geq m > 0$ throughout the interval for some m. Deduce boundedness of $1/f$.

8. If $f, g\colon [a,b] \to \mathbb{R}$ are continuous and $f(x) < g(x)$ for every x, prove that there exists $\eta > 0$ such that $f(x) + \eta \le g(x)$ for every x. Give a counterexample on $(0,1)$.
9. Prove that a continuous function on an interval which never vanishes has the same sign throughout that interval. Identify where the interval hypothesis enters the proof.
10. Let $f\colon [0,1] \to [0,1]$ be continuous and decreasing, where decreasing may be read as nonincreasing. Prove that f has exactly one fixed point.
11. Apply bisection to $f(x) = x^2 - 2$ on $[1,2]$ and write down the first three retained half-intervals. How many bisections ensure that the midpoint of the retained interval differs from its contained root by at most 10^{-3}?
12. Explain why $f(x) = \sin x$ attains a minimum and maximum on $[0, 2\pi]$, and determine its range there. Could the extreme value theorem alone tell you the numerical values of those extrema?
13. Prove uniform continuity of $f(x) = 1/(1+x)$ on $[0, +\infty)$ by an explicit estimate. Explain why Heine–Cantor on one closed bounded interval does not directly give this conclusion on the whole domain.
14. Prove that $\sin(1/x)$ is not uniformly continuous on $(0,1)$, even though it is bounded there. In contrast, prove uniform continuity of $\sin x$ on $\mathbb{R}$ by using a trigonometric identity to control the output difference.
15. Prove uniform continuity of $x^{1/3}$ on $\mathbb{R}$ and show that it is unbounded there. What does this example say about a possible implication from uniform continuity to boundedness on an arbitrary domain?

    > ***Hint.*** For real u, v, use $u^2 + uv + v^2 \ge (u-v)^2/4$ to compare $\left|u^3 - v^3\right|$ with $|u-v|^3$.

16. Suppose f and g are uniformly continuous on the same domain D. Prove that $f+g$ is uniformly continuous. If both functions are bounded, prove that fg is uniformly continuous. Show that the boundedness assumption cannot simply be omitted from the product statement.
17. Let $f\colon [a,b] \to \mathbb{R}$ be continuous, and let (x_n) be a sequence in $[a,b]$ satisfying $f(x_n) \to \max_{x\in[a,b]} f(x)$. Prove that some subsequence of (x_n) converges to a point where f attains its maximum. Must the whole sequence converge? Give an example.
18. Construct examples showing that each of the following is possible: a bounded continuous function on an open interval that has no maximum; an unbounded function on a closed bounded interval; a continuous function on a disconnected closed bounded domain that omits a value between two attained values. State which hypothesis is absent in each example.

Supplement Exercises

S1. Let $f\colon [a,b] \to \mathbb{R}$ be continuous and suppose $f(a) = f(b)$. For each positive integer n, prove that there exists $x \in [a, b - (b-a)/n]$ such that

$$f\left(x + \frac{b-a}{n}\right) = f(x).$$

> ***Hint.*** Consider the continuous difference function and the sum of its values at equally spaced inputs.

S2. Let $f\colon [a,b] \to [a,b]$ satisfy $|f(x) - f(y)| \le q\,|x-y|$ for some $0 \le q < 1$. Prove that f has a unique fixed point c. If $x_{n+1} = f(x_n)$ with $x_0 \in [a,b]$, prove $|x_n - c| \le q^n\,|x_0 - c|$ and deduce convergence. Treat $q = 0$ separately if needed.

S3. Suppose $f\colon (a,b) \to \mathbb{R}$ is uniformly continuous, where $a < b$ are finite. Prove that f has a unique continuous extension to $[a,b]$.

> ***Hint.*** First show that f sends Cauchy sequences to Cauchy sequences; then compare two sequences approaching the same endpoint.

S4. Conversely, if $f\colon (a,b) \to \mathbb{R}$ extends continuously to $[a,b]$, prove that it is uniformly continuous on (a,b). Use this criterion to decide uniform continuity of $\sin(1/x)$ and $x\sin(1/x)$ on $(0,1)$.

S5. Prove that every uniformly continuous function on a bounded interval is bounded, even when the interval is open.

> ***Hint.*** Divide the interval into finitely many pieces short enough for the output tolerance 1.

S6. Suppose $f\colon \mathbb{R} \to \mathbb{R}$ is continuous and has finite limits as $x \to +\infty$ and as $x \to -\infty$. Prove that f is uniformly continuous on $\mathbb{R}$.

> ***Hint.*** Use tail control together with Heine–Cantor on a central closed interval, taking care of pairs near the boundaries of that interval.

S7. Prove that a continuous periodic function $f\colon \mathbb{R} \to \mathbb{R}$ is bounded and uniformly continuous on $\mathbb{R}$.

> ***Hint.*** If $T > 0$ is a period, use boundedness on $[0, T]$ and uniform continuity on an interval slightly longer than one period.

S8. Let $f\colon [a, b] \to \mathbb{R}$ be continuous and suppose its maximum is attained at exactly one point c. Prove that every maximizing sequence converges to c. Contrast this with a continuous function whose maximum is attained at more than one point.

S9. Let $f, g\colon [a, b] \to \mathbb{R}$ be continuous, with f strictly increasing and g strictly decreasing. Suppose $f(a) < g(a)$ and $f(b) > g(b)$. Prove that their graphs meet at exactly one point. Then give an example with both functions strictly increasing and the same endpoint inequalities for which there are at least three intersections.

S10. Let $f\colon [0, 1] \to \mathbb{R}$ be continuous. Prove that for every $\varepsilon > 0$ there is a continuous function P that is linear on each of finitely many subintervals and satisfies

$$|f(x) - P(x)| < \varepsilon \qquad (0 \le x \le 1).$$

> ***Hint.*** Use uniform continuity to choose a partition and let P join the function values at consecutive partition points by line segments.

Chapter 1 Problem-Solving Toolkit

This toolkit summarizes how to choose among the ideas developed in Chapter 1. It is not a list of automatic substitutions: domain, direction of approach, and theorem hypotheses must be settled before a calculation begins. A reliable solution usually has two layers—first identify the structure, then justify the chosen transformation or theorem.

A decision route

Step 1. Fix the object and domain. Is the input a sequence index or a real variable? For a function, record the actual domain and whether the approach is two-sided, one-sided, or at infinity.

Step 2. Try the structurally simplest operation. Use known finite-limit laws only when their hypotheses hold. If direct substitution produces an indeterminate form, regard that form as a signal to compare rates, not as an answer.

Step 3. Expose the small or dominant quantity. Factor, rationalize, normalize by the largest scale, match the argument of a standard limit, or take logarithms for a positive power.

Step 4. Choose the proof mechanism. Use a direct ε-argument for quantitative local control, squeeze for domination, monotonicity plus boundedness for recursive sequences, or a sequence criterion to detect incompatible approaches.

Step 5. Check the logical hinge. Verify nonzero denominators, eventual signs, transformed domains, punctures in compositions, and every hypothesis of a global continuity theorem.

Step 6. State exactly what has been proved. Separate existence from uniqueness, boundedness from attainment, convergence from a merely possible fixed point, and pointwise continuity from uniform continuity.

Sequence patterns

Signal	First move	What must be checked
Rational expression in n	Divide by the highest relevant power of n.	The normalized denominator has a nonzero limit; lower-order terms really vanish.
Difference of comparable roots	Factor a difference of powers; for square roots, multiply by the conjugate.	Root domains and signs, especially when extracting $\sqrt{x^2} = \lvert x \rvert$ on a tail.
Bounded oscillation times a shrinking factor	Take absolute values and squeeze by a positive majorant.	The oscillatory factor has one bound valid for every sufficiently late index.
Recursive sequence	Find an invariant interval, then prove monotonicity and boundedness.	Solve the fixed-point equation only after convergence has been established; discard roots outside the invariant interval.
Sum or average with a growing number of terms	Split it into a fixed head and a uniformly controlled tail.	A finite-sum limit law cannot be applied directly when the number of summands depends on n.
Several positive growth scales	Normalize by the largest plausible scale and use $\ln n \ll n^{\alpha} \ll a^n \ll n! \ll n^n$.	The restrictions $\alpha > 0$, $a > 1$, and any cancellation between leading terms.

Proof and disproof templates

- **Direct limit proof.** Work backward from the desired output error to discover a sufficient input restriction, then write the proof forward with the quantifiers in their prescribed order.
- **Negating convergence.** Fix one tolerance $\varepsilon_0 > 0$ and produce bad terms or inputs arbitrarily far out or arbitrarily close to the limiting point. The fixed tolerance is the essential feature.
- **Sequential disproof.** For a sequence, find two subsequences with different limits. For a function, find two admissible input sequences approaching the same point whose output sequences have incompatible limits.
- **Uniform-continuity disproof.** Choose pairs x_n, y_n whose input distance tends to 0 while the output distance remains at least one fixed positive number.
- **Closed-interval theorem.** Write the function, domain, continuity statement, and relevant endpoint or range data before naming the theorem. Then state only the conclusion supplied by that theorem.

Function-limit and continuity patterns

Signal	First move	What must be checked
Algebraic $0/0$ or cancellation	Factor, use an exact identity, or rationalize before applying limit laws.	Cancellation is valid on the punctured domain; a remaining denominator stays away from zero.
Standard small trigonometric, logarithmic, or exponential factors	Match each argument with its denominator and use a proved equivalent in products or quotients.	Equivalent infinitesimals cannot be substituted term by term inside a difference with leading cancellation.
Positive varying power $F(x)^{G(x)}$	Rewrite it as $\exp(G(x)\ln F(x))$.	The base is positive on the relevant domain and the single exponent $G\ln F$ has been evaluated.
Change of variables $t=\phi(x)$	Translate the limit into the variable t and write its exact new approach.	Side reversal, tail-to-zero direction, transformed domain, and the puncture condition for composition.
Competing infinite quantities	Establish eventual signs, normalize, compare, or transform the expression.	Forms such as $\infty-\infty$ and $0\cdot\infty$ have no arithmetic meaning by themselves.
Suspected failure of a function limit	Compare one-sided limits or construct two admissible input sequences.	One bad sequence disproves a limit; finitely many successful sequences never prove the universal assertion.
Existence of a root or intersection	Build a continuous auxiliary difference and locate a sign change on an interval.	Continuity and the interval are explicit; uniqueness requires a separate order or injectivity argument.
Extrema or uniform control	Check the closed bounded interval hypotheses, then use the appropriate global theorem.	Attainment is stronger than boundedness; a direct estimate may yield sharper uniform information.

> ***Stop signs.*** Pause before making any of the following conclusions.
> - A fixed-point equation supplies possible limits of a recurrence, not convergence.
> - The intermediate value theorem supplies existence, not uniqueness.
> - Boundedness near a point does not supply a limit, and unboundedness does not supply a signed infinite limit.
> - The value $f(a)$ does not affect a punctured limit, but it is essential for continuity at a.
> - Equivalent infinitesimals are safe in the stated product and quotient setting, not across arbitrary addition or subtraction.
> - A theorem name never replaces verification of its domain and hypotheses.

Before finishing a solution, check the domain once more, record every one-sided direction, identify the theorem or estimate that justifies the last passage to the limit, and make sure the conclusion answers exactly the question asked.

Chapter 2 Basic Concepts of Calculus

Limits describe nearby behaviour; derivatives measure its rate of change. We first turn secant slopes into derivatives, develop rules for combining them, and interpret the result as a local linear model. Higher derivatives then describe how that model changes with its base point. The later part of this chapter will connect instantaneous change with accumulation through integration.

2.1 Derivatives

知识点概要

- ❏ *Interpret a derivative as an instantaneous rate and a tangent slope.*
- ❏ *Use the difference quotient, including one-sided and endpoint questions.*
- ❏ *Explain why differentiability implies continuity and why the converse fails.*
- ❏ *Compute derivatives from limits and distinguish a derivative from a symmetric quotient.*

A limit tells us where values are heading. To measure how fast they change, we must compare a change in output with the change in input that produced it. This comparison connects motion, geometry, and the local behaviour of a function.

2.1.1 Rates and tangents

Example 2.1 From average to instantaneous velocity A particle has position $s(t) = t^2 + 2t$ metres at time $t \geq 0$ seconds. Find its average velocity from time 1 to time $1 + h$, where $h > 0$, and its instantaneous velocity at time 1.

Solution *The displacement is $s(1+h) - s(1) = 4h + h^2$ metres. The average velocity is therefore $4 + h$ metres per second. As $h \to 0^+$, these averages tend to 4 metres per second. The same limit results from earlier times, using $-1 < h < 0$. We call this common limit the instantaneous velocity at 1. The vanishing time interval is not set equal to zero: we first form a quotient for $h \neq 0$, and then take a limit.*

For the graph $y = f(x)$, let $P = (a, f(a))$ and $Q = (a + h, f(a + h))$ with $h \neq 0$. The secant line(割线) through them has slope

$$\frac{f(a+h) - f(a)}{h}.$$

If these slopes approach a finite number m, the limiting line through P has equation $y - f(a) = m(x - a)$. It is the tangent line(切线) at P in the sense used here. A tangent need not meet a curve at only one point; its defining property is this limiting slope.

In **Figure** 2.1, the second point changes while the base point remains fixed. The formula $2 + h \to 2$, rather than the picture alone, verifies the limiting slope.

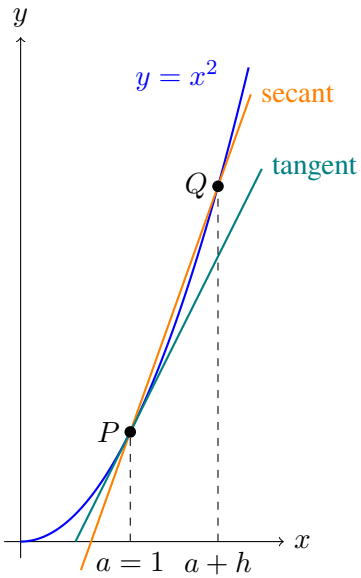


Figure 2.1: For $f(x) = x^2$ at $a = 1$, the secant slope is $2 + h$ and the tangent slope is 2. The displayed secant uses $h = 0.8$.

2.1.2 Derivatives and one-sided derivatives

Definition 2.1

Let f be defined on an open interval containing a. It is differentiable (可导) at a if the finite real limit

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} = \lim_{x \to a} \frac{f(x) - f(a)}{x - a} \tag{2.1}$$

exists. This number is the derivative (导数) of f at a. On an open interval where this holds at every point, $x \mapsto f'(x)$ is the derivative function (导函数).

♣

The substitution $x = a + h$ explains the equality of the two limits. The quotient uses the actual value $f(a)$, unlike a punctured limit of $f(x)$. Changing that value can destroy differentiability. The number $f'(a)$ and the function f' are different objects. To find $f'(a)$, we fix a throughout the limiting process and vary h. Only after that limit has been taken may we let the base point range over an interval to obtain the function f'. In particular, $\lim_{x \to a} f'(x)$ is a different question from the existence of $f'(a)$: it concerns slopes at nearby points, rather than secants anchored at a. For $y = f(x)$, the notations $\mathrm{d}y/\mathrm{d}x$ and $\mathrm{d}f/\mathrm{d}x$ also denote the derivative function; at this stage they are derivative symbols, not fractions of finite increments.

A constant has derivative 0, since its difference quotient vanishes. For $f(x) = mx + b$, that quotient equals m for every nonzero increment. These examples explain why a derivative measures slope, rather than the height of the graph.

The limit in (2.1) has the following precise meaning: for every $\varepsilon > 0$ there is $\delta > 0$ such that

$$0 < |h| < \delta \quad \Longrightarrow \quad \left| \frac{f(a+h) - f(a)}{h} - f'(a) \right| < \varepsilon.$$

The same δ must work for all sufficiently small positive and negative increments, not just a chosen sequence. Multiplying by $|h|$ gives the equivalent estimate

$$\left| f(a+h) - f(a) - f'(a)h \right| < \varepsilon \, |h| \, .$$

Thus a convergent secant slope says more than a small change in height: the error in predicting that change by $f'(a)h$ becomes small even after comparison with the size of the input change. We will organize this observation

into the concept of a differential in Section 2.3.

Example 2.2 Changing one value changes the derivative question Let $F(x) = x^2$ for $x \neq 0$ and $F(0) = 1$. Determine whether $F'(0)$ exists, and compare this with the derivatives at nonzero points.

Solution *At a nonzero point a, a sufficiently small neighbourhood avoids 0, so the local formula is simply x^2. Its difference quotient is $2a+h$, hence $F'(a) = 2a$. At 0, however, the definition uses the assigned value $F(0) = 1$:*

$$\frac{F(h) - F(0)}{h} = \frac{h^2 - 1}{h} = h - \frac{1}{h}.$$

This tends to $-\infty$ from the right and $+\infty$ from the left, so no finite derivative exists. The nearby derivatives tend to zero:

$$\lim_{x \to 0,\, x \neq 0} F'(x) = 0.$$

This limit cannot supply the missing derivative at 0. The secants through $(0, 1)$ are different from the secants through $(0, 0)$.

Definition 2.2

If f is defined on $[a, a + \delta)$ for some $\delta > 0$, its right derivative (右导数) is the finite limit

$$f'_+(a) = \lim_{h \to 0^+} \frac{f(a + h) - f(a)}{h},$$

when it exists. The left derivative (左导数) $f'_-(a)$ is defined with $h \to 0^-$ on a left neighbourhood that includes a. ♣

At an interior point, the two-sided limit criterion gives

$$f'(a) \text{ exists} \quad \Longleftrightarrow \quad f'_-(a),\ f'_+(a) \text{ exist and are equal.}$$

At a boundary point of a closed interval we explicitly use the available one-sided derivative. We do not infer a derivative on an absent side. For example, the right derivative of x at 0 on $[0, +\infty)$ is 1.

Example 2.3 A root and its endpoint Use the definition to differentiate $f(x) = \sqrt{x}$ at $a > 0$. Does it have a finite right derivative at 0?

> ***Idea.*** The numerator is a difference of roots. Multiplying by the conjugate turns it into the input increment, which can then cancel before taking the limit.

Solution *For $0 < |h| < a$,*

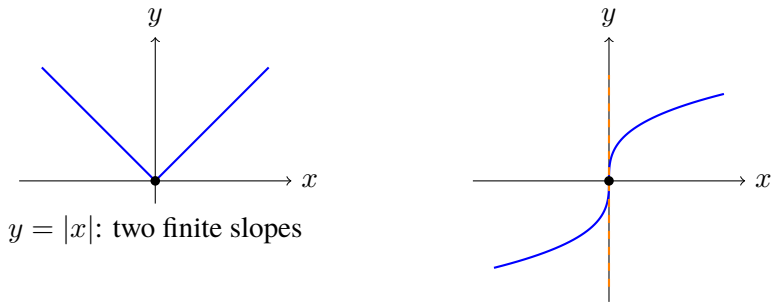
$$\frac{\sqrt{a + h} - \sqrt{a}}{h} = \frac{1}{\sqrt{a + h} + \sqrt{a}} \longrightarrow \frac{1}{2\sqrt{a}}.$$

Continuity of the square root was established in Section 1.5. At 0, the right quotient is $1/\sqrt{h} \to +\infty$. Thus the finite right derivative does not exist. The unbounded slope is often described geometrically as a vertical tangent, but is not a real derivative.

Example 2.4 A corner and a vertical tangent Determine differentiability at 0 of $|x|$ and $x^{1/3}$.

Solution *For $|x|$, the quotient $|h|/h$ equals -1 on the left and 1 on the right. Both one-sided derivatives exist, but they disagree. For the real cube root, $h^{1/3}/h = 1/|h|^{2/3} \to +\infty$ from either side. Neither function has a finite derivative at 0. The first failure is a disagreement of finite slopes; the second is their unbounded size.*

The two graphs in **Figure** 2.2 are continuous at the origin. Their difference quotients, rather than continuity alone, distinguish the two failures.



$y = |x|$: two finite slopes

$y = x^{1/3}$: vertical tangent

Figure 2.2: A corner and a vertical tangent fail the finite-derivative definition for different reasons. The cube-root curve is plotted parametrically as (t^3, t).

2.1.3 Differentiability and continuity

Convergence of slopes should prevent a jump in height. The following result identifies the precise estimate behind this expectation.

Proposition 2.1

If f is differentiable at a, then for all sufficiently small h,

$$f(a+h) - f(a) = h\big(f'(a) + \eta(h)\big), \qquad \eta(h) \to 0.$$

In particular, f is continuous at a, and there are $C, \delta > 0$ such that $|f(a+h) - f(a)| \le C\,|h|$ for $|h| < \delta$.

> ***Proof Strategy.*** Continuity asks for a small output difference; differentiability controls that difference divided by h. Multiply the bounded quotient back by the small increment. The estimate is centred at a and does not compare every pair of nearby points.

Proof For $h \neq 0$, set $\eta(h) = [f(a+h) - f(a)]/h - f'(a)$ and set $\eta(0) = 0$. The definition gives $\eta(h) \to 0$ and the asserted identity, including $h = 0$. Choose δ so that $|\eta(h)| < 1$ for $|h| < \delta$. The triangle inequality then gives

$$|f(a+h) - f(a)| \le |h|\,\big(\big|f'(a)\big| + |\eta(h)|\big) \le \big(\big|f'(a)\big| + 1\big)\,|h|\,.$$

Take $C = |f'(a)| + 1 > 0$. Given a desired bound $\varepsilon > 0$ on the output change, it suffices to require $|h| < \min(\delta, \varepsilon/C)$. The output change is then less than ε, which is exactly continuity at a. □

Continuity is necessary, but $|x|$ shows it is not sufficient. Nor does the proposition assert that f' is continuous: it assumes a derivative only at one point. The increment representation will also make the chain rule safe when an inner increment vanishes.

The two factors h and $\eta(h)$ play different roles. The first is the input increment; the second is the error in the secant slope. Both tend to zero, so the error $h\eta(h)$ in the predicted output is negligible compared with h itself. For example, h^2 has this property because $h^2/h = h \to 0$, whereas $|h|$ does not because $|h|/h$ has opposite signs on the two sides. In the notation of Definition 1.24, the proposition says

$$f(a+h) - f(a) = f'(a)h + o(h).$$

The symbol $o(h)$ represents a remainder with the stated limiting property; it is not a number that has been discarded or set equal to zero. If $f'(a) = 0$, the entire output change is $o(h)$, but it can still be nonzero for every $h \neq 0$.

Example 2.5 A tangent can cross its curve Find the tangent to $f(x) = x^3$ at 0, and decide whether the curve stays on one side of it near that point.

Solution *The anchored quotient is $h^3/h = h^2 \to 0$, so $f'(0) = 0$ and the tangent is $y = 0$. For $h > 0$, the curve lies above this line, and for $h < 0$ it lies below it. Nevertheless, its vertical error from the tangent is h^3, and $\left|h^3\right| / |h| = h^2 \to 0$. Crossing the tangent is fully compatible with the defining slope limit. Also, $f'(0) = 0$ does not say that the function is constant near 0.*

Example 2.6 Matching values before slopes Find $A, B \in \mathbb{R}$ for which

$$f(x) = \begin{cases} x^2, & x \le 1, \\ Ax + B, & x > 1 \end{cases}$$

is differentiable at 1.

> ***Strategy.*** Two pieces must first meet at the same height. Only after that condition holds can their slopes describe a tangent to the glued graph.

Solution *Continuity requires $A + B = 1$. Indeed, the actual value and the left limit are both 1, whereas the right limit is $A + B$. For $h < 0$, the quotient is $[(1 + h)^2 - 1]/h = 2 + h \to 2$. For $h > 0$, it is $[A(1 + h) + B - 1]/h = (A + B - 1)/h + A$. If the constant numerator $A + B - 1$ is nonzero, this latter quotient cannot have a finite limit. Under continuity this is A, so equality of the one-sided derivatives requires $A = 2$ and $B = -1$. Conversely, these values give quotients tending to 2 on both sides and therefore suffice. Matching the derivatives of the separate formulas alone would miss the condition on B.*

Example 2.7 Oscillation can survive continuity Set $f(0) = g(0) = 0$ and, for $x \neq 0$, let $f(x) = x^2 \sin(1/x)$ and $g(x) = x \sin(1/x)$. Check continuity and differentiability at 0.

> ***Idea.*** Both oscillations have amplitudes tending to zero. For differentiability, divide the amplitude by $|x|$ and ask whether the remaining oscillation is still suppressed.

Solution *The bounds $|f(x)| \le x^2$ and $|g(x)| \le |x|$ give continuity. The quotient for f has absolute value at most $|h|$, so $f'(0) = 0$. The quotient for g is $\sin(1/h)$. Along $h_n = (\pi/2 + 2\pi n)^{-1}$ it equals 1, and along $k_n = (3\pi/2 + 2\pi n)^{-1}$ it equals -1. Both sequences tend to 0, so $g'(0)$ does not exist.*

The estimate for f is anchored at 0. It does not promise that every secant joining two nearby positive inputs has slope close to $f'(0) = 0$. Indeed, writing $f(x) = x\eta(x)$ with $\eta(x) \to 0$ gives, for distinct nonzero x, y,

$$\left|\frac{f(x) - f(y)}{x - y}\right| \le \max\{|\eta(x)|, |\eta(y)|\} \frac{|x| + |y|}{|x - y|}.$$

The first factor is small, but the second can be large when two same-side inputs are much closer to each other than to the origin. For this oscillatory f, choosing nearby positive inputs at opposite sine phases can in fact produce secant slopes tending to $2/\pi$, rather than 0. The supplement exercises ask for the explicit construction and for the contrasting case $x > 0 > y$, where $|x - y| = |x| + |y|$ removes this difficulty. Thus the location of the two sampling points is part of the mathematical question, not a detail that can be ignored.

2.1.4 Computing from the definition

For a positive integer n, the binomial identity gives

$$\frac{(a + h)^n - a^n}{h} = na^{n-1} + \sum_{k=2}^{n} \binom{n}{k} a^{n-k} h^{k-1} \longrightarrow na^{n-1}.$$

For $n = 1$ the sum is empty and the derivative is 1. For $n \ge 2$ the formula also holds at $a = 0$. The constant power $x^0 = 1$ is handled separately.

Example 2.8 Two ways to differentiate sine Prove that $(\sin x)' = \cos x$ for every $x \in \mathbb{R}$, with angles measured in radians.

> ***Idea.*** The known limit compares $\sin h$ with h. Rewrite a sine difference so that its small argument is h or $h/2$, leaving the base point inside a continuous factor.

Solution ***Method 1: separate the increments.*** *The addition formula gives*

$$\frac{\sin(a+h)-\sin a}{h} = \sin a\,\frac{\cos h - 1}{h} + \cos a\,\frac{\sin h}{h}.$$

The second quotient tends to 1. *For the first,* $\cos h - 1 = -2\sin^2(h/2)$ *and* $|\sin(h/2)| \le |h|/2$ *imply* $|(\cos h - 1)/h| \le |h|/2 \to 0$. *The limit is* $\cos a$.

Solution ***Method 2: factor the difference.*** *The identity* $\sin(a+h) - \sin a = 2\cos(a+h/2)\sin(h/2)$ *gives*

$$\frac{\sin(a+h)-\sin a}{h} = \cos(a+h/2)\frac{\sin(h/2)}{h/2} \longrightarrow \cos a.$$

Continuity of cosine and the fundamental sine limit justify the two factors.

Both methods use only trigonometric identities and established limits. The first displays the two contributions to the increment; the second is shorter because the difference is already factored. The use of radians is essential for the sine limit to have value 1. Applying the cosine addition formula in the same way gives $(\cos x)' = -\sin x$.

The exponential limits already proved in (1.6) also give, for $b > 0$,

$$\frac{b^{a+h}-b^a}{h} = b^a\frac{b^h-1}{h} \longrightarrow b^a \ln b.$$

In particular $(e^x)' = e^x$, and the case $b = 1$ is constant. This argument does not use differentiation to establish the exponential limit.

2.1.5 Methods and further remarks

Start by identifying the base point, the available sides, and the actual function value. At a regular algebraic point, factor or rationalize the increment before taking a limit. At a joining point, first impose continuity, then compare the two difference quotients. For oscillation, estimate the quotient itself: small function values do not guarantee small slopes. When an increment has been written as $mh + r(h)$, test $r(h)/h$, not just $r(h)$. The former tests the predicted slope; the latter tests only a change in height. To disprove a quotient limit, two explicit sequences with different limiting values suffice. When both endpoints of a secant move, compare their separation with their distances from the fixed base point before trying to use a derivative at that point.

Example 2.9 What a symmetric measurement misses If $f'(a)$ exists, compute $\lim_{h\to 0}[f(a+h) - f(a-h)]/(2h)$. Is the converse true?

Solution *Split the quotient as*

$$\frac{1}{2}\frac{f(a+h)-f(a)}{h} + \frac{1}{2}\frac{f(a-h)-f(a)}{-h}.$$

Each term tends to $f'(a)/2$. *The limit is* $f'(a)$. *The converse fails for* $f(x) = |x|$ *at* 0*: the symmetric quotient is identically zero, but the ordinary derivative does not exist. Symmetric sampling can cancel a corner.*

> ***Looking Ahead.*** Using a fresh limit for every expression would be inefficient. The next section derives rules for combining differentiable functions. Later, the mean value theorem will relate derivatives throughout an interval to changes between its endpoints; a derivative at one point alone does not supply that global conclusion.

Exercise 2.1

1. Write the definition of $f'(a)$ in both increment and point forms. Explain which variable is fixed and why the limit must be finite.
2. For $f(x) = 1/x$, use the definition to find $f'(a)$ for $a \neq 0$, including the restriction on the increment.
3. Evaluate $\lim_{h\to 0}[(2+h)^5 - 32]/h$ by recognizing a derivative.
4. Find the tangent line to $y = x^3$ at $x = -1$, using the derivative established from the binomial formula.
5. For $f(x) = |x-2|$, find the two one-sided derivatives at 2 and explain their geometric meaning.
6. A position is $s(t) = 3t^2 - t$ metres for $t \geq 0$ seconds. Find the average velocity on $[1, 1+h]$ and the instantaneous velocity at 1.
7. Use the definition to differentiate $\cos x$; identify exactly which two limits are needed.
8. If $f'(a) = m$, evaluate $\lim_{h\to 0}[f(a+2h) - f(a-3h)]/h$.
9. Prove from the definition that $f(x) = |x|^{3/2}$ is differentiable at 0, and find its derivative there.
10. On $[0,1]$, determine the finite one-sided derivatives of $f(x) = \sqrt{x(1-x)}$ at both endpoints.
11. Suppose f is differentiable at a and $f(a) = 0$. Prove that $\lim_{x\to a} f(x)^2/(x-a) = 0$.
12. A function near 0 satisfies $f(0) = 3$ and $f(h) = 3 - 2h + h\eta(h)$, where $\eta(h) \to 0$. Prove that $f'(0) = -2$.
13. Explain why $f(h) = 3 - 2h + o(1)$ alone, with $f(0) = 3$, does not imply the conclusion in the preceding problem. Give a continuous counterexample.
14. Find all A, B making $f(x) = e^x$ for $x \leq 0$ and $f(x) = A\sin x + B$ for $x > 0$ differentiable at 0.
15. Set $f(0) = 0$ and $f(x) = x/(1 + e^{1/x})$ for $x \neq 0$. Prove continuity at 0 and determine the one-sided derivatives.
16. Prove or disprove: if $|f(a+h) - f(a)| \leq C\,|h|$ near 0, then $f'(a)$ exists. Also explain why the reverse implication is valid.
17. Suppose f is defined near a and, for $|h| < \rho$, satisfies $|f(a+h) - f(a) - mh| \leq Kh^2$, where $m \in \mathbb{R}$, $K \geq 0$ and $\rho > 0$. Find $f'(a)$ and give a δ guaranteeing that the difference quotient is within ε of this value. Include $K = 0$.
18. A position $s(t)$ is measured in metres with time t in seconds, and $s'(120) = v$. The same record in minutes is $S(u) = s(60u)$. Use a difference quotient to find $S'(2)$, state its units, and explain why the physical velocity has not changed.
19. Let f be defined near 0 with finite one-sided derivatives $f'_-(0) = \ell$ and $f'_+(0) = r$. Find the unique $c \in \mathbb{R}$ for which $g(x) = f(x) + c\,|x|$ is differentiable at 0, and find $g'(0)$.
20. Suppose f is continuous near 0, $f(0) = 0$, and $f(1/n) = 1/n^2$ for every positive integer n. If $f'(0)$ exists, what must it be? Must it exist? For the second question, construct a continuous counterexample and verify the failure using two sequences of increments.

    > ***Optional Hint.*** A sine factor $\sin(\pi/x)$ vanishes at every measured input. Its amplitude can still leave an oscillation in the difference quotient.

21. Suppose f, g are differentiable at a, $f(a) = g(a)$, and $f(x) \geq g(x)$ for all x sufficiently near a. Prove directly from the signs of the two one-sided difference quotients that $f'(a) = g'(a)$. Does equality of the slopes conversely force one graph to remain above the other? Give an example.

Supplement Exercises

S1. For $\alpha > 0$, set $f_\alpha(0) = 0$ and $f_\alpha(x) = |x|^\alpha \sin(1/x)$ for $x \neq 0$. Determine exactly when $f'_\alpha(0)$ exists, with proofs at all threshold cases.

S2. Let D be 1 on $\mathbb{Q}$ and 0 on $\mathbb{R} \setminus \mathbb{Q}$. Prove that $F(x) = x^2 D(x)$ is differentiable at 0 but discontinuous at every nonzero point.

S3. Suppose f is differentiable at 0 and $f(0) = 0$. Prove that $|f(x)|$ is differentiable at 0 if and only if $f'(0) = 0$.

S4. Suppose $f(0) = 0$ and $\lim_{h\to 0}[f(2h) - f(h)]/h = L$. Give a counterexample to $f'(0) = L$ without continuity. Then prove the implication if f is continuous at 0.

> ***Optional Hint.*** For the second part, telescope $f(x) - f(x/2^n)$ and control each difference by the same estimate at $x/2^k$.

S5. Let $f'(0) = m$. Prove that $[f(x_n) - f(y_n)]/(x_n - y_n) \to m$ when $x_n > 0 > y_n$ tend to 0. Is this still guaranteed for two positive sequences? Test $f(x) = x^2 \sin(1/x)$, $f(0) = 0$.

S6. If f is differentiable at a, prove that $\lim_{h\to 0} \sup_{|t|\le 1} |f(a + ht) - f(a) - f'(a)ht| \,/\, |h| = 0$. Explain why a supremum is appropriate when continuity away from a has not been assumed.

S7. For $\beta > 0$, determine differentiability at 0 of $F_\beta(x) = \cos(|x|^\beta)$, including the threshold $\beta = 1/2$.

S8. Construct two functions continuous at 0 and not differentiable there whose sum is differentiable there, and two such functions whose product is differentiable there. Verify every claim by difference quotients.

S9. Define $f(x) = x$ for $x \le 0$, and $f(x) = 1/n$ for $1/(n+1) < x \le 1/n$, $n \ge 1$. Determine continuity and differentiability at 0 on the domain $(-\infty, 1]$, using bounds uniform on each step.

S10. Let f be differentiable at 0, with $f(0) = 0$ and $f'(0) = m$. Prove that $\sum_{k=1}^{n} f(k/n^2) \to m/2$ without using integrals.

S11. Let u, v be differentiable at a with $u(a) = v(a)$, and set $F(x) = \max\{u(x), v(x)\}$ near a. Prove that the right derivative of F is $\max\{u'(a), v'(a)\}$ and its left derivative is $\min\{u'(a), v'(a)\}$. Deduce exactly when F is differentiable at a.

S12. Suppose $f'(0) = m$ and $x_n, y_n \to 0$, with $x_n \ne y_n$. Prove that the secant quotients tend to m if there is a fixed C such that $|x_n| + |y_n| \le C\,|x_n - y_n|$ for every n. Include sequences with a zero endpoint, and deduce the result for $y_n = \lambda x_n$, where $\lambda \ne 1$ is fixed and $x_n \ne 0$. Use the oscillatory example to show that some control of the point separation is needed for a guarantee valid for every differentiable f.

2.2 Differentiation Rules

知识点概要

- ❑ *Derive the arithmetic rules and apply them with their domain restrictions.*
- ❑ *Prove the chain rule without dividing by an increment that might be zero.*
- ❑ *Differentiate inverse and elementary functions on their correct branches.*
- ❑ *Choose between direct calculation, simplification, and logarithmic differentiation.*

A complicated formula is usually assembled from simpler functions. If we understand how differentiation interacts with that assembly, a few basic derivatives will generate many others. The rules in this section are consequences of the definition; their hypotheses remain part of every calculation.

2.2.1 Arithmetic rules

Theorem 2.1

Suppose f, g are differentiable at a and $c, d \in \mathbb{R}$. Then $cf + dg$ and fg are differentiable at a, with

$$(cf + dg)'(a) = cf'(a) + dg'(a), \qquad (fg)'(a) = f'(a)g(a) + f(a)g'(a).$$

If also $g(a) \neq 0$, then f/g is defined near a, is differentiable at a, and

$$(f/g)'(a) = \frac{f'(a)g(a) - f(a)g'(a)}{g(a)^2}.$$

♡

The product rule contains two contributions: change the first factor while holding the second fixed, and then change the second. There is no rule that the derivative of a product equals the product of the derivatives. To see the two contributions before taking a limit, write $\Delta f = f(a+h) - f(a)$ and $\Delta g = g(a+h) - g(a)$. Ordinary multiplication gives the exact identity

$$\Delta(fg) = g(a)\Delta f + f(a)\Delta g + \Delta f\, \Delta g.$$

The last term records the simultaneous change of both factors. By Proposition 2.1, $|\Delta f| \leq C_1 |h|$ and $|\Delta g| \leq C_2 |h|$ near 0. Consequently $|\Delta f\, \Delta g/h| \leq C_1 C_2 |h| \to 0$. This term is present in the exact increment, but makes no contribution to the limiting first-order rate.

> ***Proof Strategy.*** Add and subtract $f(a)g(a+h)$ in the product difference. This creates two known difference quotients. For a reciprocal, put the two terms over a common denominator; continuity keeps that denominator nonzero.

Proof For $h \neq 0$, the quotient of $cf + dg$ is

$$c\frac{f(a+h) - f(a)}{h} + d\frac{g(a+h) - g(a)}{h}.$$

The two finite limits exist by hypothesis, and the limit laws give $cf'(a) + dg'(a)$. For the product,

$$\frac{f(a+h)g(a+h) - f(a)g(a)}{h} = \frac{f(a+h) - f(a)}{h}\, g(a+h) + f(a)\frac{g(a+h) - g(a)}{h}.$$

Since $g(a+h) \to g(a)$ by Proposition 2.1, the required limit follows. When $g(a) \neq 0$, continuity gives $|g(a+h)| > |g(a)|/2$ for all sufficiently small h. Then

$$\frac{1/g(a+h) - 1/g(a)}{h} = -\frac{[g(a+h) - g(a)]/h}{g(a+h)g(a)} \longrightarrow -\frac{g'(a)}{g(a)^2}.$$

Apply the product rule to $f(1/g)$:

$$(f/g)'(a) = \frac{f'(a)}{g(a)} - \frac{f(a)g'(a)}{g(a)^2} = \frac{f'(a)g(a) - f(a)g'(a)}{g(a)^2}.$$

The same continuity estimate ensures that this quotient function is defined on a whole neighbourhood of a, as the definition of its derivative requires. □

Example 2.10 Two changing factors Differentiate $F(x) = x^2 \sin x$ on $\mathbb{R}$.

Solution *Both factors are differentiable everywhere by Section 2.1. Thus* $F'(x) = 2x \sin x + x^2 \cos x$. *The first term records the changing quadratic factor; the second records the changing sine factor. The incorrect expression* $2x \cos x$ *records neither contribution correctly.*

Repeated multiplication now yields the usual polynomial rule. The reciprocal rule extends $(x^n)' = nx^{n-1}$ to negative integers, on $x \neq 0$. It also gives, wherever the denominators do not vanish,

$$(\tan x)' = \frac{1}{\cos^2 x}, \quad (\cot x)' = -\frac{1}{\sin^2 x}, \quad (\sec x)' = \sec x \tan x, \quad (\csc x)' = -\csc x \cot x.$$

For example, $(\tan x)' = (\cos^2 x + \sin^2 x)/\cos^2 x$. The other three follow by the same quotient or reciprocal rule.

2.2.2 The chain rule

For a composition $f(g(x))$, the input first changes g, and that change becomes the input of f. The two rates must therefore be evaluated at different points.

Theorem 2.2 (Chain Rule)

Suppose g is defined near a and differentiable at a. Suppose f is defined on an open interval containing $b = g(a)$ and differentiable at b. Then $f \circ g$ is defined near a, differentiable at a, and

$$(f \circ g)'(a) = f'(g(a))g'(a).$$

♡

The outer derivative is evaluated at $g(a)$, not at a. For example, in $f(u) = \sin u$ and $g(x) = x^2$, the base input a first becomes the intermediate input a^2. The outer slope is therefore $\cos(a^2)$, and the inner slope is $2a$. Their product $2a\cos(a^2)$ describes output change per unit change of the original input. The conclusion also holds when $g'(a) = 0$. The familiar cancellation suggested by $\mathrm{d}y/\mathrm{d}x = (\mathrm{d}y/\mathrm{d}u)(\mathrm{d}u/\mathrm{d}x)$ is useful for remembering the order, but does not prove the theorem.

> ***Proof Strategy.*** Trying to factor the quotient by dividing by $g(a+h)-g(a)$ fails whenever that increment is zero. Instead, extend the outer difference quotient at its missing point by its limiting value. The extended function is continuous there, and its defining identity remains true even for a zero increment.

Proof Define

$$q(v) = \begin{cases} [f(b+v) - f(b)]/v, & v \neq 0, \\ f'(b), & v = 0. \end{cases}$$

Differentiability of f says exactly that q is continuous at 0. Only this continuity at 0 is needed; continuity of f' on an interval is not assumed. For every sufficiently small v, including $v = 0$, one has $f(b+v) - f(b) = vq(v)$. At $v = 0$ the identity is $0 = 0$, so defining $q(0)$ by the derivative introduces no additional restriction. Continuity of g ensures that its nearby values stay in the domain interval of f and that $v(h) = g(a+h) - g(a) \to 0$. For $h \neq 0$,

$$\frac{f(g(a+h)) - f(g(a))}{h} = q(v(h))\frac{g(a+h) - g(a)}{h}.$$

The factors tend to $f'(b)$ and $g'(a)$. In particular, $q(v(h)) \to q(0)$ even if $v(h)$ is zero along infinitely many increments, because q has been defined and is continuous there. Both factors have finite limits, so their product has the asserted limit. No division by $v(h)$ was used. □

Example 2.11 Zero inner increments are allowed Let $g(0) = 0$ and $g(x) = x^2 \sin^2(1/x)$ for $x \neq 0$. Find the derivative of $F(x) = e^{g(x)}$ at 0.

Solution ***Method 1: use the chain rule.*** *Since $|g(h)/h| \leq |h|$, $g'(0) = 0$. The exponential is differentiable at $g(0) = 0$, so the chain rule gives $F'(0) = e^0 \cdot 0 = 0$. For $h = 1/(n\pi)$, $n \geq 1$, the inner increment is exactly zero. Thus a proof based on cancelling that increment would be invalid for this example, although the chain rule itself applies.*

Solution ***Method 2: estimate the original quotient.*** *The fundamental exponential limit gives $(e^v - 1)/v \to 1$ as $v \to 0$. Hence there is $\delta_0 > 0$ such that $|e^v - 1| \leq 2|v|$ whenever $|v| < \delta_0$; at $v = 0$ the inequality also holds. For sufficiently small $h \neq 0$, $0 \leq g(h) \leq h^2 < \delta_0$, and therefore*

$$\left|\frac{F(h) - F(0)}{h}\right| \leq 2\frac{|g(h)|}{|h|} \leq 2|h| \longrightarrow 0.$$

Thus $F'(0) = 0$ directly, including increments for which $g(h) = 0$.

The chain rule gives the answer immediately after its hypotheses are checked. The direct estimate explains how the small inner change suppresses the entire output quotient and independently verifies the zero-increment case.

Example 2.12 A zero inner derivative does not replace the hypotheses With inner function $g(x) = x^2$, compare differentiability at 0 of $F(x) = |x^2|$ and $H(x) = \sqrt{x^2}$.

> ***Idea.*** Both inner changes have size h^2. Ask what each outer function does to that size before dividing the final output change by h. The outer absolute value preserves the size, whereas a square root enlarges it to $|h|$.

Solution *Although $u \mapsto |u|$ is not differentiable at 0, $F(x) = x^2$ on $\mathbb{R}$. Its quotient at 0 is h, so $F'(0) = 0$. On the other hand, $H(x) = |x|$ has quotient $|h|/h$ and is not differentiable at 0. Here the outer square root is defined only for $u \geq 0$ and has no finite right derivative at 0, so it does not satisfy the outer hypotheses of the chain rule either. In both cases $g'(0) = 0$, but the two composites have different outcomes. If a hypothesis fails, the theorem supplies no conclusion; the composite must be examined from its own definition or a valid simplification. An undefined outer derivative cannot be multiplied by 0 as if it were a real number.*

For an ordinary nested expression, work from the outside inward. For instance,

$$\left(\sin(e^{x^2})\right)' = \cos(e^{x^2})\, e^{x^2}\, 2x.$$

Each derivative has been evaluated at the input of its own layer.

2.2.3 Inverse functions

If a graph is read in the opposite direction, its input and output exchange roles. We expect a nonzero slope to become its reciprocal. Existence of an inverse and its continuity must be supplied before using this expectation.

Theorem 2.3

Let f be continuous and strictly monotone on an open interval I. Let $a \in I$, $b = f(a)$, and suppose f is differentiable at a with $f'(a) \neq 0$. Then the inverse $g = f^{-1}$ is differentiable at b, and

$$g'(b) = \frac{1}{f'(a)} = \frac{1}{f'(g(b))}.$$

♡

> ***Idea.*** Write a nearby output as $y = f(x)$. The inverse quotient is the reciprocal of the original quotient. Continuity of the inverse is what ensures that $y \to b$ also means $x \to a$.

Proof The image $f(I)$ is an interval by the intermediate value theorem. Choose $a_- < a < a_+$ in I. For an increasing f, $f(a_-) < b < f(a_+)$, and every value between these endpoints is attained by continuity. For a decreasing f, the inequalities are reversed, with the same conclusion. In either case, b is an interior point of the inverse's domain $f(I)$. The inverse is continuous by the inverse-continuity theorem of Section 1.5. For $y \neq b$ sufficiently near b, put $x = g(y)$. Injectivity gives $x \neq a$, and continuity gives $x \to a$. The original quotient $[f(x) - f(a)]/(x - a)$ tends to the nonzero number $f'(a)$. For x sufficiently near a, its absolute value is at least $|f'(a)|/2$, so taking its reciprocal is justified. Therefore

$$\frac{g(y) - g(b)}{y - b} = \frac{x - a}{f(x) - f(a)} \longrightarrow \frac{1}{f'(a)}.$$

The nonzero limiting denominator justifies the reciprocal limit. □

The notation f^{-1} means an inverse function, not $1/f$. The nonzero derivative condition cannot be omitted: $f(x) = x^3$ is a continuous strictly increasing bijection of $\mathbb{R}$, but its inverse has no finite derivative at 0.

The interval hypotheses do a different job: they establish that there is an inverse to differentiate. A derivative at just one point cannot supply those hypotheses.

Example 2.13 A nonzero derivative without a local inverse Set $q(0) = 0$ and $q(x) = x + 2x^2 \sin(1/x)$ for $x \neq 0$. Show that $q'(0) = 1$, but that q is not injective on any open interval containing 0.

> ***Strategy.*** A small oscillation in the function can leave an oscillation of fixed size after differentiation. To exclude injectivity, use the earlier fact that a continuous injective function on an interval must be strictly monotone. For a monotone function, the sign of every existing derivative follows directly from the signs of its difference quotients. We need only this necessary condition, not the later theorem inferring monotonicity from derivatives throughout an interval.

Solution *At the origin,*

$$\frac{q(h) - q(0)}{h} = 1 + 2h \sin(1/h) \longrightarrow 1.$$

Away from it the differentiation rules give

$$q'(x) = 1 + 4x \sin(1/x) - 2\cos(1/x).$$

Thus q is differentiable, hence continuous, on $\mathbb{R}$. For $x_n = 1/(2\pi n)$, $n \geq 1$, we have $q'(x_n) = -1$. Every open interval containing 0 contains all sufficiently late x_n. On such an interval q cannot be nondecreasing: a nondecreasing function has nonnegative difference quotients at an interior point, for both signs of the increment, and hence a nonnegative derivative whenever it exists. Likewise, q cannot be nonincreasing, because $q'(0) = 1 > 0$. By Proposition 1.17, continuity and injectivity would force one of these monotonicity alternatives. Therefore no such interval admits an inverse of the restriction of q.

The positive derivative still controls comparisons with the fixed value $q(0)$: sufficiently small positive h have $q(h) > q(0)$, and sufficiently small negative h have $q(h) < q(0)$. It does not order every pair of inputs on either side.

Example 2.14 An inverse without an explicit formula Let $f(x) = x + x^3$ on $\mathbb{R}$ and $g = f^{-1}$. Prove that the inverse exists and find $g'(2)$.

Solution *For $v > u$,*

$$f(v) - f(u) = (v - u)(1 + u^2 + uv + v^2) > 0,$$

since $u^2 + uv + v^2 = (u + v/2)^2 + 3v^2/4 \geq 0$. Also $f(x) \to \pm\infty$ as $x \to \pm\infty$; continuity and the intermediate

value theorem give range $\mathbb{R}$. Since $f(1) = 2$ and $f'(1) = 1 + 3 = 4$, the inverse theorem gives $g'(2) = 1/4$. The number 2 is the input of g and the output of f. Thus we evaluate f' at $g(2) = 1$, rather than at 2. Solving the cubic for a general output would add work without providing information needed for this derivative at one point. This proof of monotonicity uses algebra, not the later derivative-sign theorem.

2.2.4 Elementary derivatives and their domains

The exponential derivative is already known. Applying the inverse theorem to e^x gives

$$(\ln x)' = \frac{1}{x} \quad (x > 0).$$

For $x < 0$, the chain rule applied to $\ln(-x)$ yields $(\ln|x|)' = 1/x$ as well. For a real α and $x > 0$, the definition $x^\alpha = e^{\alpha \ln x}$ gives

$$(x^\alpha)' = \alpha x^{\alpha-1}.$$

For $\alpha = 0$ this is the constant rule. For negative bases, arbitrary real powers are not defined by this formula. Integer powers retain their earlier domains, and real odd roots can be checked on each side using their inverse functions.

On the standard inverse-trigonometric branches,

$$(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}, \quad (\arccos x)' = -\frac{1}{\sqrt{1-x^2}} \quad (-1 < x < 1),$$
$$(\arctan x)' = \frac{1}{1+x^2} \qquad (x \in \mathbb{R}).$$

Indeed, if $u = \arcsin x \in (-\pi/2, \pi/2)$, then $\cos u > 0$ and $\cos u = \sqrt{1-x^2}$; the inverse theorem gives the first formula. For $u = \arccos x \in (0, \pi)$, $\sin u > 0$ and $(\cos u)' = -\sin u$, giving the second. For $u = \arctan x \in (-\pi/2, \pi/2)$, $\sec^2 u = 1 + \tan^2 u = 1 + x^2$, giving the third. Continuity and strict monotonicity on these branches come from the elementary trigonometric functions and their inverses. The endpoint values of $\arcsin$ and $\arccos$ lie outside the derivative formulas. The inverse of $\cot$ restricted to $(0, \pi)$ is denoted by arccot. On this branch, $(\cot u)' = -\csc^2 u = -(1 + \cot^2 u)$, so $(\operatorname{arccot} x)' = -1/(1 + x^2)$ for every real x. For example, $\operatorname{arccot} 0 = \pi/2$ and its derivative there is -1.

For convenience, the main families are collected below.

Function	Derivative	Domain of the displayed formula
b^x	$b^x \ln b$	$b > 0,\ x \in \mathbb{R}$
$\log_b x$	$1/(x \ln b)$	$b > 0,\ b \neq 1,\ x > 0$
x^α	$\alpha x^{\alpha-1}$	$x > 0,\ \alpha \in \mathbb{R}$
$\sin x,\ \cos x$	$\cos x,\ -\sin x$	$x \in \mathbb{R}$
$\tan x,\ \sec x$	$\sec^2 x,\ \sec x \tan x$	$\cos x \neq 0$
$\cot x,\ \csc x$	$-\csc^2 x,\ -\csc x \cot x$	$\sin x \neq 0$

The logarithm formula also follows from $\log_b x = \ln x / \ln b$. This table records conclusions with hypotheses; it is not a substitute for checking the domain of a composite expression.

Separating e^x into its odd and even parts produces a pair of functions that differentiate into each other. Their algebraic identity describes the hyperbola $v^2 - u^2 = 1$, just as sine and cosine describe a circle.

Example 2.15 Hyperbolic functions The hyperbolic sine and cosine(双曲正弦与双曲余弦)are $\sinh x = (e^x - e^{-x})/2$ and $\cosh x = (e^x + e^{-x})/2$. Find their derivatives and that of $\tanh x = \sinh x / \cosh x$.

Solution *The chain and sum rules give* $(\sinh x)' = \cosh x$ *and* $(\cosh x)' = \sinh x$. *Direct expansion gives* $\cosh^2 x - \sinh^2 x = 1$, *and* $\cosh x > 0$. *The quotient rule therefore gives* $(\tanh x)' = 1/\cosh^2 x$ *on* $\mathbb{R}$. *Unlike the circular functions, differentiating cosine here does not introduce a minus sign. At* 0, $\sinh 0 = 0$ *and* $\cosh 0 = 1$, *so these derivatives also describe their simplest local behaviour.*

2.2.5 Choosing a representation

Example 2.16 A variable base and exponent Differentiate $y = x^x$ for $x > 0$ in two ways.

> ***Strategy.*** The ordinary power rule keeps the exponent fixed, and the exponential rule keeps the base fixed. Here neither is fixed. Taking a logarithm turns the two varying roles into the product $x \ln x$.

Solution ***Method 1: rewrite before differentiating.*** *Since* $x^x = e^{x \ln x}$, *differentiability follows from already proved rules, and*

$$y' = e^{x \ln x}(\ln x + 1) = x^x(\ln x + 1).$$

Solution ***Method 2: logarithmic differentiation.*** *The positive differentiable function just identified satisfies* $\ln y = x \ln x$. *Differentiating gives* $y'/y = \ln x + 1$, *hence the same answer.*

Logarithmic differentiation(对数求导法)means differentiating a logarithmic identity and then recovering y'. The first method also establishes differentiability without assuming it in advance. The second is often easier for long products, but positivity or the use of $\ln|y|$ must be justified.

Example 2.17 Logarithms can exclude legitimate points Find the derivative of $y = (x-1)^2(x+2)^3$ on all of $\mathbb{R}$, using direct and logarithmic differentiation.

Solution ***Method 1: product rule.*** *On all of* $\mathbb{R}$,

$$y' = 2(x-1)(x+2)^3 + 3(x-1)^2(x+2)^2 = (x-1)(x+2)^2(5x+1).$$

Solution ***Method 2: logarithmic differentiation off the zeros.*** *For* $x \neq 1, -2$,

$$\ln|y| = 2\ln|x-1| + 3\ln|x+2|, \qquad \frac{y'}{y} = \frac{2}{x-1} + \frac{3}{x+2}.$$

Multiplication by y *yields the expression above on that domain. At* 1, *the definition gives* $y(1+h)/h = h(3+h)^3 \to 0$. *At* -2, *it gives* $y(-2+h)/h = (h-3)^2h^2 \to 0$. *Thus both excluded points have derivative* 0.

The logarithmic route organizes factors efficiently; the direct route automatically includes their zeros. A cancellation in the final formula does not by itself prove differentiability at points excluded earlier.

Example 2.18 The branch matters Determine differentiability of

$$F(x) = \arcsin(2x\sqrt{1-x^2}), \qquad -1 \le x \le 1.$$

Find the derivative where it exists, and check both endpoints one-sidedly.

> ***Idea.*** Write $x = \sin t$ on the principal interval. The inner expression becomes $\sin(2t)$, but $\arcsin(\sin(2t))$ returns a principal value, not always $2t$. The points where $2t$ leaves that interval determine the cases.

Solution ***Method 1: resolve the branches.*** *With* $t = \arcsin x \in [-\pi/2, \pi/2]$ *and* $c = 1/\sqrt{2}$,

$$F(x) = \begin{cases} -\pi - 2\arcsin x, & -1 \le x \le -c, \\ 2\arcsin x, & -c \le x \le c, \\ \pi - 2\arcsin x, & c \le x \le 1. \end{cases}$$

The identities follow because each displayed angle lies in $[-\pi/2, \pi/2]$ *and has sine* $\sin(2t)$. *More explicitly, if* $-\pi/4 \le t \le \pi/4$, *then* $2t$ *is already a principal angle. If* $\pi/4 \le t \le \pi/2$, *the angle* $\pi - 2t$ *lies in* $[0, \pi/2]$

and has the same sine. If $-\pi/2 \le t \le -\pi/4$, the angle $-\pi - 2t$ lies in $[-\pi/2, 0]$ and also has the same sine. Since sine is injective on the principal interval, these are exactly the returned values. Thus $F' = 2/\sqrt{1-x^2}$ for $|x| < c$, and $F' = -2/\sqrt{1-x^2}$ for $c < |x| < 1$. At c the side slopes are $2\sqrt{2}$ and $-2\sqrt{2}$; at $-c$ they are $-2\sqrt{2}$ and $2\sqrt{2}$. The adjacent formulas agree in value at each joining point, and each branch extends differentiably across that point as an individual formula. Their derivatives therefore give the corresponding anchored one-sided quotients. Neither point is differentiable.

Solution ***Method 2: chain rule on the open pieces.*** *For $-1 < x < 1$ with $|x| \neq c$, put $u = 2x\sqrt{1-x^2}$. Then*

$$u' = \frac{2(1-2x^2)}{\sqrt{1-x^2}}, \qquad \sqrt{1-u^2} = \left|1-2x^2\right|.$$

The chain rule gives $F' = 2(1-2x^2)/[\sqrt{1-x^2}\,|1-2x^2|]$, the same piecewise answer. At $\pm c$, the outer derivative is unavailable, so the branch calculation is still needed.

For completeness, at $x = 1$ write $x = \cos s$ with $s \to 0^+$. The branch formula gives $F(x) = 2s$ and $F(1) = 0$, so $[F(x) - F(1)]/(x-1) = 2s/(\cos s - 1) \to -\infty$. Indeed, $\cos s - 1 = -2\sin^2(s/2)$, so this quotient equals $-s/\sin^2(s/2)$; since $\sin(s/2)/(s/2) \to 1$, its magnitude grows like $4/s$. At $x = -1$, write $x = -\cos s$; the quotient is $-2s/(1-\cos s) \to -\infty$. Neither endpoint has a finite one-sided derivative. The chain-rule computation is compact inside each branch; the trigonometric representation detects the folds and settles the excluded points.

2.2.6 Methods and further remarks

Inspect the outermost operation before differentiating. Sums suggest linearity, products suggest the product rule, and nesting suggests the chain rule. A variable exponent suggests a logarithm or an exponential representation. For an inverse derivative, first find the corresponding original point. Establish the inverse on a neighbourhood before using the reciprocal-slope formula; a nonzero derivative at one point supplies only part of its hypotheses.

Keep the original domain visible while simplifying. Check zeros before dividing by a function or taking its logarithm. For inverse trigonometric expressions, record the principal-value interval before cancelling an inverse with a function. The rules give sufficient hypotheses: a composition can be differentiable even when a factor function fails to be differentiable, as $|x^2| = x^2$ shows. Conversely, a zero inner derivative alone supplies no guarantee, as Example 2.12 shows. When a formal calculation produces a nonexistent derivative or a zero denominator, return to an actual function identity, an anchored quotient, or an explicit estimate.

Exercise 2.2

1. Differentiate $3x^5 - 2/x + \sqrt{x}$ on its real domain.
2. Differentiate $(x^2+1)\cos x$ and explain the two product-rule terms.
3. Differentiate $(2x+1)/(x^2-1)$, specifying the excluded points.
4. Differentiate $e^{\sin(3x)}$, displaying its three layers.
5. Differentiate $\ln(1+x^2)$ and $\ln|x^2-1|$ on their respective domains.
6. Differentiate $\arctan(2x)$ and $\arccos(x/2)$ wherever the finite derivatives exist.
7. Differentiate 5^{x^2} and $(1+x)^{\sqrt{2}}$, using the positive-base convention for the latter. If the latter is extended to $x = -1$ by the value 0, determine its right derivative there.
8. Find the derivative of $x^{\sin x}$ for $x > 0$ and explain why neither the fixed-power nor fixed-base rule alone suffices.

9. Differentiate $F(x) = \sqrt{1+\sqrt{1+x}}$ for $x > -1$. Check the endpoint -1 directly.
10. Let g be the inverse of $f(x) = x^3 + x + 1$ on $\mathbb{R}$. Justify existence and compute $g'(1)$.
11. Compute $[\sin(\sin(\sin x))]'$ and indicate the evaluation point of every cosine factor.
12. Find the tangent line to $y = \ln x$ parallel to $y = 2x + 1$.
13. Derive the formula for $(\csc x)'$ directly from the reciprocal rule and state its domain.
14. Differentiate $(x+1)^4/(x^2+1)$ by direct and logarithmic methods. Check the point omitted by the logarithm.
15. Decide whether f^2 differentiable at a implies f differentiable there. Give a counterexample and explain what changes if f is continuous near a and $f(a) \neq 0$.
16. Prove that if differentiable f is even on $\mathbb{R}$, then f' is odd; state and prove the corresponding assertion for odd f.
17. Determine every point where $F(x) = |x^2 - 4x|$ is differentiable and compute its derivative there.
18. For $H(x) = \sec x/(1+\tan x)$, determine its original domain, simplify, and differentiate. Explain whether the simplified formula extends the original domain.
19. Suppose f is differentiable near 1, g is differentiable near 0, and $f(1) = 2$, $f'(1) = 3$, $g(0) = 1$, $g'(0) = -2$. Find the derivative at 0 of $H(x) = f(g(x))/(1+g(x)^2)$, explaining the evaluation point of f'.
20. Prove that $f(x) = \ln x + x$ is a continuous strictly increasing bijection from $(0, +\infty)$ to $\mathbb{R}$, using known monotonicity of the logarithm rather than a derivative-sign theorem. Find $(f^{-1})'(1)$ without solving explicitly for the inverse.
21. Determine the real domain of $H(x) = \ln\sqrt{(x-1)^2}$ and differentiate it on every component of that domain. Explain why replacing $\sqrt{(x-1)^2}$ by $x - 1$ would lose valid inputs.
22. Let $f(x) = x^2$ and $g(x) = e^x$. Find $(f \circ g)'(0)$ and $(g \circ f)'(0)$, and explain why exchanging the order of the layers changes the answer.
23. Determine the original domain and derivative of $H(x) = \sqrt{(x^2-1)^2}/(x^2-1)$. Decide whether it can be extended continuously at 1 or -1. Explain how zero derivatives throughout its domain can coexist with these failures of extension.

Supplement Exercises

S1. Suppose f, g are defined near a, f is differentiable at a, and $f(a) \neq 0$. Prove that fg is differentiable at a if and only if g is. Give a counterexample when $f(a) = 0$.

S2. Suppose f is differentiable at 0, $f(0) = 0$, $f'(0) \neq 0$, and g is continuous at 0, $g(0) = 0$. Prove that differentiability of $f \circ g$ at 0 forces differentiability of g at 0, without assuming that f is monotone near 0.

> ***Optional Hint.*** Write $f(v) = vq(v)$ with $q(v) \to f'(0) \neq 0$, including $v = 0$.

S3. Let $f\colon \mathbb{R} \to \mathbb{R}$ satisfy $f(x+y) = f(x)f(y)$ and be differentiable at 0. Prove that $f'(x) = f'(0)f(x)$ for all x, including the possibility $f \equiv 0$.

S4. Determine and compare derivatives of $\arctan x + \arctan(1/x)$ for $x \neq 0$. Find the value on each half-line by a trigonometric identity, without using the theorem that a zero derivative implies constancy.

S5. Find $F'(x)$ for $F(x) = (\sin x)^x$ on every interval where $\sin x > 0$. Explain why replacing this domain by all nonzero x is invalid.

S6. Suppose f is defined near 0 and has a finite right derivative $f'_+(0)$. Prove that $F(x) = f(|x|)$ is differentiable at 0 if and only if $f'_+(0) = 0$. Construct such an f that has no two-sided derivative at 0, and explain why its behaviour for negative inputs does not affect F.

S7. Suppose f is differentiable at a, $f(a) = 0$, and $f'(a) \neq 0$. Prove that f has no other zero sufficiently near a. Explain why this does not assert monotonicity on a neighbourhood.

S8. Prove the derivative formula for a product of n differentiable functions, and derive the logarithmic derivative formula when all factors are nonzero.

S9. For $\alpha > 0$, let $F(x) = x^2$ for $x \leq 0$ and $F(x) = A\ln(1+x) + B + x^{\alpha}\sin(1/x)$ for $x > 0$. Determine all A, B, α for which F is differentiable at 0. Among those cases, determine exactly when F' is continuous at 0, checking both threshold exponents.

S10. For a differentiable $f\colon \mathbb{R} \to \mathbb{R}$, differentiate $F(x) = f(x)/(1 + f(x)^2)$. Show algebraically that $|F(x)| \leq 1/2$, and identify equality cases without using derivative tests for extrema.

S11. Suppose $f\colon \mathbb{R} \to \mathbb{R}$ is differentiable and $f(f(x)) = x$ for every x. Prove that f' is nowhere zero and that at any fixed point a its derivative is either 1 or -1. Give an example realizing each possibility. If f is strictly increasing, prove $f(x) = x$ for every x by comparing $f(x)$ with x; do not use a derivative-sign theorem.

S12. Find the piecewise linear formula for $F(x) = \arccos(\cos x)$ on $[0, 2\pi]$ and extend it by 2π-periodicity to $\mathbb{R}$. Determine every point of differentiability and the derivative there, and check the one-sided derivatives at every integer multiple of π. Compare this branch method with a chain-rule calculation on the open pieces.

2.3 Differentials and Linear Approximation

知识点概要

- ❏ *Compare infinitesimals and control remainders on the scale required by a calculation.*
- ❏ *Characterize differentiability by a unique linear part of the increment.*
- ❏ *Distinguish a differential, an actual increment, and their error.*
- ❏ *Use linear approximation and check its accuracy by a justified estimate.*
- ❏ *Explain first-differential invariance under a change of variable.*
- ❏ *Differentiate implicit and parametric relations with existence and branch conditions stated.*

The derivative gives a slope, but a slope can also predict an output. Near a known point, how well does the tangent line reproduce the function? This question leads from a quotient limit to a linear approximation, and gives a precise meaning to a differential.

2.3.1 Infinitesimals and comparison of errors

An approximation becomes useful only when we know what its error is small compared with. For a derivative, the relevant comparison is with the input increment, not with the value of the function. We first recall the language from Definition 1.23 and Definition 1.24, now using the approach $h \to 0$.

A function $\alpha(h)$ is an infinitesimal along this approach when $\alpha(h) \to 0$. For example, h, h^2, and $\sin h$ are infinitesimals as $h \to 0$. The fixed number 10^{-6} is not one: however small it looks, its limit as $h \to 0$ is still 10^{-6}. An infinitesimal need not be positive, monotone, or nonzero; it is a varying quantity described by a limiting process. The identically zero function is allowed. The approach matters: $1/x$ is an infinitesimal as $x \to +\infty$, but not as $x \to 0$.

To compare two functions, first choose a reference scale $\beta(h)$ which is nonzero for all sufficiently small $h \neq 0$. Then the following statements have distinct meanings:

$$\alpha = o(\beta) \quad \Longleftrightarrow \quad \alpha(h)/\beta(h) \to 0,$$
$$\alpha \sim \beta \quad \Longleftrightarrow \quad \alpha(h)/\beta(h) \to 1.$$

If both are infinitesimals, the first is a higher-order comparison and the second is equivalence. Our convention for *same order* is $\alpha/\beta \to c$ with $0 < |c| < \infty$. Thus $3h$ and h have the same order, but are not equivalent; $-h$ is also of the same order as h, with a different sign. The phrase "higher order" means smaller relative size, not a larger value. The order need not be an integer: for $h \to 0^+$, $h^{3/2}$ is smaller than h but larger than h^2 in this relative sense.

We will also use a bound that does not require a quotient limit. The notation $\alpha(h) = O(\beta(h))$ means that there are constants $C, \delta > 0$ such that

$$0 < |h| < \delta \quad \Longrightarrow \quad |\alpha(h)| \leq C\,|\beta(h)|\,.$$

This is called big-O notation(大 O 记号); it asserts bounded relative size, not equivalence and not necessarily the same order in the convention above. The corresponding little-o condition can be written

$$\text{for every } \varepsilon > 0 \text{ there is } \delta > 0: \quad 0 < |h| < \delta \Longrightarrow |\alpha(h)| \leq \varepsilon\,|\beta(h)|\,.$$

The distinction is between one fixed bound C and an arbitrarily small bound ε. Thus $o(\beta)$ implies $O(\beta)$, but not conversely. Here $o(1)$ means convergence to zero, while $O(1)$ means boundedness near the limiting point.

Example 2.19 All errors vanish, but on different scales As $h \to 0$, compare the five errors below with the input increment h. Which are $o(h)$, and which are only known to be $O(h)$?

Solution *For a two-sided comparison it is convenient to use absolute values:*

Error $r(h)$	$\lvert r(h)\rvert \,/\, \lvert h\rvert$	*Conclusion*
h^2	$\lvert h\rvert \to 0$	$o(h)$
$\lvert h\rvert^{3/2}$	$\sqrt{\lvert h\rvert} \to 0$	$o(h)$
$\lvert h\rvert$	1	$O(h)$, *not* $o(h)$
$h\sin(1/h)$	$\lvert\sin(1/h)\rvert \le 1$	$O(h)$, *not* $o(h)$
$\sqrt{\lvert h\rvert}$	$1/\sqrt{\lvert h\rvert} \to +\infty$	*not* $O(h)$

For the oscillating row, along $h_n = (\pi/2 + 2\pi n)^{-1}$ *the relative magnitude is* 1*, so it does not tend to zero. Along* $k_n = (n\pi)^{-1}$ *it is* 0*, so a bound alone does not even supply a quotient limit. All five errors tend to zero, but only the first two disappear relative to the input scale.*

Here $r(h) = o(h)$ and $r(h) = o(|h|)$ are equivalent statements, since both say $|r(h)| \,/\, |h| \to 0$. By contrast, $h \sim |h|$ is false for a two-sided approach, because equivalence retains signs.

> ***Working with remainders.*** To use a symbol such as $o(h)$ safely, temporarily give its hidden remainder a name and write it as $h\eta(h)$ with $\eta(h) \to 0$. This turns a formal-looking calculation into an identity with ordinary functions whose limits can be checked.

For $h \neq 0$, $r = o(h)$ is equivalent to $r(h) = h\eta(h)$ with $\eta(h) = r(h)/h \to 0$. Set $r(0) = \eta(0) = 0$ if an identity at zero is needed. For example, if $r_i = h\eta_i$ and $\eta_i \to 0$, then

$$r_1 + r_2 = h(\eta_1 + \eta_2) = o(h), \qquad cr_1 = h(c\eta_1) = o(h) \quad (c \in \mathbb{R}).$$

If $u = O(h)$ and $v = O(h)$, their bounds give $|uv| \le C_1C_2h^2$, so $uv = O(h^2) = o(h)$. These statements concern a fixed finite number of terms; they do not justify an interchange with an infinite sum. Two occurrences of $o(h)$ can denote different remainders. Their difference is still $o(h)$, but cannot generally be set equal to zero.

An equivalent form of $\alpha \sim \beta$ is $\alpha = \beta + o(\beta)$, obtained by subtracting 1 from α/β. This form displays the information retained by the approximation. Products and quotients can often use equivalent factors, with the nonzero conditions in Proposition 1.15. In a sum or difference, however, leading terms may cancel; then the original error scale may be too coarse.

Example 2.20 Choose the precision before replacing terms Evaluate, using current limit tools,

$$\lim_{h\to0} \frac{\sin(2h) + \ln(1+3h)}{\sqrt{1+h} - 1}, \qquad \lim_{h\to0} \frac{\tan h - \sin h}{h^3}.$$

> ***Strategy.*** In the first numerator the linear terms add to a nonzero multiple of h, so first-order remainders are sufficient. In the second numerator both terms have the same leading part h, which cancels. We need an identity exposing the remaining scale, rather than deleting those two equal-looking leading terms.

Solution ***First limit, Method 1: retain a named error scale.*** *The fundamental limits give*

$$\sin(2h) = 2h + o(h), \qquad \ln(1+3h) = 3h + o(h), \qquad \sqrt{1+h} - 1 = \tfrac{1}{2}h + o(h).$$

For sufficiently small h*, both* $1 + 3h$ *and* $1 + h$ *are positive. Adding the first two expansions and dividing numerator and denominator by* $h \neq 0$ *gives*

$$\frac{5 + o(1)}{1/2 + o(1)} \longrightarrow 10.$$

The denominator tends to $1/2$*, so it stays nonzero near the limiting point.*

Solution ***First limit, Method 2: expose the basic quotients.*** *Rationalizing the denominator gives*

$$\frac{\sin(2h)+\ln(1+3h)}{\sqrt{1+h}-1}=\left(2\frac{\sin(2h)}{2h}+3\frac{\ln(1+3h)}{3h}\right)(\sqrt{1+h}+1).$$

The parenthesis tends to $2+3=5$ *and the last factor tends to* 2*, again giving* 10*. This route shows every quotient explicitly; the remainder route is more convenient for combining several known first-order models.*

Solution ***Second limit: resolve the cancellation first.*** *Near* 0*,* $\cos h\neq 0$ *and*

$$\tan h-\sin h=\frac{\sin h(1-\cos h)}{\cos h}.$$

Therefore

$$\frac{\tan h-\sin h}{h^3}=\frac{\sin h}{h}\,\frac{1-\cos h}{h^2}\,\frac{1}{\cos h}\longrightarrow 1\cdot\frac{1}{2}\cdot 1=\frac{1}{2}.$$

The middle limit follows from $1-\cos h=2\sin^2(h/2)$ *and the sine limit. The weaker statements* $\tan h=h+o(h)$ *and* $\sin h=h+o(h)$ *would give only a difference* $o(h)$*, which says nothing decisive after division by* h^3*.*

2.3.2 The linear part of an increment

The previous comparisons explain why an approximation criterion must specify a scale. We want to predict an output from its known value at a base point and one constant rate of change. Merely asking for an error that tends to zero does not determine that rate.

Example 2.21 Which line is selected by the error condition? For $f(x)=3+2x+x^2$, compare the candidate lines $L_B(x)=3+Bx$, $B\in\mathbb{R}$, near $a=0$. Which have error tending to zero, and which have error $o(h)$ at $x=h$?

Solution *Every candidate passes through* $(0,3)$*, and*

$$f(h)-L_B(h)=(2-B)h+h^2\longrightarrow 0$$

for every fixed B*. Thus the unscaled error cannot select one line. For* $h\neq 0$*, however,*

$$\frac{f(h)-L_B(h)}{h}=2-B+h\longrightarrow 2-B.$$

The error is $o(h)$ *exactly when* $B=2$*. The line* L_2 *matches the first-order change, while a different slope leaves an error of the same scale as the input. The choice is asymptotic at the fixed centre, not a claim of best approximation over an arbitrary interval.*

This suggests the definition: separate the actual increment into a linear prediction and a remainder that is negligible compared with the input increment.

Definition 2.3

Let f *be defined near* a*. We say that* f *admits a differential* (可微) *at* a *if there is* $A\in\mathbb{R}$ *such that*

$$f(a+h)-f(a)=Ah+r(h),\qquad r(h)=o(h)\quad(h\to 0).$$

The map $h\mapsto Ah$ *is its differential* (微分) *at* a*, written* $\mathrm{d}f_a(h)$*.* ♣

The differential is a linear function of the input increment, with the base point held fixed. The remainder is negligible relative to that input increment. The coefficient A is allowed to be zero. At a fixed point, linearity means additivity and compatibility with multiplication by a real number:

$$\mathrm{d}f_a(h+k)=\mathrm{d}f_a(h)+\mathrm{d}f_a(k),\qquad \mathrm{d}f_a(\lambda h)=\lambda\mathrm{d}f_a(h).$$

Indeed, these are just $A(h+k)=Ah+Ak$ and $A\lambda h=\lambda Ah$. The actual increment $f(a+h)-f(a)$ need not satisfy these identities. The map $h\mapsto Ah$ can be evaluated for any real h; its role as a reliable approximation

concerns sufficiently small increments for which $a + h$ is in the domain. The remainder condition means that for every prescribed relative tolerance $\varepsilon > 0$, the error is at most $\varepsilon\,|h|$ for all sufficiently small h. The coefficient A is fixed before h varies; allowing a new A for each increment would merely reproduce secant slopes and would not define one linear model. In English, both this property and existence of the derivative are called differentiability; the theorem below proves their equivalence in one variable.

Theorem 2.4

A function has a derivative at a if and only if it admits a differential there. In that case the coefficient is unique and

$$\mathrm{d}f_a(h) = f'(a)h.$$

Consequently, the unique affine function through $(a, f(a))$ whose error is $o(x - a)$ is

$$L_a(x) = f(a) + f'(a)(x - a).$$

♡

> ***Proof Strategy.*** To identify a linear coefficient, divide the approximation error by the input increment. The remainder disappears and leaves the coefficient as a limit. Two different slopes would leave a nonzero constant in this normalized error, so uniqueness should follow from the same operation.

Proof Suppose first that $f'(a)$ exists. Define the actual error of its candidate linear model by

$$r(h) = f(a + h) - f(a) - f'(a)h, \qquad r(0) = 0.$$

For $h \neq 0$, $r(h)/h$ is exactly the difference quotient minus $f'(a)$, so it tends to 0. This proves existence of the differential with $A = f'(a)$, including when that derivative is zero.

Conversely, suppose an increment representation $f(a + h) - f(a) = Ah + r(h)$ holds with $r(h) = o(h)$. Division by nonzero h gives

$$\frac{f(a + h) - f(a)}{h} = A + \frac{r(h)}{h} \longrightarrow A.$$

This is precisely the derivative definition, so $f'(a) = A$.

To see uniqueness directly in the error language, suppose $Ah + r_A(h) = Bh + r_B(h)$ with both remainders $o(h)$. Then

$$|A - B| \leq \frac{|r_A(h)|}{|h|} + \frac{|r_B(h)|}{|h|}.$$

If $A \neq B$, take h small enough that each term on the right is less than $|A - B|/3$. The resulting inequality is impossible; hence $A = B$. An affine function through $(a, f(a))$ has the form $f(a) + B(x - a)$. Its error is $o(x - a)$ precisely when the same argument gives $B = f'(a)$. □

The function L_a is called the linearization(线性化)at a, although it is generally affine rather than linear in x. Its change $L_a(a + h) - L_a(a) = f'(a)h$ is linear in h. In classical notation, take $\mathrm{d}x = h$ and write $\mathrm{d}y = f'(a)\mathrm{d}x$. If the base point varies, the formula is written $\mathrm{d}y = f'(x)\mathrm{d}x$. This notation records two inputs: the point x and the proposed increment $\mathrm{d}x$. For instance, with $f(x) = x^2$, $\mathrm{d}f_1(0.01) = 0.02$ but $\mathrm{d}f_3(0.01) = 0.06$. Changing the base point changes the linear map; changing the increment evaluates the same map at a new input. The independent variable has differential $\mathrm{d}x = h$ because the identity function $x \mapsto x$ has derivative 1. We choose this differential to equal the actual input increment Δx. This does not force the dependent differential $\mathrm{d}y$ to equal Δy. Here $\mathrm{d}x$ is an ordinary real increment, not an additional kind of number; statements about infinitesimals refer to the process in which such increments tend to zero. For $\mathrm{d}x \neq 0$, the ratio of these differentials equals $f'(x)$, but the ratio of actual finite increments need not do so.

Example 2.22 Exact change versus linear change For $f(x) = x^3$ at $a = 2$, calculate $\Delta f = f(2+h) - f(2)$, $\mathrm{d}f_2(h)$, and the error.

Solution *Expansion gives*

$$\Delta f = 12h + 6h^2 + h^3, \qquad \mathrm{d}f_2(h) = 12h, \qquad \Delta f - \mathrm{d}f_2(h) = 6h^2 + h^3.$$

For $h \neq 0$, the normalized error is

$$\frac{\Delta f - \mathrm{d}f_2(h)}{h} = 6h + h^2 \longrightarrow 0.$$

The coefficient 12 *stays fixed while h varies; neither the term $6h^2$ nor the term h^3 is part of the differential. For $h = 0.1$, the actual change is* 1.261 *and the differential is* 1.2*. The approximation error is* 0.061*, not zero. At $h = -0.1$, the actual change is -1.141, the differential is -1.2, and their difference is* 0.059*. Both directions use the same linear map $h \mapsto 12h$.*

Example 2.23 A zero differential is a valid differential Compare $f(x) = x^2$ and $g(x) = |x|$ at 0 as candidates for a linear approximation.

Solution *For f, the choice $A = 0$ leaves remainder $h^2 = o(h)$, so $\mathrm{d}f_0 = 0$. This zero map predicts no* first-order *change; the actual change is still $h^2 > 0$ for $h \neq 0$. Its absolute error tends to zero and its error relative to $|h|$ tends to zero, but its error relative to the actual nonzero change is $h^2/h^2 = 1$. Thus differentiability need not make the linear prediction relatively accurate as an estimate of the output* change *when the derivative vanishes. Dividing the error by this zero differential would make no sense; the definition divides by the input increment instead. For g, an identity $|h| = Ah + o(h)$ would force $A = 1$ from positive h and $A = -1$ from negative h. More explicitly, for $h > 0$ the normalized error is $1 - A$; for $h < 0$ it is $-1 - A$. Both would have to tend to zero with the same A. No single linear map works on both sides.*

When $f'(a) \neq 0$, the situation is different:

$$\frac{f(a+h) - f(a)}{f'(a)h} = 1 + \frac{r(h)/h}{f'(a)} \longrightarrow 1.$$

Then the differential is an equivalent infinitesimal to the actual change, which is nonzero for sufficiently small $h \neq 0$. At a zero derivative, the universally valid statement is instead $\Delta f = o(h)$; one must not write $\Delta f \sim 0$. Also, $o(h)$ need not be $O(h^2)$: the error $|h|^{3/2}$ is $o(h)$, but its ratio to h^2 is $1/\sqrt{|h|}$ and is unbounded. Differentiability controls a first-order scale without promising a quadratic error bound.

In **Figure** 2.3, both changes start from the same height $f(a)$. The differential ends on the tangent line, while the actual change ends on the curve. The vertical gap is the approximation error. For this parabola, $\Delta f = 2h + h^2$, $\mathrm{d}f = 2h$, and $r(h)/h = h \to 0$. The positive increment in the figure is chosen large enough to make the gap visible; the limit concerns both signs of shrinking h.

There is another useful geometric reading of the same estimate. Use a positive scale t and write $x = a + tz$, where $|z| \leq 1$ describes a fixed viewing window. After subtracting the base height and dividing both coordinate scales by t, the rescaled graph has height

$$\frac{f(a+tz) - f(a)}{t} = f'(a)z + \frac{r(tz)}{t}.$$

Given $\varepsilon > 0$, if t is sufficiently small then $|r(tz)| \leq \varepsilon\,|tz|$ throughout this window, including $z = 0$. Thus $|r(tz)|\,/t \leq \varepsilon\,|z| \leq \varepsilon$. The rescaled graph approaches the line of slope $f'(a)$ over the whole fixed window.

The formulas in **Figure** 2.4 verify the picture exactly: the parabola's deviation from $2z$ is at most t, while $|tz|\,/t = |z|$ never becomes one line. Making a graph look small without keeping track of both scales would not establish differentiability.

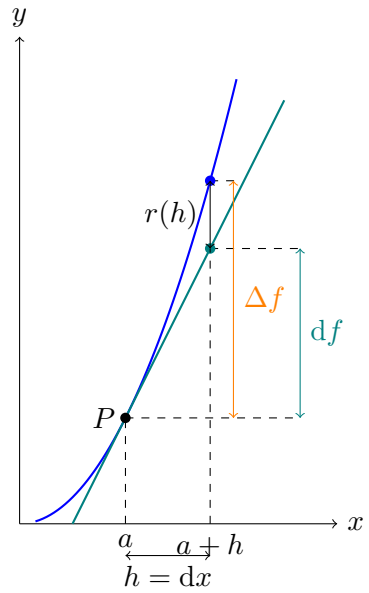
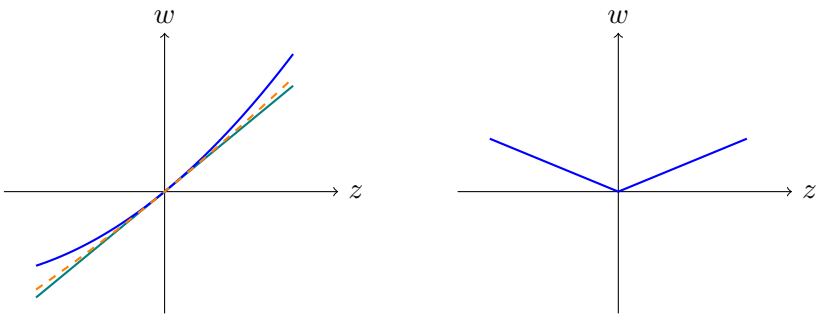


Figure 2.3: For $f(x) = x^2$ at $a = 1$, the same increment $h = 0.8$ gives actual change $2h + h^2$ and tangent change $2h$. Their difference is $r(h) = h^2$.



parabola: $w = 2z + tz^2$ corner: $w = |z|$

Figure 2.4: Rescaling both axes. Left: $f(x) = x^2$, $a = 1$, with $t = 0.6$ (blue), $t = 0.15$ (orange), and the limiting line $w = 2z$ (teal). Right: for $f(x) = |x|$ at 0, every positive scale gives the same corner.

2.3.3 Differential rules and change of variable

Multiplying the derivative rules by $\mathrm{d}x$ gives

$$\begin{aligned} \mathrm{d}(u+v) &= \mathrm{d}u + \mathrm{d}v, \\ \mathrm{d}(uv) &= u\,\mathrm{d}v + v\,\mathrm{d}u, \\ \mathrm{d}(u/v) &= \frac{v\,\mathrm{d}u - u\,\mathrm{d}v}{v^2} \quad (v \neq 0). \end{aligned}$$

Here u, v are differentiable functions of the same independent variable, and all values are taken at the same base point. These are identities between linear changes, not exact formulas for finite increments. For example, $\Delta(uv) = u\Delta v + v\Delta u + \Delta u\Delta v$ also contains a product of increments.

Proposition 2.2

Suppose $u = g(x)$ is differentiable at x and $y = f(u)$ is differentiable at $u = g(x)$. Then

$$\mathrm{d}y = f'(u)\,\mathrm{d}u, \qquad \mathrm{d}u = g'(x)\,\mathrm{d}x.$$

Thus the first formula has the same form whether u is the independent variable or itself depends differentiably on x.

Proof **Method 1: use the derivative chain rule.** The chain rule gives $\mathrm{d}y = f'(g(x))g'(x)\mathrm{d}x = f'(u)\mathrm{d}u$. The identity includes $g'(x) = 0$ and uses no division by $\mathrm{d}u$. □

> ***A second proof through the error terms.*** The two successive models have remainders measured against different inputs. To compose them, express the outer error in terms of the inner increment, then compare that increment with the original h. The inner increment is $O(h)$, which is exactly the bound needed to transfer the outer error to the original scale.

Proof **Method 2: compose the linear models directly.** Fix an original base point a, put $b = g(a)$, and write $B = g'(a)$ and $A = f'(b)$. For small h let $v = g(a+h) - g(a)$. By the differential characterizations,

$$v = Bh + h\eta(h), \qquad f(b+v) - f(b) = Av + v\theta(v),$$

where $\eta(h) \to 0$ and $\theta(v) \to 0$. Set $\eta(0) = \theta(0) = 0$; both identities then include zero increments. The first identity implies $v \to 0$ and $v = O(h)$. Substitution into the second gives the exact identity

$$f(g(a+h)) - f(g(a)) = ABh + h\big(A\eta(h) + B\theta(v) + \eta(h)\theta(v)\big).$$

All three terms in parentheses tend to zero, including along increments for which $v = 0$. Thus the composite has differential $h \mapsto ABh$. This is $A\,\mathrm{d}u$ with $\mathrm{d}u = Bh$, which proves the claimed invariance. □

The first proof is shorter once the chain rule is available. The second proves the same rule from remainders and explains why composing first-order models is legitimate; it uses the already proved derivative/differential equivalence, not the chain rule itself.

This is the invariance of the first differential(一阶微分的形式不变性). The intermediate differential means the linearized change $g'(x)\mathrm{d}x$, not the exact change $g(x+\mathrm{d}x) - g(x)$.

Example 2.24 An intermediate differential is not its exact increment Let $y = \sin u$ and $u = x^2$. For an input increment $h = \mathrm{d}x$, compare Δu, $\mathrm{d}u$ and $\mathrm{d}y$ at $x = 1$ and at $x = 0$. Verify $\mathrm{d}y$ both directly and through u.

Solution ***Method 1: retain both stages.*** *At $x = 1$, the intermediate base value is $u = 1$, and*

$$\Delta u = (1+h)^2 - 1 = 2h + h^2, \qquad \mathrm{d}u = 2h, \qquad \mathrm{d}y = \cos(1)\,\mathrm{d}u = 2\cos(1)h.$$

The actual final change is $\Delta y = \sin(1+2h+h^2) - \sin 1$. *The outer differentiability expansion yields*

$$\Delta y - \mathrm{d}y = \cos(1)h^2 + (2h+h^2)\theta(2h+h^2) = o(h),$$

because $\theta(v) \to 0$ *and* $(2h+h^2)/h$ *stays bounded. The term* $\cos(1)h^2$ *explains what would be added if one replaced* $\mathrm{d}u$ *by* Δu *in the differential formula.*

At $x = 0$, $\Delta u = h^2$, *but* $\mathrm{d}u = 0$ *and* $\mathrm{d}y = 0$. *The actual change is* $\sin(h^2)$, *and* $\left|\sin(h^2)\right| / |h| \le |h| \to 0$. *Thus the zero differential is correct, although the intermediate and final actual changes need not vanish.*

Solution ***Method 2: use the composite directly.*** *The derivative of* $y = \sin(x^2)$ *is* $2x\cos(x^2)$, *so* $\mathrm{d}y = 2x\cos(x^2)h$. *Evaluation at* $x = 1$ *and* $x = 0$ *gives the same two answers. The direct route is short; the staged route distinguishes the two error sources and the roles of* h, Δu *and* $\mathrm{d}u$.

The next section will explain why second differentials require more care under nonlinear substitution.

2.3.4 Approximation and error

The linear approximation (线性近似) $f(a+h) \approx f(a) + f'(a)h$ has error $o(h)$ as $h \to 0$. It does not by itself specify how small the error is at a particular nonzero h. A numerical guarantee needs a further estimate.

For a concrete reason, consider $f_M(x) = Mx^2$, where $M > 0$. Every member has the same data $f_M(0) = f'_M(0) = 0$, and for each fixed M its error from the zero linearization is $o(h)$, since $Mh^2/h = Mh \to 0$. At a prescribed $h_0 \neq 0$, however, the error Mh_0^2 can be arbitrarily large as M varies. There is no common numerical error bound determined by these two point data alone. This does not contradict differentiability: the small neighbourhood required for a tolerance may depend on the function, hence on M. In the following root example, exact algebra supplies the additional information that those point data do not contain.

Example 2.25 A root approximation with a proved error bound Approximate $\sqrt{4.04}$ using the centre $a = 4$. Prove the sign and a numerical upper bound for the error without Taylor's theorem.

> ***Strategy.*** The nearby perfect square gives an easy value and derivative. To certify the approximation, compare the proposed answer with the root by a conjugate identity; this exposes a squared small increment.

Solution ***Method 1: construct the linear model.*** *For* $f(x) = \sqrt{x}$, $f(4) = 2$ *and* $f'(4) = 1/4$. *With* $h = 0.04$, *the linear model gives* $L_4(4.04) = 2 + 0.04/4 = 2.01$. *Differentiability justifies the asymptotic approximation, but not yet a numerical tolerance.*

Solution ***Method 2: certify by exact algebra.*** *For* $h > -4$ *let* $s = \sqrt{4+h}$. *Since* $s - 2 = h/(s+2)$,

$$2 + \frac{h}{4} - s = \frac{h}{4} - \frac{h}{s+2} = \frac{h(s-2)}{4(s+2)} = \frac{h^2}{4(s+2)^2}.$$

The error is nonnegative. For $h = 0.04$ *one has* $s > 2$, *so*

$$0 < 2.01 - \sqrt{4.04} < \frac{0.04^2}{64} = 0.000025.$$

The differential selects the model and applies to many functions. The exact algebra supplies information specific to this root: the error's sign and a rigorous bound at the chosen increment. The general theory of quantitative remainder bounds will be developed later.

Relative changes require a nonzero reference value. If $f(a) \neq 0$, then

$$\frac{\Delta f}{f(a)} = \frac{f'(a)}{f(a)}h + o(h).$$

For $a > 0$ and $f(x) = x^\alpha$, this predicts a relative output change of approximately $\alpha h/a$. Here we divided by the fixed, nonzero reference value $f(a)$, not by the change Δf or the differential. If $f(a) = 0$, this relative change

is undefined even though the absolute linear approximation can still be valid. The sign records the direction of change; an absolute error estimate discards it.

Example 2.26 Propagation of a radius change A sphere has radius $r = 10$ cm, which increases by 0.02 cm. Estimate the change in its volume $V = 4\pi r^3/3$ and its relative change. Compare with the exact polynomial calculation.

Solution ***Method 1: differentials.*** *Here* $\mathrm{d}r = 0.02$ *cm, so* $\mathrm{d}V = 4\pi r^2 \mathrm{d}r = 8\pi$ *cm*^{3} *and* $\mathrm{d}V/V = 3\mathrm{d}r/r = 0.006 = 0.6\%$.

Solution ***Method 2: the exact relative increment.*** *Write* $\epsilon = \Delta r/r = 0.002$. *Then*

$$\frac{\Delta V}{V} = (1+\epsilon)^3 - 1 = 3\epsilon + 3\epsilon^2 + \epsilon^3 = 0.006012008.$$

The linear estimate underestimates the relative change by 0.000012008. *Multiplying by* $V = 4000\pi/3$ *gives the exact volume error.*

The differential isolates the leading sensitivity, while the polynomial identity measures the neglected terms. If 0.02 cm were a bound on measurement error of unknown sign, the first-order magnitude estimate would still be 0.6%, but a guaranteed bound would also include the higher terms.

2.3.5 Implicit and parametric relations

An equation involving x and y may describe a curve without displaying y as a function of x. An implicit function(隐函数)on an interval is a function $y = y(x)$ satisfying that equation throughout the interval. The equation can have several branches or fail to define such a function. Differentiating the equation computes a derivative of a differentiable branch; it does not establish that the branch exists.

Example 2.27 Two routes to a circle's tangent Find the tangent line to $x^2 + y^2 = 4$ at $(1, \sqrt{3})$, and justify the branch used.

Solution ***Method 1: the explicit branch.*** *Near* $x = 1$, *the upper branch is* $y = \sqrt{4 - x^2}$. *The radicand is positive, so the rules already proved show that it is differentiable. Its derivative is* $-x/\sqrt{4-x^2}$, *giving slope* $-1/\sqrt{3}$. *The tangent is* $y - \sqrt{3} = -(x-1)/\sqrt{3}$.

Solution ***Method 2: differentiate the relation.*** *For the differentiable branch just verified, the identity gives* $2x + 2yy' = 0$. *Since* $y = \sqrt{3} \neq 0$ *at the point,* $y' = -x/y = -1/\sqrt{3}$. *The same tangent can be written* $x + \sqrt{3}\,y = 4$.

The explicit method supplies existence and the chosen sign. The implicit method avoids carrying the radical through the calculation. At $(2, 0)$ the division by y fails; that failure calls for a separate geometric or parametric analysis, not an infinite real value for y'.

Example 2.28 Existence, branch choice, and differentiability are separate Compare the relations $x^2 + y^2 = 0$, $y^2 = x$, and $y^2 = x^2$ at $(0, 0)$. Which admit real function branches near the point, and what can be said about their derivatives there?

> ***Strategy.*** Before differentiating a relation, inspect the real solutions for each nearby input. Then choose an actual function from those solutions and test its difference quotient. Different choices can satisfy the same equation while having different slopes, or no slope at all.

Solution *For* $x^2 + y^2 = 0$, *both squares must vanish. The solution set consists of one point, so no real branch is defined on an open interval containing* $x = 0$.

For $y^2 = x$, *no real solution exists for negative* x. *On the one-sided domain* $x \geq 0$, *the branches* $y = \sqrt{x}$ *and* $y = -\sqrt{x}$ *are continuous, but their right difference quotients at* 0 *are* $1/\sqrt{h}$ *and* $-1/\sqrt{h}$. *Neither has a*

finite right derivative. Having a one-sided branch is therefore weaker than having a two-sided differentiable branch.

For $y^2 = x^2$, the functions $y = x$, $y = -x$, and $y = |x|$ are all continuous branches on $\mathbb{R}$. The first two have derivatives 1 and -1 at 0; the third has left derivative -1 and right derivative 1, so it is not differentiable there. If a chosen branch is differentiable, differentiating the equation gives $2yy' = 2x$. At $(0,0)$ this says only $0 = 0$: it determines no slope. For the absolute-value branch, differentiating at 0 would be unjustified in the first place. Thus a formal differentiated equation is a necessary relation for an existing differentiable branch, not a proof of its existence or differentiability.

The circle example supplied a differentiable branch explicitly before using its equation. These contrasts explain why that first step was part of the solution.

A parametric curve(参数曲线)is described by two coordinate functions $x = X(t)$, $y = Y(t)$ of a parameter t. To view it locally as a graph over x, suppose X is continuous and strictly monotone near t_0, differentiable there at t_0, and $X'(t_0) \neq 0$. If Y is differentiable at t_0, the inverse and chain rules give

$$\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{x=X(t_0)} = \frac{Y'(t_0)}{X'(t_0)}.$$

These assumptions provide a local inverse for X. A nonzero derivative at just one point, without a local invertibility hypothesis, is not being used as a substitute for them. The function q in Example 2.13 shows exactly why it cannot serve as such a substitute for the coordinate function X.

Example 2.29 A parameter is not the horizontal coordinate For $X(t) = t^2$, $Y(t) = t^3$, $t > 0$, find the slope and the tangent at $t = 2$ in two ways.

Solution ***Method 1: retain the parameter.*** *The map X is strictly increasing for $t > 0$, and $X' = 2t \neq 0$. Hence $\mathrm{d}y/\mathrm{d}x = 3t^2/(2t) = 3t/2$. At $t = 2$ the point is $(4, 8)$ and the slope is 3, giving $y - 8 = 3(x - 4)$.*

Solution ***Method 2: eliminate the parameter.*** *The positive branch gives $t = \sqrt{x}$, so $y = x^{3/2}$ for $x > 0$. Differentiation yields $(3/2)\sqrt{x}$, the same slope.*

Elimination is especially transparent here because the inverse is simple. The parameter formula remains useful when elimination would produce a complicated equation. At $t = 0$, the ratio would be $0/0$; it cannot determine the endpoint slope without a separate limit.

2.3.6 Methods and further remarks

For an approximation, choose a centre where both the function value and the derivative are known, and record the actual increment from that centre. For an infinitesimal calculation, specify the approach and the nonzero reference scale before comparing terms. Use named remainders to combine approximations, and check whether the leading terms cancel before dividing by a smaller scale. Keep three quantities distinct: the actual change, the linear prediction, and their difference. To prove differentiability, estimate the remainder relative to the input increment. To guarantee a numerical tolerance, provide an explicit bound instead of relying on the symbol $\approx$.

For implicit and parametric calculations, first identify the branch and the independent variable. Check every denominator before using a derivative ratio. Treat existence of a real branch, the choice among branches, and differentiability of the chosen branch as separate checks. First-differential invariance follows from composition of linear changes; it does not assert that finite increments obey the same exact identities.

> ***Looking Ahead.*** If a linear model is not accurate enough, a natural next question is how its slope changes. This leads to second and higher derivatives. Turning those derivatives into polynomial approximations

with general error bounds will require the later Taylor theory.

Exercise 2.3

1. State precisely what $r(h) = o(h)$ means and check whether h^2, $|h|$, and $|h|^{3/2}$ satisfy it.
2. Find $\mathrm{d}y$ for $y = \ln(1 + x^2)$, specifying the role of x and $\mathrm{d}x$.
3. Find the linearization of $\sin x$ at $a = \pi/6$.
4. For $f(x) = x^2$ at $a = 3$ and $h = -0.1$, compute Δf, $\mathrm{d}f$, and the exact error.
5. Give first-order approximations near $h = 0$ for e^h, $\ln(1 + h)$, and $(1 + h)^\alpha$, $\alpha \in \mathbb{R}$.
6. Find the differential and linearization at $a = 0$ of $f(x) = 1/(1 + x)$.
7. Prove that $1/(1 + h) - (1 - h) = h^2/(1 + h)$ for $h > -1$. Use it to bound the error when $|h| \le 0.1$.
8. Approximate $\sqrt[3]{27.03}$ by linearization at 27. State what accuracy information has and has not been established.
9. For $y = \ln u$, $u = 1 + x^2$, compute $\mathrm{d}y$ in two ways and check $x = 0$.
10. A square's side length changes from 5 cm to $5 + \delta$ cm, $|\delta| \le 0.01$. Estimate the relative area change and give a rigorous bound for the difference from that estimate.
11. Find the tangent to the upper branch of $x^2 + 4y^2 = 8$ at $(2, 1)$ by explicit and implicit differentiation.
12. For $x = e^t$, $y = t^2$, find $\mathrm{d}y/\mathrm{d}x$ and the tangent at $t = 0$; verify by eliminating t.
13. If $f'(a) \neq 0$, prove that $f(a+h) - f(a) \sim f'(a)h$. Explain why the hypothesis cannot simply be dropped.
14. Prove that the differential of f at a point, if it exists, is the unique linear map with remainder $o(h)$, without appealing to uniqueness of limits as a named theorem.
15. Suppose $f(h) = c + Ah + h\eta(h)$ with $\eta(h) \to 0$ and $f(0) = c$. Determine for which constants B the line $c + Bh$ has error $o(h)$; decide whether the error must be $O(h^2)$.
16. For $x = t^3$, $y = t^2$, $t \in \mathbb{R}$, determine whether y is a function of x and whether it has a finite derivative at $x = 0$. Explain why the parameter ratio cannot be used there.
17. Evaluate $\lim_{h\to 0}[\sin(3h) - 3\sin h]/h^3$ using a trigonometric identity and the fundamental sine limit. Explain precisely why replacing both sine terms by their first-order equivalents cannot determine this limit.
18. Let f be differentiable at a. Prove that $\mathrm{d}f_a(h + k) = \mathrm{d}f_a(h) + \mathrm{d}f_a(k)$. Give a polynomial example showing that the actual increments based at a need not satisfy this equality. For any fixed $s, t \in \mathbb{R}$, prove that
$$f(a + (s + t)h) - f(a + sh) - f(a + th) + f(a) = o(h) \quad (h \to 0),$$
including $s = 0$, $t = 0$ and $s + t = 0$.
19. Suppose $r(h) = o(h)$ as $h \to 0$. Prove $e^{r(h)} = 1 + o(h)$ and $\ln(1 + r(h)) = o(h)$, checking the logarithm's domain and allowing $r(h) = 0$ at infinitely many increments.
20. For $f(x) = x^2$ and then $f(x) = x^3$, compute the exact linearization remainder $r_a(h) = f(a + h) - f(a) - f'(a)h$. In each case decide whether one $\delta > 0$ can ensure $|r_a(h)| \le \varepsilon\,|h|$ simultaneously for every $a \in \mathbb{R}$ and $|h| < \delta$. What changes if the centres satisfy $|a| \le M$, with $M \ge 0$ fixed?
21. A rectangle's sides change from $2, 3$ cm to $2 + h, 3 - h$ cm, with $|h| < 1$ and h measured in centimetres. Regard its area as a function of the single input h. Find its differential at $h = 0$, the exact area change, and the error of the linear prediction. Repeat the comparison for sides $2 + h, 2 - h$ and explain why a zero differential can coexist with a nonzero area change.

22. Describe all continuous functions $y\colon (-1,1) \to \mathbb{R}$ satisfying $y(x)^2 = x^4$. Show that every such branch is differentiable at 0 with the same differential there, even when the branch signs on the two half-intervals differ.

> ***Optional Hint.*** For $x \neq 0$, the continuous quotient $y(x)/x^2$ takes only the values 1 and -1; use the intermediate value theorem on each half-interval.

23. If f is differentiable at a, find $\lim_{h\to 0} |f(a+h) - f(a)| \,/\, |h|$. Conversely, prove that a value 0 for this limit forces a zero differential at a, whereas a positive finite value need not force any differential.

Supplement Exercises

S1. Let $a \neq 0$ and let f be differentiable at a with $b = f(a) \neq 0$. Prove that the ratio of relative output change to relative input change tends to $K_f = af'(a)/b$. If f has a differentiable local inverse g at b and $f'(a) \neq 0$, prove that the corresponding factor $K_g = bg'(b)/a$ equals $1/K_f$. Check the factors for $f(x) = x^3$ at $a = 2$, and explain why the assumptions about zero values matter.

S2. Construct a function differentiable at 0 whose linearization error is not $O(|h|^{1+\epsilon})$ for any $\epsilon > 0$. State the bound being disproved and verify both differentiability and its failure for every exponent.

> ***Optional Hint.*** On $0 < |x| < e^{-1}$, try $f(x) = x/\ln|x|$ and set $f(0) = 0$. To disprove each bound, test $h_n = e^{-n}$ and use a binomial lower bound for $(e^{\epsilon})^n$.

S3. Suppose f is defined near a. Prove that there is a line $A + B(x-a)$ with error $o(x-a)$ as $x \to a$, $x \neq a$, if and only if assigning $f(a) = A$ makes the resulting function differentiable at a with derivative B. Explain the role of the original value at a.

S4. For $a > 0$ and $a + h > 0$, derive an exact formula for $\sqrt{a+h} - \sqrt{a} - h/(2\sqrt{a})$. Use it to prove a uniform error bound $C_a h^2$ when $|h| \leq a/2$, and specify C_a.

S5. Suppose $f(a+h) = f(a) + Ah + o(h)$ and $g(a+h) = g(a) + Bh + o(h)$. Prove the product rule using these expansions, including why the product of the increments is negligible relative to h.

S6. Let $f'(0) = 0$, $f(0) = 0$, and g be any bounded function near 0. Define $H(0) = 0$ and $H(x) = f(x)g(x)$ for $x \neq 0$. Prove that H admits a zero differential at 0, without assuming continuity of g.

S7. Suppose F has a differentiable local inverse G near $b = F(a)$, and $F'(a) \neq 0$. Starting from the chain rule, explain why a small output perturbation k gives $G(b+k) = a + k/F'(a) + o(k)$. Compare this with the forward model.

S8. The positive branch of $xy + y^2 = 2$ near $(1,1)$ is explicitly $y = (-x + \sqrt{x^2+8})/2$. Compute the linearization at 1 both explicitly and implicitly. Verify that the branch is differentiable before using the equation.

S9. For $x = t^2$, $y = t^4$, $t \in \mathbb{R}$, show that the image is a single graph $y = x^2$ on $x \geq 0$, even though $t \mapsto x$ is not injective near 0. Determine its right derivative at 0 and explain the failure of the formal $0/0$ ratio.

S10. Suppose f is differentiable near a, $f(a) = 0$, $f'(a) \neq 0$, and f' is continuous at a. The tangent at a nearby point x meets the horizontal axis at $N(x) = x - f(x)/f'(x)$. Prove that this expression is defined for all sufficiently near x and that $[N(x) - a]/(x-a) \to 0$ as $x \to a$, $x \neq a$. Interpret this as an improvement from one tangent correction; no convergence claim about repeated corrections is requested.

S11. Let f be differentiable at a, and suppose $u(h) = h + o(h)$. Prove $f(a+u(h)) - f(a+h) = o(h)$ without assuming that f' is continuous, or even defined away from a. Explain why a separate expansion of each value about the fixed centre works, while applying a derivative formula at the moving centre $a+h$ would require unavailable information.

S12. Let F be continuous and strictly monotone on an open interval containing a, with $F(a) = b$. Suppose that for all sufficiently small h, $F(a+h) = b + mh + r(h)$, where $m \neq 0$ and $|r(h)| \leq Ch^2$, $C \geq 0$. For the inverse G, prove that for all sufficiently small k,

$$\left| G(b+k) - a - \frac{k}{m} \right| \leq \frac{4C}{|m|^3} k^2.$$

Check $C = 0$ and $k = 0$ separately, and explain how inverse continuity permits use of the original local remainder estimate.

2.4 Higher Derivatives and Higher Differentials

知识点概要

- ❏ *Distinguish repeated derivatives from continuous derivatives.*
- ❏ *Prove derivative patterns and apply Leibniz's formula.*
- ❏ *Compute implicit and parametric second derivatives.*
- ❏ *Define higher differentials and track substitution terms.*

The slope of a graph generally changes as the base point moves. For a moving particle, the derivative of position is velocity, and the derivative of velocity is acceleration. We therefore apply differentiation to the derivative function itself. This requires more than information about a single tangent.

2.4.1 Repeated differentiation and regularity

Definition 2.4

If f' is defined on a neighbourhood of a and is differentiable at a, the second derivative(二阶导数)*is*

$$f''(a) = (f')'(a) = \lim_{h\to 0} \frac{f'(a+h) - f'(a)}{h}.$$

Recursively, for $n \geq 2$, the nth derivative(高阶导数)*is $f^{(n)} = (f^{(n-1)})'$ wherever this derivative is defined. We use $f^{(0)} = f$ and $f^{(1)} = f'$.*

♣

The notation $f^{(n)}$ means repeated differentiation, not the power f^n. At a point, existence of $f''(a)$ presupposes first derivatives at all nearby points; existence of $f'(a)$ alone does not. The numerator of its defining quotient is a change in *slope*, not a change in the original height. We again fix a and compare that change with the input increment h. By applying the first-order increment representation to the function f', we obtain

$$f'(a+h) - f'(a) = f''(a)h + o(h).$$

This is a first-order model for the *derivative function*. It does not yet assert a second-order remainder formula for f; that is a different statement requiring its own proof. The Leibniz notation is $\mathrm{d}^n y/\mathrm{d}x^n$.

For $s(t) = t^3$, one obtains velocity $s'(t) = 3t^2$ and acceleration $s''(t) = 6t$. If position is measured in metres and time in seconds, their units are metres per second and metres per second squared. For an affine function the second derivative is zero, expressing its constant slope.

On an open interval I, the notation $C^k(I)$ means the functions whose derivatives through order k exist and are continuous. Here $C^0(I)$ means continuous functions, and $C^\infty(I)$ means that this holds for every nonnegative integer k. Existence of a highest derivative does not automatically include its continuity. If derivatives through order k exist throughout an open interval, every derivative of order less than k is continuous there, since it is itself differentiable. The additional requirement in C^k concerns continuity of the highest derivative as well. In particular, the implications $C^2 \implies C^1 \implies C^0$ follow from the definitions, while the converse implications need not hold.

Example 2.30 Two derivatives, but not three For $f(x) = |x|^3$, find f' and f'', and decide whether $f'''(0)$ exists.

Solution *For $x > 0$, $f = x^3$; for $x < 0$, $f = -x^3$. At 0, the quotient $|h|^3/h$ tends to zero. For $h > 0$ it equals h^2, and for $h < 0$ it equals $-h^2$; both tend to zero. Thus $f'(x) = 3x|x|$ on $\mathbb{R}$. This single formula combines $3x^2$ on the positive half-line, $-3x^2$ on the negative half-line, and the separately computed value 0 at the origin.*

At 0, $[f'(h) - f'(0)]/h = 3\,|h| \to 0$. *Away from* 0, *differentiation gives* $f''(x) = 6\,|x|$, *and this formula also holds at* 0. *The quotient for* $f'''(0)$ *is* $6\,|h|\,/h$, *whose side limits are* -6 *and* 6. *Therefore* $f \in C^2(\mathbb{R})$ *but* $f'''(0)$ *does not exist. Membership in* C^2 *also includes the continuity checks:* f, $3x\,|x|$ *and* $6\,|x|$ *are all continuous on* $\mathbb{R}$. *We have not obtained the value of a new derivative at* 0 *by substituting into a formula proved only for* $x \neq 0$*; each order at the joining point was computed by its own difference quotient.*

Example 2.31 A derivative need not be continuous Revisit $f(0) = 0$, $f(x) = x^2 \sin(1/x)$ for $x \neq 0$. Compute f' and decide whether $f''(0)$ exists.

Solution *Example 2.7 established* $f'(0) = 0$ *directly. For* $x \neq 0$, *the rules give* $f'(x) = 2x \sin(1/x) - \cos(1/x)$. *Along* $x_n = 1/(2\pi n)$ *its value is* -1, *so* f' *is not continuous at* 0. *Since differentiability would imply continuity,* f' *cannot be differentiable at* 0: $f''(0)$ *does not exist. This separates differentiability on* $\mathbb{R}$ *from membership in* $C^1(\mathbb{R})$.

This same calculation also refutes the tempting rule "differentiate $o(x)$ to obtain $o(1)$." The function just used satisfies $f(x) = o(x)$ as $x \to 0$, but $f'(1/(2\pi n)) = -1$, so $f'(x)$ does not tend to zero. An estimate of function values controls a different object from the difference quotients needed to differentiate those values.

Example 2.32 Two derivatives can exist without a continuous second derivative Set $g(0) = 0$ and $g(x) = x^4 \sin(1/x)$ for $x \neq 0$. Show that g is twice differentiable everywhere but does not belong to $C^2(\mathbb{R})$.

> ***Idea.*** An extra factor x^2 suppresses enough oscillation for one more derivative at the origin. After differentiating twice away from the origin, check whether a term of fixed amplitude remains. The origin still needs a new difference quotient at each order; a formula valid only for nonzero inputs cannot assign those derivatives.

Solution *At* 0, $|g(h)/h| \leq |h|^3 \to 0$, *so* $g'(0) = 0$. *For* $x \neq 0$, *the product and chain rules give*

$$g'(x) = 4x^3 \sin(1/x) - x^2 \cos(1/x).$$

Consequently

$$\frac{g'(h) - g'(0)}{h} = 4h^2 \sin(1/h) - h \cos(1/h) \longrightarrow 0,$$

so $g''(0) = 0$ *as well. Away from the origin,*

$$g''(x) = 12x^2 \sin(1/x) - 6x \cos(1/x) - \sin(1/x).$$

Along $x_n = (\pi/2 + 2\pi n)^{-1}$ *this tends to* -1, *whereas* $g''(0) = 0$. *Thus* g'' *exists everywhere but is not continuous at* 0. *The first derivative is continuous there, as is also visible from its displayed formula; existence of the next derivative already implies this lower-order continuity.*

The three examples distinguish separate boundaries: $|x|^3$ is C^2 but has no third derivative at 0; $x^2 \sin(1/x)$ is differentiable but not C^1; and the last function is twice differentiable but not C^2. The supplement exercises examine both the exponent threshold in the last construction and the different failure in which a remainder is $o(h^2)$ but no second derivative exists at all.

2.4.2 Patterns and Leibniz's formula

Repeated computation is useful for discovering a pattern, but a few correct derivatives do not prove the formula for all orders. The following formulas are proved by differentiating the claimed expression once and

checking the initial case:

$$\frac{\mathrm{d}^n}{\mathrm{d}x^n} e^{ax} = a^n e^{ax},$$
$$\frac{\mathrm{d}^n}{\mathrm{d}x^n} \sin(ax+b) = a^n \sin(ax+b+n\pi/2),$$
$$\frac{\mathrm{d}^n}{\mathrm{d}x^n} (x-c)^{-1} = (-1)^n n!(x-c)^{-n-1}.$$

The first two hold for $n \geq 1$ and all real parameters; if $a = 0$, the functions are constant. The last holds for $n \geq 0$ and $x \neq c$. For example, differentiating its order-n expression multiplies by $-n-1$, yielding $(-1)^{n+1}(n+1)!(x-c)^{-n-2}$, exactly the next case. The base case $n = 0$ is the original function. For the sine formula, differentiation advances the phase by $\pi/2$ and contributes another factor a; its first case is the chain rule. The exponential formula is verified in the same way.

More generally, repeated use of the power rule on $x > 0$ gives

$$\frac{\mathrm{d}^n}{\mathrm{d}x^n} x^\alpha = \alpha(\alpha-1)\cdots(\alpha-n+1)x^{\alpha-n} \quad (n \geq 1).$$

When α is a nonnegative integer, derivatives above its degree are zero; this polynomial statement holds on all of $\mathbb{R}$, including 0. For $n \geq 1$ and $x > 0$, $(\ln x)^{(n)} = (-1)^{n-1}(n-1)!/x^n$, starting with the first derivative and then using the reciprocal-power pattern.

Theorem 2.5 (Leibniz's Formula)

Let f, g have derivatives through order n on an open interval I, where $n \in \mathbb{N}$. Then

$$(fg)^{(n)}(x) = \sum_{k=0}^{n} \binom{n}{k} f^{(k)}(x) g^{(n-k)}(x), \qquad x \in I.$$

♡

Each of the n differentiations acts on either the first factor or the second. The coefficient counts the ways of choosing which k differentiations act on f. For $n = 2$, the formula is $f''g + 2f'g' + fg''$.

> ***Proof Strategy.*** After one more differentiation, a term contributes once by differentiating its f factor and once by differentiating its g factor. Terms with the same final orders then combine by Pascal's identity. This explains the binomial coefficients instead of asking us to memorize them.

Proof The case $n = 0$ is $fg = fg$. Assume the order-n identity holds for functions with n derivatives, and take f, g with $n+1$ derivatives. Differentiate the finite sum using the product rule:

$$(fg)^{(n+1)} = \sum_{k=0}^{n} \binom{n}{k} f^{(k+1)} g^{(n-k)} + \sum_{k=0}^{n} \binom{n}{k} f^{(k)} g^{(n+1-k)}.$$

The extreme terms are $f^{(n+1)}g$ and $fg^{(n+1)}$. To see the indexing explicitly, set $j = k+1$ in the first sum and $j = k$ in the second. They become

$$\sum_{j=1}^{n+1} \binom{n}{j-1} f^{(j)} g^{(n+1-j)} \quad \text{and} \quad \sum_{j=0}^{n} \binom{n}{j} f^{(j)} g^{(n+1-j)}.$$

The $j = n+1$ term appears only in the first sum, and the $j = 0$ term only in the second; both coefficients are 1. For $1 \leq j \leq n$, the coefficient of $f^{(j)}g^{(n+1-j)}$ is $\binom{n}{j-1} + \binom{n}{j} = \binom{n+1}{j}$. This gives the order-$n+1$ identity and completes the induction. Continuity of the highest derivatives is not required. □

Example 2.33 A product whose higher derivatives truncate Find $(x^2 e^x)^{(n)}$ for every $n \geq 0$.

> ***Strategy.*** Only the first three derivative orders of the polynomial factor can contribute. Alternatively, differentiation preserves the form e^x times a quadratic, so we can track just that quadratic.

Solution ***Method 1: Leibniz's formula.*** *For $n \geq 2$, only $k = 0, 1, 2$ survive, giving*

$$\binom{n}{0}x^2e^x + \binom{n}{1}(2x)e^x + \binom{n}{2}(2)e^x.$$

The last coefficient is $2\binom{n}{2} = n(n-1)$, so

$$(x^2e^x)^{(n)} = e^x\big(x^2 + 2nx + n(n-1)\big).$$

Direct checking shows the same expression for $n = 0, 1$. Indeed, these cases give x^2e^x and $e^x(x^2 + 2x)$, respectively. Treating them separately avoids referring to polynomial derivative orders that are absent from their shorter Leibniz sums.

Solution ***Method 2: an invariant form and induction.*** *At $n = 0$ the formula is correct. If the order-n polynomial is $P_n = x^2 + 2nx + n(n-1)$, differentiating e^xP_n gives $e^x(P_n + P_n')$. But $P_n + P_n' = x^2 + 2(n+1)x + n(n+1) = P_{n+1}$. Thus the expression holds for all n.*

Leibniz is direct when one factor eventually vanishes under differentiation. The second method reveals a family of forms preserved by the operation, and can be useful even before a general product formula is recognized.

Example 2.34 Simplify before repeating a quotient rule Find the nth derivative of $R(x) = 1/(1-x^2)$ for $x \neq \pm 1$ and $n \geq 0$.

Solution *The decomposition $R(x) = \frac{1}{2}[(1-x)^{-1} + (1+x)^{-1}]$ reduces the problem to two reciprocal linear factors. The power and chain rules give*

$$R^{(n)}(x) = \frac{n!}{2}\left(\frac{1}{(1-x)^{n+1}} + \frac{(-1)^n}{(1+x)^{n+1}}\right).$$

The original poles remain excluded. Repeated quotient differentiation would be valid but would obscure this simple pattern.

Example 2.35 Find values without finding the whole derivative For $y = \arctan x$, determine $y^{(n)}(0)$ for every $n \geq 1$ without using a power series.

> ***Idea.*** An explicit formula for every derivative at every point is more than the problem asks. The elementary relation $(1+x^2)y' = 1$ closes under differentiation because the polynomial factor has only two nonzero higher derivative orders. At 0 the relation becomes a short recurrence.

Solution *The formula $y' = 1/(1+x^2)$ shows by repeated rules that y has derivatives of all orders on $\mathbb{R}$. Differentiating once gives $(1+x^2)y'' + 2xy' = 0$, so $y''(0) = 0$. For $n \geq 2$, apply Leibniz to the order-n derivative of $(1+x^2)y' = 1$:*

$$(1+x^2)y^{(n+1)} + 2nx\,y^{(n)} + n(n-1)y^{(n-1)} = 0.$$

At 0 this yields $y^{(n+1)}(0) = -n(n-1)y^{(n-1)}(0)$. Starting from $y'(0) = 1$ and $y''(0) = 0$, induction gives the first few values

$$y'''(0) = -2, \quad y^{(4)}(0) = 0, \quad y^{(5)}(0) = 24, \quad y^{(6)}(0) = 0.$$

The recurrence changes the derivative order by two, so the odd and even orders form separate chains. The even chain starts at zero and remains zero. For the odd chain, if $y^{(2m-1)}(0) = (-1)^{m-1}(2m-2)!$ for $m \geq 1$, the recurrence with $n = 2m$ gives

$$y^{(2m+1)}(0) = -(2m)(2m-1)(-1)^{m-1}(2m-2)! = (-1)^m(2m)!.$$

Together with the initial value $y'(0) = 1$, this proves

$$y^{(2m)}(0) = 0 \quad (m \geq 1), \qquad y^{(2m+1)}(0) = (-1)^m(2m)! \quad (m \geq 0).$$

The recurrence establishes the result without a Taylor expansion.

2.4.3 Implicit and parametric second derivatives

Differentiating an implicit identity again requires differentiating every occurrence of the dependent function. For a parametric curve, the second derivative with respect to x is obtained by applying $(1/X'(t))\mathrm{d}/\mathrm{d}t$ to the first derivative. Indeed, if a quantity along the curve is written $Q(t)$, then as a function of x it is $Q(t(x))$. The inverse and chain rules give $\mathrm{d}Q/\mathrm{d}x = Q'(t)/X'(t)$. For the second derivative we apply this to the whole first-derivative expression $Q(t) = Y'(t)/X'(t)$, rather than to Y alone. Under a local invertibility assumption on X, and with twice differentiable coordinates and $X' \neq 0$, this gives

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \frac{1}{X'}\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{Y'}{X'}\right) = \frac{Y''X' - Y'X''}{(X')^3}.$$

It is generally not Y''/X''. The quotient rule in the intermediate step gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{Y'}{X'}\right) = \frac{Y''X' - Y'X''}{(X')^2};$$

the additional division by X' changes the denominator to $(X')^3$. If X' is nonzero throughout the local parameter interval, the inverse rule applies at every point there and justifies this second use of the chain rule.

Example 2.36 Second derivatives from two descriptions Find y'' for the upper circle branch $y = \sqrt{4 - x^2}$, $-2 < x < 2$, and for $x = t^2$, $y = t^3$, $t > 0$. Check each answer by its alternative description.

Solution ***Circle: implicit calculation.*** *The branch is twice differentiable by the elementary rules and its positive radicand. Differentiating $x^2+y^2 = 4$ twice gives $2+2(y')^2+2yy'' = 0$. The first differentiation is $2x+2yy' = 0$. In the second, y and y' both depend on x, so $(yy')' = (y')^2 + yy''$ by the product rule. Since $y' = -x/y$ and $y > 0$,*

$$y'' = -\frac{1 + (x/y)^2}{y} = -\frac{4}{y^3}.$$

Circle: explicit check. *Differentiating $-x(4-x^2)^{-1/2}$ gives $-(4-x^2)^{-1/2}-x^2(4-x^2)^{-3/2} = -4/(4-x^2)^{3/2}$, the same answer.*

Solution ***Parameter calculation.*** *The first derivative is $3t/2$. Differentiating it with respect to t and dividing by $2t$ gives $y'' = 3/(4t)$. The intermediate derivative $3/2$ is the rate of change of the slope per unit t; division by $2t = \mathrm{d}x/\mathrm{d}t$ converts it to change of slope per unit x.* ***Elimination check.*** *Since $y = x^{3/2}$ for $x > 0$, the second derivative is $3/(4\sqrt{x}) = 3/(4t)$. The condition $t > 0$ both selects the branch and makes the division valid.*

The implicit description retains the geometry of the constraint. Elimination is an efficient check when a simple explicit branch is available. The parameter calculation keeps track of the variable with respect to which the derivative is being taken.

2.4.4 Higher differentials

For a function $y = f(x)$ of an independent variable x, the first differential is $f'(x)\mathrm{d}x$. To repeat the operation, we hold the increment $\mathrm{d}x$ constant while differentiating the coefficient with respect to x. One can make this convention concrete by choosing a real increment h once and writing $\mathrm{d}x = h$. The function of the base point is then $x \mapsto f'(x)h$. Its differential for that same fixed increment is

$$\big(f'(x)h\big)'h = f''(x)h^2.$$

Thus the second differential measures the first-order change of the first-differential coefficient as the base point moves, with the chosen input increment held fixed.

Definition 2.5

For an independent variable x and an n-times differentiable function f, the nth differential(高阶微分) is

$$\mathrm{d}^n y = f^{(n)}(x)(\mathrm{d}x)^n.$$

It is obtained recursively by applying d to the preceding expression, keeping $\mathrm{d}x$ fixed. ♣

For example, $y = x^3$ gives $\mathrm{d}y = 3x^2\mathrm{d}x$, $\mathrm{d}^2y = 6x(\mathrm{d}x)^2$, and $\mathrm{d}^3y = 6(\mathrm{d}x)^3$. In this convention $\mathrm{d}^2x = 0$, whereas $(\mathrm{d}x)^2$ need not be zero. The former differentiates $\mathrm{d}x = h$, which is held constant; the latter is the ordinary square h^2 of that chosen increment. Neither is the finite second difference of a function.

Example 2.37 Three different second-order objects For $f(x) = x^3$ at the base point $a = 1$, with $\mathrm{d}x = h$, compute d^2y, $(\mathrm{d}y)^2$ and the forward second difference

$$\Delta_h^2 f(1) = f(1+2h) - 2f(1+h) + f(1).$$

Explain what each operation does.

Solution *Since $f'(1) = 3$ and $f''(1) = 6$, the differentials give*

$$\mathrm{d}y = 3h, \qquad \mathrm{d}^2y = 6h^2, \qquad (\mathrm{d}y)^2 = 9h^2.$$

The notation d^2y means applying the differential operation twice with its fixed-increment convention; squaring $\mathrm{d}y$ instead squares the first derivative's coefficient as well. For the finite difference, evaluate the original function at three points:

$$\begin{aligned}\Delta_h^2 f(1) &= (1 + 6h + 12h^2 + 8h^3) - 2(1 + 3h + 3h^2 + h^3) + 1 \\ &= 6h^2 + 6h^3.\end{aligned}$$

It differs from d^2y by $6h^3$, although after division by h^2 it tends to the same value 6 as $h \to 0$. This conclusion was obtained from the exact cubic identity, not from an assumed general finite-difference theorem. Likewise, the exact first increment is $3h + 3h^2 + h^3$: its quadratic term is $\frac{1}{2}\mathrm{d}^2y$, not d^2y. General polynomial approximation formulas and their remainder conditions will be established later.

Suppose now that $y = f(u)$, $u = g(x)$, with both functions twice differentiable on the relevant open intervals and x independent. Differentiate $\mathrm{d}y = f'(u)\mathrm{d}u$ once more:

$$\mathrm{d}^2y = f''(u)(\mathrm{d}u)^2 + f'(u)\mathrm{d}^2u, \qquad \mathrm{d}^2u = g''(x)(\mathrm{d}x)^2. \tag{2.2}$$

Indeed, first differentiate $f(g(x))$ to get $f'(g(x))g'(x)$. Differentiating this product once more gives

$$(f \circ g)''(x) = \underbrace{f''(g(x))g'(x)}_{\text{derivative of } f'(g(x))}\, g'(x) + f'(g(x))g''(x).$$

Multiplication by $(\mathrm{d}x)^2$, together with $(\mathrm{d}u)^2 = (g'(x))^2(\mathrm{d}x)^2$ and $\mathrm{d}^2u = g''(x)(\mathrm{d}x)^2$, proves (2.2). Unlike the independent increment $\mathrm{d}x$, the coefficient of $\mathrm{d}u = g'(x)\mathrm{d}x$ varies with x. The extra term is why the simple first-differential invariance does not extend unchanged to order two. For an affine substitution $g(x) = ax + b$, $g'' = 0$ and that particular extra term vanishes.

Example 2.38 The missing second-order term Take $y = u^2$, $u = x^2$. Compute d^2y directly in x, and then by the substitution formula.

Solution ***Method 1: direct differentiation.*** *Since $y = x^4$, $\mathrm{d}^2y = 12x^2(\mathrm{d}x)^2$.*

Solution ***Method 2: preserve both terms.*** *Here $\mathrm{d}u = 2x\mathrm{d}x$ and $\mathrm{d}^2u = 2(\mathrm{d}x)^2$. Thus*

$$\mathrm{d}^2y = 2(\mathrm{d}u)^2 + 2u\,\mathrm{d}^2u = 8x^2(\mathrm{d}x)^2 + 4x^2(\mathrm{d}x)^2.$$

This agrees with the direct answer. Keeping only $2(\mathrm{d}u)^2$ loses the changing rate of u itself. For example, at

$x = 1$ *the incomplete expression would be* $8(\mathrm{d}x)^2$*, while the correct value is* $12(\mathrm{d}x)^2$*. At* $x = 0$ *both happen to vanish, so testing only that point would fail to reveal the omitted term.*

The direct method is a useful check. The two-term expression explains structurally where the additional contribution comes from.

2.4.5 Methods and further remarks

For high-order calculations, inspect the expression before differentiating repeatedly. A polynomial factor suggests a truncated Leibniz sum. A rational expression with simple linear factors suggests decomposition. A stable family such as an exponential times a polynomial suggests induction within that family. If only values at one point are requested, a low-order differential identity may yield a recurrence more efficiently than a global formula.

At a special point, use the recursive definition to compute each new derivative. Do not substitute that point into a formula proved only off the point. For parameter derivatives, write the operator $(1/X')\mathrm{d}/\mathrm{d}t$ explicitly. For higher differentials, identify which variable is independent before treating an increment as constant. Distinguish differentiation of a quantity from squaring it, and distinguish both operations from evaluating the original function at several separated points. An estimate such as $r(h) = o(h)$ describes function values on a specified scale; it does not permit differentiating the remainder without additional hypotheses and a new argument. The explicit derivative in Example 2.31 shows this failure, and Example 2.32 shows why existence and continuity of the highest derivative must be checked separately.

> ***Looking Ahead.*** Higher derivatives provide coefficients for more accurate local models. A formula involving those coefficients does not yet come with a remainder estimate. Taylor's theorem will later explain when the corresponding polynomial approximates the function and how its error is controlled.

Exercise 2.4

1. For $f(x) = x^5 - 2x^2 + 1$, find f'', $f^{(5)}$, and $f^{(6)}$.
2. Find the 2026th derivative of $\sin x$ and the nth derivative of $\cos(3x)$ for $n \geq 1$.
3. Prove by induction that $(e^{-2x})^{(n)} = (-2)^n e^{-2x}$ for $n \geq 0$.
4. Find $(1/(2-x))^{(n)}$ for $x \neq 2$ and $n \geq 0$; check the sign at the first two orders.
5. Find d^2y and d^3y for $y = \ln x$, $x > 0$, with x independent.
6. A particle has position $s(t) = t^4 - 2t^2$ metres. Find velocity and acceleration, including their units.
7. Find $(x \sin x)^{(n)}$ for $n \geq 1$ and explain why the Leibniz sum has only two terms.
8. Find $(1/(x^2-4))^{(n)}$ for $x \neq \pm 2$, $n \geq 0$, by decomposition into linear factors.
9. For $x = t^3$, $y = t^2$, $t > 0$, compute $\mathrm{d}^2y/\mathrm{d}x^2$ in two ways.
10. If a twice differentiable branch satisfies $y^2 + xy = 2$, derive a formula for y'' where $2y + x \neq 0$. Evaluate it at $(1, 1)$.
11. For $y = \sin u$, $u = x^2$, compute d^2y using both terms in (2.2).
12. Show that $f(x) = x\,|x|$ is C^1 on $\mathbb{R}$ but has no second derivative at 0.
13. Let $y = e^x \cos x$. Verify that $y'' - 2y' + 2y = 0$ and use it to give a recurrence for $y^{(n)}(0)$.
14. Find A, B, C so that $f(x) = Ax^2 + Bx + C$ for $x \leq 0$ and $f(x) = e^x + \sin x$ for $x > 0$ is C^2 on $\mathbb{R}$.
15. Prove that existence of $f''(a)$ implies continuity of f' at a. Give an example showing that continuity of f' at a does not imply existence of $f''(a)$.

16. For $f(x) = x\,|x|$, compute $[f(h) - 2f(0) + f(-h)]/h^2$ and compare with existence of $f''(0)$. Explain why the symmetric second quotient cannot define the ordinary second derivative.
17. Suppose f' is defined near a and, for $0 < |h| < \rho$, satisfies $f'(a+h) = f'(a) + Lh + h^2\cos(1/h)$. Find $f''(a)$ by its definition and identify which function is being linearly approximated.
18. Let f, g be twice differentiable near a, with $f(a) = 1$, $f'(a) = 2$, $f''(a) = 3$, $g(a) = 4$, $g'(a) = -1$, $g''(a) = 2$. Find $(fg)''(a)$ and explain the coefficient of the mixed term.
19. For a twice differentiable position $s(t)$ and constant $c > 0$, let $S(u) = s(cu)$. Find $S''(u)$ in terms of s'', and explain the factor c^2 by applying the chain rule twice. In particular, compare acceleration in metres per second squared with metres per minute squared.
20. Suppose f is even and has derivatives through order k on $\mathbb{R}$. Prove that $f^{(j)}$ is even for even j and odd for odd j, $0 \le j \le k$, and deduce which derivatives at 0 vanish. For any given integer $k \ge 1$, construct a polynomial that is not even but has every odd-order derivative through order k equal to zero at 0.
21. For $x = t + t^3$, $y = (t + t^3)^2$, $t \in \mathbb{R}$, justify a local inverse for the horizontal coordinate and find $\mathrm{d}^2y/\mathrm{d}x^2$. Verify the result both from the parameter formula and by eliminating the parameter. Why is $Y''(0)/X''(0)$ unusable even though the correct second derivative exists?

Supplement Exercises

S1. For $m \in \mathbb{N}_{>0}$, find the largest k such that $f(x) = |x|^{2m+1}$ belongs to $C^k(\mathbb{R})$. Prove the endpoint order by the recursive definition.
S2. For $\alpha \ge 4$, set $f_\alpha(0) = 0$ and $f_\alpha(x) = |x|^\alpha \sin(1/x)$ for $x \ne 0$. Prove that $f_\alpha''(0) = 0$ and determine exactly which parameters give continuity of f_α'' at 0. Recover Example 2.32 at the threshold and justify the two-sided estimates for noninteger α.
S3. Find $(x^m e^{ax})^{(n)}$ for $m, n \in \mathbb{N}$ and $a \in \mathbb{R}$. State a formula that also covers $a = 0$ without an undefined power 0^0.
S4. Derive the third derivative of $f(g(x))$ assuming both functions are three times differentiable. Interpret the three groups of terms and check with $f(u) = u^2$, $g(x) = x^2$.
S5. Suppose f is twice differentiable and has a twice differentiable inverse g near $b = f(a)$, with $f'(a) \ne 0$. Derive $g''(b)$ in terms of $f'(a)$ and $f''(a)$.
S6. For $y = (\arcsin x)^2$ on $(-1, 1)$, prove $(1 - x^2)y'' - xy' = 2$ by direct differentiation, without dividing by y' at 0. Use Leibniz's formula to compute all derivatives at 0.
S7. Let P be a polynomial of degree at most m. Prove $(P(x)e^x)^{(n)} = e^x \sum_{k=0}^{\min(m,n)} \binom{n}{k} P^{(k)}(x)$. Explain what this says about the dependence on n for fixed x.
S8. For a three-times differentiable parametrization with a local inverse for X and $X' \ne 0$ on the interval, derive $\mathrm{d}^3y/\mathrm{d}x^3$ by applying $(1/X')\mathrm{d}/\mathrm{d}t$ to the second-derivative formula.
S9. Let $y = f(u)$, $u = ax + b$, where a, b are constants and f has n derivatives. Prove $\mathrm{d}^n y = f^{(n)}(u)(\mathrm{d}u)^n$ for independent x and $n \ge 1$, including $a = 0$. Explain why a nonlinear substitution generally needs extra terms.
S10. Set $f(0) = 0$ and $f(x) = x^3 \sin(1/x)$ for $x \ne 0$. Determine whether $f''(0)$ exists. Compare with the approximation $f(h) = o(h^2)$ and explain why a small function remainder alone does not license differentiating that remainder twice.
S11. For each real $\alpha > 0$, determine whether $f_\alpha(x) = |x|^\alpha$ has a second derivative at 0 and whether it belongs to $C^2(\mathbb{R})$. Treat $\alpha = 1$ and $\alpha = 2$ explicitly, using the recursive derivative definition at 0 rather than substitution into a formula proved only for $x \ne 0$.

S12. Let g be the inverse on $\mathbb{R}$ of $f(x) = x + x^3$. Justify that g has at least five continuous derivatives, and compute $g^{(j)}(0)$ for $1 \leq j \leq 5$ without a Taylor expansion.

> ***Optional Hint.*** Use $g(y) + g(y)^3 = y$. Oddness removes the even derivatives at 0; repeated product differentiation finds the remaining ones without solving the cubic.

2.5 Antiderivatives and Indefinite Integrals

知识点概要

- ❑ *Recover a function from its derivative.*
- ❑ *Distinguish a primitive from the full family.*
- ❑ *Track intervals and arbitrary constants.*
- ❑ *Recognise algebraic forms and derivatives.*
- ❑ *Use initial data and check piecewise joins.*
- ❑ *Separate existence, verification and uniqueness.*

Differentiation starts with a quantity and finds its rate of change. A different problem arises when a measuring device records velocity: can we recover position? Even perfect knowledge of velocity cannot tell us where the particle started. Thus reversing differentiation should produce a family of functions, with extra information needed to select one member. The higher derivatives of the previous section lead to the same question: acceleration must be reversed twice to recover position.

2.5.1 Recovering a function from its derivative

Definition 2.6

Let I be a nonempty open interval and let $f\colon I \to \mathbb{R}$. A differentiable function $F\colon I \to \mathbb{R}$ is an antiderivative(原函数) *of f on I if*

$$F'(x) = f(x) \qquad \text{for every } x \in I.$$

♣

This is an identity on an interval, not a condition at just one point. The function whose derivative is prescribed is F; the prescribed derivative is f. For example, x^2 is an antiderivative of $2x$ on $\mathbb{R}$, as are $x^2 + 1$ and $x^2 - 5$. The derivative of a constant is zero, so adding any real constant to an antiderivative gives another one. In contrast, $x^2 + x$ is not an antiderivative of $2x$: its derivative is $2x + 1$. For a closed interval, we will explicitly require continuity to its endpoints and the derivative identity in its interior; endpoint derivative claims, when needed, will be one-sided.

Example 2.39 A rate and one position Find a differentiable position function $s\colon \mathbb{R} \to \mathbb{R}$ satisfying $s'(t) = 2t$ and $s(1) = 3$.

Solution *Since $(t^2)' = 2t$, try $s(t) = t^2 + C$. The condition at $t = 1$ gives $1 + C = 3$, so $s(t) = t^2 + 2$. Its derivative and initial value verify both requirements. This proves that an answer exists; the next subsection will prove that there are no other answers.*

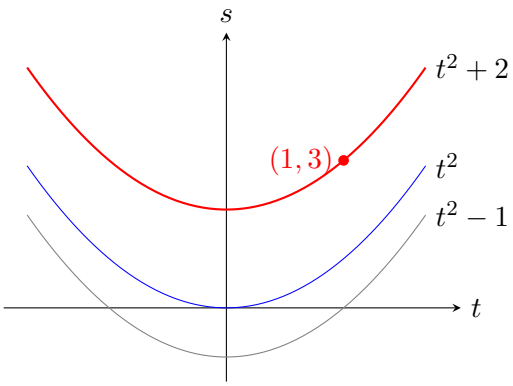


Figure 2.5: Vertical translations have the same slope at each input. A prescribed position selects one translation.

The graphs in **Figure** 2.5 explain why a constant must be allowed. To show that this accounts for *all* possibilities, however, we need a global statement: a function with zero derivative everywhere on an interval is constant there. A zero derivative at only one point does not suffice, as x^2 at 0 already shows.

2.5.2 Why the constant is the only freedom

The mean value theorem will later supply a general relation between derivatives and endpoint changes. Here we prove just the consequence needed for antiderivatives, using the extreme value theorem already established in Section 1.6.

Lemma 2.1

If H is differentiable on an open interval I and $H' = 0$ on I, then H is constant on I.

♡

> ***Proof Strategy.*** If two values increased, subtracting a line of small positive slope would preserve that increase between the endpoints, while making the derivative strictly negative everywhere. Such a function initially moves below its left endpoint value, yet ends above it. Its minimum on the segment must therefore occur inside, where the two-sided difference quotient forces the derivative to be zero. This is the contradiction we want; the small linear tilt creates an interior extreme point.

Proof First note that a differentiable function q at an interior local minimum c has $q'(c) = 0$. For small $h > 0$, the quotient $[q(c + h) - q(c)]/h$ is nonnegative; for small $h < 0$ it is nonpositive. If the two-sided derivative exists, its value is therefore both nonnegative and nonpositive.

Take any $a < b$ in I. Suppose first that $H(b) > H(a)$, and choose the explicit positive slope

$$\lambda = \frac{H(b) - H(a)}{2(b - a)}.$$

Set $q(x) = H(x) - \lambda(x - a)$ on $[a, b]$. Then $q(b) - q(a) = [H(b) - H(a)]/2 > 0$ and $q'(x) = -\lambda$. The difference quotient at a tends to $-\lambda$, so there is $\delta > 0$ such that

$$0 < h < \min\{\delta, b - a\} \quad \Longrightarrow \quad \frac{q(a + h) - q(a)}{h} < -\frac{\lambda}{2} < 0.$$

Multiplication by the positive h gives $q(a + h) < q(a) < q(b)$. This uses only the anchored quotient at a, not a theorem about monotonicity on the whole interval. The continuous function q attains a minimum on $[a, b]$ by Theorem 1.26. Neither endpoint can be a minimizer, so a minimizer c lies in (a, b). The first paragraph gives $q'(c) = 0$, contradicting $q'(c) = -\lambda$. If $H(b) < H(a)$, apply the proved impossibility of an increase to $-H$. Thus $H(a) = H(b)$ for every pair in I, which is constancy. □

Theorem 2.6

If F is one antiderivative of f on an open interval I, then all antiderivatives of f on I are exactly

$$G(x) = F(x) + C, \qquad C \in \mathbb{R}.$$

♡

Proof Every $F + C$ has derivative f. Conversely, if $G' = f$, the derivative rules give $(G - F)' = 0$ on I. Apply Lemma 2.1 to $G - F$. □

Definition 2.7

When f has an antiderivative F on an interval I, its indefinite integral(不定积分) *denotes the family of all its antiderivatives, written*

$$\int f(x)\,\mathrm{d}x = F(x) + C, \qquad C \in \mathbb{R}.$$

The function f is the integrand(被积函数)*; x is the variable of integration.* ♣

The symbol C is a number independent of x, and ranges over all real numbers. It is not an unspecified function of x. For example, $\int 2x\,\mathrm{d}x = x^2 + C$ describes infinitely many functions, while $x^2 + 2$ is one member. Consequently, prescribing $F(x_0) = y_0$ at one point $x_0 \in I$ gives the unique member $F(x) + y_0 - F(x_0)$. This completes the uniqueness claim in Example 2.39. The notation is suggestive of an inverse operation, but differentiation loses a constant:

$$\frac{\mathrm{d}}{\mathrm{d}x}(F(x) + C) = f(x), \qquad \int F'(x)\,\mathrm{d}x = F(x) + C.$$

Example 2.40 Two intervals, two constants Describe all antiderivatives of $1/x$ on $\mathbb{R} \setminus \{0\}$, with differentiability required at every point of this domain.

Solution *On $x > 0$, $(\ln x)' = 1/x$. On $x < 0$, the chain rule gives $(\ln(-x))' = (-1)/(-x) = 1/x$. Thus all the answers are*

$$F(x) = \begin{cases} \ln(-x) + C_-, & x < 0, \\ \ln x + C_+, & x > 0, \end{cases} \qquad C_-, C_+ \in \mathbb{R}.$$

The theorem applies separately to each interval, and there is no point of the domain joining them. For instance, $C_- = 0$, $C_+ = 1$ gives a valid answer that cannot be written with one common constant relative to $\ln|x|$ on the entire punctured line.

Accordingly, the usual formula $\int x^{-1}\,\mathrm{d}x = \ln|x| + C$ is interpreted on an interval avoiding 0. An initial value on the positive half-line does not determine the negative-half-line constant.

2.5.3 Basic antiderivatives and linearity

Reversing the derivative formulas of Section 2.2 gives the following table. Each row specifies one antiderivative; add C to describe the family on any interval inside the stated domain.

Integrand	One antiderivative	Domain or condition
x^α	$x^{\alpha+1}/(\alpha+1)$	$x > 0$, $\alpha \neq -1$
$1/x$	$\ln\lvert x\rvert$	$x \neq 0$
e^x	e^x	$x \in \mathbb{R}$
a^x	$a^x/\ln a$	$a > 0$, $a \neq 1$
$\cos x$	$\sin x$	$x \in \mathbb{R}$
$\sin x$	$-\cos x$	$x \in \mathbb{R}$
$\sec^2 x$	$\tan x$	$\cos x \neq 0$
$\csc^2 x$	$-\cot x$	$\sin x \neq 0$
$\sec x \tan x$	$\sec x$	$\cos x \neq 0$
$\csc x \cot x$	$-\csc x$	$\sin x \neq 0$
$1/(1+x^2)$	$\arctan x$	$x \in \mathbb{R}$
$1/\sqrt{1-x^2}$	$\arcsin x$	$-1 < x < 1$

The power rule is valid on larger real domains in particular cases. For a nonnegative integer exponent it holds on all of $\mathbb{R}$; for a negative integer exponent it holds away from 0, with exponent -1 handled by the logarithm. Other real powers require the branch conventions established earlier, so $x > 0$ is the uniform safe domain for arbitrary real α. For the excluded exponential parameter $a = 1$, the integrand is 1 and an antiderivative is x. The case $\alpha = -1$ cannot be obtained by dividing by $\alpha + 1 = 0$. All the table entries are checked by differentiation, including the minus signs in the sine and cotangent rows. No theorem about a definite integral is being used here.

Proposition 2.3

Suppose $F' = f$ and $G' = g$ on the same open interval I, and let $\alpha, \beta \in \mathbb{R}$. Then $\alpha F + \beta G$ is an antiderivative of $\alpha f + \beta g$, and

$$\int (\alpha f(x) + \beta g(x))\,\mathrm{d}x = \alpha F(x) + \beta G(x) + C.$$

♠

Proof The derivative of $\alpha F + \beta G$ is $\alpha f + \beta g$ by the derivative rules. The family theorem supplies all remaining answers. This includes $\alpha = \beta = 0$, when the answer is the family of constants. □

Writing the answer in terms of chosen representatives avoids a small notational trap: multiplying the entire family $F + C$ by zero produces only the zero function, whereas $\int 0\,\mathrm{d}x$ is the family of *all* constants. Linearity of candidates and a final arbitrary constant give the intended meaning even in degenerate cases. There is no corresponding rule obtained by simply multiplying antiderivatives. For example, the product of the candidates x and x for 1 and 1 is x^2, whose derivative is $2x$, not 1.

Example 2.41 Different formulas for the same family Compare $\arcsin x + C$ and $-\arccos x + C$ on $(-1, 1)$, and $\ln(3x) + C$ and $\ln x + C$ on $(0, +\infty)$.

Solution ***Method 1: identities.*** *Since $\arcsin x + \arccos x = \pi/2$ on the principal branches, the first two representatives differ by $\pi/2$. Since $\ln(3x) = \ln 3 + \ln x$, the logarithmic representatives differ by $\ln 3$. Allowing an arbitrary constant therefore gives the same family in each comparison; the value denoted by C need not be the same in two displays.*

> ***An alternative test.*** A complicated identity need not be guessed first. If the derivatives agree on an interval, their difference has already been forced to be constant; one convenient input determines that constant.

Solution ***Method 2: derivatives and a value.*** *The derivatives of $\arcsin x$ and $-\arccos x$ are both $1/\sqrt{1 - x^2}$ on $(-1, 1)$. Their difference is constant by the lemma, and at 0 it is $\pi/2$. The derivatives of $\ln(3x)$ and $\ln x$ are both $1/x$ for $x > 0$; at 1 their difference is $\ln 3$. This recovers both comparisons without first using the identities.*

The identity route is shortest when the identity is familiar. The derivative route is more transferable, but its interval and branch hypotheses are essential.

2.5.4 Recognising a derivative and imposing data

Basic integration is often a problem of representation: can the integrand be rewritten as a linear combination of derivatives we know? Before reaching for a new technique, inspect factors, quotients, and elementary identities.

Example 2.42 Simplification does not erase the original domain Find all antiderivatives of $(x^3+1)/(x+1)$ on its domain.

> ***Idea.*** The numerator vanishes at -1, which suggests a factor $x+1$. Dividing removes the complicated-looking quotient; the removed point must still be remembered when describing all answers.

Solution *For $x \neq -1$,*

$$\frac{x^3+1}{x+1} = x^2 - x + 1.$$

Thus a representative is $P(x) = x^3/3 - x^2/2 + x$, and differentiation checks the identity. All antiderivatives on the original domain are $P + C_-$ on $(-\infty, -1)$ and $P + C_+$ on $(-1, +\infty)$, with independent constants. If instead the integrand is extended to the polynomial value 3 at -1 and an antiderivative is required on all of $\mathbb{R}$, only one constant is allowed. These are different domain specifications.

Example 2.43 A product with two useful representations Evaluate $\int \sin x \cos x \, \mathrm{d}x$ on $\mathbb{R}$ by two methods and compare the results.

> ***Idea.*** One factor is the derivative of the other, so differentiating a square may recreate the product. Alternatively, a product of sine and cosine is a single double-angle sine, which belongs to the basic table after accounting for its inner derivative.

Solution ***Method 1: recognise a derivative.*** *The chain rule gives $(\frac{1}{2}\sin^2 x)' = \sin x \cos x$. Hence the answer is $\frac{1}{2}\sin^2 x + C$.*

Solution ***Method 2: change the trigonometric form.*** *Since $\sin x \cos x = \frac{1}{2}\sin(2x)$ and $(-\frac{1}{4}\cos(2x))' = \frac{1}{2}\sin(2x)$, the answer is $-\frac{1}{4}\cos(2x) + C$. The identity $\sin^2 x = (1 - \cos(2x))/2$ shows that the first representative exceeds the second by $1/4$. Both computations are valid on all of $\mathbb{R}$.*

The first method uses the paired factor and its derivative; the second uses frequency reduction. Neither invokes the substitution theorem, which will be developed systematically in the next chapter. Here the proposed answers are justified directly by the chain rule.

Example 2.44 Joining primitives across a change of formula Let $f(x) = x$ for $x < 0$ and $f(x) = 2x$ for $x \geq 0$. Find an antiderivative F on $\mathbb{R}$ satisfying $F(0) = 1$.

> ***Idea.*** On each open half-line the table finds the candidates, but differentiability at the join is an additional requirement. First align their heights using continuity; then inspect the difference quotient at the join itself.

Solution ***Method 1: match the pieces.*** *The half-line formulas are $x^2/2 + C_-$ and $x^2 + C_+$. Continuity at 0 and $F(0) = 1$ require $C_- = C_+ = 1$. For the resulting function,*

$$\frac{F(h) - F(0)}{h} = \begin{cases} h/2, & h < 0, \\ h, & h > 0, \end{cases}$$

which tends to $0 = f(0)$. The derivative identity on both half-lines follows from the table, so this is an antiderivative on all of $\mathbb{R}$. Its uniqueness follows from Theorem 2.6.

Solution ***Method 2: verify one global formula.*** *Consider*

$$F(x) = 1 + \frac{3}{4}x^2 + \frac{1}{4}x\,|x|\,.$$

For $x < 0$ this reduces to $1 + x^2/2$, and for $x > 0$ to $1 + x^2$, giving the prescribed derivatives. At 0, its difference quotient is $3h/4 + |h|/4$, which tends to 0. Also $F(0) = 1$. Thus the single formula meets all requirements.

The piecewise method discovers the matching constants naturally. The global formula packages the same answer compactly, but it does not remove the need to verify differentiability where the absolute value changes form.

Example 2.45 Two reversals require two constants Suppose $s''(t) = 6t$ on $\mathbb{R}$, $s(0) = 2$, and $s'(0) = -1$. Determine s.

Solution *First $s'(t) = 3t^2 + C_1$, so $C_1 = -1$. Reversing differentiation again gives $s(t) = t^3 - t + C_2$, so $C_2 = 2$. The function $t^3 - t + 2$ satisfies the second derivative identity and both data. The family theorem at each stage proves uniqueness. Before imposing data, the general solution is $t^3 + C_1 t + C_2$; the first constant becomes a linear term after the second reversal.*

Example 2.46 Existence is a separate question Show that $f(x) = 0$ for $x < 0$ and $f(x) = 1$ for $x \geq 0$ has no antiderivative on $\mathbb{R}$. Does this mean that every function with an antiderivative must be continuous?

> ***Idea.*** The derivative is prescribed separately on two intervals, so determine the only possible primitives there. Continuity fixes their meeting height; their incompatible one-sided slopes then rule out a derivative at the join.

Solution *An antiderivative would have the form C_- for $x < 0$ and $x + C_+$ for $x > 0$. Differentiability at 0 implies continuity there, forcing $C_- = C_+ = F(0)$. The left difference quotient at 0 is then 0 and the right difference quotient is 1, a contradiction.*

The proposed general conclusion about continuity is false. In Example 2.31, the differentiable function $G(0) = 0$, $G(x) = x^2 \sin(1/x)$ for $x \neq 0$ has a discontinuous derivative. That derivative nevertheless has the antiderivative G, by definition. Thus a jump may obstruct a primitive, but discontinuity by itself is not a decisive test.

2.5.5 Methods and further remarks

Start with the domain. Determine the intervals on which the integrand is defined before simplifying. An algebraically removable hole need not be removed in the problem; different intervals may carry different constants. Check real powers, logarithms and inverse-trigonometric branches explicitly.

Look for an already known derivative. Factor or divide rational expressions, use elementary identities, and inspect a factor accompanied by its derivative. Recognition produces a candidate; differentiating it proves the claim. For instance, on an interval where $g \neq 0$, the chain rule verifies $(\ln|g|)' = g'/g$. This is a useful recognition pattern, not permission to ignore zeros of g.

Separate existence, the family, and normalization. One verified candidate proves existence. The zero-derivative lemma proves that it gives the complete family on an interval. An initial value then fixes the constant. For piecewise data, match heights and independently check slopes at the joins.

Compare answers by subtraction or differentiation. Equivalent formulas may differ by a constant. If differentiating their difference gives zero, first check that the comparison takes place on one interval, then evaluate at a convenient point.

> ***Looking Ahead.*** We have not yet proved that every continuous function has an antiderivative. The next section defines definite integrals as limits of sums, and Section 2.7 will use a variable upper bound to construct an antiderivative of any continuous function. Existence will not require guessing an elementary formula. Systematic substitution and integration by parts belong to the next chapter.

Exercise 2.5

1. Explain the difference between one antiderivative and an indefinite integral. For $f(x) = 3x^2$ on $\mathbb{R}$, give two antiderivatives and a differentiable function that is not an antiderivative.

2. Find all antiderivatives of $4x^3 - 2x + 5$ on $\mathbb{R}$ and verify your answer by differentiation.
3. Evaluate $\int (3\sqrt{x} + 2/x^2)\,\mathrm{d}x$ on $(0, +\infty)$.
4. Evaluate $\int (2\cos x - 3\sin x)\,\mathrm{d}x$ on $\mathbb{R}$.
5. Find F on $(0, +\infty)$ with $F'(x) = 1/x$ and $F(1) = 4$.
6. Find F on $(-1, 1)$ with $F'(x) = 1/\sqrt{1 - x^2}$ and $F(0) = -2$.
7. For every real parameter a, find all antiderivatives of e^{ax} on $\mathbb{R}$. Treat $a = 0$ separately.
8. Find all antiderivatives of $(x^4 - 1)/(x^2 - 1)$ on its original domain. Which answers extend to differentiable functions on all of $\mathbb{R}$?
9. Compute $\int (\sec^2 x + \tan^2 x)\,\mathrm{d}x$ on $(-\pi/2, \pi/2)$ using a trigonometric identity, and verify the result.
10. Determine all $a, b \in \mathbb{R}$ for which $e^x(ax + b)$ is an antiderivative of xe^x on $\mathbb{R}$.
11. Let F, G be antiderivatives of the same function on an open interval I. If $F(u) = G(u)$ at one $u \in I$, prove $F = G$ on I. Explain why the statement need not hold on a union of two disjoint open intervals.
12. Find y on $\mathbb{R}$ with $y''(x) = 2$, $y(0) = 1$ and $y(2) = 5$.
13. Find the primitive F with $F(0) = 0$ of the continuous function $f(x) = 2x$ for $x < 0$, $f(x) = \sin x$ for $x \geq 0$. Verify the joining-point derivative directly.
14. Suppose f and g each have an antiderivative on an interval. Must the product of chosen antiderivatives be an antiderivative of fg? Give a counterexample and derive the correct derivative of that product.
15. For $a, b, c \in \mathbb{R}$, let $f(x) = a$ for $x < 0$, $f(0) = c$, and $f(x) = b$ for $x > 0$. Determine exactly when f has an antiderivative on $\mathbb{R}$.
16. On an open interval I, suppose $F' = f$, f is continuous, and $f(x) \neq 0$ everywhere. Decide whether $F^2/2$ is an antiderivative of f; determine whether any such pair (f, F) can satisfy this requirement throughout I.
17. Let $F' = f$ on an open interval I and let $a, b \in \mathbb{R}$. Find the derivative of $aF + b$ and describe all antiderivatives of af on I. In the case $a = 0$, explain why multiplying the family $F + C$ by 0 does not by itself describe that full answer.
18. Find all antiderivatives of $f(x) = x\,|x|$ on $\mathbb{R}$. Check the derivative at 0 directly, and determine whether the antiderivatives belong to $C^2(\mathbb{R})$.
19. For prescribed $A, B \in \mathbb{R}$, determine exactly when a differentiable F on $\mathbb{R}$ can satisfy $F'(x) = 2x + 1$, $F(0) = A$, and $F(1) = B$. Find every solution in the compatible case and explain why one initial value and two arbitrary values are different requirements.
20. Find all antiderivatives of $1/(x \ln x)$ on its full real domain, using differentiation to verify a candidate. Does the condition $F(e) = 0$ determine the value $F(e^{-1})$? Explain the role of each logarithmic domain restriction and each independent constant.
21. Let F be differentiable on an open interval I. Find an antiderivative of FF' by recognizing a derivative. If F is nowhere zero on I, also find an antiderivative of F'/F^2. Explain why zeros of F cause no problem in the first question but must be excluded in the second.

Supplement Exercises

S1. Let f be continuous on $\mathbb{R}$, let $T > 0$, and suppose $f(x + T) = f(x)$. Assume that an antiderivative F is given. Prove that $F(x + T) - F(x)$ is constant in x, and prove that one antiderivative is T-periodic if and only if all of them are. Give one example of each possibility.

S2. Let f be continuous and even on $\mathbb{R}$, and suppose $F' = f$, $F(0) = 0$. Prove that F is odd by comparing $F(x)$ with $-F(-x)$. If instead f is odd, determine the parity of the normalized F and prove your claim.

S3. Classify all twice differentiable functions F on $\mathbb{R}$ satisfying $F'' = 0$. Then classify all k times differentiable functions with $F^{(k)} = 0$, for an integer $k \geq 1$, without Taylor's theorem.

S4. A function F is differentiable on $\mathbb{R} \setminus \{0\}$, continuous at 0, and satisfies $F'(x) = 3x^2$ for $x \neq 0$. Prove that F is differentiable at 0 and determine $F'(0)$. Show by an example why continuity at 0 cannot be omitted.

S5. Let f have antiderivatives F_- on $(-2, 1)$ and F_+ on $(0, 3)$. Show how to change one additive constant so that they define a single antiderivative on $(-2, 3)$. Prove that the result is unique after specifying its value at 0.

S6. For real a, b, find an antiderivative on $\mathbb{R}$ of $\cos(ax)\cos(bx)$ by a product-to-sum identity. Separate $a = b$, $a = -b$, and $a = b = 0$ from the generic case and verify all formulas.

S7. Let P be a polynomial and suppose $F' = P$ on $\mathbb{R}$ and F is periodic with a positive period. Prove that $P = 0$ and F is constant.

> ***Optional Hint.*** First integrate the polynomial term by term. A nonconstant polynomial is unbounded along a sequence $x_0 + nT$, while periodicity fixes its values there.

S8. Define $G(0) = 0$ and $G(x) = x^2 \sin(1/x^2)$ for $x \neq 0$. Find G' everywhere and show that this derivative has an antiderivative but is unbounded on every neighbourhood of 0. Use a specific sequence to prove unboundedness.

S9. Find the unique function F on $\mathbb{R}$ such that $F'(x) + F(x) = 1$ for all x and $F(0) = 0$.

> ***Optional Hint.*** Compute $(e^x F)'$ to recognise the left side; verify your candidate.

S10. Let $f(x) = x$ for $x \leq 0$ and $f(x) = x^2$ for $x > 0$. Find its primitive with $F(0) = 0$. Prove $F \in C^1(\mathbb{R})$, test $F''(0)$, and state the required matching condition.

S11. Find all differentiable functions on $\mathbb{R} \setminus \{0\}$ satisfying $xF'(x) - F(x) = x^2$. Determine which have continuous extensions to 0, and which extensions are differentiable there.

> ***Optional Hint.*** The numerator in the quotient rule for $F(x)/x$ is already present in the equation. Keep separate constants on the two half-lines before examining the missing point.

S12. Find all differentiable pairs $F, G\colon \mathbb{R} \to \mathbb{R}$ satisfying $F' = G$ and $G' = -F$. Prove that $F^2 + G^2$ is constant, and show that the values $F(0), G(0)$ uniquely determine the pair.

> ***Optional Hint.*** Differentiate $F(x)\cos x - G(x)\sin x$ and $F(x)\sin x + G(x)\cos x$. The paired derivative relations suggest combinations whose varying terms cancel.

2.6 Definite Integrals

知识点概要

- ❏ *Build totals from sampled contributions.*
- ❏ *Use arbitrary tags and a shrinking mesh.*
- ❏ *Distinguish signed integrals from areas.*
- ❏ *Establish or disprove integrability.*
- ❏ *Apply linearity, order and interval splitting.*
- ❏ *Compute from sums and control errors.*

If a quantity changes at a constant rate c during a time interval of length $b-a$, its total change is $c(b-a)$. For a varying rate $f(t)$, a short time interval suggests the approximation "one sampled rate times the elapsed time." Adding these contributions gives an approximation to total change. The central question is whether increasingly fine measurements settle to a unique value, independently of where the samples are taken. This construction will define a number from a function on an interval. Its connection with antiderivatives is a theorem to be proved in the next section, not part of the definition.

2.6.1 Accumulation and tagged partitions

Definition 2.8

Let $a < b$. A partition(分割) *of $[a,b]$ is a finite ordered list*

$$P: \quad a = x_0 < x_1 < \cdots < x_n = b.$$

Write $\Delta x_i = x_i - x_{i-1}$ and define its mesh(网格宽度) *by $\|P\| = \max_{1\le i\le n} \Delta x_i$. A choice of* tags (标记点) *$\xi_i \in [x_{i-1}, x_i]$ gives the tagged Riemann sum*

$$S(f;P,\xi) = \sum_{i=1}^{n} f(\xi_i)\Delta x_i.$$

♣

The widths need not be equal, and a tag may be an endpoint. The mesh measures the least resolved part of the whole interval: many tiny pieces do not compensate for one piece of fixed width. For a nonnegative height function, the summands are rectangle areas. For velocity, their units are distance, but their signs record direction; a negative velocity contributes a negative change of position.

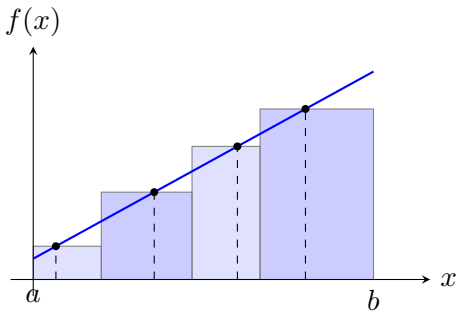


Figure 2.6: Unequal widths and freely chosen tags are allowed. Each rectangle contributes its sampled height times its own width.

2.6.2 The Riemann integral

Definition 2.9

A bounded function $f\colon [a,b]\to\mathbb{R}$*, with* $a<b$*, is* Riemann integrable (黎曼可积) *if there is a real number* I *such that for every* $\varepsilon>0$ *there is* $\delta>0$ *satisfying*

$$\|P\|<\delta \quad\Longrightarrow\quad |S(f;P,\xi)-I|<\varepsilon$$

for every partition P *and every permitted choice of tags* ξ*. The number* I *is the* definite integral (定积分)*, denoted* $\displaystyle\int_a^b f(x)\,\mathrm{d}x$.

♣

The order of the quantifiers matters: δ may depend on $f,[a,b],\varepsilon$, but cannot be chosen separately for a favourable partition or set of tags. The value is unique. Indeed, if I,J both satisfied the definition, take a partition finer than both corresponding mesh bounds; then $|I-J|<2\varepsilon$ by comparison with its one sum. Letting ε be arbitrarily small gives $I=J$. The integration variable is a dummy variable: $\int_a^b f(x)\,\mathrm{d}x=\int_a^b f(t)\,\mathrm{d}t$. Unlike an indefinite integral, this is one real number, with no arbitrary constant. Our proper Riemann integral is defined for bounded functions on a finite closed interval; singular or infinite-interval integrals require a later definition.

Example 2.47 A constant rate Use the definition to find $\int_a^b c\,\mathrm{d}x$, where $a<b$ and $c\in\mathbb{R}$.

Solution *Every tagged sum is* $\sum c\Delta x_i=c(b-a)$*, so its error from this value is exactly zero. Hence the integral is* $c(b-a)$*. This includes negative* c*: the integral is signed accumulation, whereas the geometric rectangle area is* $|c|\,(b-a)$.

Example 2.48 An integral from arbitrary tags Prove directly that $\int_a^b x\,\mathrm{d}x=(b^2-a^2)/2$ for $a<b$.

> ***Idea.*** An arbitrary sampled height is awkward, but the midpoint height has a useful algebraic feature: multiplying it by the width gives a difference of squares. Compare every tag with this midpoint, so that the remaining errors are controlled by the largest width.

Solution ***Method 1: telescope and bound every tag.*** *Put* $m_i=(x_{i-1}+x_i)/2$*. Then*

$$\sum m_i\Delta x_i=\frac{1}{2}\sum(x_i^2-x_{i-1}^2)=\frac{b^2-a^2}{2},$$

and

$$\left|\sum(\xi_i-m_i)\Delta x_i\right|\le\frac{1}{2}\sum(\Delta x_i)^2\le\frac{b-a}{2}\|P\|.$$

Choosing $\delta=2\varepsilon/(b-a)$ *proves the definition.*

Solution ***Method 2: use a convenient sequence after existence is known.*** *For* n *equal widths* $d=(b-a)/n$ *and right tags* $a+id$*, the sum is*

$$\sum_{i=1}^{n}(a+id)d=a(b-a)+\frac{(b-a)^2(n+1)}{2n}\longrightarrow\frac{b^2-a^2}{2}.$$

Here $1+\cdots+n=n(n+1)/2$*, obtainable by pairing the sum in reverse order. The existence proved in Method 1 ensures that this limit is the integral.*

The arbitrary-tag bound proves existence and supplies an error estimate; equal-width sums are simpler for evaluating an integral whose existence has already been established. A single successful sequence alone does not prove integrability.

Example 2.49 Why the number of pieces is insufficient For $f(x) = x^2$ on $[0, 1]$, keep $[0, 1/2]$ as one piece and divide $[1/2, 1]$ into n equal pieces. On the latter pieces use identical tags in two sums, but on the first piece use 0 in one sum and $1/2$ in the other. Can the growing number of pieces force these two sums towards the same value?

Solution *Their difference is always* $[(1/2)^2 - 0^2](1/2) = 1/8$. *The number of pieces tends to infinity, but the mesh remains* $1/2$. *The definition requires the mesh, not merely the reciprocal of the number of pieces, to tend to zero. This example does not say that* x^2 *is nonintegrable; it identifies a sequence of partitions to which the definition's guarantee does not apply.*

2.6.3 How integrability is established

Checking every tag individually can be replaced by trapping all possible sums between two bounds. For a bounded function, set

$$m_i = \inf_{[x_{i-1},x_i]} f, \qquad M_i = \sup_{[x_{i-1},x_i]} f, \qquad L(f,P) = \sum m_i \Delta x_i, \quad U(f,P) = \sum M_i \Delta x_i.$$

These are the *lower and upper sums*(下和与上和), also called Darboux sums. An infimum or supremum need not be attained; this is why they are used instead of minimum and maximum. For example, $f(x) = x$ on $[0, 1)$ with $f(1) = 0$ has supremum 1 on $[0, 1]$, but no value equal to 1. Nevertheless, tags sufficiently near 1 from the left give values as close to that supremum as desired. Every tagged sum lies between $L(f, P)$ and $U(f, P)$. Adding partition points can only increase L and decrease U, since an infimum on a smaller interval cannot decrease and a supremum cannot increase. The union of two sets of partition points is a common refinement. For $R = P \cup Q$, the full comparison is

$$L(f,P) \le L(f,R) \le U(f,R) \le U(f,Q).$$

Thus every lower sum is at most every upper sum, even if the two original partitions are unrelated by refinement.

Theorem 2.7

A bounded function f on $[a, b]$ is Riemann integrable if and only if, for every $\varepsilon > 0$, there is a partition P with

$$U(f,P) - L(f,P) < \varepsilon.$$

In that case every lower sum is at most the integral and every upper sum is at least the integral. ♡

> ***Proof Strategy.*** Small upper-minus-lower sums express agreement between all sampling choices on a selected partition. The extra issue is to pass from selected partitions to every sufficiently fine partition. Only pieces crossing the finitely many selected dividing points can cause trouble; their total length tends to zero with the new mesh. Keep the three roles separate: P is chosen once for a small Darboux gap; Q is any sufficiently fine partition whose tags must be controlled; $Q^* = P \cup Q$ links the two.

Proof **From arbitrary-tag convergence to a small gap.** Suppose first that the tagged definition holds with value I. Choose a partition P with mesh below the bound for an error $\varepsilon/4$. For an arbitrary $\tau > 0$, the definition of the supremum allows a tag ξ_i in each cell such that

$$0 \le M_i - f(\xi_i) < \frac{\tau}{b-a}.$$

Multiply by the widths and add:

$$0 \le U(f,P) - S(f;P,\xi) < \sum_i \frac{\tau}{b-a}\Delta x_i = \tau.$$

Every such tagged sum is less than $I+\varepsilon/4$, hence $U(f,P) < I+\varepsilon/4+\tau$. Since τ is arbitrary, $U(f,P) \le I+\varepsilon/4$. Choosing tags near each infimum similarly gives $L(f,P) \ge I-\varepsilon/4$. Therefore $U(f,P)-L(f,P) \le \varepsilon/2 < \varepsilon$.

Find the only possible value. Conversely, define the finite numbers

$$\underline{I} = \sup_P L(f,P), \qquad \overline{I} = \inf_P U(f,P).$$

The common-refinement comparison gives $\underline{I} \le \overline{I}$. For any fixed partition P, we also have $\overline{I} \le U(f,P)$ and $\underline{I} \ge L(f,P)$, so

$$0 \le \overline{I} - \underline{I} \le U(f,P) - L(f,P).$$

The hypothesis makes the right side arbitrarily small, forcing equality of the two numbers. Call their common value I. Consequently $L(f,P) \le I \le U(f,P)$ for each P.

Control every sufficiently fine measurement. Let $|f| \le M$. If $M = 0$, the conclusion is immediate; otherwise fix $\varepsilon > 0$ and choose P with gap less than $\varepsilon/2$. Let r be the number of its interior dividing points. Both P and r are now fixed; they do not depend on the new partition or its tags. For any other partition Q, insert those points to obtain Q^*. Only cells of Q containing a dividing point of P in their interior are split. There are at most r such cells, even if some contain several dividing points, and their total length is at most $r\|Q\|$. On every unchanged cell keep its tag; choose any tags on the split cells. The old and new sums differ by at most $2Mr\|Q\|$, because each of the two contributions has absolute value at most M times the total affected length. Since Q^* refines P, its sum lies between $L(f,P)$ and $U(f,P)$, and hence within $\varepsilon/2$ of I. Writing ξ^* for the chosen refined tags, this gives

$$\begin{aligned} |S(f;Q,\xi) - I| &\le |S(f;Q,\xi) - S(f;Q^*,\xi^*)| + |S(f;Q^*,\xi^*) - I| \\ &< 2Mr\|Q\| + \varepsilon/2. \end{aligned}$$

Choose $\delta = \varepsilon/[4M(r+1)]$. If $\|Q\| < \delta$, the last bound is less than ε, including when $r = 0$. This one δ works for every Q and every original set of tags, as required by the definition. The final bounds follow from the construction of I. □

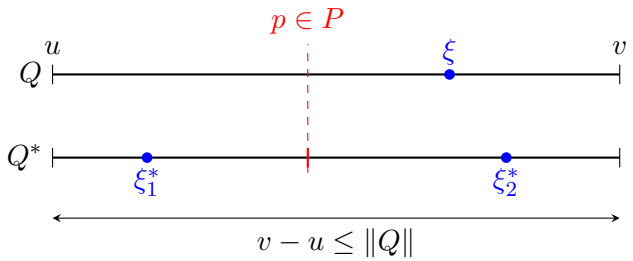


Figure 2.7: One cell crossing a fixed point of P is split. The old contribution and the sum of the new contributions each have magnitude at most $M(v-u)$; their difference is at most $2M(v-u)$. Unchanged cells keep their tags.

Figure 2.7 isolates the source of the refinement error. It is the total length of the changed cells that matters, not the number of tags on the rest of the interval.

Theorem 2.8

Each of the following functions on a finite closed interval is Riemann integrable:

1. *a continuous function;*
2. *a monotone function with finite endpoint values;*
3. *a bounded function continuous except possibly at finitely many points.*

Changing finitely many values of an integrable function preserves integrability and the integral. ♡

> ***Proof Strategy.*** For continuous functions, fine cells make the oscillation small everywhere. For finitely many exceptional points, that mechanism can fail near a bad point. Use a different small quantity there: put all bad points inside intervals of small total length, using boundedness to control their heights. The two error budgets are "small oscillation times ordinary length" and "bounded oscillation times small length."

Proof A continuous function is bounded and uniformly continuous by the results of Section 1.6. For any $\eta > 0$, take a mesh small enough that values in one cell differ by less than η. Then $U - L \le \eta(b-a)$, which proves the first assertion.

For a nondecreasing function, its endpoint values bound it, and

$$U - L = \sum \big(f(x_i) - f(x_{i-1})\big)\Delta x_i \le \|P\|\big(f(b) - f(a)\big).$$

The criterion proves integrability. Apply this to $-f$ for a nonincreasing function.

For the third assertion, let $|f| \le M$. If $M = 0$, there is nothing to prove; if there are no exceptional points, the first assertion applies. Otherwise fix $\varepsilon > 0$. Enclose the finitely many exceptional points in short closed subintervals, placing an interior exceptional point in the interior of an enclosure and using a one-sided enclosure for an exceptional endpoint. Choose all non-global enclosure endpoints outside the finite exceptional set, and merge overlaps if necessary. Their total length ℓ can be chosen to satisfy

$$\ell < \min\left\{b - a, \frac{\varepsilon}{4M}\right\}.$$

The remaining gaps, together with their boundary points, are finitely many closed intervals containing no exceptional point. On each of these intervals f is uniformly continuous. Set $\eta = \varepsilon/[2(b-a)]$ and take the minimum of the finitely many positive mesh bounds needed for oscillation at most η on a remaining cell. Choose a partition containing all enclosure endpoints, and subdivide the remaining intervals to meet this mesh bound. The cells within the enclosures have total contribution at most $2M\ell$ to $U - L$; all other cells have total contribution at most $\eta(b-a)$. Consequently

$$U - L \le 2M\ell + \eta(b-a) < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

The criterion proves integrability. Choosing the enclosure boundaries at continuous points ensures that the remaining closed intervals really satisfy the continuity hypothesis used above.

Finally, suppose f, g differ only at N points and $|g - f| \le A$ everywhere. For any tagged partition, at most $2N$ cells have a tag among these points (an interior endpoint can tag two adjacent cells). Therefore $|S(g; P, \xi) - S(f; P, \xi)| \le 2NA\|P\|$. If the sums for f tend to I, so do those for g. □

Here a piecewise continuous function, with finite one-sided limits at finitely many breaks, satisfies the third condition. A finite jump is therefore allowed. The third condition is more general: it does not require a one-sided limit at an exceptional point.

Example 2.50 Bounded oscillation at an endpoint is allowed Set $f(0) = 0$ and $f(x) = \sin(1/x)$ for $0 < x \le 1$. Prove that f is Riemann integrable, although its right limit at 0 does not exist.

Solution ***Method 1: use the finite-exception theorem.*** *The bound $|f| \le 1$ holds everywhere, and f is con-*

tinuous at every point of $(0,1]$ *relative to* $[0,1]$. *Only* 0 *can be exceptional, so Theorem 2.8 applies. Along* $x_n = (\pi/2 + 2\pi n)^{-1}$ *the values are* 1, *and along* $y_n = (3\pi/2 + 2\pi n)^{-1}$ *they are* -1. *Thus there is no right limit at* 0.

> ***A direct error budget.*** The oscillation near 0 never becomes small. Instead, make the single cell containing that endpoint short, then use ordinary continuity on the rest.

Solution ***Method 2: construct a small Darboux gap.*** *Given* $\varepsilon > 0$, *choose* $0 < \eta < \min(1, \varepsilon/4)$. *The cell* $[0,\eta]$ *contributes at most* $2\eta < \varepsilon/2$ *to* $U - L$. *On* $[\eta, 1]$, *uniform continuity gives a finite partition with oscillation on every cell less than* $\varepsilon/2$. *Those cells contribute at most* $(\varepsilon/2)(1-\eta) < \varepsilon/2$. *Combining the partitions gives gap less than* ε, *so the Darboux criterion proves integrability directly.*

The first route uses a reusable theorem; the second displays how the troublesome oscillation is contained without estimating it pointwise. Neither route evaluates the integral or treats it as an improper integral.

Boundedness is still essential: continuity away from a single point does not make $1/x$ on $(0,1]$ a proper Riemann integrand on $[0,1]$. Nor is boundedness alone enough, as the next example shows.

Example 2.51 Two sampling choices that never agree Define $D(x) = 1$ for rational x and $D(x) = 0$ for irrational x, on $[0,1]$. Show that D is not Riemann integrable.

Solution ***Method 1: incompatible tagged sums.*** *Every interval of positive length contains rational and irrational numbers. For any partition, choosing all rational tags gives a sum of* 1, *whereas choosing all irrational tags gives a sum of* 0. *These choices remain possible at arbitrarily small mesh and cannot converge to a common number.*

Solution ***Method 2: a persistent Darboux gap.*** *Every cell has infimum* 0 *and supremum* 1, *so* $L(D,P) = 0$ *and* $U(D,P) = 1$ *for every partition. The gap never becomes small, contradicting Theorem 2.7.*

The tagged argument directly contradicts the definition; the Darboux argument packages the same obstruction into a criterion useful even when explicit tag choices are cumbersome.

Example 2.52 Evaluating a continuous function from sums Find $\int_0^1 x^2\,\mathrm{d}x$ without using an antiderivative theorem.

Solution ***Method 1: right-endpoint sums.*** *Continuity proves integrability. Equal-width right sums give*

$$\frac{1}{n}\sum_{i=1}^{n}\left(\frac{i}{n}\right)^2 = \frac{n(n+1)(2n+1)}{6n^3} \longrightarrow \frac{1}{3}.$$

The identity $\sum_{i=1}^{n} i^2 = n(n+1)(2n+1)/6$ *follows by induction: it is true for* $n = 1$, *and adding* $(n+1)^2$ *to the right side produces* $(n+1)(n+2)(2n+3)/6$. *Thus the integral is* $1/3$.

Solution ***Method 2: squeeze all sums by Darboux bounds.*** *For the equal partition, monotonicity gives*

$$U_n = \frac{1}{3} + \frac{1}{2n} + \frac{1}{6n^2}, \qquad L_n = U_n - \frac{1}{n} = \frac{1}{3} - \frac{1}{2n} + \frac{1}{6n^2}.$$

The gap is $1/n$, *so the criterion independently proves existence. The integral lies between these bounds for every* n, *forcing its value to be* $1/3$.

The first route uses continuity as an existence theorem; the second obtains existence and computable upper and lower bounds together.

Example 2.53 An isolated spike Let $p(1/2) = 7$ and $p(x) = 0$ for all other $x \in [0,1]$. Find its integral and explain why the spike has no effect.

Solution ***Method 1: control arbitrary tags.*** *At most two cells can use* $1/2$ *as a tag, so* $0 \le S(p;P,\xi) \le 14\|P\|$. *All sums therefore tend to* 0.

Solution ***Method 2: isolate its support.*** *Choose partition points* $1/2 - \eta$ *and* $1/2 + \eta$, *with* $0 < \eta < 1/2$. *The lower sum is* 0; *the upper sum is* 14η, *because only the interval enclosing* $1/2$ *has nonzero supremum. The*

Darboux criterion gives integrability, and the integral lies between 0 *and* 14η *for every such* η*, hence is* 0.

The first argument gives a mesh-dependent error for any sampling scheme. The second explains geometrically how a high but isolated value can be confined to an arbitrarily short interval.

2.6.4 Algebra, order and splitting intervals

Theorem 2.9

Let f, g *be Riemann integrable on* $[a, b]$*, where* $a < b$*, and let* $\alpha, \beta \in \mathbb{R}$*. Then* $\alpha f + \beta g$*,* fg*, and* $|f|$ *are integrable. Moreover,*

$$\int_a^b (\alpha f + \beta g)\,\mathrm{d}x = \alpha \int_a^b f\,\mathrm{d}x + \beta \int_a^b g\,\mathrm{d}x,$$

$$f \le g \quad \Longrightarrow \quad \int_a^b f\,\mathrm{d}x \le \int_a^b g\,\mathrm{d}x,$$

$$\left| \int_a^b f\,\mathrm{d}x \right| \le \int_a^b |f|\,\mathrm{d}x.$$

If $a < c < b$*, the restrictions to* $[a, c]$ *and* $[c, b]$ *are integrable and*

$$\int_a^b f\,\mathrm{d}x = \int_a^c f\,\mathrm{d}x + \int_c^b f\,\mathrm{d}x.$$

Conversely, integrability on both adjacent intervals implies integrability on their union and the same identity.

♡

Proof Linearity holds for each tagged sum, and the sum limit laws prove the first identity and the asserted linear-combination integrability. Pointwise order gives order of tagged sums because widths are positive; order is preserved in their limits.

To check the other integrability claims without presupposing their existence, write $\omega_J(f) = \sup_J f - \inf_J f$ for the oscillation on a cell J. If $|f| \le M$ and $|g| \le N$, the triangle inequality gives

$$\omega_J(|f|) \le \omega_J(f), \qquad \omega_J(fg) \le M\omega_J(g) + N\omega_J(f).$$

For the second inequality use $|f(u)g(u) - f(v)g(v)| \le M\,|g(u) - g(v)| + N\,|f(u) - f(v)|$, then take suprema over $u, v \in J$. Take a common refinement of partitions with small gaps for f, g and apply the Darboux criterion. Finally $-|f| \le f \le |f|$ and order imply the absolute-value inequality.

Insert c into a partition with small Darboux gap for f on $[a, b]$. Each of the two restricted gaps is nonnegative and at most the total gap, proving integrability of both restrictions. Conversely, concatenate partitions with small gaps on the two pieces; their gaps add, proving integrability on the union. For partitions containing c, a tagged sum is exactly the sum of the two restricted tagged sums. Taking fine such partitions gives the displayed identity by uniqueness of the integral. □

Product integrability is not a rule for evaluating products. In general,

$$\int_a^b f(x)g(x)\,\mathrm{d}x \ne \left(\int_a^b f(x)\,\mathrm{d}x \right) \left(\int_a^b g(x)\,\mathrm{d}x \right).$$

On $[0, 1]$, $f = g = x$ gives $1/3$ on the left and $1/4$ on the right.

For equal or reversed bounds we define

$$\int_a^a f(x)\,\mathrm{d}x = 0, \qquad \int_b^a f(x)\,\mathrm{d}x = -\int_a^b f(x)\,\mathrm{d}x \quad (a < b).$$

With these conventions, splitting at any third point in an interval of integrability is valid in any order of the three points. To see this, order the three points, apply the proved adjacent-interval identity, and reverse or rearrange

terms as required. Pointwise order, however, preserves the direction of integral inequalities only for increasing bounds; reversing the bounds reverses the inequality.

2.6.5 Signed accumulation and estimates

For continuous $f \geq 0$, we interpret $\int_a^b f(x)\,\mathrm{d}x$ as the area under its graph. For a function that changes sign, the integral is area above the axis minus area below it. The total geometric area is $\int_a^b |f(x)|\,\mathrm{d}x$. This signed-area interpretation agrees with rectangles and triangles and with the sum construction; it must not be used to assume integrability of an arbitrary bounded function.

Example 2.54 Displacement and total area Compute $\int_0^3 (x-1)\,\mathrm{d}x$ and $\int_0^3 |x-1|\,\mathrm{d}x$ by geometry and by the established integral formulas.

Solution ***Method 1: signed triangles.*** *On* $[0,1]$ *the graph lies below the axis and forms a triangle of area* $1/2$. *On* $[1,3]$ *the positive triangle has area* 2. *Thus the signed integral is* $2 - 1/2 = 3/2$, *while the absolute-value integral is* $2 + 1/2 = 5/2$.

Solution ***Method 2: linearity and splitting.*** *The constant and affine examples give* $\int_0^3 (x-1)\,\mathrm{d}x = 9/2 - 3 = 3/2$. *For absolute value, use* $1 - x$ *on* $[0,1]$ *and* $x - 1$ *on* $[1,3]$*:*

$$\int_0^1 (1-x)\,\mathrm{d}x + \int_1^3 (x-1)\,\mathrm{d}x = \left(1 - \frac{1}{2}\right) + \left(\frac{9-1}{2} - 2\right) = \frac{5}{2}.$$

The geometric route displays cancellation; the algebraic route makes the sign split explicit and remains useful when a graph is harder to read.

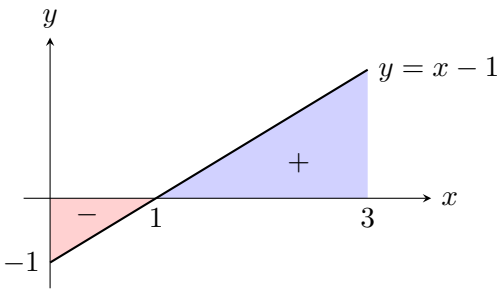


Figure 2.8: The two lobes add for total area but subtract for signed accumulation.

If $m \leq f \leq M$ on $[a,b]$ with $a < b$, order and the constant example give

$$m(b-a) \leq \int_a^b f(x)\,\mathrm{d}x \leq M(b-a).$$

In particular, $|f| \leq K$ implies $\left|\int_a^b f(x)\,\mathrm{d}x\right| \leq K(b-a)$. For arbitrary order of the limits, this becomes $\left|\int_a^b f(x)\,\mathrm{d}x\right| \leq K\,|b-a|$.

Proposition 2.4

If f *is continuous and nonnegative on* $[a,b]$, $a < b$, *and* $f(c) > 0$ *at some* $c \in [a,b]$, *then* $\int_a^b f(x)\,\mathrm{d}x > 0$. ♠

Proof Continuity gives a subinterval of positive length inside $[a,b]$, possibly one-sided at an endpoint, on which $f \geq f(c)/2$. Its integral is at least $f(c)/2$ times that positive length. The remaining integrals are nonnegative by order and splitting, so the full integral is positive. □

Continuity at a positive point is the mechanism turning a positive height into a contribution over positive length. An isolated positive spike does not have that consequence.

Example 2.55 Useful bounds without an explicit evaluation Show that $1/2 < \int_0^1 (1+x^2)^{-1}\,dx < 1$ without evaluating the integral.

> ***Idea.*** The denominator lies between 1 and 2, so reciprocal bounds give a first estimate. For strict inequalities, examine the nonnegative gaps from these two constant bounds; each gap is positive on an interval, not just at an isolated point.

Solution *The integrand is continuous, so it is integrable, and* $1/2 \le (1+x^2)^{-1} \le 1$. *The continuous gap* $(1+x^2)^{-1} - 1/2$ *is positive at* 0, *and the continuous gap* $1-(1+x^2)^{-1}$ *is positive at* 1. *By Proposition 2.4, both gap integrals are strictly positive. Linearity gives the required strict bounds.*

2.6.6 Methods and further remarks

Separate existence from evaluation. First use continuity, monotonicity, finitely many discontinuities, or a small Darboux gap to justify integrability. Then choose convenient tags to compute its value. One convergent sequence of sums is evidence about that sequence; it is not an integrability theorem.

Choose a useful comparison. For direct definitions, compare arbitrary tags with telescoping midpoint values or trap them by upper and lower sums. Track errors with the mesh, not merely the number of cells. For finite changes, count affected cells and bound their total length. For finitely many discontinuities, allocate separate tolerances to the total length of exceptional cells and to the oscillation on the remaining cells.

Use structure before calculation. Linearity handles sums, splitting handles sign changes and formula changes, and order provides bounds without a primitive. Keep increasing bounds while applying order; introduce a minus sign when reversing them. To prove a strict inequality, identify a continuous positive gap on a subinterval of positive length.

> ***Looking Ahead.*** The integral has been built entirely from sampled sums. Next we will let an endpoint move and show that the resulting accumulation function has derivative equal to the original continuous rate. That construction will justify evaluating definite integrals with antiderivatives and will connect the two meanings of the integral sign.

Exercise 2.6

1. For $P = (0, 1/4, 3/4, 1)$ and tags $(0, 1/2, 1)$, compute the mesh and the sum for $f(x) = 2x + 1$.
2. Explain the quantifiers in the Riemann definition. Which choices must one δ control?
3. Compute $\int_{-2}^{3} 4\,dx$, $\int_{3}^{-2} 4\,dx$, and $\int_{2}^{2} 4\,dx$.
4. Use linearity and the affine example to compute $\int_{-1}^{2} (3x+2)\,dx$.
5. Compute lower and upper sums for $f(x) = x$ on $[0,1]$ with n equal subintervals. Find their difference.
6. A bounded function on $[0,2]$ equals 3 except at three points, where it has arbitrary finite values. Find its integral and justify existence.
7. Explain why D in Example 2.51 is not integrable, although sums using only irrational tags are all zero.
8. Using sums, show $\int_0^2 x^2\,dx = 8/3$. State the existence theorem used.
9. Let $f = 2$ on $[0,1)$ and $f = -1$ on $[1,3]$. Compute $\int_0^3 f(x)\,dx$ and $\int_0^3 |f(x)|\,dx$.
10. Show $1/3 < \int_0^1 (2+x)^{-1}\,dx < 1/2$ without finding a primitive.
11. If f, g are integrable on $[a,b]$ and $|f-g| \le \eta$, prove $\left|\int_a^b f(x)\,dx - \int_a^b g(x)\,dx\right| \le \eta(b-a)$, where $a < b$ and $\eta \ge 0$.

12. Suppose f is nondecreasing on $[a,b]$. Prove that for a partition of mesh h, both its upper and lower sums approximate the integral with error at most $h(f(b)-f(a))$.
13. Compute $\int_{-1}^{2}|x|\,\mathrm{d}x$ and $\left|\int_{-1}^{2}x\,\mathrm{d}x\right|$, and explain the strict inequality between them.
14. Let f be integrable on $[0,2]$ and suppose
$$\int_0^2 f(x)\,\mathrm{d}x=3,\qquad \int_1^2 f(x)\,\mathrm{d}x=5.$$
Find $\int_0^1 f(x)\,\mathrm{d}x$ and $\int_2^0 f(x)\,\mathrm{d}x$. Can f be nonnegative throughout $[0,2]$?
15. Show that a continuous nonnegative f on $[a,b]$, $a<b$, has integral zero if and only if f is identically zero. Explain why continuity cannot simply be deleted.
16. If an integrable f on $[a,b]$ satisfies $m\le f\le M$, prove $m\int_a^b g(x)\,\mathrm{d}x\le\int_a^b f(x)g(x)\,\mathrm{d}x\le M\int_a^b g(x)\,\mathrm{d}x$ for every nonnegative integrable g.
17. On $[0,1]$, find continuous f,g for which $\int_0^1 f(x)g(x)\,\mathrm{d}x\neq(\int_0^1 f(x)\,\mathrm{d}x)(\int_0^1 g(x)\,\mathrm{d}x)$, and find nonzero continuous f with $\int_0^1 f(x)\,\mathrm{d}x=0$. Verify both examples from already established formulas.
18. A bounded function f is integrable on $[0,1]$ with integral I. A tagged approximation uses a partition of mesh tending to zero, but up to N sampled values are replaced by erroneous values whose differences from the true values are at most A, with N,A fixed. Prove that the approximations still tend to I.
19. Let $f(x)=x$ for $0\le x<1$ and $f(1)=0$. Find its supremum on $[0,1]$ and decide whether it is attained. For $P=(0,1/2,1)$, compute $L(f,P)$ and $U(f,P)$. Find the integral and explain why the endpoint change can alter a coarse Darboux bound while leaving the integral unchanged.
20. Fix $0<\alpha<1$, and let $f(x)=0$ for $x<\alpha$ and $f(x)=1$ for $x\ge\alpha$ on $[0,1]$. Find its integral and prove that every tagged sum on the equal partition into n cells has error at most $1/n$. Include the case when α is itself a partition point.
21. For $f(x)=x$ on $[0,1]$, take $P=(0,1/2,1)$ and $Q=(0,1/4,1)$. Compute the lower and upper sums for P, Q, and $R=P\cup Q$. Verify the comparison $L(f,P)\le L(f,R)\le U(f,R)\le U(f,Q)$, and explain why it does not require P to refine Q.
22. Let f,g be Riemann integrable on $[a,b]$, $a<b$. Prove that $\max\{f,g\}$ and $\min\{f,g\}$ are integrable and that the sum of their integrals equals $\int_a^b f(x)\,\mathrm{d}x+\int_a^b g(x)\,\mathrm{d}x$. Express maximum and minimum using $f+g$ and $|f-g|$ before invoking integrability properties.
23. A Riemann integrable f is sampled on partitions P_n with meshes $h_n\to0$. At most N_n sample values are replaced, each with error at most a fixed $A\ge0$. Prove that $N_nh_n\to0$ guarantees the corrupted sums still converge to the integral. Give an example with $N_n\to\infty$ where the guarantee applies, and a counterexample showing that $h_n\to0$ alone need not suffice.

Supplement Exercises

S1. Suppose $|f(x)-f(y)|\le L|x-y|$ on $[a,b]$, with $L\ge0$. Prove integrability and the error bound $\left|S(f;P,\xi)-\int_a^b f(x)\,\mathrm{d}x\right|\le L(b-a)\|P\|$ for every tagged partition.
S2. For the same hypothesis, prove that the composite midpoint sum has error at most $L(b-a)\|P\|/4$. Work on one cell first; do not use Taylor's theorem.
S3. Prove by reflected partitions that, for integrable f on $[-a,a]$, $a>0$, the function $g(x)=f(-x)$ is integrable and $\int_{-a}^{a}g(x)\,\mathrm{d}x=\int_{-a}^{a}f(x)\,\mathrm{d}x$. Deduce the integral of an odd function and the twice-half-interval formula for an even function.
S4. Let f be continuous on $[a,b]$, $a<b$. Prove that $\left|\int_a^b f(x)\,\mathrm{d}x\right|=\int_a^b|f(x)|\,\mathrm{d}x$ if and only if f has one sign throughout the interval (zero values are allowed).

S5. Define $f(1/n) = 1/n$ for positive integers n, and set $f(x) = 0$ at every other $x \in [0,1]$. Prove directly that f is integrable with integral zero, although it differs from zero at infinitely many points.

> ***Optional Hint.*** Separate finitely many large spikes from a tail whose heights are uniformly small.

S6. Define $T(x) = 1/q$ when $x = p/q \in [0,1]$ is in lowest terms with $q > 0$, and $T(x) = 0$ when x is irrational. Prove its integral is zero by bounding the contribution of the finitely many rationals with denominator at most a chosen integer.

S7. Let $a < b$ and let f be continuous on $[a,b]$ and suppose $\int_u^v f(x)\,\mathrm{d}x = 0$ for every $a \le u < v \le b$. Prove $f = 0$ without differentiating an integral. Is the conclusion true for all Riemann integrable functions?

S8. If bounded functions f, g satisfy $f = g$ except at finitely many points, prove that f is integrable if and only if g is. Explain why replacing "finitely many" by "countably many" fails, using a specific pair.

S9. Let f be continuous and nonnegative on $[0,1]$, and let $\int_0^1 f(x)\,\mathrm{d}x = A$. If $f \ge m > 0$ on a subinterval of length ℓ, prove $m\ell \le A$. For prescribed $A > 0$ and $m > 0$, construct a continuous nonnegative function of integral A whose maximum exceeds m. Explain how a tall triangular spike can satisfy the width bound.

S10. For continuous f, g on $[a,b]$, with $a < b$, prove $\left(\int_a^b f(x)g(x)\,\mathrm{d}x\right)^2 \le \left(\int_a^b f(x)^2\,\mathrm{d}x\right)\left(\int_a^b g(x)^2\,\mathrm{d}x\right)$. Treat $\int_a^b g(x)^2\,\mathrm{d}x = 0$ separately, and characterize equality when $\int_a^b g(x)^2\,\mathrm{d}x > 0$.

> ***Optional Hint.*** Integrate the nonnegative square $(f - tg)^2$ and choose a useful real t.

S11. Suppose every f_n is Riemann integrable on $[a,b]$, $a < b$, and $f\colon [a,b] \to \mathbb{R}$ satisfies

$$\sup_{x \in [a,b]} |f_n(x) - f(x)| \longrightarrow 0.$$

Prove that f is bounded and Riemann integrable, and that $\int_a^b f_n(x)\,\mathrm{d}x \to \int_a^b f(x)\,\mathrm{d}x$.

> ***Optional Hint.*** Choose one sufficiently accurate f_n before choosing its partition. Bound the Darboux gap for f by that for f_n plus twice the uniform error times $b - a$.

S12. Suppose f is Riemann integrable on $[a,b]$, $a < b$, and $|f(x)| \ge c > 0$ everywhere. Prove that $1/f$ is integrable by comparing its oscillation on a cell with that of f. Show by an example that the weaker assumption $f(x) > 0$ at every point does not suffice, even if f has only one discontinuity.

2.7 The Fundamental Theorem of Calculus

知识点概要

- ❏ *Interpret and locate average values.*
- ❏ *Distinguish continuous and differentiable accumulation.*
- ❏ *Prove both parts of the fundamental theorem.*
- ❏ *Check primitive and endpoint hypotheses.*
- ❏ *Differentiate integrals with moving limits.*
- ❏ *Apply net change and evaluate sum limits.*

The two preceding sections introduced quite different objects: an indefinite integral is a family of functions with a prescribed derivative, whereas a definite integral is a number obtained from sampled sums. The bridge is to let one integration endpoint vary. The resulting function records accumulated change up to the current input. Increasing that input by a short amount adds a short strip whose average height should be close to the current rate. We will make this statement precise, then use it to replace a limit of many small contributions by a difference of two endpoint values.

2.7.1 Average values

For $a < b$ and an integrable f on $[a, b]$, its *average value*(平均值)is

$$f_{\text{av}} = \frac{1}{b-a}\int_a^b f(t)\,\mathrm{d}t.$$

It is the constant height that gives the same signed accumulation on an interval of the same length. For a constant function it returns that constant. For $f(t) = t$ on $[0, 2]$, the integral is 2 and the average is 1, rather than the integral itself. For a rate measured in metres per second, the integral has units of metres, while the average retains units of metres per second.

Theorem 2.10 (Mean value theorem for integrals)

If f is continuous on $[a, b]$ and $a < b$, then there is $c \in [a, b]$ such that

$$\int_a^b f(t)\,\mathrm{d}t = f(c)(b-a).$$

♡

> ***Proof Strategy.*** Integral order places the average between the smallest and largest values of the function. To turn this numerical bound into an actual sampled value, we need the function to attain every intermediate height. Thus extrema provide the bounds and continuity fills the gap between them.

Proof By the extreme value theorem, f has a minimum m and a maximum M on $[a, b]$. Integral order gives $m \le f_{\text{av}} \le M$. The intermediate value theorem on the interval joining a minimizer and a maximizer gives a point c with $f(c) = f_{\text{av}}$. If $m = M$, every point works. □

This is a mean value theorem for *integrals*, proved using continuity; the later differential mean value theorem is a different result. The average need not occur at the midpoint, and the point c need not be unique. For a continuous example, $f(x) = x^2$ on $[0, 1]$ has average $1/3$, attained at $c = 1/\sqrt{3}$; its midpoint value is only $f(1/2) = 1/4$. On $[-1, 1]$ the same function again has average $1/3$, now attained at both $c = \pm 1/\sqrt{3}$. To verify the latter integral using only the preceding section, reflect each tagged partition of $[0, 1]$ into $[-1, 0]$. The widths and squared heights are unchanged, so the two half-interval integrals are equal; their sum is $2/3$, and division by the interval length 2 gives the average. The theorem guarantees attainment of the average, not a preferred or

unique input.

Example 2.56 An average that is never attained Let $f(t) = 0$ for $0 \le t < 1$ and $f(t) = 2$ for $1 \le t \le 2$. Find its average on $[0, 2]$. Does it equal a value of f?

Solution *The function is integrable by finite-piece splitting, and its integral is* $0 \cdot 1 + 2 \cdot 1 = 2$. *Thus its average is* 1, *but its only values are* 0 *and* 2. *Integrability guarantees a well-defined average; continuity supplies the intermediate-value property used to attain it.*

2.7.2 Accumulation functions

Fix $x_0 \in [a, b]$ and let f be integrable on $[a, b]$. The function

$$A(x) = \int_{x_0}^{x} f(t)\,\mathrm{d}t, \qquad x \in [a, b],$$

is an *accumulation function*(累积函数). The letter t is bound by the integral; x is the free input that moves the upper endpoint. For $x < x_0$, the orientation convention is part of this definition. In particular, $A(x_0) = 0$ and interval splitting gives, for any $x, y \in [a, b]$,

$$A(y) - A(x) = \int_{x}^{y} f(t)\,\mathrm{d}t.$$

This identity expresses changes of accumulation in terms of the newly included or removed interval.

Proposition 2.5

If f is Riemann integrable on $[a, b]$ and $|f| \le M$, then its accumulation function satisfies

$$|A(y) - A(x)| \le M\,|y - x| \qquad (x, y \in [a, b]).$$

In particular, A is continuous, even if f is not.

♠

Proof Apply the absolute integral estimate to the preceding splitting identity, reversing the bounds when necessary. The bound tends to zero with $|y - x|$, proving continuity. □

The estimate is called a Lipschitz bound: input changes produce at most a fixed multiple as much output change. It requires bounded integrability of f, not continuity of f. Differentiability of the accumulation is a separate question.

2.7.3 Differentiating accumulation

Theorem 2.11 (Fundamental theorem, differentiation of accumulation)

Let f be Riemann integrable on $[a, b]$ and continuous at an interior point $x \in (a, b)$. Then

$$\frac{\mathrm{d}}{\mathrm{d}x} \int_{x_0}^{x} f(t)\,\mathrm{d}t = f(x).$$

If f is continuous on all of $[a, b]$, the accumulation is an antiderivative on (a, b), continuous on $[a, b]$, with one-sided derivatives $f(a)$ and $f(b)$ at the endpoints.

♡

> ***Proof Strategy.*** The difference quotient is the average of f over the short interval between x and $x + h$. Subtract $f(x)$ *inside* that average. Continuity makes every height difference on the short interval small, so its integral is bounded by "small height times interval length." Division by the same length leaves a small error. This works for both signs of h and does not require an explicit formula for the accumulation.

Proof For $h \neq 0$ with $x + h \in [a, b]$, splitting and the constant integral give

$$\frac{A(x+h) - A(x)}{h} - f(x) = \frac{1}{h} \int_x^{x+h} \big(f(t) - f(x)\big)\, \mathrm{d}t.$$

Given $\varepsilon > 0$, continuity at x gives $\delta > 0$ such that $|f(t) - f(x)| < \varepsilon$ whenever $t \in [a, b]$ and $|t - x| < \delta$. For $0 < |h| < \delta$, every point between x and $x + h$ satisfies this bound. The absolute integral estimate, with the bounds ordered if $h < 0$, gives

$$\left| \frac{A(x+h) - A(x)}{h} - f(x) \right| \leq \frac{1}{|h|}\, \varepsilon\, |h| = \varepsilon.$$

Thus $A'(x) = f(x)$. At an endpoint the same estimate uses only the allowed sign of h, proving the one-sided statements. □

Alternative proof when f is continuous throughout Apply Theorem 2.10 to the closed interval with endpoints $x, x + h$. There is c_h between them with $\int_x^{x+h} f(t)\, \mathrm{d}t = f(c_h)h$; when $h < 0$, this follows by reversing the increasing-bound formula. The difference quotient is $f(c_h)$. Since $|c_h - x| \leq |h|$, continuity gives $f(c_h) \to f(x)$. □

The short-strip estimate needs continuity only at the differentiation point. The mean-value route is shorter when continuity on every nearby interval is already known, but its sampled point depends on h and cannot be chosen in advance.

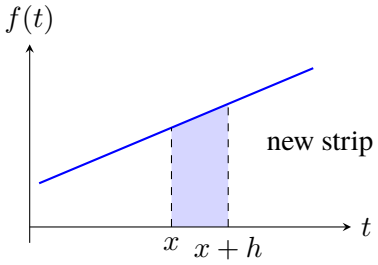


Figure 2.9: For $h > 0$, the added accumulation is a strip of width h. Its average height tends to $f(x)$; negative h removes a strip and is handled by orientation.

Corollary 2.1

Every continuous function on an open interval I has an antiderivative there. For any $x_0 \in I$, all its antiderivatives are

$$F(x) = \int_{x_0}^{x} f(t)\, \mathrm{d}t + C, \qquad C \in \mathbb{R}.$$

♡

Proof For each $x \in I$, the closed segment joining x_0 and x lies in I, where f is continuous and therefore integrable. Near any chosen x, apply the theorem on a slightly larger closed segment inside I containing x_0 and that neighbourhood. It gives derivative $f(x)$. The complete family follows from Theorem 2.6. □

This is an existence construction, even when the table of elementary antiderivatives does not produce a usable expression.

Example 2.57 A function defined by its accumulation Let $E(x) = \int_0^x e^{-t^2}\, \mathrm{d}t$ on $\mathbb{R}$. Find $E'(x)$, $E(0)$, and $\lim_{x \to 0} E(x)/x$.

Solution *The integrand is continuous on every finite interval, so E is well-defined and $E'(x) = e^{-x^2}$. Equal bounds give $E(0) = 0$. By the derivative definition, $\lim_{x \to 0} E(x)/x = E'(0) = 1$. These facts require no guessed elementary expression for E and no general rule for interchanging derivatives and integrals.*

Solution ***A direct second route to the limit.*** *For $x \neq 0$ and t between 0 and x, we have $e^{-x^2} \leq e^{-t^2} \leq 1$.*

The quotient $E(x)/x$ is the average on that segment: for $x < 0$, both the integral and denominator reverse sign. Therefore $e^{-x^2} \le E(x)/x \le 1$ for either sign of x, and the squeeze theorem gives the limit 1. This route displays an explicit bound; the derivative route obtains the limit immediately once E' is known.

Example 2.58 What can happen at a discontinuity On $[-1,1]$, let $h(t) = 0$ for $t < 0$ and $h(t) = 1$ for $t \ge 0$. Let $p(0) = 1$ and $p(t) = 0$ elsewhere. Find $H(x) = \int_0^x h(t)\,\mathrm{d}t$ and $P(x) = \int_0^x p(t)\,\mathrm{d}t$, and examine their derivatives at 0.

Solution *Finite changes and constant integrals give $H(x) = 0$ for $x \le 0$ and $H(x) = x$ for $x \ge 0$. It is continuous, but the left and right difference quotients at zero are 0 and 1, so $H'(0)$ does not exist. The isolated spike has zero integral on each segment, so $P(x) = 0$ everywhere. Thus $P'(0) = 0$, even though $p(0) = 1$. At a discontinuity of an integrable input, the accumulation can fail to be differentiable, or it can be differentiable with derivative different from the assigned input value. The theorem's continuity hypothesis supplies a guarantee; it is not claimed to be a necessary condition for the accumulation to have a derivative.*

2.7.4 Evaluating an integral at the endpoints

Theorem 2.12 (Fundamental theorem, Newton–Leibniz formula)

Suppose f is continuous on $[a,b]$, $a < b$, and F is continuous on $[a,b]$ and differentiable on (a,b) with $F' = f$ there. Then

$$\int_a^b f(x)\,\mathrm{d}x = F(b) - F(a) = [F(x)]_a^b .$$

The same formula holds with equal or reversed bounds by the orientation convention. ♡

> ***Proof Strategy.*** The accumulation function gives one primitive whose starting value is known. Any proposed primitive has the same derivative, so their difference is constant on the interior. Continuity carries this constant relation to the endpoints, where subtraction eliminates it.

Proof Set $A(x) = \int_a^x f(t)\,\mathrm{d}t$. The preceding theorem gives $A' = f$ on (a,b), and the Lipschitz estimate gives continuity on $[a,b]$. Thus $(F - A)' = 0$ on (a,b), and Lemma 2.1 gives $F - A = C$ there. By continuity the same identity holds at a and b. Since $A(a) = 0$, we have $C = F(a)$ and $A(b) = F(b) - F(a)$, as asserted. Reversing endpoints negates both sides; equal endpoints make both zero. □

The additive constant cancels: $[F + C]_a^b = F(b) - F(a)$. The formula changes the evaluation method, not the definition of the definite integral. For example, $\int_0^1 x^2\,\mathrm{d}x = [x^3/3]_0^1 = 1/3$, agreeing with the sum calculation of the previous section.

Example 2.59 Endpoint evaluation compared with a sum limit Evaluate $\int_0^1 e^{2x}\,\mathrm{d}x$ by two methods.

Solution ***Method 1: verify a primitive and use endpoints.*** *The chain rule gives $(e^{2x}/2)' = e^{2x}$. The integrand and primitive are continuous on $[0,1]$, so*

$$\int_0^1 e^{2x}\,\mathrm{d}x = \left[\frac{e^{2x}}{2}\right]_0^1 = \frac{e^2 - 1}{2}.$$

Solution ***Method 2: a geometric sum.*** *Continuity establishes integrability. Put $r_n = e^{2/n}$. The right-endpoint sum is*

$$\frac{1}{n}\sum_{k=1}^{n} r_n^k = \frac{r_n(r_n^n - 1)}{n(r_n - 1)} = \frac{e^{2/n}(e^2 - 1)}{n(e^{2/n} - 1)}.$$

The finite geometric formula follows by multiplying the sum by $r_n - 1$ and telescoping. Since $(e^u - 1)/u \to 1$ as $u \to 0$, its denominator tends to 2, while $e^{2/n} \to 1$. The limit is $(e^2 - 1)/2$.

Endpoint evaluation avoids a sum identity and limit calculation. The sum route ties the answer back to sampled accumulation and independently verifies the value using only continuity and the exponential derivative at zero.

Example 2.60 A candidate can be checked without a new integration technique Use a candidate of the form $F(x) = e^x(a\sin x + b\cos x)$ to evaluate $\int_0^{\pi/2} e^x \cos x \, dx$.

> ***Idea.*** Differentiating either $e^x \sin x$ or $e^x \cos x$ stays within their two-function span. Choose coefficients so that the sine terms cancel and the cosine coefficient becomes one. This finite coefficient problem suggests a primitive without requiring an integration-by-parts formula.

Solution *The product rule gives* $F' = e^x[(a-b)\sin x + (a+b)\cos x]$. *Thus* $a - b = 0$ *and* $a + b = 1$, *giving* $a = b = 1/2$. *This verifies the candidate everywhere. The endpoint formula yields*

$$\int_0^{\pi/2} e^x \cos x \, dx = \left[\frac{e^x(\sin x + \cos x)}{2}\right]_0^{\pi/2} = \frac{e^{\pi/2} - 1}{2}.$$

Example 2.61 Two invalid endpoint shortcuts Explain why $[-1/x]_{-1}^{1} = -2$ does not evaluate a proper integral of $1/x^2$ across 0. Also let $f = 0$ on $[0, 1]$ and set $F = 0$ on $[0, 1)$, $F(1) = 5$. Why is $F(1) - F(0)$ not the integral of f?

Solution *The function* $1/x^2$ *is undefined at* 0 *and unbounded nearby; assigning any value at* 0 *would not make it a bounded proper Riemann integrand on* $[-1, 1]$. *The proposed primitive also fails to extend continuously across* 0. *Neither the integral definition nor the theorem's hypotheses are met. The negative result would additionally contradict nonnegative-integrand order if a proper integral existed. A later theory of improper integrals will treat singularities with separate limits.*

In the second example, $F' = 0 = f$ *on* $(0, 1)$, *but* F *is not continuous at* 1. *The actual integral is* 0, *whereas the endpoint difference is* 5. *Interior derivative agreement does not license arbitrary endpoint values.*

We can now compare three properties that must not be identified: continuity of the input, integrability of the input, and existence of a primitive. Use the spike p of Example 2.58, and define on $\mathbb{R}$

$$G(0) = H(0) = 0, \qquad G(x) = x^2 \sin(1/x), \qquad H(x) = x^2 \sin(1/x^2) \quad (x \neq 0).$$

The table concerns Riemann integrability on $[-1, 1]$ and the primitive identity at every point of $(-1, 1)$.

Table 2.1: Continuity supplies both guarantees, but neither integrability nor the existence of a primitive implies the other.

Input	Continuous	Riemann integrable	Has a primitive
$2x$	yes	yes	yes: x^2
p	no	yes	no
G'	no	yes	yes: G
H'	no	no	yes: H

Here are the checks behind **Table** 2.1. The spike has integral zero by finite-value invariance. If $F' = p$ on $(-1, 1)$, the zero-derivative lemma would make F constant separately on $(-1, 0)$ and $(0, 1)$. Continuity at 0 would force both constants to equal $F(0)$. Then $F'(0) = 0$, contradicting $p(0) = 1$. Thus an integral can ignore one assigned value that the primitive identity cannot ignore.

For G, the already verified derivative is $G'(0) = 0$ and $G'(x) = 2x\sin(1/x) - \cos(1/x)$ for $x \neq 0$. It is bounded on $[-1, 1]$ and continuous except at 0, so it is integrable by Theorem 2.8; its primitive is G by

construction. For H, the anchored quotient $H(h)/h$ tends to zero, while off 0 the rules give

$$H'(x) = 2x\sin(1/x^2) - \frac{2}{x}\cos(1/x^2).$$

Along $x_n = (2\pi n)^{-1/2}$, $n \geq 1$, this equals $-2/x_n \to -\infty$. Hence H' has a primitive but fails the boundedness required for a proper Riemann integral. These examples explain both failed implications in the table. The continuity hypothesis in our endpoint theorem is a sufficient condition; the theorem does not say that every discontinuous input prevents an endpoint formula. The supplement exercises verify such a formula for the particular bounded derivative G' by working first on intervals away from 0.

2.7.5 Moving limits and applications

Proposition 2.6

Let f be continuous on an open interval I, and let u, v be differentiable functions on an open interval J whose values lie in I. Then

$$\frac{\mathrm{d}}{\mathrm{d}x}\int_{u(x)}^{v(x)} f(t)\,\mathrm{d}t = f(v(x))v'(x) - f(u(x))u'(x), \qquad x \in J.$$

No ordering assumption between $u(x)$ and $v(x)$ is required. ♠

Proof Fix $c \in I$ and set $A(z) = \int_c^z f(t)\,\mathrm{d}t$. Splitting gives the integral as $A(v(x)) - A(u(x))$, including reversed and coincident limits. The fundamental theorem and chain rule give the stated derivative. □

Example 2.62 Bounds that cross Differentiate $B(x) = \int_x^{x^2} \sqrt{1+t^2}\,\mathrm{d}t$ on $\mathbb{R}$.

Solution *The integrand is continuous on $\mathbb{R}$, and both bounds are differentiable. Hence*

$$B'(x) = 2x\sqrt{1+x^4} - \sqrt{1+x^2}.$$

Although the bounds agree at $x = 0$ and $x = 1$, no case split is needed. In fact $B(0) = B(1) = 0$, but $B'(0) = -1$ and $B'(1) = \sqrt{2}$. Equality of the bounds at a single input gives a zero value, not necessarily a zero derivative.

The proposition assumes that the integrand is a function of the integration variable alone. If the free input also occurs inside the integrand, that additional dependence must be accounted for. The following example does so using algebra and already proved rules.

Test the issue first with the simplest input $f \equiv 1$:

$$\int_0^x (x-t)\,\mathrm{d}t = x^2 - \frac{x^2}{2} = \frac{x^2}{2}.$$

Its derivative is x, whereas substituting $t = x$ into the integrand gives zero. The missing contribution comes from changing the height $x - t$ at every old point of the interval, not just adding the new endpoint strip.

Example 2.63 A parameter inside the integrand For continuous $f\colon \mathbb{R} \to \mathbb{R}$, define $K(x) = \int_0^x (x-t)f(t)\,\mathrm{d}t$. Find K' and K'', and check the values at 0.

> ***Idea.*** Substituting the upper bound into $(x-t)f(t)$ would give zero and miss the change of the integrand on the entire old interval. Expand the factor $x - t$: for each fixed x, linearity separates the free parameter from two ordinary accumulation functions.

Solution *Write $A(x) = \int_0^x f(t)\,\mathrm{d}t$ and $C(x) = \int_0^x tf(t)\,\mathrm{d}t$. Then $K = xA - C$. Since $A' = f$ and $C' = xf(x)$, the product rule gives*

$$K'(x) = \underbrace{A(x)}_{\text{free parameter}} + \underbrace{xf(x) - xf(x)}_{\text{endpoint terms cancel}} = \int_0^x f(t)\,\mathrm{d}t, \qquad K''(x) = f(x).$$

Equal bounds give $K(0) = K'(0) = 0$. Every step is valid also for $x < 0$ by oriented linearity. Only continuity of f is needed; no derivative f' or general rule for differentiation under the integral sign has been used.

For a continuously differentiable Q, we obtain the *net change principle* (净变化原理)

$$Q(b) - Q(a) = \int_a^b Q'(t)\,\mathrm{d}t.$$

The word "net" matters when the rate changes sign. For one-dimensional motion with continuous velocity v, displacement is $\int_a^b v(t)\,\mathrm{d}t$, whereas distance travelled is $\int_a^b |v(t)|\,\mathrm{d}t$. On an interval where v has a fixed sign, the absolute endpoint change equals the speed integral; splitting at sign changes adds these distances. The signed lobes in **Figure** 2.8 can now be read as the velocity record for the next example: negative area is backward displacement, while both lobe areas contribute to distance.

Example 2.64 Recovering position and distinguishing distance A particle has velocity $v(t) = t - 1$ for $0 \le t \le 3$ and position $s(0) = 4$. Find its position, displacement and distance travelled during this interval.

Solution ***Method 1: integrate the rate and the speed.*** *The unique position is $s(t) = 4 + \int_0^t (u-1)\,\mathrm{d}u = 4 + t^2/2 - t$. Displacement is $\int_0^3 (t-1)\,\mathrm{d}t = 3/2$. Since velocity changes sign at 1, distance is*

$$\int_0^1 (1-t)\,\mathrm{d}t + \int_1^3 (t-1)\,\mathrm{d}t = \frac{1}{2} + 2 = \frac{5}{2}.$$

Solution ***Method 2: position changes on the two legs.*** *Reversing differentiation and using $s(0) = 4$ gives the same $s(t) = 4 + t^2/2 - t$. For $0 \le a < b \le 1$, the net-change identity and $v \le 0$ imply $s(b) \le s(a)$; for $1 \le a < b \le 3$ they imply $s(b) \ge s(a)$. Thus the two legs have endpoints $s(0) = 4$, $s(1) = 7/2$, $s(3) = 11/2$. Their lengths add to $|7/2 - 4| + |11/2 - 7/2| = 5/2$, whereas the final minus initial position is $3/2$.*

The speed integral works directly from a rate record. Endpoint changes work once direction is known on each leg; taking only the absolute final displacement would lose the reversal.

To read this example as a graph of accumulation, subtract the initial position:

$$A(x) = s(x) - 4 = \int_0^x (t-1)\,\mathrm{d}t = \frac{x^2}{2} - x = \frac{(x-1)^2}{2} - \frac{1}{2}.$$

It decreases from 0 to $-1/2$ on $[0,1]$ and then increases on $[1,3]$. These comparisons follow from the integral signs in $A(v) - A(u) = \int_u^v (t-1)\,\mathrm{d}t$; they do not require the later derivative-sign theorem. The displayed square also directly verifies the minimum at $x = 1$. On $(1,2)$ the current rate is positive, but the accumulated value is still negative. At $x = 2$ the earlier negative contribution has just been cancelled: $A(2) = 0$, although the rate is 1.

In **Figure** 2.10, a positive rate tells us how the lower graph changes, not whether that graph lies above its horizontal axis.

Example 2.65 Recognising a sum as an integral Find $\lim\limits_{n\to\infty} \sum\limits_{k=1}^{n} \dfrac{1}{2n+k}$, and give an error estimate.

> ***Idea.*** Factor n from the denominator to expose a width $1/n$ and a sampled height depending on k/n. The endpoints k/n identify the interval $[0,1]$. Monotonicity then supplies a second interpretation as lower sums, allowing a quantitative bound.

Solution ***Method 1: recognise and evaluate the Riemann sum.*** *The sum is $\frac{1}{n}\sum_{k=1}^{n}(2 + k/n)^{-1}$ for the continuous function $f(x) = 1/(2+x)$ on $[0,1]$. Therefore its limit is*

$$\int_0^1 \frac{\mathrm{d}x}{2+x} = [\ln(2+x)]_0^1 = \ln\frac{3}{2}.$$

The primitive is valid because $2 + x > 0$ on the full interval.

Solution ***Method 2: squeeze by monotone rectangles.*** *The function is decreasing, so the given right sum R_n*

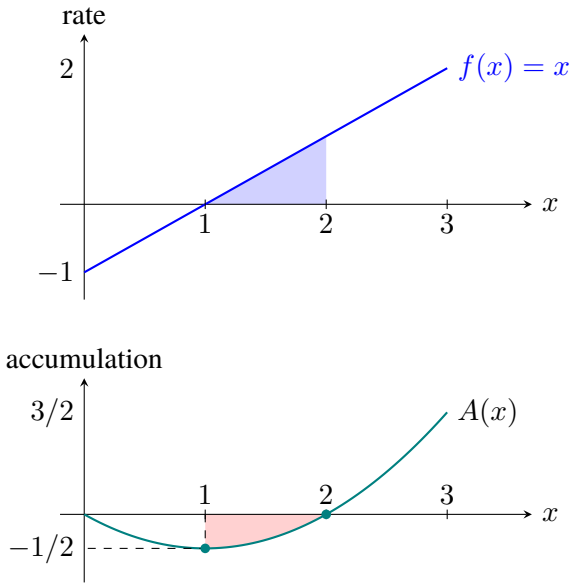


Figure 2.10: The inputs are vertically aligned. On the highlighted interval $(1, 2)$, the rate is positive while the accumulation is negative and increasing. The sign of $f(x)$ describes current change; the sign of $A(x)$ also depends on the earlier history and chosen base point.

is the lower sum. The left sum L_n exceeds it by $[f(0) - f(1)]/n = 1/(6n)$. Consequently

$$0 \le \ln \frac{3}{2} - R_n \le \frac{1}{6n}.$$

This both forces the same limit and gives a guaranteed approximation error for each n.

Recognition identifies the limiting object quickly; upper and lower comparisons additionally locate every finite approximation relative to the answer.

2.7.6 Methods and further remarks

Identify which direction of the theorem is needed. For an accumulation, use a short-interval difference quotient to obtain a derivative. For a definite integral, first verify a primitive, then subtract its endpoint values. The first direction proves existence; the second accelerates evaluation.

Check hypotheses over the whole interval. Continuous input ensures integrability and the derivative identity. Bounded integrable input alone ensures continuous accumulation, but says less about its derivative at discontinuities. Use **Table** 2.1 to distinguish this guarantee from the existence of a primitive. A candidate primitive must extend continuously to evaluation endpoints; a singularity inside the interval cannot be skipped by endpoint arithmetic.

Track free inputs, dummy variables and orientation. For two moving bounds, express the integral as a difference of two accumulations and apply the chain rule. Coincident or crossed bounds need no new formula. When x also appears inside the integrand, expose that dependence by legitimate algebra, as in Example 2.63; the endpoint-only rule is insufficient.

Choose a representation suited to the question. Primitive recognition makes exact evaluation efficient, geometric splitting explains signed versus total change, and Riemann-sum recognition converts sampled data into an integral. Preserve an estimate from upper and lower sums when a finite approximation matters. When reading an accumulation graph, keep current rate, change in accumulation, and accumulated value distinct; changing the base point shifts the last of these by a constant.

> ***Looking Ahead.*** The chain and product rules have already helped us verify useful candidates. The next chapter will develop systematic substitution and integration by parts, followed by further integration techniques, applications and numerical approximation. A later treatment of improper integrals will introduce the separate limits needed for singularities and infinite intervals. Those extensions require their own hypotheses; the proper integral and the fundamental theorem established here remain their starting point.

Exercise 2.7

1. Find the average of $f(x) = 2x$ on $[0, 2]$ and a point where it is attained.
2. For $A(x) = \int_1^x (1 + t^4)\,\mathrm{d}t$, find $A'(x)$ and $A(1)$.
3. Find $A(x) = \int_0^x \cos t\,\mathrm{d}t$ and verify both its derivative and its value at 0.
4. Evaluate $\int_{-1}^{2}(3x^2 - 2)\,\mathrm{d}x$ and check a primitive by differentiation.
5. Evaluate $\int_1^e x^{-1}\,\mathrm{d}x$, stating why the logarithmic primitive is valid on the interval.
6. Evaluate $\int_0^1 (1 + x^2)^{-1}\,\mathrm{d}x$ and compare with the bounds in Example 2.55.
7. Compute $\int_2^0 (1 + x)\,\mathrm{d}x$ in two ways: reverse the limits, and use the oriented endpoint formula.
8. Find the average of x^2 on $[0, 3]$ and all points of the interval attaining it.
9. Differentiate $\int_0^{x^3} e^{-t^2}\,\mathrm{d}t$ on $\mathbb{R}$.
10. Differentiate $\int_{\sin x}^{\cos x} t^2\,\mathrm{d}t$ on $\mathbb{R}$; explain why crossed bounds cause no exception.
11. Find a, b so that $(ax + b)e^x$ is a primitive of $(x + 2)e^x$, and evaluate $\int_0^1 (x + 2)e^x\,\mathrm{d}x$.
12. On $[-1, 1]$, let $f(t) = -1$ for $t < 0$ and $f(t) = 2$ for $t \geq 0$. Find $A(x) = \int_0^x f(t)\,\mathrm{d}t$ and determine its continuity and differentiability at 0.
13. A particle has $s'(t) = 2t - 2$ on $[0, 3]$ and $s(0) = 1$. Find $s(3)$, displacement and distance travelled, justifying the split in the distance integral.
14. Express $\lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} \sqrt{1 - (k/n)^2}$ as an integral and evaluate it geometrically.
15. For continuous f on $\mathbb{R}$, let $K(x) = \int_0^x (x - t)f(t)\,\mathrm{d}t$. Prove that K is the unique twice differentiable function with $K'' = f$ and $K(0) = K'(0) = 0$.
16. A continuous f on $[0, 1]$ has $\int_0^1 f(x)\,\mathrm{d}x = 0$. Must $A(x) = \int_0^x f(t)\,\mathrm{d}t$ be identically zero? Give a verified counterexample. What extra conclusion follows if $f \geq 0$?
17. Find the unique differentiable $F\colon \mathbb{R} \to \mathbb{R}$ satisfying $F'(x) = 2x + \int_0^x \cos t\,\mathrm{d}t$ and $F(0) = 3$. Verify the identity without leaving an unevaluated integral.
18. Find $\lim_{n\to\infty} \sum_{k=1}^{n} n/(n^2 + k^2)$ and bound the error from this limit by $1/(2n)$ using monotone sums.
19. Let $f(x) = \sin x$ on $[0, 2\pi]$ and $A(x) = \int_0^x f(t)\,\mathrm{d}t$. Find A and show that it is nonnegative everywhere, even though it decreases on $[\pi, 2\pi]$. Identify its maximum and where it is attained, using an explicit formula or integral signs rather than an unproved derivative test.
20. For continuous f on an open interval I and $a, b \in I$, compare $A_a(x) = \int_a^x f(t)\,\mathrm{d}t$ and $A_b(x) = \int_b^x f(t)\,\mathrm{d}t$. Find their difference and derivatives, and explain why they describe the same rate but need not have the same sign at a given input.
21. Evaluate $J(x) = \int_x^{x+1} (t - x)\,\mathrm{d}t$ on $\mathbb{R}$ and find $J'(x)$. Derive the same derivative by writing $J(x) = \int_x^{x+1} t\,\mathrm{d}t - x\int_x^{x+1} 1\,\mathrm{d}t$ and applying the proved rules. Identify the term that an endpoint-only calculation would miss.
22. Let f be continuous near a and let $k \in \mathbb{R}$ be fixed. Evaluate $\lim_{h\to 0} h^{-1} \int_a^{a+kh} f(t)\,\mathrm{d}t$. Justify the answer for both signs of h, for $k < 0$, and for $k = 0$, and give both an accumulation-derivative argument and a direct integral estimate.

23. Suppose f is continuous on $[0,1]$ and $F(x)=\int_0^{x^2} f(t)\,\mathrm{d}t$ for $-1<x<1$. Prove that F is differentiable, including at 0. If F' vanishes throughout $(-1,1)$, prove $f=0$ on all of $[0,1]$, explaining separately how the endpoint values are determined.

Supplement Exercises

S1. Let f be continuous on $\mathbb{R}$ with positive period T. Prove that $F(x)=\int_0^x f(t)\,\mathrm{d}t$ is T-periodic if and only if $\int_0^T f(t)\,\mathrm{d}t=0$. Show that a continuous nonnegative periodic f with a periodic primitive must be identically zero.

S2. Let f be integrable on $[a,b]$, and suppose its finite one-sided limits $f(c-)$ and $f(c+)$ exist at an interior point c. Show that the left and right derivatives of $A(x)=\int_a^x f(t)\,\mathrm{d}t$ at c are these limits. Deduce exactly when $A'(c)$ exists, and explain why the assigned value $f(c)$ is irrelevant here.

S3. Suppose f is continuous on $[a,b]$ and w is continuous, nonnegative and not identically zero there, with $a<b$. Prove that $\int_a^b f(x)w(x)\,\mathrm{d}x=f(c)\int_a^b w(x)\,\mathrm{d}x$ for some $c\in[a,b]$. Give a continuous sign-changing weight for which such a c need not exist.

S4. For $E(x)=\int_0^x e^{-t^2}\,\mathrm{d}t$, prove that E is odd and strictly increasing. Show $xe^{-x^2}<E(x)<x$ for $x>0$, using integral order rather than a series expansion.

S5. Let f be continuous on $\mathbb{R}$ and define $H(x)=\int_0^x (x-t)^2 f(t)\,\mathrm{d}t$. Find H', H'', H''' and their initial data through order two, by expanding the kernel and applying only the fundamental theorem and product rule.

S6. Let f be continuous on $\mathbb{R}$ and suppose $\int_0^x f(t)\,\mathrm{d}t=xf(x)$ for every real x. Prove that f is constant. Do not assume at the outset that f is differentiable.

S7. Find all continuous $f\colon\mathbb{R}\to\mathbb{R}$ satisfying $f(x)=1+\int_0^x f(t)\,\mathrm{d}t$. Prove existence and uniqueness by first deriving differentiability and then differentiating $e^{-x}f(x)$.

S8. Let f be continuous on an open interval containing c. Prove $\lim_{h\to0}\frac{1}{2h}\int_{c-h}^{c+h} f(t)\,\mathrm{d}t=f(c)$, including $h<0$. Construct a bounded function with a jump at c for which the same symmetric limit exists although the corresponding accumulation is not differentiable at c.

S9. For $\beta>0$, set $H_\beta(0)=0$ and $H_\beta(x)=x^2\sin(1/|x|^\beta)$ for $x\neq0$. Prove differentiability on $\mathbb{R}$, and determine exactly when H_β' is bounded on $[-1,1]$ and when it is continuous at 0. In every unbounded case, give a specific sequence and explain why the expression $\int_{-1}^{1} H_\beta'(x)\,\mathrm{d}x$ is not a proper Riemann integral, despite the existence of the primitive. Compare the threshold cases with **Table** 2.1.

S10. Let f be continuous on $\mathbb{R}$, and fix $h>0$. Set $M_h(x)=h^{-1}\int_x^{x+h} f(t)\,\mathrm{d}t$. Find $M_h'(x)$ and prove that M_h is constant if and only if f is h-periodic. For periodic f, identify the constant and explain why it need not be zero.

S11. Let $G(0)=0$ and $G(x)=x^2\sin(1/x)$ for $x\neq0$, and put $f=G'$. Prove that $A(x)=\int_0^x f(t)\,\mathrm{d}t$ equals $G(x)$ for $-1\le x\le1$. Deduce that $A'(0)=f(0)$ even though f has no limit at 0.

> ***Optional Hint.*** First apply the endpoint theorem between x and a nonzero point c on the same side of 0. Then let $c\to0$, using boundedness of f to control the omitted integral and continuity of G to control its omitted value.

S12. For real constants a, λ, find all continuous $f\colon \mathbb{R} \to \mathbb{R}$ satisfying

$$f(x) = a + \lambda \int_0^x (x - t) f(t)\, \mathrm{d}t \qquad (x \in \mathbb{R}).$$

Treat $\lambda > 0$, $\lambda = 0$, and $\lambda < 0$ separately; prove existence and uniqueness using only this chapter's tools.

> ***Optional Hint.*** First derive $f'' = \lambda f$ and the initial data from the integral equation. For the difference of two candidates, positive $\lambda = r^2$ suggests differentiating $e^{rx}(f' - rf)$; negative $\lambda = -r^2$ suggests the nonnegative expression $(f')^2 + r^2 f^2$. Verify that a solution of the derivative equation with those data also satisfies the original integral equation.

Chapter 2 Problem-Solving Toolkit

The chapter connects three questions: how a function changes at a point, how well a linear expression describes that change, and how changes accumulate over an interval. Choosing a tool begins by deciding which of these questions is being asked. A symbolic formula for a derivative does not settle its value at an excluded point, and a formula for a primitive does not automatically justify evaluation across a singularity.

A decision route

Step 1. Fix the object and the point or interval. Is the task to find $f'(a)$, approximate an increment, recover a function from its rate, or evaluate an accumulated quantity? Write the original domain before applying a formula.

Step 2. Distinguish a formula point from a joining point. Inside a smooth formula's domain, use the differentiation rules. At a piecewise join or a newly assigned value, begin with continuity and the difference quotient. Compute one-sided quotients when needed.

Step 3. State the precision of an approximation. The identity $f(a+h) = f(a) + f'(a)h + o(h)$ controls the error relative to h at a fixed base point. If leading terms cancel, retain the remainders or find an exact identity; the linear approximation alone may not determine the requested limit.

Step 4. For an integral, separate existence from evaluation. A Riemann integral is defined by sums; a primitive is defined by a derivative. Check the assumptions of the appropriate FTC statement before connecting them.

Step 5. Check the answer in the direction opposite to its construction. Differentiate a proposed primitive, substitute initial data, compare a sum with its mesh width, or bound an integral by its integrand. Keep exact identities, limiting statements and numerical approximations distinct.

Signals, first moves and conditions

Signal	First move	Check
Piecewise definition at a	Form $(f(a+h)-f(a))/h$.	Use the assigned value $f(a)$ and both admissible sides.
Nested functions	Differentiate from the outside inward.	Each derivative is evaluated at the correct inner argument.
Positive product or variable power	Take logarithms first.	Positivity on an interval; restore the original function after differentiation.
Implicit or inverse relation	Differentiate the known relation.	Existence of the local inverse and a nonzero derivative where required.
Small input change	Write the linear term and a named remainder.	Base point fixed; error is $o(h)$, not necessarily $O(h^2)$.
Repeated derivatives	Look for a cycle, falling powers, or a product pattern.	Index range, factorials and chain-rule factors.
Primitive with initial data	Find a primitive family, then fix its constant.	Separate constants on disconnected intervals.
Moving integral endpoints	Write a difference of accumulations.	Continuity at the relevant endpoints and their chain-rule factors.
Riemann-sum limit	Identify Δx and sample points.	The entire sum matches one fixed integrand and interval.

A joining-point check also tests regularity

Example 2.66 Differentiable does not mean continuously differentiable Revisit the function from Example 2.7: $f(0)=0$ and $f(x)=x^2\sin(1/x)$ for $x\neq 0$. Find its derivative everywhere and determine whether that derivative is continuous at zero.

> ***Use different tools for the two questions.*** The quotient at zero contains a shrinking factor h, but differentiating the oscillation away from zero produces $1/x^2$. That factor cancels the small amplitude x^2. Thus the derivative's continuity is a new question, not a consequence of the first difference-quotient limit.

Solution *At zero,*

$$\left|\frac{f(h)-f(0)}{h}\right| = |h\sin(1/h)| \le |h| \longrightarrow 0,$$

so $f'(0)=0$. *For* $x\neq 0$, *the product and chain rules give*

$$f'(x) = 2x\sin(1/x) - \cos(1/x).$$

Along $x_n = 1/(2\pi n)$ *and* $y_n = 1/((2n+1)\pi)$, $n\ge 1$, *these derivatives equal* -1 *and* 1, *respectively. Both sequences tend to zero, so* f' *has no limit there. The function is differentiable on* $\mathbb{R}$ *but is not* C^1 *near zero.*

This revisits the regularity distinction from Section 2.4 to connect the definition, differentiation rules and sequential criterion for limits. The punctured derivative formula alone cannot define $f'(0)$.

An average near a vanishing denominator

Example 2.67 Estimate first, differentiate where the quotient is defined Let f be continuous on an open interval containing zero and put $A(x) = x^{-1}\int_0^x f(t)\,\mathrm{d}t$ for $x \neq 0$ in that interval. Extend A continuously to zero and find $A'(x)$ for $x \neq 0$.

> ***Compare an average with a constant average.*** The integral of $f(0)$ over the oriented interval is $xf(0)$. Subtract it before estimating: the length $|x|$ then cancels the denominator, leaving only the oscillation of f near zero.

Solution *For either sign of x,*

$$|A(x) - f(0)| = \left|\frac{1}{x}\int_0^x (f(t) - f(0))\,\mathrm{d}t\right| \leq \sup_{|t|\leq|x|} |f(t) - f(0)| \longrightarrow 0,$$

where $|x|$ is small enough that the symmetric interval lies in the domain. Thus set $A(0) = f(0)$. For $x \neq 0$, the FTC and quotient rule give

$$A'(x) = \frac{xf(x) - \int_0^x f(t)\,\mathrm{d}t}{x^2} = \frac{f(x) - A(x)}{x}.$$

The displayed quotient is not a formula at $x = 0$. Indeed, for $f(t) = |t|$, one obtains $A(x) = |x|/2$, whose continuous extension is not differentiable there. For $f(t) = 1 + t$, instead $A(x) = 1 + x/2$, including at zero.

Continuity is enough to extend the average; it is not enough to differentiate that extension. The two basic examples locate the precise limit of the conclusion.

Proof and verification habits

- **Keep the base point in a derivative quotient.** Changing only $f(a)$ changes every quotient anchored there. For practice, return to Example 2.6 and the joining-point exercises in Section 2.1.
- **Give the remainder a name.** Write $r(h) = h\eta(h)$ with $\eta(h) \to 0$ before composing or subtracting approximations. Compare the two precision requirements in Example 2.20.
- **Check all dependence on the external variable.** An endpoint rule does not differentiate an extra x inside the integrand. The algebraic decomposition in Example 2.63 and the moving-parameter exercises in Section 2.7 supply current-tool routes.
- **Use an estimate when an exact formula is unnecessary.** Integral order and the bound $|\int_a^b f| \leq |b - a| \max |f|$ can settle signs, continuity and small-interval limits. An estimate must hold on the whole interval to which it is applied.
- **Keep domain components separate.** A primitive on either side of a pole does not justify an integral across it. Recall Example 2.61 before using endpoint evaluation.

> ***Looking Ahead.*** The chain and product rules will lead to substitution and integration by parts in the next chapter. Mean value theorems and Taylor formulas will later provide additional global estimates and higher-order local models. Those later tools are not needed for this Toolkit's arguments.

Chapter 3 Techniques and Applications of Integration

The fundamental theorem turns accumulation into an endpoint calculation when a useful primitive is available. We now learn to recognize structures that produce such primitives, construct integrals from geometric and physical models, and estimate them when exact evaluation is inconvenient. Throughout, a method is justified by its domain, hypotheses and interpretation.

3.1 Substitution

知识点概要

- ❑ *Reverse the chain rule.*
- ❑ *Recognize and complete a differential.*
- ❑ *Choose an inverse branch on its domain.*
- ❑ *Change definite bounds with orientation.*
- ❑ *Use reflection and periodicity.*
- ❑ *Verify the result by differentiation.*

The fundamental theorem reduces an integral to finding a primitive, but the primitive table cannot list every composite expression. The chain rule explains how to enlarge that table: a complicated expression may become a familiar one when treated as a single variable. The useful change is suggested by both the expression and the differential attached to it.

3.1.1 Reversing the chain rule

A *substitution*(换元)replaces one variable by a specified function of another, transforming the integrand and its differential together. For example, $u = 1 + x^2$ gives $\mathrm{d}u = 2x\,\mathrm{d}x$. This last equality abbreviates $u'(x) = 2x$; it does not allow the factor $2x$ to be discarded.

Theorem 3.1 (Substitution for primitives)

Let g be differentiable on an open interval I, let $g(I)$ lie in an open interval J, and let $F' = f$ on J. Then $F \circ g$ is a primitive of $f(g(x))g'(x)$ on I:

$$\int f(g(x))g'(x)\,\mathrm{d}x = F(g(x)) + C.$$

♡

Proof The chain rule gives $(F \circ g)'(x) = F'(g(x))g'(x) = f(g(x))g'(x)$. The primitive-family theorem supplies all primitives on I. □

The notation $\int f(u)\,\mathrm{d}u$ describes the simpler primitive problem, after which $u = g(x)$ must be restored. No inverse of g is needed for this theorem: composing a known primitive is legitimate even when g takes the same value several times. For instance, $g(x) = x^2$ gives a primitive of $2xe^{x^2}$ on all of $\mathbb{R}$. In contrast, replacing $\int e^{x^2}\,\mathrm{d}x$ by $\int e^u\,\mathrm{d}u$ loses the missing differential factor.

> ***Choosing a variable.*** Look first at an expression that is costly to expand, such as a power, logarithm, or denominator. Differentiate it mentally and compare the result with the remaining factors. A constant mismatch is harmless; a variable mismatch must be expressed in the new variable or dealt with by a different method.

Example 3.1 A composite power Evaluate $\int x(1+x^2)^5\,\mathrm{d}x$.

Solution *Set $u = 1+x^2$, so $x\,\mathrm{d}x = \frac{1}{2}\mathrm{d}u$. Then*

$$\int x(1+x^2)^5\,\mathrm{d}x = \frac{1}{2}\int u^5\,\mathrm{d}u = \frac{(1+x^2)^6}{12} + C.$$

Differentiating returns the integrand. Expansion would also work, but the substitution preserves the structure and shortens the calculation.

3.1.2 Completing a derivative and handling leftover factors

Example 3.2 A logarithmic derivative Evaluate $\int (x+1)/(x^2+2x+3)\,\mathrm{d}x$.

Solution *The denominator has derivative $2(x+1)$ and is $(x+1)^2+2>0$. Hence*

$$\int \frac{x+1}{x^2+2x+3}\,\mathrm{d}x = \frac{1}{2}\ln(x^2+2x+3) + C.$$

In general, $\int g'/g = \ln|g| + C$ on each interval where $g \neq 0$. Here positivity makes the absolute value unnecessary.

Example 3.3 One factor remains Evaluate $\int x^3\sqrt{1+x^2}\,\mathrm{d}x$.

Solution *With $u = 1+x^2$, only $x\,\mathrm{d}x$ is absorbed; the remaining x^2 becomes $u-1$. Thus*

$$\begin{aligned}\int x^3\sqrt{1+x^2}\,\mathrm{d}x &= \frac{1}{2}\int (u-1)u^{1/2}\,\mathrm{d}u\\ &= \frac{1}{2}\int (u^{3/2}-u^{1/2})\,\mathrm{d}u\\ &= \frac{1}{5}u^{5/2} - \frac{1}{3}u^{3/2} + C\\ &= \frac{1}{5}(1+x^2)^{5/2} - \frac{1}{3}(1+x^2)^{3/2} + C.\end{aligned}$$

This illustrates a useful test: after substitution, every factor must be written in the new variable.

Example 3.4 Translate the factor that is left over Evaluate $\int x/\sqrt{1+x}\,\mathrm{d}x$ on $x>-1$.

> ***Which expression should become the variable?*** The root suggests either naming its argument or naming the root itself. The numerator is not the derivative of either expression, but it can be expressed in either new variable. A substitution can be useful without an exact derivative match at first sight.

Solution ***Method 1: use the argument.*** *Set $u = 1+x>0$. Then $x = u-1$ and $\mathrm{d}x = \mathrm{d}u$, so*

$$\begin{aligned}\int \frac{x}{\sqrt{1+x}}\,\mathrm{d}x &= \int \frac{u-1}{\sqrt{u}}\,\mathrm{d}u = \int (u^{1/2}-u^{-1/2})\,\mathrm{d}u\\ &= \frac{2}{3}u^{3/2} - 2u^{1/2} + C = \frac{2}{3}(1+x)^{3/2} - 2\sqrt{1+x} + C.\end{aligned}$$

The two power terms arise from the entire numerator, not from replacing only its x by u.

Solution ***Method 2: use the root.*** *Set $t = \sqrt{1+x}>0$, so $x = t^2-1$ and $\mathrm{d}x = 2t\,\mathrm{d}t$. Then*

$$\int \frac{x}{\sqrt{1+x}}\,\mathrm{d}x = \int \frac{t^2-1}{t}\,2t\,\mathrm{d}t = 2\int (t^2-1)\,\mathrm{d}t = \frac{2}{3}t^3 - 2t + C.$$

Restoring t gives precisely the first answer.

The first change uses fractional powers; the second gives a polynomial and a nontrivial differential. Both respect $x>-1$. Differentiation gives $\sqrt{1+x} - 1/\sqrt{1+x} = x/\sqrt{1+x}$, checking both calculations at once.

Example 3.5 A logarithm inside a rational expression Evaluate $\int \ln x/[x(1+(\ln x)^2)]\,\mathrm{d}x$ for $x>0$.

Solution *The factor* $\mathrm{d}x/x$ *accompanies* $\ln x$*, so set* $u=\ln x$*. Since* $\mathrm{d}u=\mathrm{d}x/x$*, the transformed integral is* $\int u/(1+u^2)\,\mathrm{d}u$*. Now put* $v=1+u^2>0$*, so* $\mathrm{d}v=2u\,\mathrm{d}u$*:*

$$\int \frac{u}{1+u^2}\,\mathrm{d}u=\frac{1}{2}\int\frac{\mathrm{d}v}{v}=\frac{1}{2}\ln v+C=\frac{1}{2}\ln(1+(\ln x)^2)+C.$$

The positive domain comes from $\ln x$*, not from the final outer logarithm alone.*

3.1.3 Inverse changes and branches

Sometimes the original integrand does not contain the needed differential. We instead choose $x=\phi(t)$ to simplify it. If ϕ is a continuously differentiable bijection from an open interval K to I, with $\phi'\neq 0$, then

$$\int f(x)\,\mathrm{d}x \quad \text{becomes} \quad \int f(\phi(t))\phi'(t)\,\mathrm{d}t.$$

If $H'(t)=f(\phi(t))\phi'(t)$, the inverse derivative rule shows that $H(\phi^{-1}(x))$ has derivative $f(x)$. These sufficient conditions make restoration to x unambiguous. A *branch*(分支)here means a restriction on which the change has one inverse, such as $t>0$ for $x=t^2$ on $x>0$.

Example 3.6 Rationalizing a square root Evaluate $\int \mathrm{d}x/(1+\sqrt{x})$ for $x>0$.

Solution *Use* $x=t^2$ *with* $t>0$*. Then* $\sqrt{x}=t$ *and* $\mathrm{d}x=2t\,\mathrm{d}t$*, so*

$$\int\frac{\mathrm{d}x}{1+\sqrt{x}}=2\int\left(1-\frac{1}{1+t}\right)\mathrm{d}t=2\sqrt{x}-2\ln(1+\sqrt{x})+C.$$

If we had chosen $t<0$*, the correct identity would be* $\sqrt{x}=|t|=-t$*. The equation* $x=t^2$ *alone does not justify replacing* $\sqrt{x}$ *by* t*. The displayed primitive extends continuously to* $x=0$*, but its verification above takes place on the open interval* $x>0$*.*

Keep the original domain even when algebra cancels a factor. For example, rewriting $x/x=1$ does not give a value to the original quotient at $x=0$. Constants of integration may be chosen independently on disconnected intervals, as in Section 2.5.

3.1.4 Definite substitution and orientation

Theorem 3.2 (Change of variable in a definite integral)

Let $\alpha<\beta$*, let* f *be continuous on an interval* J*, and let* $\phi\in C^1([\alpha,\beta])$ *with* $\phi([\alpha,\beta])\subset J$*. Then*

$$\int_\alpha^\beta f(\phi(t))\phi'(t)\,\mathrm{d}t=\int_{\phi(\alpha)}^{\phi(\beta)} f(x)\,\mathrm{d}x.$$

Here endpoints use continuous extensions, and integrals with reversed limits are oriented as in Section 2.6.

♡

Proof Let m,M be the minimum and maximum of ϕ on the compact interval. If $m=M$, the change is constant and both integrals are zero. Otherwise extend $f|_{[m,M]}$ continuously to $\mathbb{R}$ by the constant values $f(m)$ to the left and $f(M)$ to the right. The FTC gives a differentiable primitive F of this extension on $\mathbb{R}$, with $F'=f$ throughout the image. Thus the ordinary chain rule is valid even where ϕ reaches an endpoint of J. The chain rule and the evaluation part of the FTC give

$$\int_\alpha^\beta f(\phi(t))\phi'(t)\,\mathrm{d}t=F(\phi(\beta))-F(\phi(\alpha))=\int_{\phi(\alpha)}^{\phi(\beta)} f(x)\,\mathrm{d}x.$$

□

Unlike an inverse change for an arbitrary primitive, this formula does not require ϕ to be injective or ϕ' to be nonzero. On a return trip, the derivative changes orientation and the contributions cancel. For a substitution $u = g(x)$, attach $g(a)$ to the lower endpoint a and $g(b)$ to the upper endpoint b; do not silently sort the new bounds. Once all variables and bounds are transformed, a definite computation needs no substitution back.

Example 3.7 A return trip Evaluate $\int_{-1}^{1} x\cos(x^2)\,\mathrm{d}x$.

Solution ***Method 1: the composite differential.*** *The theorem with $f(u) = \cos u$ and $g(x) = x^2$ gives*

$$\int_{-1}^{1} x\cos(x^2)\,\mathrm{d}x = \frac{1}{2}\int_{1}^{1} \cos u\,\mathrm{d}u = 0.$$

This is valid because the original integrand is exactly $\frac{1}{2}f(g)g'$ and g is C^1 everywhere on the interval.

Solution ***Method 2: reflection.*** *The integrand is odd. Substituting $x = -t$ on $[-1, 0]$ gives the negative of the integral on $[0, 1]$, so their sum is zero.*

The first route handles any continuous outer function in place of cosine. The second reveals the cancellation directly. Equal transformed endpoints alone do not justify a change: $u = \tan x$ on $[0, \pi]$, for example, fails to be continuous at $\pi/2$.

Example 3.8 Remove a parameter before differentiating Let f be continuous on $\mathbb{R}$ and define

$$H(x) = \int_{0}^{x} tf(x^2 - t^2)\,\mathrm{d}t \qquad (x \in \mathbb{R}).$$

Find $H'(x)$ without assuming that f is differentiable.

> ***Distinguish the two variables.*** For the integral, x is fixed and t varies. The derivative of $x^2 - t^2$ with respect to t is $-2t$, already present up to a constant. This change can remove x from the integrand, leaving it only in an endpoint where the FTC applies.

Solution *For each fixed x, put $u = x^2 - t^2$. At $t = 0$ the new endpoint is x^2; at $t = x$ it is 0. Thus*

$$H(x) = -\frac{1}{2}\int_{x^2}^{0} f(u)\,\mathrm{d}u = \frac{1}{2}\int_{0}^{x^2} f(u)\,\mathrm{d}u.$$

For $x < 0$ this follows by reversing the original interval before applying the definite substitution theorem; the displayed oriented identity is unchanged. It also holds at $x = 0$, where both integrals vanish. If $F(s) = \int_0^s f(u)\,\mathrm{d}u$, the FTC gives $F' = f$, hence

$$H'(x) = \frac{1}{2}F'(x^2)\,2x = xf(x^2)$$

for every real x, including 0. For example, $f(u) = 1 + u$ gives $H(x) = x^2/2 + x^4/4$, whose derivative is $x + x^3$.

Differentiating the original integrand formally would introduce f', which need not exist. The substitution proves differentiability of H using only continuity of f.

3.1.5 Reflection and periodicity

For continuous f on $[a, b]$, the substitution $x = a + b - t$ yields

$$\int_{a}^{b} f(x)\,\mathrm{d}x = \int_{a}^{b} f(a + b - x)\,\mathrm{d}x.$$

Adding the two expressions pairs values at equal distances from the midpoint. In particular, an odd integrand on $[-a, a]$ has integral zero, and an even one has twice its integral on $[0, a]$.

> ***A definite integral without a primitive.*** If direct integration looks difficult, ask whether the interval is preserved by a simple change. Compute the reflected expression before seeking its primitive. A sum such

> as $f(x)+f(a+b-x)$ may be much simpler than either summand, and integral linearity then determines the answer.

Example 3.9 Pairing a logarithm Compute $I=\int_0^{\pi/4}\ln(1+\tan x)\,dx$.

Solution ***Method 1: reflection algebra.*** *On this interval* $\tan x\in[0,1]$ *and*

$$1+\tan(\pi/4-x)=\frac{2}{1+\tan x}.$$

Thus $f(x)+f(\pi/4-x)=\ln 2$ *for* $f(x)=\ln(1+\tan x)$. *Integrating gives* $2I=(\pi/4)\ln 2$, *hence* $I=\pi\ln 2/8$.

Solution ***Method 2: complementary areas.*** *The graph increases from* 0 *to* $\ln 2$. *The region between this graph and the line* $y=\ln 2$ *has height* $\ln 2-f(x)=f(\pi/4-x)$. *Horizontal reflection gives it the same area as the region below the graph. The two regions fill a rectangle of area* $(\pi/4)\ln 2$, *so each has half that area.*

Both routes use the same pointwise identity. Algebra extends to signed integrands; the area explanation uses the nonnegative heights of this example.

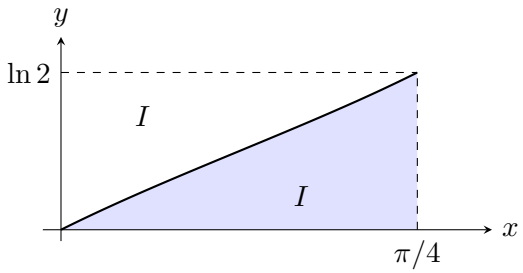


Figure 3.1: Reflection identifies the area above the graph with the area below it.

Proposition 3.1 (A full period has a fixed integral)

If f *is continuous on* $\mathbb{R}$ *and* $f(x+T)=f(x)$ *for some* $T>0$, *then*

$$\int_c^{c+T}f(x)\,dx=\int_0^T f(x)\,dx\qquad(c\in\mathbb{R}).$$

Proof By the FTC the left side, as a function of c, has derivative $f(c+T)-f(c)=0$. It is therefore constant; its value at $c=0$ is the right side. □

Example 3.10 Whole periods and a remainder Compute $\int_0^{5\pi/2}|\sin x|\,dx$.

Solution *The integrand has period* π, *and* $\int_0^{\pi}\sin x\,dx=2$. *Two full periods contribute* 4; *the remaining interval has the same integral as* $[0,\pi/2]$ *and contributes* 1. *The answer is* 5.

Example 3.11 A substitution reveals the symmetry Compute $I=\int_0^1 x^3/[x^4+(1-x^2)^2]\,dx$.

> ***Look for complementary expressions.*** The denominator pairs x^2 with $1-x^2$, rather than x with $1-x$. Set $t=x^2$ first. The remaining numerator becomes t, whose reflected partner is $1-t$; their sum is constant.

Solution ***Method 1: substitute, then reflect.*** *The denominator is positive. With* $t=x^2$,

$$I=\frac12\int_0^1\frac{t}{t^2+(1-t)^2}\,dt=\frac12\int_0^1\frac{1-t}{t^2+(1-t)^2}\,dt.$$

Adding and using $u=2t-1$ *gives*

$$I=\frac14\int_0^1\frac{dt}{t^2+(1-t)^2}=\frac14\int_{-1}^1\frac{du}{1+u^2}=\frac{\pi}{8}.$$

Solution ***Method 2: split off a derivative.*** *Put $D = x^4+(1-x^2)^2$, so $D' = 4x(2x^2-1)$ and $x^3 = D'/8+x/2$. Since $D = ((2x^2-1)^2+1)/2$, direct differentiation verifies the primitive*

$$\frac{1}{8}\ln D + \frac{1}{4}\arctan(2x^2-1).$$

At both endpoints $D = 1$, so the logarithms cancel and the answer is again $\pi/8$.

Reflection avoids an unnecessary logarithm in the definite calculation. The second route also supplies an indefinite primitive; both use only substitution and known derivatives.

3.1.6 Methods summary

A preliminary substitution may reveal a symmetry hidden in the original variable. Compare a paired definite calculation with a primitive calculation before choosing the longer route.

First simplify algebraically, then compare a candidate inner function with its derivative. Write the change, its differential, the transformed integrand, and its domain together. For primitives, restore the original variable and differentiate the answer; for definite integrals, transform both bounds with their orientation. Inverse branches control square-root signs, whereas the direct composite-differential formula needs no injectivity. For a difficult definite integral, inspect reflection and full periods before calculating a primitive. Trigonometric identities and systematic rationalization will be developed in Section 3.4; they are not needed for the arguments above.

For indefinite integrals, give the answer on each connected open interval of the original domain and include an arbitrary constant.

Exercise 3.1

1. Evaluate the related integrals:
   (a) $\int (2x+3)(x^2+3x+4)^4\,dx$;
   (b) $\int (2x+3)/(x^2+3x+4)\,dx$;
   (c) $\int (2x+3)/\sqrt{x^2+3x+4}\,dx$.
2. Evaluate (a) $\int \cos x/(2+\sin x)\,dx$, (b) $\int \cos x/(2+\sin x)^2\,dx$, and (c) $\int \cos x\sqrt{2+\sin x}\,dx$. Explain why the same substitution works on all of $\mathbb{R}$.
3. Evaluate (a) $\int e^{2x}/(1+e^{2x})\,dx$, (b) $\int e^{2x}/(1+e^{2x})^2\,dx$, and (c) $\int e^{2x}\sqrt{1+e^{2x}}\,dx$. Track the factor introduced by differentiating e^{2x}.
4. Evaluate $\int dx/(x\ln x)$ and state its domain.
5. Evaluate $\int x/\sqrt{9-x^2}\,dx$.
6. Evaluate $\int \sin\sqrt{x}/\sqrt{x}\,dx$ for $x>0$.
7. Compute $\int_0^1 2x/(1+x^2)\,dx$.
8. Evaluate (a) $\int x^3/(1+x^2)^2\,dx$, (b) $\int x^5/(1+x^2)^2\,dx$, and (c) $\int x^3/(1+x^2)^3\,dx$. Explain which factor becomes part of the differential and how the remaining powers change.
9. Evaluate $\int dx/(1+\sqrt{1+x})$ for $x>-1$.
10. For $x>0$, evaluate (a) $\int (\ln x)^2/x\,dx$, (b) $\int \ln x/(x\sqrt{1+(\ln x)^2})\,dx$, and (c) $\int 1/[x(1+(\ln x)^2)]\,dx$. Identify both layers when two substitutions are used.
11. Compute $\int_0^{\pi/2} \sin x\cos x/(1+\sin^2 x)\,dx$.
12. Compute $\int_1^0 e^x/(1+e^x)\,dx$.
13. If f is continuous on $[1,4]$, express $\int_1^4 f(x)\,dx$ using $f(3t+1)$ on $[0,1]$.
14. Compute $\int_0^{\pi} \cos x\, e^{\sin x}\,dx$ and explain why equal new bounds are valid here.

15. For $a > 0$, reduce $\int_{-a}^{a} (x^3 + \cos x)/(1 + x^2)\,\mathrm{d}x$ to $[0, a]$ and bound its absolute value by $2 \arctan a$.
16. Let $a > 0$ and $f > 0$ be continuous on $[0, a]$. Compute $\int_0^a f(x)/(f(x) + f(a - x))\,\mathrm{d}x$.
17. Compute $\int_{\pi}^{4\pi} |\sin x|\,\mathrm{d}x$.
18. Compute $\int_0^{\pi} x \sin x/(1 + \cos^2 x)\,\mathrm{d}x$. Pair x with $\pi - x$.
19. A proposed computation changes $u = 1/x$ in $\int_{-1}^{1} x^{-2}\,\mathrm{d}x$. Explain why this is not a proper definite-integral computation.
20. Evaluate $\int x/\sqrt{x^2}\,\mathrm{d}x$ separately for $x > 0$ and $x < 0$. Explain the role of $\sqrt{x^2} = |x|$.

Supplement Exercises

S1. Evaluate $\int (1 + x)e^x/(1 + xe^x)\,\mathrm{d}x$ on its original domain.
S2. Evaluate $\int (1 + x)/[x(1 + xe^x)]\,\mathrm{d}x$. Split the fraction before choosing a variable.
S3. If $a > 0$ and $f \in C([0, a^2])$, prove $\int_0^a xf(x^2)\,\mathrm{d}x = \frac{1}{2}\int_0^{a^2} f(u)\,\mathrm{d}u$.
S4. If f is continuous on an open interval containing $[a, b]$, prove

$$\lim_{h \to 0} \frac{1}{h} \int_a^b \big(f(x + h) - f(x)\big)\,\mathrm{d}x = f(b) - f(a).$$

Do not assume f is differentiable; shift the interval first.
S5. If $f \in C([a, b])$ and $f(a + b - x) = f(x)$, prove $\int_a^b (x - (a + b)/2)f(x)\,\mathrm{d}x = 0$.
S6. Compute $\int_{-2}^{2} x \ln(1 + e^x)\,\mathrm{d}x$ by pairing opposite arguments.
S7. Let f be continuous and T-periodic. Prove that $G(x) = \int_0^x f(t)\,\mathrm{d}t$ is T-periodic if and only if $\int_0^T f = 0$.
S8. For the same f, prove $\lim_{L \to +\infty} L^{-1} \int_0^L f = T^{-1} \int_0^T f$. Split L into an integer number of periods and a bounded remainder.
S9. If $f > 0$ is continuous on $[0, 1]$, compute $\int_0^{\pi/2} f(\sin x)/(f(\sin x) + f(\cos x))\,\mathrm{d}x$.
S10. Prove $\int_0^1 \sqrt{1 - x^2}\,\mathrm{d}x = \pi/4$ by $x = \sin t$. To integrate $\cos^2 t$ without a new primitive formula, compare it with $\sin^2 t$ by reflection and add the two integrals.
S11. Let g be continuous on $[0, 1]$ and satisfy $g(t) = g(1 - t)$. Prove $\int_0^1 x^3 g(x^2)\,\mathrm{d}x = \frac{1}{4}\int_0^1 g(t)\,\mathrm{d}t$. Apply this to reduce $\int_0^1 x^3\sqrt{1 + (2x^2 - 1)^2}\,\mathrm{d}x$ to one quarter of $\int_0^1 \sqrt{1 + u^2}\,\mathrm{d}u$, and prove bounds $1/4$ and $\sqrt{2}/4$. Exact evaluation will become available after Section 3.4.
S12. Let $a < b$ and let $\phi : [a, b] \to [a, b]$ be continuously differentiable and satisfy $\phi(a) = b$, $\phi(b) = a$ and $\phi(\phi(x)) = x$. For continuous f on $[a, b]$, prove

$$\int_a^b f(x)\,\mathrm{d}x = \frac{1}{2}\int_a^b \big(f(x) - f(\phi(x))\phi'(x)\big)\,\mathrm{d}x.$$

For $0 < a < 1$, apply this with $b = 1/a$, $\phi(x) = 1/x$ and $f(x) = 1/(1 + x^2)$. Explain why reflection has weight 1 whereas reciprocal pairing needs the derivative factor $1/x^2$.
S13. Let $f \in C(\mathbb{R})$ and let $m \geq 1$ be an integer. For $H_m(x) = \int_0^x t^{m-1} f(x^m - t^m)\,\mathrm{d}t$, prove $H_m(x) = m^{-1}\int_0^{x^m} f(u)\,\mathrm{d}u$ and find H_m' on $\mathbb{R}$. Explain why odd and even m give the same differentiation rule, although x^m has different signs.
S14. Suppose $g \in C([0, 1])$ and $g(1 - x) = -g(x)$. (a) Prove $\int_0^1 g = 0$ and $\int_0^1 x^2 g = \int_0^1 xg$. (b) Compute $\int_0^1 x^2 \ln((1 + x)/(2 - x))\,\mathrm{d}x$ in terms of $J = \int_0^1 x \ln((1 + x)/(2 - x))\,\mathrm{d}x$ without finding a primitive. (c) More generally, prove $\int_0^1 p(x)g(x)\,\mathrm{d}x = \frac{1}{2}\int_0^1 (p(x) - p(1 - x))g(x)\,\mathrm{d}x$ for every polynomial p.

3.2 Integration by Parts

知识点概要

- ❑ *Reverse the product rule.*
- ❑ *Predict the new integrand.*
- ❑ *Repeat a simplifying choice.*
- ❑ *Solve a returning integral equation.*
- ❑ *Use boundaries and accumulation kernels.*
- ❑ *Derive reduction formulas.*

Substitution is effective when one factor is the derivative of an inner expression. For products such as x^2e^x, neither factor supplies such a change. The product rule suggests another trade: differentiate the polynomial, which becomes simpler, and integrate the exponential, whose form is preserved.

3.2.1 The formula and the choice of factors

Theorem 3.3 (Integration by parts)

If $u, v \in C^1([a,b])$, then

$$\int_a^b u(x)v'(x)\,\mathrm{d}x = [u(x)v(x)]_a^b - \int_a^b u'(x)v(x)\,\mathrm{d}x.$$

On an open interval, if $u, v \in C^1$ and the indicated primitives exist, the corresponding primitive formula is

$$\int uv'\,\mathrm{d}x = uv - \int u'v\,\mathrm{d}x.$$

♡

Proof Integrate $(uv)' = u'v+uv'$ and apply the FTC. For the primitive form, if $H' = u'v$, then $(uv-H)' = uv'$; add one arbitrary constant to obtain the family. □

Integration by parts (分部积分) is often written $\int u\,\mathrm{d}v = uv - \int v\,\mathrm{d}u$. The notation means $\mathrm{d}v = v'(x)\,\mathrm{d}x$ and $\mathrm{d}u = u'(x)\,\mathrm{d}x$. It trades the integral of uv' for that of $u'v$; it is useful only if the latter is simpler or enters a solvable relation. For xe^x, taking $u = x$, $v = e^x$ changes the remaining product to e^x. Reversing these choices instead produces a higher-degree polynomial times e^x.

> ***Predict before calculating.*** Choose the factor to differentiate by asking what its derivative removes: polynomial degree, a logarithm, or an inverse function. Choose the factor to integrate by asking whether its primitive stays manageable. Write down the proposed $u'v$ before applying the formula; a valid formula can still be an unhelpful choice.

3.2.2 Repeated integration by parts

Example 3.12 A decreasing polynomial degree Evaluate $\int x^2e^x\,\mathrm{d}x$.

Solution ***Method 1: repeat the trade.*** *Taking $u = x^2$, $v = e^x$, and then $u = x$, $v = e^x$, gives*

$$\begin{aligned}\int x^2e^x\,\mathrm{d}x &= x^2e^x - 2\int xe^x\,\mathrm{d}x\\ &= x^2e^x - 2(xe^x - e^x) + C\\ &= e^x(x^2 - 2x + 2) + C.\end{aligned}$$

Each differentiation lowers the polynomial degree, so the process stops.

Solution ***Method 2: determine a polynomial.*** *Because $(e^xP)' = e^x(P + P')$, look for a quadratic $P = Ax^2 + Bx + D$ with $P + P' = x^2$. Expanding gives $P + P' = Ax^2 + (B + 2A)x + (D + B)$. Matching the*

coefficients of $x^2, x, 1$ *gives* $A = 1$, $B + 2A = 0$, $D + B = 0$. *Thus* $P = x^2 - 2x + 2$, *giving the same primitive directly.*

Repeated parts explains why a polynomial suffices. The coefficient method compresses the calculation for polynomial times exponential integrands; it would not automatically simplify an arbitrary factor.

For a polynomial P of degree m and a nonzero constant a, repeated parts terminates after at most $m + 1$ trades. The derivative test is still essential: signs alternate and each integration of e^{ax} introduces a factor $1/a$.

3.2.3 Logarithms and inverse functions

A lone logarithm or inverse function can be viewed as a product with 1. Its derivative may be algebraic even when its primitive is not obvious.

Example 3.13 A hidden product Evaluate $\int \arctan x \, \mathrm{d}x$.

Solution *Use* $u = \arctan x$, $v = x$:

$$\int \arctan x \, \mathrm{d}x = x \arctan x - \int \frac{x}{1 + x^2} \, \mathrm{d}x = x \arctan x - \frac{1}{2} \ln(1 + x^2) + C.$$

Differentiating the product produces $\arctan x + x/(1 + x^2)$; *the logarithm cancels the extra term.*

Example 3.14 Lower a logarithmic power Evaluate $\int x(\ln x)^2 \, \mathrm{d}x$ for $x > 0$.

> ***Choose a measure of complexity.*** Differentiating $(\ln x)^2$ lowers its logarithmic power and introduces $1/x$. Integrating the other factor x supplies $x^2/2$, so that denominator cancels one power. A second application should therefore finish the calculation.

Solution ***Method 1: two reductions of the logarithm.*** *Choose* $u = (\ln x)^2$ *and* $v = x^2/2$. *Then* $u' = 2 \ln x/x$, *and*

$$\int x(\ln x)^2 \, \mathrm{d}x = \frac{x^2}{2}(\ln x)^2 - \int x \ln x \, \mathrm{d}x.$$

For the remaining integral, choose $u = \ln x$, $v = x^2/2$:

$$\int x \ln x \, \mathrm{d}x = \frac{x^2}{2} \ln x - \frac{1}{2} \int x \, \mathrm{d}x = \frac{x^2}{2} \ln x - \frac{x^2}{4} + C.$$

Combining these identities with one final arbitrary constant gives

$$\frac{x^2}{2}(\ln x)^2 - \frac{x^2}{2} \ln x + \frac{x^2}{4} + C.$$

Solution ***Method 2: turn logarithmic degree into polynomial degree.*** *Set* $t = \ln x$, *so* $x = e^t$ *and* $x \, \mathrm{d}x = e^{2t} \, \mathrm{d}t$. *Repeated parts now gives*

$$\begin{aligned} \int t^2 e^{2t} \, \mathrm{d}t &= \frac{1}{2} t^2 e^{2t} - \int t e^{2t} \, \mathrm{d}t \\ &= \frac{1}{2} t^2 e^{2t} - \frac{1}{2} t e^{2t} + \frac{1}{4} e^{2t} + C. \end{aligned}$$

Restoring t *gives the same formula.*

Direct parts is shorter; logarithmic substitution relates the entire family of log-power integrals to polynomial times exponential integrals. Both routes stay on $x > 0$, and differentiation cancels all terms except $x(\ln x)^2$.

Example 3.15 The boundary term matters Compute $\int_1^e (\ln x)/x^2 \, \mathrm{d}x$.

Solution *Set* $u = \ln x$, $v = -1/x$. *Both are* C^1 *on* $[1, e]$. *Then*

$$\int_1^e \frac{\ln x}{x^2} \, \mathrm{d}x = \left[-\frac{\ln x}{x} \right]_1^e + \int_1^e \frac{\mathrm{d}x}{x^2} = -\frac{1}{e} + 1 - \frac{1}{e} = 1 - \frac{2}{e}.$$

The positive answer agrees with the sign of the integrand. Here evaluating the boundary term immediately is simpler than carrying an indefinite primitive through the calculation.

Example 3.16 Choose the constant in the primitive Compute $I = \int_0^1 \ln(1+x^2)\,\mathrm{d}x$ in two ways.

> ***Use the endpoints as well as the integrand.*** Differentiating the logarithm removes it. A primitive of 1 can be x or $x-1$. The latter vanishes at the upper endpoint, while the logarithm vanishes at the lower endpoint, so both boundary contributions disappear.

Solution ***Method 1: the usual primitive.*** *Taking $u = \ln(1+x^2)$ and $v = x$ gives*

$$I = \ln 2 - 2\int_0^1 \frac{x^2}{1+x^2}\,\mathrm{d}x = \ln 2 - 2 + \frac{\pi}{2}.$$

Here $x^2/(1+x^2) = 1 - 1/(1+x^2)$.

Solution ***Method 2: a primitive normalized at the endpoint.*** *Taking $v = x-1$ makes $[uv]_0^1 = 0$, so*

$$I = 2\int_0^1 \frac{x - x^2}{1+x^2}\,\mathrm{d}x = \ln 2 - 2 + \frac{\pi}{2}.$$

The first numerator term is a logarithmic derivative; the second is handled as above.

Changing v by a constant changes both terms of the parts identity, which cancel in the total. Endpoint normalization is especially useful for nested integrals and estimates; it is a choice within the same formula, not a new integration rule.

3.2.4 Returning integrals and equations

Differentiation and integration of sine, cosine and exponentials preserve their types. Consequently a second application of parts may return to the original integral. This is useful when the resulting equation has a nonzero coefficient. For clarity, first work with fixed definite integrals; arbitrary constants must not be treated as unrelated unknowns at every step.

Example 3.17 A cycle and a coefficient system Find a primitive of $e^{ax}\sin(bx)$ for real a, b with $a^2+b^2 > 0$.

Solution ***Method 1: two integrations by parts.*** *For $a \neq 0$ define $I(x) = \int_0^x e^{at}\sin(bt)\,\mathrm{d}t$ and $J(x) = \int_0^x e^{at}\cos(bt)\,\mathrm{d}t$. Parts gives*

$$aI = e^{ax}\sin(bx) - bJ, \qquad aJ = e^{ax}\cos(bx) - 1 + bI.$$

Eliminating J yields

$$(a^2+b^2)I = e^{ax}(a\sin(bx) - b\cos(bx)) + b.$$

Therefore the primitive family is

$$\frac{e^{ax}}{a^2+b^2}(a\sin(bx) - b\cos(bx)) + C.$$

If $a = 0$, then $b \neq 0$ and direct integration gives $-\cos(bx)/b + C$, the same formula. The case $b = 0$, $a \neq 0$ gives the constant family for the zero integrand.

Solution ***Method 2: a stable form under differentiation.*** *Seek $F = e^{ax}(A\sin(bx) + B\cos(bx))$. Then*

$$F' = e^{ax}\big((aA - bB)\sin(bx) + (bA + aB)\cos(bx)\big).$$

Solving $aA - bB = 1$, $bA + aB = 0$ gives $A = a/(a^2+b^2)$ and $B = -b/(a^2+b^2)$. This verifies the formula, including $a = 0$ and $b = 0$ separately as above.

The cycle derives the answer from the parts formula; the second route predicts a small class of candidate primitives. When $b = 0$, matching the unused sine coefficient is just a sufficient construction, not a uniqueness claim for A. If both parameters vanish, the integrand is zero and the displayed quotient is inapplicable; its primitives are constants.

Example 3.18 Substitution is the simpler choice Evaluate $\int xe^{x^2}\,\mathrm{d}x$ and compare a parts attempt.

Solution *The inner derivative $2x$ is already present, so $u = x^2$ gives $\frac{1}{2}e^{x^2} + C$. Parts with $u = e^{x^2}$, $v = x^2/2$ instead gives*

$$\int xe^{x^2}\,\mathrm{d}x = \frac{x^2}{2}e^{x^2} - \int x^3e^{x^2}\,\mathrm{d}x,$$

which raises the polynomial degree. The identity is valid, but it does not exploit the easiest structure.

3.2.5 Accumulation kernels

Example 3.19 An integral of an accumulated quantity For continuous f on $[a, b]$, prove

$$\int_a^b \left(\int_a^x f(t)\,\mathrm{d}t\right)\mathrm{d}x = \int_a^b (b-x)f(x)\,\mathrm{d}x.$$

> ***Remove the nested integral.*** The inner integral is a function $F(x)$ whose derivative is known by the FTC. Differentiate this factor in parts. Choose the primitive of 1 as $x - b$, so one boundary factor vanishes at b while $F(a) = 0$ removes the other.

Solution ***Method 1: parts with vanishing boundaries.*** *Put $F(x) = \int_a^x f(t)\,\mathrm{d}t$. Since $F' = f$,*

$$\int_a^b F(x)\,\mathrm{d}x = [(x-b)F(x)]_a^b - \int_a^b (x-b)f(x)\,\mathrm{d}x.$$

The boundary term is zero, proving the result.

Solution ***Method 2: compare derivatives of two accumulations.*** *For a variable upper endpoint y, write*

$$H(y) = \int_a^y (y-t)f(t)\,\mathrm{d}t = y\int_a^y f(t)\,\mathrm{d}t - \int_a^y tf(t)\,\mathrm{d}t.$$

The product rule and FTC give $H'(y) = \int_a^y f(t)\,\mathrm{d}t = F(y)$. Also $H(a) = 0$. The FTC now gives $H(b) = \int_a^b F(y)\,\mathrm{d}y$.

The first method is shorter; the second explains why the factor $b - x$ counts the remaining accumulation length. Both arguments stay within one-variable calculus and require no interchange of two integral signs.

Example 3.20 Integrate an inverse without solving a cubic Let $f(t) = t + t^3$ for $0 \le t \le 1$, and let $g : [0, 2] \to [0, 1]$ be its inverse. Compute $\int_0^2 g(y)\,\mathrm{d}y$.

> ***Use the known direction of the relation.*** An explicit formula for g would require solving a cubic. The relation $y = f(t)$ is already known and has positive derivative. Substitution turns the inverse into t, and parts then transfers the derivative from f.

Solution ***Method 1: substitution and parts.*** *Since $f' = 1 + 3t^2 > 0$, its continuous inverse exists on the stated interval. With $y = f(t)$,*

$$\begin{aligned}\int_0^2 g(y)\,\mathrm{d}y &= \int_0^1 tf'(t)\,\mathrm{d}t \\ &= [tf(t)]_0^1 - \int_0^1 f(t)\,\mathrm{d}t \\ &= 2 - \left[\frac{t^2}{2} + \frac{t^4}{4}\right]_0^1 = \frac{5}{4}.\end{aligned}$$

Directly integrating $t(1 + 3t^2)$ gives the same $1/2 + 3/4$.

Solution ***Method 2: complementary areas.*** *In the rectangle $0 \le t \le 1$, $0 \le y \le 2$, the graph $y = f(t)$ divides the area into two parts. The part below the graph has area $\int_0^1 f = 3/4$. At height y, the part above the graph*

extends horizontally from $t = 0$ to $t = g(y)$, so its area is $\int_0^2 g$. Their sum is the rectangle's area 2, proving the result.

The algebraic identity works with oriented integrals beyond this nonnegative picture. The geometric route explains the complementary areas; it does not require a formula for the inverse.

Example 3.21 Cancellation gives an estimate For $f \in C^1([a,b])$, $a < b$, and real $k \neq 0$, bound $I_k = \int_a^b f(x)\sin(kx)\,\mathrm{d}x$ by a quantity tending to zero as $|k| \to \infty$.

> ***Integrate the oscillation.*** The direct bound $|I_k| \le \int_a^b |f|$ discards all cancellation and does not depend on k. A primitive of $\sin(kx)$ contains $1/k$; parts transfers a derivative to the slowly varying factor and exposes that small factor.

Solution *Parts gives*

$$I_k = \left[-\frac{f(x)\cos(kx)}{k}\right]_a^b + \frac{1}{k}\int_a^b f'(x)\cos(kx)\,\mathrm{d}x.$$

Thus

$$|I_k| \le \frac{|f(a)| + |f(b)| + \int_a^b |f'(x)|\,\mathrm{d}x}{|k|}.$$

The numerator is finite and independent of k, proving $I_k \to 0$. If f vanishes at both endpoints, only the integral of $|f'|$ remains. For $f(x) = x(1-x)$ on $[0,1]$, it equals $1/2$, giving $|I_k| \le 1/(2|k|)$.

An estimate can reveal a feature that an exact primitive hides. The differentiability hypothesis justifies this argument; it is not asserted to be necessary for oscillatory cancellation.

3.2.6 Reduction formulas

A *reduction formula*(递推公式)relates a family of integrals to members with smaller integer indices. The formula is useful only with base cases and a range of indices that actually decreases.

Example 3.22 Wallis integrals Let $I_n = \int_0^{\pi/2} \sin^n x\,\mathrm{d}x$ for integers $n \ge 0$. Find a recurrence and evaluate the family.

Solution *The base cases are $I_0 = \int_0^{\pi/2} 1\,\mathrm{d}x = \pi/2$ and $I_1 = [-\cos x]_0^{\pi/2} = 1$. For $n \ge 2$, take $u = \sin^{n-1} x$, $v = -\cos x$. At $\pi/2$, the factor $\cos x$ is zero; at 0, the factor $\sin^{n-1} x$ is zero because $n \ge 2$. Also $u' = (n-1)\sin^{n-2} x \cos x$. Thus the boundary term is zero and*

$$\begin{aligned} I_n &= (n-1)\int_0^{\pi/2} \sin^{n-2} x \cos^2 x\,\mathrm{d}x \\ &= (n-1)(I_{n-2} - I_n). \end{aligned}$$

Hence $I_n = \frac{n-1}{n} I_{n-2}$, and iteration gives

$$I_{2m} = \frac{\pi}{2}\prod_{j=1}^{m}\frac{2j-1}{2j}, \qquad I_{2m+1} = \prod_{j=1}^{m}\frac{2j}{2j+1} \quad (m \ge 0).$$

An empty product is 1, so the formulas include both base cases. For example, $I_4 = 3\pi/16$ and $I_5 = 8/15$.

The index falls by two; parity determines which base case is reached. The same recurrence, combined with $0 \le \sin x \le 1$, will also yield a product limit in the supplementary exercises.

3.2.7 Methods summary

For definite integrals, a constant in the chosen primitive can simplify boundary terms. Integrating an oscillating factor can also expose a small frequency factor and give a useful bound.

Predict $u'v$ before applying parts. Differentiate a factor whose complexity falls and integrate one whose form remains manageable. Repeat when a degree decreases, solve an equation when the original integral returns, and retain both base cases for a recurrence. For definite integrals, check the hypotheses on the closed interval and evaluate every boundary term with its sign. For nested integrals, the FTC makes the accumulated quantity a natural factor to differentiate. Finally, compare with substitution: a composite derivative already present usually deserves attention first.

Primitive answers are understood on the original connected open domains.

Exercise 3.2

1. Evaluate (a) $\int xe^{2x}\,dx$, (b) $\int x^2e^{2x}\,dx$, and (c) $\int xe^{-x}\,dx$. Compare the effect of polynomial degree with that of the exponential coefficient.
2. Evaluate (a) $\int x\cos x\,dx$, (b) $\int x\sin(2x)\,dx$, and (c) $\int (x+1)\cos(2x)\,dx$. Check signs and frequency factors by differentiation.
3. Evaluate $\int \ln x\,dx$ for $x>0$.
4. Evaluate $\int \arcsin x\,dx$ for $|x|<1$.
5. For $x>0$, evaluate (a) $\int x\ln x\,dx$, (b) $\int x^2\ln x\,dx$, and (c) $\int x^2(\ln x)^2\,dx$. Which complexity decreases at each application of parts?
6. Compute (a) $\int_0^1 xe^x\,dx$, (b) $\int_0^1 (1-x)e^x\,dx$, and (c) $\int_0^1 x(1-x)e^x\,dx$. Record the boundary terms before simplifying.
7. Compute $\int_0^{\pi} x\sin x\,dx$.
8. Evaluate $\int x^3e^x\,dx$.
9. Evaluate $\int x^2\cos x\,dx$.
10. For $x>0$, evaluate (a) $\int(\ln x)^2\,dx$, (b) $\int(\ln x)^2/x\,dx$, and (c) $\int(\ln x)^2/x^2\,dx$. Decide first whether substitution or parts lowers the complexity.
11. Evaluate $\int(\arctan x)/x^2\,dx$ for $x\neq 0$.
12. Compute $\int_1^e x^2\ln x\,dx$.
13. Evaluate $\int e^{2x}\cos(3x)\,dx$ by two methods.
14. Compute $\int_0^{\pi} x^2\sin x\,dx$.
15. Evaluate I_4 and I_5 in Example 3.22 from their base cases.
16. For $f\in C^1([a,b])$, express $\int_a^b xf'(x)\,dx$ using $f(a)$, $f(b)$ and $\int_a^b f$.
17. For $f\in C^1([0,1])$, simplify $\int_0^1(1-x)f'(x)\,dx$.
18. For $\int xe^x\,dx$, compare $u=x, v=e^x$ with $u=e^x, v=x^2/2$. Write both resulting identities and explain which is useful.
19. Compute $\int_1^e \ln x/(1+x)^2\,dx$. Use $1/[x(1+x)]=1/x-1/(1+x)$ after parts.
20. Let $J_m(x)=\int_1^x(\ln t)^m\,dt$ for $x>0$ and integers $m\geq 0$. Find J_0, derive a recurrence for $m\geq 1$, and compute $J_2(e)$.

Supplement Exercises

S1. For a polynomial P of degree m and $a\neq 0$, prove

$$\int P(x)e^{ax}\,dx = e^{ax}\sum_{k=0}^{m}\frac{(-1)^kP^{(k)}(x)}{a^{k+1}}+C.$$

Compare repeated parts with differentiation of the proposed answer.

S2. Evaluate $\int\cos(\ln x)\,dx$ for $x>0$ by changing $t=\ln x$ and then solving a cycle.

S3. Evaluate $\int xe^x \sin x \, \mathrm{d}x$.

S4. For integers $m, n \geq 0$, prove $\int_0^1 x^m (1-x)^n \, \mathrm{d}x = m!n!/(m+n+1)!$. Reduce n by parts, treating $n = 0$ separately.

S5. Prove $\lim_{m\to\infty} \prod_{j=1}^{m} 4j^2/(4j^2-1) = \pi/2$. Compare $I_{2m+1}, I_{2m}, I_{2m-1}$ and use the recurrence to squeeze their ratios.

S6. Compute $\int_0^{\pi} x \sin^5 x \, \mathrm{d}x$ by reflection and the recurrence.

S7. For $f \in C^1([a,b])$ and $\lambda \neq 0$, prove

$$\left| \int_a^b f(x) \sin(\lambda x) \, \mathrm{d}x \right| \leq \frac{|f(a)| + |f(b)| + \int_a^b |f'(x)| \, \mathrm{d}x}{|\lambda|}.$$

Explain why the integrals tend to zero as $|\lambda| \to \infty$. Then prove the corresponding estimate for $\cos(\lambda x)$, identifying its different boundary term.

S8. If $f \in C^1([a,b])$ and $f(a) = f(b) = 0$, prove $\int_a^b ff' = 0$ and $\int_a^b xff' = -\frac{1}{2}\int_a^b f^2$.

S9. If $f \in C^2([0,1])$ and $f(0) = f(1) = 0$, prove $\int_0^1 ff'' = -\int_0^1 (f')^2 \leq 0$ and determine the equality case.

S10. For continuous f on $[a,b]$, put $K_n(x) = \frac{1}{n!}\int_a^x (x-t)^n f(t) \, \mathrm{d}t$ for integers $n \geq 0$. Prove $K_0' = f$, $K_n' = K_{n-1}$ for $n \geq 1$, and $K_n(a) = 0$. Expand the polynomial in $x - t$ and use the FTC; do not assume a general parameter-differentiation rule.

S11. Suppose $f \in C^1([0,1])$, $f(0) = 0$ and $|f'| \leq M$, where $M \geq 0$. Prove $\int_0^1 f = \int_0^1 (1-x)f'(x) \, \mathrm{d}x$ and $|\int_0^1 f| \leq M/2$. If also $f(1) = 0$, improve the bound to $M/4$ by estimating $|f(x)|$ from each endpoint. Show that $1/4$ cannot be decreased for all such C^1 functions.

> ***Optional hint.*** For sharpness with $M = 1$ and $\varepsilon > 0$, replace the corner of $\min(x, 1-x)$ by $f_\varepsilon(x) = \sqrt{1/4 + \varepsilon^2} - \sqrt{(x - 1/2)^2 + \varepsilon^2}$, then estimate its difference from the tent function uniformly.

S12. Let $f \in C^2([a,b])$ with $f(a) = f(b) = 0$. Prove for $k \neq 0$ that

$$\left| \int_a^b f(x) \sin(kx) \, \mathrm{d}x \right| \leq \frac{|f'(a)| + |f'(b)| + \int_a^b |f''(x)| \, \mathrm{d}x}{k^2}.$$

If $a = 0$, $b = \pi$ and k is a nonzero integer, remove both derivative endpoint terms from the bound. Explain which boundary values make this possible.

S13. For an integer $m \geq 1$, let g_m be the inverse of $f_m(t) = t + t^m$ on $[0,1]$. (a) Find $\int_0^2 g_m(y) \, \mathrm{d}y$. (b) Find $\int_0^2 g_m(y)^2 \, \mathrm{d}y$ by substitution. (c) Check both answers when $m = 1$, and explain why solving for g_m is unnecessary.

S14. For real a and $x > 0$, evaluate $\int x^a (\ln x)^2 \, \mathrm{d}x$. Treat $a = -1$ separately and verify the other formula by differentiating. For $a \neq -1$, compare direct repeated parts with $x = e^t$; explain the source of each denominator $a + 1$.

3.3 Integration of Rational Functions

知识点概要

- ❑ *Divide before decomposing.*
- ❑ *Read a template from the factors.*
- ❑ *Determine and verify coefficients.*
- ❑ *Complete the square.*
- ❑ *Reduce repeated quadratics.*
- ❑ *Preserve domains and parameter cases.*

A quotient of polynomials may have no visible composite derivative. Factoring its denominator reveals simpler fractions whose primitives are logarithms, powers or arctangents. The aim is an algebraic identity before any integration takes place.

3.3.1 Division and the original domain

A *rational function*(有理函数)has the form $P(x)/Q(x)$, where P, Q are real polynomials and Q is not the zero polynomial. Its original domain consists of the real x with $Q(x) \neq 0$. A fraction is *proper*(真分式)when $\deg P < \deg Q$ (including the zero numerator). Polynomial division writes any rational function as

$$\frac{P}{Q} = S + \frac{R}{Q}, \qquad \deg R < \deg Q.$$

The polynomial S is integrated term by term; decomposition is needed only for the proper remainder. For example, $1/(x+1)$ is proper, whereas $x^2/(x+1) = x - 1 + 1/(x+1)$ is not. If factors cancel, the simplified expression agrees with the original only on its original domain. Each connected open interval between excluded points has its own arbitrary constant.

Example 3.23 Divide first Evaluate $\int x^3/(x^2+1)\,\mathrm{d}x$.

Solution *Since* $x^3 = x(x^2+1) - x$,

$$\int \frac{x^3}{x^2+1}\,\mathrm{d}x = \int \left(x - \frac{x}{x^2+1}\right)\mathrm{d}x = \frac{x^2}{2} - \frac{1}{2}\ln(1+x^2) + C.$$

The denominator is positive on $\mathbb{R}$.

3.3.2 The complete partial-fraction form

An *irreducible quadratic*(不可约二次式)is a real quadratic with no real root. For a monic quadratic $q = x^2+px+r$, this means $p^2-4r<0$, or equivalently $q = (x+p/2)^2+a^2$ with $a>0$. Thus x^2+1 is irreducible, whereas $x^2-1=(x-1)(x+1)$ is not. We use the standard algebraic fact that a real polynomial factors into real linear factors and irreducible quadratics.

Theorem 3.4 (Partial fractions)

Suppose a proper rational function has denominator

$$Q = c\prod_i (x-r_i)^{m_i}\prod_j q_j(x)^{n_j},$$

where $c \neq 0$, m_i, n_j *are positive integers, the* r_i *are distinct, and the* q_j *are distinct monic irreducible quadratics. Then there are real constants giving the identity*

$$\frac{P}{Q} = \sum_i \sum_{k=1}^{m_i} \frac{A_{ik}}{(x-r_i)^k} + \sum_j \sum_{k=1}^{n_j} \frac{B_{jk}x + C_{jk}}{q_j(x)^k}$$

on the original domain. These coefficients are unique. ♡

This is a *partial-fraction decomposition*(部分分式分解). Each numerator has degree strictly below that of its underlying factor. Every power from 1 to the multiplicity must be allowed: $1/(x-1)^2$, for instance, cannot be represented by a multiple of $1/(x-1)$ alone. Likewise a constant numerator over an irreducible quadratic would omit integrands such as $x/(x^2+1)$.

Algebraic mechanism Absorb c into P. For coprime polynomial factors A, B, the Euclidean algorithm gives $UA + VB = 1$. Multiplying by $P/(AB)$ splits the fraction into $PU/B + PV/A$. Division reduces the numerators to degrees below their denominators; any polynomial parts cancel because the original fraction is proper and all remaining fractions tend to zero as $|x| \to \infty$. Repeat for the distinct factor powers.

For a single factor p of degree d, repeated polynomial division gives $R = R_0 + R_1 p + \cdots + R_{m-1}p^{m-1}$ when $\deg R < md$, with $\deg R_k < d$. Dividing by p^m supplies all the powers and the prescribed numerators. For uniqueness, subtract two decompositions and multiply by Q. Reduction modulo a chosen factor p isolates its highest-denominator numerator times a polynomial coprime to p. The Euclidean identity therefore makes that numerator divisible by p; its smaller degree forces it to be zero. Remove that term and repeat for lower powers and all factors. □

> ***Let the denominator set the template.*** Divide first, then factor completely over the reals. Before calculating a coefficient, write every permitted denominator power and numerator degree. Multiplying by the common denominator turns the task into a polynomial identity. Roots of linear factors isolate some coefficients; comparison of coefficients finds the rest and checks the whole identity.

3.3.3 Linear factors

For a real r,

$$\int \frac{\mathrm{d}x}{x-r} = \ln|x-r| + C, \qquad \int \frac{\mathrm{d}x}{(x-r)^k} = -\frac{1}{(k-1)(x-r)^{k-1}} + C \quad (k \geq 2).$$

Both formulas apply separately on $x < r$ and $x > r$. The absolute value in the logarithm is necessary on the negative branch.

Example 3.24 Two ways to find coefficients Evaluate $\int (3x+1)/[(x-1)(x+2)]\,\mathrm{d}x$.

Solution ***Method 1: evaluate the polynomial identity at roots.*** *Write* $3x+1 = A(x+2)+B(x-1)$. *At* $x=1$ *this gives* $4 = 3A$; *at* $x=-2$ *it gives* $-5 = -3B$. *Hence*

$$\int \frac{3x+1}{(x-1)(x+2)}\,\mathrm{d}x = \frac{4}{3}\ln|x-1| + \frac{5}{3}\ln|x+2| + C.$$

Evaluating at the roots is legitimate in the multiplied polynomial identity, not in the original undefined fractions.

Solution ***Method 2: compare coefficients.*** *The same identity gives* $A+B=3$ *and* $2A-B=1$, *so* $A=4/3$, $B=5/3$. *Integrating the resulting two simple fractions gives the displayed answer.*

Root evaluation is shorter for distinct linear factors. Coefficient comparison also works when roots do not isolate all unknowns. The original domain has three intervals, separated by -2 and 1.

Example 3.25 A repeated linear factor Evaluate $\int 1/[x(x-1)^2]\,\mathrm{d}x$.

Solution *Use* $A/x + B/(x-1) + D/(x-1)^2$. *After multiplication,* $1 = A(x-1)^2 + Bx(x-1) + Dx$. *At* 0 *and* 1 *we obtain* $A = D = 1$, *and the quadratic coefficient gives* $B = -1$. *Thus*

$$\int \frac{\mathrm{d}x}{x(x-1)^2} = \ln|x| - \ln|x-1| - \frac{1}{x-1} + C.$$

Substitution into the polynomial identity verifies all coefficients. The point $x=1$ *requires both powers in the template, even though the coefficient of one power could have vanished for a different numerator.*

Example 3.26 A repeated root also supplies derivative information Evaluate $\int(2x^2+1)/[(x-1)^2(x+2)]\,\mathrm{d}x$ on its original domain.

> ***Do not stop after substituting the roots.*** At a repeated root, substituting into the multiplied identity isolates only the highest pole coefficient. Differentiating that polynomial identity can isolate the next coefficient without expanding every term. This is differentiation of an identity of polynomials, not evaluation of a singular fraction.

Solution ***Method 1: values and a derivative.*** *Use the full template*

$$\frac{2x^2+1}{(x-1)^2(x+2)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{D}{x+2}.$$

Multiplication gives

$$2x^2+1 = A(x-1)(x+2) + B(x+2) + D(x-1)^2.$$

At $x=1$ this says $3=3B$, so $B=1$; at $x=-2$ it says $9=9D$, so $D=1$. Differentiate the identity and set $x=1$:

$$4x = A(2x+1) + B + 2D(x-1), \qquad 4 = 3A+1.$$

Thus $A=1$ and the answer is

$$\ln|x-1| - \frac{1}{x-1} + \ln|x+2| + C.$$

The three original domain intervals are separated by -2 and 1.

Solution ***Method 2: coefficient comparison.*** *Expansion of the same identity gives*

$$A+D=2, \qquad A+B-2D=0, \qquad -2A+2B+D=1.$$

From $D=2-A$ and $B=4-3A$, the last equation becomes $10-9A=1$, so $A=1$ and $B=D=1$. Integrating the same three terms gives the preceding primitive.

Root values with derivatives exploit multiplicity; coefficient comparison supplies a uniform method even when such evaluations do not simplify the system. Recombining the fractions, or differentiating the primitive, is an independent check of the signs.

3.3.4 Irreducible quadratic factors

For $q = x^2+px+r = (x+p/2)^2+a^2$, write

$$Bx + C = \frac{B}{2}q' + \left(C - \frac{Bp}{2}\right).$$

The first part gives a logarithm; the second uses

$$\int \frac{\mathrm{d}x}{(x+p/2)^2+a^2} = \frac{1}{a}\arctan\frac{x+p/2}{a} + C.$$

Here $q>0$, so $\ln q$ needs no absolute value.

Example 3.27 A derivative plus a remainder Evaluate $\int(3x+4)/(x^2+2x+5)\,\mathrm{d}x$.

Solution *Let $q = x^2+2x+5 = (x+1)^2+4$. Since $3x+4 = \frac{3}{2}q' + 1$,*

$$\int \frac{3x+4}{q}\,\mathrm{d}x = \frac{3}{2}\ln q + \frac{1}{2}\arctan\frac{x+1}{2} + C.$$

For the second part, $t=(x+1)/2$ gives $\mathrm{d}x = 2\,\mathrm{d}t$ and $q = 4(1+t^2)$, hence $\int \mathrm{d}x/q = \frac{1}{2}\arctan t + C$. Differentiating the logarithm gives $(3x+3)/q$ and differentiating the arctangent term gives $1/q$, recovering $3x+4$.

Example 3.28 A linear and a quadratic factor Evaluate $\int 1/[x(x^2+1)]\,\mathrm{d}x$.

Solution *The template is* $A/x + (Bx+D)/(x^2+1)$. *The identity* $1 = A(x^2+1) + (Bx+D)x$ *gives* $A = 1$, $B = -1$, $D = 0$. *Therefore*

$$\int \frac{\mathrm{d}x}{x(x^2+1)} = \ln|x| - \frac{1}{2}\ln(1+x^2) + C, \qquad x \neq 0.$$

Allowing a linear numerator was necessary even though its constant term turned out to be zero.

Example 3.29 A mixed denominator in a definite integral Compute $I = \int_0^1 (x^2+2)/[(x+1)(x^2+1)]\,\mathrm{d}x$.

> ***Separate the algebra from endpoint evaluation.*** The linear factor needs a constant numerator, while the irreducible quadratic needs a linear numerator. Once the coefficients are found, that linear numerator splits into a logarithmic derivative and an arctangent term.

Solution *Write*

$$\frac{x^2+2}{(x+1)(x^2+1)} = \frac{A}{x+1} + \frac{Bx+D}{x^2+1}.$$

Then $x^2+2 = A(x^2+1) + (Bx+D)(x+1)$, *so*

$$A + B = 1, \qquad B + D = 0, \qquad A + D = 2.$$

The solution is $A = 3/2$, $B = -1/2$, $D = 1/2$. *There is no pole on* $[0,1]$*; in particular both logarithm arguments below are positive. Therefore*

$$\begin{aligned} I &= \frac{3}{2}\int_0^1 \frac{\mathrm{d}x}{x+1} - \frac{1}{2}\int_0^1 \frac{x\,\mathrm{d}x}{x^2+1} + \frac{1}{2}\int_0^1 \frac{\mathrm{d}x}{x^2+1} \\ &= \left[\frac{3}{2}\ln(x+1) - \frac{1}{4}\ln(x^2+1) + \frac{1}{2}\arctan x\right]_0^1 \\ &= \frac{5}{4}\ln 2 + \frac{\pi}{8}. \end{aligned}$$

The coefficient $-1/4$ *includes the extra factor* $1/2$ *from differentiating* x^2+1.

A decomposition that omitted Bx would give an inconsistent coefficient system. Checking the algebra before integrating identifies such a mistake earlier than a final derivative check.

3.3.5 Repeated quadratics

The derivative part of $(Bx+C)/q^k$ is still elementary: for $k \geq 2$, $\int q'/q^k = -q^{1-k}/(k-1) + C$. After shifting the remaining quadratic, only integrals of $(u^2+a^2)^{-k}$ remain.

Example 3.30 A quadratic reduction formula For $a > 0$ and integers $n \geq 1$, put $J_n(x) = \int_0^x (t^2+a^2)^{-n}\,\mathrm{d}t$. Derive a recurrence and evaluate $\int (1+x^2)^{-2}\,\mathrm{d}x$.

> ***Create the same denominator with a derivative.*** Differentiating $x/(x^2+a^2)^{n-1}$ produces denominator power n and a numerator linear in x^2. Replace x^2 by $(x^2+a^2) - a^2$; this separates the nth integral from the $(n-1)$st.

Solution *The base case is* $J_1(x) = a^{-1}\arctan(x/a)$. *For* $n \geq 2$,

$$\left(\frac{x}{(x^2+a^2)^{n-1}}\right)' = \frac{2(n-1)a^2}{(x^2+a^2)^n} - \frac{2n-3}{(x^2+a^2)^{n-1}}.$$

Integrating from 0 *to* x *yields*

$$J_n(x) = \frac{x}{2(n-1)a^2(x^2+a^2)^{n-1}} + \frac{2n-3}{2(n-1)a^2}J_{n-1}(x).$$

There are no arbitrary constants in this fixed-integral recurrence. For $a = 1$, $n = 2$,

$$\int \frac{\mathrm{d}x}{(1+x^2)^2} = \frac{x}{2(1+x^2)} + \frac{1}{2}\arctan x + C.$$

The recurrence terminates at J_1 and applies on all of $\mathbb{R}$. Together with division and partial fractions, it shows that every rational function has primitives expressible through rational functions, logarithms and arctangents on each original domain interval. It is usually more efficient to exploit a numerator derivative than to expand every term mechanically.

Example 3.31 Reciprocal structure before decomposition Find primitives of $(1+x^2)/(x^4+3x^2+1)$ and $(1-x^2)/(x^4+3x^2+1)$.

> ***Divide a symmetric quartic by its middle power.*** Dividing the denominator by x^2 gives x^2+3+x^{-2}. This is both $(x-1/x)^2+5$ and $(x+1/x)^2+1$. The two proposed numerators, divided by x^2, match the derivatives of these expressions up to sign.

Solution ***Method 1: reciprocal substitutions.*** *For $x \neq 0$, put $u = x - 1/x$ in the first integral and $v = x + 1/x$ in the second:*

$$\int \frac{1+x^2}{x^4+3x^2+1}\,\mathrm{d}x = \frac{1}{\sqrt{5}}\arctan\frac{x-1/x}{\sqrt{5}} + C, \qquad \int \frac{1-x^2}{x^4+3x^2+1}\,\mathrm{d}x = -\arctan(x+1/x) + C.$$

These are composite-derivative identities; $v' = 0$ at $x = \pm 1$ causes no difficulty. The substitutions introduce an artificial hole at 0, where the original integrands are continuous. For the first formula choose constants $-\pi/(2\sqrt{5})$ on $x < 0$ and $\pi/(2\sqrt{5})$ on $x > 0$; for the second choose $-\pi/2$ and $\pi/2$ respectively. Set both resulting functions to zero at 0. Each is a global primitive: on either half-line it differs from $\int_0^x f(t)\,\mathrm{d}t$ by a constant, and its zero limit at 0 makes that constant zero. The FTC then gives the derivative at 0.

Solution ***Method 2: factor over the reals.*** *Let $\alpha = (3+\sqrt{5})/2$ and $\beta = (3-\sqrt{5})/2$, so $\alpha+\beta = 3$, $\alpha\beta = 1$, and the denominator is $(x^2+\alpha)(x^2+\beta)$. For $c = 1$ or $c = -1$,*

$$\frac{1+cx^2}{(x^2+\alpha)(x^2+\beta)} = \frac{A}{x^2+\alpha} + \frac{B}{x^2+\beta}, \qquad A = \frac{c\alpha - 1}{\alpha - \beta}, \quad B = \frac{1-c\beta}{\alpha-\beta}.$$

Indeed $A + B = c$ and $A\beta + B\alpha = 1$. A global primitive, zero at the origin, is

$$\frac{A}{\sqrt{\alpha}}\arctan\frac{x}{\sqrt{\alpha}} + \frac{B}{\sqrt{\beta}}\arctan\frac{x}{\sqrt{\beta}}.$$

Differentiation checks the answer on all of $\mathbb{R}$.

The reciprocal changes are shorter and expose the numerator's role. Factorization gives a single global formula without matching constants; equality follows because both global primitives have the same derivative and value at zero.

3.3.6 Parameters and domains

Example 3.32 A parameter changes the factor type Find primitives of $1/(x^2+\lambda)$ for every real λ.

Solution *If $\lambda > 0$, the denominator is irreducible, and the answer is $\lambda^{-1/2}\arctan(x/\sqrt{\lambda}) + C$ on $\mathbb{R}$. If $\lambda = 0$, it is x^2, giving $-1/x + C$ separately on the two sides of 0. If $\lambda < 0$, put $a = \sqrt{-\lambda} > 0$ and decompose*

$$\frac{1}{x^2-a^2} = \frac{1}{2a}\left(\frac{1}{x-a} - \frac{1}{x+a}\right).$$

The answer is $\frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| + C$ on each interval separated by $-a, a$. The sign of λ determines both the algebra and the domain; the positive-parameter formula cannot simply be used at $\lambda = 0$.

3.3.7 Methods summary

Before decomposing a reciprocal polynomial, compare the numerator with the derivatives of $x - 1/x$ and $x + 1/x$. Any artificial hole introduced by such a change requires matching constants if a global primitive is wanted.

Keep the original excluded points. Divide an improper fraction, factor its denominator, and write the full partial-fraction template. Determine coefficients in a polynomial identity and verify that identity before integrating. For quadratics, split off the denominator derivative and complete the square. Reduce repeated quadratic powers to the arctangent base case. Finally differentiate the primitive and check parameters, signs and constants on every connected domain interval.

Give primitive answers on the original connected open domains.

Exercise 3.3

1. Evaluate (a) $\int x^2/(x^2+1)\,\mathrm{d}x$, (b) $\int x^4/(x^2+1)\,\mathrm{d}x$, and (c) $\int x^3/(x+1)\,\mathrm{d}x$. State the original domains before division.
2. Evaluate $\int (2x+1)/(x^2+x+1)\,\mathrm{d}x$.
3. Evaluate (a) $\int 1/[x(x+1)]\,\mathrm{d}x$, (b) $\int 1/[x(x+2)]\,\mathrm{d}x$, and (c) $\int 1/[(x+1)(x+2)]\,\mathrm{d}x$. State every excluded point before decomposing.
4. Evaluate (a) $\int 1/(x-2)^2\,\mathrm{d}x$, (b) $\int x/(x-2)^2\,\mathrm{d}x$, and (c) $\int x/(x-2)^3\,\mathrm{d}x$. Use $x = (x-2)+2$ to predict when a logarithm occurs.
5. Evaluate $\int 1/(x^2+4)\,\mathrm{d}x$.
6. Compute $\int_0^1 1/[(x+1)(x+2)]\,\mathrm{d}x$.
7. Compute $\int_0^1 x/(x^2+2)\,\mathrm{d}x$.
8. Evaluate $\int (x^3+1)/(x^2-1)\,\mathrm{d}x$ and state every original domain interval.
9. Evaluate $\int (x+3)/[(x-1)(x+2)]\,\mathrm{d}x$.
10. Evaluate $\int (x^2+1)/[x(x-1)^2]\,\mathrm{d}x$.
11. Evaluate (a) $\int (2x+5)/(x^2+2x+5)\,\mathrm{d}x$, (b) $\int (2x+2)/(x^2+2x+5)^2\,\mathrm{d}x$, and (c) $\int 1/(x^2+2x+5)^2\,\mathrm{d}x$. Explain why only the last integral requires the repeated-quadratic reduction.
12. Evaluate $\int 1/[x^2(x+1)]\,\mathrm{d}x$.
13. Evaluate (a) $\int (x+1)/(x^2+1)^2\,\mathrm{d}x$, (b) $\int x/(x^2+1)^3\,\mathrm{d}x$, and (c) $\int (x^2-1)/(x^2+1)^2\,\mathrm{d}x$. Decide which terms need the quadratic recurrence and which are already derivatives of rational functions.
14. Compute $\int_0^1 (1+x^2)^{-2}\,\mathrm{d}x$.
15. Write the complete template for a proper fraction with denominator $(x-1)^3(x^2+1)^2$. Explain both numerator degrees.
16. For $a > 0$, decompose $1/(x^2-a^2)$ and integrate it, specifying the intervals.
17. Compute $\int_0^1 1/(x^2+2x+2)\,\mathrm{d}x$.
18. Explain why division, partial fractions and the quadratic recurrence suffice for every rational function. Identify the resulting types of terms.
19. Evaluate $\int (x^4+1)/(x^2+1)^2\,\mathrm{d}x$. Simplify the numerator before using the recurrence.
20. Compute $\int_{-1}^1 (x^3+2x^2+1)/(x^2+1)\,\mathrm{d}x$ using parity before decomposition.

Supplement Exercises

S1. Supply the details of the uniqueness argument in Theorem 3.4: multiply a zero decomposition by its denominator and reduce modulo each underlying factor, starting with its highest power.

S2. Evaluate $\int 1/(1+x^4)\,\mathrm{d}x$ by factoring into $x^2+\sqrt{2}x+1$ and $x^2-\sqrt{2}x+1$. Verify the decomposition.

S3. For $0<a<1$, compute $\int_a^{1/a} 1/(1+x^4)\,\mathrm{d}x$. First pair x with $1/x$, then use $u=x-1/x$ in the sum.

S4. For $\lambda>0$, evaluate $F(\lambda)=\int_0^1 1/(x^2+\lambda)\,\mathrm{d}x$ and prove $F(\lambda)\to+\infty$ as $\lambda\downarrow 0$. For $0<\lambda\le 1$, estimate the contribution from $[0,\sqrt{\lambda}]$.

S5. For $(x^2+1)/(x^4+3x^2+1)$, use $u=x-1/x$ on $x\neq 0$ to obtain local primitives. Choose constants so they extend to a primitive on $\mathbb{R}$. Use the FTC to justify its derivative at 0.

S6. Evaluate $\int (1+x^2)^{-3}\,\mathrm{d}x$ using the recurrence, and differentiate your answer.

S7. Integrate $(x^2+c)/(1+x^2)^2$ for real c. For which c does the arctangent term in the reduction formula vanish?

S8. Evaluate $\int 1/[x^3(1+x)]\,\mathrm{d}x$, retaining every required power.

S9. On $[-1,1]$ define $f_c(x)=(x^2-1)/(x-1)$ when $x\neq 1$ and $f_c(1)=c\in\mathbb{R}$. Prove that f_c is Riemann integrable and compute its integral for every c. Estimate how changing one endpoint value affects a tagged sum.

S10. Compute $\int_0^1 (x^5-x)/(1+x^2)^3\,\mathrm{d}x$ by substituting $u=1+x^2$. Check the sign of your result from the integrand.

S11. For $c>-2$, prove $x^4+cx^2+1>0$ on $\mathbb{R}$. Find primitives of $(1+x^2)/(x^4+cx^2+1)$ on $x\neq 0$ by $u=x-1/x$ and match constants to obtain a primitive on $\mathbb{R}$. What fails when $c=-2$? Identify the genuine poles.

S12. For $a\neq 0$ and $x>\max(0,-a)$, verify that a primitive of $1/[x(x+a)]$ is $F_a(x)=a^{-1}\ln(x/(x+a))$. Prove $F_a(x)\to -1/x$ as $a\to 0$ for each fixed $x>0$. Explain why taking the limit of the two logarithm coefficients separately is misleading.

S13. Let P be a polynomial of degree at most 2 and let $r\neq s$ be real. In the decomposition $P(x)/[(x-r)^2(x-s)]=A/(x-r)+B/(x-r)^2+D/(x-s)$, prove

$$B=\frac{P(r)}{r-s},\qquad A=\frac{P'(r)}{r-s}-\frac{P(r)}{(r-s)^2},\qquad D=\frac{P(s)}{(s-r)^2}.$$

Apply this to $P(x)=x^2+2$, $r=0$, $s=-1$, and integrate on every domain interval.

S14. For real a,b,c, find constants A,B,D such that

$$\frac{ax^2+bx+c}{(1+x^2)^2}=\left(\frac{Ax+B}{1+x^2}\right)'+\frac{D}{1+x^2}.$$

Integrate the identity and determine exactly when its arctangent coefficient vanishes. Compare this method with using the quadratic recurrence after splitting the numerator.

3.4 Techniques for Trigonometric and Related Integrals

知识点概要

- ❑ *Use parity to reserve a differential.*
- ❑ *Reduce powers and products.*
- ❑ *Rationalize with a local half-angle chart.*
- ❑ *Choose branches for quadratic radicals.*
- ❑ *Use a hyperbolic alternative.*
- ❑ *Rationalize fractional powers and exponentials.*

An identity can change an integral's structure before a substitution is chosen. For trigonometric powers, the identities $\sin^2 x + \cos^2 x = 1$ and $\sec^2 x = 1 + \tan^2 x$ make one derivative available while turning the remaining factors into a polynomial. For radicals, the same identities suggest a change that removes the square root. The angle range is part of the change, since $\sqrt{\cos^2 \theta} = |\cos \theta|$.

3.4.1 Powers of sine and cosine

Consider $\int \sin^m x \cos^n x \, \mathrm{d}x$, where m, n are nonnegative integers. The aim is to reserve the derivative of one trigonometric function and express every other factor in that same function. The identity $\sin^2 x + \cos^2 x = 1$ converts even powers, so the parity of each exponent determines what can be reserved.

An odd sine power: $m = 2k + 1$, $k \geq 0$. Separate one sine factor and set $u = \cos x$, so $\mathrm{d}u = -\sin x \, \mathrm{d}x$. For any integer $n \geq 0$,

$$\begin{aligned} \int \sin^{2k+1} x \cos^n x \, \mathrm{d}x &= \int (1 - \cos^2 x)^k \cos^n x \sin x \, \mathrm{d}x \\ &= -\int (1 - u^2)^k u^n \, \mathrm{d}u. \end{aligned}$$

The transformed integrand is a polynomial. Expanding by the binomial theorem and integrating term by term gives the finite formula

$$-\sum_{r=0}^{k} (-1)^r \binom{k}{r} \frac{\cos^{n+2r+1} x}{n + 2r + 1} + C.$$

In particular, $k = 0$ needs no expansion: one sine factor is already the required differential.

An odd cosine power: $n = 2j + 1$, $j \geq 0$. Now reserve $\cos x \, \mathrm{d}x$ and use $u = \sin x$. For any integer $m \geq 0$,

$$\begin{aligned} \int \sin^m x \cos^{2j+1} x \, \mathrm{d}x &= \int \sin^m x (1 - \sin^2 x)^j \cos x \, \mathrm{d}x \\ &= \int u^m (1 - u^2)^j \, \mathrm{d}u \\ &= \sum_{r=0}^{j} (-1)^r \binom{j}{r} \frac{\sin^{m+2r+1} x}{m + 2r + 1} + C. \end{aligned}$$

For instance, $m = 2, j = 1$ gives $\sin^3 x/3 - \sin^5 x/5 + C$. If both exponents are odd, either substitution works; expanding the smaller of $(1 - u^2)^k$ and $(1 - u^2)^j$ usually requires fewer terms. Neither substitution requires an inverse sine or cosine: the chain rule verifies the resulting primitive on all of $\mathbb{R}$.

Two even powers: $m = 2k$, $n = 2j$, $k, j \geq 0$. There is no unpaired factor to supply either differential. Instead, the double-angle identities give

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \qquad \cos^2 x = \frac{1 + \cos 2x}{2}.$$

Consequently,

$$\int \sin^{2k} x \cos^{2j} x \, \mathrm{d}x = \frac{1}{2^{k+j}} \int (1 - \cos 2x)^k (1 + \cos 2x)^j \, \mathrm{d}x.$$

Expand the polynomial in $\cos 2x$. When $k + j > 0$, its powers have degree at most $k + j$, strictly below the original total degree $2(k + j)$. An odd power of cosine is handled by reserving one cosine factor; an even power is reduced again by the double-angle identity. The degrees decrease until only constants and first powers remain, so this procedure terminates after finitely many steps. Remember the frequency factor: if $y = cx$ with $c \neq 0$, then $\mathrm{d}x = \mathrm{d}y/c$. The exceptional case $k = j = 0$ is simply $\int 1 \, \mathrm{d}x = x + C$.

Example 3.33 Reserve the odd factor Evaluate $\int \sin^3 x \cos^2 x \, \mathrm{d}x$.

Solution *Write* $\sin^3 x = (1 - \cos^2 x) \sin x$ *and set* $u = \cos x$*:*

$$-\int (1 - u^2) u^2 \, \mathrm{d}u = \frac{\cos^5 x}{5} - \frac{\cos^3 x}{3} + C.$$

This substitution is a composite-derivative calculation and does not require cosine to be injective on the primitive's domain.

Example 3.34 Two even powers Evaluate $\int \sin^2 x \cos^2 x \, \mathrm{d}x$.

Solution *Since* $\sin^2 x \cos^2 x = \frac{1}{4} \sin^2 2x = \frac{1}{8}(1 - \cos 4x)$*, the answer is* $x/8 - \sin 4x/32 + C$*. The average term survives; replacing even powers only by a derivative would miss it.*

Example 3.35 A square root retains an absolute value Compute $\int_0^{\pi} \sqrt{1 + \cos 2x} \, \mathrm{d}x$.

> ***Simplification is not permission to drop a sign.*** The double-angle identity changes the root to $\sqrt{2 \cos^2 x}$. The sign of cosine changes halfway through the interval, whereas the original integrand is nonnegative everywhere. This identifies the point at which a formula must change.

Solution ***Method 1: split at the sign change.*** *Since* $\sqrt{2 \cos^2 x} = \sqrt{2} |\cos x|$*,*

$$\begin{aligned} \int_0^{\pi} \sqrt{1 + \cos 2x} \, \mathrm{d}x &= \sqrt{2} \left(\int_0^{\pi/2} \cos x \, \mathrm{d}x - \int_{\pi/2}^{\pi} \cos x \, \mathrm{d}x \right) \\ &= \sqrt{2}\big((1 - 0) - (0 - 1)\big) = 2\sqrt{2}. \end{aligned}$$

Solution ***Method 2: reflect the nonnegative integrand.*** *The identity* $|\cos(\pi - x)| = |\cos x|$ *makes the two half-interval integrals equal. On* $[0, \pi/2]$ *cosine is nonnegative, so the answer is* $2\sqrt{2} \int_0^{\pi/2} \cos x \, \mathrm{d}x = 2\sqrt{2}$*.*

The split records each sign explicitly; reflection avoids repeating the second calculation. Replacing the root by $\sqrt{2} \cos x$ on the whole interval would incorrectly give zero.

> ***Save one differential, convert the rest.*** The parity of an exponent indicates whether one sine or cosine factor can be removed without leaving an odd power behind. Reserve that factor first, then rewrite everything else in the proposed variable. When both powers are even, frequency reduction supplies a constant term and simpler oscillations instead.

3.4.2 Tangent, secant and products

We use $\sec x = 1/\cos x$ on $\cos x \neq 0$. Consider $\int \sec^n x \tan^m x \, \mathrm{d}x$ for nonnegative integers m, n, on a fixed open interval where $\cos x \neq 0$. The two useful derivative factors are

$$(\tan x)' = \sec^2 x, \qquad (\sec x)' = \sec x \tan x,$$

and the identity $\sec^2 x = 1 + \tan^2 x$ converts the factors left over.

A positive even secant power: $n = 2k$, $k \geq 1$. Reserve $\sec^2 x\,\mathrm{d}x$ and put $u = \tan x$. Whatever the parity of m,

$$\begin{aligned}\int \sec^{2k} x \tan^m x\,\mathrm{d}x &= \int (1+\tan^2 x)^{k-1} \tan^m x \sec^2 x\,\mathrm{d}x \\ &= \int (1+u^2)^{k-1} u^m\,\mathrm{d}u \\ &= \sum_{r=0}^{k-1} \binom{k-1}{r} \frac{\tan^{m+2r+1} x}{m+2r+1} + C.\end{aligned}$$

Thus an even secant power of at least two always gives a polynomial integral.

Example 3.36 Tangent and secant powers Evaluate $\int \tan^3 x \sec^4 x\,\mathrm{d}x$.

Solution *Reserve* $\sec^2 x\,\mathrm{d}x$ *and replace the other* $\sec^2 x$ *by* $1+\tan^2 x$. *Thus* $u = \tan x$ *gives* $\int u^3(1+u^2)\,\mathrm{d}u = \tan^4 x/4 + \tan^6 x/6 + C$. *The answer applies separately between successive zeros of* $\cos x$.

First reduce a pure odd secant power. Write $I_r = \int \sec^r x\,\mathrm{d}x$. As in Section 3.2, we suppress additive constants while deriving a relation between primitives, and restore one arbitrary constant in the final answer.

> ***Make the integral return with a useful coefficient.*** For $\sec^{2k+1} x$, reserving $\sec^2 x\,\mathrm{d}x$ leaves an odd power of secant, not a polynomial in tangent. Integration by parts uses that same factor differently: integrating it gives $\tan x$, while differentiating the remaining secant power supplies a second tangent. The resulting $\tan^2 x = \sec^2 x - 1$ produces the original integral and one with exponent two lower.

For $k \geq 1$, take $u = \sec^{2k-1} x$ and $\mathrm{d}v = \sec^2 x\,\mathrm{d}x$. Then $v = \tan x$ and $\mathrm{d}u = (2k-1)\sec^{2k-1} x \tan x\,\mathrm{d}x$, so

$$\begin{aligned}I_{2k+1} &= \sec^{2k-1} x \tan x - (2k-1)\int \sec^{2k-1} x \tan^2 x\,\mathrm{d}x \\ &= \sec^{2k-1} x \tan x - (2k-1)\big(I_{2k+1} - I_{2k-1}\big).\end{aligned}$$

Moving the returning term to the left and dividing by $2k$ gives

$$I_{2k+1} = \frac{\sec^{2k-1} x \tan x}{2k} + \frac{2k-1}{2k} I_{2k-1}, \qquad k \geq 1. \tag{3.1}$$

Each repetition lowers the odd exponent by two, ending at I_1. For example, $I_5 = \sec^3 x \tan x/4 + 3I_3/4$, and the next step reduces I_3 to I_1. To establish that base case, observe that the derivative of $\sec x + \tan x$ contains the same factor. Since $(\sec x + \tan x)' = \sec x(\sec x + \tan x)$ and $(\sec x + \tan x)(\sec x - \tan x) = 1$,

$$\int \sec x\,\mathrm{d}x = \ln|\sec x + \tan x| + C.$$

The logarithm's argument never vanishes on a domain interval, but it can be negative.

Example 3.37 An odd secant power Evaluate $\int \sec^3 x\,\mathrm{d}x$ on $\cos x \neq 0$.

Solution *Parts with* $u = \sec x$, $v = \tan x$ *gives*

$$\int \sec^3 x\,\mathrm{d}x = \sec x \tan x - \int \sec x \tan^2 x\,\mathrm{d}x = \sec x \tan x - \int \sec^3 x\,\mathrm{d}x + \int \sec x\,\mathrm{d}x.$$

Choosing fixed primitives before rearranging, or checking the resulting derivative, yields

$$\frac{1}{2}\big(\sec x \tan x + \ln|\sec x + \tan x|\big) + C.$$

The returning integral is useful because its coefficient becomes 2.

An odd secant power with an even tangent power. Let $n = 2k+1$ and $m = 2j$, where $k, j \geq 0$. Use $\tan^2 x = \sec^2 x - 1$ to remove all tangent factors:

$$\begin{aligned}\int \sec^{2k+1} x \tan^{2j} x \, dx &= \int \sec^{2k+1} x (\sec^2 x - 1)^j \, dx \\ &= \sum_{r=0}^{j} (-1)^{j-r} \binom{j}{r} I_{2(k+r)+1}.\end{aligned}$$

Every term is a pure odd secant integral, so (3.1) completes the calculation. For a concrete specialization,

$$\begin{aligned}\int \sec^3 x \tan^2 x \, dx = I_5 - I_3 &= \frac{1}{4} \sec^3 x \tan x - \frac{1}{4} I_3 \\ &= \frac{1}{4} \sec^3 x \tan x - \frac{1}{8} \sec x \tan x - \frac{1}{8} \ln |\sec x + \tan x| + C.\end{aligned}$$

The case $j = 0$ is the pure odd power already treated; when $k = j = 0$, the base case alone suffices.

An odd secant power with an odd tangent power. Let $n = 2k+1$ and $m = 2j+1$, where $k, j \geq 0$. Now one secant and one tangent can be reserved together. With $u = \sec x$,

$$\begin{aligned}\int \sec^{2k+1} x \tan^{2j+1} x \, dx &= \int \sec^{2k} x (\sec^2 x - 1)^j \sec x \tan x \, dx \\ &= \int u^{2k} (u^2 - 1)^j \, du \\ &= \sum_{r=0}^{j} (-1)^{j-r} \binom{j}{r} \frac{\sec^{2(k+r)+1} x}{2(k+r)+1} + C.\end{aligned}$$

For instance, $\int \sec^3 x \tan^3 x \, dx = \sec^5 x/5 - \sec^3 x/3 + C$. This route requires no secant reduction formula. More generally, when $m = 2j+1$ and $n \geq 1$, the same reserved differential gives $\int u^{n-1}(u^2-1)^j \, du$. Thus if n is positive and even while m is odd, both tangent and secant substitutions work; choose the polynomial requiring fewer expanded terms. These are chain-rule substitutions, so zeros of $\tan x$ need not be excluded merely because the secant change is not locally invertible there.

The boundary case $n = 0$. Zero is even, but no secant factor is available to reserve. For $T_m = \int \tan^m x \, dx$ and $m \geq 2$, write

$$\tan^m x = \tan^{m-2} x \sec^2 x - \tan^{m-2} x.$$

The first term is integrated with $u = \tan x$, giving

$$T_m = \frac{\tan^{m-1} x}{m-1} - T_{m-2}.$$

The base primitives are $T_0 = x$ and $T_1 = -\ln |\cos x|$, up to constants. If both original powers are zero, the integrand is simply 1 and its primitive $x + C$ is valid on all of $\mathbb{R}$.

Products of different frequencies. For products of different frequencies, the addition formulas give, for example,

$$2 \sin A \cos B = \sin(A+B) + \sin(A-B), \qquad 2 \cos A \cos B = \cos(A+B) + \cos(A-B).$$

Adding the formulas for $A+B$ and $A-B$ proves these identities. They turn a product into a sum of single-frequency primitives, including a constant when a frequency vanishes.

Example 3.38 A denominator and its derivative Evaluate $\int (2 \sin x + 3 \cos x)/(\sin x + \cos x) \, dx$ on its original domain.

> ***Solve a small coefficient problem.*** Differentiating $D = \sin x + \cos x$ gives $D' = \cos x - \sin x$. These two expressions allow independent control of the sine and cosine coefficients. Try $AD + BD'$ before applying a general rationalizing change.

Solution ***Method 1: linear combination.*** *The identity* $2\sin x + 3\cos x = \frac{5}{2}D + \frac{1}{2}D'$ *gives*

$$\frac{5}{2}x + \frac{1}{2}\ln|\sin x + \cos x| + C.$$

This works on each interval separated by $x = -\pi/4 + k\pi$, $k \in \mathbb{Z}$.

Solution ***Method 2: tangent substitution.*** *On an interval with* $\cos x \neq 0$, *put* $t = \tan x$. *Then*

$$\frac{2t+3}{(t+1)(1+t^2)} = \frac{1}{2(t+1)} + \frac{-t/2 + 5/2}{1+t^2}.$$

Integration gives $\frac{1}{2}\ln|t+1| - \frac{1}{4}\ln(1+t^2) + \frac{5}{2}\arctan t + C$. *Here* $\arctan(\tan x) = x - k\pi$ *on each tangent chart, and the logarithms combine to* $\frac{1}{2}\ln|\sin x + \cos x|$. *Thus the local answers agree with Method 1 up to constants.*

The first route is shorter and remains valid at zeros of $\cos x$ where $D \neq 0$. The tangent route is systematic but introduces chart boundaries that the original problem did not have.

3.4.3 The half-angle rationalization

Some integrands are rational expressions in both $\sin x$ and $\cos x$ with no convenient reserved differential. The *tangent half-angle substitution*(万能三角代换) $t = \tan(x/2)$ gives

$$\sin x = \frac{2t}{1+t^2}, \qquad \cos x = \frac{1-t^2}{1+t^2}, \qquad \mathrm{d}x = \frac{2\,\mathrm{d}t}{1+t^2}.$$

The formulas follow from the double-angle identities and the derivative of tangent. They turn any rational expression in sine and cosine into a rational function of t. This is a local chart: on $((2k-1)\pi, (2k+1)\pi)$, $k \in \mathbb{Z}$, its inverse is $x = 2\arctan t + 2k\pi$. An original denominator must still be nonzero.

Example 3.39 A rational half-angle integral Compute $\int_0^{\pi/2} 1/(2+\sin x)\,\mathrm{d}x$.

Solution *The interval lies in one chart, and its transformed endpoints are* $0, 1$. *Therefore*

$$\int_0^{\pi/2} \frac{\mathrm{d}x}{2+\sin x} = \int_0^1 \frac{\mathrm{d}t}{t^2+t+1} = \frac{2}{\sqrt{3}}\left[\arctan\frac{2t+1}{\sqrt{3}}\right]_0^1 = \frac{\pi}{3\sqrt{3}}.$$

The completed square is $(t+1/2)^2 + 3/4$, *and the angles are* $\pi/3$ *and* $\pi/6$.

If an interval crosses an odd multiple of π, the chart has a pole even when the original integrand is continuous there. Work on the two adjacent finite subintervals and take the limits of their original integrals, or match constants in local primitives. Continuity justifies these limits by the integral estimate from Section 2.6. A single change that ignores the pole is invalid; equal transformed endpoints do not imply zero integral. We do not assign an integral to an unbounded transformed interval here.

3.4.4 Quadratic radicals and branches

Completing the square reduces quadratic radicals to the following forms, with $a > 0$. For $\sqrt{a^2 - x^2}$ choose $x = a\sin\theta$, $-\pi/2 < \theta < \pi/2$, so the root is $a\cos\theta > 0$. For $\sqrt{a^2 + x^2}$ choose $x = a\tan\theta$ on the same angle range, giving $a\sec\theta > 0$. For an exterior radical $\sqrt{x^2 - a^2}$, positive and negative x require separate branches. The range decision must precede removal of an absolute value.

Example 3.40 A circular radical in two ways Evaluate $\int \sqrt{a^2 - x^2}\,\mathrm{d}x$ for $a > 0$, $|x| < a$.

Solution ***Method 1: trigonometric substitution.***

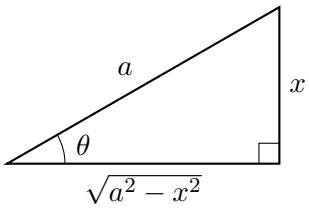


Figure 3.2: For $0 < x < a$, the triangle records $\sin\theta = x/a$ and the positive root. For $x < 0$, use the signed angle, not a negative side length.

With $x = a\sin\theta$ on the stated range, the integral is $a^2\int\cos^2\theta\,\mathrm{d}\theta$. Using $\cos^2\theta = (1+\cos 2\theta)/2$ gives

$$a^2\int\cos^2\theta\,\mathrm{d}\theta = \frac{a^2}{2}\theta + \frac{a^2}{4}\sin 2\theta + C.$$

Here $\theta = \arcsin(x/a)$ and $\sin 2\theta = 2\sin\theta\cos\theta = 2x\sqrt{a^2-x^2}/a^2$. Consequently the primitive is

$$\frac{1}{2}\left(x\sqrt{a^2-x^2} + a^2\arcsin\frac{x}{a}\right) + C.$$

Solution ***Method 2: parts and a returning integral.*** *Choose a primitive I of the root. Parts gives, up to a constant, $I = x\sqrt{a^2-x^2} + \int x^2/\sqrt{a^2-x^2}\,\mathrm{d}x$. Since $x^2 = a^2 - (a^2-x^2)$, the latter is $a^2\arcsin(x/a) - I$ up to a constant. Thus $2I = x\sqrt{a^2-x^2} + a^2\arcsin(x/a) + C$, yielding the same family.*

The first route exposes the circular identity; the second uses only the known arcsine derivative and parts. Both require the interior domain for derivative calculations. The integrand and displayed primitive extend continuously to the endpoints, so the FTC evaluates proper definite integrals there.

Example 3.41 A quotient of complementary radicals Evaluate $\int\sqrt{(1+x)/(1-x)}\,\mathrm{d}x$ on $-1 < x < 1$.

> ***Make the two factors share one root.*** Both $1+x$ and $1-x$ are positive on the given interval. Multiplying numerator and denominator inside the square root can turn the quotient into $(1+x)/\sqrt{1-x^2}$. Alternatively, naming the entire root makes x a rational expression in the new variable.

Solution ***Method 1: split after an algebraic identity.*** *Positivity gives*

$$\sqrt{\frac{1+x}{1-x}} = \frac{1+x}{\sqrt{1-x^2}}.$$

Consequently

$$\begin{aligned}\int\sqrt{\frac{1+x}{1-x}}\,\mathrm{d}x &= \int\frac{\mathrm{d}x}{\sqrt{1-x^2}} + \int\frac{x\,\mathrm{d}x}{\sqrt{1-x^2}}\\ &= \arcsin x - \sqrt{1-x^2} + C.\end{aligned}$$

In the second integral, $u = 1-x^2$ gives $-\frac{1}{2}\int u^{-1/2}\,\mathrm{d}u$.

Solution ***Method 2: rationalize the whole root.*** *Set $t = \sqrt{(1+x)/(1-x)} > 0$. Solving $t^2(1-x) = 1+x$ gives*

$$x = \frac{t^2-1}{t^2+1}, \qquad \mathrm{d}x = \frac{4t}{(1+t^2)^2}\,\mathrm{d}t.$$

This maps $(0,\infty)$ increasingly onto $(-1,1)$. The transformed integral is

$$\begin{aligned}4\int\frac{t^2}{(1+t^2)^2}\,\mathrm{d}t &= 4\int\left(\frac{1}{1+t^2} - \frac{1}{(1+t^2)^2}\right)\mathrm{d}t\\ &= 2\arctan t - \frac{2t}{1+t^2} + C,\end{aligned}$$

using the quadratic reduction from Section 3.3. Now $2t/(1+t^2) = \sqrt{1-x^2}$ and $\arcsin x = 2\arctan t - \pi/2$, since the right side lies in $(-\pi/2,\pi/2)$ and has sine x. The displayed primitives differ by $\pi/2$.

The algebraic split is shorter for this numerator. Whole-root rationalization is more systematic for rational expressions in the same root. The angle range is needed when comparing inverse-trigonometric formulas.

3.4.5 A hyperbolic alternative

For a plus sign under the radical, an identity with a difference of squares is useful. Define the *hyperbolic sine and cosine*(双曲正弦与双曲余弦)by

$$\sinh t = \frac{e^t - e^{-t}}{2}, \qquad \cosh t = \frac{e^t + e^{-t}}{2}.$$

Direct algebra and differentiation give $\cosh^2 t - \sinh^2 t = 1$, $(\sinh t)' = \cosh t$, and $(\cosh t)' = \sinh t$. Also $\cosh t > 0$; for instance $\sinh 0 = 0$ and $\cosh 0 = 1$. The positive derivative makes $\sinh$ strictly increasing, by integrating the derivative on each interval. Its limits at the two ends of $\mathbb{R}$ are $\pm\infty$, so it has an inverse on $\mathbb{R}$.

Example 3.42 Trigonometric and hyperbolic routes Evaluate $\int 1/\sqrt{a^2+x^2}\,\mathrm{d}x$ for $a > 0$.

Solution ***Method 1: tangent substitution.*** *Set $x = a\tan\theta$, $|\theta| < \pi/2$. The integral becomes $\int \sec\theta\,\mathrm{d}\theta$. Since $\sec\theta + \tan\theta = (\sqrt{a^2+x^2}+x)/a > 0$,*

$$\int \frac{\mathrm{d}x}{\sqrt{a^2+x^2}} = \ln\frac{x+\sqrt{a^2+x^2}}{a} + C.$$

Solution ***Method 2: hyperbolic substitution.*** *Set $x = a\sinh t$. Then the positive root is $a\cosh t$ and $\mathrm{d}x = a\cosh t\,\mathrm{d}t$, so the integral is $t + C$. Writing $z = e^t > 0$, the equation $2x/a = z - z^{-1}$ has the positive solution $z = (x+\sqrt{x^2+a^2})/a$. Thus t is the same logarithm.*

The tangent route stays with circular functions; the hyperbolic route cancels the radical immediately. Both cover every real x, and the logarithm argument is positive even when $x < 0$.

Example 3.43 An exterior radical on both branches Evaluate $\int 1/[x\sqrt{x^2-a^2}]\,\mathrm{d}x$ for $a > 0$, $|x| > a$.

> ***Try the root itself before choosing an angle.*** Here the radical's derivative is $x/\sqrt{x^2-a^2}$. The extra reciprocal x can be absorbed by $x^2 = u^2 + a^2$. This keeps the two signs of x separate without requiring two angle formulas.

Solution ***Method 1: use the radical as variable.***

For $u = \sqrt{x^2-a^2} > 0$, $\mathrm{d}u = x\,\mathrm{d}x/u$. On either domain interval,

$$\frac{\mathrm{d}x}{x\sqrt{x^2-a^2}} = \frac{\mathrm{d}u}{u^2+a^2},$$

so a primitive is $a^{-1}\arctan(\sqrt{x^2-a^2}/a) + C$. On the negative branch u decreases with x; the signed differential already records this.

Solution ***Method 2: a reciprocal circular change.***

Let $w = a/|x| \in (0,1)$. The derivative of $-a^{-1}\arcsin(a/|x|)$ is

$$\frac{(|x|)'}{|x|\sqrt{x^2-a^2}} = \frac{1}{x\sqrt{x^2-a^2}},$$

because $(|x|)'/|x| = 1/x$ on both intervals. This gives another primitive.

The angle identity $\arctan(\sqrt{x^2-a^2}/a) + \arcsin(a/|x|) = \pi/2$ shows that the two answers differ by $\pi/(2a)$. The first route avoids inverse-function sign bookkeeping; the second makes the role of $|x|$ explicit.

Example 3.44 A conjugate pair as a new variable Evaluate $\int 1/(x+\sqrt{1+x^2})\,\mathrm{d}x$ on $\mathbb{R}$.

> ***Use a constant product.*** The denominator and its conjugate satisfy $(\sqrt{1+x^2}+x)(\sqrt{1+x^2}-x) = 1$. Multiplication removes the denominator; alternatively, naming one factor t makes the other $1/t$, so their sum and difference express both the root and x.

Solution ***Method 1: rationalize and use parts.*** *The integrand equals* $\sqrt{1+x^2}-x$. *Writing* $r=\sqrt{1+x^2}$, *we have* $(xr)'=2r-1/r$. *Using the plus-root primitive already proved gives*

$$\frac{1}{2}\bigl(xr+\ln(x+r)-x^2\bigr)+C.$$

Since $r>|x|$, *the logarithm's argument is positive for every real* x.

Solution ***Method 2: rationalize both variables.*** *Set* $t=x+r>0$. *The conjugate identity gives*

$$x=\frac{t-t^{-1}}{2},\qquad r=\frac{t+t^{-1}}{2},\qquad \mathrm{d}x=\frac{1+t^{-2}}{2}\,\mathrm{d}t.$$

The inverse formula has positive derivative on $(0,\infty)$ *and ranges over* $\mathbb{R}$. *Thus the integral becomes*

$$\frac{1}{2}\int(t^{-1}+t^{-3})\,\mathrm{d}t=\frac{1}{2}\ln t-\frac{1}{4t^2}+C.$$

Because $xr-x^2=(1-t^{-2})/2$, *the two displayed primitives differ by* $1/4$.

The conjugate split is fastest here. The new variable is an instance of an *Euler substitution*(欧拉代换); it converts every rational expression in x and $\sqrt{1+x^2}$ into a rational expression in t, with original denominator restrictions retained.

3.4.6 Other rationalizing changes

Fractional powers become integer powers if we choose a common denominator of their exponents. For positive x, setting $x=t^q$, $t>0$, makes each $x^{p/q}$ a power of t. The positivity specifies the inverse branch. Real odd roots can also be treated on negative intervals, with their actual real-root definition.

Example 3.45 Two fractional powers at once Evaluate $\int 1/[\sqrt{x}(1+\sqrt[3]{x})]\,\mathrm{d}x$ for $x>0$.

Solution *The exponents have denominators* $2,3$, *so put* $x=t^6$, $t>0$. *Then* $\sqrt{x}=t^3$, $\sqrt[3]{x}=t^2$, *and* $\mathrm{d}x=6t^5\,\mathrm{d}t$. *The integral becomes*

$$\int\frac{6t^5}{t^3(1+t^2)}\,\mathrm{d}t=6\int\left(1-\frac{1}{1+t^2}\right)\mathrm{d}t=6(t-\arctan t)+C.$$

Return to $t=x^{1/6}$ *to obtain* $6(x^{1/6}-\arctan(x^{1/6}))+C$. *Both roots have been rationalized, rather than removing one and leaving the other.*

A rational expression in e^x becomes rational in $t=e^x>0$, since $\mathrm{d}x=\mathrm{d}t/t$. More generally, $t=e^{ax}$ works for $a\neq 0$ with $\mathrm{d}x=\mathrm{d}t/(at)$.

Example 3.46 Rationalization or a direct split Evaluate $\int 1/(1+e^x)\,\mathrm{d}x$.

Solution ***Method 1: exponential substitution.*** *With* $t=e^x>0$, $\int \mathrm{d}t/[t(1+t)]=\ln t-\ln(1+t)+C=x-\ln(1+e^x)+C$.

Solution ***Method 2: split the numerator.*** *The identity* $1/(1+e^x)=1-e^x/(1+e^x)$ *gives the same answer by one logarithmic derivative.*

The split is shorter here; rationalization extends to general rational expressions in e^x. Differentiating either answer gives the original positive integrand.

3.4.7 Methods summary

The methods in this section change the expression of an integral while preserving its value or primitive family on the stated domain. The decisive object is the whole differential $f(x)\,\mathrm{d}x$. For an inverse change $x=\phi(t)$, it becomes $f(\phi(t))\phi'(t)\,\mathrm{d}t$; simplifying $f(\phi(t))$ alone is not enough. Alternatively, recognizing $F(g(x))g'(x)$ allows direct composition of a primitive, without requiring g to be one-to-one.

From trigonometric expressions to polynomials or rational functions. A polynomial is a rational function with denominator 1. It is the simplest target because its terms can be integrated immediately; a general rational function is treated by division and partial fractions from Section 3.3. For example, reserving the odd sine factor in $\sin^{2k+1} x \cos^n x \, \mathrm{d}x$ leaves the polynomial $-(1-u^2)^k u^n$ in $u = \cos x$. When no such factor is available, the half-angle formulas express both sine and cosine rationally in $t = \tan(x/2)$, and $\mathrm{d}x = 2\,\mathrm{d}t/(1+t^2)$ is rational as well. Sums, products and quotients of rational expressions remain rational wherever the denominators are nonzero. This is why the change works for a rational expression in both sine and cosine, rather than only for one specially chosen numerator. The local chart and the original denominator restrictions still have to be checked.

From hyperbolic functions to exponentials and rational functions. The definitions earlier in this section imply

$$\cosh t + \sinh t = e^t, \qquad \cosh t - \sinh t = e^{-t}.$$

Thus $z = e^t > 0$ gives

$$\sinh t = \frac{z - z^{-1}}{2}, \qquad \cosh t = \frac{z + z^{-1}}{2}, \qquad \mathrm{d}t = \frac{\mathrm{d}z}{z}.$$

Consequently a rational expression $R(\sinh t, \cosh t)\,\mathrm{d}t$ becomes

$$R\left(\frac{z - z^{-1}}{2}, \frac{z + z^{-1}}{2}\right)\frac{\mathrm{d}z}{z},$$

which is rational in z on its original domain with $z > 0$. The same principle explains $z = e^{ax}$ for rational expressions in e^{ax}: for $a \neq 0$, the differential is $\mathrm{d}x = \mathrm{d}z/(az)$. Negative integer powers of z cause no difficulty; they are rational functions, although they are not polynomials.

A quadratic radical connects the different coordinates. Let $a > 0$ and $r = \sqrt{a^2 + x^2}$. The circular change $x = a\tan\theta$, $|\theta| < \pi/2$, gives $r = a\sec\theta > 0$. The hyperbolic change $x = a\sinh t$ gives $r = a\cosh t > 0$ on all of $\mathbb{R}$. Following the second change by $z = e^t$ yields

$$x = \frac{a}{2}(z - z^{-1}), \qquad r = \frac{a}{2}(z + z^{-1}), \qquad \mathrm{d}x = \frac{a}{2}(1 + z^{-2})\,\mathrm{d}z.$$

Adding the first two formulas gives $z = (x + r)/a$. Thus the earlier Euler coordinate $u = x + r$ is exactly az: the hyperbolic and exponential changes together explain the same rational parametrization obtained from the conjugate identity $(r + x)(r - x) = a^2$. The circular coordinate is related by $z = \tan\theta + \sec\theta$ on the chosen angle range. These are exact identities between different coordinates for the same radical. Their usefulness depends on the differential: in Example 3.42, the hyperbolic change gives $\mathrm{d}x/r = \mathrm{d}t$ immediately, while the circular change gives $\sec\theta\,\mathrm{d}\theta$. If the remaining expression is more complicated, the Euler coordinate makes both x, r and the differential rational at once.

Rational powers and the limits of the method. For finitely many rational powers x^{p_i/q_i} on $x > 0$, choose a positive common multiple q of the positive denominators q_i and set $x = t^q$, $t > 0$. Then $x^{p_i/q_i} = t^{qp_i/q_i}$ has an integer exponent and $\mathrm{d}x = qt^{q-1}\,\mathrm{d}t$. Every rational expression in these powers is therefore rational in t, including its differential. The common choice is what removes all roots in Example 3.45 at once.

These transformations apply to specified algebraic combinations; they do not turn every mixture of elementary functions into a rational one. For example, $t = e^x$ changes $\mathrm{d}x/(x + e^x)$ into

$$\frac{\mathrm{d}t}{t(\ln t + t)},$$

which still contains a logarithm. This calculation does not decide whether some other method succeeds; it shows why every remaining factor must be inspected.

> ***Choose a simpler expression, then check the change.*** First try a visible derivative, an algebraic split, or a linear combination of a denominator and its derivative. For powers, reserve a differential; for products, reduce frequencies; for a residual odd secant power, reduce its exponent. Use a general rationalizing change when these shorter routes do not finish the calculation. Track the differential, original domain, inverse branch and transformed bounds, then differentiate the answer on each original domain interval. Different coordinates may produce different-looking primitives; after their domains are matched, their derivatives reveal whether they differ only by a constant.

Exercise 3.4

1. Evaluate (a) $\int \sin^5 x \, dx$, (b) $\int \sin^3 x \cos^4 x \, dx$, and (c) $\int \sin^4 x \cos^3 x \, dx$. Explain how parity selects the differential in each case.
2. Evaluate $\int \sin x \cos^3 x \, dx$.
3. Evaluate $\int \tan x \sec^2 x \, dx$.
4. Evaluate the following on intervals where $\cos x \neq 0$.
   (a) $\int \tan^2 x \sec^2 x \, dx$, (b) $\int \tan x \sec^3 x \, dx$, and (c) $\int \tan^2 x \sec^4 x \, dx$. State whether $\tan x$ or $\sec x$ is the more direct variable.
5. Evaluate (a) $\int \sin 2x \cos 3x \, dx$, (b) $\int \cos 2x \cos 3x \, dx$, and (c) $\int \cos^2(3x) \, dx$. Explain why the equal-frequency case has a linear term.
6. Compute $\int_0^{\pi/2} \cos^3 x \, dx$.
7. Compute $\int_0^{\pi} \sin^2 x \, dx$.
8. Evaluate $\int \cos^4 x \, dx$.
9. Evaluate $\int \tan^2 x \, dx$.
10. Derive $\int \sec x \, dx$ by multiplying numerator and denominator by $\sec x + \tan x$. Explain why the logarithm uses an absolute value.
11. Compute $\int_0^{\pi/2} 1/(2 + \cos x) \, dx$ by half-angle substitution, and also compare it with Example 3.39 by reflection.
12. Evaluate and state the original domains: (a) $\int 1/\sqrt{9 - x^2} \, dx$, (b) $\int x/\sqrt{9 - x^2} \, dx$, and (c) $\int 1/(9 + x^2) \, dx$. Compare the scale factors and the different roles of the signs under the denominator.
13. For $a > 0$, evaluate $\int x^2/\sqrt{a^2 - x^2} \, dx$ on $|x| < a$.
14. For $a > 0$, evaluate $\int \sqrt{x^2 + a^2} \, dx$ by parts and Example 3.42.
15. For $a > 0$, evaluate $\int 1/[x^2\sqrt{x^2 - a^2}] \, dx$ on both exterior intervals.
16. Evaluate $\int 1/(\sqrt{x} + \sqrt[4]{x}) \, dx$ for $x > 0$.
17. Evaluate (a) $\int e^{2x}/(1 + e^x) \, dx$, (b) $\int e^x/(1 + e^{2x}) \, dx$, and (c) $\int 1/(e^x + e^{-x}) \, dx$. Compare the transformed rational functions and explain why (b) and (c) have the same answer.
18. Compute $\int_0^{2\pi} 1/(2 + \cos x) \, dx$ using symmetry and finite half-angle subintervals. Explain why the chart pole at π is harmless for the original integrand.
19. Evaluate $\int 1/\sqrt{x^2 - 4x + 13} \, dx$.
20. Evaluate $\int 1/[\sqrt{1 - x^2}(1 + x^2)] \, dx$ for $|x| < 1$ by $x = \sin\theta$ followed by $t = \tan\theta$.

Supplement Exercises

S1. Explain an algorithm for $\int \sin^m x \cos^n x \, dx$ for integers $m, n \geq 0$. When $m = 2r + 1$, give the finite polynomial primitive in $\cos x$; explain how the other parity cases terminate.

S2. For integers $n \geq 2$, on a fixed interval with $\cos x \neq 0$, derive

$$\int \sec^n x \, dx = \frac{\sec^{n-2} x \tan x}{n-1} + \frac{n-2}{n-1} \int \sec^{n-2} x \, dx \quad (n \geq 2).$$

Specify the base primitives and verify the recurrence by differentiation.

S3. Compute $\int_0^{\pi/2} \sin^3 x/(\sin x + \cos x) \, dx$ by reflection and the sum-of-cubes identity.

S4. For $a, b > 0$, compute $\int_0^{\pi/2} 1/(a^2 \cos^2 x + b^2 \sin^2 x) \, dx$. Use $t = \tan x$ below the endpoint, then continuity of the original integral.

S5. For real a, b with $a > |b|$, compute $\int_0^{2\pi} 1/(a + b \cos x) \, dx$. Explain the role of this condition and of splitting the half-angle chart.

S6. For $a > 0$, evaluate $\int 1/\sqrt{x^2 - a^2} \, dx$ separately for $x > a$ and $x < -a$ using $x = \pm a \cosh t$, $t > 0$. Combine the answers with a sign-safe logarithm.

S7. Starting from local half-angle primitives of $1/(2 + \cos x)$, choose constants on all charts to obtain a global primitive on $\mathbb{R}$. Justify differentiability at the chart poles by the FTC.

S8. Evaluate $\int 1/[(\sqrt[3]{x})^2(1 + \sqrt[3]{x})] \, dx$ for real $x \neq 0, -1$. Explain why a real cube-root substitution allows negative branches.

S9. Compute $\int_0^1 \sqrt{1 - x^2}/(1 + x^2) \, dx$. After $x = \sin\theta$, split $\cos^2\theta/(1 + \sin^2\theta)$ into a constant and a multiple of $1/(1 + \sin^2\theta)$.

S10. Evaluate $\int (1 + e^x)/(1 + e^{2x}) \, dx$. Rationalize and identify a logarithm, an arctangent and a linear term.

S11. Let $a, b, p, q \in \mathbb{R}$ with $a^2 + b^2 > 0$. Evaluate $\int (p \sin x + q \cos x)/(a \sin x + b \cos x) \, dx$ on each original domain interval. Find exactly when the logarithmic term vanishes, and check the formula when $a = 0$ or $b = 0$.

S12. For $a > 0$, set $t = x + \sqrt{x^2 + a^2}$. Prove that this is a differentiable bijection $\mathbb{R} \to (0, \infty)$ with differentiable inverse, and express both x and the radical rationally in t. Use it to evaluate $\int 1/(x + \sqrt{x^2 + a^2})^2 \, dx$. Verify the result by differentiation and explain why $t = 0$ is never reached.

S13. For $a < b$, evaluate $\int \sqrt{(x - a)/(b - x)} \, dx$ on (a, b) in two ways: (a) put $x = a + (b - a)\sin^2\theta$, $0 < \theta < \pi/2$; (b) put $t = \sqrt{(x - a)/(b - x)}$ and use rational integration. Compare the resulting primitives, including their angle ranges.

S14. Let $n \geq 1$ be an integer. (a) Compute $\int_0^{\pi} \sqrt{1 + \cos(2nx)} \, dx$. (b) Compute $\int_0^{\pi} x\sqrt{1 + \cos(2nx)} \, dx$ using reflection. (c) Explain why replacing the square root by $\sqrt{2}\cos(nx)$ is invalid, even though the squared expressions agree.

3.5 Applications of Definite Integrals

知识点概要

- ❏ *Build an integral from small pieces.*
- ❏ *Compare vertical, horizontal and polar areas.*
- ❏ *Choose slices, washers or shells.*
- ❏ *Measure length and surface area.*
- ❏ *Account for force, depth and lift.*
- ❏ *Locate centers of mass.*

A definite integral is useful only after its integrand has been identified. For each model, specify the object, the small piece, its approximate contribution and the limiting sum. A strip contributes area, a slice contributes volume, and a weighted slice can contribute mass or work. Units and geometry often reveal a missing factor before calculation begins. All integrals in this section are proper Riemann integrals.

3.5.1 Area between graphs

Let L, U be continuous on $[a, b]$ with $L \le U$. The region $a \le x \le b$, $L(x) \le y \le U(x)$ has *area*(面积)

$$A = \int_a^b (U(x) - L(x))\,\mathrm{d}x.$$

Indeed, inner and outer rectangles bound the region on each small interval. Uniform continuity makes the difference of the bounding sums tend to zero; their common limit is the displayed integral. The integrand is the vertical strip's height. If the region is instead $c \le y \le d$, $\ell(y) \le x \le r(y)$, horizontal strips give $A = \int_c^d (r - \ell)\,\mathrm{d}y$. Both descriptions compute the same geometric area through bounding rectangles.

Area is nonnegative, whereas $\int_a^b f$ is signed accumulation. The area between $y = f(x)$ and the axis is $\int_a^b |f(x)|\,\mathrm{d}x$. When two curves cross, find the crossings and split where their order changes.

Example 3.47 Two directions for the same region Find the area enclosed by $y = x^2$ and $y = x$.

Solution ***Method 1: vertical strips.*** *The intersections are* $x = 0, 1$, *and* $x^2 \le x$ *there. Thus* $A = \int_0^1 (x - x^2)\,\mathrm{d}x = 1/6$.

Solution ***Method 2: horizontal strips.*** *For* $0 \le y \le 1$, *the region has* $y \le x \le \sqrt{y}$. *Hence* $A = \int_0^1 (\sqrt{y} - y)\,\mathrm{d}y = 2/3 - 1/2 = 1/6$.

Both directions need one integral. The inequalities describing the region, rather than a curve's name, determine which function is subtracted.

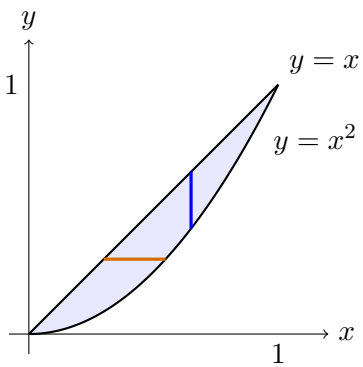


Figure 3.3: The blue segment measures $x - x^2$; the orange segment measures $\sqrt{y} - y$. A thin strip adds the corresponding $\mathrm{d}x$ or $\mathrm{d}y$.

Example 3.48 One direction avoids a split Find the area enclosed by $y^2 = 2x$ and $x - y = 4$.

> ***Inspect the endpoints of a strip.*** The parabola is already a function of y: its left boundary is $x = y^2/2$. The line supplies a right boundary $x = y + 4$. A vertical strip encounters two different lower boundaries, so horizontal strips should simplify the setup.

Solution ***Method 1: horizontal strips.*** *The intersections satisfy* $y^2/2 = y + 4$, *giving* $y = -2, 4$. *For* $-2 \le y \le 4$, *the line is to the right:*

$$A = \int_{-2}^{4} \left(y + 4 - \frac{y^2}{2} \right) \mathrm{d}y = \left[\frac{y^2}{2} + 4y - \frac{y^3}{6} \right]_{-2}^{4} = \frac{40}{3} - \left(-\frac{14}{3} \right) = 18.$$

Solution ***Method 2: vertical strips.*** *The region projects onto* $0 \le x \le 8$. *For* $0 \le x \le 2$, *its full height is* $2\sqrt{2x}$. *For* $2 \le x \le 8$, *its lower boundary is the line* $y = x - 4$:

$$A = \int_0^2 2\sqrt{2x}\,\mathrm{d}x + \int_2^8 \left(\sqrt{2x} - (x - 4) \right) \mathrm{d}x = \frac{16}{3} + \frac{38}{3} = 18.$$

The second route verifies the region's geometry; the first is shorter because its boundary descriptions do not change.

3.5.2 Polar area

Polar coordinates (极坐标) describe a point by $x = r\cos\theta$, $y = r\sin\theta$, with $r \ge 0$. The origin has every angle; other points have angles differing by multiples of 2π. For example, $r = a$ describes a circle of radius a, and a fixed angle describes a ray.

Suppose $r = r(\theta) \ge 0$ is continuous on $[\alpha, \beta]$, with $0 < \beta - \alpha \le 2\pi$, and the region consists of $0 \le s \le r(\theta)$ along each ray, counted once. A sector of radius r and angle $\Delta\theta$ has area $\frac{1}{2}r^2\Delta\theta$. Minimum and maximum radii on each angular interval bound the region by sectors. Uniform continuity then gives

$$A = \frac{1}{2}\int_\alpha^\beta r(\theta)^2\,\mathrm{d}\theta.$$

For two continuous radial boundaries $0 \le r_{\text{in}} \le r_{\text{out}}$, subtract their squared radii. The angle is in radians. Tracing a boundary twice or allowing signed radii without tracking their rays can count area twice.

Example 3.49 A single petal For $a > 0$, find the area of the petal $r = a\sin 3\theta$ on $0 \le \theta \le \pi/3$.

Solution *The radius is nonnegative, vanishes at both ends and is positive inside. The intervening rays are distinct, so this is one petal:*

$$A_{\text{petal}} = \frac{a^2}{2}\int_0^{\pi/3} \sin^2 3\theta\,\mathrm{d}\theta = \frac{a^2}{4}\left[\theta - \frac{\sin 6\theta}{6} \right]_0^{\pi/3} = \frac{\pi a^2}{12}.$$

Rotating this petal through $2\pi/3$ *and* $4\pi/3$ *gives two disjoint congruent petals (apart from the origin), with total area* $\pi a^2/4$. *No extra traversal is needed.*

3.5.3 Volumes by slices and revolution

For the regular solids considered here, *volume* (体积) is obtained as the limit of thin prism approximations. If a slice perpendicular to the x-axis has continuous area $A(x)$, these sums give

$$V = \int_a^b A(x)\,\mathrm{d}x.$$

Continuous graph boundaries justify the bounding approximations as in the area construction. A slice's area has units of length squared, so multiplying by its thickness gives length cubed. The same formula applies along another coordinate.

Rotating a region about an axis produces a *solid of revolution*(旋转体). Perpendicular slices are disks, with area πR^2, or *washers*(圆环截面), with area $\pi(R^2 - r^2)$. The radii are distances to the rotation axis, not automatically the values of named functions. If a vertical strip at $x > 0$ has height $h(x)$ and rotates about the y-axis, it gives a *cylindrical shell*(柱壳). A thin radial interval of thickness Δx has volume $\pi((x+\Delta x)^2 - x^2)h(x)$, whose leading term is $2\pi x h(x)\Delta x$. Thus $V = 2\pi \int_a^b x h(x)\,\mathrm{d}x$ for a region on one side of the axis with distinct strips producing disjoint radial layers. For a shifted axis, use its distance in place of x.

Example 3.50 Shells and washers Rotate $0 \le y \le x^2$, $0 \le x \le 1$, about the y-axis. Find the volume.

Solution ***Method 1: shells.*** *Radius x, height x^2 and thickness $\mathrm{d}x$ give $V = 2\pi \int_0^1 x^3\,\mathrm{d}x = \pi/2$.*

Solution ***Method 2: washers.*** *At height $0 \le y \le 1$, the original region has $\sqrt{y} \le x \le 1$. The outer radius is 1 and inner radius is $\sqrt{y}$: $V = \pi \int_0^1 (1 - y)\,\mathrm{d}y = \pi/2$.*

Shells keep the original graph variable; washers expose the hole around the axis. Both are valid because every radial layer or cross-section is counted once.

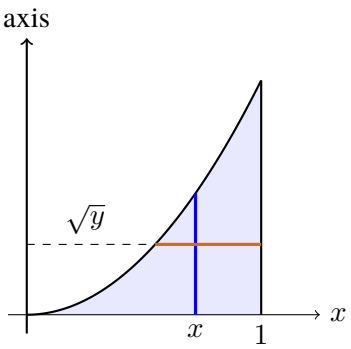


Figure 3.4: A vertical strip gives a shell of height x^2. A horizontal strip gives a washer with inner radius $\sqrt{y}$ and outer radius 1.

Example 3.51 Overlapping rotations Rotate the rectangle $[-1, 1] \times [0, 1]$ about the y-axis. Is $2\pi \int_{-1}^1 |x|\,\mathrm{d}x$ its volume?

Solution *The solid is a cylinder of radius 1 and height 1, so its volume is π. The proposed integral is 2π, since the strips at x and $-x$ generate the same shell. Either use only $0 \le x \le 1$, or take the union of the generated cross-sections. Adding contributions is valid only for pieces with disjoint interiors.*

Example 3.52 An axis through the region Rotate the triangle $0 \le y \le x \le 1$ about the line $y = 1/2$. Find the volume of the resulting solid, counting overlapping rotations only once.

> ***Rotate a whole segment before choosing radii.*** A vertical segment below the axis leaves a hole; a segment that reaches the axis does not. When the segment crosses the axis, its two sides can generate the same disk. The outer radius is the larger endpoint distance, not a signed difference of the endpoints.

Solution ***Method 1: washers followed by disks.*** *For $0 \le x \le 1/2$, the segment $0 \le y \le x$ lies below the axis. Its outer radius is $1/2$ and inner radius is $1/2 - x$, so $A(x) = \pi(1/4 - (1/2 - x)^2) = \pi(x - x^2)$. For $1/2 \le x \le 1$, it contains the axis. The lower endpoint is at distance $1/2$ and the upper endpoint at distance $x - 1/2 \le 1/2$, so $A(x) = \pi/4$. Thus*

$$V = \pi \int_0^{1/2} (x - x^2)\,\mathrm{d}x + \frac{\pi}{4} \int_{1/2}^1 \mathrm{d}x$$
$$= \pi \left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^{1/2} + \frac{\pi}{8} = \frac{\pi}{12} + \frac{\pi}{8} = \frac{5\pi}{24}.$$

Solution ***Method 2: take the union at each shell radius.*** *For $0 \le r \le 1/2$, the horizontal levels $y = 1/2 - r$ and $y = 1/2 + r$ rotate to the same radius. The triangle contains the intervals*

$$[1/2 - r, 1] \quad \textit{and} \quad [1/2 + r, 1]$$

of x at those levels. The second interval lies inside the first; their union has length $1/2 + r$. Therefore

$$V = 2\pi \int_0^{1/2} r(1/2 + r)\,\mathrm{d}r = 2\pi \left[\frac{r^2}{4} + \frac{r^3}{3}\right]_0^{1/2} = \frac{5\pi}{24}.$$

Perpendicular slices reveal where the hole disappears. Shells are also valid after overlapping horizontal intervals have been replaced by their union. Adding both lengths would instead give $\pi/4$, overcounting by $\pi/24$.

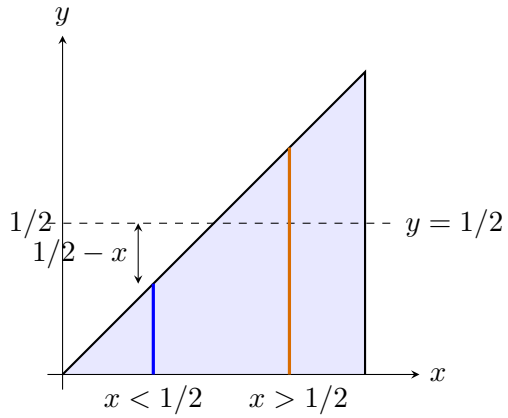


Figure 3.5: The blue segment generates a washer. The orange segment crosses the axis and generates a disk; rotating its upper part adds no new volume.

Example 3.53 A solid not obtained by rotation A solid has base $x^2 + y^2 \le 1$ in the plane. Its cross-sections perpendicular to the x-axis are equilateral triangles whose sides are the corresponding base chords. Find its volume.

Solution *At $x \in [-1, 1]$, the side is $2\sqrt{1 - x^2}$. The triangle's area is $\frac{\sqrt{3}}{4}(2\sqrt{1 - x^2})^2 = \sqrt{3}(1 - x^2)$. Hence $V = \sqrt{3}\int_{-1}^{1}(1 - x^2)\,\mathrm{d}x = 4\sqrt{3}/3$. The cross-section's shape, rather than a rotation formula, determines the integrand.*

Example 3.54 The napkin-ring problem A cylindrical hole of radius r, $0 \le r < R$, is bored along a diameter of a sphere of radius R. Show that the remaining volume depends only on its height $H = 2\sqrt{R^2 - r^2}$, and find it.

> ***Subtract areas before integrating.*** At height y, the outer squared radius is $R^2 - y^2$ and the inner squared radius is r^2. Their difference depends on R and r only through $R^2 - r^2 = (H/2)^2$. This cancellation suggests comparing slices with those of a smaller sphere.

Solution ***Method 1: washers.*** *Put $h = H/2$. The solid exists for $-h \le y \le h$, where its annular area is $\pi(h^2 - y^2)$. Therefore*

$$V = \pi \int_{-h}^{h} (h^2 - y^2)\,\mathrm{d}y = \frac{4\pi h^3}{3} = \frac{\pi H^3}{6}.$$

At $r = 0$ this recovers the sphere's volume; as $r \uparrow R$, the height and volume tend to zero.

Solution ***Method 2: equal cross-sections.*** *A sphere of radius h has disk area $\pi(h^2 - y^2)$ at the same height. The two solids therefore have equal volumes by the slice formula, although one slice is an annulus and the other a disk. The sphere's disk integral gives $4\pi h^3/3$.*

The first calculation proves the formula; the comparison explains its independence from the original radius. Equal outer radii alone would not suffice: the relevant quantity is the complete cross-sectional area.

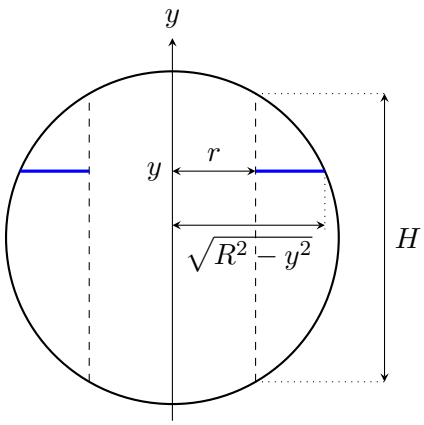


Figure 3.6: A meridian section of the drilled sphere. The blue segments rotate into one annulus. At each height use the outer radius at that same height and the fixed inner radius r.

3.5.4 Arc length

The distance between two points is the Euclidean norm $\|(u, v)\| = \sqrt{u^2 + v^2}$ of their difference. For a continuous parametrized curve $\gamma(t) = (x(t), y(t))$ on $[a, b]$, its *arc length*(弧长)is the supremum of the polygonal lengths

$$\sum_i \|\gamma(t_i) - \gamma(t_{i-1})\|$$

over all partitions $a = t_0 < \cdots < t_n = b$. A curve with finite length is called rectifiable. This definition measures the entire parametrized traversal; repeated passages contribute repeatedly. A line segment has its endpoint distance as length.

Theorem 3.5 (Length of a continuously differentiable curve)

For $\gamma \in C^1([a, b]; \mathbb{R}^2)$,

$$L = \int_a^b \|\gamma'(t)\| \, \mathrm{d}t = \int_a^b \sqrt{x'(t)^2 + y'(t)^2} \, \mathrm{d}t.$$

In particular, the graph $y = f(x)$ with $f \in C^1([a, b])$ has $L = \int_a^b \sqrt{1 + f'(x)^2} \, \mathrm{d}x$. ♡

Here C^1 for a parametrized curve means that both coordinate functions have continuous first derivatives.

> ***Compare chords with local velocity.*** The FTC expresses each displacement as an accumulated velocity. Uniform continuity says that velocity is nearly constant on a short interval. Its norm times the interval width should therefore approximate a chord, with an error controlled uniformly across the partition.

Proof Integrate vectors componentwise. The elementary inequality $\| \int v\| \le \int \|v\|$ follows by taking the scalar projection onto the direction of $\int v$ and using $u \cdot v \le \|u\|\|v\|$; if $\int v = 0$ it is immediate. The latter inequality in two dimensions follows from $(u_1 v_2 - u_2 v_1)^2 \ge 0$. By the FTC each chord is at most the integral of $\|\gamma'\|$ on its interval. Every polygonal length is therefore at most the displayed integral.

For $\delta \geq 0$, let $\omega(\delta) = \sup\{\|\gamma'(s)-\gamma'(t)\| : s,t \in [a,b],\ |s-t| \leq \delta\}$. Uniform continuity gives $\omega(\delta) \to 0$. For a partition of mesh at most δ, the FTC and the triangle inequality give

$$\big|\|\gamma(t_i) - \gamma(t_{i-1})\| - \|\gamma'(t_{i-1})\|(t_i - t_{i-1})\big| \leq \omega(\delta)(t_i - t_{i-1}).$$

Summing, polygonal lengths differ from the indicated Riemann sums by at most $\omega(\delta)(b-a)$. Along finer partitions they converge to the integral, so the supremum is also at least that integral. For a graph, take $\gamma(x) = (x, f(x))$. □

The proof allows $\gamma' = 0$ at isolated points, including endpoints. For finitely many C^1 pieces, split and add their lengths. If a graph's derivative is unbounded at an endpoint, the stated graph theorem does not apply there; a C^1 parametrization may resolve the issue.

Example 3.55 Simplify the speed first Find the length of $y = x^3/6 + 1/(2x)$ for $1 \leq x \leq 2$.

Solution *Here $y' = x^2/2 - 1/(2x^2)$ and $1 + y'^2 = (x^2/2 + 1/(2x^2))^2$. The latter factor is positive, so*

$$L = \int_1^2 \left(\frac{x^2}{2} + \frac{1}{2x^2}\right) \mathrm{d}x = \left[\frac{x^3}{6} - \frac{1}{2x}\right]_1^2 = \frac{17}{12}.$$

Expanding the root without looking for this square would obscure an elementary integral.

Example 3.56 A cycloid with endpoint cusps For $a > 0$, find the length of $x = a(t-\sin t)$, $y = a(1-\cos t)$, $0 \leq t \leq 2\pi$.

Solution *The speed is $a\sqrt{(1-\cos t)^2 + \sin^2 t} = 2a|\sin(t/2)| = 2a\sin(t/2)$ on this interval. Hence $L = 2a\int_0^{2\pi} \sin(t/2)\,\mathrm{d}t = 8a$. Both coordinate derivatives vanish at the endpoints, but remain continuous. Thus the parameter theorem includes the cusps; a graph derivative formula would be awkward there.*

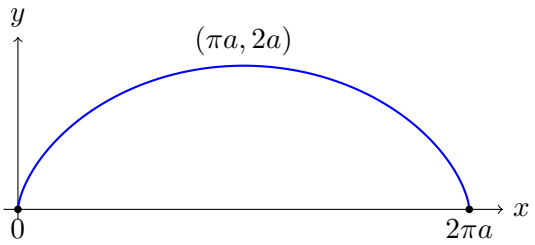


Figure 3.7: One cycloid arch. The parametrization has continuous velocity even where the endpoint tangent has a cusp. Coordinates are scaled by a.

3.5.5 Surface area of revolution

Rotate a nonnegative graph $y = f(x)$, $f \in C^1([a,b])$, about the x-axis. Its *surface area*(旋转曲面面积) is defined by the limit of the areas of frustums obtained from an inscribed polygonal graph. A segment of length Δs with endpoint radii r_0, r_1 generates a frustum of lateral area $\pi(r_0+r_1)\Delta s$. Uniform continuity of f, together with the chord estimates in the length proof, makes these sums converge to

$$S = 2\pi \int_a^b f(x)\sqrt{1+f'(x)^2}\,\mathrm{d}x.$$

The factor $2\pi f$ is circumference, so $(2\pi f)\,\mathrm{d}s$ has units of area. The formula includes only the generated lateral surface, not flat end caps. For a C^1 curve with $y(t) \geq 0$ and $x(t)$ strictly monotone, the same frustum construction gives

$$S = 2\pi \int_\alpha^\beta y(t)\sqrt{x'(t)^2+y'(t)^2}\,\mathrm{d}t.$$

These hypotheses ensure that the surface patches are counted once. Repeated or overlapping generating curves need a separate geometric check.

Example 3.57 The sphere without an endpoint derivative problem Find the surface area of the sphere of radius $a > 0$.

Solution *Rotate the upper semicircle, parametrized by $x = a\cos t$, $y = a\sin t$, $0 \leq t \leq \pi$. Here x is strictly decreasing, $y \geq 0$, and the speed is a: $S = 2\pi a^2 \int_0^\pi \sin t\,\mathrm{d}t = 4\pi a^2$. The graph $y = \sqrt{a^2-x^2}$ has an unbounded derivative at $\pm a$; the C^1 parametrization avoids using the graph theorem outside its hypotheses.*

3.5.6 Work and hydrostatic force

If a scalar force $F(x)$ acts along the positive direction of displacement, its *work*(功) from a to b is the limit of sums $F(\xi_i)\Delta x_i$: $W = \int_a^b F(x)\,\mathrm{d}x$. A negative force contribution produces negative work. For example, a constant force F_0 over a distance d does work $F_0 d$. In SI units, newton times metre is joule. For lifting fluid, we assume constant mass density ρ, constant gravitational acceleration g, and negligible friction; a piece of volume ΔV weighs $\rho g \Delta V$. Its required work is weight times its lifting distance.

Example 3.58 Pumping from a cylinder A vertical cylinder of radius 1 metre contains water from height $y = 0$ to $y = 2$ metres. Water is pumped to an outlet at height 3 metres. Express the work in terms of ρ, g.

Solution *A horizontal slice has volume approximately $\pi\,\mathrm{d}y$, weight $\rho g\pi\,\mathrm{d}y$ and lift $3-y$. Thus $W = \rho g\pi \int_0^2 (3-y)\,\mathrm{d}y = 4\pi\rho g$ joules when all lengths are entered in metres and ρ, g in SI units. The tank height alone is not the lifting distance; upper slices travel less far.*

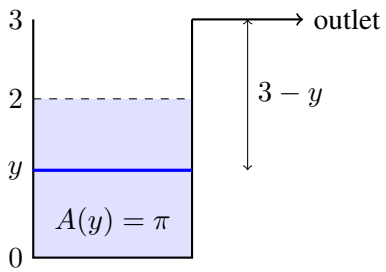


Figure 3.8: Use a horizontal fluid slice. Its cross-sectional area and its lift are different factors in the work integral. Heights are in metres.

For a fluid at rest with constant weight density $\gamma = \rho g$, the *hydrostatic pressure*(静水压强)at depth y below its horizontal surface is $p(y) = \gamma y$. This follows by balancing the weight of a vertical fluid column against its base area. We measure pressure relative to atmospheric pressure. For a vertical plate at depths $0 \le c < d$, with continuous width $w(y) \ge 0$, each horizontal strip has area $w(y)\,\mathrm{d}y$. The total force on one face is $F = \gamma \int_c^d y w(y)\,\mathrm{d}y$. Pressure has units force per area; multiplying by strip area restores force.

Example 3.59 A plate widening with depth A vertical triangular plate extends from depth 0 to 2 metres and has horizontal width $w(y) = y$ metres. Find the hydrostatic force.

Solution *The strip contribution is $\gamma y \cdot y\,\mathrm{d}y$. Therefore $F = \gamma \int_0^2 y^2\,\mathrm{d}y = 8\gamma/3$ newtons in SI units. Depth and width both vary; using one endpoint pressure for the whole triangle is not exact.*

3.5.7 Mass, moments and centers

A continuous nonnegative *linear density*(线密度)$\rho(x)$ on a rod $[a, b]$ assigns mass approximately $\rho(x)\,\mathrm{d}x$ to a small segment. Its mass and first moment are $M = \int_a^b \rho(x)\,\mathrm{d}x$ and $J = \int_a^b x\rho(x)\,\mathrm{d}x$. If $M > 0$, its *center of mass*(质心)is $\bar{x} = J/M$. This is the mass-weighted mean position and satisfies the balance identity $\int_a^b (x - \bar{x})\rho(x)\,\mathrm{d}x = 0$. For a uniform rod, $\bar{x} = (a + b)/2$; if $M = 0$, this quotient does not define a center.

Example 3.60 A nonuniform rod Find the center of mass of a rod on $[0, 2]$ with density $\rho(x) = 1 + x$.

Solution $M = \int_0^2 (1 + x)\,\mathrm{d}x = 4$ *and* $J = \int_0^2 (x + x^2)\,\mathrm{d}x = 14/3$. *Thus* $\bar{x} = 7/6$, *to the right of the midpoint, as expected from the increasing density.*

The *areal density*(面密度)is mass per unit area. For a uniform thin *lamina*(薄片)of constant areal density $\mu > 0$ between $L(x)$ and $U(x)$, its mass is μA; a rectangle of area A_0 has mass μA_0. A thin vertical rectangle has mean height $(U + L)/2$. Consequently its moments about the axes are

$$M_y = \mu \int_a^b x(U - L)\,\mathrm{d}x, \qquad M_x = \frac{\mu}{2} \int_a^b (U^2 - L^2)\,\mathrm{d}x.$$

The center is $(\bar{x}, \bar{y}) = (M_y/M, M_x/M)$. For uniform density this is the geometric *centroid*(形心); the factor μ cancels. The subscripts name the axes: the moment about the y-axis uses the distance coordinate x. Signed coordinates in a moment express position, not negative mass.

Example 3.61 The centroid of the lens Find the centroid of $x^2 \le y \le x$, $0 \le x \le 1$.

Solution *Its area is $A = 1/6$. With density set to 1, $M_y = \int_0^1 x(x - x^2)\,\mathrm{d}x = 1/12$ and $M_x = \frac{1}{2}\int_0^1 (x^2 - x^4)\,\mathrm{d}x = 1/15$. Thus $(\bar{x}, \bar{y}) = (1/2, 2/5)$. At $\bar{x} = 1/2$, the height $2/5$ lies between $1/4$ and $1/2$, consistent with the picture.*

Example 3.62 A centroid computes a volume For continuous $L \le U$ on $[a, b]$, $0 \le a < b$, let the region between their graphs have positive area A and centroid coordinate $\bar{x}$. Prove that its volume of revolution about the y-axis is $V = 2\pi\bar{x}A$. Apply the result to the disk $(x - R)^2 + y^2 \le r^2$, where $R > r > 0$.

Solution ***Method 1: first moment and shells.*** *The region lies on one side of the axis, so distinct vertical strips generate disjoint radial layers. The shell formula and the centroid definition give*

$$V = 2\pi \int_a^b x(U - L)\,\mathrm{d}x = 2\pi A\bar{x}.$$

For the disk, $A = \pi r^2$ and $\bar{x} = R$ by symmetry (the centered first moment is an odd integral). The generated solid, a torus(圆环体)*, therefore has volume $2\pi^2 R r^2$. This is the volume form of the Pappus centroid rule for the stated class of regions: area times the distance traveled by the centroid.*

Solution ***Method 2: direct calculation for the disk.*** *The shell height is $2\sqrt{r^2 - (x - R)^2}$, so $u = x - R$ gives*

$$V = 4\pi \int_{-r}^{r} (R + u)\sqrt{r^2 - u^2}\,\mathrm{d}u = 4\pi R \cdot \frac{\pi r^2}{2} = 2\pi^2 R r^2.$$

The odd term integrates to zero; the remaining integral is the area of a semicircle.

The centroid method reuses geometric information, while the direct calculation verifies that information through symmetry. The one-sided assumption prevents overlapping shells; a signed first moment cannot count the volume of a union when the region crosses the axis.

For a C^1 wire parametrized once by $\gamma(t)$ with continuous nonnegative density $\rho(t)$ per unit length, write $\mathrm{d}m = \rho(t)\|\gamma'(t)\|\,\mathrm{d}t$ as shorthand for its contribution to a mass sum. Then

$$\begin{aligned} M &= \int_\alpha^\beta \rho(t)\|\gamma'(t)\|\,\mathrm{d}t, \\ \bar{x} &= \frac{1}{M}\int_\alpha^\beta x(t)\rho(t)\|\gamma'(t)\|\,\mathrm{d}t, \\ \bar{y} &= \frac{1}{M}\int_\alpha^\beta y(t)\rho(t)\|\gamma'(t)\|\,\mathrm{d}t \end{aligned}$$

when $M > 0$. For instance, a uniform line segment has its midpoint as center. A wire's center generally differs from that of a filled region with the same boundary. A uniform circular wire has its center at the circle's center, because the sine and cosine moments over a full period vanish. This point lies off the wire: a center of mass is a balance position, not necessarily an occupied point.

Example 3.63 Pumping from a tank with changing width A tank is formed by rotating $y = x^2$, $0 \le x \le 2$, about the y-axis and filling its interior up to height 4 metres. Water has constant density ρ kilograms per cubic metre and is pumped to an outlet at height 5 metres. Find the work, with constant gravitational acceleration g and no friction.

> ***Use a height for which each slice has one lifting distance.*** A horizontal slice at height y has radius $\sqrt{y}$, not y^2. Its volume per unit height is πy; its lifting distance is $5 - y$. The same information can also be summarized by the fluid's mass and mean height.

Solution ***Method 1: integrate slice work.*** *The slice contribution is $(\rho g)\cdot(\pi y\,\mathrm{d}y)\cdot(5-y)$. Hence, using metre coordinates,*

$$\begin{aligned} W &= \rho g \pi \int_0^4 y(5-y)\,\mathrm{d}y \\ &= \rho g \pi \left[\frac{5}{2}y^2 - \frac{1}{3}y^3\right]_0^4 = \frac{56\pi}{3}\rho g \quad \textit{joules}. \end{aligned}$$

The radius, slice thickness and lift supply four powers of length; multiplying by weight per unit volume gives force times length.

Solution ***Method 2: total weight and mean lift.*** *The fluid's mass and height moment, computed by horizontal slices, are*

$$M = \rho\pi\int_0^4 y\,\mathrm{d}y = 8\pi\rho, \qquad J = \rho\pi\int_0^4 y^2\,\mathrm{d}y = \frac{64\pi\rho}{3}.$$

Its mean height is $\bar{y} = J/M = 8/3$ metres. Since lifting work is linear in the distance, the total is

$$g(5M - J) = Mg(5 - \bar{y}) = 8\pi\rho g \cdot \frac{7}{3} = \frac{56\pi}{3}\rho g.$$

This use of a mean height follows directly from the two slice integrals; it does not treat all the fluid as actually lying at one height. The answer lies between Mg and $5Mg$, the bounds obtained from lift distances between 1 and 5 metres.

3.5.8 Methods summary

Compare complete cross-sectional areas to explain geometric invariance. For a region on one side of the rotation axis, the centroid can summarize a shell integral; check nonoverlap before using this shortcut.

Describe the region or object before integrating. Choose strips whose boundaries remain simple, and split wherever their order changes. For polar area, count each ray once. For volume, write a cross-sectional area or a shell's radius and height; check for overlap. For length and surface area, simplify the speed and verify the curve hypotheses. For work and pressure, distinguish displacement, lift, depth and width, with their units. For mass and centers, form the small mass first, then the moments, and divide only when total mass is positive. An independent geometric estimate or a second slicing direction provides a useful check.

Exercise 3.5

1. Find the areas enclosed by (a) $y = 2x$ and $y = x^2$; (b) $y = x$ and $y = x^3$; (c) $y = |x|$ and $y = x^2$. Find every intersection and count all bounded components; split wherever the order changes.
2. Find the area enclosed by $y = x^2$ and $y = 4x - x^2$.
3. Find the area between $y = \sin x$ and the axis on $[-\pi, \pi]$, and compare it with the signed integral.
4. Find the area inside $r = 2\cos\theta$ for $-\pi/2 \le \theta \le \pi/2$. Identify the circle in Cartesian coordinates.
5. Rotate $0 \le y \le x$, $0 \le x \le 1$, about (a) the x-axis, (b) the y-axis, and (c) the line $y = -1$. Find each volume, specifying the radii and whether the slices are disks or washers.
6. Rotate $0 \le y \le \sqrt{x}$, $0 \le x \le 4$, about the x-axis. Find the volume.
7. Find the mass and center of a rod on $[0, 1]$ for densities (a) $2 - x$, (b) $1 + 2x$, and (c) $3x^2$. Compare each center with the midpoint and the direction in which mass is concentrated.
8. A force acts along a line, with x measured in metres. Find its work from $x = 0$ to $x = 2$ for (a) $F(x) = 3 + x$, (b) $F(x) = 3 - x$, and (c) $F(x) = x - 1$, in newtons. For (c), compare signed work with $\int_0^2 |F(x)|\,\mathrm{d}x$ and explain the difference.
9. Find the area enclosed by $x = y^2$ and $x = 2 - y$.
10. Rotate $x^2 \le y \le 2x$, $0 \le x \le 2$, about the y-axis. Compute the volume by both shells and washers.
11. A solid has unit-disk base and square cross-sections perpendicular to the x-axis, with side equal to the base chord. Find its volume.
12. Find the length of $y = \ln(\cos x)$ on $[0, \pi/3]$.
13. Find the length of $x = 3\cos t$, $y = 3\sin t$, $0 \le t \le \pi$.
14. Rotate each graph about the x-axis: (a) $y = x$, $0 \le x \le 1$; (b) $y = 2x$, $0 \le x \le 1$; (c) $y = 1 + x$, $0 \le x \le 1$. Find the generated lateral surface area, excluding flat end caps, and compare with the cone or frustum formula.
15. A vertical cylinder of radius 2 metres holds fluid up to height 3 metres above its bottom. Pump it to height 4 metres. Express the work using constant ρ, g.
16. A vertical rectangular plate has width 3 metres and extends from depth 1 to 4 metres. Find the hydrostatic force in terms of weight density γ.
17. Find the mass and center of a rod on $[0, 1]$ with density x^2.
18. Find the centroid of $0 \le y \le 1 - x$, $0 \le x \le 1$, from its moments.
19. Find the area of the petal $r = \cos 2\theta$ on $[-\pi/4, \pi/4]$. Form a four-petal region by rotating this petal by $\pi/2, \pi, 3\pi/2$; find its total area. Explain why the interiors are disjoint.

20. Rotate $[-2,1]\times[0,3]$ about the y-axis. Find the volume and quantify the error in adding shells from both sides without removing overlap.
21. Find the center of a uniform wire forming the upper semicircle of radius $a>0$. Compare its height with the midpoint of the radius.
22. A lifting force is $F(y)=k/(R+y)^2$ for $0\le y\le h$, with $k,R,h>0$. Find the work and compare it with kh/R^2, the constant initial-force estimate.

Supplement Exercises

S1. If $f\in C^1([a,b])$ with $f'>0$, prove $\int_a^b f(x)\,\mathrm{d}x+\int_{f(a)}^{f(b)} f^{-1}(y)\,\mathrm{d}y=bf(b)-af(a)$. Use substitution and parts, and explain the rectangle-area interpretation when $a,f(a)\ge 0$.

S2. Rotate $0\le y\le x$, $0\le x\le 1$, about $x=2$. Compute the volume by shells and by washers; write the radii explicitly.

S3. For $x=a\cos t$, $y=a\sin t$, $0\le t\le 4\pi$, compute the parametrized length. Compare it with the circumference of the geometric image and explain the difference.

S4. For a C^1 curve on $[a,b]$, prove $\|\gamma(b)-\gamma(a)\|\le L\le (b-a)\max_{[a,b]}\|\gamma'\|$. Explain the two estimates geometrically.

S5. Rotate $x=a\cos t$, $y=a\sin t$, $\alpha\le t\le\beta$, where $0\le\alpha<\beta\le\pi$. Show that the resulting spherical band has area $2\pi aH$, where $H=a(\cos\alpha-\cos\beta)$ is its axial height. Exclude flat end caps.

S6. For a plate of width $w(y)\ge 0$ and positive area A, prove $F=\gamma A\bar{y}$, where $\bar{y}$ is its centroid depth. Apply it to $w(y)=H-y$, $0\le y\le H$, $H>0$.

S7. A tank has horizontal cross-sectional area $A(y)=\pi(1+y)^2$ square metres for $0\le y\le 1$ metre. Pump all its fluid to height 2 metres. Find the work and bound it using the smallest and largest lifting distances.

S8. For $p>0$, find the centroid of $0\le y\le x^p$, $0\le x\le 1$. State why the endpoint is allowed even if a boundary derivative is unbounded.

S9. If $\rho\ge 0$ is continuous on $[a,b]$, $a<b$, and not identically zero, prove $a<\bar{x}<b$. Show why nonnegativity is essential by using $\rho(x)=1-9x/5$ on $[0,1]$ as algebraic weights.

S10. A lamina occupies $0\le y\le x$, $0\le x\le 1$, with areal density $\mu(x)=1+x$, constant along each vertical strip. Derive its mass and two moments from strips and find its center.

S11. The drilled sphere in Example 3.54 has constant density $\rho>0$. Find its mass, centroid and second moment of the axial coordinate $J=\int y^2\,\mathrm{d}m$, where a slice has mass $\rho A(y)\,\mathrm{d}y$. Express all answers using H and ρ. Explain why this second moment does not describe distance from the axis of rotation.

S12. Rotate $x^2\le y\le x$, $0\le x\le 1$, about the line $x=-d$, $d\ge 0$. Compute its volume using (a) the centroid and (b) shells, and verify agreement. Why can the same signed-distance calculation about $x=1/2$ not give the volume of the geometric union?

S13. Rotate $0\le y\le x\le 1$ about $y=c$, where $0<c<1$. Show that its cross-sectional area is

$$A_c(x)=\begin{cases}\pi(2cx-x^2), & 0\le x\le c,\\ \pi\max\{c,x-c\}^2, & c\le x\le 1.\end{cases}$$

Find the volume separately for $0<c\le 1/2$ and $1/2\le c<1$. Check agreement at $c=1/2$, and the limits as $c\downarrow 0$ and $c\uparrow 1$.

S14. A tank has continuous cross-sectional area $A(y)\ge 0$ on $[0,H]$, positive volume, constant density $\rho>0$, and outlet at height $B\ge H>0$. (a) Prove $W=Mg(B-\bar{y})$ from slice sums, where $M=\rho\int_0^H A(y)\,\mathrm{d}y$ and $\bar{y}=\rho M^{-1}\int_0^H yA(y)\,\mathrm{d}y$. (b) Evaluate W when $A(y)=ky^p$ with $k>0$, $p\ge 0$ (use $A\equiv k$ when $p=0$). (c) At $B=H$, explain why the work remains positive even though the top slice has zero lift.

3.6 Numerical Approximation of Integrals

知识点概要

- ❏ *Form rules from sampled values.*
- ❏ *Derive midpoint and trapezoid sums.*
- ❏ *Derive Simpson's weights and exactness.*
- ❏ *Prove convergence and error bounds.*
- ❏ *Choose a certified step count.*
- ❏ *Separate discretization, data and arithmetic errors.*

The FTC guarantees a primitive for a continuous integrand, but does not always provide a convenient expression or decimal value. Measurements may give only samples. A *quadrature rule*(数值求积公式)replaces an integral by a weighted sum of function values. For $I = \int_a^b f(x)\,\mathrm{d}x$ and a rule Q, define the signed error $E = I - Q$ and the absolute error $|E|$. An *error certificate*(误差保证)is a proved bound $|I - Q| \le B$ under checked assumptions; it gives the enclosure $Q - B \le I \le Q + B$. A decimal approximation by itself is not a certificate. We assume $a < b$ throughout.

3.6.1 Midpoint and trapezoid rules

For an integer $n \ge 1$, partition $[a, b]$ uniformly into n intervals: $h = (b-a)/n$, $x_i = a + ih$, and $m_i = (x_{i-1} + x_i)/2$. The *composite midpoint rule*(复合中点公式)uses rectangles:

$$M_n = h \sum_{i=1}^{n} f(m_i).$$

Replacing the graph on each interval by the straight line through its endpoint values gives the *composite trapezoid rule*(复合梯形公式):

$$T_n = h\left(\frac{f(x_0)}{2} + \sum_{i=1}^{n-1} f(x_i) + \frac{f(x_n)}{2}\right).$$

The formula is a signed integral of the approximating lines even when values are negative. Both rules integrate every affine function exactly; for a constant c, their positive weights sum to $b-a$, giving $c(b-a)$. They are not exact for every quadratic.

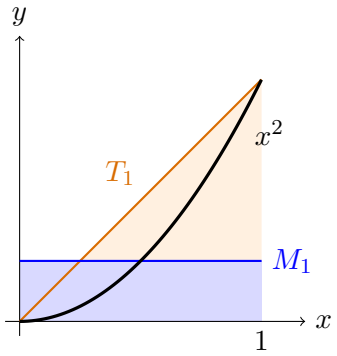


Figure 3.9: On one interval for $f(x) = x^2$, the midpoint rectangle has area $1/4$, the trapezoid area $1/2$, and the integral $1/3$. The error signs will follow from derivative kernels.

3.6.2 Simpson's rule and polynomial exactness

On a pair of adjacent intervals, centered at m with endpoints $m-h, m+h$, fit a quadratic $p(m+s) = A + Bs + Ds^2$ to the three sampled values. Then $A = f(m)$ and $2A + 2Dh^2 = f(m-h) + f(m+h)$. Integrating the polynomial gives

$$\int_{-h}^{h} p(m+s)\,\mathrm{d}s = 2hA + \frac{2}{3}Dh^3 = \frac{h}{3}\big(f(m-h) + 4f(m) + f(m+h)\big).$$

Thus, for even $n \geq 2$, the *composite Simpson rule*(复合辛普森公式)is

$$S_n = \frac{h}{3}\left(f(x_0) + 4\sum_{\substack{1\leq i\leq n-1\\ i \text{ odd}}} f(x_i) + 2\sum_{\substack{2\leq i\leq n-2\\ i \text{ even}}} f(x_i) + f(x_n)\right).$$

An empty sum is zero. The rule is exact for quadratics by its construction. It is also exact for cubics: around each panel center, the additional cubic term is odd, so both its integral and its weighted sum vanish. A rule has *degree of exactness*(代数精度)d if it integrates every polynomial of degree at most d exactly, but fails for at least one polynomial of degree $d+1$. Midpoint and trapezoid have degree 1; Simpson has degree 3 (a fourth power is a counterexample). Indeed, on $[-1,1]$ the integral of t^4 is $2/5$, whereas one Simpson panel gives $2/3$. Exactness concerns a class of functions, not a universal comparison of accuracy.

Example 3.64 Three rules on a quadratic Apply the three rules to $\int_0^1 x^2\,\mathrm{d}x$ with $n = 2$.

Solution *Here $h = 1/2$:*

$$M_2 = \frac{1}{2}\left(\frac{1}{16} + \frac{9}{16}\right) = \frac{5}{16}, \quad T_2 = \frac{1}{4}\left(0 + \frac{1}{2} + 1\right) = \frac{3}{8}, \quad S_2 = \frac{1}{6}(0 + 1 + 1) = \frac{1}{3}.$$

The integral is $1/3$. The signed errors are $1/48$, $-1/24$ and 0. Simpson's exactness explains its zero error; the other signs reflect the graph's upward curvature.

Example 3.65 A sample table for a logarithm Approximate $\int_1^2 1/x\,\mathrm{d}x$ with $n = 2$.

Solution *The endpoint samples are $1, 2/3, 1/2$; the midpoint samples are $4/5, 4/7$. The step is $h = 1/2$, so the weights give*

$$M_2 = \frac{1}{2}\left(\frac{4}{5} + \frac{4}{7}\right) = \frac{24}{35},$$
$$T_2 = \frac{1}{2}\left(\frac{1}{2} + \frac{2}{3} + \frac{1}{4}\right) = \frac{17}{24},$$
$$S_2 = \frac{1}{6}\left(1 + 4\cdot\frac{2}{3} + \frac{1}{2}\right) = \frac{25}{36}.$$

These are approximately 0.685714, 0.708333 and 0.694444; the exact integral is $\ln 2$. The fractions describe the rules exactly. The closeness of their decimals to each other is not yet an error bound.

3.6.3 Convergence for continuous functions

For continuous f on $[a,b]$ and $\delta \geq 0$, define its *modulus of continuity*(连续模)

$$\omega_f(\delta) = \sup\{|f(x) - f(y)| : x, y \in [a,b],\ |x-y| \leq \delta\}.$$

It measures the largest oscillation over distances at most δ. Uniform continuity gives $\omega_f(\delta) \to 0$ as $\delta \downarrow 0$. For $f(x) = x$, it is $\min(\delta, b-a)$.

Proposition 3.2 (Convergence of the three rules)

For continuous f, $M_n, T_n \to I$ as $n \to \infty$, and $S_n \to I$ through even n. For all applicable n, $|I - Q_n| \le (b-a)\omega_f(2h)$.

♠

Proof On each interval or Simpson panel, the rule has positive weights summing to its width. Compare every sampled value with $f(x)$ throughout that piece. Their arguments differ by at most h for midpoint/trapezoid pieces and by at most $2h$ for Simpson panels. Integrating the weighted comparison bounds a piece's error by its width times $\omega_f(2h)$. Sum the widths to obtain the estimate, then use uniform continuity. □

Convergence needs no derivative, but does not by itself specify how large n must be for a chosen accuracy.

3.6.4 Derivative bounds with proved constants

Theorem 3.6 (Composite error bounds)

If $f \in C^2([a,b])$ and $D_2 \ge \max|f''|$, then

$$|I - M_n| \le \frac{(b-a)h^2 D_2}{24}, \qquad |I - T_n| \le \frac{(b-a)h^2 D_2}{12}.$$

If $f \in C^4([a,b])$, $D_4 \ge \max|f^{(4)}|$, and n is even, then

$$|I - S_n| \le \frac{(b-a)h^4 D_4}{180}.$$

When $f'' \ge 0$, $M_n \le I \le T_n$. When $f^{(4)} \ge 0$, $I \le S_n$.

♡

> ***Find a kernel, then sum the local budgets.*** A rule is exact on low-degree polynomials, so its error should depend on the first derivative order it does not reproduce. Rather than invoke a later Taylor theorem, construct a function whose integrations by parts generate the integral minus the sampled values. The kernel's sign gives the error's sign, and its integral gives the constant. Adding local bounds makes the dependence on h explicit.

Proof First take one interval $[u,v]$ of width w, midpoint m. Two integrations by parts give the trapezoid identity

$$\int_u^v f - \frac{w}{2}(f(u)+f(v)) = -\frac{1}{2}\int_u^v (t-u)(v-t)f''(t)\,\mathrm{d}t.$$

Indeed the kernel vanishes at both endpoints, its derivatives there are $-w/2, w/2$, and its second derivative is 1. Its absolute integral is $w^3/12$.

For the midpoint, integrating twice on the two halves gives

$$\frac{1}{2}\int_u^m (t-u)^2 f''(t)\,\mathrm{d}t = \frac{w^2}{8}f'(m) - \frac{w}{2}f(m) + \int_u^m f,$$
$$\frac{1}{2}\int_m^v (v-t)^2 f''(t)\,\mathrm{d}t = -\frac{w^2}{8}f'(m) - \frac{w}{2}f(m) + \int_m^v f.$$

Their sum is $\int_u^v f - wf(m)$. The positive kernels have total integral $w^3/24$. These identities give the local bounds and signs. Summing n intervals with $w = h$ gives the two C^2 estimates.

For Simpson, first normalize a panel to $[-1,1]$. Its exactness through degree 3 suggests four integrations by parts. On $[0,1]$, seek a quartic kernel with fourth derivative 1 and a triple zero at 1: $K(t) = (t-1)^3(t+c)/24$. An even extension needs $K'(0) = 0$, which forces $c = 1/3$. Thus define

$$K(t) = -\frac{(1-|t|)^3(1+3|t|)}{72}.$$

On each side of 0, $K^{(4)} = 1$. At ± 1, K, K', K'' vanish, and $K'''(-1) = -1/3$, $K'''(1) = 1/3$. The functions

K, K', K'' are continuous at 0, while $K'''(0-) = 2/3$ and $K'''(0+) = -2/3$. Four integrations by parts on $[-1,0]$ and $[0,1]$ therefore give, for $F \in C^4([-1,1])$,

$$\int_{-1}^{1} K(t)F^{(4)}(t)\,\mathrm{d}t = \int_{-1}^{1} F(t)\,\mathrm{d}t - \frac{F(-1)+4F(0)+F(1)}{3}.$$

The continuous boundary terms cancel at 0; the jump of K''' supplies $-4F(0)/3$. Also $K \le 0$ and $\int_{-1}^{1}|K| = \frac{2}{72}\int_0^1 (1-t)^3(1+3t)\,\mathrm{d}t = 1/90$. With $x = m+ht$ and $F(t) = f(m+ht)$, the local error is at most $D_4h^5/90$. There are $n/2$ panels, giving $(b-a)D_4h^4/180$. The kernel's sign gives the final sign assertion. □

The constants require the stated smoothness on the entire closed interval. D_2, D_4 are derivative bounds, distinct from the midpoint sums M_n. A safe upper bound for a derivative is sufficient; it need not be its exact maximum.

Example 3.66 Allocate intervals where curvature is larger Let $f(x) = x^2$ on $[-1,0]$ and $f(x) = 4x^2$ on $[0,1]$. Approximate $I = \int_{-1}^{1} f$ using one trapezoid on the left half and two on the right, and give a certified error bound.

> ***Check smoothness on the pieces you actually use.*** The second derivatives are 2 and 8; at zero the second derivative does not exist. The global C^2 theorem is therefore unavailable, but its local version applies separately on each half. More intervals on the right should reduce the larger curvature contribution.

Solution *The left contribution is* $\frac{1}{2}(f(-1)+f(0)) = 1/2$. *On the right the nodes* $0, 1/2, 1$ *have values* $0, 1, 4$, *so its two-trapezoid contribution is*

$$\frac{1/2}{2}(0+2\cdot 1+4) = \frac{3}{2}.$$

Thus $Q = 2$. *The separate error bounds are* $2/12 = 1/6$ *and* $8/(12\cdot 2^2) = 1/6$. *Both errors have the overestimate sign, giving*

$$0 \le Q - I \le \frac{1}{3}, \qquad \frac{5}{3} \le I \le 2.$$

Here the exact value is $I = 1/3 + 4/3 = 5/3$, *so the bound is attained. Using two intervals on the left and one on the right would instead give a bound* $2/(12\cdot 2^2) + 8/12 = 17/24$ *with the same total number of intervals.*

This comparison explains how a mesh can respond to known local curvature. It does not assume that comparing two computed answers alone certifies either error.

Example 3.67 Reuse trapezoid samples to obtain Simpson's rule From trapezoid values for $f(x) = 1/x$ on $[1,2]$, derive a Simpson approximation without new samples and give a proved error bound.

> ***Compare coefficients before estimating errors.*** Refining a trapezoid mesh retains its old nodes and inserts its midpoints. Subtracting one copy of the coarse rule from four copies of the fine rule changes the shared-node weights into Simpson's pattern. This is an exact identity of rules; no guessed asymptotic error expansion is needed.

Solution *For any* $n \ge 1$, *separate the even and odd nodes of the mesh with* $2n$ *intervals. Writing* $h = (b-a)/n$ *gives* $T_{2n} = (T_n + M_n)/2$ *and therefore*

$$\frac{4T_{2n} - T_n}{3} = \frac{T_n + 2M_n}{3} = S_{2n}.$$

Indeed each old interval contributes $h(f(x_{i-1}) + 4f(m_i) + f(x_i))/6$. *For the stated integrand,* $T_1 = 3/4$ *and* $T_2 = 17/24$, *so the combination is* $25/36$. *Since* $f^{(4)}(x) = 24/x^5$ *lies between* 0 *and* 24,

$$0 \le \frac{25}{36} - \ln 2 \le \frac{24}{180\cdot 2^4} = \frac{1}{120}.$$

Thus $247/360 \le \ln 2 \le 25/36$ *follows without using a decimal value of* $\ln 2$.

Using different mesh sizes to improve an approximation is called *extrapolation*(外推). Here the combi-

nation is exactly Simpson's rule, so its established C^4 estimate supplies the guarantee. The smaller difference $|T_{2n} - T_n|$ by itself supplies no such guarantee, and continuity alone does not imply fourth-order accuracy.

3.6.5 Step counts and data uncertainty

To guarantee absolute error at most $\varepsilon > 0$, sufficient choices satisfy

$$n \geq \sqrt{\frac{D_2(b-a)^3}{24\varepsilon}} \quad \text{(midpoint)}, \qquad n \geq \sqrt{\frac{D_2(b-a)^3}{12\varepsilon}} \quad \text{(trapezoid)},$$

and $n \geq \left(D_4(b-a)^5/(180\varepsilon)\right)^{1/4}$ for Simpson, rounded up to an even integer at least 2. For a strict error target, a non-strict bound equal to the target is insufficient.

Example 3.68 Certified step counts Choose sufficient step counts for $\int_0^1 e^{-x^2}\,\mathrm{d}x$ with absolute error at most 10^{-5}.

Solution *Differentiation gives* $f'' = (4x^2 - 2)e^{-x^2}$ *and* $f^{(4)} = (16x^4 - 48x^2 + 12)e^{-x^2}$. *On* $[0,1]$, $|f''| \leq 2$ *and* $|f^{(4)}| \leq 16 + 48 + 12 = 76$. *The three sufficient bounds become* $1/(12n^2)$, $1/(6n^2)$ *and* $76/(180n^4)$. *Solving these inequalities requires*

$$n^2 \geq \frac{100000}{12}, \qquad n^2 \geq \frac{100000}{6}, \qquad n^4 \geq \frac{7600000}{180}.$$

For midpoint, $91^2 = 8281 < 100000/12 < 92^2$*; for trapezoid,* $129^2 = 16641 < 100000/6 < 130^2$. *For Simpson,* $14^4 = 38416 < 7600000/180 < 16^4 = 65536$. *Thus the least permitted integers according to these bounds are respectively* 92, 130 *and the even integer* 16. *These are guaranteed choices from the chosen derivative bounds, not claims about the smallest* n *for the actual error. Each computed rule value* Q *then places the integral in* $[Q - 10^{-5}, Q + 10^{-5}]$ *before data or arithmetic errors are added.*

Suppose each sampled value has absolute uncertainty at most δ. For a positive-weight rule $Q = \sum w_j f(t_j)$, with $\sum w_j = b - a$, the data contribution obeys $|\sum w_j(\widetilde{f}(t_j) - f(t_j))| \leq (b-a)\delta$. If the quadrature bound is B and the weighted arithmetic has an additional absolute error at most η, the total certificate is

$$|I - \widetilde{Q}| \leq B + (b-a)\delta + \eta.$$

Discretization, input uncertainty and arithmetic rounding are separate sources of error. Increasing n reduces the first bound but need not reduce the other two.

Example 3.69 A data budget A rule on an interval of width 4 has quadrature error at most $5 \cdot 10^{-4}$. Each sample is uncertain by at most 10^{-4}, and weighted arithmetic is exact. Bound the total error.

Solution *The data error is at most* $4 \cdot 10^{-4}$ *because the positive weights sum to* 4. *The total is at most* $9 \cdot 10^{-4}$. *Reporting only the quadrature bound would omit a comparable source of uncertainty.*

Example 3.70 A finite identity gives a numerical certificate Obtain a rational enclosure for $\ln 2$ of width less than $1/6000$, without using an infinite series or a decimal value of $\ln 2$.

> ***Choose an interval where the remainder is small.*** The substitution $x = (1+t)/(1-t)$ maps $[0, 1/3]$ to $[1, 2]$. It changes $\mathrm{d}x/x$ into $2\,\mathrm{d}t/(1-t^2)$. On the shorter interval, a finite geometric identity leaves a high power of t times a bounded positive factor.

Solution *First,*

$$\ln 2 = 2\int_0^{1/3} \frac{\mathrm{d}t}{1-t^2}.$$

The exact algebraic identity

$$\frac{1}{1-t^2} = 1 + t^2 + t^4 + \frac{t^6}{1-t^2}$$

gives $\ln 2 = L + R$, *where*

$$L = 2\left[t + \frac{t^3}{3} + \frac{t^5}{5}\right]_0^{1/3} = \frac{842}{1215}, \qquad R = 2\int_0^{1/3} \frac{t^6}{1-t^2}\,\mathrm{d}t.$$

Since $1 \le 1/(1-t^2) \le 9/8$ *there,*

$$0 < R \le \frac{9}{4}\int_0^{1/3} t^6\,\mathrm{d}t = \frac{1}{6804} < \frac{1}{6000}.$$

Therefore $842/1215 < \ln 2 \le 842/1215 + 1/6804$ *is the required enclosure. The strict positivity follows because the remainder integrand is positive away from zero.*

The certificate comes from a finite identity and an integral inequality. More terms in that identity give smaller remainders, but no theorem about infinite power series is being assumed. Compared with sampled quadrature, this method uses extra algebraic structure specific to the integrand.

3.6.6 What refinement does not prove

Comparing Q_n with Q_{2n} is useful for diagnosing change, but their difference alone cannot certify the error. The rules might both miss a narrow feature, or a function might vanish at every sampled point. Derivative bounds or another proved estimate are needed for a guarantee.

Example 3.71 A corner defeats polynomial exactness Apply the rules with $n = 2$ to $\int_{-1}^{1} |x|\,\mathrm{d}x$.

Solution *The exact integral is* 1. *The midpoint samples are* $1/2, 1/2$, *so* $M_2 = 1$. *The endpoint samples are* $1, 0, 1$, *so* $T_2 = 1$ *and* $S_2 = 2/3$. *The function is continuous, hence all three rules converge under refinement. It is not* C^2 *at* 0, *so the derivative bounds do not apply; cubic exactness does not make Simpson universally more accurate. Splitting at the corner permits exact affine integration on each piece.*

Example 3.72 Invisible to a finite collection of samples Can agreement between coarse and fine quadrature rules guarantee accuracy for every continuous function?

Solution *Let* $r_1, \ldots, r_k$ *be all distinct nodes used by both rules on* $[a, b]$. *The polynomial* $p(x) = \prod_{j=1}^{k}(x - r_j)^2$ *is nonnegative and vanishes at every node, so both rules return zero. It is positive on some subinterval, hence* $\int_a^b p > 0$ *by the positivity property. Multiplying by an arbitrary positive constant makes the actual error as large as desired while preserving exact agreement. This does not contradict the derivative estimates: scaling also enlarges the derivative bounds.*

3.6.7 Methods summary

Shared trapezoid samples can reproduce Simpson weights exactly. A refinement difference becomes a certified estimate only after including a proved remainder or another valid error bound.

Specify the interval, mesh and sample locations before evaluating a rule. Use an even number of intervals for Simpson. Exactness helps check formulas; smoothness and derivative bounds provide quantitative guarantees. Sum local error budgets and round step counts in the direction required by the tolerance. Add sample and arithmetic uncertainties to the quadrature error. Split at known nonsmooth points when possible. Use refinement comparisons as evidence of change, and call a result certified only when a proved bound covers its assumptions and error sources.

Exercise 3.6

1. Compute M_4 on $[0,1]$ for (a) x^2, (b) $2x^2+1$, and (c) x^2-x. Use exactness on affine functions and linearity to avoid repeating all samples.
2. Compute T_4 on $[0,1]$ for (a) x^2 (the original integral in the preceding exercise), (b) $2x^2+1$, and (c) x^2-x. Find the signed error in each case and compare its sign with the second derivative.
3. Compute S_2 on $[0,1]$ for (a) x^3, (b) x^3-2x+1, and (c) x^4. Compare with the exact integrals and explain the first two exact answers and the third nonzero error.
4. Compute M_2 for $\int_0^4 \sqrt{x}\,\mathrm{d}x$.
5. Compute T_2 for $\int_0^{\pi/2} \cos x\,\mathrm{d}x$.
6. Compute S_4 for $\int_1^3 1/x\,\mathrm{d}x$.
7. Which positive integers n are permitted for composite Simpson? Write all three rules' weights for $[0,1]$, $n=2$, and check their sums.
8. Using $|f''|\le 1$, choose a sufficient midpoint count for $\int_0^{\pi/2} \cos x\,\mathrm{d}x$ with error at most 10^{-4}.
9. Choose a sufficient trapezoid count for $\int_0^1 e^x\,\mathrm{d}x$ with error at most 10^{-4}.
10. Using $f^{(4)}=24/x^5$, choose a sufficient Simpson count for $\int_1^2 1/x\,\mathrm{d}x$ with error at most 10^{-6}.
11. Prove $M_n = 1/3 - 1/(12n^2)$ and $T_n = 1/3 + 1/(6n^2)$ for x^2 on $[0,1]$. State S_n for even n.
12. Determine three weights at $-1,0,1$ by requiring exactness for $1,t,t^2$. Show that the resulting rule is also exact for t^3.
13. A positive-weight rule on an interval of width 3 has quadrature bound 0.001. Bound its total error when (a) each sample has uncertainty 0.002 and arithmetic is exact; (b) each sample has uncertainty 0.0002 and arithmetic has error at most 0.0001; (c) each sample has uncertainty $\delta\ge 0$ and arithmetic is exact. For (c), find the largest δ for which the stated bounds certify total error at most 0.002.
14. Prove $M_n \le I \le T_n$ when $f\in C^2$ and $f''\ge 0$, using the kernels. Explain why the signs reverse if $f''\le 0$.
15. At $x=0,1,2,3,4$, the sampled values are $2,3,5,4,1$. (a) Compute T_4 and S_4. (b) Compute T_2 using only the even-indexed samples and check $(4T_4-T_2)/3=S_4$. (c) If every sample is uncertain by at most 0.01, bound the resulting change in each rule, without claiming a bound for the discretization error.
16. Use the trapezoid rule for $f(x)=x^2$ on nodes $0,1/4,1$. Compute its error and compare it with the sum of the two local derivative bounds.
17. If $|E_n|\le 1/(6n^2)$, choose the least n certified to give $|E_n|<1/600$. Explain why equality in a bound does not certify a strict target.
18. For $f(x)=\sqrt{x}$ on $[0,1]$, explain why continuity guarantees convergence but the stated C^2 error bounds are unavailable.
19. On an interval of width 2, suppose $|f''|\le 3$ and $|f^{(4)}|\le 5$. For error at most 10^{-4}, compare sufficient n and numbers of distinct sampled values for the three rules.
20. A computed value is 0.7500, with quadrature, data and arithmetic error bounds 0.002, 0.001 and 0.0001. Give a certified interval for the integral.

Supplement Exercises

S1. If $|f(x)-f(y)|\le L|x-y|$, prove $|I-M_n|\le L(b-a)h/4$ and $|I-T_n|\le L(b-a)h/2$ by integrating the distances to the sample points.

S2. For nodes $a = x_0 < \cdots < x_n = b$ with $h_i = x_i - x_{i-1}$ and $|f''| \le D_2$, derive the nonuniform trapezoid rule and prove $|E| \le (D_2/12)\sum h_i^3 \le D_2(b-a)(\max h_i)^2/12$.

S3. Show that the constants $1/24$, $1/12$ and $1/180$ cannot be decreased for these universal derivative bounds. Use a quadratic for midpoint/trapezoid and t^4 on $[-1,1]$ with one Simpson panel.

S4. Prove the sharper continuity bounds $|I - M_n| \le (b-a)\omega_f(h/2)$ and $|I - T_n| \le (b-a)\omega_f(h)$. Compare them with the Simpson bound using $\omega_f(2h)$.

S5. For nondecreasing f on $[a,b]$, prove $|I - M_n|, |I - T_n| \le h(f(b) - f(a))$ using upper and lower rectangle sums, without derivatives.

S6. Prove $M_n = 2T_{2n} - T_n$ and $S_{2n} = (4T_{2n} - T_n)/3$. If shared samples have uncertainty at most δ, explain why the data bound for the second expression is $(b-a)\delta$, rather than needing the larger bound obtained by treating the two trapezoid errors independently.

S7. If $|Q_{2n} - Q_n| \le 10^{-5}$ and a separate proved estimate gives $|I - Q_{2n}| \le 2 \cdot 10^{-6}$, bound $|I - Q_n|$. Identify which information turns a refinement comparison into a certificate.

S8. For $p(x) = x^2(x - 1/2)^2(x-1)^2$ on $[0,1]$, compute T_1, T_2, S_2 and $\int_0^1 p$. Scale p to produce any prescribed positive integral without changing the sample values.

S9. Compute $\int_0^1 |x - 1/3|\,\mathrm{d}x$ by splitting at the corner. Explain why midpoint and trapezoid with $n = 6$, and Simpson with $n = 6$, are exact even though the global C^2 hypothesis fails.

S10. Suppose $f \in C^4([a,b])$ and $f^{(4)} \ge 0$. For even n, prove $I = S_n$ if and only if f is a polynomial of degree at most 3 on $[a,b]$. Use strict negativity of the panel kernel inside each panel, then the FTC.

S11. Let $f \in C^4([a,b])$ with $|f^{(4)}| \le D_4$. Use $S_{2n} = (4T_{2n} - T_n)/3$ to prove

$$\left| I - T_{2n} - \frac{T_{2n} - T_n}{3} \right| \le \frac{D_4(b-a)^5}{2880n^4}.$$

Deduce an explicit bound for $|I - T_{2n}|$ involving the observed difference and this remainder. Explain why deleting the remainder is not justified by this argument.

S12. Let f be C^2 on each of two intervals of lengths $\ell_1, \ell_2 > 0$. Use composite trapezoids separately on these intervals, with bounds $|f''| \le D_1, D_2$, where $D_1, D_2 > 0$. For positive integers n_1, n_2 with fixed sum $N \ge 2$, prove the error bound $D_1\ell_1^3/(12n_1^2) + D_2\ell_2^3/(12n_2^2)$. First allowing real positive n_i, find the minimizing ratio n_1/n_2 by differentiation and the FTC sign test. This objective minimizes a guaranteed error bound, not necessarily the actual error. Then explain how to choose the integer minimum by testing neighboring admissible integers (and an endpoint if the real minimum is outside $[1, N-1]$).

S13. For an integer $m \ge 0$, set $L_m = 2\sum_{j=0}^{m}[(2j+1)3^{2j+1}]^{-1}$. Use a finite geometric identity to prove

$$0 < \ln 2 - L_m \le \frac{9}{4(2m+3)3^{2m+3}}.$$

Find the least m for which this bound is less than 10^{-6}. Explain why no exchange of an infinite sum with an integral is involved.

S14. On $[-1,1]$, let $f(x) = ax^2$ for $x \le 0$ and $f(x) = bx^2$ for $x \ge 0$, where $a, b > 0$. Use n_1 and n_2 trapezoid intervals on the two halves. Prove the exact identity $Q - I = a/(6n_1^2) + b/(6n_2^2)$. For $a = 1, b = 4, n_1 + n_2 = 4$, compare all three choices of positive integers and select the most accurate one. Why is it legitimate to use the half-interval error identities when $a \ne b$?

Chapter 3 Problem-Solving Toolkit

An integration problem can ask for a primitive, a number, a geometric or physical quantity, or a controlled approximation. These goals can favor different representations of the same expression. A definite integral may yield to symmetry even when its primitive is cumbersome; an application may be simple only after the correct slices have been chosen.

A decision route

Step 1. State the goal and original domain. For a primitive, identify each connected interval. For a definite integral, check continuity or the precise integrability and FTC assumptions. For an application, describe the quantity and its units before selecting a formula.

Step 2. Simplify before committing to a technique. Divide polynomials, complete squares, use an identity, inspect a numerator derivative, or pair reflected arguments. Do not expand a useful composite structure merely because expansion is possible.

Step 3. Predict the transformed problem. For substitution, translate every factor and the differential. For parts, write the proposed $u'v$ and explain what becomes simpler or what equation will close. For partial fractions, write the complete template before finding coefficients.

Step 4. Track branches, bounds and multiplicity. A new variable may reverse orientation or have a pole that the original integrand lacks. A rotated region may count the same shell twice. A parametrized curve may trace the same image repeatedly.

Step 5. Choose the form of verification. Differentiate a primitive; compare independent slicing methods; check positivity, units and limiting cases; or give a numerical interval with a proved total error bound.

Recognizing an efficient first move

Signal	First move	Condition or stopping test
Composite expression with a nearby derivative	Name the inner expression.	Translate all leftover factors; keep the differential multiplier.
A reflected interval and paired expressions	Compute $f(x)+f(a+b-x)$.	Reflection must preserve the actual interval and domain.
Polynomial times exponential or trigonometric factor	Differentiate the polynomial in parts.	Degree decreases, or a finite cycle gives a solvable equation.
Logarithm or inverse function	Differentiate that factor in parts.	The new integral must be simpler; evaluate boundary terms with signs.
Rational function	Divide, factor, write all pole powers.	Numerator degrees, repeated factors and original holes.
Trigonometric powers	Reserve a differential or reduce frequencies.	Every remaining factor belongs to the new variable.
Quadratic radical	Complete the square; compare a direct split and substitution.	Choose the angle or root branch before removing absolute values.
Geometric or physical quantity	Identify one strip, slice or mass element.	Distances, units, nonoverlap and appropriate smoothness.
A requested decimal accuracy	Find derivative bounds or another remainder estimate.	Include data and arithmetic errors; refinement agreement alone is insufficient.

Combining methods without losing the structure

Example 3.73 Substitution changes which factor should be differentiated Evaluate $\int x^3\ln(1+x^2)\,dx$ on $\mathbb{R}$.

> ***Try one structural change before a long calculation.*** The logarithm suggests $u=1+x^2$, while $x^3\,dx = x^2(x\,dx)$ leaves a linear factor $u-1$. The result is a polynomial times a logarithm, a natural parts problem. Direct parts is also possible, but its remainder is a rational function requiring division.

Solution ***Method 1: substitute, then integrate by parts.*** *With $u=1+x^2$, the integral becomes $\frac{1}{2}\int(u-1)\ln u\,du$. Parts gives*

$$\int u\ln u\,du=\frac{u^2}{2}\ln u-\frac{u^2}{4}+C,\qquad \int \ln u\,du=u\ln u-u+C.$$

Therefore, after restoring u, one primitive is

$$\left(\frac{u^2}{4}-\frac{u}{2}\right)\ln u-\frac{u^2}{8}+\frac{u}{2},\qquad u=1+x^2.$$

The direct composite-derivative substitution does not require $x\mapsto 1+x^2$ to be one-to-one.

Solution ***Method 2: parts first, then polynomial division.*** *Taking the logarithm as the differentiated factor gives*

$$\int x^3\ln(1+x^2)\,dx=\frac{x^4}{4}\ln(1+x^2)-\frac{1}{2}\int\frac{x^5}{1+x^2}\,dx.$$

Since $x^5/(1+x^2) = x^3 - x + x/(1+x^2)$*, the answer becomes*

$$\frac{x^4-1}{4}\ln(1+x^2) - \frac{x^4}{8} + \frac{x^2}{4} + C.$$

Expanding the first answer shows that it differs from the second by $3/8$ before the arbitrary constants are chosen. Substitution makes the logarithmic structure clearer; direct parts keeps the original variable and uses rational division. Both are valid on all of $\mathbb{R}$ because $1 + x^2$ is positive.

Recovering from an unhelpful choice

A correct identity need not move the problem forward. If a substitution leaves both old and new variables, express the old factors in the new variable before proceeding; if this produces more complicated radicals, reconsider the change. If parts raises every relevant degree, compare it with an inner derivative already present, as in Example 3.18. If a rationalization introduces a pole, retain the original integral and split into finite intervals or match local constants, as in Example 3.31. Two local primitives can differ by different constants on different domain components.

For transfer practice, use the grouped calculations in Section 3.1. Then try the log-power parameter problem in Section 3.2 and the rational-derivative ansatz in Section 3.3. For each, state what feature determines the first move before calculating.

Model and error audits

- **Area and volume.** Write the lower and upper boundaries, or the whole rotated radial interval. If an axis crosses the region, use unions as in Example 3.52; subtracting two squared signed heights is not automatically a washer formula.
- **Length, surface and mass.** Decide whether the question concerns a parametrized traversal or its geometric image. For surface area, distinguish the generated surface from end caps. For a center, divide moments by a positive total mass, not by a coordinate length or boundary length of the wrong object.
- **Work and force.** A pump slice uses weight times lift; a pressure slice uses depth times width times thickness. The mean-lift check in Example 3.63 is available because lifting work is linear in height.
- **Numerical output.** Record the rule, nodes, smoothness on each piece, derivative bounds and final enclosure. Reuse shared samples when possible, but keep a proved remainder in an extrapolation estimate. Compare Example 3.66 with Example 3.72: knowledge of the function, not agreement of samples alone, supplies the guarantee.
- **Final answer.** Restore the original variable for a primitive and include its arbitrary constant. For a definite integral, report the value with its sign and units when applicable. For an estimate, distinguish a sufficient mesh count from the unknown smallest count that happens to work.

The strongest check is often adapted to the problem: a derivative for symbolic integration, geometry for a model, and an explicit inequality for a numerical result.

Chapter 4 Mean Value Theorems and Taylor's Formula

A derivative describes change at one point. We now ask how such local information controls a finite change, an approximation error, or the shape of an entire graph. The mean value theorems connect endpoint changes with a derivative at an intermediate point. Taylor's formula replaces a function by a polynomial while keeping track of what has been discarded. These tools then turn derivative signs into tests for monotonicity, extrema and convexity.

An intermediate point usually cannot be calculated explicitly. Its value is that it lies in a known interval, where we can bound the relevant derivative. A recurring proof strategy is to subtract a simple model until an auxiliary function has equal endpoint values or several vanishing derivatives. The choice of that model will explain both the formula and its hypotheses.

4.1 Rolle's and Lagrange's Mean Value Theorems

知识点概要

- ❏ *Relate an instantaneous rate to a finite change.*
- ❏ *Understand the hypotheses of Rolle's theorem.*
- ❏ *Turn derivative bounds into estimates.*
- ❏ *Build auxiliary functions from a target equation.*
- ❏ *Count zeros by repeated differentiation.*
- ❏ *Distinguish continuity from the intermediate value property of a derivative.*

A derivative describes a function near one point. Can this local information control the change between two distant points? The mean value theorem supplies the bridge: the average rate on a closed interval occurs as an instantaneous rate at an interior point. Its proof begins with the simpler situation in which there is no net change.

4.1.1 An interior extreme point has a horizontal tangent

Before looking for a horizontal tangent, we need a condition that forces one. A function may attain its largest value on an interval at an endpoint, where its graph need not turn. An extreme value at an interior point is different.

Definition 4.1

Let f be defined on an open interval I and let $c \in I$. We say that f has a local maximum (极大值) *at c if some $\delta > 0$ satisfies*

$$(c-\delta, c+\delta) \subset I, \qquad f(x) \le f(c) \quad \text{whenever } |x-c| < \delta.$$

A local minimum (极小值) *is defined with the opposite inequality. Either is a* local extremum (局部极值) .

The comparison is only with nearby points, and equality is allowed. For example, x^2 has a local minimum at 0, while $-x^2$ has a local maximum there. A constant function has both at every interior point. Neither definition requires differentiability. The systematic distinction between local and absolute extrema will be developed in Section 4.5.

This necessary condition is Fermat's theorem(费马定理).

Theorem 4.1 (Fermat's theorem)

If f has a local extremum at an interior point c of its domain and is differentiable at c, then $f'(c) = 0$. ♡

> ***Look at the signs on the two sides.*** At a local minimum, the numerator $f(c+h)-f(c)$ is nonnegative on both sides. Dividing by h changes its sign when we cross c. A two-sided derivative must be compatible with both signs, which leaves only zero.

Proof Suppose first that c is a local minimum. For sufficiently small $h > 0$, the difference quotient is nonnegative, whereas for sufficiently small $h < 0$ it is nonpositive. Taking the corresponding one-sided limits gives $f'(c) \geq 0$ and $f'(c) \leq 0$. Thus $f'(c) = 0$. Apply this argument to $-f$ for a local maximum. □

Example 4.1 What Fermat's theorem does and does not say Compare x^2, x^3, and $|x|$ near 0, and $f(x) = x$ on $[0, 1]$.

Solution *The function x^2 has a minimum and derivative zero at 0. The function x^3 also has derivative zero there, but takes both negative and positive values arbitrarily close to 0; it has no extremum there. Thus derivative zero is not sufficient for an extremum. The function $|x|$ has a minimum at 0, but its one-sided derivatives are -1 and 1, so no derivative exists. Finally, x attains its minimum on $[0, 1]$ at 0, although its right derivative is 1. This does not contradict Fermat's theorem: 0 is not an interior point of that domain.*

A point where a derivative exists and is zero is called a *stationary point*(驻点). Fermat's theorem identifies a necessary condition for a differentiable interior extremum, not a complete test for extrema.

4.1.2 Equal endpoint values: Rolle's theorem

If a continuous graph starts and ends at the same height, either it stays at that height or it makes an excursion above or below it. In the latter case, an attained extreme value lies inside the interval. This turns a statement about two endpoint values into a statement about an interior derivative.

Rolle's theorem(罗尔定理)gives the corresponding equal-endpoint statement.

Theorem 4.2 (Rolle's theorem)

Let $a < b$. Suppose that f is continuous on $[a, b]$, differentiable on (a, b), and $f(a) = f(b)$. Then there is $c \in (a, b)$ such that $f'(c) = 0$. ♡

> ***Force an extreme value away from the endpoints.*** The extreme value theorem gives a maximum and a minimum, but initially either might be at an endpoint. Equal endpoint values remove this difficulty: if some value differs from that common value, the corresponding maximum or minimum is strictly beyond both endpoint values.

Proof By Theorem 1.26, f attains a minimum m and a maximum M on $[a, b]$. If $m = M$, then f is constant, and any $c \in (a, b)$ works. Otherwise, since $f(a) = f(b)$, at least one of m, M differs from the common endpoint value. A point where that extreme value is attained lies in (a, b). Fermat's theorem gives derivative zero there. □

Example 4.2 The Rolle point need not be unique Find all Rolle points for $f(x) = x^3 - x$ on $[-1, 1]$.

Solution *The polynomial is continuous and differentiable, and $f(-1) = f(1) = 0$. Since $f'(x) = 3x^2 - 1$, the required points are $c = \pm 1/\sqrt{3}$, both in $(-1, 1)$. Rolle's theorem guarantees existence, but does not assert uniqueness or identify a particular point in advance.*

Example 4.3 Three separate hypotheses Show that each hypothesis of Rolle's theorem has a role.

Solution *On* $[-1,1]$*, the function* $|x|$ *is continuous and has equal endpoint values, but its derivative is never zero where it exists; differentiability fails at* 0*. On* $[0,1]$*, define* $f(x)=x$ *for* $x<1$ *and* $f(1)=0$*. Then* $f(0)=f(1)$ *and* $f'=1$ *on* $(0,1)$*, but continuity fails at the right endpoint. Finally,* $f(x)=x$ *on* $[0,1]$ *is continuous and differentiable but has unequal endpoint values and no zero derivative. These examples remove one hypothesis at a time.*

Endpoint differentiability is not among the hypotheses. Continuity at an endpoint means continuity from within $[a,b]$. Also, no continuity of f' is assumed.

4.1.3 Subtracting the secant: Lagrange's theorem

When the endpoints have different heights, subtract the straight line through them. The residual then vanishes at both endpoints. A horizontal tangent to the residual corresponds to a tangent to the original graph parallel to the secant.

The resulting identity is Lagrange's mean value theorem(拉格朗日中值定理).

Theorem 4.3 (Lagrange's mean value theorem)

Let $a<b$*, and suppose that* f *is continuous on* $[a,b]$ *and differentiable on* (a,b)*. Then there exists* $c\in(a,b)$ *such that*

$$f(b)-f(a)=f'(c)(b-a). \tag{4.1}$$

♡

The quotient $(f(b)-f(a))/(b-a)$ is the average rate of change. The theorem says that this rate equals an instantaneous rate $f'(c)$. In particular, it applies to a differentiable position function even when the velocity is not continuous.

> ***Let the desired derivative determine the auxiliary function.*** We want $f'(c)-(f(b)-f(a))/(b-a)=0$. Subtracting a linear function with the secant slope produces exactly this derivative. Choose its constant term so that the residual also vanishes at the left endpoint; the choice of slope then makes it vanish at the right endpoint as well.

Proof Set

$$\ell(x)=f(a)+\frac{f(b)-f(a)}{b-a}(x-a), \qquad h(x)=f(x)-\ell(x).$$

The function h is continuous on $[a,b]$, differentiable on (a,b), and $h(a)=h(b)=0$. Rolle's theorem gives $c\in(a,b)$ with

$$0=h'(c)=f'(c)-\frac{f(b)-f(a)}{b-a}.$$

Rearranging proves the assertion. □

Figure 4.1 shows the construction on a particular graph. The proof works for every function with the stated hypotheses, regardless of how many times it bends.

An equivalent increment form is

$$f(x+h)-f(x)=hf'(x+\theta h), \qquad 0<\theta<1,$$

provided $h\neq 0$ and f satisfies the hypotheses on the segment with endpoints $x, x+h$. This holds for negative as well as positive h. The number θ generally depends on both x and h; it is not a universal constant.

Example 4.4 An endpoint where the derivative is not finite Apply the mean value theorem to $f(x)=\sqrt{x}$ on $[0,4]$ and find the point c.

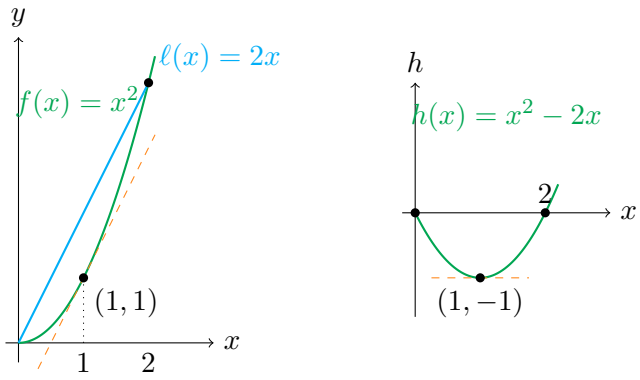


Figure 4.1: On $[0, 2]$, the secant to x^2 has slope 2, as does the tangent at 1. Subtracting the secant produces equal endpoint values and a horizontal tangent at the same point.

Solution *Continuity holds on the closed interval, and* $f'(x) = 1/(2\sqrt{x})$ *exists for* $0 < x < 4$. *The secant slope is* $(2 - 0)/4 = 1/2$. *The mean value equation therefore becomes*

$$\frac{1}{2\sqrt{c}} = \frac{1}{2} \quad \Longleftrightarrow \quad \sqrt{c} = 1 \quad \Longleftrightarrow \quad c = 1.$$

This point belongs to $(0, 4)$, *as required. The absence of a finite right derivative at* 0 *is irrelevant to the theorem.*

4.1.4 From derivative bounds to finite changes

The unknown point c is often useful precisely because we know where it lies. If a derivative is bounded throughout the interval, its value at c obeys the same bounds.

Corollary 4.1

Suppose that f *is continuous on* $[a, b]$ *and differentiable on* (a, b). *If* $m \le f'(x) \le M$ *for every* $x \in (a, b)$, *then*

$$m(b - a) \le f(b) - f(a) \le M(b - a).$$

In particular, $f' \ge 0$ *on an interval implies that* f *is nondecreasing there;* $f' > 0$ *implies that it is strictly increasing. The corresponding statements with reversed signs hold as well.* ♡

Proof Insert the derivative bounds into (4.1). For the monotonicity statements, apply that identity to any two points $x < y$ in the interval. Their difference $y - x$ is positive, so the sign of the derivative at the intermediate point determines the sign of $f(y) - f(x)$. □

Taking $m = M = 0$ recovers Lemma 2.1, proved independently when antiderivatives were introduced. Taking $m = -M$ shows that $|f'| \le M$ gives

$$|f(y) - f(x)| \le M\,|y - x|\,.$$

An estimate of this form, holding for every pair in an interval, is called a *Lipschitz bound*(利普希茨界)with constant M. It controls all pairs of points by the same constant; the definition of differentiability at a single point does not by itself supply such a bound on an entire interval.

Example 4.5 Existence and uniqueness have different jobs Prove that $x + e^x = 0$ has exactly one real solution.

Solution ***Method 1: intermediate values and Rolle.*** *For* $f(x) = x + e^x$, *we have* $f(-1) = -1 + e^{-1} < 0$ *and*

$f(0) = 1 > 0$. Continuity gives a zero in $(-1, 0)$. If there were two distinct real zeros, Rolle's theorem would give a point between them where $f' = 0$. But $f'(x) = 1 + e^x > 0$ everywhere, a contradiction.

Solution ***Method 2: compare finite changes.*** *The same endpoint signs give existence. For any $x < y$, the mean value theorem gives*

$$f(y) - f(x) = (1 + e^c)(y - x) > 0$$

for some $c \in (x, y)$. Thus f is strictly increasing and can take the value zero at most once.

The first method rules out two zeros directly. The second proves the stronger fact that every value is taken at most once. Neither replaces the separate continuity argument establishing existence.

Example 4.6 Two ways to bound a logarithm Prove that, for $x > 0$,

$$\frac{x}{1+x} < \ln(1+x) < x.$$

Solution ***Method 1: sample the derivative.*** *Apply the mean value theorem to $\ln(1 + t)$ on $[0, x]$. For some $c \in (0, x)$,*

$$\ln(1+x) = \frac{x}{1+c}.$$

Since $1/(1 + x) < 1/(1 + c) < 1$, multiplication by $x > 0$ proves both inequalities.

Solution ***Method 2: compare accumulated rates.*** *The fundamental theorem gives $\ln(1+x) = \int_0^x (1+t)^{-1}\,\mathrm{d}t$. Throughout $(0, x)$, the integrand lies strictly between $1/(1 + x)$ and 1. Integrating yields the claimed strict inequalities. For example, the difference in the upper bound is*

$$x - \ln(1+x) = \int_0^x \left(1 - \frac{1}{1+t}\right)\mathrm{d}t = \int_0^x \frac{t}{1+t}\,\mathrm{d}t > 0.$$

Strictness follows from continuity and positivity on a small interval around $x/2$. For the lower bound, use the positive difference $(1 + t)^{-1} - (1 + x)^{-1}$ on $(0, x)$ in the same way.

The first route only needs differentiability of the original function. The integral route uses a continuous integrand and makes accumulation explicit. One cannot replace an arbitrary differentiable function by the integral of its derivative without checking the hypotheses of the fundamental theorem.

Example 4.7 A bound that works for every pair Prove that $|\sin x - \sin y| \le |x - y|$ for all real x, y.

Solution ***Method 1: a derivative bound.*** *If $x = y$, the statement is immediate. Otherwise, the mean value theorem gives $\sin x - \sin y = (x - y)\cos c$ for some c strictly between them. Taking absolute values and using $|\cos c| \le 1$ proves the result.*

Solution ***Method 2: a trigonometric identity.*** *Using the elementary bound $|\sin u| \le |u|$ and the difference identity,*

$$|\sin x - \sin y| = 2\left|\cos\frac{x+y}{2}\sin\frac{x-y}{2}\right| \le 2\left|\frac{x-y}{2}\right| = |x-y|\,.$$

The identity gives a short proof special to sine. The derivative argument applies to any function with a bounded derivative. The constant 1 cannot be lowered: setting $y = 0$ and letting $x \to 0$ would otherwise contradict $\sin x/x \to 1$.

4.1.5 Auxiliary functions and zeros

Rolle's theorem also controls the number of zeros. If $x_1 < \cdots < x_n$ are distinct zeros of a differentiable function, then applying Rolle on each $[x_i, x_{i+1}]$ produces $n - 1$ distinct zeros of its derivative. Repeated application gives the following useful principle. Here $C^n([a, b])$, for an integer $n \ge 1$, means that f is continuous

on $[a, b]$, has derivatives through order n on (a, b), and each of those derivatives has a continuous extension to $[a, b]$. Derivatives at endpoints, when used, are understood from within the interval.

Proposition 4.1

Let $n \geq 1$ be an integer. If f is n times continuously differentiable on $[a, b]$ and has $n + 1$ distinct zeros there, then $f^{(n)}$ vanishes at some point of (a, b). Consequently, if $f^{(n)}$ never vanishes, f has at most n distinct zeros on $[a, b]$.

Proof After one application on consecutive zero intervals, f' has at least n distinct interior zeros. Repeating this argument gives at least $n - 1$ zeros of f'', and eventually one zero of $f^{(n)}$. The final statement is the contrapositive. □

Continuity of all derivatives is a convenient sufficient hypothesis here. The repeated argument only needs continuity on each closed interval and differentiability on its interior at the stage where Rolle is applied.

Example 4.8 Find zeros by first finding a primitive Let $n \geq 1$ be an integer and let $a_1, \ldots, a_n$ be real numbers. Prove that

$$q(x) = \sum_{k=1}^{n} a_k \cos(kx)$$

has a zero in $(0, \pi)$.

> ***The endpoint condition may belong to a different function.*** The endpoint values of q need not agree, so applying Rolle directly to q does not match the problem. Instead, the desired equation $q(c) = 0$ suggests writing q as a derivative. Each cosine has a sine primitive, and all those sine terms vanish at both 0 and π.

Solution ***Method 1: apply Rolle to a primitive.*** *Set*

$$Q(x) = \sum_{k=1}^{n} \frac{a_k}{k} \sin(kx).$$

This finite sum is continuous on $[0, \pi]$ and differentiable inside. Since $\sin(k\pi) = \sin 0 = 0$ for every integer k, we have $Q(0) = Q(\pi) = 0$. Termwise differentiation of the finite sum gives

$$Q'(x) = \sum_{k=1}^{n} \frac{a_k}{k} k \cos(kx) = q(x).$$

Rolle's theorem supplies $c \in (0, \pi)$ with $q(c) = Q'(c) = 0$.

Solution ***Method 2: use the zero integral.*** *The fundamental theorem gives*

$$\int_0^{\pi} q(x)\,\mathrm{d}x = \sum_{k=1}^{n} \frac{a_k}{k} \big(\sin(k\pi) - \sin 0\big) = 0.$$

If the continuous function q had no zero in $(0, \pi)$, the intermediate value theorem would force it to have a fixed strict sign there. Continuity gives the same weak sign at the endpoints, and a small closed interval around any interior point gives a strictly positive lower bound for $|q|$. Its integral would then have that strict sign, contradicting the displayed equality. Thus an interior zero exists.

The first route uses equal endpoint values of a primitive. The second expresses the same cancellation as zero total signed area and applies to any continuous function with zero integral. Neither method requires the coefficients to be nonzero; if they all vanish, every interior point is a zero.

Example 4.9 Count roots without solving for all of them Prove that $2^x = x^2$ has exactly three distinct real solutions: one in $(-1, 0)$, and the two solutions 2 and 4.

> ***Use different arguments for the lower and upper counts.*** Endpoint signs and direct substitution can exhibit three roots. To exclude a fourth, differentiate the difference three times: the quadratic part disappears, leaving a derivative of fixed sign. Repeated Rolle converts four hypothetical roots into a zero of that derivative.

Solution *Put* $F(x) = 2^x - x^2$. *We have*

$$F(-1) = \frac{1}{2} - 1 = -\frac{1}{2} < 0, \qquad F(0) = 1 > 0, \qquad F(2) = 4 - 4 = 0, \qquad F(4) = 16 - 16 = 0.$$

By continuity, there is at least one zero in $(-1, 0)$*; together with* $2, 4$*, this gives three distinct zeros. On the other hand,*

$$F'(x) = (\ln 2)2^x - 2x, \qquad F''(x) = (\ln 2)^2 2^x - 2, \qquad F'''(x) = (\ln 2)^3 2^x > 0.$$

If four distinct real zeros existed, enclose them in a finite closed interval. The function is C^3 *there, so Proposition 4.1 would give a zero of* F'''*, a contradiction. There are therefore exactly three zeros, including exactly one in* $(-1, 0)$.

The negative solution needs no elementary formula for this argument. The same upper-count method applies whenever a sufficiently high derivative of a difference has no zeros.

Example 4.10 Recognizing a derivative after multiplication Let f be continuous on $[a, b]$, differentiable on (a, b), and satisfy $f(a) = f(b) = 0$. For a given real λ, prove that some $c \in (a, b)$ satisfies $f'(c) = \lambda f(c)$.

> ***Build the target expression into a derivative.*** The target is a zero of $f' - \lambda f$. The product rule gives $(wf)' = wf' + w'f$. To make this a nonzero factor times $f' - \lambda f$, choose $w'/w = -\lambda$, hence $w = e^{-\lambda x}$. The endpoint zeros survive this multiplication.

Solution *Set* $F(x) = e^{-\lambda x} f(x)$. *It has the continuity and differentiability required by Rolle's theorem, and* $F(a) = F(b) = 0$. *Thus*

$$0 = F'(c) = e^{-\lambda c}\big(f'(c) - \lambda f(c)\big)$$

at some $c \in (a, b)$. *The exponential factor is positive, so the desired identity follows.*

The factor is chosen from the target derivative equation, not guessed from the endpoint data alone. When $\lambda = 0$, this is Rolle's theorem itself.

Example 4.11 An intermediate point that moves with the interval Evaluate $\lim_{n\to\infty} n\big(\arctan(n+1) - \arctan n\big)$.

Solution ***Method 1: trap the sampled derivative.*** *For* $f(t) = \arctan t$*, the derivative* $f'(t) = 1/(1+t^2)$ *is positive and becomes small on intervals far to the right. For each positive integer* n*, the mean value theorem gives* $c_n \in (n, n+1)$ *such that*

$$0 < n\big(\arctan(n+1) - \arctan n\big) = \frac{n}{1 + c_n^2} \le \frac{n}{1 + n^2} \longrightarrow 0.$$

Here $c_n > n > 0$ *implies* $1 + c_n^2 > 1 + n^2$*; the interval location supplies a bound independent of the unknown choice of* c_n*. The squeeze theorem gives the limit* 0.

Solution ***Method 2: bound an integral.*** *By the fundamental theorem,*

$$0 < n \int_n^{n+1} \frac{\mathrm{d}t}{1 + t^2} \le \frac{n}{1 + n^2} \longrightarrow 0.$$

The integral method is available because this derivative is continuous. Both methods use the same interval bound; neither needs an explicit formula for c_n. In general, a point selected by a theorem for each n must not be treated as fixed while n changes.

4.1.6 Further result: a derivative cannot jump over a value

The derivative need not be continuous, so the intermediate value theorem for continuous functions cannot simply be applied to it. Nevertheless, derivatives have an intermediate value property. The proof again creates an interior extremum, this time by subtracting a line of a prescribed slope.

This intermediate-value property is Darboux's theorem(达布定理).

Theorem 4.4 (Darboux's theorem for derivatives)

Let f be differentiable on an open interval containing $[a,b]$, with $a<b$. If k lies strictly between $f'(a)$ and $f'(b)$, then some $c\in(a,b)$ satisfies $f'(c)=k$. ♡

> ***Prescribe the slope of the line to subtract.*** To find $f'(c)=k$, seek a stationary point of $F(x)=f(x)-kx$. If $F'(a)<0<F'(b)$, values just to the right of a lie below $F(a)$, and values just to the left of b lie below $F(b)$. Neither endpoint can minimize F on the segment; an interior minimum supplies the desired point.

Proof First suppose $f'(a)<k<f'(b)$ and set $F(x)=f(x)-kx$. The inequality $F'(a)<0$ implies $F(a+h)<F(a)$ for all sufficiently small $h>0$. Likewise, $F'(b)>0$ implies $F(b-h)<F(b)$ for all sufficiently small $h>0$. The continuous function F attains a minimum on $[a,b]$, and the preceding inequalities exclude both endpoints. At an interior minimizer c, Fermat's theorem gives $F'(c)=0$, hence $f'(c)=k$. If $f'(a)>k>f'(b)$, apply the proved case to $-f$ and $-k$. □

This does not imply continuity of f'. For example, $f(x)=x^2\sin(1/x)$ for $x\neq 0$, with $f(0)=0$, is differentiable at 0 with $f'(0)=0$, but

$$f'(x)=2x\sin(1/x)-\cos(1/x)\qquad(x\neq 0)$$

has no limit there. It oscillates instead of making a jump. The proof of Darboux's theorem uses continuity of f, not continuity of f'.

4.1.7 Methods and Further Remarks

A finite difference $f(b)-f(a)$ suggests the mean value theorem, especially when derivative bounds are simpler than the original expression. An existence statement for f' suggests Rolle after a suitable construction: subtract a line for a prescribed slope, or multiply by a factor when the target contains both f and f'. Several zeros suggest applying Rolle between consecutive zeros and then repeating. If the target is a zero of an expression rather than a derivative, first ask whether that expression has a primitive with equal endpoint values. For an exact root count, exhibit enough roots by signs or substitution, then obtain a matching upper bound from derivatives.

Check the entire closed segment for continuity and its interior for differentiability. A theorem that guarantees a point does not usually locate it or make it unique. In a limit problem, record the interval containing each selected point and use that interval to justify its limiting behavior. For equations, separate existence from uniqueness; a derivative test alone does not guarantee that a zero exists.

Exercise 4.1

1. Find all points supplied by Rolle's theorem for (a) $f(x)=x^2-4x+3$ on $[1,3]$; (b) $f(x)=\sin x$ on $[0,\pi]$; (c) $f(x)=x(x-1)(x-2)$ on $[0,2]$.
2. Find all mean value points for (a) $f(x)=x^3$ on $[0,2]$; (b) $f(x)=1/x$ on $[1,4]$; (c) $f(x)=\ln x$ on $[1,e]$; (d) $f(x)=e^x$ on $[-h,h]$, where $h>0$.

3. Decide whether the hypotheses of the mean value theorem hold for (a) $\sqrt{x}$ on $[0,1]$; (b) $|x|^{2/3}$ on $[-1,1]$. In each case, determine whether the conclusion holds.
4. For each statement, give a counterexample: (a) every stationary point is a local extremum; (b) a continuous function with equal endpoint values must have a zero derivative somewhere inside; (c) a differentiable function on (a,b) with assigned equal endpoint values must have a zero derivative inside.
5. A position function s is continuous on $[0,12]$ and differentiable on $(0,12)$, with $s(0)=5$ and $s(12)=185$, in metres and seconds. Prove that the velocity equals $15\,\mathrm{m/s}$ at some instant. Explain why the assertion concerns velocity rather than total distance travelled.
6. Suppose f is differentiable on $\mathbb{R}$ and $f'(x)=3$ everywhere. If $f(-2)=7$, find $f(x)$ and justify why no other function is possible.
7. Prove $|\arctan x-\arctan y|\le|x-y|$ for all real x,y, and show that the constant 1 is optimal. For a fixed $a>0$, also prove $|\ln x-\ln y|\le|x-y|/a$ for all $x,y\ge a$, and show that the constant $1/a$ is optimal on this domain.
8. Prove that $(b-a)/b<\ln(b/a)<(b-a)/a$ when $0<a<b$.
9. Prove that $x^3+2x-5=0$ has exactly one real root, and locate it in $(1,2)$.
10. Let f be differentiable on an interval I, with $f'\ge m>0$, and suppose $f(r)=0$ for $r\in I$. Prove $|x-r|\le|f(x)|/m$ for $x\in I$. Explain how a small residual $|f(x)|$ controls root error.
11. Let f,g be continuous on $[a,b]$, differentiable on (a,b), satisfy $f(a)=g(a)$, and obey $f'\le g'$ there. Prove $f(x)\le g(x)$ for $x\in[a,b]$.
12. Suppose f is continuous on $[a,b]$, differentiable except possibly at one interior point d, and $f'(x)=0$ wherever the derivative is required to exist. Prove that f is constant.
13. Prove that a differentiable function on an interval whose derivative is nowhere zero has at most one zero. Is the converse true? Give a counterexample if not.
14. Suppose f is continuous on $[1,4]$, differentiable inside, $f(1)=3$, and $2\le f'\le 5$. Find the sharp interval in which $f(4)$ must lie, and give examples attaining its endpoints.
15. Let $f\in C^3([a,b])$ have four distinct zeros in $[a,b]$. Prove that f''' has a zero in (a,b), identifying the disjoint intervals used at each step.
16. If two continuous functions on $[a,b]$, differentiable inside, have equal values at both endpoints, prove that their graphs have parallel tangents at some common interior input.
17. For $f(x)=x^2$ when $0\le x\le 1$ and $f(x)=Ax+B$ when $1<x\le 2$, find all A,B for which the hypotheses of the mean value theorem hold on $[0,2]$. Then find every mean value point.
18. Suppose f is continuous on $[a,b]$, differentiable on (a,b), and $\lim_{x\to a^+}f'(x)=L\in\mathbb{R}$. Prove that the right derivative of f at a exists and equals L.
19. For $f(x)=x^2\sin(1/x)$ when $x\neq 0$, $f(0)=0$, prove that f is differentiable everywhere but f' is discontinuous at 0. Explain why the mean value theorem still applies on $[0,1]$.
20. Let f be differentiable on $\mathbb{R}$ and suppose $f'(x+T)=f'(x)$ for a fixed $T>0$. Prove that $f(x+T)-f(x)$ is constant in x. Show that boundedness of f on $\mathbb{R}$ forces this constant to be zero.

Supplement Exercises

S1. Let $f\in C^n([a,b])$, where $n\ge 1$, and let p be a polynomial of degree at most n with leading coefficient A for x^n (possibly $A=0$). If $f-p$ vanishes at $n+1$ distinct points of $[a,b]$, prove that $f^{(n)}(c)=n!A$ for some $c\in(a,b)$.

S2. For an integer $n\ge 1$, set $P(x)=x^n(1-x)^n$. Prove that $P^{(n)}$ has exactly n distinct zeros in $(0,1)$.

> ***Optional Hint.*** For $k < n$, both endpoints remain zeros of $P^{(k)}$. Count the interior zeros gained at each differentiation, then use the degree of $P^{(n)}$.

S3. Suppose f is differentiable on $\mathbb{R}$ and has the same finite limit at $+\infty$ and $-\infty$. Prove that $f'(c) = 0$ for some real c. Do not apply Rolle directly to an interval with infinite endpoints.

S4. Let $f \in C^2([a,b])$, $f(a) = f(b) = 0$, and $|f''| \le M$. Prove that

$$|f(x)| \le \frac{M}{2}(x-a)(b-x) \quad (a \le x \le b).$$

> ***Optional Hint.*** Fix an interior x and subtract a multiple of $(t-a)(t-b)$ that makes the difference vanish at a, x, b.

S5. Let $f\colon [a,b] \to [a,b]$ be continuous, differentiable on (a,b), and satisfy $|f'| \le q < 1$, where $0 \le q < 1$. Prove that f has exactly one fixed point r. For $x_{n+1} = f(x_n)$, prove $|x_n - r| \le q^n\,|x_0 - r|$ for $n \ge 1$.

S6. Let f be continuous on $[0,+\infty)$, differentiable on $(0,+\infty)$, and satisfy $f'(x) \ge \lambda f(x)$ for a real constant λ. Prove $f(x) \ge f(0)e^{\lambda x}$ for $x \ge 0$.

S7. Use Darboux's theorem to prove that a nowhere-zero derivative on an interval has a constant sign, and hence that its original function is strictly monotone. Explain why continuity of the derivative was unnecessary.

S8. If f is differentiable on $\mathbb{R}$ and $f'(x)$ is an integer for every x, prove that $f(x) = mx + b$ for some integer m and real b.

S9. Suppose f is differentiable and bounded on $\mathbb{R}$, and f' is nondecreasing. Prove that f is constant.

S10. Suppose f is differentiable on $(0,+\infty)$ and $f'(x) \to L \in \mathbb{R}$ as $x \to +\infty$. Prove $f(x)/x \to L$ using a fixed large starting point and the mean value theorem, without L'Hôpital's rule.

S11. Define $f(0) = 0$ and $f(x) = x + 2x^2 \sin(1/x)$ for $x \neq 0$. Prove $f'(0) = 1$, but show that f is not increasing on any interval containing 0 as an interior point. Explain why positivity of a derivative at one point differs from positivity throughout an interval.

S12. Let $\lambda_1 < \cdots < \lambda_n$ be real numbers and $a_1, \dots, a_n$ be nonzero real coefficients. Prove that $\sum_{j=1}^{n} a_j e^{\lambda_j x}$ has at most $n-1$ distinct real zeros.

> ***Optional Hint.*** Multiply by $e^{-\lambda_1 x}$ and differentiate. One term disappears; use induction and Rolle between any finitely many alleged zeros.

S13. Let I be an open interval, let $c \in I$, and let $f \in C^n(I)$ for an integer $n \ge 1$. Suppose a sequence of distinct zeros of f in $I \setminus \{c\}$ converges to c. Prove that $f^{(k)}(c) = 0$ for every $0 \le k \le n$, where $f^{(0)} = f$. Show by an example that the converse is false for every fixed n.

> ***Optional Hint.*** Continuity first gives $f(c) = 0$. Apply Rolle between c and each approaching zero, and repeat for the derivatives.

S14. Let $P(x) = \prod_{j=1}^{n}(x - r_j)$, where $n \ge 2$ and $r_1 < \cdots < r_n$ are real numbers. For every real λ, prove that $P'(x) = \lambda P(x)$ has exactly one root in each interval (r_j, r_{j+1}). Show that there is one additional root in $(r_n, +\infty)$ if $\lambda > 0$, one additional root in $(-\infty, r_1)$ if $\lambda < 0$, and no additional root if $\lambda = 0$. Verify that none of the r_j is a root of this equation.

> ***Optional Hint.*** Away from the r_j, the product rule gives $P'/P = \sum_{j=1}^{n}(x - r_j)^{-1}$. Determine its derivative and its one-sided limits on every interval of its domain.

4.2 Cauchy's Mean Value Theorem

知识点概要

- *Compare two finite changes at one common point.*
- *Distinguish a cross-product identity from a quotient.*
- *Interpret a change of parameter geometrically.*
- *Bound a ratio by comparing derivatives.*
- *Connect derivative ratios with weighted averages.*

Lagrange's theorem compares the change in f with the change in the input x. Sometimes another quantity $g(x)$ is the more useful scale. We then want to compare $f(b)-f(a)$ with $g(b)-g(a)$. Applying Lagrange separately to f and g gives two intermediate points, which need not agree. Cauchy's mean value theorem (柯西中值定理) obtains both derivatives at one point.

4.2.1 Why a common point matters

Example 4.12 Two mean value points cannot simply be identified For $f(t)=t^2$ and $g(t)=t^3$ on $[0,1]$, compare the separate Lagrange identities with the proposed identity

$$\frac{f(1)-f(0)}{g(1)-g(0)}=\frac{f'(c)}{g'(c)}.$$

Solution *The separate identities require $2c_f=1$ and $3c_g^2=1$, hence $c_f=1/2$ and $c_g=1/\sqrt{3}$. Dividing those identities gives $1=f'(c_f)/g'(c_g)$, with different arguments. For a common point $c\in(0,1)$, we instead have*

$$\frac{f'(c)}{g'(c)}=\frac{2c}{3c^2}=\frac{2}{3c}.$$

The cancellation is legal because $c>0$. Equating this quotient to the endpoint quotient 1 gives $3c=2$, hence $c=2/3$. This point lies in $(0,1)$ but is neither of the two separate points. The existence of a common point needs its own proof.

Theorem 4.5 (Cauchy's mean value theorem)

Let $a<b$, and let f,g be continuous on $[a,b]$ and differentiable on (a,b). There is $c\in(a,b)$ such that

$$\big(g(b)-g(a)\big)f'(c)=\big(f(b)-f(a)\big)g'(c). \tag{4.2}$$

If, in addition, $g'(x)\neq 0$ for every $x\in(a,b)$, then $g(b)\neq g(a)$ and the identity can be written as

$$\frac{f(b)-f(a)}{g(b)-g(a)}=\frac{f'(c)}{g'(c)}. \tag{4.3}$$

♡

The first statement involves no division. The extra hypothesis in the second statement guarantees that both denominators are nonzero. Taking $g(x)=x$ recovers Lagrange's theorem.

> ***Match the derivative and the endpoint condition at once.*** Cross-multiplying the desired equality suggests a linear combination of f' and g' with constant coefficients. Integrate that combination formally to obtain a linear combination of f and g. Subtract their values at a so that the auxiliary function van-

> ishes there; the cross-product coefficients then make it vanish at b as well. This is a construction by the derivative rules, not an assumption that arbitrary derivatives are integrable.

Proof Define

$$H(x) = \big(g(b) - g(a)\big)\big(f(x) - f(a)\big) - \big(f(b) - f(a)\big)\big(g(x) - g(a)\big).$$

It is continuous on $[a, b]$, differentiable on (a, b), and $H(a) = H(b) = 0$. Rolle's theorem gives $H'(c) = 0$ at some $c \in (a, b)$, which is (4.2). Under the additional hypothesis, $g(a) = g(b)$ would contradict Rolle's theorem applied to g. Thus $g(b) - g(a) \neq 0$, and $g'(c) \neq 0$ as well. Dividing yields (4.3). □

Example 4.13 A cross-product identity can hold where a quotient is undefined Take $f(t) = t^2$ and $g(t) = t^3$ on $[-1, 1]$. Show why the quotient conclusion cannot be used here.

Solution *The endpoint increments are* $f(1) - f(-1) = 0$ *and* $g(1) - g(-1) = 2$. *The cross-product identity becomes* $4c = 0$, *so its only possible point is* $c = 0$. *At that point both derivatives vanish. For* $c \neq 0$, *the derivative quotient is* $2/(3c)$, *which never equals the endpoint quotient* 0. *Thus there is no point at which the quotient identity is both defined and true. The nonvanishing hypothesis on* g' *cannot simply be dropped from the theorem.*

The hypothesis is sufficient, rather than necessary for every particular pair of functions. For example, if $f = g = t^3$ on $[-1, 1]$, both quotients equal 1 at every nonzero interior point even though $g'(0) = 0$. Each application must check its own assumptions; an occasionally valid conclusion does not make a general theorem stronger.

4.2.2 The geometry of two changing coordinates

Consider the curve whose horizontal and vertical coordinates are

$$X = g(t), \qquad Y = f(t), \qquad a \le t \le b.$$

This is a *parametric representation*(参数表示): the parameter t specifies the two coordinates together. At a point with $g'(c) \neq 0$, the tangent slope is $f'(c)/g'(c)$. Indeed, as $h \to 0$,

$$\frac{f(c+h) - f(c)}{g(c+h) - g(c)} = \frac{[f(c+h) - f(c)]/h}{[g(c+h) - g(c)]/h} \longrightarrow \frac{f'(c)}{g'(c)}.$$

The denominator is nonzero for sufficiently small nonzero h because its quotient tends to $g'(c) \neq 0$. Cauchy's theorem says that at some interior parameter value this tangent is parallel to the chord joining the two endpoint positions.

In **Figure** 4.2, $g'(0) = 0$ at an endpoint causes no problem: the theorem requires nonvanishing only for $0 < t < 1$. The curve interpretation explains the same-point requirement; a tangent uses the simultaneous changes of its two coordinates.

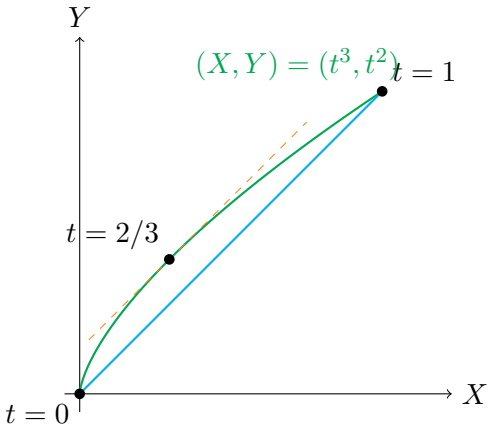


Figure 4.2: The endpoint chord and the tangent at $t = 2/3$ both have slope 1. The horizontal coordinate is t^3, so the slope is a ratio of coordinate derivatives.

4.2.3 Choosing a useful scale for comparison

Corollary 4.2

Under the quotient hypotheses of Cauchy's theorem, if

$$m \le \frac{f'(t)}{g'(t)} \le M \qquad (a < t < b),$$

then

$$m \le \frac{f(b) - f(a)}{g(b) - g(a)} \le M.$$

Strict bounds on the derivative quotient give corresponding strict bounds on the increment quotient. ♡

Proof The increment quotient equals the derivative quotient at a point in (a, b). □

No multiplication by an unspecified sign was used. It is the quotient itself that is compared with m and M. To obtain an inequality for the numerator alone, the sign of $g(b) - g(a)$ must then be taken into account.

Example 4.14 The logarithmic mean lies between its inputs For $0 < a < b$, prove

$$a < \frac{b - a}{\ln b - \ln a} < b.$$

The middle expression is called the *logarithmic mean*(对数平均数)of a, b.

Solution ***Method 1: use logarithmic change as the scale.*** *Apply Cauchy's theorem to $f(t) = t$ and $g(t) = \ln t$ on $[a, b]$. Since $g'(t) = 1/t > 0$, there is $c \in (a, b)$ with*

$$\frac{b - a}{\ln b - \ln a} = \frac{1}{1/c} = c.$$

The desired bounds follow from the location of c.

Solution ***Method 2: bound the denominator by integration.*** *The continuous function $1/t$ lies strictly between $1/b$ and $1/a$ on (a, b). Thus*

$$\frac{b - a}{b} < \int_a^b \frac{\mathrm{d}t}{t} = \ln b - \ln a < \frac{b - a}{a}.$$

All quantities are positive; taking reciprocals and multiplying by $b - a$ gives the result.

Cauchy's theorem identifies the mean as a sampled input. The integral proof emphasizes that the denominator accumulates reciprocal input values. Symmetry in a, b follows by changing the signs of numerator and denominator together.

Example 4.15 Bounds without computing the intermediate point For $0 < x \le 1$, prove

$$1 < \frac{\arctan x}{\ln(1+x)} \le \frac{1+\sqrt{2}}{2}.$$

> ***Choose functions whose values vanish at the same endpoint.*** Both numerator and denominator vanish at 0. Their derivative quotient is rational, so it is easier to bound than the original quotient. The selected point lies in $(0, x)$, where the interval restriction supplies the sign of $t - t^2$.

Solution *Use $f(t) = \arctan t$, $g(t) = \ln(1+t)$ on $[0, x]$. Cauchy's theorem gives $c \in (0, x) \subset (0, 1)$ such that*

$$\frac{\arctan x}{\ln(1+x)} = \frac{1+c}{1+c^2}.$$

This exceeds 1 because $c - c^2 > 0$. Put $A = (1+\sqrt{2})/2$. The identities $2A(\sqrt{2}-1) = 1$ and $A(\sqrt{2}-1)^2 = A-1$ give the square completion

$$At^2 - t + (A-1) = A\big(t - (\sqrt{2}-1)\big)^2.$$

For every real t,

$$A(1+t^2) - (1+t) = A\big(t - (\sqrt{2}-1)\big)^2 \ge 0.$$

Since $1 + t^2 > 0$, the desired upper bound follows.

The square completion proves the needed bound directly; it does not depend on a later maximum test.

More generally, for every $x \in (a, b]$, the same theorem on $[a, x]$ yields a possibly different $c_x \in (a, x)$ with

$$\frac{f(x) - f(a)}{g(x) - g(a)} = \frac{f'(c_x)}{g'(c_x)}, \tag{4.4}$$

provided g' does not vanish on (a, b). In particular, $c_x \to a$ as $x \to a^+$. This exact identity on a shrinking interval is the starting point for the next section's limit theorem; no limit theorem has been used to prove it. Before passing to a limit, the same identity can also determine how a quotient varies.

Example 4.16 Use a sampled derivative to control a quotient Prove that $q(x) = \sin x/x$ is strictly decreasing on $(0, \pi)$, and deduce

$$\frac{2}{\pi} < \frac{\sin x}{x} < 1 \qquad \left(0 < x < \frac{\pi}{2}\right).$$

> ***Compare the quotient with the derivative ratio at its endpoint.*** Direct differentiation gives $q'(x) = (x\cos x - \sin x)/x^2$. Its sign is not supplied by the signs of sine and cosine alone. Cauchy's theorem compares $\sin x/x$ with $\cos c$ at an earlier point $c < x$; the decrease of cosine then supplies exactly the missing comparison with $\cos x$.

Solution ***Method 1: use Cauchy's common point.*** *Fix $0 < x < \pi$ and apply Cauchy's theorem to $f(t) = \sin t$ and $g(t) = t$ on $[0, x]$. The functions are continuous and differentiable, and $g' = 1 \neq 0$. For some $c \in (0, x)$,*

$$q(x) = \frac{\sin x - \sin 0}{x - 0} = \frac{\cos c}{1} = \cos c.$$

Since $-\sin t < 0$ on $(0, \pi)$, Corollary 4.1 shows that cosine strictly decreases there. Consequently $q(x) = \cos c > \cos x$, so

$$q'(x) = \frac{x\cos x - \sin x}{x^2} = \frac{\cos x - q(x)}{x} < 0.$$

The same corollary makes q strictly decreasing on $(0, \pi)$. For $0 < x < \pi/2$, comparison with $\pi/2$ gives $q(x) > q(\pi/2) = 2/\pi$. Also $0 < c < \pi$ gives $q(x) = \cos c < 1$, proving the upper bound.

Solution ***Method 2: express the derivative numerator as an integral.*** *Let $H(x) = x\cos x - \sin x$. The product rule gives*

$$H'(x) = \cos x - x\sin x - \cos x = -x\sin x, \qquad H(0) = 0.$$

The fundamental theorem therefore yields

$$H(x) = -\int_0^x t\sin t\,\mathrm{d}t < 0 \qquad (0 < x < \pi).$$

The integrand is positive inside the integration interval, so the inequality is strict. Thus $q' = H/x^2 < 0$, *and comparison with* $q(\pi/2)$ *gives the lower bound as before. For the upper bound, write* $q(x) = x^{-1}\int_0^x \cos t\,\mathrm{d}t$ *and use* $\cos t < 1$ *for* $0 < t < x < \pi$.

The first method replaces a quotient by an earlier value of its derivative ratio. The second finds an exact integral for the sign-determining numerator. It uses the continuity needed for the fundamental theorem, while the common-point argument suggests a more general comparison for merely differentiable functions.

4.2.4 Weighted averages from two primitives

An ordinary average treats equal lengths equally. A positive *weight*(权函数) $w(t)$ lets some portions of an interval contribute more strongly. For continuous u, w on $[a, b]$ with $w > 0$, the *weighted average*(加权平均值) of u is

$$\frac{\int_a^b u(t)w(t)\,\mathrm{d}t}{\int_a^b w(t)\,\mathrm{d}t}.$$

The denominator is positive and normalizes the total weight. A constant $u = K$ has weighted average K for every such weight. The case $w = 1$ is the ordinary integral average from Section 2.7.

Proposition 4.2

If u, w *are continuous on* $[a, b]$, $a < b$, *and* $w(t) > 0$ *there, then some* $c \in (a, b)$ *satisfies*

$$\int_a^b u(t)w(t)\,\mathrm{d}t = u(c)\int_a^b w(t)\,\mathrm{d}t.$$

> ***Make the unwanted weight cancel after differentiation.*** The numerator and denominator are accumulations over the same interval. Use them as endpoint increments of their accumulation functions. Differentiation recovers uw and w, whose quotient is exactly u.

Proof Set $F(x) = \int_a^x u(t)w(t)\,\mathrm{d}t$ and $G(x) = \int_a^x w(t)\,\mathrm{d}t$. The fundamental theorem gives $F' = uw$ and $G' = w > 0$ in (a, b); both functions are continuous on $[a, b]$. Cauchy's theorem now gives

$$\frac{F(b) - F(a)}{G(b) - G(a)} = \frac{F'(c)}{G'(c)} = u(c).$$

The increments are the indicated integrals, proving the identity. □

Example 4.17 A mean shifted by a positive weight Compute the weighted average of $u(t) = t$ with weight $w(t) = e^t$ on $[0, 1]$, and prove that it equals $u(c)$ for some $0 < c < 1$ by two approaches.

Solution ***Method 1: Cauchy applied to primitives.*** *Take* $F(t) = (t-1)e^t$ *and* $G(t) = e^t$. *Then* $F' = te^t$ *and* $G' = e^t > 0$. *Their endpoint increments are* $F(1) - F(0) = 1$ *and* $G(1) - G(0) = e - 1$. *Cauchy's theorem gives*

$$\frac{\int_0^1 te^t\,\mathrm{d}t}{\int_0^1 e^t\,\mathrm{d}t} = \frac{1}{e-1} = \frac{F'(c)}{G'(c)} = c \qquad (0 < c < 1).$$

Solution ***Method 2: positivity of accumulated weights.*** *Integration by parts gives the numerator step by step:*

$$\int_0^1 te^t\,\mathrm{d}t = \left[te^t\right]_0^1 - \int_0^1 e^t\,\mathrm{d}t = e - (e-1) = 1.$$

The denominator is $\int_0^1 e^t\,\mathrm{d}t = e - 1 > 0$. *Because* $0 < t < 1$ *inside the interval and the weight is positive,*

$$0 < \int_0^1 te^t\,\mathrm{d}t < \int_0^1 e^t\,\mathrm{d}t.$$

Both inequalities are strict by continuity and positivity on an interior subinterval. Consequently $c = 1/(e-1)$ *belongs to* $(0,1)$, *and* $u(c) = c$ *is the average.*

The first proof selects the point through derivatives of primitives. The second places the average between the values of u by positivity. For a general continuous u, the latter approach bounds the average between its minimum and maximum and then uses intermediate values.

Example 4.18 Control measured in a nonlinear scale Let f, g be continuous on $[a,b]$ and differentiable on (a,b). Suppose $g' > 0$ and $|f'| \le Mg'$ there, where $M \ge 0$. Prove, for every $x \in [a,b]$, that

$$|f(x) - f(a)| \le M\big(g(x) - g(a)\big).$$

Solution ***Method 1: compare a derivative quotient.*** *The assertion is equality when* $x = a$. *For* $x > a$, *Lagrange's theorem gives* $g(x) > g(a)$. *Cauchy's theorem on* $[a,x]$ *and the hypothesis give*

$$\left|\frac{f(x) - f(a)}{g(x) - g(a)}\right| = \left|\frac{f'(c)}{g'(c)}\right| \le M.$$

Multiplying by the positive denominator proves the result.

Solution ***Method 2: compare two ordinary changes.*** *Both* $(Mg + f)'$ *and* $(Mg - f)'$ *are nonnegative. The derivative-sign consequence of Lagrange's theorem gives*

$$M\big(g(x) - g(a)\big) \pm \big(f(x) - f(a)\big) \ge 0.$$

The two inequalities together are exactly the claimed absolute-value bound.

The quotient method is convenient when a derivative ratio is already present. The second method avoids division and reveals the two comparisons hidden inside the absolute value. Both require only the stated differentiability, not continuity of the derivatives.

An additive bound controls a difference $f(b) - f(a)$. For a positive quantity, its ratio $f(b)/f(a)$ measures relative growth instead. Logarithms convert this ratio into a difference; taking a logarithm of the input also makes power functions the natural comparison models.

Example 4.19 Relative derivative bounds give power bounds Let $0 < a < b$ and let f be positive and continuous on $[a,b]$, and differentiable on (a,b). Suppose real constants $m \le M$ satisfy

$$m \le \frac{xf'(x)}{f(x)} \le M \qquad (a < x < b).$$

Prove that

$$\left(\frac{b}{a}\right)^m \le \frac{f(b)}{f(a)} \le \left(\frac{b}{a}\right)^M.$$

> ***Choose logarithms to match a relative change.*** The derivative of $\ln f$ is f'/f, while the derivative of $\ln x$ is $1/x$. Their derivative quotient is therefore xf'/f, precisely the quantity bounded in the hypothesis. Alternatively, the equality model $xf' = pf$ suggests comparing f with x^p by differentiating f/x^p.

Solution ***Method 1: compare two logarithmic changes.*** *Positivity of* f *and* x *makes* $F(x) = \ln f(x)$ *and* $G(x) = \ln x$ *continuous on* $[a,b]$ *and differentiable inside, with* $G'(x) = 1/x > 0$. *Cauchy's theorem gives* $c \in (a,b)$ *with*

$$\frac{\ln f(b) - \ln f(a)}{\ln b - \ln a} = \frac{f'(c)/f(c)}{1/c} = \frac{cf'(c)}{f(c)} \in [m, M].$$

Since $\ln(b/a) > 0$*, multiplying by it preserves both inequalities:*

$$m\ln(b/a) \le \ln\big(f(b)/f(a)\big) \le M\ln(b/a).$$

The exponential function is strictly increasing, so exponentiating proves the required bounds.

Solution ***Method 2: divide by the comparison powers.*** *Because* $x > 0$ *and* $f(x) > 0$*, the hypotheses give*

$$\left(\frac{f(x)}{x^m}\right)' = x^{-m-1}\big(xf'(x) - mf(x)\big) \ge 0,$$

$$\left(\frac{f(x)}{x^M}\right)' = x^{-M-1}\big(xf'(x) - Mf(x)\big) \le 0.$$

The derivative-sign corollary on $[a, b]$ *yields*

$$\frac{f(b)}{b^m} \ge \frac{f(a)}{a^m}, \qquad \frac{f(b)}{b^M} \le \frac{f(a)}{a^M}.$$

Rearrange and divide by $f(a) > 0$ *to obtain the same result.*

Both arguments allow negative m or M and need no continuity of f'. The logarithmic route explains why a derivative ratio controls a multiplicative change. The power-comparison route is useful when the comparison model is already known. The bounds are sharp: $f(x) = Cx^m$ and $f(x) = Cx^M$, with $C > 0$, attain the lower and upper bounds, respectively.

4.2.5 Methods and Further Remarks

A quotient of two endpoint increments suggests Cauchy's theorem. Choose the two functions from those increments, check continuity on the full segment, and prove that every denominator you divide by is nonzero. Do not replace two independently chosen Lagrange points by a common point.

Once the common point is obtained, its location often matters more than its exact value. Bounds for f'/g' transfer directly to the quotient of changes. For integral ratios with continuous integrands, accumulation functions can turn the same argument into a statement about averages. A nonnegative weight supports comparison; a weight that changes sign can destroy it. To study an anchored quotient, compare it with the derivative ratio at the moving endpoint and inspect the quotient derivative. For positive multiplicative quantities, try logarithmic changes; for power comparison, also test the derivative of f/x^p.

Exercise 4.2

1. Apply Cauchy's theorem to (a) $f(x) = \ln x$, $g(x) = x^2$ on $[1, 2]$; (b) $f(x) = \sqrt{x}$, $g(x) = \ln x$ on $[1, 4]$. Find each common point and verify that it lies in the required interval.
2. (a) Apply the quotient form to $f(x) = \sin x$, $g(x) = \cos x$ on $[0, \pi/2]$ and find the common point. Explain why $g'(0) = 0$ does not invalidate this application. (b) Do the same for $f(t) = e^t$, $g(t) = e^{-t}$ on $[-h, h]$, where $h > 0$.
3. For $f(x) = x$, $g(x) = x^2$ on $[-1, 1]$, find the points satisfying the cross-product identity. Explain why the endpoint quotient cannot be formed.
4. Suppose the quotient hypotheses hold and $f(a) = f(b)$. Explain why its conclusion is consistent with Rolle's theorem, and why the theorem still does not assert that f is constant.
5. For $1 \le a < b \le 2$, prove

$$\frac{e}{2} \le \frac{e^b - e^a}{b^2 - a^2} \le \frac{e^2}{4}.$$

6. Let $p, q > 0$ and $0 < a < b$. Bound $(b^p - a^p)/(b^q - a^q)$ between the smaller and larger of $(p/q)a^{p-q}$ and $(p/q)b^{p-q}$. Determine when the two bounds coincide, and prove strictness when they do not.
7. Suppose f, g are continuous on $[0, 1]$, differentiable on $(0, 1)$, $f(0) = g(0) = 0$, $g' \neq 0$, and $2 \leq f'/g' \leq 5$ there. Prove $2 \leq f(x)/g(x) \leq 5$ for $0 < x \leq 1$, including justification that the quotient exists.
8. Compute the weighted average with weight $w(t) = 1 + t$ on $[0, 1]$ for (a) $u(t) = t^2$; (b) $u(t) = t^3$; (c) $u(t) = (t - 1/2)^2$. In each case find every $c \in (0, 1)$ at which $u(c)$ equals that average, and decide whether it is unique.
9. Give functions for which g' vanishes at an interior point but the quotient conclusion of Cauchy's theorem is nevertheless true at another point. Explain why this does not justify removing the nonvanishing hypothesis in general.
10. Suppose f, g are differentiable on an open interval, g never vanishes, and $f'g - fg' = 0$ everywhere. Prove $f = Kg$ for some constant K. Distinguish this quotient-derivative identity from Cauchy's quotient of increments.
11. Suppose f is continuous on $[0, 2]$, differentiable inside, and $f(2) = 3f(0)$. Prove that some $c \in (0, 2)$ satisfies $(1 + c)f'(c) = f(c)$.
12. If $f' = \lambda g'$ on (a, b) and f, g are continuous on $[a, b]$, prove $f(b) - f(a) = \lambda(g(b) - g(a))$. Is a nonvanishing assumption on g' needed here?
13. For $f(t) = t^2$, $g(t) = t^4$ on $[0, 1]$, compute the two individual Lagrange points and the Cauchy point. Compare all three.
14. Take $u(t) = t$ and $w(t) = 8t - 3$ on $[0, 1]$. Compute $\int_0^1 uw / \int_0^1 w$ and show that it is not a value of u on $[0, 1]$. Identify the failed weight assumption.

Supplement Exercises

S1. Suppose f, g are continuous on $[a, b]$, differentiable inside, $f(a) = g(a) = 0$, and $g' > 0$. If f'/g' is strictly increasing, prove that f/g is strictly increasing on (a, b).

> ***Optional Hint.*** First compare $f(x)/g(x)$ with $f'(x)/g'(x)$ using a Cauchy point strictly between a and x. Then differentiate f/g.

S2. Show that the converse of the preceding assertion is false, even with nondecreasing in place of strictly increasing. Use $g(x) = x$ and $f(x) = x^2 + x \sin x$ on $[0, \pi]$. Verify the required behavior of both f/g and f'/g' explicitly.

S3. Let u, w be continuous on $[a, b]$, $w > 0$, and suppose u has Lipschitz constant L. With $m = (a + b)/2$, prove

$$\left| \frac{\int_a^b u(t)w(t)\,\mathrm{d}t}{\int_a^b w(t)\,\mathrm{d}t} - u(m) \right| \leq \frac{L}{2}(b - a).$$

S4. Suppose f, g are differentiable on $\mathbb{R}$, have finite limits $f_\pm, g_\pm$ at $\pm\infty$, satisfy $g' \neq 0$, and $g_+ \neq g_-$. Prove that some real c satisfies

$$\frac{f_+ - f_-}{g_+ - g_-} = \frac{f'(c)}{g'(c)}.$$

> ***Optional Hint.*** Subtract a suitable multiple of g from f to obtain equal limits at the two ends. Reduce to a finite interval as in the generalized Rolle argument.

S5. Let g be differentiable on an open interval I and $g' > 0$ there. Prove that its inverse on $g(I)$ is differentiable, with derivative $1/g'(g^{-1}(y))$, using difference quotients and continuity of the inverse. For a differentiable

function f on I and points $a < b$ in I, derive Cauchy's quotient formula on $[a, b]$ by applying Lagrange's theorem to $f \circ g^{-1}$. Explain why this argument does not require continuity of g'.

S6. Let f, g be continuous on $[a, b]$, differentiable inside, and suppose $g' \neq 0$. Prove that some $c \in (a, b)$ satisfies

$$\frac{f(a) - f(c)}{g(c) - g(b)} = \frac{f'(c)}{g'(c)}.$$

> ***Optional Hint.*** The product $(f(x) - f(a))(g(x) - g(b))$ vanishes at both endpoints. Check the denominators after applying Rolle.

S7. Let $f, g \in C^1([a, b])$, with $g' \geq 0$ and $g(b) > g(a)$. Suppose a continuous function h satisfies $f' = hg'$. Prove that $(f(b) - f(a))/(g(b) - g(a)) = h(c)$ for some $c \in (a, b)$. Explain why this statement need not assert that $f'(c)/g'(c)$ is defined.

S8. Let $a = x_0 < x_1 < \cdots < x_n = b$, and suppose g is strictly increasing. Express $(f(b)-f(a))/(g(b)-g(a))$ as a weighted average of the n quotients

$$\frac{f(x_j) - f(x_{j-1})}{g(x_j) - g(x_{j-1})}.$$

Prove that the full quotient lies between the smallest and largest of these quotients, and characterize equality.

S9. Suppose f, g are differentiable on an open interval, $g' > 0$, and $m \leq M$. Prove the equivalence between $m \leq f'/g' \leq M$ everywhere and

$$m \leq \frac{f(y) - f(x)}{g(y) - g(x)} \leq M \quad \text{for every } x < y.$$

Give an example showing that strict inequalities for all increment quotients need not force strict inequalities for every derivative quotient.

S10. Suppose f is continuous on $[0, 1]$, differentiable inside, $f(0) = f(1) = 0$, and $f(1/2) = 1$. For each real λ, prove that some $c \in (0, 1)$ satisfies

$$f'(c) - \lambda\big(f(c) - c\big) = 1.$$

> ***Optional Hint.*** Compare the values of $e^{-\lambda x}(f(x) - x)$ at $0, 1/2, 1$ and force an interior extremum.

S11. Let u, w be continuous on $[a, b]$, where $a < b$, let u be strictly increasing, and suppose $w > 0$. Define, for $a < x \leq b$,

$$A(x) = \frac{\int_a^x u(t)w(t)\,\mathrm{d}t}{\int_a^x w(t)\,\mathrm{d}t}.$$

Prove that $u(a) < A(x) < u(x)$, that A is strictly increasing, and that $A(x) \to u(a)$ as $x \to a^+$.

S12. Let $f, g \in C^2([a, b])$, where $a < b$, and suppose

$$f(a) = g(a) = f'(a) = g'(a) = 0, \qquad g''(t) > 0 \quad (a < t < b).$$

For every $x \in (a, b]$, prove that $g(x) > 0$ and that some $\xi \in (a, x)$ satisfies

$$\frac{f(x)}{g(x)} = \frac{f''(\xi)}{g''(\xi)}.$$

Deduce that bounds $m \leq f''/g'' \leq M$ on (a, b) imply $m \leq f/g \leq M$ on $(a, b]$.

> ***Optional Hint.*** First prove $g' > 0$ away from a. Apply Cauchy to f, g on $[a, x]$, then to f', g' on a smaller interval ending at the selected point.

4.3 L'Hôpital's Rule

知识点概要

- ❑ *Recognize indeterminate forms and check one-sided hypotheses.*
- ❑ *Derive a limit rule from Cauchy's mean value theorem.*
- ❑ *Convert products, differences and powers into suitable quotients.*
- ❑ *Compare growth rates and recognize when the rule gives no answer.*
- ❑ *Use the FTC to compare integrals with simple asymptotic models.*

Cauchy's theorem compares a ratio of increments with a ratio of derivatives at one common point. If the increments both tend to zero, can that comparison determine their relative size? This is the purpose of L'Hôpital's rule(洛必达法则). It is a sufficient test for a limit, not a permission to differentiate every quotient.

4.3.1 What an indeterminate form leaves undecided

The notation $0/0$ describes the separate limits of a numerator and denominator; it is not division of two numbers. We call it an *indeterminate form*(未定式) because those two limits alone do not determine the quotient limit. The same distinction applies to ∞/∞: the signs and relative rates still matter.

Example 4.20 The same form, different outcomes Determine the behavior as $x \to 0^+$ of

$$\frac{x^2}{x}, \qquad \frac{x}{x^2}, \qquad \frac{x\sin(1/x)}{x}.$$

Solution *Both numerator and denominator tend to zero in every quotient. After cancellation the expressions are* x, $1/x$, *and* $\sin(1/x)$. *Their outcomes are respectively* 0, $+\infty$, *and no limit. For the last claim, the sequences* $(\pi/2+2\pi n)^{-1}$ *and* $(3\pi/2+2\pi n)^{-1}$ *give values* 1 *and* -1. *The form records missing information rather than a common answer.*

By contrast, a quotient with numerator tending to 2 and denominator tending to 3 is already determined by the quotient law. A nonzero finite number divided by a quantity tending to zero is also not a $0/0$ form; its one-sided signs must be examined directly.

4.3.2 The zero-over-zero rule

Theorem 4.6 (L'Hôpital's Rule: Zero-over-Zero Form)

Let f, g *be differentiable on* $(a, a+\delta)$, *where* $a \in \mathbb{R}$ *and* $\delta > 0$. *Suppose that* $g'(x) \neq 0$ *on this interval and*

$$\lim_{x\to a^+} f(x) = \lim_{x\to a^+} g(x) = 0.$$

If $f'(x)/g'(x)$ *tends to* $L \in \mathbb{R} \cup \{-\infty, +\infty\}$ *as* $x \to a^+$, *then* $g(x) \neq 0$ *on the interval and*

$$\lim_{x\to a^+} \frac{f(x)}{g(x)} = L.$$

The analogous statement holds at a left-hand endpoint. ♡

There is no requirement that f or g be defined, let alone differentiable, at the limiting point. The nonzero

condition is on g' throughout a one-sided neighborhood, not just at a single point. The conclusion concerns f/g; the expression f'/g' is not the derivative of that quotient.

> ***Proof Strategy.*** The zero limits supply the endpoint values that Cauchy's theorem needs. After assigning both endpoint values to be zero, a quotient of function values becomes a quotient of increments. The resulting intermediate point lies between a and x, so it approaches a along with x.

Proof Set $f(a) = g(a) = 0$ for this proof. These extensions are continuous at a. For $a < x < a + \delta$, Rolle's theorem shows that $g(x) \neq g(a)$: equality would force g' to vanish inside (a, x). By Theorem 4.5, there is $\xi_x \in (a, x)$ such that

$$\frac{f(x)}{g(x)} = \frac{f(x) - f(a)}{g(x) - g(a)} = \frac{f'(\xi_x)}{g'(\xi_x)}.$$

Since $a < \xi_x < x$, we have $\xi_x \to a^+$. For finite L, an ε bound on the derivative quotient near a therefore gives the same bound on $f(x)/g(x)$. For $L = +\infty$, every prescribed lower bound on the derivative quotient gives that lower bound on $f(x)/g(x)$; the case $L = -\infty$ uses upper bounds. The left-hand proof uses the interval $[x, a]$ instead. □

A two-sided conclusion follows by applying the corresponding one-sided statements separately and obtaining the same limit. If the two sides disagree, there is no two-sided limit.

Example 4.21 A second-order logarithmic cancellation Evaluate

$$\lim_{x \to 0} \frac{\ln(1 + x) - x}{x^2}.$$

> ***Strategy.*** The linear parts cancel in the numerator. One differentiation removes that cancellation and exposes a factor x. An alternative is to express the difference as an integral of its derivative, then rescale the interval to a fixed one.

Solution ***Method 1: differentiate once, then simplify.*** *On $(-1, 1) \setminus \{0\}$ the functions are differentiable and $(x^2)' = 2x \neq 0$. Both original limits are zero, while*

$$\frac{1/(1 + x) - 1}{2x} = -\frac{1}{2(1 + x)} \longrightarrow -\frac{1}{2}.$$

The rule on each side gives the required limit $-1/2$. There is no need for another differentiation after the factor is canceled.

Solution ***Method 2: the FTC and a fixed interval.*** *For $|x| < 1$, the FTC and $t = xs$ give*

$$\ln(1 + x) - x = -\int_0^x \frac{t}{1 + t}\, \mathrm{d}t = -x^2 \int_0^1 \frac{s}{1 + xs}\, \mathrm{d}s.$$

This substitution also holds for negative x, with the oriented integral taking care of the sign. For $|x| \leq 1/2$, the difference between $s/(1 + xs)$ and s has absolute value at most $2\,|x|\, s^2$. The integral therefore tends to $\int_0^1 s\, \mathrm{d}s = 1/2$. The limit is again $-1/2$.

The first route is shorter. The second provides an exact error representation and a bound, which can be useful when a numerical estimate is also required.

Example 4.22 Repeated use requires repeated checks Evaluate $\lim_{x \to 0} (x - \sin x)/x^3$.

Solution *For the original $0/0$ quotient, differentiate numerator and denominator separately:*

$$\frac{(x - \sin x)'}{(x^3)'} = \frac{1 - \cos x}{3x^2}.$$

This new quotient is still of type $0/0$. Its numerator and denominator give the next derivative quotient

$$\frac{(1 - \cos x)'}{(3x^2)'} = \frac{\sin x}{6x} = \frac{1}{6}\frac{\sin x}{x} \longrightarrow \frac{1}{6}.$$

Each denominator derivative, $3x^2$ and $6x$, is nonzero off zero, and both applications have differentiable functions on each side. First the second application proves that $(1-\cos x)/(3x^2) \to 1/6$; the first application then proves the requested limit. One could apply the rule a third time to $\sin x/(6x)$ and obtain $\cos x/6 \to 1/6$. Using the known sine limit saves that step. The reasoning runs backward from the last verified limit, even when the calculation is written forward.

4.3.3 Infinite denominators and limits at infinity

For a zero-over-zero limit at $+\infty$, set $t = 1/x$. The functions $F(t) = f(1/t)$ and $G(t) = g(1/t)$ tend to zero as $t \to 0^+$, and

$$\frac{F'(t)}{G'(t)} = \frac{-t^{-2}f'(1/t)}{-t^{-2}g'(1/t)} = \frac{f'(1/t)}{g'(1/t)}.$$

The chain rule transfers all hypotheses to Theorem 4.6. The substitution $x = -1/t$ treats $-\infty$. For a denominator of unbounded size there is a different proof, because no finite endpoint value can be assigned.

Theorem 4.7 (L'Hôpital's Rule: Infinite-Denominator Form)

Let f, g be differentiable on a one-sided neighborhood of a finite endpoint, or on a tail approaching $+\infty$ or $-\infty$. Suppose $g' \neq 0$ there, $|g(x)| \to +\infty$, and $f'(x)/g'(x) \to L \in \mathbb{R} \cup \{-\infty, +\infty\}$ along that approach. Then

$$\frac{f(x)}{g(x)} \longrightarrow L.$$

No separate assumption that f tends to infinity is needed. ♡

> ***Proof Strategy.*** Cauchy's theorem still compares increments, but now one endpoint must stay inside the domain. Choose that fixed endpoint close enough to the limiting end that all derivative quotients beyond it are controlled. After division by $g(x)$, the fixed values $f(c)$ and $g(c)$ disappear. The intermediate point need not approach the limiting end; it only needs to remain where the derivative estimate holds.

Proof We treat $x \to a^+$ and $x \to +\infty$ together. Choose an interior point c so that the desired bound on f'/g' holds between c and the limiting end. For x farther toward that end, Cauchy's theorem gives

$$Q_x := \frac{f(x)-f(c)}{g(x)-g(c)} = \frac{f'(\xi_x)}{g'(\xi_x)}$$

at some point between c and x. The denominator of Q_x is nonzero by Rolle's theorem. For sufficiently advanced x, also $g(x) \neq 0$, and the exact identity is

$$\frac{f(x)}{g(x)} = Q_x\left(1 - \frac{g(c)}{g(x)}\right) + \frac{f(c)}{g(x)}. \tag{4.5}$$

If L is finite, choose c so that $|Q_x - L| < \varepsilon$ throughout the remaining neighborhood or tail. Subtracting L in (4.5) gives

$$\left|\frac{f(x)}{g(x)} - L\right| \le \varepsilon\left|1 - \frac{g(c)}{g(x)}\right| + \frac{|f(c) - Lg(c)|}{|g(x)|}.$$

The factor on the right tends to 1 and the final term tends to zero. For example, choosing the derivative error initially less than $\varepsilon/4$, then making the factor at most 2 and the final term less than $\varepsilon/2$, proves the limit definition.

If $L = +\infty$, fix $B > 0$ and choose c so that $Q_x > 2(B+1)$. Eventually $1 - g(c)/g(x) > 1/2$ and

$|f(c)/g(x)| < 1$. Equation (4.5) then gives $f(x)/g(x) > B$. For $L = -\infty$, apply this argument to $-f$. The other directions use exactly the same interval between a fixed interior c and x. □

Example 4.23 Failure of the test is not failure of the limit Evaluate $\lim_{x\to+\infty}(x + \sin x)/x$ and explain why the derivative quotient does not settle it.

Solution *Since* $|\sin x/x| \le 1/x$, *the original quotient tends to* 1. *The derivative quotient is* $1 + \cos x$, *which has values* 2 *at* $2\pi n$ *and* 0 *at* $(2n + 1)\pi$. *It has no limit. Thus the theorem's hypothesis fails while its possible conclusion happens to hold. One must return to another method, here the bounded perturbation estimate.*

Example 4.24 Logarithmic, polynomial and exponential growth Prove that, for every real $a > 0$ and integer $m \ge 0$,

$$\lim_{x\to+\infty} \frac{\ln x}{x^a} = 0, \qquad \lim_{x\to+\infty} \frac{x^m}{e^x} = 0.$$

Solution *For the first quotient,* $x^a \to +\infty$, *its derivative* ax^{a-1} *is positive, and*

$$\frac{1/x}{ax^{a-1}} = \frac{1}{ax^a} \longrightarrow 0.$$

For $m = 0$, *the second assertion is* $e^{-x} \to 0$. *For* $m \ge 1$, *repeated application to polynomial-over-exponential quotients ends with* $m!/e^x \to 0$. *At every step the denominator and its derivative are* $e^x > 0$ *and tend to infinity. Working back through those applications gives the result. For a real power* x^r *with* $r \ge 0$, *choose an integer* $m \ge r$ *and use* $0 \le x^r/e^x \le x^m/e^x$ *for* $x \ge 1$.

4.3.4 Products, differences and variable powers

The rule applies to quotients in the preceding theorems. Other indeterminate expressions must first be transformed using identities valid on their actual domains. For a product of a small and a large factor, either $uv = u/(1/v)$ or $uv = v/(1/u)$ may be useful. Choose the version whose derivatives are simpler, and check that its denominator is nonzero. For a difference of two large terms, combine fractions or rationalize before differentiating. For $u(x)^{v(x)}$ with $u(x) > 0$, use

$$u(x)^{v(x)} = \exp\big(v(x)\ln u(x)\big).$$

The forms 1^∞, 0^0 and ∞^0 refer to limiting base and exponent, not to arithmetic rules for those symbols.

Example 4.25 A difference of reciprocal quantities Evaluate

$$\lim_{x\to1}\left(\frac{1}{\ln x} - \frac{1}{x-1}\right).$$

> ***Strategy.*** Neither term has a finite limit, so subtracting separate limits loses their cancellation. The common denominator turns that cancellation into a zero-over-zero quotient. After differentiating, look for a factor before applying the rule again.

Solution *For* $x > 0$, $x \ne 1$, *the difference is*

$$\frac{x - 1 - \ln x}{(x-1)\ln x}.$$

The numerator and denominator tend to zero. The denominator derivative is $\ln x + (x-1)/x$; *it has the sign of* $x - 1$, *because both summands do. The derivative quotient equals*

$$\frac{(x-1)/x}{\ln x + (x-1)/x} = \frac{1}{x\,\ln x/(x-1) + 1} \longrightarrow \frac{1}{2},$$

where $\ln x/(x-1) \to 1$ *is the derivative of* $\ln$ *at* 1. *L'Hôpital's rule on both sides gives* $1/2$.

Example 4.26 Taking logarithms keeps the domain visible Evaluate $\lim_{x\to0^+} x^x$ and, for fixed $a, b \in \mathbb{R}$, $\lim_{x\to0}(1 + ax)^{b/x}$.

Solution ***Method 1: logarithms.*** *For the first limit,*

$$x \ln x = \frac{\ln x}{1/x} \longrightarrow 0,$$

because the denominator tends to infinity in absolute value, its derivative is $-1/x^2 \neq 0$*, and the derivative quotient is* $-x \to 0$*. Continuity of the exponential gives* $x^x \to 1$*. For the second,* $1 + ax > 0$ *for all sufficiently small* $|x|$*. The logarithm of the expression is* $b \ln(1 + ax)/x$*. Its limit is* ab*, by one zero-over-zero application (also valid for* $a = 0$*). Thus the requested limit is* e^{ab}*.*

Solution ***Method 2 for the second limit: a basic exponential limit.*** *If* $a \neq 0$*, put* $t = ax$*. Then* $(1 + ax)^{b/x} = \big((1+t)^{1/t}\big)^{ab}$*. The inner positive expression tends to* e*, so continuity of the fixed real power gives* e^{ab}*. If* $a = 0$*, the expression is identically* 1*.*

The second route reuses a known fundamental limit. The logarithmic route is more flexible when the base is a complicated positive function. A negative base with arbitrary real exponents is not automatically a real-valued function and cannot be handled by this transformation.

Example 4.27 A base approaching one hides a second-order change Evaluate

$$\lim_{x \to 0} \left(\frac{\sin x}{x} \right)^{1/x^2}$$

without Taylor's formula.

> ***Strategy.*** The base tends to one, so taking its logarithm exposes a small difference that the large exponent can magnify. After differentiating, combine $\cot x - 1/x$ before differentiating again. Alternatively, the already known limit of $(\sin x - x)/x^3$ measures exactly how far the base is from one.

Solution ***Method 1: differentiate the logarithm and simplify.*** *For* $0 < |x| < \pi$*, the base* $u(x) = \sin x/x$ *is positive. Since* $u(x) \to 1$*, the logarithm of the required expression is the* $0/0$ *quotient*

$$L(x) = \frac{\ln(\sin x/x)}{x^2}.$$

Its derivative quotient simplifies to

$$\frac{\cot x - 1/x}{2x} = \frac{x \cos x - \sin x}{x^3} \frac{x}{2 \sin x}.$$

To evaluate the first factor, its numerator and denominator both tend to zero, and

$$\frac{(x \cos x - \sin x)'}{(x^3)'} = \frac{-x \sin x}{3x^2} \longrightarrow -\frac{1}{3}.$$

The denominator derivative $3x^2$ *is nonzero off zero, so this factor tends to* $-1/3$ *by the rule on each side. The second factor tends to* $1/2$*. Now the derivative quotient for* L *has limit* $-1/6$*; its denominator derivative* $2x$ *is also nonzero on each side. One more justified application gives* $L(x) \to -1/6$*. Exponentiating gives the requested limit* $e^{-1/6}$*.*

Solution ***Method 2: reuse the known residual.*** *The derivative of* $\ln u$ *at* $u = 1$ *gives* $\ln u/(u - 1) \to 1$*. With that quotient extended to have value* 1 *at* $u = 1$*, we may write*

$$L(x) = \frac{\ln u(x)}{u(x) - 1} \frac{\sin x - x}{x^3}.$$

The second factor tends to $-1/6$ *by Example 4.22. Thus* $L(x) \to -1/6$ *and the original power again tends to* $e^{-1/6}$*.*

The first route derives the cancellation limit while computing the logarithm. The second is shorter when the small change in the base has already been measured. In both routes, the final exponential is taken only after the logarithmic limit is established.

To describe the leading size of an integral, it is often enough to find a simpler function whose ratio to it

tends to one. This extends the notation for equivalent infinitesimals to general scales. For a specified approach, we write $F(x) \sim G(x)$ and call F, G *asymptotically equivalent*(渐近等价)if G is eventually nonzero and $F/G \to 1$. Equivalently, $(F-G)/G \to 0$: the error is small relative to the comparison scale, although neither function need tend to zero. For example, $x^2 + x \sim x^2$ as $x \to +\infty$, since their ratio is $1 + 1/x \to 1$, even though their absolute difference tends to infinity.

Example 4.28 An integral whose antiderivative is not needed Let $F(x) = \int_0^x e^{t^2}\,\mathrm{d}t$ for $x > 0$. Show that

$$F(x) \sim \frac{e^{x^2}}{2x} \qquad (x \to +\infty).$$

Solution *The FTC gives $F'(x) = e^{x^2}$. The comparison $e^{x^2}/(2x) \to +\infty$ follows from the exponential growth estimate with $u = x^2$. To differentiate the comparison, write it as $\frac{1}{2}x^{-1}e^{x^2}$ and use the product rule:*

$$\left(\frac{e^{x^2}}{2x}\right)' = -\frac{e^{x^2}}{2x^2} + \frac{2xe^{x^2}}{2x} = e^{x^2}\left(1 - \frac{1}{2x^2}\right).$$

This derivative is positive for $x > 1$. Consequently the derivative quotient tends to 1:

$$\frac{F'(x)}{\left(e^{x^2}/(2x)\right)'} = \frac{1}{1 - 1/(2x^2)} \longrightarrow 1.$$

The infinite-denominator rule proves the equivalence. Also $F(x) \geq x \to +\infty$, although the theorem does not require this separate check. An asymptotic formula has been obtained without finding an elementary antiderivative.

Example 4.29 Both endpoints of an integral move For $x > 0$, let $I(x) = \int_x^{2x} e^{-t^2}\,\mathrm{d}t$. Prove that

$$I(x) \sim \frac{e^{-x^2}}{2x} \qquad (x \to +\infty).$$

> ***Strategy.*** The interval becomes longer, but its integrand becomes much smaller. An upper bound must first determine which effect wins before the quotient can be classified as $0/0$. The FTC then replaces the integral by contributions from its two moving endpoints; the faster exponential decay at $2x$ will disappear in the ratio.

Solution *For $x \leq t \leq 2x$ we have $0 < e^{-t^2} \leq e^{-x^2}$, so*

$$0 < I(x) \leq xe^{-x^2} \longrightarrow 0$$

by Example 4.24, with $u = x^2$. Also $D(x) = e^{-x^2}/(2x) \to 0$. Set $A(u) = \int_0^u e^{-t^2}\,\mathrm{d}t$. Then $I(x) = A(2x) - A(x)$, and the FTC and chain rule give

$$I'(x) = 2e^{-4x^2} - e^{-x^2}, \qquad D'(x) = -e^{-x^2}\left(1 + \frac{1}{2x^2}\right) < 0.$$

Both functions are differentiable for $x > 0$, and

$$\frac{I'(x)}{D'(x)} = \frac{1 - 2e^{-3x^2}}{1 + 1/(2x^2)} \longrightarrow 1.$$

The zero-over-zero rule at infinity therefore gives $I(x)/D(x) \to 1$. Every integral here has finite endpoints; no improper-integral theorem is needed.

4.3.5 Methods and further remarks

Begin by finding the form using ordinary limit laws. Use a direct identity, a known equivalent, or a bound if it already resolves the problem. If L'Hôpital's rule is appropriate, state the approach and verify differentiability, the nonzero denominator derivative, and the derivative-ratio limit. For a repeated application, check those conditions again on the new quotient. Stop as soon as another elementary limit law applies. For a positive power

with base approaching one, first ask whether the difference of the base from one has already been estimated. For an integral, use a bound to establish the form, then apply the FTC with every endpoint contribution and its chain-rule factor.

An indeterminate form is necessary for the zero-over-zero version, but by itself does not guarantee the derivative ratio exists. For example, differentiating $(1+x)/x$ as $x \to 0^+$ would produce 1, although the original quotient tends to $+\infty$. The original numerator did not tend to zero, and the denominator did not tend to infinity. Conversely, Example 4.23 shows that a missing derivative-ratio limit gives no verdict on the original quotient.

For a sequence indexed by integers, differentiation is not directly defined. One may instead construct a differentiable function of a real variable, prove its limit, and then restrict to integer inputs. Such an extension must actually agree with the sequence; it is not an automatic extra operation on sequences.

> ***Looking Ahead.*** Repeated cancellations often force repeated differentiation. Taylor's formula will organize the same local information into a polynomial with a controlled remainder. It will also explain how to choose the needed order before doing the calculation. None of the proofs in this section assumes that formula.

Exercise 4.3

1. For each expression as $x \to 0^+$, identify its form and evaluate the limit without L'Hôpital if possible: $\sin x/x$, $(1+x)/x$, $x/(1+x)$, $x\ln x$, and $1/x - 1/\sin x$.
2. Explain which hypothesis is missing in the false computation below, and find the correct limit:
$$\lim_{x\to 0}\frac{1+x^2}{2+x^2} = \lim_{x\to 0}\frac{2x}{2x} = 1.$$
3. Evaluate $\lim_{x\to 0}(e^{3x}-1)/\sin(2x)$ and $\lim_{x\to 0}\arctan(2x)/\ln(1+3x)$, checking the domains.
4. Evaluate $\lim_{x\to 0}(\tan x - x)/x^3$. Simplify a derivative quotient before deciding whether to differentiate again. Also evaluate $\lim_{x\to 0}(x-\arctan x)/x^3$ and $\lim_{x\to 0}(\arcsin x - x)/x^3$, stating a common neighborhood on which the expressions are defined.
5. Evaluate $\lim_{x\to 0}(e^x - 1 - x)/x^2$ and $\lim_{x\to 0}(e^x - 1 - x - x^2/2)/x^3$.
6. Evaluate the four limits
$$\lim_{x\to+\infty}\frac{(\ln x)^2}{x}, \quad \lim_{x\to+\infty}\frac{x^3}{e^{2x}},$$
$$\lim_{x\to+\infty}\frac{\ln\ln x}{\ln x}, \quad \lim_{x\to+\infty}\frac{(\ln x)^3}{\sqrt{x}}.$$
Explain how the substitution $u = \ln x$ simplifies the growth comparison.
7. Evaluate $\lim_{x\to 0^+} x^a \ln x$ for every real $a > 0$. Explain why the restriction on a matters.
8. Evaluate $\lim_{x\to 1}(x^a - 1)/(x^b - 1)$ for real a and $b \neq 0$, with $x > 0$. Include the case $a = 0$.
9. Evaluate $\lim_{x\to 0}(1/\sin x - 1/x)$ and $\lim_{x\to 0}(1/x - 1/(e^x - 1))$. Explain why each difference must first be combined.
10. Evaluate $\lim_{x\to 0}(\cos x)^{1/x^2}$ and $\lim_{x\to+\infty}(1+2/x)^{3x}$. State where the bases are positive.
11. Evaluate $\lim_{x\to 0^+}(\sin x)^x$ and $\lim_{x\to 0^+} x^{\sin x}$. In addition, evaluate the following two-sided limits at zero, checking positivity of the bases:
$$\lim_{x\to 0}\left(\frac{e^x-1}{x}\right)^{1/x}, \qquad \lim_{x\to 0}(1+x^2)^{1/[x\ln(1+x)]}.$$

12. Determine the right- and left-hand limits of $\ln(1+x)/x^2$ at 0. Does the two-sided limit exist in the extended real sense?
13. For $f(x) = x^2\sin(1/x)$ and $g(x) = x$ when $x \neq 0$, determine $\lim_{x\to 0} f(x)/g(x)$. Prove that $f'(x)/g'(x)$ has no limit, and explain the logical significance.
14. Evaluate $\lim_{x\to+\infty}(\sqrt{x^2+x} - x)$ in two ways: rationalization and a suitable quotient for L'Hôpital.
15. Find the real a for which $\lim_{x\to 0}[\ln(1+x) - ax]/x^2$ is finite, and find that limit. Describe what happens for every other a on each side.
16. Let f be continuous at 0, differentiable for $0 < |x| < \delta$, and suppose $f'(x) \to L \in \mathbb{R}$ as $x \to 0$. Prove that $f'(0)$ exists and equals L. Is continuity at 0 dispensable?
17. Let $F(x) = \int_0^x \cos(t^2)\,dt$. Evaluate $\lim_{x\to 0}[F(x) - x]/x^5$ using the FTC and current limit tools.
18. Suppose f is differentiable for $x > 1$ and $f'(x) \to \ell \in \mathbb{R}$ as $x \to +\infty$. Prove $f(x)/x \to \ell$ without assuming that $f(x)$ has a limit.
19. Give an example of differentiable f, g on $(0,1)$ such that $f, g \to 0$ at 0^+ but f/g has no limit. For your example, identify the precise failed hypothesis of Theorem 4.6.
20. Let $a > 0$ and define a real-variable extension of $n[(1+a/n)^{1/a} - 1]$, $n \geq 1$. Use it to determine the sequence limit and explain why differentiation was applied to a function, not to the integer index.

Supplement Exercises

S1. Prove, for $p > 0$, that $\int_1^x t^p \ln t\,dt \sim x^{p+1}\ln x/(p+1)$ as $x \to +\infty$, without first evaluating the integral.
S2. Suppose f is differentiable on $(A, +\infty)$ and $f(x) \to \ell \in \mathbb{R}$. Prove that if $f'(x)$ has an extended real limit, that limit is 0. Give an example showing that the existence of the limit of f' cannot be dropped.
S3. Let f be twice differentiable at a in the recursive sense of Section 2.4. Prove, without Taylor's formula,
$$\lim_{h\to 0} \frac{f(a+h) + f(a-h) - 2f(a)}{h^2} = f''(a).$$
Explain why your proof does not require f'' at points other than a.
S4. Suppose f, g are differentiable on $(0, \delta)$, tend to zero at 0^+, and $g' \neq 0$ there. If $m \leq f'/g' \leq M$, prove $m \leq f/g \leq M$. Use this to obtain a quotient bound even when f'/g' does not converge.
S5. For fixed $a, b \in \mathbb{R}$, evaluate $\lim_{x\to 0}[(1+ax)e^{-ax}]^{b/x^2}$, restricting to a neighborhood where $1 + ax > 0$. Include $a = 0$ and negative b.
S6. Let f be continuous at 0, differentiable on $(0, \delta)$, and $f(0) = 0$. Suppose $f'(x) \to +\infty$ as $x \to 0^+$. Prove $f(x)/x \to +\infty$ and explain why this does not give a finite right derivative. Give a concrete example.
S7. Let f, g be differentiable near 0, with $f(x) \to c \neq 0$, $g(x) \to 0$, $g'(x) \neq 0$ for $0 < x < \delta$, and $f'(x)/g'(x) \to L \in \mathbb{R}$. Prove $(f(x) - c)/g(x) \to L$ as $x \to 0^+$. Explain why this conclusion cannot generally be substituted for a limit of f/g.
S8. For $a, b > 0$, evaluate $\lim_{x\to+\infty} \ln(1+x^a)/\ln(1+x^b)$ in two ways: L'Hôpital and factoring out the dominant power inside each logarithm.
S9. Suppose f is differentiable on $(0, +\infty)$ and $f'(x)/f(x) \to a > 0$ as $x \to +\infty$, with $f(x) > 0$. Prove $\ln f(x)/x \to a$, and then prove $f(x) \to +\infty$. Is $f(x)/e^{ax} \to 1$ forced? Give a counterexample if not.
S10. Let $F(x) = \int_0^x e^{t^2}\,dt$ for $x > 0$. Prove
$$\lim_{x\to+\infty} \frac{\ln F(x)}{x^2} = 1, \qquad \lim_{x\to+\infty} F(x)^{1/x^2} = e.$$
Give one solution using Example 4.28, and another by applying L'Hôpital twice after taking logarithms.

S11. For a fixed $c > 0$, set $J_c(x) = \int_x^{x+c/x} e^{-t^2}\,\mathrm{d}t$ for $x > 0$. Determine

$$\lim_{x\to+\infty} \frac{J_c(x)}{e^{-x^2}/(2x)}.$$

Compare the answer with Example 4.29, and explain why the moving upper endpoint changes the leading coefficient.

S12. Let $m, n \geq 1$ be integers, $f \in C^m$ and $g \in C^n$ near 0. Suppose $f^{(j)}(0) = 0$ for $0 \leq j < m$ and $g^{(j)}(0) = 0$ for $0 \leq j < n$, while $A = f^{(m)}(0) \neq 0$ and $B = g^{(n)}(0) \neq 0$. Use repeated L'Hôpital, without Taylor's formula, to prove that $g(x) \neq 0$ for sufficiently small $x \neq 0$ and

$$\frac{f(x)}{g(x)} \sim \frac{A\,n!}{B\,m!} x^{m-n} \qquad (x \to 0).$$

Determine the two-sided limit when $m \geq n$, and both one-sided limits when $m < n$. In the latter case, state precisely when a two-sided extended real limit exists.

4.4 Taylor's Formula

知识点概要

- *Build a local polynomial by matching derivatives at a chosen center.*
- *Distinguish a small-order remainder from a numerical error bound.*
- *Combine finite expansions without losing the required order.*
- *Use remainder formulas to certify approximations and understand their limits.*

The derivative replaces a function near a by its tangent line:

$$f(a+h) = f(a) + f'(a)h + o(h).$$

When the linear error matters, a quadratic term may capture the next part of the change. Taylor's formula (泰勒公式)continues this idea, matching more derivatives while keeping an explicit statement about the error. The polynomial and the remainder belong together: a polynomial by itself says neither how close it is nor on what interval it is useful.

4.4.1 From a tangent line to a local polynomial

Write $P(a+h) = c_0 + c_1 h + \cdots + c_n h^n$. The requirement $P^{(k)}(a) = f^{(k)}(a)$ gives $k!c_k = f^{(k)}(a)$. Thus derivative matching forces every coefficient; there is no free choice once the center and order are fixed.

Definition 4.2 (Taylor Polynomial and Remainder)

Let $n \geq 0$ be an integer, and suppose f has derivatives through order n at a. Its Taylor polynomial of order n at a (n 阶泰勒多项式) *is*

$$T_{n,a}(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k, \qquad f^{(0)} = f.$$

The difference $R_{n,a}(x) = f(x) - T_{n,a}(x)$ is its remainder (余项). *The center $a = 0$ gives the* Maclaurin polynomial (麦克劳林多项式).

♣

The order is the highest power allowed; the actual degree can be smaller when a coefficient vanishes. For instance, the order-4 polynomial for an odd function at 0 may have degree 3. The identity $f = T_{n,a} + R_{n,a}$ is true by definition. The content of a Taylor theorem is a useful description of $R_{n,a}$ under stated differentiability assumptions.

Example 4.30 Matching successive features of the exponential Find the Taylor polynomials of orders 0, 1 and 2 for e^x at 0. Explain what information each retains.

Solution *Since every derivative of e^x equals e^x, its value and all its derivatives at 0 are 1. The polynomials are 1, $1+x$, and $1+x+x^2/2$. The first matches the value, the second also matches the slope, and the third also matches the second derivative. The factor $1/2$ is necessary because $(cx^2)'' = 2c$. The theorem below will give respective errors $o(1)$, $o(x)$ and $o(x^2)$. These are local order statements, not yet error bounds at a specified nonzero input.*

4.4.2 Taylor's formula with Peano remainder

Recall from Section 2.3 that $r(h) = o(h^n)$ means $r(h)/h^n \to 0$ as $h \to 0$, with $h \neq 0$. Equivalently, for every $\varepsilon > 0$, sufficiently small $|h|$ gives $|r(h)| \leq \varepsilon |h|^n$. The neighborhood may depend on ε. This statement does not supply a numerical constant or an explicit neighborhood by itself.

Theorem 4.8 (Taylor's Formula with Peano Remainder)

Let $n \geq 1$ be an integer. Suppose f has derivatives through order $n-1$ on an open interval about a, and $f^{(n-1)}$ is differentiable at a. For $n = 1$, this last condition simply says that f is differentiable at a. Then

$$f(a+h) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!} h^k + o(h^n) \qquad (h \to 0).$$

The term $o(h^n)$ is called the Peano remainder(佩亚诺余项).

The theorem requires the nth derivative at the center, but not at every nearby point, and does not require its continuity. The lower derivatives must exist nearby so that the last derivative at a has its recursive meaning. For $n = 0$, the corresponding formula $f(a+h) = f(a) + o(1)$ is exactly continuity at a.

> ***Proof Strategy.*** Subtract the polynomial whose derivatives have been chosen to cancel the low-order behavior. Differentiating this residual lowers both the desired order and the number of available derivatives by one. An induction therefore reduces the remainder limit to the derivative definition. L'Hôpital's rule transfers that lower-order limit back to the original residual; it is not being used to differentiate an already known little-o identity.

Proof For $n = 1$, the assertion is the definition of $f'(a)$. Suppose $n \geq 2$ and the assertion holds for order $n-1$. Set $r(h) = f(a+h) - T_{n,a}(a+h)$. The induction hypothesis applies to f': it has derivatives through order $n-2$ nearby, and its derivative of order $n-1$ exists at a. Consequently

$$r'(h) = f'(a+h) - \sum_{k=1}^{n} \frac{f^{(k)}(a)}{(k-1)!} h^{k-1} = o(h^{n-1}).$$

Here r is differentiable for small nonzero h, and $r(h) \to 0$ by continuity of f at a. The derivative quotient for $r(h)/h^n$ is

$$\frac{r'(h)}{nh^{n-1}} \longrightarrow 0.$$

The denominator derivative is nonzero for $h \neq 0$. Applying Theorem 4.6 on each side of zero yields $r(h)/h^n \to 0$. This completes the induction. □

The coefficients can also be recognized from the approximation property alone. If two polynomials P, Q of degree at most n both approximate $f(a+h)$ with error $o(h^n)$, then $P(h) - Q(h) = o(h^n)$. If their difference were nonzero, let $c_m h^m$ be its lowest nonzero term. Dividing by h^m would give a nonzero limit c_m, whereas $o(h^n)/h^m \to 0$ for $m \leq n$. This contradiction proves uniqueness. When the theorem's differentiability hypotheses hold, any such polynomial must therefore be the Taylor polynomial.

Remark An expansion does not, by itself, permit differentiation of its remainder. As emphasized in Section 2.4, $r(h) = o(h^n)$ need not imply $r'(h) = o(h^{n-1})$. The proof above first establishes an expansion for the actual derivative using the induction hypothesis. It never differentiates a little-o term formally. Likewise, recovering $f^{(n)}(a)$ from a coefficient requires the differentiability that identifies the expansion as Taylor's formula.

4.4.3 Computing and combining local expansions

Example 4.31 A finite collection of standard expansions Derive the following formulas as $h \to 0$, for each fixed positive integer n and fixed real α:

$$e^h = \sum_{k=0}^{n} \frac{h^k}{k!} + o(h^n),$$

$$\sin h = \sum_{\substack{j\geq 0 \\ 2j+1\leq n}} \frac{(-1)^j h^{2j+1}}{(2j+1)!} + o(h^n),$$

$$\cos h = \sum_{\substack{j\geq 0 \\ 2j\leq n}} \frac{(-1)^j h^{2j}}{(2j)!} + o(h^n),$$

$$\ln(1+h) = \sum_{k=1}^{n} \frac{(-1)^{k-1}}{k} h^k + o(h^n),$$

$$(1+h)^\alpha = \sum_{k=0}^{n} \binom{\alpha}{k} h^k + o(h^n).$$

Here $\binom{\alpha}{0} = 1$ and $\binom{\alpha}{k} = \alpha(\alpha-1)\cdots(\alpha-k+1)/k!$ for $k \geq 1$.

Solution *The derivatives of e^h are all e^h. The sine and cosine derivatives cycle with period four, giving their zero coefficients and alternating signs. For $k \geq 1$,*

$$\frac{\mathrm{d}^k}{\mathrm{d}h^k} \ln(1+h) = \frac{(-1)^{k-1}(k-1)!}{(1+h)^k},$$

and the kth derivative of $(1+h)^\alpha$ is $\alpha(\alpha-1)\cdots(\alpha-k+1)(1+h)^{\alpha-k}$. Evaluate these derivatives at 0 and apply Theorem 4.8. For the last two formulas take $h > -1$, so the logarithm and arbitrary real power are defined on an open interval about zero. These are finite formulas for fixed n; no passage to an infinite sum has been made.

Example 4.32 Changing the center Expand $\ln x$ through order 3 at $a = 2$.

Solution ***Method 1: derivatives at the center.*** *The first three derivatives are $1/x$, $-1/x^2$, and $2/x^3$. Writing $h = x - 2$ gives*

$$\ln x = \ln 2 + \frac{h}{2} - \frac{h^2}{8} + \frac{h^3}{24} + o(h^3).$$

The center is 2, so the powers are powers of $x - 2$, not powers of x.

Solution ***Method 2: reuse the expansion at zero.*** *For $|h| < 2$, $\ln(2+h) = \ln 2 + \ln(1 + h/2)$. Substituting $h/2$ into the logarithmic formula gives the same result, since $o((h/2)^3) = o(h^3)$.*

Direct derivatives work at any center where they are available. A simple change of variable is shorter when it reduces the function to a standard expansion.

Before multiplying or substituting expansions, choose the final order. For fixed polynomials and bounded functions, the ordinary rules from Section 2.3 give $O(1)o(h^n) = o(h^n)$ and $h^j o(h^k) = o(h^{j+k})$. When the requested remainder is $o(h^n)$, discard only terms that are themselves $o(h^n)$. For composition, one must also track the size of the inner variable: $u = O(h^2)$ makes $o(u^2) = o(h^4)$, provided $u \to 0$ (with the remainder set to zero at $u = 0$). Indeed the latter remainder is $u^2\varepsilon(u)$ with $\varepsilon(u) \to 0$, and u^2/h^4 is bounded.

Example 4.33 An expansion obtained by integration Find the order-5 expansion of $\arctan x$ at 0 and give an explicit bound on its error.

> ***Strategy.*** The derivative $1/(1+x^2)$ has a finite algebraic identity with an exact remainder. Integrating that identity gives both the coefficients and the error without computing five derivatives. The derivative recurrence from Section 2.4 provides an independent coefficient check.

Solution ***Method 1: a finite geometric identity and the FTC.*** *The exact identity*

$$\frac{1}{1+t^2} = 1 - t^2 + t^4 - \frac{t^6}{1+t^2}$$

gives, after integration from 0 *to* x,

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \int_0^x \frac{t^6}{1+t^2}\,\mathrm{d}t.$$

The absolute value of the final integral is at most $|x|^7/7$, *on either side of zero. It is therefore* $O(x^7)$ *and in particular* $o(x^5)$.

Solution ***Method 2: derivative matching.*** *The recurrence in Example 2.35 gives* $f'(0) = 1$, $f''(0) = 0$, $f'''(0) = -2$, $f^{(4)}(0) = 0$, *and* $f^{(5)}(0) = 24$ *for* $f = \arctan$. *Dividing by the corresponding factorials and applying Theorem 4.8 yields* $x - x^3/3 + x^5/5 + o(x^5)$.

The derivative route confirms the Taylor coefficients. The integral route is more informative here because its exact remainder gives a numerical error bound as well.

Example 4.34 Multiplication with a target order Expand $e^x \sin x$ through order 4 and find its fourth derivative at 0.

Solution ***Method 1: multiply finite expansions.*** *Use*

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + o(x^4), \qquad \sin x = x - \frac{x^3}{6} + o(x^4).$$

Collect the terms by total degree:

degree	1	2	3	4
coefficient	1	1	$\frac{1}{2} - \frac{1}{6} = \frac{1}{3}$	$\frac{1}{6} - \frac{1}{6} = 0.$

Thus their product is

$$e^x \sin x = x + x^2 + \frac{x^3}{3} + 0x^4 + o(x^4).$$

In degree 4, *the contributions* $x^3 x/6$ *and* $x(-x^3/6)$ *cancel. Every omitted polynomial term has degree at least* 5, *and products involving the displayed remainders are* $o(x^4)$ *because the other factors are bounded near zero. The function is smooth, so uniqueness of its Taylor polynomial gives* $(e^x \sin x)^{(4)}(0) = 4! \cdot 0 = 0$.

Solution ***Method 2: the Leibniz rule.*** *At* 0, *all derivatives of* e^x *equal* 1, *while the derivatives of* $\sin x$ *of orders* $0, 1, 2, 3, 4$ *have values* $0, 1, 0, -1, 0$. *The fourth derivative of the product is therefore*

$$\sum_{j=0}^{4} \binom{4}{j} (e^x)^{(4-j)}(0)(\sin x)^{(j)}(0) = 4 - 4 = 0.$$

Multiplication supplies the entire local polynomial. Leibniz is efficient when only a particular derivative is wanted and the individual derivative values have a simple pattern.

Example 4.35 Division by matching a product Find the order-5 expansion of $\tan x$ at zero in two ways.

> ***Strategy.*** Because $\cos x$ tends to a nonzero number, an expansion for $\tan x$ can be checked by multiplying it by $\cos x$ and recovering $\sin x$ to the required order. Oddness suggests trying only odd powers. An alternative is to use the exact differential identity $(\tan x)' = 1 + \tan^2 x$ to obtain the derivatives at zero.

Solution ***Method 1: determine the coefficients by multiplication.*** *Seek a polynomial* $P(x) = x + Ax^3 + Bx^5$. *The standard finite expansions give*

$$\begin{aligned}\cos x\, P(x) &= \left(1 - \frac{x^2}{2} + \frac{x^4}{24} + o(x^4)\right)(x + Ax^3 + Bx^5) \\ &= x + \left(A - \frac{1}{2}\right)x^3 + \left(B - \frac{A}{2} + \frac{1}{24}\right)x^5 + o(x^5).\end{aligned}$$

Here $P(x) = O(x)$ *raises the order of the cosine remainder to* $o(x^5)$. *To match* $\sin x = x - x^3/6 + x^5/120 + o(x^5)$, *solve*

$$A - \frac{1}{2} = -\frac{1}{6}, \qquad B - \frac{A}{2} + \frac{1}{24} = \frac{1}{120}.$$

This gives $A = 1/3$ *and* $B = 1/120 + 1/6 - 1/24 = 2/15$. *The matching proves* $\sin x - \cos x\, P(x) = o(x^5)$. *For sufficiently small* $|x|$, *we have* $\cos x \geq 1/2$, *so division by* $\cos x$ *preserves that remainder order. Therefore*

$$\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + o(x^5).$$

Solution ***Method 2: differentiate an exact identity.*** *Let* $y(x) = \tan x$, *which is smooth near zero and satisfies* $y' = 1 + y^2$. *Successive differentiations of this exact identity give*

$$\begin{aligned}y'' &= 2yy', \\ y''' &= 2(y')^2 + 2yy'', \\ y^{(4)} &= 6y'y'' + 2yy''', \\ y^{(5)} &= 6(y'')^2 + 8y'y''' + 2yy^{(4)}.\end{aligned}$$

Since $y(0) = 0$ *and* $y'(0) = 1$, *these formulas give*

$$y''(0) = 0, \quad y'''(0) = 2, \quad y^{(4)}(0) = 0, \quad y^{(5)}(0) = 16.$$

Taylor's formula with Peano remainder now gives

$$y(x) = x + \frac{2}{3!}x^3 + \frac{16}{5!}x^5 + o(x^5),$$

which is the same expansion. Only exact differential identities were differentiated, not an asymptotic remainder.

The product method avoids high derivatives and works for any quotient whose denominator stays away from zero. The second method is useful when an exact differential identity makes the derivative values easy to generate.

Example 4.36 The denominator determines the needed order Evaluate $\lim_{x\to 0}(\sin x - x\cos x)/x^3$ in two ways.

Solution ***Method 1: expand only far enough.*** *Use* $\sin x = x - x^3/6 + o(x^3)$ *and* $\cos x = 1 - x^2/2 + o(x^2)$. *Multiplication by* x *raises the second remainder to* $o(x^3)$. *Thus the numerator is* $x^3/3 + o(x^3)$, *and the limit is* $1/3$. *Keeping only the equivalents* $\sin x \sim x$ *and* $\cos x \sim 1$ *would lose the surviving cubic term.*

Solution ***Method 2: L'Hôpital followed by a basic limit.*** *The original quotient is* $0/0$, *and its denominator derivative* $3x^2$ *is nonzero off zero. The numerator derivative simplifies to* $x\sin x$. *Hence the derivative quotient is* $\sin x/(3x) \to 1/3$. *The rule on both sides proves the same result.*

Here the derivative cancellation makes the second route shorter. The Taylor route exposes the coefficient responsible for the answer and adapts directly to more complicated combinations.

Example 4.37 Composition and a high derivative Find the order-4 expansion of $f(x) = e^{\cos x}$ at zero, and compute $f^{(4)}(0)$.

> ***Strategy.*** The outer exponential must be expanded about the limiting inner value 1. Separate that constant by writing $e^{\cos x} = e\,e^{u}$, where $u = \cos x - 1 = O(x^2)$. Because each factor u contributes at least two powers of x, only the quadratic expansion in u can affect order 4.

Solution *We have*

$$u = -\frac{x^2}{2} + \frac{x^4}{24} + o(x^4), \qquad u^2 = \frac{x^4}{4} + o(x^4).$$

Since $e^u = 1 + u + u^2/2 + o(u^2)$ *and* $o(u^2) = o(x^4)$,

$$\begin{aligned} e^{\cos x} &= e\left[1 + \left(-\frac{x^2}{2} + \frac{x^4}{24}\right) + \frac{1}{2}\frac{x^4}{4}\right] + o(x^4) \\ &= e\left[1 - \frac{x^2}{2} + \left(\frac{1}{24} + \frac{1}{8}\right)x^4\right] + o(x^4) \\ &= e\left(1 - \frac{x^2}{2} + \frac{x^4}{6}\right) + o(x^4). \end{aligned}$$

The function has derivatives of every order, so the coefficient of x^4 *is* $f^{(4)}(0)/4!$. *It follows that* $f^{(4)}(0) = 4e$. *We identify a Taylor coefficient by uniqueness; we do not differentiate the remainder four times.*

4.4.4 Lagrange remainder and certified approximation

Peano's formula answers a local comparison question as $h \to 0$. To guarantee an error at a particular input, we need control over an entire interval. For the next theorem, $f \in C^n(J)$ on a closed interval J means that its derivatives through order n in the interior extend continuously to the endpoints, with the endpoint derivatives understood one-sidedly. For $n = 0$ this means continuity on J. This is the closed-interval convention corresponding to the open-interval notation of Section 2.4.

Theorem 4.9 (Taylor's Formula with Lagrange Remainder)

Let $n \geq 0$ *be an integer and* $x \neq a$. *Suppose* $f \in C^n(J)$ *on the closed interval* J *with endpoints* a, x, *and* $f^{(n+1)}$ *exists in its interior. Then there is* ξ *strictly between* a *and* x *such that*

$$f(x) = T_{n,a}(x) + \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-a)^{n+1}.$$

The final term is the Lagrange remainder(拉格朗日余项).

> ***Proof Strategy.*** The Taylor residual and its first n derivatives vanish at a, but the residual need not vanish at x. Subtract a multiple of $(t-a)^{n+1}$ chosen to create that additional zero. Repeated Rolle applications then force an $(n+1)$st derivative to vanish at an interior point. The added term survives exactly that many differentiations, producing the desired remainder coefficient.

Proof Let $K = [f(x) - T_{n,a}(x)]/(x-a)^{n+1}$ and set

$$H(t) = f(t) - T_{n,a}(t) - K(t-a)^{n+1}.$$

We have $H(x) = 0$ and $H^{(j)}(a) = 0$ for $0 \leq j \leq n$. Rolle's theorem gives a zero c_1 of H' strictly between a and x. If $n \geq 1$, $H'(a) = 0$, so another application between a and c_1 gives a zero c_2 of H''. Continue in this way. The stated continuity and differentiability assumptions justify every application, including the last one to $H^{(n)}$. There is a point ξ strictly between a and x where

$$0 = H^{(n+1)}(\xi) = f^{(n+1)}(\xi) - (n+1)!K.$$

Substitution of K proves the formula. For $n = 0$ only the first Rolle application is needed. The same construction works when $x < a$, with the interval endpoints reversed. □

For $n = 0$, this is Lagrange's mean value theorem. For higher n, the point ξ may depend on x, on n, and on the function. It is not generally the midpoint. The existence statement becomes a bound when $\left|f^{(n+1)}(t)\right| \le M$ throughout the segment:

$$|f(x) - T_{n,a}(x)| \le \frac{M}{(n+1)!}\,|x-a|^{n+1}. \tag{4.6}$$

The bound must cover every possible ξ, not just the center.

Example 4.38 A sine approximation with a certificate Approximate $\sin(1/2)$ by $P_5(1/2)$, where $P_5(x) = x - x^3/6 + x^5/120$, and give an error less than $2 \cdot 10^{-6}$.

Solution *The order-6 Taylor polynomial for sine is still P_5, because the sixth derivative at zero is zero. The seventh derivative is $-\cos t$, whose absolute value is at most 1 on $[0, 1/2]$. Using the order-6 Lagrange formula gives*

$$\left|\sin\frac{1}{2} - \left(\frac{1}{2} - \frac{1}{48} + \frac{1}{3840}\right)\right| \le \frac{(1/2)^7}{7!} = \frac{1}{645120} < 2 \cdot 10^{-6}.$$

The approximating number is $1841/3840$. Moreover $-\cos\xi < 0$ on this segment, so the polynomial is an upper bound. The improved order comes from a zero Taylor coefficient together with the order-6 theorem, not from silently replacing the remainder by the next nonzero term of an infinite series.

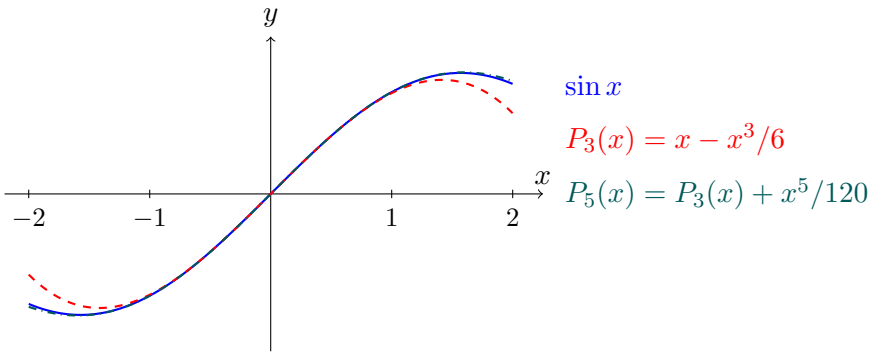


Figure 4.3: Both polynomials match sine near zero. The order-5 polynomial remains closer on this plotted interval; a guaranteed error on a specified interval comes from a remainder bound, not from the picture alone.

Example 4.39 Canceling the leading error in a central slope Let $h > 0$, suppose $f \in C^5([a-h, a+h])$, and assume $\left|f^{(5)}\right| \le M$ on this interval. For $0 < t \le h$, define the central slope

$$D_t = \frac{f(a+t) - f(a-t)}{2t}.$$

Find a linear combination of $D_{h/2}$ and D_h that cancels the leading error, and prove that

$$Q_h = \frac{4D_{h/2} - D_h}{3} \quad \text{satisfies} \quad \left|Q_h - f'(a)\right| \le \frac{Mh^4}{288}.$$

> ***Strategy.*** The symmetric difference cancels even powers in the two Taylor polynomials, leaving an error beginning with h^2 after division by $2h$. Halving h multiplies that term by $1/4$. Choose weights that preserve the derivative term and cancel this predictable error, then carry the remainder bounds through the same combination.

Solution *Apply the order-4 Lagrange formula separately at $a+h$ and $a-h$:*

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2}f''(a) + \frac{h^3}{6}f'''(a) + \frac{h^4}{24}f^{(4)}(a) + R_+,$$

$$f(a-h) = f(a) - hf'(a) + \frac{h^2}{2}f''(a) - \frac{h^3}{6}f'''(a) + \frac{h^4}{24}f^{(4)}(a) + R_-.$$

The uniform fifth-derivative bound gives $|R_\pm| \le Mh^5/120$. *Subtracting and dividing by* $2h$ *yields*

$$D_h = f'(a) + \frac{h^2}{6} f'''(a) + E_h, \qquad |E_h| = \left|\frac{R_+ - R_-}{2h}\right| \le \frac{Mh^4}{120}.$$

The same bound applies on the smaller interval, so

$$D_{h/2} = f'(a) + \frac{h^2}{24} f'''(a) + E_{h/2}, \qquad |E_{h/2}| \le \frac{Mh^4}{1920}.$$

For $\alpha D_{h/2} + \beta D_h$ *to preserve* $f'(a)$ *and cancel the displayed second-order error, we require*

$$\alpha + \beta = 1, \qquad \frac{\alpha}{4} + \beta = 0.$$

Thus $\alpha = 4/3$ *and* $\beta = -1/3$. *The remaining error obeys*

$$|Q_h - f'(a)| \le \frac{4|E_{h/2}| + |E_h|}{3} \le \frac{Mh^4}{3}\left(\frac{4}{1920} + \frac{1}{120}\right) = \frac{Mh^4}{288}.$$

This is a sufficient error certificate; no optimality of its constant is asserted.

The cancellation is the same extrapolation idea used for numerical integration in Section 3.6. Here the fifth-derivative bound makes the improved approximation quantitative; existence of $f'(a)$ alone would not justify this error estimate.

Example 4.40 Logarithmic bounds and the sign of the error Prove, for $0 \le x \le 1/2$, that

$$x - \frac{x^2}{2} \le \ln(1+x) \le x - \frac{x^2}{2} + \frac{x^3}{3}.$$

Solution ***Method 1: an exact integral remainder.*** *Integrating* $1/(1+t) = 1 - t + t^2/(1+t)$ *gives*

$$\ln(1+x) = x - \frac{x^2}{2} + \int_0^x \frac{t^2}{1+t}\,\mathrm{d}t.$$

For $0 \le t \le x$, *the integrand lies between* 0 *and* t^2. *The remainder is therefore between* 0 *and* $x^3/3$, *which proves both inequalities.*

Solution ***Method 2: Lagrange remainders.*** *For* $x > 0$, *the order-*2 *formula gives a positive remainder* $x^3/[3(1+\xi)^3]$, *because the third derivative of* $\ln(1+t)$ *is* $2/(1+t)^3$. *The order-*3 *formula gives a negative remainder* $-x^4/[4(1+\eta)^4]$. *The points* ξ, η *lie in* $(0, x)$ *and need not coincide. These signs prove the lower and upper bounds; equality holds at* $x = 0$.

The integral route supplies both bounds in one identity and in fact works for every $x \ge 0$. The derivative route applies to many functions whose antiderivative representation does not simplify so conveniently.

4.4.5 Integral remainder and the limits of finite formulas

When the next derivative is continuous, the error has an exact integral representation. This stronger hypothesis is needed for the version of the FTC and integration by parts already established in this course.

Proposition 4.3 (Integral Remainder)

Let $n \ge 0$ *be an integer. If* $f \in C^{n+1}(J)$ *on the closed interval joining* a *and* x, *then*

$$f(x) = T_{n,a}(x) + \frac{1}{n!}\int_a^x f^{(n+1)}(t)(x-t)^n\,\mathrm{d}t.$$

The integral is oriented in the usual way when $x < a$.

Proof For $n = 0$, the formula is the FTC. For $n \geq 1$, integration by parts gives

$$\frac{1}{n!}\int_a^x f^{(n+1)}(t)(x-t)^n\,\mathrm{d}t = -\frac{f^{(n)}(a)}{n!}(x-a)^n + \frac{1}{(n-1)!}\int_a^x f^{(n)}(t)(x-t)^{n-1}\,\mathrm{d}t.$$

Repeating until the final integral is $\int_a^x f'(t)\,\mathrm{d}t = f(x) - f(a)$ yields the assertion. Continuity of the derivatives and the C^1 factors justify each integration by parts on the segment. □

If $\left|f^{(n+1)}(t)\right| \leq M$ on the segment, taking absolute values in the integral gives (4.6). For $x < a$, reverse the integration limits first. The integral formula can also retain information about the sign or variation of the derivative that a single maximum bound discards. The three forms serve different purposes: Peano gives local order under weak pointwise differentiability, Lagrange gives an intermediate-point formula and error certificate, and the integral form uses stronger continuity to give an exact accumulated error.

A finite Taylor theorem keeps n fixed while $x \to a$ or while an error is estimated on a chosen segment. Letting $n \to \infty$ is a separate question: one must prove that the remainders tend to zero on the inputs in question. Existence of derivatives of every order alone does not settle it.

Example 4.41 A smooth function invisible to every Taylor polynomial Define $f(0) = 0$ and $f(x) = e^{-1/x^2}$ for $x \neq 0$. Show that f has derivatives of every order at 0, all equal to zero, although $f(x) > 0$ for $x \neq 0$.

> ***Optional Extension.*** The exponential growth estimate of Section 4.3 says that e^{-u} defeats every fixed power of u as $u \to +\infty$. Here $u = 1/x^2$. Every differentiation away from zero adds only powers of $1/x$, so the exponential factor can still force both continuity and a zero derivative at the center.

Solution *For each integer $m \geq 0$, putting $u = 1/x^2$ gives*

$$|x|^{-m}\,e^{-1/x^2} = u^{m/2}e^{-u} \longrightarrow 0.$$

Away from zero, induction with the chain and product rules shows that $f^{(k)}(x) = P_k(1/x)e^{-1/x^2}$ for a polynomial P_k. Start with $P_0 = 1$. Differentiating such an expression again gives the same form with another polynomial. Suppose the kth derivative has already been extended to zero with value 0. The displayed decay shows that this extension is continuous and that

$$\lim_{h\to 0}\frac{P_k(1/h)e^{-1/h^2} - 0}{h} = 0.$$

Thus its derivative at zero exists and is zero; away from zero it is the expression for $k+1$. Starting from continuity of f, this induction proves all assertions. Every Taylor polynomial at zero is consequently the zero polynomial, but the function is positive elsewhere.

4.4.6 Methods and further remarks

Choose the center from the limiting input or the convenient nearby value, and write $h = x - a$ before expanding. Next choose the order from the question: a denominator h^m usually requires all terms through degree m, while an absolute error requires a bound on the next derivative over the full segment. In a product, keep every pair of terms whose total degree is relevant. In a composition, determine the order of the inner change before deciding how many outer terms are needed. For a quotient, match a product and check that division by the denominator preserves the remainder order.

Subtraction is where careless equivalents are most dangerous. Two leading terms may cancel, leaving the first term that an overly short expansion omitted. For example, knowing $\sin x \sim x$ gives no cubic coefficient for $\sin x - x$. Write remainders explicitly until the final division or coefficient identification is justified.

For numerical work, record the polynomial, the interval, a uniform derivative bound, and the resulting error together. A plot helps explain local contact but cannot replace this certificate. For high derivatives, use smoothness and uniqueness to read a coefficient; do not repeatedly differentiate an asymptotic identity. When approximations at two scales have the same leading error term with known scale factors, choose weights to cancel it and combine the remainder bounds explicitly. The sign of the first nonzero Taylor term will later help study local extrema in Section 4.5; that application uses no additional term hidden in the remainder.

Exercise 4.4

1. Suppose $f(0)=2$, $f'(0)=-1$, $f''(0)=6$, and $f'''(0)=12$. Find $T_{3,0}$ and explain the factorials in its coefficients. State precise hypotheses under which $f(x)-T_{3,0}(x)=o(x^3)$.
2. Decide which of x^3, x^2, $x^2\sin x$, and $x^2\sin(1/x)$ are $o(x^2)$ as $x\to 0$. Which are $O(x^2)$? Justify each answer from the definitions.
3. Find the order-3 expansion of e^{2x} and the order-5 expansion of $\sin(3x)$ at zero.
4. Find the order-4 expansion of $\ln x$ at 1 and the order-2 expansion of $\sqrt{x}$ at 4. State a neighborhood on which each function is defined.
5. Find the order-4 expansion of $(1+x)^{-2}$ at zero in two ways: derivative matching and multiplication of two finite expansions of $(1+x)^{-1}$. Also expand $(1-2x)^{-1}$ through order 3 and $(1+x+x^2)^{-1}$ through order 5 at zero. Look for an exact algebraic identity before using repeated derivatives.
6. Expand $e^x\cos x$ through order 4 at zero and find its fourth derivative there. Check the derivative by the Leibniz rule.
7. Expand $\ln(1+x)/(1+x)$ through order 3 at zero. Keep the remainder order visible during multiplication. Also expand $(e^x-1)/x$ and $x/(e^x-1)$ through order 3, extending each function continuously at zero. Explain how division by x determines the expansion order needed for the numerator.
8. Expand $\sin(x^2)$ through order 6 at zero. Explain why an order-3 expansion of the outer sine suffices.
9. Expand $\ln(\cos x)$ through order 4 at zero, on a neighborhood where $\cos x>0$. Also expand $\ln(1+\sin x)$ and $\sqrt{1+\tan x}$ through order 3 at zero. State neighborhoods on which both compositions are real-valued and track the inner remainder order.
10. Evaluate $\lim_{x\to 0}[\ln(1+x)-\sin x+x^2/2]/x^3$. Explain why first-order equivalents do not determine this limit.
11. Evaluate $\lim_{x\to 0}(e^x-\cos x-\sin x)/x^2$ by Taylor's formula and by L'Hôpital's rule. Compare the work required.
12. Find all real a,b for which $\lim_{x\to 0}(e^x-1-ax-bx^2)/x^3$ is finite, and find its value.
13. Determine $\lim_{x\to 0}\big((1+x)^{1/x}/e-1\big)/x$. State the domain and track the order after taking logarithms.
14. Find the fourth derivative at zero of $e^{\sin x}$ by composition of finite expansions. Explain why the coefficient may be used without differentiating a little-o term.
15. Approximate $e^{1/10}$ with its order-3 polynomial at zero. Give a rigorous Lagrange error bound in terms of $e^{1/10}$, including the sign of the error.
16. Approximate $\sin(1/5)$ by $1/5-(1/5)^3/6$. Prove that the absolute error is at most $1/375000$ by choosing the appropriate Taylor order.
17. Approximate $\ln(6/5)$ by the order-3 polynomial for $\ln(1+x)$ at $x=1/5$. Prove that the polynomial is an upper bound and that its error is at most $1/2500$.

18. Let $h \neq 0$ be sufficiently small. Suppose $f \in C^4$ near a and $\left|f^{(4)}\right| \leq M$ on $[a-|h|, a+|h|]$. Prove
$$\left|\frac{f(a+h)-2f(a)+f(a-h)}{h^2} - f''(a)\right| \leq \frac{Mh^2}{12}.$$

19. For an integer $n \geq 1$, show that $r(h) = |h|^{n+1/2}$ is $o(h^n)$ but is not $O(h^{n+1})$. Explain why a Peano remainder cannot automatically be replaced by an error of the next integer order.

20. Find $T_{3,1}$ for $f(x) = x^4$ and its exact remainder. Verify the Lagrange formula, identifying a possible fourth derivative value.

21. Use a finite geometric identity and integration to prove, for $n \geq 1$ and $0 \leq x \leq 1/2$,
$$\ln(1+x) = \sum_{k=1}^{n} \frac{(-1)^{k-1}x^k}{k} + (-1)^n \int_0^x \frac{t^n}{1+t}\,\mathrm{d}t.$$
Deduce the sign and an upper bound for the absolute error.

22. For $P_3(x) = x - x^3/6$, find an explicit symmetric interval on which $|\sin x - P_3(x)| \leq 10^{-4}$ is guaranteed by a Lagrange bound. Explain why the visual closeness in **Figure** 4.3 is not itself that guarantee.

Supplement Exercises

S1. Define $r(0) = 0$ and $r(x) = x^3 \sin(1/x^2)$ for $x \neq 0$. Show that $r = o(x^2)$ and that r is differentiable everywhere, but r' is not $o(x)$. Relate this to the warning about differentiating remainders.

S2. Suppose f has derivatives through order 3 near a, and $f^{(4)}(a)$ exists. Prove
$$\lim_{h\to 0} \frac{f(a+2h)-4f(a+h)+6f(a)-4f(a-h)+f(a-2h)}{h^4} = f^{(4)}(a).$$
Identify the cancellations of lower-degree terms.

S3. Let $f \in C^2([a,b])$ and let L be the affine function agreeing with f at a, b. For each $x \in (a,b)$, prove that $f(x) - L(x) = f''(\xi)(x-a)(x-b)/2$ for some $\xi \in (a,b)$. Construct an auxiliary function with three zeros and derive a uniform error bound when $|f''| \leq M$.

S4. Suppose $f \in C^2(\mathbb{R})$, $|f| \leq M$, and $|f''| \leq N$, where $M, N > 0$. Prove $|f'| \leq \sqrt{2MN}$ by applying Taylor's formula at $x+h$ and $x-h$ and then choosing $h > 0$. Discuss separately what happens if $M = 0$ or $N = 0$.

S5. Define $g(0) = 0$ and $g(x) = e^{-1/x^2}\sin(1/x)$ for $x \neq 0$. Prove that g has derivatives of all orders at zero, all equal to zero, and has infinitely many nonzero zeros approaching zero. Explain why the argument must control difference quotients at zero, not just derivatives away from zero.

S6. Suppose f is twice differentiable at a in the recursive sense and $f''(a) \neq 0$. For each sufficiently small $h \neq 0$, choose $0 < \theta_h < 1$ such that $f(a+h) - f(a) = hf'(a+\theta_h h)$. Prove $\theta_h \to 1/2$. Show by an example that the nonzero second-derivative assumption cannot simply be omitted.

S7. Suppose f is twice differentiable at zero in the recursive sense. Prove
$$\int_0^x f(t)\,\mathrm{d}t = f(0)x + \frac{f'(0)}{2}x^2 + \frac{f''(0)}{6}x^3 + o(x^3).$$
Justify integration of the remainder by an ε bound valid for every t between 0 and x, including negative x.

S8. For $n \geq 0$, derive a finite expansion of $\arctan x$ through degree $2n+1$ with an exact integral remainder. Prove that its absolute value is at most $|x|^{2n+3}/(2n+3)$ and determine its sign for $x > 0$.

S9. Using $e < 3$, find the smallest n for which the Lagrange bound $3/(n+1)!$ certifies that the order-n Taylor polynomial for e^x approximates it within 10^{-8} for every $x \in [-1,1]$. Explain the difference between a sufficient certified degree and a claim that no lower-degree polynomial could attain that accuracy.

S10. Suppose f has derivatives through order 2 near a, and $f'''(a)$ exists. Prove

$$\lim_{h\to 0}\frac{f(a+h)-f(a)-hf'(a+h/2)}{h^3}=\frac{f'''(a)}{24}.$$

Use separate valid expansions for f and f'; do not differentiate a remainder.

S11. Find the order-5 expansion of $\arcsin x$ at zero in two ways. First use $\sin(\arcsin x)=x$, oddness, and uniqueness of the finite Taylor polynomial to determine its coefficients. Then independently integrate an expansion of $(1-x^2)^{-1/2}$ from 0 to x. Justify smoothness and the treatment of the remainder in each approach.

S12. Let $f\in C^2$ near a, and set $w(t)=6t^2-6t+1$. Prove that

$$\lim_{h\to 0}\frac{1}{h^2}\int_0^1 w(t)f(a+ht)\,\mathrm{d}t=\frac{f''(a)}{60}.$$

Compute the three moments $\int_0^1 t^j w(t)\,\mathrm{d}t$ for $j=0,1,2$ to identify the cancellations. Justify the integral remainder estimate uniformly for $0\le t\le 1$, including negative h.

4.5 Monotonicity and Extrema

知识点概要

- ❑ *Convert derivative information on an interval into comparisons of function values.*
- ❑ *Distinguish stationary points, local extrema, and absolute extrema.*
- ❑ *Choose a first-, second-, or higher-order test and recognize its limitations.*
- ❑ *Solve optimization problems by checking the feasible domain and all candidates.*

A derivative describes change near a point, whereas an optimization problem compares values at many points. The mean value theorem connects these two kinds of information. Taylor's formula gives a second connection: the first nonzero term of a local expansion can determine whether nearby values lie above or below a given value. We now organize these tools into methods for studying functions and finding their extrema.

4.5.1 Derivative signs on an interval

Recall from Definition 1.10 that a nondecreasing function satisfies $f(x)\le f(y)$ whenever $x<y$ in its interval of definition. Strict increase requires the strict inequality for every such pair. The distinction matters even when the function is differentiable.

Theorem 4.10

Let I be an open interval and let $f\colon I\to\mathbb{R}$ be differentiable. Then f is nondecreasing on I if and only if $f'(x)\ge 0$ for every $x\in I$. Similarly, f is nonincreasing if and only if $f'\le 0$ on I. If $f'>0$ everywhere on I, then f is strictly increasing; if $f'<0$ everywhere, it is strictly decreasing. The last two sufficient conditions are not necessary. ♡

> ***Proof Strategy.*** To compare arbitrary values $f(x)$ and $f(y)$ with $x<y$, use the mean value theorem on the interval joining those two points. Conversely, monotonicity determines the sign of every difference quotient, including those with negative increments. Taking a limit preserves a non-strict inequality, but need not preserve a strict one.

Proof If $f' \geq 0$, then for $x < y$ the mean value theorem gives

$$f(y) - f(x) = f'(\xi)(y - x) \geq 0$$

for some $\xi \in (x, y)$. The same argument gives strict positivity when $f' > 0$. Conversely, if f is nondecreasing, then $(f(x + h) - f(x))/h \geq 0$ for every sufficiently small nonzero h. Its limit is $f'(x) \geq 0$. Apply these arguments to $-f$ for the decreasing cases. The example below disproves necessity of strict positivity. □

The result also controls a closed or half-open interval when f is continuous at each included endpoint and differentiable in its interior. Indeed, one may apply the mean value theorem directly between any two points of that interval. No derivative at an included endpoint is required.

Example 4.42 Strict increase with a horizontal tangent Show that $f(x) = x^3$ is strictly increasing on $\mathbb{R}$, even though $f'(0) = 0$.

Solution ***Method 1: compare values.*** *If $x < y$, then*

$$y^3 - x^3 = (y - x)(x^2 + xy + y^2) > 0,$$

since $x^2 + xy + y^2 = (x + y/2)^2 + 3y^2/4$ vanishes only when $x = y = 0$.

Solution ***Method 2: use interval information.*** *The derivative $3x^2$ is positive on each side of 0. Thus f is strictly increasing on each half-line, including 0 by continuity. If $x < 0 < y$, then $x^3 < 0 < y^3$, which joins the two comparisons.*

The first method proves the claim directly from its definition. The second shows why one horizontal tangent does not create a constant segment. By contrast, a constant function has nonnegative derivative but is not strictly increasing.

Proposition 4.4

A differentiable function on an open interval is strictly increasing if and only if its derivative is nonnegative everywhere and the function is constant on no nondegenerate subinterval. ♠

Proof Necessity follows from the preceding theorem and the definition of strict increase. For sufficiency, f is nondecreasing. If $x < y$ and $f(x) = f(y)$, then $f(x) \leq f(t) \leq f(y)$ for every $t \in [x, y]$, making f constant on $[x, y]$. This contradicts the hypothesis. □

Example 4.43 Why the interval cannot be omitted The function $f(x) = -1/x$ has derivative $1/x^2 > 0$ wherever it is defined. It is strictly increasing on each of $(-\infty, 0)$ and $(0, +\infty)$. It is not increasing on their union: $-1 < 1$, but $f(-1) = 1 > f(1) = -1$. There is no interval in the domain joining these two points, so the mean value theorem cannot compare them.

Example 4.44 Sharper bounds from monotone differences Prove that, for $x \geq 1$,

$$\frac{2(x-1)}{x+1} \leq \ln x \leq \frac{1}{2}\left(x - \frac{1}{x}\right),$$

with both inequalities strict when $x > 1$. Compare them with the bounds in Example 4.6.

> ***Strategy.*** Both proposed bounds agree with $\ln x$ at 1. Subtract each bound from the logarithm in the appropriate order; an increasing difference would then be positive to the right of 1. For a second route, the repeated combination $(x-1)/(x+1)$ suggests using it as a new variable in the logarithmic integral.

Solution ***Method 1: compare derivatives of the gaps.*** *Set*

$$L(x) = \ln x - \frac{2(x-1)}{x+1}, \qquad U(x) = \frac{1}{2}\left(x - \frac{1}{x}\right) - \ln x.$$

Both functions vanish at 1*, and*

$$L'(x) = \frac{1}{x} - \frac{4}{(x+1)^2} = \frac{(x-1)^2}{x(x+1)^2},$$
$$U'(x) = \frac{1}{2} + \frac{1}{2x^2} - \frac{1}{x} = \frac{(x-1)^2}{2x^2}.$$

The denominators are positive, and both derivatives are positive for $x > 1$*. Hence* $L(x) > L(1) = 0$ *and* $U(x) > U(1) = 0$ *there. At* $x = 1$ *both desired inequalities are equalities.*

Solution ***Method 2: bound an integral after substitution.*** *Suppose* $x > 1$ *and put* $u = (x-1)/(x+1)$*, so* $0 < u < 1$*. In* $\ln x = \int_1^x \mathrm{d}s/s$*, use* $s = (1+t)/(1-t)$*. Since* $\mathrm{d}s/s = 2\,\mathrm{d}t/(1-t^2)$*, we obtain*

$$\ln x = 2\int_0^u \frac{\mathrm{d}t}{1-t^2}.$$

For $0 < t < u$*, the integrand lies strictly between* 1 *and* $1/(1-u^2)$*. Its continuity makes the integrated inequalities strict, giving*

$$2u < \ln x < \frac{2u}{1-u^2} = \frac{(x-1)(x+1)}{2x} = \frac{1}{2}\left(x - \frac{1}{x}\right).$$

This is the substitution used for $\ln 2$ *in Example 3.70, now with a variable endpoint. When* $x = 1$*, both bounds and* $\ln x$ *equal* 0*, completing this proof as well.*

The derivative method starts from proposed bounds and verifies their gaps. The integral method explains how one substitution produces both bounds at once. The earlier mean value estimates, with their variable replaced by $x-1$, are $(x-1)/x < \ln x < x-1$. For $x > 1$, both new bounds are closer to the logarithm, since

$$\frac{2(x-1)}{x+1} - \frac{x-1}{x} = \frac{(x-1)^2}{x(x+1)} > 0, \qquad (x-1) - \frac{1}{2}\left(x - \frac{1}{x}\right) = \frac{(x-1)^2}{2x} > 0.$$

4.5.2 Local extrema and candidate points

Absolute extrema compare a value with every value on a specified domain, as in Definition 1.28. A local comparison uses only nearby points. Recall from Definition 4.1 that a *local extremum* refers to an interior point c of the domain: for some $\delta > 0$, either $f(x) \le f(c)$ for all $|x-c| < \delta$ (a local maximum), or the reversed inequality holds (a local minimum). The extremum is *strict* if the inequality is strict whenever $x \neq c$. Endpoints will be included in the search for absolute extrema, but will not be called local extrema under this convention.

An interior point c with $f'(c) = 0$ is a *stationary point*. A *critical point*(临界点)is an interior point of the domain where either $f'(c) = 0$ or $f'(c)$ does not exist. A point outside the domain is neither kind of point. By Theorem 4.1, every differentiable local extremum is stationary. Thus every local extremum is a critical point, but a critical point is only a candidate.

Example 4.45 Local and absolute comparisons in the same problem For $f(x) = (x^2-1)^2$ on $[-2,2]$, identify the stationary points and determine which of them give local or absolute extrema. Also check the endpoints.

Solution *Since* $f'(x) = 4x(x-1)(x+1)$*, the stationary points are* $-1, 0, 1$*. At* ± 1 *the value is* 0*, and* $f(x) > 0$ *nearby except at the point itself; these are strict local minima and absolute minima. For* $0 < |x| < 1$*, we have* $(1-x^2)^2 < 1 = f(0)$*, so* 0 *is a strict local maximum. It is not an absolute maximum, because* $f(\pm 2) = 9$*. On* $[-2,2]$*,* $0 \le x^2 \le 4$*, so* $|x^2 - 1| \le 3$*; hence* 9 *is indeed the absolute maximum, attained at both endpoints.*

The word "local" changes the set of values being compared. The familiar examples x^3 and $|x|$ from the discussion of Fermat's theorem still explain why neither a zero derivative nor differentiability is a complete test for an extremum.

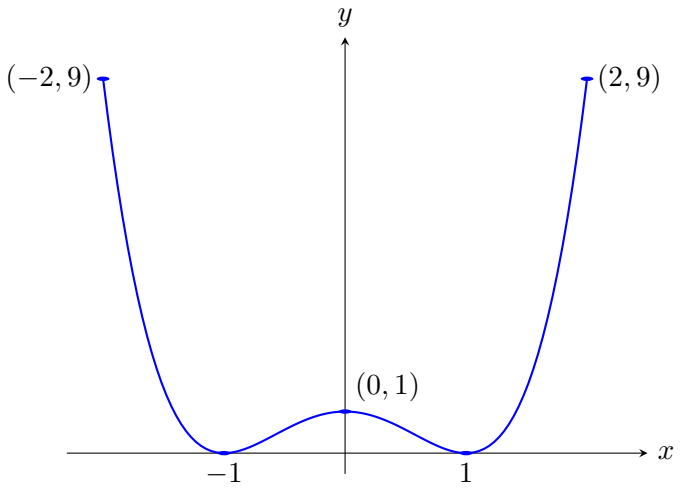


Figure 4.4: A local maximum can be lower than endpoint values. The two interior minima are also absolute minima.

4.5.3 Tests for local extrema

Theorem 4.11 (First derivative test)

Suppose f is continuous on $(c-\delta, c+\delta)$ and differentiable there except possibly at c. If $f' < 0$ on $(c-\delta, c)$ and $f' > 0$ on $(c, c+\delta)$, then c is a strict local minimum. Reversing the signs gives a strict local maximum. If f' is positive on both intervals, or negative on both intervals, then c is not a local extremum.

♡

Proof In the minimum case, the monotonicity theorem and continuity at c give $f(x) > f(c)$ on both sides. The maximum case follows by applying this to $-f$. If both signs are positive, values immediately to the left are smaller than $f(c)$ and values immediately to the right are larger; the negative case reverses these comparisons. □

The test needs sign information on whole neighboring intervals. It makes no assertion when the derivative oscillates in every such interval. A single value of $f'(c)$ cannot replace interval information.

Example 4.46 A singular point must remain on the candidate list Determine the local and absolute extrema of $f(x) = x^{2/3}(x-2)$ on $[-1, 3]$, where $x^{2/3} = (\sqrt[3]{x})^2$.

> ***Strategy.*** Fractional powers are continuous at 0 even when their derivative is not. Separate the zero of the derivative's numerator from the point where its denominator vanishes; both can affect the graph.

Solution *For $x \neq 0$,*

$$f'(x) = \frac{2}{3}x^{-1/3}(x-2) + x^{2/3} = \frac{2(x-2)+3x}{3x^{1/3}} = \frac{5x-4}{3x^{1/3}}.$$

At 0, the difference quotient is

$$\frac{f(h)-f(0)}{h} = h^{-1/3}(h-2),$$

which tends to $+\infty$ from the left and $-\infty$ from the right. Thus 0 is a critical point even though it is not a zero of the numerator in the derivative formula. The signs are

x	$(-1,0)$	$(0,4/5)$	$(4/5,3)$
$5x-4$	$-$	$-$	$+$
$3x^{1/3}$	$-$	$+$	$+$
$f'(x)$	$+$	$-$	$+$

Thus 0 is a strict local maximum and $4/5$ is a strict local minimum. The candidates for absolute extrema, with their values, are

$$f(-1)=-3, \qquad f(0)=0, \qquad f(4/5)=-\frac{6}{5}\left(\frac{4}{5}\right)^{2/3}, \qquad f(3)=3^{2/3}.$$

Since $-6/5 < f(4/5) < 0$, the absolute minimum is -3 at -1, and the absolute maximum is $3^{2/3}$ at 3. Continuity at 0 justifies joining the two derivative-sign intervals there.

Theorem 4.12 (Second derivative test)

Suppose f' exists in a neighborhood of c, $f'(c)=0$, and $f''(c)$ exists. If $f''(c)>0$, then c is a strict local minimum; if $f''(c)<0$, it is a strict local maximum. If $f''(c)=0$, this test gives no conclusion. ♡

Proof If $f''(c)>0$, then

$$\frac{f'(c+h)-f'(c)}{h}>0$$

for all sufficiently small nonzero h. Since $f'(c)=0$, this says $f'(c+h)<0$ on the left and $f'(c+h)>0$ on the right. Apply the first derivative test. Replace f by $-f$ for the other case. □

This proof does not assume that f'' is continuous near c. It uses the derivative of f' at the one point c.

More generally, suppose Taylor's formula gives

$$f(c+h)-f(c)=Ah^m+o(h^m), \qquad A\neq 0.$$

For sufficiently small $h\neq 0$, the coefficient $A+o(1)$ has the same sign as A. If m is even, c is a strict local minimum for $A>0$ and a strict local maximum for $A<0$. If m is odd, the difference changes sign across c, so there is no extremum. This is the *higher derivative test*(高阶导数判别法)when $A=f^{(m)}(c)/m!$ and all lower positive-order derivatives vanish. It requires a valid Taylor expansion; a formal polynomial without a controlled remainder is not enough.

Example 4.47 When the second derivative test is inconclusive Classify 0 for x^4, $-x^4$, and x^5. Then show that e^x-1-x has a strict minimum at 0.

Solution *For all three powers, the first two derivatives vanish at 0. Their exact leading terms show, respectively, a strict minimum, a strict maximum, and no extremum. For the exponential expression, Taylor's formula gives*

$$e^x-1-x=\frac{x^2}{2}+o(x^2),$$

so 0 is a strict local minimum.

Solution ***A global alternative for the exponential expression.*** *Its derivative is e^x-1, negative for $x<0$ and positive for $x>0$. It therefore decreases up to 0 and increases thereafter. Its minimum 0 is absolute on $\mathbb{R}$.*

Taylor's formula answers a local question efficiently. The derivative signs answer the stronger global question in this example.

Example 4.48 A parameter changes the critical-point pattern For each real a, classify the local and absolute extrema of $f_a(x)=x^4-2ax^2$ on $\mathbb{R}$.

Solution ***Method 1: derivative signs.*** *We have $f_a'(x) = 4x(x^2 - a)$. If $a < 0$, its sign is the sign of x, so 0 is the unique absolute minimum. If $a = 0$, the same conclusion holds, although the second derivative at 0 vanishes. In these two cases the minimum value is $f_a(0) = 0$. If $a > 0$, the critical points are $-\sqrt{a}, 0, \sqrt{a}$. The signs from left to right are $-, +, -, +$. Hence 0 is a strict local maximum, while $\pm\sqrt{a}$ are strict local minima. Their values are*

$$f_a(0) = 0, \qquad f_a(\pm\sqrt{a}) = a^2 - 2a^2 = -a^2.$$

The sign pattern shows that the two minima are absolute. For every real a, $f_a(x) \to +\infty$ as $|x| \to \infty$, so no absolute maximum exists.

Solution ***Method 2: complete a square.*** *For $a \le 0$, $x^4 - 2ax^2 \ge 0$, with equality only at $x = 0$. To rule out other local extrema without differentiating, put $u = x^2$. For $0 \le u < v$,*

$$(v^2 - 2av) - (u^2 - 2au) = (v - u)(v + u - 2a) > 0.$$

Thus the expression increases strictly with x^2, giving decrease on the negative half-line and increase on the positive half-line. For $a > 0$,

$$f_a(x) = (x^2 - a)^2 - a^2 \ge -a^2,$$

with equality precisely at $x = \pm\sqrt{a}$. Also, for $0 < |x| < \sqrt{2a}$ we have $f_a(x) = x^2(x^2 - 2a) < 0 = f_a(0)$, proving the strict local maximum at 0. The function $(u - a)^2$ decreases for $0 \le u \le a$ and increases for $u \ge a$. Combining this order with that of $u = x^2$ on each half-line gives the same four monotonicity intervals as in Method 1; there are no further local extrema. The completed square also tends to $+\infty$ as $|x| \to \infty$, excluding an absolute maximum.

The derivative method locates all changes of behavior. Completing the square proves the global lower bound and its equality cases directly. The value $a = 0$ must be checked separately, because three critical points merge there.

4.5.4 Absolute extrema and optimization

For a continuous function on $[a, b]$, the extreme value theorem guarantees both absolute extrema. An extremum must occur at an endpoint or at an interior critical point. When the critical points form a finite list, evaluate f at every candidate and compare the resulting values. If there are infinitely many candidates, the same necessity statement remains true, but it is not a finite algorithm; additional structure is needed.

On an open or unbounded domain, existence must be justified separately. An endpoint limit may give a supremum or infimum that is not attained. A critical point is not automatically an absolute optimum merely because it is the only one found.

Example 4.49 A feasible interval before differentiation Among rectangles of fixed perimeter $P > 0$, find the largest area and the dimensions that attain it.

> ***Strategy.*** The perimeter constraint makes one side depend on the other. Write that relation first; it gives both a one-variable area function and the interval on which the dimensions are positive.

Solution ***Method 1: differentiation.*** *If the sides are x and $P/2 - x$, then $0 < x < P/2$ and*

$$A(x) = x(P/2 - x), \qquad A'(x) = P/2 - 2x.$$

Thus A increases until $x = P/4$ and decreases thereafter. Its absolute maximum is $P^2/16$, attained only by the square with side $P/4$. The limiting area at either excluded endpoint is 0 and cannot improve the maximum.

Solution ***Method 2: a direct bound.*** *Completing the square gives*

$$A(x) = \frac{P^2}{16} - \left(x - \frac{P}{4}\right)^2 \le \frac{P^2}{16}.$$

Equality holds exactly at $x = P/4$.

The bound is shorter here because the objective is quadratic. Differentiation applies to much wider classes of objectives, but still requires the same feasible-domain check.

Example 4.50 An open box and an exact maximum certificate Squares of side x are cut from the four corners of a square sheet of side 12. The remaining sides are folded up to form a box without a lid. Find the largest possible volume and the dimensions that attain it.

> ***Strategy.*** Cutting changes both base dimensions as well as the height. Positive dimensions determine the feasible interval before differentiation. Once a candidate maximum is found, subtract the volume from its candidate value; a repeated factor at the stationary point may turn the comparison into an exact nonnegative identity.

Solution ***Method 1: derivative signs on the feasible interval.*** *The height is* x *and each base side is* $12 - 2x$, *so* $0 < x < 6$. *Therefore*

$$V(x) = x(12 - 2x)^2,$$
$$V'(x) = (12 - 2x)^2 - 4x(12 - 2x) = (12 - 2x)(12 - 6x).$$

The first factor is positive throughout the feasible interval. The second is positive for $0 < x < 2$ *and negative for* $2 < x < 6$. *Hence* V *increases up to* 2 *and decreases thereafter, and its unique absolute maximum is*

$$V(2) = 2 \cdot 8^2 = 128.$$

The optimal box has base 8 *by* 8 *and height* 2. *At either excluded endpoint, the limiting volume is* 0; *neither endpoint represents a box with positive dimensions.*

Solution ***Method 2: factor the gap from the proposed maximum.*** *Expansion and factorization give*

$$128 - V(x) = 128 - 144x + 48x^2 - 4x^3 = 4(x - 2)^2(8 - x).$$

Since $8 - x > 0$ *for* $0 < x < 6$, *this proves* $V(x) \le 128$ *throughout the feasible domain, with equality exactly at* $x = 2$.

The derivative method locates the maximizing cut without knowing its value in advance. The factorization certifies the resulting global bound and its unique equality case in one calculation.

Example 4.51 Why the arithmetic mean minimizes squared error Given real measurements $a_1, \ldots, a_n$, where $n \ge 1$, find the real t minimizing $Q(t) = \sum_{j=1}^{n} (t - a_j)^2$.

Solution ***Method 1: derivative signs.*** *Put* $\bar{a} = n^{-1} \sum_{j=1}^{n} a_j$. *Then* $Q'(t) = 2n(t - \bar{a})$ *is negative below* $\bar{a}$ *and positive above it. Hence* $\bar{a}$ *is the unique absolute minimizer on* $\mathbb{R}$.

Solution ***Method 2: separate the mean.*** *Write* $t - a_j = (t - \bar{a}) + (\bar{a} - a_j)$ *and expand:*

$$Q(t) = n(t - \bar{a})^2 + 2(t - \bar{a}) \sum_{j=1}^{n} (\bar{a} - a_j) + \sum_{j=1}^{n} (\bar{a} - a_j)^2.$$

The middle term vanishes because $\sum_{j=1}^{n} (\bar{a} - a_j) = n\bar{a} - \sum_{j=1}^{n} a_j = 0$. *Thus*

$$Q(t) = \sum_{j=1}^{n} (\bar{a} - a_j)^2 + n(t - \bar{a})^2.$$

The second term is nonnegative and vanishes only at $t = \bar{a}$.

The second identity quantifies the penalty for choosing another center: it is exactly $n(t-\bar{a})^2$. Neither proof requires a compact domain.

4.5.5 Methods and further remarks

A useful order of work is: determine the domain, simplify the derivative, partition the domain at its zeros and singular points, establish signs on each whole interval, and then classify the candidates. Factorization usually reveals signs more reliably than testing many numerical values. A test point determines the sign on an interval only after one knows that the derivative has no zero or sign change there; continuity of the derivative or its Darboux property can supply this justification.

For an inequality, move all terms to one side and ask whether the resulting function can be compared with a convenient reference value by monotonicity. For a local extremum with several vanishing derivatives, seek the first nonzero Taylor term. For a global minimum, also look for an exact nonnegative decomposition: it may prove existence, uniqueness, and an error estimate at once.

The logical order should remain visible: *Fermat identifies candidates; a sign or Taylor test classifies them; comparison over the whole domain establishes an absolute optimum.* Supplementary problems below explore why the sign-neighborhood tests are sufficient conditions rather than necessary descriptions of every local extremum.

Exercise 4.5

1. Find the intervals of strict increase and decrease of (a) x^3-3x^2; (b) $x/(1+x^2)$; (c) x^2e^{-x}; (d) $(\ln x)/\sqrt{x}$ on its natural domain.
2. A differentiable function satisfies $f'(x) = (x+2)(x-1)^2$ on $\mathbb{R}$. Determine its stationary points and classify its local extrema without finding a formula for f.
3. Decide whether each statement is true, giving a proof or counterexample: (a) $f' \geq 0$ implies strict increase; (b) strict increase implies $f' \geq 0$ wherever the derivative exists; (c) $f'(c) = 0$ implies a local extremum. All interval claims concern differentiable functions on an open interval.
4. Find the local and absolute extrema of $f(x) = 1/(1+x^2)$ on $[-1,2]$. State explicitly which endpoints must be checked.
5. For $f(x) = |x-1|$ on $[-1,3]$, identify all critical points and absolute extrema. Explain why Fermat's theorem does not exclude its interior minimum.
6. Show that $f(x) = \sqrt[3]{x}$ has a critical point at 0 but no local extremum there. Give both an order argument and a derivative-sign argument.
7. Use an appropriate test to classify 0 for (a) $x^2/(1+x^2)$; (b) x^6-x^8; (c) $\sin x - x$; (d) $\ln(1+x)-x+x^2/2$; (e) $\cos x - 1 + x^2/2$. State when the second derivative test is inconclusive.
8. Suppose $h'(x) = (x-1)^3(x+1)^2$ on $\mathbb{R}$. Find all stationary points, intervals of monotonicity, and local extrema. Is any local extremum necessarily absolute?
9. Find the absolute extrema of $f(x) = x\sqrt{4-x^2}$ on $[-2,2]$. Check the endpoints even though the derivative formula is valid only in the interior.
10. Find all local and absolute extrema of $\sin x + \cos x$ on $[0,\pi]$.
11. Prove that $x - \ln x \geq 1$ for $x > 0$, and determine the equality case. Explain why a minimum exists despite the noncompact domain.

12. Let $a > 0$. For each function, find its maximum for $x > 0$ and decide whether a minimum is attained: (a) $x^a e^{-x}$; (b) $x^a e^{-x^2}$.
13. (a) A closed circular cylinder has fixed volume $V > 0$. Find the radius and height minimizing its total surface area, including both bases. Justify a global minimum over the positive radii. (b) Repeat the problem for a cylinder with a base but no top, and compare the optimal height-to-radius ratios.
14. Find the points on $y = x^2$ nearest to $(0, 1)$. Minimize the squared distance and explain why this has the same minimizers as distance.
15. Prove that for $x > 0$,
$$\frac{x}{1+x^2} < \arctan x < x.$$
Use monotonicity of suitable differences, and discuss equality at $x = 0$.
16. For each $b \in \mathbb{R}$, determine the number of distinct real solutions of $x^3 - 3x = b$. Include the cases $b = \pm 2$.
17. Suppose f is continuous on $[a, b]$, differentiable on (a, b), and $f' > 0$ except at finitely many points where $f' = 0$. Prove that f is strictly increasing on $[a, b]$.
18. Determine $a, b \in \mathbb{R}$ so that $x^3 + ax^2 + bx$ has stationary points at -1 and 1. Classify them and decide whether either is absolute on $\mathbb{R}$.
19. If f' exists near an interior local minimum c and $f''(c)$ exists, prove that $f''(c) \geq 0$. Is $f'(c) = f''(c) = 0$ sufficient for a local minimum?
20. Construct a polynomial whose only stationary points are $-1, 0, 1$, with a strict local maximum at -1, no extremum at 0, and a strict local minimum at 1. Explain how the desired sign pattern determines a suitable derivative.

Supplement Exercises

S1. Define $f(0) = 0$ and $f(x) = x + x^2 \sin(1/x^2)$ for $x \neq 0$. Prove that $f'(0) = 1$, but that f is not monotone on any neighborhood of 0.

S2. Define $g(0) = 0$ and $g(x) = x^4(2 + \sin(1/x))$ for $x \neq 0$. Show that 0 is a strict local minimum, although g' takes both signs arbitrarily close to 0 on its right. What does this say about the converse of the first derivative test?

S3. Let $F(x) = \int_0^x t^2 \sin^2 t \, \mathrm{d}t$ for $x \in \mathbb{R}$. Prove that F is strictly increasing, even though F' vanishes at infinitely many points.

S4. For weights $w_j > 0$ and real data a_j $(1 \leq j \leq n)$, minimize $\sum_{j=1}^n w_j (t - a_j)^2$ over $\mathbb{R}$. Obtain an exact decomposition around the minimizing value, and explain what fails if weights of both signs are allowed.

S5. For each $a \in \mathbb{R}$, determine the monotonicity and all absolute extrema of $f_a(x) = (1 - a)x - \ln x$ on $(0, +\infty)$. Does any value of a give a finite infimum that is not attained?

S6. Suppose y is continuous on $[a, b]$, twice differentiable on (a, b), and
$$y''(x) + p(x)y'(x) + q(x)y(x) = 0, \qquad q(x) < 0,$$
with $y(a) = y(b) = 0$. Prove that $y \equiv 0$. Deduce uniqueness of solutions with prescribed endpoint values when two solutions satisfy these hypotheses.

> ***Optional Hint.*** If y is positive somewhere, its positive absolute maximum occurs in the interior. Compare the sign of y'' there with the sign forced by the equation.

S7. For each real k, determine the number of real solutions of $e^x = kx$. For $k < 0$, treat negative solutions as well as positive ones.

S8. Find the absolute minimum of x^x on $(0, +\infty)$. Give a proof by differentiating x^x and a proof by minimizing its logarithm. Explain why the two minimization problems are equivalent.

S9. Let c be an interior local minimum of f and let G be strictly increasing on an interval containing the range of f near c. Prove that $G \circ f$ has a local minimum at c, with strictness preserved. State the corresponding conclusion for strictly decreasing G. Give an example showing that no such conclusion holds for arbitrary differentiable G.

S10. Given real data $a_1, \dots, a_n$ with $n \geq 1$, minimize $E(t) = \max_{1 \leq j \leq n} |t - a_j|$ over $\mathbb{R}$. Compare the minimizing value with that for squared error, and explain why differentiating every summand separately would miss the structure of E.

S11. For $a, b > 0$, determine the monotonicity, range, and all absolute extrema of

$$F_{a,b}(x) = \frac{x^a}{(1+x)^b}, \qquad x > 0.$$

Distinguish an attained maximum, a finite supremum that is not attained, and an unbounded range.

S12. Let f, g be continuous and nondecreasing on $[a, b]$, where $a < b$. Prove the integral inequality

$$(b-a) \int_a^b f(x)g(x)\,\mathrm{d}x \geq \left(\int_a^b f(x)\,\mathrm{d}x \right) \left(\int_a^b g(x)\,\mathrm{d}x \right).$$

Show that the inequality is strict if both functions are strictly increasing. What happens if f is nondecreasing and g is nonincreasing?

> ***Optional Hint.*** Choose c so that $f(c)$ equals the integral average of f. Compare the signs of $f(x) - f(c)$ and $g(x) - g(c)$ on the two sides of c.

4.6 Convexity and Curve Sketching

知识点概要

- ❏ *Describe convexity by chords and distinguish it from monotonicity.*
- ❏ *Relate convexity to increasing slopes and supporting tangent lines.*
- ❏ *Identify inflection points without confusing necessary and sufficient conditions.*
- ❏ *Combine limits, derivative signs, and asymptotes into a justified sketch.*
- ❏ *Use convexity to prove inequalities and estimate integrals.*

Two increasing functions can have quite different shapes. For $x > 0$, the slopes of x^2 increase as x increases, whereas those of $\ln x$ decrease. The first derivative records whether the graph rises or falls; the change of the first derivative records how the graph bends. We first express bending without derivatives, so that the definition also applies to a graph with a corner.

4.6.1 Chords and the definition of convexity

For $x < y$ and $0 \leq t \leq 1$, the point $z = (1-t)x + ty$ divides $[x, y]$ in the same ratio in which $(1-t)f(x) + tf(y)$ divides the heights of the two endpoints of the chord. Comparing this chord height with $f(z)$ gives the required definition.

Definition 4.3

A function f on an interval I is convex (凸函数（下凸）) *if*

$$f((1-t)x+ty) \le (1-t)f(x)+tf(y) \qquad (x,y \in I,\ 0 \le t \le 1).$$

It is strictly convex *if the inequality is strict whenever $x \ne y$ and $0 < t < 1$. The function is* concave (凹函数（上凸）)*, or strictly concave, if $-f$ is convex, or strictly convex, respectively.*

♣

Thus a convex graph lies below each of its chords; a concave graph lies above each chord. Here "convex" permits equality, and strict convexity is a separate property. The graph's direction of travel is a different question: x^2 is convex on $\mathbb{R}$, decreasing to the left of 0, and increasing to its right.

Example 4.52 Reading the definition directly Show that x^2 is strictly convex on $\mathbb{R}$, that $|x|$ is convex but not strictly convex, and that every affine function $ax+b$ is both convex and concave.

Solution *For the square function, subtract the graph height from the chord height:*

$$(1-t)x^2+ty^2-((1-t)x+ty)^2 = t(1-t)(x-y)^2.$$

This is nonnegative, and is positive for distinct x, y and $0 < t < 1$. The triangle inequality gives

$$|(1-t)x+ty| \le (1-t)\,|x|+t\,|y|\,,$$

proving convexity of $|x|$ without differentiating at its corner. Equality holds when $x, y > 0$, so strict convexity fails. For $ax+b$, the two sides of the convexity inequality are identical; both inequalities therefore hold.

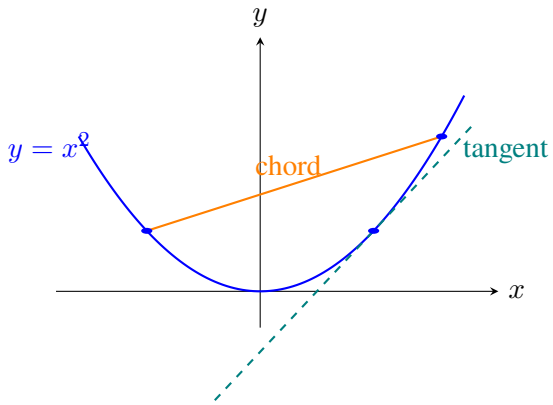


Figure 4.5: A convex differentiable graph lies below its chords and above its tangent lines. These are comparisons of heights, not claims that the function is increasing.

4.6.2 Increasing slopes and supporting tangents

Theorem 4.13 (Three descriptions of convexity)

Let f be differentiable on an open interval I. The following statements are equivalent:

1. *f is convex on I.*
2. *f' is nondecreasing on I.*
3. *At every $c \in I$, the tangent line lies below the whole graph:*

$$f(x) \ge f(c)+f'(c)(x-c) \qquad (x \in I).$$

Moreover, f is strictly convex if and only if f' is strictly increasing.

♡

> ***Proof Strategy.*** A chord comparison contains a comparison of secant slopes after division by a positive horizontal distance. Letting an endpoint of a secant approach a fixed point produces a derivative. In the other direction, the mean value theorem expresses a secant slope as an intermediate derivative. These two operations explain why the geometric definition is equivalent to an order property of f'.

Proof If f is convex and $x < z < y$, rearranging the defining inequality gives

$$\frac{f(z)-f(x)}{z-x} \le \frac{f(y)-f(x)}{y-x} \le \frac{f(y)-f(z)}{y-z}.$$

Let $z \to x^+$ in the first inequality and $z \to y^-$ in the second. Then

$$f'(x) \le \frac{f(y)-f(x)}{y-x} \le f'(y),$$

so f' is nondecreasing.

Conversely, suppose f' is nondecreasing. For $x \neq c$, the mean value theorem gives $f(x) - f(c) = f'(\xi)(x-c)$ with ξ between c and x. When $x > c$, we have $f'(\xi) \ge f'(c)$. When $x < c$, we have $f'(\xi) \le f'(c)$ and multiplication by $x - c < 0$ reverses the inequality. Both cases give the tangent inequality.

Finally, assume the tangent inequality and let $z = (1-t)x + ty$, where $0 < t < 1$. Apply it at $c = z$ to x and y, multiply the two inequalities by $1 - t$ and t, and add. The linear terms cancel, leaving $f(z) \le (1-t)f(x) + tf(y)$.

If f' is strictly increasing, the tangent inequalities used above are strict for $x \neq c$, giving strict convexity. If f is strictly convex, we already know that f' is nondecreasing. Equality $f'(x) = f'(y)$ for $x < y$ would make f' constant on $[x, y]$, and the mean value theorem would make f affine there. That would contradict strict convexity. □

A tangent line satisfying this global inequality is called a *supporting tangent*(支撑切线). The secant-slope limit argument also shows that a convex function has a supporting tangent at any point where it is differentiable, even if it has corners elsewhere. For a concave function, f' is nonincreasing and every tangent lies above the graph; replace f by $-f$ in the theorem. If the interval has included endpoints, the derivative criterion extends to them when f is continuous there. Take limits of the interior chord inequalities to obtain the endpoint inequalities. Continuity is relevant: the function $f(0) = -1$ and $f(x) = 0$ for $0 < x \le 1$ has $f' = 0$ on $(0, 1)$ but is not convex on $[0, 1]$.

Corollary 4.3 (Second derivative criterion)

If f is twice differentiable on an open interval, then f is convex there if and only if $f'' \ge 0$ there. It is concave if and only if $f'' \le 0$. The stronger conditions $f'' > 0$ and $f'' < 0$ imply strict convexity and strict concavity, respectively, but are not necessary. ♡

Proof Apply Theorem 4.10 to f' and then use the preceding theorem. □

Example 4.53 A tangent inequality in two forms Prove that $\ln x \le x - 1$ for $x > 0$, with equality only at $x = 1$.

> ***Strategy.*** The right side is the tangent line to $\ln x$ at 1, which suggests a global tangent comparison. Alternatively, move all terms to one side; the proposed equality point then becomes a candidate for the minimum of the difference.

Solution ***Method 1: use concavity.*** *Since $(\ln x)'' = -1/x^2 < 0$, the logarithm is strictly concave. Its tangent at 1 is $y = x - 1$, so the tangent inequality is strict away from 1.*

Solution ***Method 2: minimize the difference.*** *Set $h(x) = x - 1 - \ln x$. Then $h'(x) = 1 - 1/x$ is negative for $0 < x < 1$ and positive for $x > 1$. Thus h has the unique absolute minimum $h(1) = 0$.*

The first method explains the geometric origin of the bound and supplies a tangent bound at any other point in the same way. The second needs only monotonicity and is useful when no familiar convex or concave function is visible.

Example 4.54 Choose the tangent where the linear terms cancel For $a, b, c > 0$, prove

$$\frac{1}{a}+\frac{1}{b}+\frac{1}{c} \geq \frac{9}{a+b+c},$$

and determine equality.

> ***Strategy.*** The right side depends only on the sum of the three inputs. Their deviations from the arithmetic mean therefore sum to zero. A tangent taken at that mean supplies three affine lower bounds whose linear terms cancel on addition. Alternatively, multiply by the positive sum and look for pairwise differences squared.

Solution ***Method 1: sum supporting tangent inequalities.*** *The function $h(u) = 1/u$ satisfies $h''(u) = 2/u^3 > 0$ on $(0, +\infty)$. Set $r = (a+b+c)/3 > 0$. The supporting tangent at r gives*

$$\frac{1}{u} \geq \frac{1}{r} - \frac{u-r}{r^2},$$

with equality only at $u = r$ because h is strictly convex. Apply this at a, b, c and add:

$$\frac{1}{a}+\frac{1}{b}+\frac{1}{c} \geq \frac{3}{r} - \frac{a+b+c-3r}{r^2} = \frac{3}{r} = \frac{9}{a+b+c}.$$

Equality holds exactly when $a = b = c = r$.

Solution ***Method 2: an exact sum of squares.*** *Expanding and grouping reciprocal pairs gives*

$$\begin{aligned}
&(a+b+c)\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right) - 9 \\
&\quad = \left(\frac{a}{b}+\frac{b}{a}-2\right) + \left(\frac{a}{c}+\frac{c}{a}-2\right) + \left(\frac{b}{c}+\frac{c}{b}-2\right) \\
&\quad = \frac{(a-b)^2}{ab} + \frac{(a-c)^2}{ac} + \frac{(b-c)^2}{bc} \geq 0.
\end{aligned}$$

Division by $a+b+c > 0$ proves the inequality. All three nonnegative terms vanish exactly when $a = b = c$.

The tangent method extends to other convex functions whenever the deviations from a chosen mean cancel. The algebraic identity uses no differentiation and gives the exact gap for these three reciprocals.

Example 4.55 Strict convexity with a zero second derivative The function $f(x) = x^4$ is strictly convex on $\mathbb{R}$ although $f''(0) = 0$. Indeed, $f'(x) = 4x^3$ is strictly increasing, so Theorem 4.13 applies. The value $f''(0) = 0$ neither creates an affine segment nor changes the sign of f''. It is therefore neither an obstruction to strict convexity nor, by itself, evidence of an inflection point.

4.6.3 Inflection points and changes of bending

Definition 4.4

Let f be continuous at an interior point c of its domain. We call $(c, f(c))$ an inflection point (拐点) *if, for some $\delta > 0$, f is strictly convex on one of $(c-\delta, c)$ and $(c, c+\delta)$ and strictly concave on the other.* ♣

This convention requires a genuine change of bending at a point on the graph and excludes affine pieces. The derivative at c need not exist. In particular, an inflection point may have a vertical tangent. If f'' exists on the two punctured intervals and has strictly opposite signs there, continuity at c proves that c is an inflection abscissa. This is a sufficient test, not the definition.

Proposition 4.5

If c is an inflection abscissa and $f''(c)$ exists, then $f''(c) = 0$.

♠

Proof Existence of $f''(c)$ means that f' exists near c and is continuous at c. On one side of c, f' is increasing; on the other, it is decreasing, or the reverse. Continuity joins these comparisons at c, making f' have a local maximum or minimum there. Fermat's theorem applied to f' gives $(f')'(c) = 0$. □

Consequently, zeros of f'' and points where f'' fails to exist form a candidate list, subject to membership in the domain and continuity. Every candidate still needs a two-sided bending check. A first-derivative sign change concerns an extremum of f; a second-derivative sign change concerns an extremum of its slope. These are different tests.

Example 4.56 Four different outcomes at zero Decide whether 0 is an inflection abscissa for x^3, x^4, $\sqrt[3]{x}$, and $1/x$.

Solution *For x^3, the second derivative $6x$ changes from negative to positive, so $(0,0)$ is an inflection point. For x^4, the second derivative $12x^2$ is positive on both sides, so there is no inflection point at 0 despite $f''(0) = 0$. For $\sqrt[3]{x}$, when $x \neq 0$,*

$$f'(x) = \frac{1}{3x^{2/3}}, \qquad f''(x) = -\frac{2}{9x^{5/3}}.$$

The graph is strictly convex to the left and strictly concave to the right. It is continuous at 0, so $(0,0)$ is an inflection point, with a vertical tangent. Finally, $(1/x)'' = 2/x^3$ changes sign across 0, but 0 is outside the domain. There is no graph point there and hence no inflection point.

4.6.4 Asymptotes and complete sketches

Derivative signs control finite intervals. Limits at excluded boundary points and at infinity describe the ends of the graph, and an asymptote gives a simple line against which to compare them.

Definition 4.5

The line $x = a$ is a vertical asymptote (铅直渐近线) *if at least one of the one-sided limits of $f(x)$ at a is $+\infty$ or $-\infty$. The line $y = L$ is a* horizontal asymptote (水平渐近线) *at $+\infty$ if $f(x) \to L$ as $x \to +\infty$. The line $y = mx + b$, with $m \neq 0$, is an* oblique asymptote (斜渐近线) *at $+\infty$ if*

$$f(x) - (mx + b) \longrightarrow 0 \qquad (x \to +\infty).$$

The corresponding definitions at $-\infty$ use that end separately.

♣

Thus an asymptote describes a limiting difference, not a barrier that the graph cannot cross. If an oblique asymptote exists, division by x and subtraction give

$$m = \lim_{x\to+\infty} \frac{f(x)}{x}, \qquad b = \lim_{x\to+\infty} (f(x) - mx).$$

Conversely, finite limits of these forms with $m \neq 0$ prove that the line is an oblique asymptote. The first limit alone is insufficient: $x + \sqrt{x}$ has ratio to x tending to 1, but its difference from $x + b$ tends to $+\infty$ for every constant b.

Example 4.57 Find a line before differentiating Find the asymptotes of $f(x) = (x^2 + 1)/(x - 1)$ and determine on which side of its oblique asymptote the graph lies.

Solution ***Method 1: divide the polynomials.*** *The identity*

$$f(x) = x + 1 + \frac{2}{x - 1}$$

shows that $y = x + 1$ is an oblique asymptote at both ends. The graph is below it for $x < 1$ and above it for $x > 1$. As $x \to 1^-$ the function tends to $-\infty$, and as $x \to 1^+$ it tends to $+\infty$; thus $x = 1$ is a vertical asymptote. There are no horizontal asymptotes.

Solution ***Method 2: compute slope and intercept limits.*** *At either infinite end,*

$$m = \lim \frac{x^2+1}{x(x-1)} = 1, \qquad b = \lim \left(\frac{x^2+1}{x-1} - x \right) = \lim \frac{x+1}{x-1} = 1.$$

Hence the oblique asymptote is $y = x + 1$. Subtracting this line gives $2/(x-1)$, from which the side and vertical-limit conclusions follow as before.

Polynomial division provides both the line and the exact error in one step. The two-limit method applies more generally when no division identity is available.

A complete sketch is a record of compatible facts. Begin with the domain and any symmetry; then locate intercepts and boundary limits. Use f' for monotonicity and extrema, and f'' for bending and inflection points. Place the resulting points and asymptotes before drawing the arcs. A graph drawn by a device can suggest features, but it cannot establish that a small extremum or a remote branch is absent.

Example 4.58 An increasing curve with three inflection points Sketch $f(x) = x^3/(1+x^2)$, including all extrema, inflection points, and asymptotes.

> ***Strategy.*** The denominator is always positive, so there is no domain break. Subtracting x identifies the behavior at infinity before any sign chart is made. The first derivative will have a squared factor, so its zero may leave monotonicity unchanged; the second derivative will decide the changes of bending separately.

Solution *The domain is $\mathbb{R}$, the function is odd, and its only intercept is $(0,0)$. Division gives*

$$f(x) = x - \frac{x}{1+x^2},$$

so $y = x$ is an oblique asymptote at both ends. The graph lies above the line when $x < 0$ and below it when $x > 0$, crossing it at 0. Apply the quotient rule and collect the numerator before making a sign chart:

$$f'(x) = \frac{3x^2(1+x^2) - 2x^4}{(1+x^2)^2} = \frac{x^2(x^2+3)}{(1+x^2)^2}.$$

Differentiating once more and cancelling a positive denominator factor gives

$$f''(x) = \frac{(4x^3+6x)(1+x^2) - 4x(x^4+3x^2)}{(1+x^2)^3} = \frac{6x - 2x^3}{(1+x^2)^3} = \frac{2x(3-x^2)}{(1+x^2)^3}.$$

The first derivative is nonnegative and vanishes only at 0. Thus it cannot vanish on a nondegenerate interval, and Proposition 4.4 proves strict increase on $\mathbb{R}$. In particular, there are no local extrema. Its range is $\mathbb{R}$, so it has no absolute extrema either. The second-derivative signs are

x	$(-\infty, -\sqrt{3})$	$(-\sqrt{3}, 0)$	$(0, \sqrt{3})$	$(\sqrt{3}, +\infty)$
$f''(x)$	$+$	$-$	$+$	$-$
shape	*convex*	*concave*	*convex*	*concave*

Therefore the three inflection points are

$$\left(-\sqrt{3}, -\frac{3\sqrt{3}}{4}\right), \quad (0,0), \quad \left(\sqrt{3}, \frac{3\sqrt{3}}{4}\right).$$

The middle one has a horizontal tangent, but is not an extremum.

Example 4.59 A bell-shaped graph Sketch $f(x) = e^{-x^2}$ and locate the points at which its bending changes.

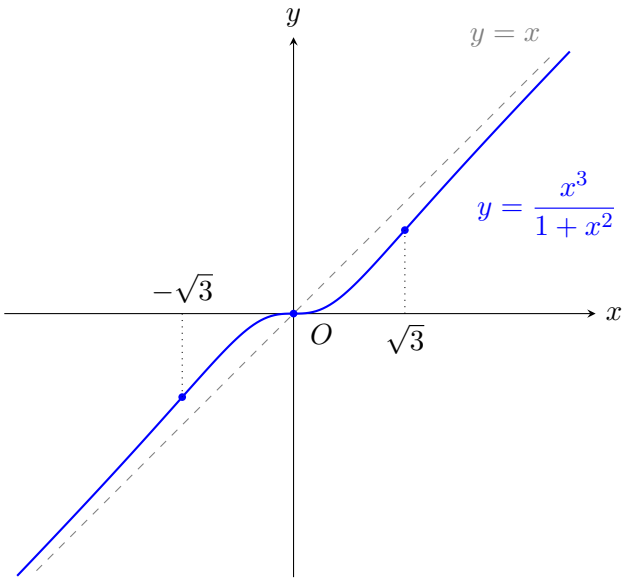


Figure 4.6: The graph is strictly increasing despite a horizontal tangent. Its three changes of bending and its crossing of the asymptote follow from separate calculations.

Solution *The function is positive and even, with* $f(0) = 1$ *and* $f(x) \to 0$ *as* $x \to \pm\infty$. *Thus* $y = 0$ *is a horizontal asymptote at both ends and the graph never meets it. Since*

$$f'(x) = -2xe^{-x^2}, \qquad f''(x) = -2e^{-x^2} + 4x^2e^{-x^2} = (4x^2 - 2)e^{-x^2},$$

the graph increases on $(-\infty, 0)$ *and decreases on* $(0, +\infty)$. *Its unique absolute maximum is* 1 *at* 0. *Its infimum is* 0, *which is not attained, so there is no absolute minimum. The second derivative is negative for* $|x| < 1/\sqrt{2}$ *and positive for* $|x| > 1/\sqrt{2}$. *Hence the graph is strictly concave in the central interval and strictly convex on the two outer intervals, with inflection points*

$$\left(-\frac{1}{\sqrt{2}}, e^{-1/2}\right), \qquad \left(\frac{1}{\sqrt{2}}, e^{-1/2}\right).$$

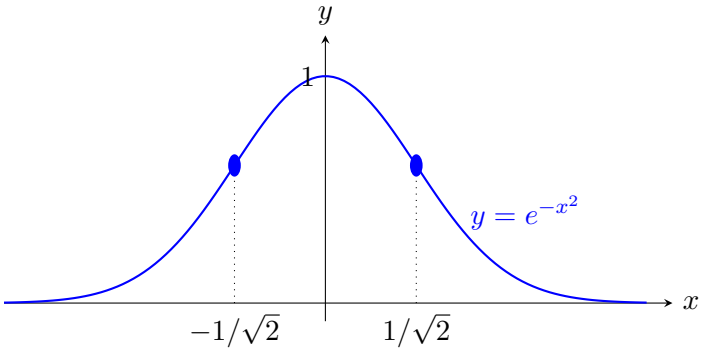


Figure 4.7: Monotonicity changes at the maximum; bending changes at two other points. The horizontal axis is approached at both ends.

Example 4.60 Two different behaviors at the same missing point Sketch $f(x) = xe^{1/x}$ on its domain. Determine its extrema, bending, range, and all asymptotes, including the side from which the graph approaches an oblique asymptote.

> ***Strategy.*** The domain splits at 0, so compute its two one-sided limits separately. At infinity, $1/x$ is small:

> Taylor's formula for e^u can identify the linear part and control the remaining difference. In the derivatives, isolate the positive exponential factor before deciding signs.

Solution *The domain is* $(-\infty,0)\cup(0,+\infty)$. *The function has the sign of* x *and no intercepts. At* 0^-, *both* x *and* $e^{1/x}$ *tend to* 0, *so* $f(x)\to 0^-$. *At* 0^+, *put* $t=1/x$; *exponential domination gives*

$$f(x)=\frac{e^t}{t}\longrightarrow+\infty \qquad (t\to+\infty).$$

Thus $x=0$ *is a vertical asymptote because of the right-hand limit, even though the left-hand limit is finite.*

At either infinite end, Taylor's formula gives

$$e^{1/x}=1+\frac{1}{x}+\frac{1}{2x^2}+o(x^{-2}),\qquad f(x)=x+1+\frac{1}{2x}+o(x^{-1}).$$

Hence $f(x)-(x+1)\to 0$, *and* $y=x+1$ *is an oblique asymptote at both ends. The strict inequality* $e^u>1+u$ *for* $u\neq 0$, *proved in Example 4.47, determines the side of approach: multiply by* $x>0$ *to obtain* $f(x)>x+1$, *and by* $x<0$ *to obtain* $f(x)<x+1$. *These inequalities hold on the whole respective branches. There are no horizontal asymptotes, since* $f(x)\to-\infty$ *at the left end and* $+\infty$ *at the right end.*

Product and chain rules give

$$f'(x)=e^{1/x}+xe^{1/x}\left(-\frac{1}{x^2}\right)=e^{1/x}\left(1-\frac{1}{x}\right),$$

$$f''(x)=e^{1/x}\left[-\frac{1}{x^2}\left(1-\frac{1}{x}\right)+\frac{1}{x^2}\right]=\frac{e^{1/x}}{x^3}.$$

Since the exponential factor is positive, the sign information is

x	$(-\infty,0)$	$(0,1)$	$(1,+\infty)$
$f'(x)$	$+$	$-$	$+$
$f''(x)$	$-$	$+$	$+$

Therefore $(1,e)$ *is a strict local minimum and the absolute minimum of the positive branch. The negative branch increases from* $-\infty$ *to its unattained supremum* 0. *The full range is*

$$(-\infty,0)\cup[e,+\infty),$$

so the function has neither an absolute maximum nor an absolute minimum on its full domain. It is strictly concave for $x<0$ *and strictly convex for* $x>0$. *Although the sign of* f'' *changes across* 0, *that point is missing from the graph; there is no inflection point.*

4.6.5 Convexity as an estimation tool

Example 4.61 Midpoint and trapezoid bounds Let f be continuous and convex on $[a,b]$, where $a<b$. Show that

$$f\left(\frac{a+b}{2}\right)\le\frac{1}{b-a}\int_a^b f(x)\,\mathrm{d}x\le\frac{f(a)+f(b)}{2}.$$

Apply the result to $f(x)=1/x$ on $[1,2]$.

> ***Strategy.*** The integral average can be bounded by integrating pointwise height comparisons. A single chord supplies the upper bound. For the lower bound without differentiability, pair points reflected about the midpoint: their midpoint is fixed even while the points move.

Solution ***Method 1: chords and reflected pairs.*** *Convexity gives*

$$f(x)\le\frac{b-x}{b-a}f(a)+\frac{x-a}{b-a}f(b).$$

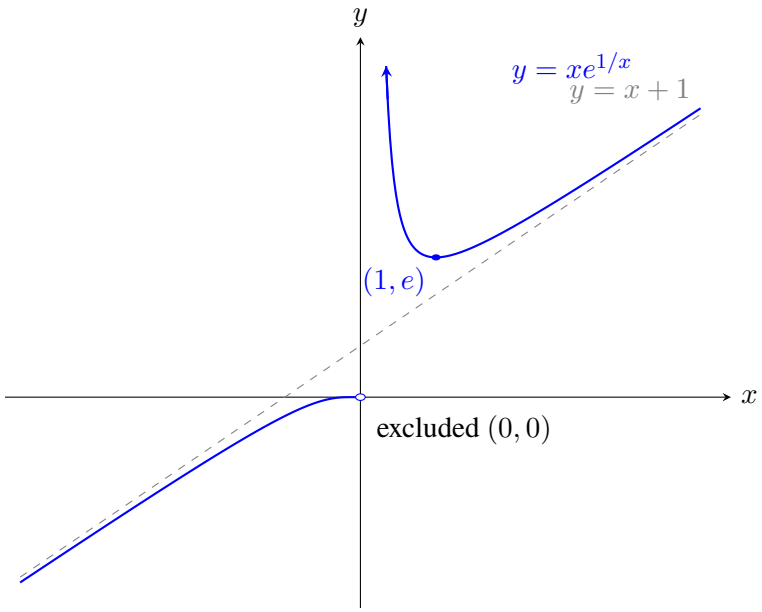


Figure 4.8: The negative branch approaches a missing point at the origin, while the positive branch tends to infinity along the vertical asymptote. The branches lie on opposite sides of the same oblique asymptote.

The two weights have the same integral:

$$\int_a^b \frac{b-x}{b-a}\,\mathrm{d}x = \frac{b-a}{2} = \int_a^b \frac{x-a}{b-a}\,\mathrm{d}x.$$

Integrating the chord inequality and dividing by $b-a>0$ gives the upper bound. Also, x and $a+b-x$ have midpoint $(a+b)/2$, so

$$2f\left(\frac{a+b}{2}\right) \le f(x) + f(a+b-x).$$

In the second integral, the substitution $u = a+b-x$ reverses the endpoints:

$$\int_a^b f(a+b-x)\,\mathrm{d}x = -\int_b^a f(u)\,\mathrm{d}u = \int_a^b f(u)\,\mathrm{d}u.$$

Thus integration of the reflected-pair inequality gives $2(b-a)f((a+b)/2) \le 2\int_a^b f(x)\,\mathrm{d}x$, which is the lower bound. For $f(x) = 1/x$, whose second derivative is $2/x^3 > 0$, these bounds give

$$\frac{2}{3} \le \ln 2 \le \frac{3}{4}.$$

In fact both inequalities are strict, since the corresponding chord and midpoint inequalities are strict away from the exceptional endpoints or midpoint.

Solution ***Method 2: integrate a supporting tangent.*** *If f is also differentiable at $c = (a+b)/2$, its supporting tangent gives*

$$f(x) \ge f(c) + f'(c)(x-c).$$

Integration cancels the linear term because

$$\int_a^b (x-c)\,\mathrm{d}x = \frac{(b-c)^2 - (a-c)^2}{2} = 0,$$

where $b-c = -(a-c)$. This proves the same lower bound. The chord argument still gives the upper bound.

The first method applies even when f has a corner at the midpoint. The second is shorter when a tangent is available and explains why centering the interval is useful. These estimates give bounds, not a formula for the exact integral.

4.6.6 Methods and common pitfalls

For convexity, identify the comparison required by the problem. Use the chord definition for weighted values, a supporting tangent for an affine bound, and the sign of f'' when differentiating twice is simpler than comparing arbitrary pairs. Keep non-strict and strict conclusions separate: $f'' \geq 0$ proves convexity, whereas strict convexity requires ruling out affine segments, or proving that f' is strictly increasing.

For an inflection point, first require a continuous point of the graph and then verify opposite bending on its two sides. Neither $f''(c) = 0$ nor a sign change across a point outside the domain is enough. For a full sketch, organize the domain, symmetry, intercepts, endpoint limits, asymptotes, first-derivative signs, and second-derivative signs before drawing. Check that all the resulting features can hold together: a strictly increasing curve cannot have a local maximum in its interior, and a claimed asymptote must satisfy the difference limit at the stated end.

Exercise 4.6

1. Use the chord definition to prove that $f(x) = x^2 + 3x - 2$ is strictly convex and that $g(x) = -x^2$ is strictly concave on $\mathbb{R}$.
2. Determine all intervals of convexity and concavity for (a) $x^3 - 3x$; (b) $\ln(1 + x^2)$; (c) $x^4 - 6x^2$; (d) xe^x. Locate their inflection points.
3. For each assertion, decide whether it is true. Prove it or give a counterexample: (a) an increasing function is convex; (b) a convex differentiable function has nonnegative derivative; (c) a strictly convex twice differentiable function has strictly positive second derivative everywhere.
4. Prove that $e^x \geq 1+x$ for all real x, with equality only at 0, using (a) a tangent inequality; (b) the derivative of the difference.
5. Show that $\sqrt{x} \leq (x + 1)/2$ for $x > 0$ by concavity. Give a second proof using a square, and determine equality.
6. Locate and classify the possible inflection points of (a) x^5; (b) x^6; (c) $x^{2/3}$. State explicitly what happens at 0 in each case.
7. Find every vertical and horizontal asymptote of $f(x) = x^2/(x^2 - 1)$, including the appropriate one-sided limits at finite boundary points.
8. Find the oblique asymptotes separately at $+\infty$ and $-\infty$ for

   $$\text{(a)} \quad f(x) = \sqrt{x^2 + 1}, \qquad \text{(b)} \quad g(x) = \sqrt{x^2 + 2x + 2}.$$

   Is each graph above or below its asymptotes? For each function, explain why there is no single oblique line valid at both ends.
9. Sketch $f(x) = x + 1/x$. Include its domain, symmetry, extrema, bending, and all asymptotes. Does the sign change of f'' at the excluded point 0 give an inflection point?
10. Sketch $f(x) = x/(1 + x^2)$. Include all extrema and inflection points, and decide whether its horizontal asymptote can be crossed.
11. Sketch $f(x) = \ln x/x$ on $(0, +\infty)$, giving its intercept, extrema, inflection point, and boundary behavior.
12. Sketch $f(x) = xe^{-x}$ on $\mathbb{R}$. Give its asymptotes at the two ends, if any, and distinguish its extremum from its inflection point.
13. Let f be differentiable and convex on an open interval, and let $a < b$ belong to that interval. Prove

    $$f'(a) \leq \frac{f(b) - f(a)}{b - a} \leq f'(b).$$

If f is strictly convex, prove that both inequalities are strict.

14. Prove that a function which is both convex and concave on an interval is affine. Do not assume differentiability.
15. (a) Use midpoint and trapezoid bounds on $[0,1]$ to bound $\int_0^1 e^{x^2}\,dx$. Justify the required convexity and decide whether the bounds are strict. (b) Obtain corresponding bounds for $\int_0^1 \sqrt{1+x}\,dx$, taking account of concavity. Decide strictness and check your bounds against the exact integral.
16. Determine all real a for which $f_a(x) = x^4 + ax^2$ is convex on $\mathbb{R}$. For the remaining parameters, find every inflection point.
17. The graph of $f(x) = x + \sin x$ is increasing. Prove that it has neither a horizontal nor an oblique asymptote at $+\infty$, despite $f(x)/x \to 1$.
18. Suppose f is twice differentiable on an interval and $f'' \geq m > 0$. Show that for any a, b in the interval,
$$f\left(\frac{a+b}{2}\right) \leq \frac{f(a)+f(b)}{2} - \frac{m(b-a)^2}{8}.$$
Identify a function for which equality holds.
19. Let f be convex on $[a,b]$ and let $c \in (a,b)$ be a local minimum. Prove directly from the chord definition that c is an absolute minimum. If f is strictly convex, prove uniqueness.
20. Construct a polynomial of the smallest possible degree which has a horizontal tangent at the inflection point $(0,0)$ and a strict local minimum at $x = 1$. Find all its inflection points and justify the degree claim.

Supplement Exercises

S1. Let f be convex on an interval. Prove the finite weighted inequality
$$f\left(\sum_{j=1}^{n} \lambda_j x_j\right) \leq \sum_{j=1}^{n} \lambda_j f(x_j), \qquad \lambda_j \geq 0, \quad \sum_{j=1}^{n} \lambda_j = 1.$$
Use induction on n. If f is strictly convex and all weights are positive, determine when equality holds.

S2. Use strict concavity of $\ln x$ to prove the weighted arithmetic–geometric mean inequality
$$a^{1-t}b^t \leq (1-t)a + tb \qquad (a, b > 0,\ 0 < t < 1).$$
Determine equality, and compare this proof with the square proof when $t = 1/2$.

S3. Let f be continuous on $[a,b]$ and twice differentiable in (a,b), with $f'' > 0$ there. Prove that f has at most two distinct zeros in $[a,b]$ by (a) strict convexity; (b) two applications of Rolle's theorem. Explain why merely assuming $f'' \geq 0$ does not suffice.

S4. A differentiable function f on an open interval satisfies $f(y) > f(x) + f'(x)(y-x)$ whenever $x \neq y$. Prove that f is strictly convex. Conversely, show that every differentiable strictly convex function satisfies this strict tangent inequality.

S5. Let f be twice differentiable on $\mathbb{R}$, with $f(0) = f'(0) = 0$ and $0 < m \leq f''(x) \leq M$ for all x. Prove
$$\frac{m}{2}x^2 \leq f(x) \leq \frac{M}{2}x^2$$
for all real x by (a) convexity and tangent bounds; (b) Taylor's formula. Compare the assumptions needed by the two proofs.

S6. Find all real a for which $f_a(x) = e^x + ax^2$ is convex on $\mathbb{R}$. For every other a, find its unique inflection point and describe the bending on each side.

S7. Let f be continuous and convex on $[a,b]$. For a partition into equal subintervals, let M_n and T_n be the

composite midpoint and trapezoid approximations to $\int_a^b f(x)\,\mathrm{d}x$. Explicitly, with $\Delta = (b-a)/n$ and $x_j = a + j\Delta$, these are

$$M_n = \Delta \sum_{j=1}^{n} f\left(\frac{x_{j-1}+x_j}{2}\right), \qquad T_n = \frac{\Delta}{2}\sum_{j=1}^{n}\big(f(x_{j-1})+f(x_j)\big).$$

Prove $M_n \le \int_a^b f \le T_n$ and, on halving every subinterval, prove $M_n \le M_{2n}$ and $T_{2n} \le T_n$.

S8. Construct a differentiable strictly increasing function on $\mathbb{R}$ with infinitely many inflection points. Verify both claims, including points where its derivative vanishes.

S9. Let f be differentiable and convex on an open interval containing 0, with $f(0) = 0$. Prove that $f(x)/x$ is nondecreasing for $x > 0$ in the interval, first by the chord definition and then by a supporting tangent. What changes if f is strictly convex?

S10. Suppose f is convex on $\mathbb{R}$ and has horizontal asymptotes at both $+\infty$ and $-\infty$. Prove that f is constant, without assuming differentiability. Show by an example that one horizontal asymptote alone does not force constancy.

S11. Suppose $f \in C^2[a,b]$, $f(a) < 0 < f(b)$, $f' > 0$, and $f'' \ge 0$ on $[a,b]$. To approximate the zero, replace the graph near an iterate by its tangent line. The tangent at x meets the horizontal axis at $x - f(x)/f'(x)$. Starting with $x_0 = b$, define the sequence of these tangent intercepts by

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \qquad (n \ge 0).$$

This is *Newton's iteration*(牛顿迭代). (a) Prove that f has a unique zero $r \in (a,b)$, that every iterate is defined, and that $r \le x_{n+1} \le x_n \le b$. Show that $x_{n+1} < x_n$ when $x_n > r$, and prove $x_n \to r$. (b) If $f' \ge m > 0$ and $0 \le f'' \le M$ on $[a,b]$, prove

$$0 \le x_{n+1} - r \le \frac{M}{2m}(x_n - r)^2.$$

(c) For $f(x) = x^2 - 2$ on $[1,2]$, calculate x_1, x_2 and identify the limit.

> ***Optional Hint.*** Evaluate the supporting tangent at the unknown zero to locate its intercept. Once the iterates are trapped, use monotone convergence and then the recurrence to identify the limit. Taylor's formula between x_n and r controls the error in part (b).

S12. Let f be convex on an interval I, and let g be convex and nondecreasing on an interval J containing $f(I)$. Prove directly from the chord definition that $g \circ f$ is convex on I. Give a counterexample if the monotonicity assumption on g is omitted. If f is strictly convex and g is strictly increasing, prove that $g \circ f$ is strictly convex, even when g is not strictly convex.

Chapter 4 Problem-Solving Toolkit

The common theme of this chapter is to recover information about a function from its derivatives. The information may be an intermediate point, a comparison of increments, an approximation with a controlled error, or a description of the whole graph. The target determines both the auxiliary function and the hypotheses that must be checked.

A decision route

Step 1. Fix the domain and direction. Record interval endpoints, excluded points, parameter restrictions, and the side from which a limit is taken. A derivative sign controls an interval; it does not compare values across a gap in the domain.

Step 2. Identify the required conclusion. Is the problem asking for existence, uniqueness, a limit, a numerical error bound, or an attained optimum? For an existence problem, write the desired equation as a derivative equal to zero whenever possible.

Step 3. Expose the useful structure. Subtract a line to equalize endpoint values, use a primitive to turn a zero integral into equal endpoints, or multiply by an integrating factor to recognize a derivative. In a limit, factor or rationalize before differentiating; take a logarithm only after proving that the base is positive.

Step 4. Choose a theorem and check its hypotheses. Specify the closed interval for a mean value theorem, the nonvanishing denominator derivative for a quotient theorem, and the derivative order needed for a Taylor remainder. These checks belong to each application, including repeated applications.

Step 5. Decide whether the control is local or uniform. A Peano remainder describes a limit near one point. A bound on a higher derivative gives an error estimate throughout a stated interval. An unspecified mean value point remains inside its own interval; its location cannot be strengthened without an argument.

Step 6. Finish the logical claim. Combine an existence argument with a root-count bound to prove uniqueness or an exact count. Compare all candidates and endpoint limits in optimization. State equality cases, attained values, and one-sided conclusions where they matter.

Constructing the right function

Signal	Construction	Essential check
Equal values or several zeros	Apply Rolle to a residual; repeat between consecutive zeros.	Continuity on each closed segment and enough derivatives inside it. Count distinct zeros; see Proposition 4.1.
Prescribed slope or chord error	Subtract the line with that slope, or the endpoint chord.	The resulting endpoint values must agree. An additional prescribed zero may require a quadratic correction.
A combination $f' - \lambda f$	Differentiate $e^{-\lambda x} f(x)$.	Check endpoint values or derivative signs after multiplication; the exponential is positive. See Example 4.10.
Two finite changes	Compare $f(b) - f(a)$ with $g(b) - g(a)$ by Cauchy's theorem.	Both derivatives are evaluated at one common point. For the quotient form, require $g' \neq 0$ throughout the open interval.
A weighted integral	Use primitives of uw and w when $w > 0$, or compare u with known bounds on the interval.	For continuous u, w, a nonnegative weight with positive total mass permits averaging; strict comparisons need positive mass where the inequality is strict.

The same construction may answer different questions. For example, dividing a positive function by x^p turns a relative derivative bound into an ordinary derivative sign; taking logarithms instead compares relative increments directly (Example 4.19). The two methods should be compared by the assumptions and information they use.

Limits, finite expansions, and error bounds

Signal	First move	What completes the argument
A quotient of type $0/0$	Simplify, then consider L'Hôpital's rule.	Check the derivative quotient and its limit. A failed derivative-quotient test gives no conclusion about the original quotient.
A denominator of unbounded magnitude	Check the infinite-denominator version of L'Hôpital's rule.	The denominator derivative must be nonzero on the relevant tail or punctured interval; state the direction of approach.
A positive base tending to 1	Take a logarithm and expose the base minus 1.	Justify the transformed limit, then exponentiate. Example 4.27 shows why simplification can avoid repeated differentiation.
Cancellation through several orders	Choose the expansion order from the denominator and retain all contributing terms.	After division, the remainder must tend to zero at the required scale. Recheck parameter values that cancel a leading coefficient.
A quotient or composition of expansions	Multiply by an unknown finite polynomial, or set the small inner increment equal to u.	After extracting known vanishing factors, the divisor must stay away from zero. The order of u determines which powers contribute. See Example 4.35.
A requested accuracy	Use a Lagrange or integral remainder with a derivative bound.	Carry the bound through every operation. Cancellation of a leading error does not by itself bound the new error; see Example 4.39.

Finite Taylor arithmetic is justified by remainder estimates, so it does not require an infinite power series. In particular, differentiate an exact identity to find coefficients; do not differentiate an unspecified $o(x^n)$ term. For a two-sided limit involving an odd power, check that the sign conclusion agrees on both sides.

From derivative signs to a global conclusion

1. **Partition the actual domain.** Mark excluded points, zeros of the relevant derivative, and points where that derivative fails to exist. A sign chart must cover every intervening interval.
2. **Assign each derivative its role.** Use f' for increase and decrease, and f'' for convexity and concavity where the derivative criteria apply. A stationary point need not be an extremum, and $f''(c) = 0$ need not give an inflection point.
3. **Check boundaries and attainment.** On a compact interval, compare endpoint and interior candidate values. On an open or unbounded domain, also compute the relevant limits and decide whether the best bound is achieved. Parameters may change both the candidates and the sign chart, as in Example 4.48.
4. **Use geometry to organize inequalities.** A convex graph lies above its supporting tangents and below its chords (Figure 4.5). Select the tangent point to make linear terms cancel, and track equality in every comparison.
5. **Verify the proposed asymptote.** Subtract the candidate line and prove that the difference tends to zero on each stated tail. A limiting slope alone does not establish a slant asymptote.

A synthesis: the error in replacing a curve by its chord

When values at two endpoints are known, the simplest approximation between them is the joining line. For $a < b$, this *linear interpolant* (线性插值函数) is

$$\ell(x) = \frac{b-x}{b-a} f(a) + \frac{x-a}{b-a} f(b), \qquad a \le x \le b.$$

It agrees with f at both endpoints and is the unique affine function with those two values. The residual $f - \ell$ therefore has two known zeros, exactly the structure used in Supplement Exercise 4 of Section 4.1. Supplement Exercise 3 of Section 4.4 already asks for its interpolation interpretation. We work out that argument here, add a second proof using convexity, and examine sharpness and a signed application.

Example 4.62 A sharp chord-error bound Let $f \in C^2([a,b])$ and suppose $|f''(t)| \le M$ on $[a,b]$, where $M \ge 0$. Prove that

$$|f(x) - \ell(x)| \le \frac{M}{2}(x-a)(b-x) \le \frac{M(b-a)^2}{8} \qquad (a \le x \le b).$$

Show that the uniform constant $1/8$ cannot be reduced, and apply the estimate to $f(x) = \ln x$ on $[1,2]$, including the sign of the error.

> ***Choosing the comparison.*** Subtracting the chord removes the endpoint values without changing the second derivative. At a fixed interior point, one adjustable multiple of $(t-a)(t-b)$ can remove one more value, creating three zeros for two applications of Rolle. Alternatively, a bound on the second derivative suggests two quadratic comparison functions with second derivatives M and $-M$. Their convexity will control all points of the interval at once.

Solution ***Method 1: three zeros and repeated Rolle.*** *Put $h = f - \ell$, so $h(a) = h(b) = 0$ and $h'' = f''$. The estimate is immediate at a and b. Fix $x \in (a,b)$ and choose*

$$A = \frac{h(x)}{(x-a)(x-b)}, \qquad H(t) = h(t) - A(t-a)(t-b).$$

Then $H(a) = H(x) = H(b) = 0$. Rolle on $[a,x]$ and $[x,b]$ gives $u \in (a,x)$ and $v \in (x,b)$ with $H'(u) = H'(v) = 0$. Rolle on $[u,v]$ gives $\xi \in (u,v)$ with

$$0 = H''(\xi) = f''(\xi) - 2A.$$

Substituting $A = f''(\xi)/2$ yields the exact representation

$$h(x) = \frac{f''(\xi)}{2}(x-a)(x-b).$$

Since $(x-a)(x-b) < 0$, taking absolute values gives $|h(x)| \le M(x-a)(b-x)/2$.

Solution ***Method 2: convex and concave comparisons.*** *Again let $h = f - \ell$, and set $q(x) = M(x-a)(b-x)/2$. Both h and q vanish at the endpoints, and $q'' = -M$. Hence*

$$(h-q)'' = f'' + M \ge 0, \qquad (h+q)'' = f'' - M \le 0.$$

The function $h - q$ is convex and lies below its zero endpoint chord, so $h - q \le 0$. The function $h + q$ is concave and lies above its zero endpoint chord, so $h + q \ge 0$. Thus $-q \le h \le q$, which is the same pointwise estimate.

Solution ***Uniform bound, sharpness, and the logarithm.*** *Completing the square gives*

$$(x-a)(b-x) = \frac{(b-a)^2}{4} - \left(x - \frac{a+b}{2}\right)^2 \le \frac{(b-a)^2}{4}.$$

This proves the uniform bound. For $M > 0$, the quadratic $f(x) = M(x-a)(b-x)/2$ has $\ell = 0$, $|f''| = M$, and equality at the midpoint. Therefore a smaller universal constant cannot work. If $M = 0$, then $f'' = 0$, so f is affine and the error is identically zero.

On $[1,2]$*, the function* $\ln x$ *is concave, and* $|(\ln x)''| = 1/x^2 \le 1$*. Its chord is* $\ell(x) = (x-1)\ln 2$*. Concavity supplies the sign, while either estimate supplies the magnitude:*

$$0 \le \ln x - (x-1)\ln 2 \le \frac{(x-1)(2-x)}{2} \le \frac{1}{8}.$$

The first inequality is strict for $1 < x < 2$*, because the logarithm is strictly concave. The upper bound is a guaranteed certificate; it is not asserted to be the exact maximum error for the logarithm.*

Both proofs use the stated C^2 hypothesis. Repeated Rolle gives an exact representation at a point that may depend on x; the convexity proof gives upper and lower comparisons simultaneously on the interval. The quadratic factor also shows why the pointwise estimate becomes smaller near either endpoint.

Checks before submitting a solution

Do not divide two separate mean value identities as if their intermediate points were the same. Do not infer the behavior of f/g merely because the derivative quotient fails to have a limit. Do not turn a Peano remainder into a numerical certificate, or turn a local derivative test into an attained global optimum without checking the domain. A complete answer states the interval, the hypotheses, the decisive estimate or construction, and precisely the conclusion they support.